

Mounting Directive for Vans



Sprinter – model series 907 (rear/all-wheel drive)

Edition AeJ2025-1a

Mercedes-Benz



Changes to the body/equipment mounting directive BR 907

With the publication of this edition of the Body/Equipment Mounting Directive (ARL) in the Bodybuilder Portal, the previous edition loses its validity. The new edition must be taken into account in its entirety from the point in time of publication. Contractual agreements have priority over this.

Previous (old) edition: AeJ2023-1a

Current (new) edition: AeJ2025-1a

The most important modifications are listed below.

Please note that this list does not contain all of the changes in full. Instead, it serves as an aid for a quick overview of the most important change topics.

Chapter	Title	Changes
1	Introduction	Clarification and addition to the information on the general handling of the Body/Equipment Mounting Directive
2.1	Advice for bodybuilder manufacturers	Adoption of contents from change information 02/2024: Information on the Technical Consultancy for Bodybuilder Manufacturers
2.2.8	WLTP	Clarification of the presentation of the certification type and reference mass, addition of emissions class Euro 6e-bis (code M6B)
3.4	Vehicle modifications	Addition to the information on checking the vehicle systems and driving assistance systems
3.13	Adhesive films on the vehicle exterior	Addition of the PARKTRONIC sensors in the representation of the impermissible areas on the vehicle front for Parking Packages (code JB6/JB7) and Moving-off information Assist (code JF7)
3.14	Vehicle markings with warning signs	Addition of new content
4.2.1	Suspension	Addition of information to the guide on suspension design
4.2.2	Permissible axle loads	Update to the overview of the maximum permissible axle loads, moved from chapter 3.2 to 4.2.2
4.2.3	Approved tire sizes	Conversion of the information to a reference to the online wheel and tire table
4.3.7, 6.2.5	Wheelbase modification	Deletion of the contents and code O03, information on contacting the Technical Consultancy and on the obligation to apply for compatibility testing
6.6.4	Shelving units and load rails in the load compartment	Addition of information on retrofitting load rails to the roof for fastening load securing equipment Information on correct lashing between load rails
7.14.5	Connecting point for driving light assist (code LA2)/rain sensor (code JF1) on vehicles without windshield	Addition of information and representation of correct positioning of the light/rain sensor on the windshield
7.14.6	Multi-purpose camera (MPC) on vehicles without windshield	Adoption of contents from change information 02/2024: Clarification and addition of information for completion by bodybuilder manufacturers

Chapter	Title	Changes
8.9	Driving assistance systems	Addition to the information on checking the vehicle systems and driving assistance systems
8.9.4	Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Side-guard Assist	Adoption of the contents from change information 02/2024, as well as further clarification and addition of the information for completion by bodybuilder manufacturers
8.9.8	Parking packages/Moving-off information Assist	Addition of information on the Moving-off information Assist (code JF7) Clarification of the specifications for the detection ranges of the camera and the sensors
8.9.9	Reversing camera	Adoption of the contents from change information 02/2024, as well as further clarification and addition of the information for completion by bodybuilder manufacturers
8.9.10	Rearview camera for digital inside mirror	Adoption of the contents from change information 02/2024, as well as further clarification and addition of the information for completion by bodybuilder manufacturers
8.19	Commissioning of control units	Addition of information on the new guide for commissioning ("Commissioning guide") Addition to the information on checking the vehicle systems and driving assistance systems
8.20	Tachograph	Addition of code EQ3: "Pre-installation for toll system" Deletion of codes J19 and JD2

Change information	2
1 Introduction	9
1.1 Aim of this Body/Equipment Mounting Directive	11
1.2 Conventions	12
1.3 Vehicle safety	13
1.4 Operational safety	14
1.4.1 Important safety notes	14
1.4.2 Diagnostics connection	15
1.4.3 Qualified specialist workshop	16
1.4.4 Modifications to the vehicle	16
1.4.5 High-voltage on-board electrical system	17
1.5 Information about copyright	17
2 General	18
2.1 Advice for bodybuilder manufacturers	18
2.1.1 Checking compatibility for a Letter of Compatibility	19
2.1.2 Differentiation between Individual and Concept Letters of Compatibility	20
2.1.3 Applying for Letters of Compatibility	20
2.1.4 Documents required when applying for Individual Letters of Compatibility	21
2.1.5 Documents required when applying for Concept Letters of Compatibility	22
2.1.6 Issuing of Letter of Compatibility	23
2.2 Product and vehicle information for bodybuilder manufacturers	24
2.2.1 Bodybuilder Portal	24
2.2.2 Information for bodybuilder manufacturers outside the sales network	25
2.2.3 Workshop Information System (WIS)	26
2.2.4 Diagnosis	26
2.2.5 CERON	27
2.2.6 European type approval (ETA) and EC Certificate of Conformity (CoC)	28
2.2.7 ADVANTAGES product information system	29
2.2.8 Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP)	29
2.3 Product safety and product liability	37
2.3.1 Product safety	37
2.3.2 Product liability	37
2.3.3 Safety-relevant features	38

2.4	Guarantee of traceability	39
2.5	Trademarks	39
2.5.1	The Mercedes-Benz trademark in interaction with external bodybuilder manufacturers	39
2.5.2	The function of the trademark	39
2.5.3	Trademark protection	39
2.5.4	Trademark rights	40
2.5.5	Advice on applying Mercedes-Benz trademarks to vans	40
2.5.6	Use of Mercedes-Benz trademarks	40
2.5.7	Brand separation/identity	41
2.5.8	Trademark communication	42
2.6	Accident prevention	43
2.7	Reprocessing (recycling)	43
2.8	Quality system	45
2.9	Technical Compliance Awareness Package	46

3	Planning of bodies.....	47
3.1	Vehicle and model designation	47
3.2	Model overview.....	49
3.3	Selecting the basic vehicle	53
3.4	Vehicle modifications	57
3.5	Dimensions and weights.....	59
3.6	Vehicle identification data	61
3.6.1	QR code rescue sticker	62
3.7	Vehicle stability.....	64
3.8	Tires	65
3.9	Threaded connections and weld joints.....	66
3.9.1	Threaded connections.....	66
3.9.2	Weld joints.....	68
3.10	Soundproofing	70
3.11	Maintenance and repair	71
3.11.1	Storing the vehicle	72
3.11.2	Maintenance and storage of batteries.....	72
3.11.3	Work before handing over the modified vehicle	73
3.12	Special equipment	74
3.13	Adhesive films on the vehicle exterior	75
3.14	Vehicle markings with warning signs.....	77

4	Technical limit values for planning78	5	Damage prevention99
4.1	Limit values for the basic vehicle78	5.1	Brake hoses, cables, and lines.....100
4.1.1	Steerability78	5.2	Welding work101
4.1.2	Maximum permissible position of the vehicle center of gravity78	5.3	Anticorrosion protection measures.....102
4.1.3	Permissible vehicle dimensions79	5.4	Painting and preservation work104
4.2	Limit values for the suspension80	5.5	Storing and handing over the vehicle105
4.2.1	Suspension80		
4.2.2	Permissible axle loads82	6	Modifications to the basic vehicle.....106
4.2.3	Approved tire sizes.....83	6.1	Suspension106
4.2.4	Turning circle diameter.....83	6.1.1	General information on the suspension.....106
4.2.5	Modifications to the axles83	6.1.2	Springs, dampers, and stabilizers108
4.2.6	Modifications to the steering.....83	6.1.3	Brake system108
4.2.7	Modifications to the brake system84	6.1.4	Air suspension110
4.2.8	Modifications to springs, spring suspension, and dampers84	6.1.5	Wheels and tires111
4.2.9	Wheel alignment84	6.1.6	Spare wheel.....111
4.3	Limit values for the body-in-white85	6.2	Body-in-white and body112
4.3.1	Modifications to the body-in-white.....85	6.2.1	General information on the body-in-white112
4.3.2	Limit values for the vehicle frame85	6.2.2	Attachments on the frame.....117
4.3.3	Lower wheel well on rear axle for closed model designations85	6.2.3	Material for the vehicle frame118
4.3.4	Lower wheel well on rear axle for open model designations86	6.2.4	Overhang extension119
4.3.5	Overhang88	6.2.5	Wheelbase modifications for open and closed model designations122
4.3.6	Attachment to the frame89	6.2.6	Modifications to the cab.....122
4.3.7	Wheelbase modifications.....89	6.2.7	Side wall, windows, doors, and flaps.....123
4.3.8	Roof and roof load90	6.2.8	Fenders and wheel wells127
4.4	Limit values for engine peripherals and drivetrain91	6.2.9	Frame end crossmember127
4.4.1	Fuel system91	6.2.10	Cowl128
4.4.2	Modifications to the engine and drivetrain ..91	6.2.11	Panel van and crewbus roof128
4.4.3	Engine cooling92	6.2.12	Cutting the cab roof and B-pillar roof bow132
4.5	Limit values for the interior.....96	6.2.13	Underfloor paneling.....134
4.5.1	Modifications in the area of the restraint systems96	6.3	Engine peripherals and drivetrain136
4.5.2	Modifications to seats96	6.3.1	Fuel system (diesel)136
4.6	Limit values for electrics and electronics97	6.3.2	SCR exhaust gas aftertreatment system139
4.6.1	Vehicle position lamps and side marker lamps97	6.3.3	Exhaust system.....141
4.6.2	Retrofitting of electrical devices97	6.3.4	Engine cooling147
4.6.3	Mobile communication systems97	6.3.5	Engine air intake147
4.6.4	CAN bus97	6.3.6	Clearance for major assemblies148
4.7	Limit values for additional assemblies.....97	6.3.7	Propeller shafts.....148
4.8	Limit values for attachments97	6.3.8	Engine speed regulation150
4.9	Limit values for the body98		

6.4	Interior.....	151	7.14	Bodies on chassis with cab base (code F28, F50)	227
6.4.1	General information	151	7.14.1	Cab base with cowl (code F50) and cab base with doors (code F28).....	227
6.4.2	Safety equipment.....	152	7.14.2	Mechanical equipment	229
6.4.3	Seats	160	7.14.3	Wiring harness connecting points.....	231
6.4.4	Reducing interior noise.....	162	7.14.4	Automatic driving lights (code LA2)/ rain sensor (code JF1)	232
6.4.5	Ventilation	163	7.14.5	Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1) on vehicles without windshield	233
6.4.6	Attachment points for load compartment trim parts on side wall and roof	164	7.14.6	Multi-purpose camera (MPC) on vehicles without windshield	236
6.5	Additional assemblies	166	7.14.7	Connecting point with omission of driver's and front passenger seat for pre-installation for seat belt warning system (code J1D/J1E) and seat occupancy recognition (code E3S)	237
6.5.1	Retrofitting an air conditioning system.....	166	7.14.8	Driver's and front passenger door connecting point.....	239
6.5.2	Auxiliary heating	167	7.14.9	Headlamp connecting point.....	240
6.5.3	Power take-offs	169	7.14.10	Connecting point for antenna switchover box for third-party antennas (code E4A).....	241
6.5.4	Retrofitting an alternator	176	7.14.11	Pre-installation for bodybuilder manufacturer interface H (code E2A) and bodybuilder manufacturer interface M (code E5M).....	241
6.5.5	Auxiliary brakes (retarders).....	176	7.14.12	Pre-installation for windshield heater (code E2F) with omission of windshield	243
6.6	Attachments	177	7.14.13	Airbag pre-installation with crash certificate for fully integrated vehicles (code SV4).....	244
6.6.1	Air deflectors	177	7.14.14	Pre-installation for a digital signal processor (DSP) box (code E3D)	244
6.6.2	Attachment above cab	177	7.14.15	Pre-installation for rear speakers (code EP7)	245
6.6.3	Roof luggage racks	178	7.14.16	Overhead control panel switch module with code F50.....	245
6.6.4	Shelves and load rails in the load compartment.....	178	7.14.17	E-Call in cab base vehicles	246
6.6.5	Loading crane	188	7.14.18	Fresh-air intake with code F50	247
6.6.6	Liftgate (cargo liftgate)	190	7.14.19	Hood contact switch with code F50	247
6.6.7	Trailer coupling	193	7.14.20	Reduced Keyless-Start detection for bodybuilder manufacturer conversion with code F50.....	248
6.6.8	Underride guard.....	198	7.14.21	Temperature sensor with code F50.....	249
7	Design of bodies.....	200			
7.1	Assembly frame	200			
7.1.1	Material quality, general	201			
7.1.2	Design	202			
7.1.3	Section dimensions and dimensioning	203			
7.1.4	Attachment to the frame	204			
7.1.5	Assembly frame as floor assembly	208			
7.2	Self-supporting bodies	208			
7.3	Modifications to the interior	209			
7.3.1	Retrofitting seats	209			
7.4	Modifications to the panel van.....	212			
7.5	Platform bodies	214			
7.6	Box bodies	214			
7.7	Refrigerated vehicles and temperature- controlled vehicles.....	215			
7.8	Vehicles for transportation of technical gases	216			
7.9	Dumper bodies	216			
7.10	Semitrailer trucks.....	218			
7.11	Recovery vehicles	222			
7.12	Torsionally stiff body types	222			
7.13	Elevating work platforms	223			

7.15	Camper vans.....	250	8.5	Lighting	286
7.15.1	Body-in-white of camper vans.....	251	8.5.1	Headlamps	286
7.15.2	Suspension of camper vans.....	252	8.5.2	Mounting additional lamps	287
7.15.3	Electrics and electronics for camper vans.....	253	8.5.3	Tail lamps.....	287
7.15.4	Connecting point for a digital signal processor (DSP) box (code E3D)	253	8.5.4	Marker lamps.....	290
7.15.5	Additional main fuses for camper vans (codes E1R and E1Y).....	254	8.5.5	Actuation of exterior lighting functions	291
7.15.6	Features for camper vans.....	255	8.5.6	Interior lamps.....	296
7.15.7	Airbags for camper vans.....	256	8.6	Mobile communication systems	299
7.16	Bodies on chassis with lowered roof.....	257	8.6.1	Devices.....	299
7.16.1	Fitting the auxiliary roof frame.....	258	8.6.2	Connection and cable routing of the antenna (radio)	299
7.16.2	Fitting the body to the auxiliary roof frame.....	258	8.7	Electronic ignition lock (EIS).....	300
7.17	Semi-integrated bodies and optional mounting of free-standing bodies	259	8.7.1	General information	300
7.18	Buses	263	8.7.2	Central locking system and post-delivery integration of doors of bodybuilder manufacturer	301
8	Electrics and electronics	265	8.8	Windows and doors.....	305
8.1	General information	265	8.8.1	Window lifters and window hinges	305
8.2	Electromagnetic compatibility (EMC)	266	8.8.2	Electric sliding door to load compartment	305
8.3	Battery.....	267	8.8.3	Windshield wipers	306
8.3.1	On-board electrical system battery	267	8.8.4	Outside mirror.....	306
8.3.2	Auxiliary battery.....	269	8.8.5	Windshield heater and rear window heater	306
8.3.3	Battery maintenance and storage	271	8.9	Driving assistance systems	307
8.3.4	Disconnecting the on-board electrical system voltage.....	271	8.9.1	Electronic Stability Program (ESP®)	308
8.4	Interfaces and electric lines	271	8.9.2	Crosswind Assist (CWA)	309
8.4.1	CAN bus and networking.....	271	8.9.3	Active Brake Assist/Active Distance Assist DISTRONIC	314
8.4.2	Electric lines and electrical fuses.....	272	8.9.4	Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist	318
8.4.3	Cable extension	273	8.9.5	High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist.....	339
8.4.4	Additional electric circuits.....	274	8.9.6	Automatic driving lights/rain sensor.....	340
8.4.5	Operating switches (pre-installation).....	274	8.9.7	Tire pressure loss warning system	342
8.4.6	Retrofitting of electrical devices	275	8.9.8	Parking packages/Moving-off information Assist.....	344
8.4.7	Alternator	276	8.9.9	Reversing camera	347
8.4.8	Current collection	277	8.9.10	Rearview camera for digital inside mirror...	364
8.4.9	Interface overview.....	281	8.10	Parameterizable special module (PSM)	372
8.4.10	Speed signal	283	8.10.1	PSM functions	375
8.4.11	Travel distance signal	283	8.10.2	PSM interfaces	376
8.4.12	Ground bolts.....	284	8.10.3	Bodybuilder manufacturer CAN	376
8.4.13	Deep discharge protection (standby mode).....	285			

8.11	Signal acquisition and actuation module SAM	377
8.12	Pre-installation for retarder.....	377
8.13	Pre-installation for liftgate	379
8.14	Continuous engine operation feature.....	379
8.15	Automatic start/stop	380
8.16	Electrical circuit diagrams	381
8.17	Telematics and Infotainment.....	381
8.17.1	Infotainment systems	382
8.17.2	Antennas	383
8.17.3	Speakers	385
8.18	Connectivity.....	386
8.19	Commissioning of control units	387
8.20	Tachograph	390
9	Calculations.....	391
9.1	Vehicle center of gravity	391
9.1.1	Determination of the center of gravity in the x-direction	391
9.1.2	Determination of the vehicle center of gravity in the z-direction	393
9.2	Location of fifth-wheel coupling.....	397

10	Technical details.....	398
10.1	Bulb ratings of tail lamps	398
10.1.1	Conventional headlamps	398
10.1.2	Pre-installation for LED headlamps with code L93.....	399
10.1.3	Conventional tail lamps	399
10.1.4	Conventional tail lamps on open model designations with code L90	400
10.1.5	LED tail lamps on open model designations with code L91	401
10.1.6	Additional lighting functions	402
10.2	Trailer coupling hole patterns	403
10.2.1	Installation dimensions, version 1	404
10.2.2	Installation dimensions, version 2.....	404
10.2.3	Installation dimensions, version 3.....	404

Index	405
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List of abbreviations.....	410
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This Body/Equipment Mounting Directive provides bodybuilder manufacturers (legal entities) and converters (natural entities), hereinafter referred to collectively as "bodybuilder manufacturers" (ABH), with important technical information that must be observed in the planning and production of a safe and roadworthy body. The attachments, bodies, installations, or modifications that are necessary for this purpose will subsequently be referred to as "body mounting work".

In the Mercedes-Benz Vans Body/Equipment Mounting Directives, different vehicle model designations and vehicle variants ex works are combined under the generic term "basic vehicle". These are listed in 3.1 Vehicle and model designation (→ page 47).

! NOTE

This Body/Equipment Mounting Directive is based on the development of the Sprinter model series 907 with rear-wheel or all-wheel drive (combustion engines) produced in Europe.

This Body/Equipment Mounting Directive and other applicable documents, any editions in foreign languages, Body/Equipment Mounting Directives for vehicles produced outside of Europe, as well as Supplemental Directives for vehicles with an electric drive, are published on the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



i With the bodybuilder manufacturer information bulletin from 35/2022, it was communicated that, as of model year 2024, selected model designations will no longer be available ex factory. For this reason, these model designations are no longer included in the present Body/Equipment Mounting Directive, See 3.2 Model overview (→ Page 49). If required, you can find information and specifications in previous Body/Equipment Mounting Directives in which the model designations are still included. These are still available in the Body/Equipment Mounting Directives Archive in the Bodybuilder Portal.

i If necessary, you can also obtain information on Body/Equipment Mounting Directives from the Technical Consultancy at the Sales department of Mercedes-Benz AG (→ page 18).

! NOTE

Due to the large number of bodybuilder manufacturers and body types, Mercedes-Benz AG cannot take into account the consequences of possible modifications to the vehicle – e.g. regarding performance, stability, load distribution, center of gravity, and handling characteristics – that may result from body mounting work. For this reason, Mercedes-Benz AG can accept no liability for accidents or injuries sustained as a result of such modifications to its vehicles. The bodybuilder manufacturer must observe all relevant country-specific laws, specifications, directives and registration requirements. They are responsible for ensuring that all of the body work that they perform complies with the information and specifications in these body/equipment mounting directives. Following any body work, the overall vehicle must remain free of faults, must be operationally safe and safe for use on the road and may not present any danger to persons or property.

If this obligation is violated in any way, the bodybuilder manufacturer shall assume full product liability.

This Body/Equipment Mounting Directive is aimed at professional bodybuilder manufacturers. As a result, this Body/Equipment Mounting Directive assumes that the bodybuilder manufacturer has suitable background knowledge.

Moreover, the bodybuilder manufacturer shall observe and adhere to the operator's manual that is valid for the vehicle.

-  You can download Operator's Manuals from the country-specific Mercedes-Benz AG websites. If necessary, contact your Mercedes-Benz service for this.

Please note that certain types of work (e.g. welding work on load-bearing parts) must only be carried out by appropriately qualified personnel. This is to prevent risk of injury and ensure that the degree of quality required for the body mounting work is attained.

NOTE

As a bodybuilder manufacturer, you must always bear in mind that only the body mounting work described in this Body/Equipment Mounting Directive is permissible. Body mounting work not described here or deviations from the specifications in this Body/Equipment Mounting Directive are generally not permitted.

If body mounting work not described here or deviations from the specifications in this Body/Equipment Mounting Directive are nevertheless deemed necessary, contact the Technical Consultancy in any case before beginning planning or execution (→ page 18). In addition, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

To supplement the Body/Equipment Mounting Directive, the bodybuilder manufacturer guide "Design Quality for Vans" has been produced and made freely accessible in the Bodybuilder Portal together with the Body/Equipment Mounting Directives for Vans. These guides are intended to aid with assessing and achieving the required design quality for body mounting work on all Mercedes-Benz Van model series. They supplement the Body/Equipment Mounting Directives for vans and contain both positive and negative examples of design quality for body mounting work. They are intended for trained professionals from bodybuilder manufacturers that carry out work on Mercedes-Benz vans and they also assume that the professionals in question have an appropriate level of technical knowledge.

NOTE

The design quality guidelines apply on a mandatory basis to single-invoice transactions (Mercedes-Benz ERG), commissioned body mounting work where Mercedes-Benz AG is the general contractor and within the "Van Solution" business unit.

In the case of so-called dual-invoice transactions (ZRG), Mercedes-Benz AG strongly recommends the implementation of this guide by the bodybuilder manufacturer.

You can download the design quality guidelines in the Bodybuilder Portal as follows:

-  Select "Vans" on the home screen (dashboard),
-  Click on the "Engineering and Information" button, and then the vehicle image of the required model series,
-  Click on ⊕.
-  Under "Technical Data", click on the "Quality of Workmanship" button and select the required language.

! NOTE

Important revisions to Body/Equipment Mounting Directives, such as updates in the period between two full updates/revisions of the Body/Equipment Mounting Directive, will in future be published as "Revisions" in the Bodybuilder Portal. Upon their publication in the Bodybuilder Portal, these "Revisions to the Body/Equipment Mounting Directive" become an integral part of the current Body/Equipment Mounting Directive or supersede the previous version of the Body/Equipment Mounting Directive and must be complied with.

Other relevant guidelines that supplement the Body/Equipment Mounting Directives can be found in the Bodybuilder Portal under "Engineering and information" at the same location as the body/equipment mounting directive.

The following supplements are available for model series 907:

- Supplemental MBAC (Mercedes-Benz Advanced Control) directive of the communication interface for camper vans.
- Supplemental Directive for eSprinter – Model Series 907 (electric drive).

1.1 Aim of this Body/Equipment Mounting Directive

This Body/Equipment Mounting Directive is divided into 10 interlinked chapters to help you find the information you require more quickly:

- 1 Introduction (→ page 9)
- 2 General (→ page 18)
- 3 Planning of bodies (→ page 47)
- 4 Technical limit values for planning (→ page 78)
- 5 Damage prevention (→ page 99)
- 6 Modifications to the basic vehicle (→ page 106)
- 7 Design of bodies (→ page 200)
- 8 Electrics and electronics (→ page 265)
- 9 Calculations (→ page 391)
- 10 Technical details (→ page 398)

Annex:

Index (→ page 405)

For further information See 2.2 Product and vehicle information for bodybuilder manufacturers (→ Page 24).

The index in PDF format is linked to help you find the information you require quickly.

The limit values selected in chapter 4 Technical limit values for planning (→ page 78) must be adhered to at all times and must form the basis of the planning.

Chapters 6 "Modifications to the basic vehicle", 7 "Design of bodies", and 8 "Electrics and electronics" are the main source of technical information contained in this Body/Equipment Mounting Directive.

i On account of the ongoing technical further development of Mercedes-Benz Van products, bodybuilder manufacturers are notified about new topics or content/updates at points between the regular publication dates within the scope of change information regarding the Body/Equipment Mounting Directive.

i As is the case in general for the Body/Equipment Mounting Directives, the change information for the Body/Equipment Mounting Directive can also be retrieved from the Mercedes-Benz Bodybuilder Portal at:

<https://bb-portal.mercedes-benz-vans.com>



If you are using a printed version of the "Body/Equipment Mounting Directive for the Sprinter – BR 907", the following should be noted: Important revisions to Body/Equipment Mounting Directives, such as updates in the period between two full updates/revisions of the Body/Equipment Mounting Directive, will in future be published as "Revisions" in the Bodybuilder Portal.

Upon its publication in the Bodybuilder Portal, the change information for the Body/Equipment Mounting Directive becomes part of the current Body/Equipment Mounting Directive or supersedes the previous version of the Body/Equipment Mounting Directive, and must be complied with.

Illustrations and schematic drawings are examples only and serve to explain the texts and tables.

! NOTE

All graphics are for illustrative purposes only and do not depict all of the technical details faithfully.

i You can obtain further information from any Mercedes-Benz Service Partner.

1.2 Conventions

The following conventions are used in this Body/Equipment Mounting Directive:

! DANGER

Danger notices draw attention to hazards that will endanger the health or life of yourself or others.

 Action

! WARNING

Warning notices draw attention to hazards that may endanger the health or life of yourself or others.

 Action

! NOTE

This note draws your attention to possible damage to the vehicle and/or other objects.

 Action

ENVIRONMENTAL INFORMATION

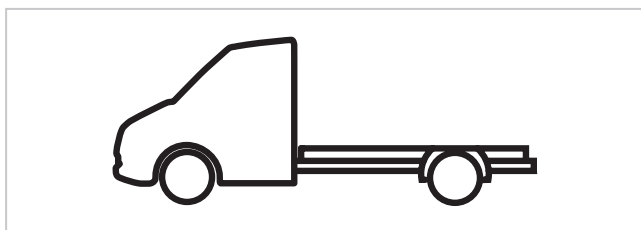
Environmental information gives you information on how to protect the environment.

 Action

i Useful notes or further information/sources of information that could be helpful to you.

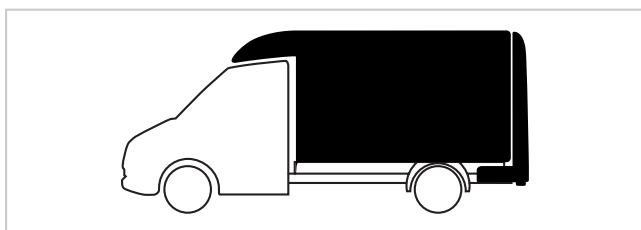
→ This symbol indicates where you can find further information about a topic.

 Action



Basic vehicle

Under this symbol, you will find information concerning the delivered basic vehicle (chassis, panel van and crewbus).



Body

Under this symbol, you will find information concerning the modification or mounting/attachment of the body by the bodybuilder manufacturer.

1.3 Vehicle safety

⚠ WARNING

Before starting to install non-MB bodies or major assemblies, please read the relevant chapters of these Body/Equipment Mounting Directives, the instructions and information from the equipment supplier, and the detailed Operator's Manual for the basic vehicle. You could otherwise fail to recognize dangers that could put yourself or others at risk.

Notes on vehicle safety

It is recommended to only use parts, major assemblies, conversion parts, or accessory parts that are suitable for the vehicle model and that have been tested by Mercedes-Benz.

If parts, major assemblies, conversion parts, or accessory parts that have not been recommended are used, ensure that the safety of the vehicle is verified without delay.

! NOTE

It is essential to ensure that you comply with national registration regulations, since body mounting work performed on the vehicle may change the vehicle type for registration purposes and may invalidate the operating permit. This applies in particular to:

- modifications that change the vehicle type as approved in the general operating permit,
- modifications that could endanger road users, or
- modifications that adversely affect exhaust gas emissions or noise levels.

Vehicle modifications by the bodybuilder manufacturer

Before starting body mounting work, the bodybuilder manufacturer must check that:

- The vehicle is suitable for the planned body,
- The vehicle model and equipment remain suitable for the operating conditions even after the body has been installed.

Vehicles delivered ex works comply with EU and EC specifications and policies/directives, UNECE regulations, and country-specific regulations. This does not apply to vehicles without CoC documents, with incomplete CoC documents, with country-specific registration documents or – depending on the vehicle equipment – in some cases also to vehicles for non-European countries.

NOTE

Following the implementation of any modifications by the bodybuilder manufacturer, vehicles must continue to satisfy EU and EC regulations and directives, UNECE regulations, and/or the national laws and regulations applicable to the location in which the vehicle is to be used.

The bodybuilder manufacturer must inform the officially recognized approval official or inspector of any modifications to the basic vehicle. Acceptance and inspection organizations will decide on compliance with legislation and regulations after any modifications have been made to the basic vehicle, and therefore on the eligibility for registration of the complete vehicle.

The bodybuilder manufacturer must ensure that the vehicle meets the registration requirements with its modifications.

NOTE

Observe all national laws, directives, and registration regulations.

Acceptance inspections conducted by official test facilities or official approvals do not inevitably guarantee compatibility with all of the functions and systems in the basic vehicle.

If you have any questions with regard to ensuring compatibility, contact the Technical Consultancy (→ page 18).

1.4 Operational safety

1.4.1 Important safety notes

WARNING

If you fail to have the prescribed service/maintenance work and any necessary repairs carried out, malfunctions or system failures could occur.

There is risk of an accident and danger to life and limb!

Always have the prescribed service/maintenance work and any necessary repairs carried out at a qualified specialist workshop.

WARNING

If you switch off the ignition while driving, safety-relevant functions will be restricted or no longer available. This can affect functions such as the power steering and brake force boosting. It will then take considerably more effort to steer and brake.

There is risk of an accident and danger to life and limb!

Do not switch off the ignition while driving.

⚠ WARNING

If you drive over obstacles at high speed or the vehicle bottoms out when driving off-road, severe impacts can occur on the underbody, tires, or wheels. This may cause both visible and invisible damage to your vehicle. You could lose control of the vehicle, or damaged components could fail.

There is risk of an accident and danger to life and limb!

Therefore, drive slowly over obstacles and avoid bottoming out the vehicle when driving off road.

If driving safety is impaired or you suspect there is damage while driving, stop safely without delay, notify a qualified specialist workshop, and have your vehicle inspected and repaired there.

1.4.2 Diagnostics connection

The diagnostics connection is used to connect diagnostic devices at a qualified specialist workshop.

⚠ WARNING

When you connect devices to a diagnostics connection in the vehicle, the functions of various vehicle systems could be impaired. This can compromise the operational safety of the vehicle.

There is risk of an accident and danger to life and limb!

Only connect devices to the diagnostics connection in the vehicle that have been approved by Mercedes-Benz for your vehicle.

⚠ WARNING

Loose devices or hanging cables from devices connected to a diagnostics connection can obstruct the space for the pedals. During sudden driving or braking maneuvers, the devices or cables could fall between the pedals. This could impede the functioning of the pedals.

There is risk of an accident and danger to life and limb!

For this reason, it is essential to ensure that no devices or cables that could move have been fastened in the driver footwell while driving.

⚠ WARNING

Objects in the driver footwell can restrict pedal travel or block a fully depressed pedal. This compromises the operational and road safety of the vehicle.

There is risk of an accident and danger to life and limb!

Stow all objects in the vehicle securely so that they cannot fall into the driver footwell. Always install floor mats securely and as specified to ensure that there is always sufficient clearance for the pedals. Do not use loose floor mats and do not place multiple floor mats on top of each other.

! NOTE

If devices are used at the diagnostics connection while the engine is off, the on-board electrical system battery can discharge.

1.4.3 Qualified specialist workshop

A qualified specialist workshop is one that possesses the necessary expertise, tools, and qualifications for the Sprinter. These enable the required work on the vehicle to be carried out properly. This applies in particular to safety-relevant work.

Pay attention to the further information in the service booklet.

It is recommended to employ the services of a specialist workshop that is qualified for the Sprinter.

Always have the following work on the vehicle carried out at a qualified specialist workshop:

- Safety-relevant work
- Service and maintenance work
- Repair work
- Modifications, installations and conversions
- Work on electrical and electronic component parts

You can get up-to-date information about your vehicle's service at any time from a qualified specialist workshop, e.g. at a Mercedes-Benz Service Partner.

1.4.4 Modifications to the vehicle

WARNING

Improper work or modifications carried out on the vehicle, e.g. the routing of cables behind trim panels, could cause the safety systems in your vehicle to stop functioning correctly. Because of this, the safety systems may not protect you or others as they are supposed to. You could lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Always have all work on the vehicle – in particular, safety-relevant work and work on safety-relevant systems and service operations – carried out at a specialist workshop that is qualified for the Sprinter. It is recommended to employ the services of a Mercedes-Benz Service Partner for this purpose.

WARNING

Improper work on the drive system or modifications to the vehicle could cause systems in your vehicle to stop functioning correctly. The driver may lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Always have work on the drive system and modifications to the vehicle, e.g. installations or conversions, carried out at a specialist workshop that is qualified for the Sprinter.

⚠ WARNING

Modifications to electronic component parts, their software or wiring can impair their functioning and/or the functioning of other networked components. Safety-relevant systems in particular may also be affected. Because of this, they may no longer function properly and/or compromise the operational safety of the vehicle.

There is risk of an accident and danger to life and limb!

Never carry out any modifications to the wiring and electronic component parts or their software. Have all work on electrical and electronic devices carried out at a qualified specialist workshop.

1.5 Information about copyright

This Body/Equipment Mounting Directive and all the text, illustrations and data it contains are protected by copyright.

If you have any queries, these can be sent in English or German by email to:

bb-portal-vans.support@mercedes-benz.com

**1.4.5 High-voltage on-board electrical system**

By nature of the design, a high-voltage on-board electrical system is not present on the Sprinter model designations in this Body/Equipment Mounting Directive.

- ① Information and specifications regarding the high-voltage on-board electrical system on the eSprinter with an electric drive can be found in the identically named chapter of the Supplemental Directive for the New eSprinter – Model Series 907 (rear wheel drive).

2.1 Advice for bodybuilder manufacturers

Technical Consultancy

The team at the "Vans Bodybuilder Manufacturer Center" is available to provide technical consultation regarding the compatibility of bodies with the basic vehicle and also to issue letters of compatibility via email in English or German:

abh-technik-vans@mercedes-benz.com



An overview of all persons to contact can be obtained by logging into the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ After logging in to the Bodybuilder Portal, click the "Contact persons" button on the home screen.
- ▶ and expand the required category by clicking on the arrow symbol (>). For example:
 - International contact persons for vans
 - International contact persons for camper vans, vans (...)
- ▶ Choose the contact person with contact details based on the given job description.

Technical advice regarding PSM

Further to the information regarding the parameterizable special module in 8.10 Parameterizable special module (PSM) (→ page 372), and in addition to your Mercedes-Benz contact person on-site, the PSM Support team is available to answer any questions regarding PSM programming.

- ▶ Select the relevant contact person(s) as described above.

Additionally, German bodybuilder manufacturers can use the chargeable "Van Parameterization Support (Management & Administration)" option in the Bodybuilder Portal by selecting "Dashboard – Vans".

Please email your queries regarding the PSM in English or German to:

psm_van@mercedes-benz.com



2.1.1 Checking compatibility for a Letter of Compatibility

Mercedes-Benz AG does not issue any body/equipment approvals or instructions for non-MB bodies. It does not issue any work instructions or other work specifications to the bodybuilder manufacturers for body mounting work on Mercedes-Benz basic vehicles.

The term "body mounting work" is used as an all-inclusive term for any type of attachments, bodies, installations, or modifications that are used by the bodybuilder manufacturer for the planning and production of a roadworthy and reliable body and its joining with the basic vehicle.

In addition to the important information and technical specifications regarding handling of the basic vehicle that are made available to bodybuilder manufacturers along with the Body/Equipment Mounting Directive, Mercedes-Benz AG only conducts inspection work for determining the compatibility of the Mercedes-Benz basic vehicle with the body of the bodybuilder manufacturer through a voluntary process.

A Letter of Compatibility (LOC) consists of a technical check of the compatibility of body mounting work to be carried out by the bodybuilder manufacturer on Mercedes-Benz AG vehicles (basic vehicle) as well as a check of the interfaces to the basic vehicle. Additionally, when any modifications are made to the vehicle, the basic design reliability for the designated vehicle model designation must be inspected.

! NOTE

The bodybuilder manufacturer must apply for a compatibility inspection if body mounting work or modifications are carried out on the basic vehicle that are not or are only partially listed in the Mercedes-Benz AG Body/Equipment Mounting Directive for the basic vehicle or that deviate from the specifications in the Body/Equipment Mounting Directive.

In addition, specifications on the mandatory application for a compatibility check must be observed if stipulated in the Body/Equipment Mounting Directive.

If this inspection work determines that the basic vehicle is compatible with the body of the bodybuilder manufacturer, Mercedes-Benz AG shall issue the bodybuilder manufacturer a Letter of Compatibility.

The scope of the inspection work does not include any inspection of the body's design, the function of the vehicle with body, or the use of the vehicle with body.

The Letter of Compatibility does not release the bodybuilder manufacturer from its obligation to conduct its own calculations, tests and overall vehicle testing, so as to ensure the operational safety, road safety and driving characteristics of the overall vehicle it has manufactured.

The bodybuilder manufacturer shall ensure that the design, production, and assembly of the body is state-of-the-art in its execution, and complies with national laws and directives, and complies with the Body/Equipment Mounting Directive of Mercedes-Benz AG valid for the basic vehicle – insofar as the Letter of Compatibility does not expressly permit any deviations hereto.

Even after the Letter of Compatibility has been issued, the bodybuilder manufacturer is still responsible for ensuring that the overall vehicle meets approval and registration requirements. With its Letter of Compatibility, Mercedes-Benz AG does not confirm that the vehicle with the body is eligible for approval and/or registration.

For information on applying for the inspection work, See 2.1.3 Applying for Letters of Compatibility (→ Page 20).

There is no legal entitlement to the issuance of a Letter of Compatibility, See 2.1.6 Issuing of Letter of Compatibility (→ Page 23).

! NOTE

Observe all national laws, directives, and registration regulations.

2.1.2 Differentiation between Individual and Concept Letters of Compatibility

Individual Letter of Compatibility

An Individual Letter of Compatibility refers to a specifically identified vehicle as per the Vehicle Identification Number (VIN/FIN). Individual Letters of Compatibility (also known as VIN LOCs) encompass standardized changes:

- Change to the permissible trailer load or tongue weight
- Change to the permissible gross mass
- Change to electrics through installation of electrical or electronic devices or change to the maximum speed

! NOTE

A technical inspection of the compatibility of body mounting work with the basic vehicle is not conducted here.

A Concept Letter of Compatibility is also required in the event that the body deviates from the Body/Equipment Mounting Directive.

Concept Letter of Compatibility

The Concept Letter of Compatibility refers to a vehicle model or a vehicle model designation and therefore to small series production, and as such is not limited to an individual vehicle. An application for an individual vehicle will also result in a Concept Letter of Compatibility if it does not relate to a standardized modification (see Individual Letter of Compatibility).

A Concept Letter of Compatibility consists of a technical check of the compatibility of body mounting work to be carried out by the bodybuilder manufacturer on Mercedes-Benz AG vehicles (basic vehicle), as well as a check of the interfaces to the basic vehicle. Additionally, when any modifications are made to the vehicle, the basic design reliability for the designated vehicle model designation must be inspected.

2.1.3 Applying for Letters of Compatibility

Applying for Individual Letters of Compatibility

To apply, log into the Bodybuilder Portal and click on the "VIN Letters of Compatibility" button.

Applying for Concept Letters of Compatibility

To apply, log into the Bodybuilder Portal and click on the "Concept Letters of Compatibility" button.

- ① If you have any questions with regard to applying for Letters of Compatibility, contact the Technical Consultancy (→ page 18).
- ① More comprehensive information about the letters and how to apply for them is available in the Bodybuilder Portal and in bodybuilder manufacturer information bulletin edition 16/2021.
- ① Should you have any queries, contact the Hotline in the Bodybuilder Portal:
<https://bb-portal.mercedes-benz-vans.com>



2.1.4 Documents required when applying for Individual Letters of Compatibility

For the purpose of assessment as part of checks undertaken for the Letter of Compatibility, the following documents are to be submitted in English or German by the bodybuilder manufacturer before starting any work on the vehicle. When requested by Mercedes-Benz AG, further documents about the intended body scope shall also be submitted.

Change to the permissible trailer load or tongue weight

The compatibility assessment is based on the VIN(s) provided.

For a change to the permissible trailer load or tongue weight, no additional documents other than the VIN are required.

Change to the permissible gross mass

A change to the permissible gross mass is only possible for a few applications, such as emergency vehicles and municipal vehicles or camper vans.

- The planned body mounting work must be represented in a body layout.
- A change to the vehicle's gross vehicle mass requires the weight balance of a real loading model, which indicates the gross vehicle mass and the resulting axle loads.
- Do not exceed the maximum permissible axle loads when increasing load capacity.

Along with the application to change the permissible gross mass (load capacity increase or reduction), the body layout and weight balance must also be submitted.

Both documents must be given a single identifier (not identical to the Vehicle Identification Number of the basic vehicle) under which they are stored in the Bodybuilder Portal database and which can become a part of the Letter of Compatibility in suitable form (e.g. drawing number). The documents must contain the company name of the bodybuilder manufacturer and the designation of the body.

Change to maximum design speed

No documents other than the VIN are required for a change to the maximum design speed.

2.1.5 Documents required when applying for Concept Letters of Compatibility

For the purpose of assessment as part of checks undertaken for the Letter of Compatibility, and depending on the extent of the intended body mounting work (e.g. equipment affecting the electrics/electronics, mechanical modifications), the following documents and drawings must be submitted by the bodybuilder manufacturer in English or German before work is started on the vehicle. For this, you will be provided with a technical concept that has to be filled out. Documents that cannot be incorporated into this are to be included as a separate attachment.

In coordination with the Technical Consultancy (→ page 18) and/or at the request of Mercedes-Benz AG, further documents about the intended body scope may also need to be submitted:

- General information about the bodybuilder manufacturer (contact persons, contact details, email, telephone number)
- Information about the basic vehicle (vehicle model designation, vehicle identification number, etc.)
- Description of the planned body:
 - Function description of the new system
 - All deviations from the corresponding Mercedes-Benz Body/Equipment Mounting Directive
 - Influence on the basic vehicle
 - How the body is attached to the vehicle
 - Usage profile, usage markets
 - List of removed and/or modified Mercedes-Benz parts (module/item number/major assembly model designation, code)
 - Weight balance of the vehicle with body in ready-to-drive state, including axle loads
 - Overall center of gravity position for complete vehicle with body, See 9.1 Vehicle center of gravity (→ Page 391)
- Design, wiring diagram:
 - Bill of materials (item, material, weight, etc.)
 - Installation instructions
 - 3D design data incl. installation location, etc.
- Interface with the basic vehicle:
 - Mechanical interface with the basic vehicle (dimensions, tolerances, welded and adhesively bonded connections, fasteners, etc.)
 - Electrical interface with the basic vehicle and wiring diagram (electrical connection, pin assignment, actuation of electrical components, etc.)
 - Safety/certification relevance (DS/DZ): Emissions specifications (FOG/VOC), information on inflammability, as well as all other certification-relevant scopes
- Tests and certifications:
 - Test reports, description of test sequence and test set-up
 - Proof of homologation
 - CE marking and/or other comparable marking depending on country-specific regulations, seat tensile strength test, etc.
- Body-specific Operator's Manual: Descriptions, function instructions and Operator's Manuals, and specifications for the body and – if made necessary by the effects of the body itself – for the basic vehicle. These instructions – in addition to the Mercedes-Benz Operator's Manual for the basic vehicle – are provided to the customer in the national languages of the target market upon vehicle delivery.

Submitting the required documentation in full will make queries on our part unnecessary and will speed up the processing time. If you have any questions, please contact the Technical Consultancy team (→ page 18).

2.1.6 Issuing of Letter of Compatibility

There is no legal entitlement to the issuance of a Letter of Compatibility. Notably, it will not be issued if the compatibility check reveals the presence of potential safety-related, emissions-related or compliance-related objections with regard to the compatibility of the body and basic vehicle.

The issuance or rejection of a Letter of Compatibility may be due to various causes:

1. A Letter of Compatibility can be rejected for economic reasons if the costs incurred by Mercedes-Benz AG (the contractor) are not covered because the volume of vehicles with body is expected to be low, production of the basic vehicle will soon be discontinued, or if excessive expenses are associated with the inspections required for acquiring the Letter of Compatibility and the outcome is uncertain. The economic reasons for rejection can be waived where the client agrees to bear the inspection costs. As a result, the Letter of Compatibility can be implemented if necessary within the framework of available capacities. The allocated time for the processing then depends on the available resources. Technical aspects are not affected by any cost acceptance.
2. A Letter of Compatibility can be rejected for technical reasons.
3. A Letter of Compatibility can be rejected for strategic reasons if the modification is unable to meet the brand values of Mercedes-Benz AG.
4. A Letter of Compatibility can be rejected if the applicant has failed to provide sufficient or verifiable technical information. Where insufficient information is repeatedly provided in response to subsequent demands, Mercedes-Benz AG reserves the right to reject or dismiss the request.
5. A Letter of Compatibility can be rejected even when the body meets the requirements of the Mercedes-Benz Body/Equipment Mounting Directives for Vans.

Also note the following:

- A previously issued Letter of Compatibility can be revoked if technical further developments on the basic vehicle and knowledge resulting from these developments or other sources results in objections with regard to the compatibility of the body with the basic vehicle.
- The Letter of Compatibility only applies to the specifically requested body and/or the specifically requested vehicle. The bodybuilder manufacturer must apply for a new compatibility inspection if further development of its body results in deviation from the verified technical status.
- Retroactive issuance of a Letter of Compatibility for vehicles that have already been completed or delivered can be refused.
- The Letter of Compatibility does not release the bodybuilder manufacturer from its obligation to conduct its own calculations, tests and overall vehicle testing so as to ensure the operational safety, road safety and driving characteristics of the overall vehicle it has manufactured.
- Even after the Letter of Compatibility has been issued, the bodybuilder manufacturer is still responsible for ensuring that the overall vehicle meets approval and registration requirements.

With an application for a Letter of Compatibility, the "Conditions for the performance of inspection work to determine the compatibility of the basic vehicle with the body" must be observed and accepted. These can be obtained from the Bodybuilder Portal or from the Technical Consultancy (→ page 18). If Mercedes-Benz AG agrees to carry out the work, the contract for carrying out the inspection work is thereby concluded. The "Conditions for the performance of inspection work to determine the compatibility of the basic vehicle with the body" are an integral part of this contract.

2.2 Product and vehicle information for bodybuilder manufacturers

As a bodybuilder manufacturer, you not only have the option to contact the Technical Consultancy directly (→ page 18), but also have additional options for obtaining detailed information regarding our products and systems.

2.2.1 Bodybuilder Portal

General information

The Bodybuilder Portal is the central communications platform between Mercedes-Benz and you, our partners in the bodybuilder manufacturer industry. The Bodybuilder Portal provides information and access to body-related topics for all model series of Mercedes-Benz vans. You can access the portal at:

<https://bb-portal.mercedes-benz-vans.com>



Letters of Compatibility can be applied for through the Bodybuilder Portal (→ page 20). In addition to your order data in connection with single-invoice transactions, the order data for dual-invoice transactions must now be retrieved in the order data system of the Bodybuilder Portal for registered bodybuilder manufacturers. Information on this can be found in bodybuilder manufacturer information bulletins via the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ❶ Older bodybuilder manufacturer information bulletins can be found in the Bodybuilder Portal in the "Model series archive" section, under "Engineering and information (Vans)".

Technology

In the "**Engineering and information**" section of the portal, you can find the relevant technical data sheets, 2D drawings (offer drawings), wiring diagrams and engine diagrams, as well as the Body/Equipment Mounting Directives and bodybuilder manufacturer information bulletins. Along with the current model series, you will also find directives for vehicle models that are no longer produced (under "Archive"). Ensure that work is only ever carried out based on the current Body/Equipment Mounting Directives.

You can also add your company and your products in the portal through a certification in the "**My Partner Status**" area, so that you can present them to us and our sales staff and, in a further step, to customers too. Compliance with the Body/Equipment Mounting Directive or a Letter of Compatibility is necessary.

This way, you will be optimally prepared to provide your customers with favorably priced, quick, and comprehensive advice. You will also be optimally prepared for feasibility studies, to plan and determine the final vehicle concept, to create offers, and to prepare designs.

Design data

After logging in to the Bodybuilder Portal, you can download 3D design data in STEP-AP214 and JT formats:

- ▶ Select "Vans" on the home screen (dashboard),
- ▶ Click on the "Engineering and Information" tile, then select the vehicle image of the required model series,
- ▶ Select "⊕ Retrieve Technology".
- ▶ Under "Technical Data", click on the "3D Data" button and select the required data.

- ❗ A change to the way in which the 3D design data is provided is planned for 2025. A description with regard to this will be published via a bodybuilder manufacturer information bulletin. Make sure to read this information. Should it result in any questions, send an email to the following address.

Whenever new 3D design data becomes available, registered bodybuilder manufacturers will be notified via bodybuilder manufacturer information bulletins, which contain a detailed description of the data download procedure. These can be found in the Bodybuilder Portal after selecting the required model series in the "Further Information" area in the "Bodybuilder Manufacturer Information Bulletin" button.

To access the 3D design data, you must have an applicable Confidentiality Agreement (GHV) specifically for access to the data of the required vehicle model series.

If such an agreement is not present and you therefore do not yet have access to the 3D data, the "Engineering and Information" button will be grayed out and will invert when clicked on. The reverse side has information on the procedure and the Confidentiality Agreement to download. As soon as the signed Confidentiality Agreement has been received via the following email address, you will obtain access to the 3D design data.

- ❗ If you have any questions or require further information about 3D design data, send an email in English or German to:

bb-3d_data-van@mercedes-benz.com



2.2.2 Information for bodybuilder manufacturers outside the sales network

Bodybuilder manufacturers outside the sales network can find aftersales information on the following topics on the Mercedes-Benz B2B Connect portal:

- Mercedes-Benz genuine parts
- Chemical products and care products
- Genuine Mercedes-Benz engine oils
- Genuine Mercedes-Benz tires
- Repair and maintenance information
- Diagnosis
- Mobile apps
- Workshop equipment

You can access the portal at:

<https://b2bconnect.mercedes-benz.com>



2.2.3 Workshop Information System (WIS)

Repair and maintenance information is available to you in Mercedes-Benz B2B Connect under "Workshop Solutions":

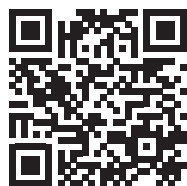
- Workshop Information System (WIS)
- Working hours (ASRA)
- Parts information
- Repair cases (XENTRY TIPS)
- Further information

In the Workshop Information System, you can find information such as the following:

- Basic data (dimensions, tightening torques)
- Function descriptions
- Wiring diagrams
- Repair instructions
- Maintenance sheets

Bodybuilder manufacturers outside the sales network can obtain access to the Workshop Information System via:

<https://b2bconnect.mercedes-benz.com>



 Further information is available from your Mercedes-Benz Service Partner or on request by emailing the Customer Support Center (CSC) in English or German:

support-b2bconnect@mercedes-benz.com



2.2.4 Diagnosis

Existing diagnosis solutions are available to you in Mercedes-Benz B2B Connect at "Workshop Solutions – Diagnosis":

- XENTRY Diagnosis System
- XENTRY Pass Thru EU
- XENTRY Scope
- XENTRY Update Service
- XENTRY Accessories
- XENTRY Flash/Diagnosis user rights

XENTRY Diagnosis Kit 4 is Mercedes-Benz's most efficient diagnosis tool. It includes the diagnostic computer (XENTRY Diagnosis Pad 2), a multiplexer (XENTRY Diagnosis VCI), and an OBD and USB connection cable. The range of functions includes, for example: quick test, actuations, display of actual and specified values, guided troubleshooting, commissioning of control units, and cross references to other systems such as XENTRY TIPS and the Workshop Information System.

XENTRY Pass Thru EU is diagnosis software for a standard PC, explicitly for independent market operators.

XENTRY Pass Thru EU also includes:

- Quick test and functional diagnosis (single-channel, thereby longer operating time)
- Basic diagnosis (in XENTRY Diagnosis only)
- Commissioning including flashing, coding, parameterization

- i** You can find further information about diagnosis solutions at:

<https://b2bconnect.mercedes-benz.com>



XENTRY Diagnosis Kit 4 can be purchased or leased by bodybuilder manufacturers.

If you have any questions about XENTRY Diagnosis Kit 4, contact your Mercedes-Benz Service Partner or the Customer Support Center (CSC) at:

support-b2bconnect@mercedes-benz.com



XENTRY can be accessed via:

<https://xentry.mercedes-benz.com/home>



2.2.5 CERON

The CERON (Certification Online) system is a platform for exchanging and providing documents on the system approvals and overall operating permits of the EC (European Community) and UNECE (Economic Commission for Europe). Here, you can find documents regarding ETA (European Type Approval), for example.

You can access the data in CERON via the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ After logging into the Bodybuilder Portal, select the "CERON" button in the dashboard.
- ▶ If you are registering for the first time, you will first need to send a message with your contact details (in English or German) to the specified email address.
- ▶ Once you have received confirmation of your registration, you can use the direct link to CERON like any other already registered user.

- i** Use of CERON is subject to a charge.

2.2.6 European type approval (ETA) and EC Certificate of Conformity (CoC)

Regulation (EU) 2018/858 and/or Directive 2007/46/EC of the European Parliament forms the basis for the approval of motor vehicles and motor vehicle trailers, as well as of systems, component parts and separate technical units for these vehicles. In this directive, regulations have also been enacted concerning approvals for vehicles that are manufactured in multiple stages, i.e. the multi-stage type approval procedure. According to this, every manufacturer involved in the construction of a vehicle is responsible for the approval of modified or added scopes in their own manufacturing stage. The manufacturer can select one of the four procedures:

- EC type approval (ETA)
- EC small series type approval
- National small series type approval
- Individual approval

CoC stands for Certificate of Conformity. This is a document that attests that certain goods – including vehicles and bodies – conform to recognized (international) standards. The meaning and purpose of this EC Certificate of Conformity is to simplify the registration process for goods on the international markets. This document is therefore required for customs clearance, particularly in the context of import and export. The manufacturer, who is in possession of an EC type approval certificate or EC small series type approval certificate, must provide every vehicle that corresponds to an approved model with a Certificate of Conformity.

The vehicle-specific data on the EC Certificate of Conformity (CoC) for the basic vehicle is available in the Bodybuilder Portal for convenient and quick processing of the body scope.

If you are planning an ETA in a subsequent stage (multi-stage type approval), a contract is required as per Regulation (EU) 2018/858 Annex IX and/or 2007/46 EC Annex XVII. The use of CERON is also regulated as part of this contract.

The manufacturer agreement concerning the multi-stage procedure can be obtained at:

multistage-van@mercedes-benz.com



You can retrieve the data for the basic vehicle for the orders assigned to you after logging in to the Bodybuilder Portal as follows:

- ▶ On the home screen (dashboard), select "Vans" and click on the "Order Data" button.

To ensure that the button is visible to you, it has to be activated in the company data in the contact person area for each contact person through "User for Order Data". You can choose between an overview and an Excel file, but not the original document of the EC Certificate of Conformity.

After retrieving the order data, you can retrieve the EC Certificate of Conformity (online information on the CoC document) for your vehicle order as follows:

- ▶ Click on the order number, click on the registration data, view the online data or download the data as an Excel file.

Note: The online data is for information purposes only and does not include the official CoC documents that are supplied when the vehicle is delivered. CoC data can change in the order due to WLTP calculations.

- ❗ More information on the topic of WLTP is available in the Bodybuilder Portal via the "WLTP Information for Vans" button and in 2.2.8 Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP) (→ page 29).
- ❗ If you have any questions about navigating within the Bodybuilder Portal, please use the specified hotline:

<https://bb-portal.mercedes-benz-vans.com>



2.2.7 ADVANTAGES product information system

You can access the new ADVANTAGES product information system via the Bodybuilder Portal.

The product information system contains information on vehicle equipment options (codes, suspension variants, overview of products, trailer loads, etc.).

- ① The "ADVANTAGES" button for access to the product information system is available in the Bodybuilder Portal under "Engineering and Information (Vans)" after selecting the vehicle model in the "Further Information" section.

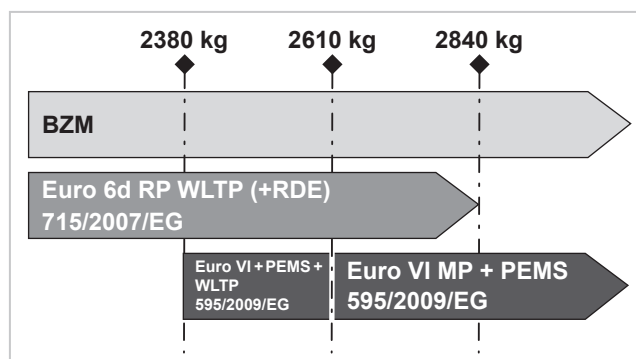
2.2.8 Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP)

WLTP relevance

In Europe, there are two different regulations governing the testing of consumption and emission values for motor vehicles:

- Regulation (EC) 715/2007 for light vehicles with a reference mass not exceeding 2840 kg. This test is performed on a roller test rig.
- Regulation (EC) 595/2009 for heavy vehicles with a reference mass of 2380 kg and over. This test is performed on a bench dynamometer.

The overlap of the fields of application produces a range in which both regulations could apply. Both regulations are binding below and above this range.



Certification type dependent on reference mass

- BZM Reference mass
- RP Roller test rig
- MP Bench dynamometer

- The **reference mass** is the mass of the vehicle in curb condition, plus an additional mass of 25 kg.
 - The **mass of the vehicle in curb condition** is the curb weight of the basic model vehicle, plus the mass of bodies or modifications.
 - The **curb weight of the basic vehicle** comprises:
 - Fuel tank 90% full
 - AdBlue® tank 100% full
 - Other operating fluids 100% full
 - Vehicle tool kit and spare wheel, where applicable
 - Driver mass of 75 kg
- i** When certified according to Regulation (EC) 715/2007 on the roller test rig, the emissions class is denoted by Arabic numerals, e.g. "Euro 6d". When certified according to Regulation (EC) 595/2009 on the bench dynamometer, the emissions class is denoted by Roman numerals, e.g. "Euro VI".

Depending on the weight variant, model series 907 vehicles fall under the Regulation (EC) 715/2007 or (EC) 595/2009.

WLTP for vehicles with Euro VI emissions class

For vehicles equipped with a homologated engine compliant with Regulation (EC) No. 595/2009 and a reference mass of less than 2610 kg, the CO₂ and consumption values that have been determined as per the WLTP procedure must be stated.

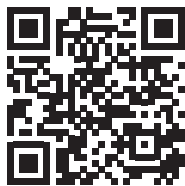
As only a maximum value certification may be used here, a calculation is not required. This is why the CO₂ values of the initial approval stage do not differ from subsequent approval stages.

CO₂ and consumption values of the initial approval stage also retain their validity in the event of any multiple-stage approval, and they can be transferred.

On vehicles with a car registration (registration type M1) that are not approved as a vehicle with a special purpose, a CO₂ value must be stated in accordance with Regulation (EU) 2019/631, irrespective of the reference mass.

An overview of the maximum reference masses is available for registered bodybuilder manufacturers in the Bodybuilder Portal at:

<https://bb-portal.mercedes-benz-vans.com>



Mercedes-Benz AG makes these values available for all relevant vehicles as from the respective start of production.

Emission classes and WLTP relevance

Code	Emissions class	Registration type	Reference mass	WLTP relevance	Remarks
MR6	Euro 6c	All	All	Yes	Individual calculation by means of the WLTP Calculator
MT2	Euro 6d-TEMP	All	All	Yes	
MT6	Euro 6d	All	All	Yes	
MT5	Euro 6e/RDE5	All	All	Yes	
M6B	Euro 6e-BIS	All	All	Yes	
MP6	Euro VI	M1	All	Yes	Indirect indication required due to CO ₂ regulation, only maximum value
		N1/N2	≤ 2610 kg	Yes	Indication in the CoC required, only maximum value
		N1/N2/ M2/M3	> 2610 kg	No	Indication in the CoC required only for N1, only maximum value; voluntary indication possible, e.g. via manufacturer's certificate up to the tested reference mass

The reference mass of the vehicle with body/modification is decisive when selecting the correct emissions class.

- ▶ The subsequent reference mass of the completed vehicle must be estimated as early as the planning stage of body/modifications. The correct emissions class must be selected depending on this.
- ❶ Subsequent changing of the basic vehicle emission type approval (e.g. from Euro 6 to Euro VI) cannot be done by Mercedes-Benz AG for a vehicle that has already been built.

Scope of application of emissions type approvals of the basic vehicle within the WLTP

Mercedes-Benz AG defines, certifies and approves an interpolation family (IP family) for each engine/drive-train variant.

Each IP family is characterized by the parameter values:

- Mass
- Aerodynamic drag/frontal area
- Rolling resistance

The minimum and maximum parameters of an IP family are based on the requirements of the market and what is technically feasible. They are limits that must never be undershot or exceeded.

The parameter limits provide space for custom body mounting work and/or special equipment items available ex works.

The emissions type approval of the basic vehicle can be transferred when the parameter limits of the applicable IP family are complied with.

If the defined parameter limits are exceeded or undershot, separate evidence must be provided and presented upon request. More information can be found in the section "Exceeding of parameter limits" (→ page 36).

It is recommended to coordinate the approval and registration requirements of a body or modification with the Technical Service and/or with the relevant type-approval authority during the planning phase.

- ❶ The emissions type approval of the basic vehicle can be transferred for all bodies or modifications where changes to the vehicle mass, the aerodynamic drag/frontal area, and the rolling resistance are within the parameter limits.
- ❶ An overview of the IP families, including the associated parameter limits of the emissions type approval, is available to registered and logged-in bodybuilder manufacturers for the planning of bodies or modifications in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



WLTP calculator – provision of CO₂ and consumption values for individual vehicles

Mercedes-Benz AG provides CO₂ and consumption values for each basic vehicle.

A calculation service (via the WLTP Calculator) for determining the CO₂ and consumption values of the completed vehicle is available for bodybuilder manufacturers in the Bodybuilder Portal. There, bodybuilder manufacturers can enter a Vehicle Identification Number (VIN) or an offer PIN to first obtain the parameters of the basic vehicle:

- Mass of the vehicle in curb condition, including all ex factory special equipment
- Frontal area
- Rolling resistance

Parameters that have changed due to bodies or modifications can be added via appropriate input fields.

The values can be entered in two different ways:

- By entering new overall values for the finished complete vehicle with body or modifications
- By entering the difference values of the individual bodies or modifications

 The WLTP Calculator can be found at:

<https://bb-portal.mercedes-benz-vans.com>



NOTE

According to Regulation (EU) 2017/1151 Annex XII Paragraph 2, the bodybuilder manufacturer is responsible for calculating the final CO₂ and consumption values, as well as for the correctness of the data.

► The parameters for the calculation and the calculation results must be documented in the Certificate of Conformity (CoC) as per Regulation (EU) 2018/858 Article 36, Section 3 and/or 2007/46/EC Annex IX, and must therefore always be determined with the utmost care.

Direct any questions about the WLTP and the WLTP Calculator to:

WLTP-calculation-service@mercedes-benz.com



VECTO simulation energy consumption

For heavy commercial vehicles with a permissible gross mass above 5000 kg, as of 01/2024, a new method of calculating the energy consumption as per Regulation (EU) 2017/2400 is necessary for the initial registration in countries with CoC documents.

As a result from the VECTO simulation instrument, the energy consumption values for model designation 907.x5x with weight code XH2 (5200 kg), XH3 (5380 kg) and XL4 (5500 kg) – except for vehicles with all-wheel drive code A4M (4×4 with torque-on-demand) – are documented in a supplementary sheet to the CoC documents.

 In the event of questions regarding VECTO, contact the Technical Consultancy (→ page 18) or send an email to the previously specified email address.

Determining WLTP-relevant parameters

To determine the parameters for bodies or modifications, the following procedure is recommended.

Mass of the vehicle in curb condition

Bodies or modifications change the mass of the basic vehicle.

 The change of mass must be calculated and taken into account.

Exceptions to this are items of equipment that are used e.g. for load securing, and which fall under the provisions of Regulation (EU) 2018/858 Annex IX and/or 2007/46/EC Annex XVII Paragraph 3.4.

The vehicle mass of vehicles with bodies or modifications, or the additional mass of bodies or modifications, should be available for each axle. This is absolutely essential in the case of mixed tires; see also the section on "Rolling resistance" (→ page 35).

- ▶ The mass can be determined via methods such as:
 - Calculation with the aid of a CAx system
 - Mathematical determination using weight tables
 - Weighing

! NOTE

The scales used for determining the mass by weighing must be calibrated and accredited. A maximum measurement tolerance of 1.5% must be complied with.

- ▶ Accreditation can be obtained from an officially recognized accreditation facility.

! NOTE

The fuel mass of a fill level of 90% must be included in the vehicle mass.

The fuel quantity is already taken into account in the basic vehicle. Changes are therefore only required when installing a different or additional fuel tank.

There is a calculation service for determining the fuel mass at a fill level of 90 % available to bodybuilder manufacturers in the Bodybuilder Portal.

Aerodynamic drag – frontal area

Bodies or modifications can change the aerodynamic drag of the basic vehicle.

- ▶ If the frontal area is changed by bodies or modifications, the change in the frontal area must be determined and taken into account.
- ▶ The frontal area can be achieved through, e.g.:
 - Calculation with the aid of a CAx system
 - Certified testing institute (laboratory, wind tunnel, etc.)

- ① Mercedes-Benz AG provides the frontal area for CAx systems for each basic model designation. Further information can be obtained at:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ When using a frontal area provided by Mercedes-Benz AG in a CAx system, ensure the following:
 - The zero position and the inclination are contained in the frontal area model.
 - Bodies or modifications must be positioned accordingly relative to the frontal area.
- ▶ If the CAx system is unable to account for the inclination of the front surface area as supplied, then the inclination about the center of the front axle must be set toward the front for the overall model:
 - Model series 907, all vehicle model designations: 1.35°
- ▶ Changes to the frontal area due to bodies and modifications are to be specified in the unit m² rounded to three decimal places.
- ① 3D design data available in the Bodybuilder Portal – See 2.2.1 Bodybuilder Portal (→ Page 24) – can also be used to determine frontal areas. For this, their inclination must also be taken into account. The bodybuilder manufacturer must ensure that the data matches the basic vehicle.

Cooling air intake at the front of the vehicle

For vehicles certified according to Regulation (EU) 2017/1151, the size of the opening surface for the engine air cooling is another relevant parameter value for the emissions type approval of vehicles.

For model series 907, this concerns emission codes MT2, MT6, MT5, M6B, See 3.1 Vehicle and model designation (→ Page 47) in combination with engine codes MU1, MU3, MU5, MU6, See 3.1 Vehicle and model designation (→ Page 47).

NOTE

The size of the intake area for the engine cooling air at the front of the vehicle is in a certified state.

Bodies or modifications that change the size of the cooling air intake at the front of the vehicle can cause the emissions type approval of the basic vehicle to cease to be generally valid.

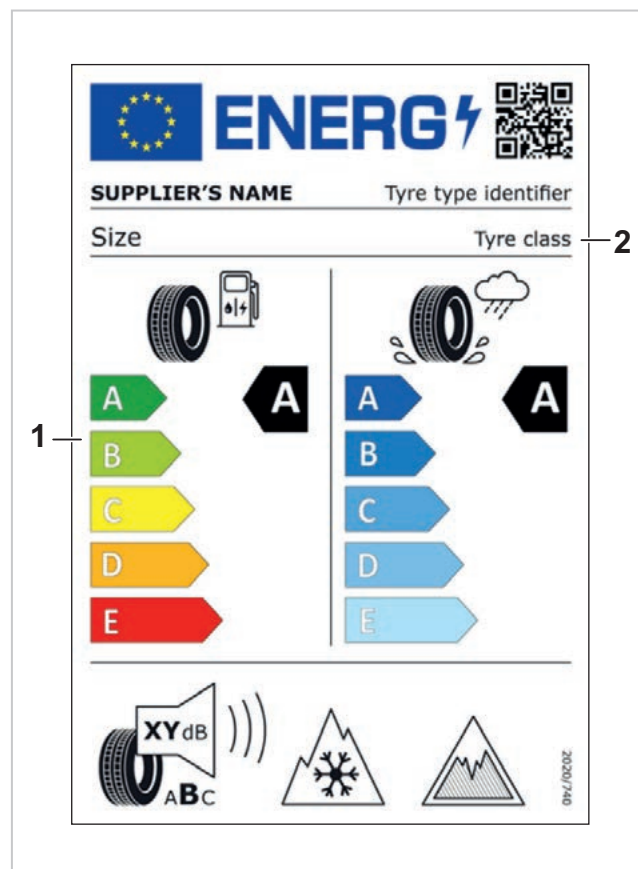
- i** For further information on changes to the cooling air opening, See 4.4.3 Engine cooling (→ Page 92).

Rolling resistance

Changes to the tires resulting from body mounting work or modifications can change the rolling resistance.

- ▶** The rolling resistance of the tires must be determined and taken into account.

Regulation (EC) 2020/740 defines the labeling of tires in terms of fuel efficiency.



EU tire label

- 1 Efficiency class
2 Tire class

The efficiency classes of tires are classified from A to E in accordance with the tire class. Class A corresponds to a low rolling resistance; class E corresponds to a high rolling resistance.

- ▶ According to the previously mentioned regulation, the efficiency class and the tire class of a tire are stated on the corresponding EU tire label.
- ▶ If a tire label is not available, it is recommended to request this information from the tire manufacturer.

It is possible to use tires of different efficiency classes on the front and rear axles if the vehicle mass per axle is known.

! NOTE

Only the wheel size groups approved by Mercedes-Benz AG are to be used (See 3.8 Tires (→ Page 65)).

! NOTE

In the case of vehicles with more than two axles, only tires with the same efficiency class are permitted within the axle pair on the rear axle.

! NOTE

No changes may be made to scopes that change the definition of the underlying IP families. This concerns, for example:

- Engine
- Transmission
- Axles
- Drive type

Exceeding of parameter limits

! NOTE

If bodies or modifications cause parameter limits (frontal area, mass, rolling resistance) to be exceeded, the emissions type approval of the basic vehicle generally ceases to be valid.

- ▶ The registration requirements and legality of the vehicle with body or modifications must be independently verified and assured by the bodybuilder manufacturer.
- ▶ The possibility of using the emissions type approval of the basic vehicle even if the parameter limits are exceeded by bodies or modifications must be clarified by the bodybuilder manufacturer with the relevant authority on a case-by-case basis.

There may be differences depending on the country and/or the vehicle licensing agency, and these may also be dependent on the type of registration being applied for:

- Multi-stage approval
- European individual approval
- National individual approval
- Other specific national regulations

i If bodies or modifications cause the parameter limits to be exceeded, it is the duty of the bodybuilder manufacturer alone to clarify the validity of emissions type approvals with the relevant authority.

i If the parameter limits of the IP family of the basic vehicle are exceeded because of bodies or modifications, please contact the following for determining/providing CO₂ and consumption values:

WLTP-calculation-service@mercedes-benz.com



2.3 Product safety and product liability

2.3.1 Product safety

Vehicle manufacturers and bodybuilder manufacturers alike must always ensure that the products they manufacture are safe when they are brought into circulation and do not present any danger to people or property. Otherwise, there may be consequences under civil, criminal, or administrative law. Every manufacturer is always liable for the product that it has manufactured.

The bodybuilder manufacturer must ensure compliance with the legal regulation 2001/95/EC on general product safety.

The bodybuilder manufacturer must also ensure compliance with other statutory regulations, insofar as these are applicable to its body mounting work:

- Functional safety in accordance with ISO 26262
- Cyber security in accordance with ISO 21434 and UN R 155
- Software Update Management System (SUMS) in accordance with ISO 24089 and UN R 156.

2.3.2 Product liability

The bodybuilder manufacturer bears responsibility for the operational and road safety of the body mounting work they perform, and in particular for:

- Testing and maintaining the operating and driving safety of the overall vehicle after the body is mounted (the body must not have a negative effect on the driving, braking, or steering characteristics of the vehicle).
- The influences of body mounting work on the basic vehicle
- Consequential damage arising from body mounting work
- Consequential damage resulting from retrofitted electrical and electronic systems
- Maintaining the functional reliability and unobstructed movement of all moving parts of the basic vehicle (e.g. axles, springs, propeller shafts, steering, gearshift linkage, hinges, etc.) after the body mounting work is complete; even in the case of diagonal torsion of the vehicle

Work carried out on or modifications to the basic vehicle or body must be entered in the maintenance booklet in the "Confirmations of the bodybuilder manufacturer" section.

2.3.3 Safety-relevant features

Component parts and systems are safety-relevant when their fault or failure could result in an immediate danger to the life and limb of road users.

An assessment of the safety relevance of the components or functions is recommended for the following work:

- Modifications to the basic vehicle
- Vehicle installations
- Interface between vehicle and body (mechanical components, electrics/electronics, power take-offs, hydraulic components, pneumatics)

A component or function must be classified as safety-relevant if one of the ten safety functions in particular is affected:

- Occupant protection in accidents
- Avoiding momentary loss of road view
- Avoiding steering failure
- Avoiding loss or partial failure of braking function
- Avoiding failure of driving function
- Avoiding uncontrolled drive
- Avoiding sudden failure of drive force
- Avoiding leakage of operating fluids/risk of fire
- Avoiding cargo/trailers/parts/bodies/semitrailers becoming loose
- Avoiding injury while driving and during alternative operation of the vehicle

Customer-related influences must be considered when evaluating safety relevance:

- Extreme operating conditions
- Wear
- Ambient conditions

Documentation

If safety relevance is identified according to the ten safety aspects, these aspects are to be appropriately marked as safety-relevant in paper and data records and the associated functions and features documented as well as the measures taken to avoid the dangers.

2.4 Guarantee of traceability

There is a possibility that body-related hazards may only be detected after the vehicle is delivered, making retroactive market measures necessary (customer information bulletins, warnings, recalls). To ensure that these measures can be implemented as efficiently as possible, it must be possible to trace the product after delivery.

It is strongly recommended that bodybuilder manufacturers file the serial number/identification number of their body together with the VIN (vehicle identification number) of the basic vehicle in their databases for this purpose and in order to allow them to use the Central Vehicle Register (ZFZR) of the Federal Office for Motor Vehicles in Germany or comparable databases in other countries. It is also recommended to store customer addresses and give subsequent acquirers the possibility to register their details.

2.5 Trademarks

2.5.1 The Mercedes-Benz trademark in interaction with external bodybuilder manufacturers

In terms of the relationship between Mercedes-Benz and bodybuilder manufacturers, it is not just product safety and product liability that are of particularly high importance, but also the usage of the trademark.

The aim of these guidelines is to explain the trademark-related interests of Mercedes-Benz for Sprinter vehicles to bodybuilder manufacturers.

The individual provisions of the guide do not release the bodybuilder manufacturer from their liability for after-market modifications to Mercedes-Benz vehicles.

2.5.2 The function of the trademark

Every company that wishes to operate a successful trademark must ensure that the characteristics of the trademark are consistently protected. This applies equally to the Mercedes-Benz Group AG trademarks.

The trademark indicates the origin of a product and allows the product to be recognized and remembered. The trademark acts as a guarantee for the quality of a product, embodying its characteristic values and identity.

2.5.3 Trademark protection

Trademark protection laws give Mercedes-Benz Group AG the exclusive right to use its trademarks.

In particular, this applies to the use of Mercedes-Benz trademarks on vehicles and vehicle parts/accessories (e.g. rims) and in communication of the brand name.

2.5.4 Trademark rights

The "Mercedes star", the "Mercedes badge" and the "Mercedes-Benz" word mark/lettering and the model series and vehicle model designations are registered trademarks of the Mercedes-Benz Group AG.



Mercedes-Benz

SPRINTER

2.5.5 Advice on applying Mercedes-Benz trademarks to vans

In the event of any questions about applying Mercedes-Benz trademarks to Mercedes-Benz vans, the Marketing Consultancy is available to assist body-builder manufacturers via email in English or German:

van-vma_marketing@mercedes-benz.com



2.5.6 Use of Mercedes-Benz trademarks

Principle

Only Mercedes-Benz trademarks may be applied to completely unchanged Mercedes-Benz vehicles.

Modification of the Mercedes-Benz trademarks on vehicles of the Mercedes-Benz brand is not permissible.

The application of Mercedes-Benz trademarks to vehicles and parts other than those of the Mercedes-Benz brand is not permissible.

Mercedes-Benz trademarks on vehicles with technical modifications

For vehicle modifications that comply with the Mercedes-Benz Body/Equipment Mounting Directives, the Mercedes-Benz trademark may remain unchanged on the vehicle.

Drastic vehicle modifications encroach significantly into the original condition of the vehicles, e.g. through interventions in the suspension, body structure, engine, brake system, and on-board electronics. These modifications are generally not covered by the Body/Equipment Mounting Directives. This can result in deterioration of the overall vehicle with associated risks that Mercedes-Benz AG is not liable for according to product liability and product safety legislation. For the technical feasibility and product safety, a Letter of Compatibility is required (→ page 19), which can be requested via the Bodybuilder Portal (→ page 20).

If the vehicle does not meet the requirements of Mercedes-Benz, a Letter of Compatibility will not be issued and Mercedes-Benz AG reserves the right to demand removal of its trademarks.

2.5.7 Brand separation/identity

Principle

Mercedes-Benz model series have specific characteristics and values as well as a brand-specific design. The modifications should be in the style of the Mercedes-Benz design characteristics.

Separating Mercedes-Benz trademarks from bodybuilder manufacturer trademarks ensures that a distinction can be made between the brands. This applies, in particular, to product origin and the resulting responsibilities.

Mercedes-Benz AG reserves the right to demand immediate removal of the Mercedes-Benz Group AG trademarks if the vehicle fails to comply with the requirements of Mercedes-Benz.

a) Trademarks at the front of the vehicle

Use of the Mercedes star

- On vehicles with genuine Mercedes-Benz cabs, the Mercedes star must be retained in the as-delivered state.
- On vehicles with a cab designed separately by the bodybuilder manufacturer, the Mercedes star may be attached centered on the front-end assembly in order to identify the basic vehicle. If the bodybuilder manufacturer trademark is also applied, it must be applied at an appropriate distance from the Mercedes star.

Use of the Mercedes badge

- The Mercedes badge signifies a genuine Mercedes-Benz design. It is only permitted to be retained on vehicles with a genuine Mercedes-Benz cab.
- The use of the Mercedes badge is not permissible on vehicles with a cab designed separately by the bodybuilder manufacturer.

Model series and vehicle model designation at front end

The Sprinter does not have any model series designations on the front of the vehicle. Model series designations may therefore not be applied to vehicles modified by the bodybuilder manufacturer.

b) Trademarks in rear area of the vehicle

Use of the Mercedes star

When the Mercedes star is applied to the rear area of vehicles whose appearance has been extensively changed by a body or other modifications, Mercedes-Benz AG reserves the right to demand immediate removal of the Mercedes star. It is recommended that the bodybuilder manufacturer should discuss the matter with Mercedes-Benz AG before application.

Bodybuilder manufacturer trademarks in combination with Mercedes-Benz trademarks

Bodybuilder manufacturers that attach their trademarks to their body must ensure that they are located at a suitable distance away from Mercedes-Benz trademarks or badges.

c) Trademarks in the interior

All interior elements (e.g. seats, interior trims, etc.) that are integrated by the bodybuilder manufacturer must not bear any Mercedes-Benz Group AG trademarks (e.g. Mercedes star or the "Mercedes-Benz" lettering).

2.5.8 Trademark communication

Use of Mercedes-Benz trademarks in communication

- The use of Mercedes-Benz trademarks in communication instruments is regulated for certified Mercedes-Benz VanPartners/Mercedes-Benz VanSolution Partners in the "Bodybuilder Brand & Design Style Guide", which is available in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com/de/GLOBAL/bodybuilder-styleguide-vans>



If necessary, contact the Marketing Consultancy for Bodybuilder Manufacturers (→ page 40).

- The bodybuilder manufacturer must be clearly identifiable as the originator of the communication in all its communication media and channels.
- When depicting modified vehicles that comply with the Mercedes-Benz guidelines and that bear our trademarks, care should be taken to ensure that any modification to the vehicle or the vehicle body made by the bodybuilder manufacturer is visible in the depiction. Only the obvious prominence of Mercedes-Benz trademarks without any visible attachment, body, or conversion is not permissible.

2.6 Accident prevention

The body and attached or installed devices must comply with all applicable laws and regulations, and with health and safety and accident prevention regulations, safety regulations, and information sheets issued by accident insurers.

The laws, standards, directives, etc. for occupational health and safety and accident prevention regulations when operating vehicles, equipment and machinery (general machinery directives) must be complied with to avoid any unsafe operating conditions during vehicle use, and all the necessary technical precautionary measures for this purpose must comply with the latest state-of-the-art technology.

NOTE

Observe all national laws, directives, and registration regulations.

The bodybuilder manufacturer shall be responsible for compliance with these laws and regulations.

2.7 Reprocessing (recycling)



ENVIRONMENTAL INFORMATION

When planning body mounting work, the following principles for environmentally compatible design and material selection should be taken into account, in particular with regard to the EU Directive on end-of-life vehicles 2000/53/EC.

Bodybuilder manufacturers must ensure that body mounting work complies with current environmental legislation and applicable regulations, in particular EU Directive 2000/53/EC concerning end-of-life vehicles and EU Directive 2003/11/EC concerning restrictions on bringing into circulation and the use of certain dangerous substances and dangerous preparations ("low flammability" and certain flame retardants) to specify the Directive 76/769/EEC.

In accordance with Article 33 of the European Chemicals Regulation REACH (Registration, Evaluation, Authorisation and Restriction of Chemicals, Regulation (EC) No. 1907/2006), there is also an obligation to provide information on substances contained in bodies, products and articles that fulfill the criteria of Article 57 and Article 59, para. 1, and whose concentration exceeds 0.1 percent by weight.

 Further information on REACH is available in English or German at:

- German:

<https://reach.mercedes-benz.com/de/home>



- English:

<https://reach.mercedes-benz.com/en/home>



For single-invoice transactions (ERG) with Mercedes-Benz AG, there is an obligation to provide information via the Bodybuilder Portal about bodies, products and articles that contain the affected substances.

In all other instances, e.g. dual-invoice transactions (ZRG), the responsibility for compliance with and documentation of the previously mentioned information requirement lies with the bodybuilder manufacturer.

The installation documentation for the modifications must be kept by the vehicle owner and, if the vehicle is to be scrapped, handed over to the dismantling company concerned at the time of vehicle handover. This is intended to ensure that converted vehicles are also processed in an environmentally responsible manner.

Avoid using materials with risk potential such as halogen additions, heavy metals, asbestos, CFC, and CHC.

- Comply with EU Directive 2000/53/EC.
- Preferably use materials that facilitate material-sensitive recycling and closed recycling loops.
- Select materials and manufacturing processes that generate only low quantities of easily recyclable waste during production.
- Only use plastics where they provide advantages in terms of cost, function, or weight.
- In the case of plastics, and composite materials in particular, only compatible substances within one material family are to be used.
- Keep the number of different types of plastics to be used for component parts that are relevant to recycling to a minimum.
- Check whether a component part can be made from recycled material or with recycled elements.
- Ensure that component parts can be dismantled easily for recycling, e.g. by means of snap-on fasteners, predetermined fracture points, easy access, or by using standard tools.
- Ensure that service fluids can be removed simply and in an environmentally responsible manner by means of drain screws, etc.
- Where possible, avoid painting or coating component parts: use dyed plastic parts instead.
- Design component parts in areas at risk from accidents in such a way that they are damage-tolerant, repairable, and easy to replace.
- Mark all plastic parts in accordance with the VDA Materials Leaflet 260 ("Components of Motor Vehicles; Identification of Materials"), e.g. "PP - GF30R".

2.8 Quality system

Worldwide competition, increased quality standards demanded by the customer from the van as a whole, national product liability laws, new organizational forms and rising cost pressures make efficient quality assurance systems a necessity in all sectors of the automotive industry.

The requirements for a quality management system of this kind are defined in DIN EN ISO 9001.

For the previously mentioned reasons, it is strongly recommended that all bodybuilder manufacturers set up and maintain a quality management system, based on the specifications in DIN EN ISO 9001, that includes these requirements, among others:

- Defining responsibilities and authorities including organizational planning
 - Description of processes and procedures, as well as their interactions
 - Conducting contractual inspections and buildability checks
 - Comprehensive development documentation including risk assessment of process and product for scopes relevant in terms of safety, regulatory conformity/compliance, and emissions (SCE).
-  For further information See 2.3.3 Safety-relevant features (→ Page 38).
- Comprehensive management of SCE scopes in production, quality assurance, and service
 - Definition of product-related quality targets and SCE-relevant scopes, here with the target value of "Zero"
 - Conducting product tests in accordance with the specified procedures stated in the test instructions
 - Policy on dealing with faulty products
 - Documenting and archiving test results
 - Ensuring that all employees have currently valid proof of the qualifications required
 - Systematic monitoring of test equipment, including ensuring the process reliability of the retrogressive risk assessment
 - Systematic identification of materials and parts
 - Conducting quality assurance measures at suppliers
 - Ensuring that the instructions for processes, work, and tests are up to date and available in all departments and at all workplaces
 - Documentation of customer-specific requirements (AQAP, QMP, FMEA, acceptance test report, measurement reports for individual parts, etc.)
 - Verification of data exchange between the contract partners as per an agreed-upon format (data format, exchange server, etc.)

2.9 Technical Compliance Awareness Package

Via the Bodybuilder Portal, you have access to the "Technical Compliance Awareness Package" for bodybuilder manufacturers.

This contains information on the technical Compliance Management System (tCMS) of Mercedes-Benz AG and our understanding of what technical compliance means in the context of collaboration with bodybuilder manufacturers, as well as the associated product development process at bodybuilder manufacturers.

To this end, we provide you as a bodybuilder manufacturer with various options for dialog and communication, which are described in more detail in the awareness package (AWP).

-  You can retrieve the information on the AWP and tCMS after logging into the Bodybuilder Portal:
 -  Select "Vans" on the home screen (dashboard),
 -  Click on the "Engineering and Information" button, then select the vehicle image of the required model series,
 -  Select "⊕ Retrieve Technology".
 -  Click on the "tCMS" button under "Technical Data".

3.1 Vehicle and model designation

- ❗ For information on the position of the identification plates, see 3.6 Vehicle identification data (→ page 61).

This Body/Equipment Mounting Directive is valid for the following vehicle model designations of model series 907 (rear-wheel and all-wheel drive):

Versions and vehicle model designations

Version	Wheelbase	Vehicle model designation by permissible gross mass			
	[mm]	3000 kg	3500 kg	4000 kg	5000 kg
Panel van (FKA)	3665	–	907,633	907,643	907,653
	4325	–	907,635	907,645	907,655
	4325 ¹⁾	–	907,637	907,647	907,657
Tourer (FKB)	3665	907,723	907,733	907,743	–
	4325	–	907,735	907,745	–
Chassis with cab (FHS) ²⁾	3665	–	907,133	907,143	907,153
	4325	–	907,135	907,145	907,155
Chassis with crewcab (FHL) ²⁾	3665	907,223	907,233	907,243	907,253
	4325	–	907,235	907,245	907,255

1) Version with extended overhang

2) Version with platform ex factory: Special equipment code P02

Explanation of vehicle model designations

Model series	Body	Permissible gross mass	Wheelbase	Steering	Version
907 633 13	907 633 13	907 633 13	907 633 13	907 633 13	907 633 13
907 – Sprinter Rear-wheel and all-wheel drive	1 – Chassis with cab (FHS)	2 – 3000 kg	1 – 3250 mm ¹⁾	1 – Left-hand drive vehicle	3 – Complete
	2 – Chassis with crewcab (FHL)	3 – 3500 kg	3 – 3665 mm	2 – Right-hand drive vehicle	
	6 – Panel van (FKA)	4 – 4000 kg	5 – 4325 mm		
	7 – Tourer (FKB)	5 – 5000 kg	7 – 4325 long overhang		
	8 – Bus ²⁾				

1) Wheelbase 1 (3250 mm) no longer available as of model year 2024

2) Bus body no longer available ex-factory from Argentina

- ❗ Brief descriptions of the new features in the vehicle can also be found in the Bodybuilder Portal under the heading "Bodybuilder Manufacturer Information Bulletin".

<https://bb-portal.mercedes-benz-vans.com>



Engine variants

Engine code	Engine	Maximum output	Model designation ¹⁾ according to permissible gross mass			
			3000 kg	3500 kg	4000 kg	5000 kg
MU1	OM654 DE20LA	84 (114)	211 CDI	311 CDI	411 CDI	511 CDI
MU3	OM654 DE20LA	110 (150)	215 CDI	315 CDI	415 CDI	515 CDI
MU5	OM654 DE20LA	125 (170)	217 CDI	317 CDI	417 CDI	517 CDI
MU6	OM654 DE20LA	140 (190)	–	319 CDI	419 CDI	519 CDI

1) Model designation is supplemented by "4×4" for vehicles with all-wheel drive and by "Tourer" for type FKB.

i Vehicles with engines OM651 and OM642 are no longer available ex factory. For this reason, this information is no longer included in the present Body/Equipment Mounting Directive. If required, you can find corresponding information and specifications in previous Body/Equipment Mounting Directives in which the engines are still included. These are still available in the Mounting Directives Archive in the Bodybuilder Portal.

Assignment of emissions class

Curb weight [kg]	Code	Emissions class
≤ 2355	MR6	Euro 6c N1 Group III/N2
	MT2	Euro 6d-TEMP N1 Group III/N2
	MT6	Euro 6d N1 Group III/N2
	MT5	Euro 6e/RDE5 N1 Group III/N2
	M6B	Euro 6e-BIS N1
> 2355 ≤ 2815	MR6	Euro 6c N1 Group III/N2
	MT2	Euro 6d-TEMP N1 Group III/N2
	MT6	Euro 6d N1 Group III/N2
	MT5	Euro 6e/RDE5 N1 Group III/N2
	M6B	Euro 6e-BIS N1
	MP6	Euro VI
> 2815	MP6	Euro VI

The required curb weight after completion of the vehicle with code MR6, MT2, MT6, MT5 and M6B is at least 1735 kg, and for vehicles with code MP6, it is at least 2356 kg.

Special codes and body versions (optional)

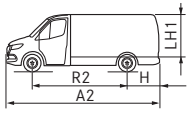
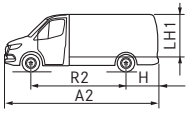
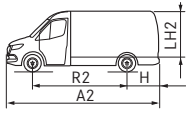
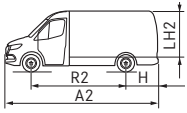
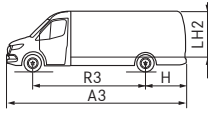
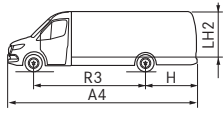
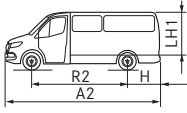
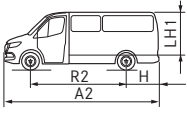
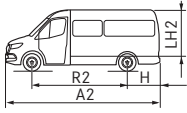
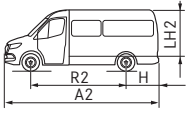
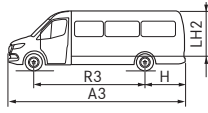
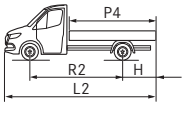
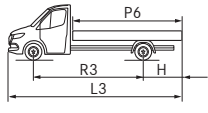
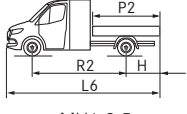
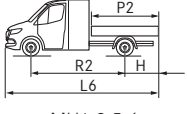
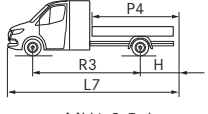
Code	Version
D03	High roof
CW2¹⁾	Omission of vehicle lowering
P02	Platform ex factory

1) Observe the information on code CW2 in 4.2.1 Suspension (→ page 80).

3.2 Model overview

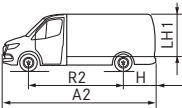
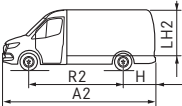
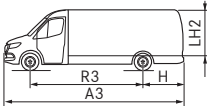
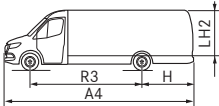
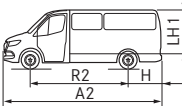
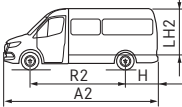
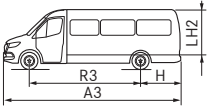
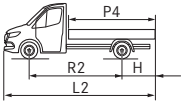
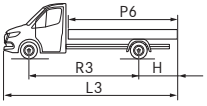
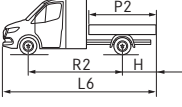
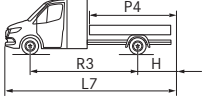
You will find detailed and additional drawings and dimensions in the 2D drawings (offer drawings) in the Bodybuilder Portal (→ page 24). Explanations of vehicle model designations and engine codes can be found in the tables (→ page 47).

The following models are available for model series 907 (rear wheel drive) (as at 03/2023):

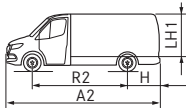
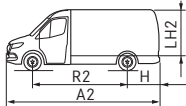
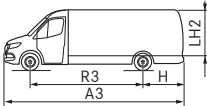
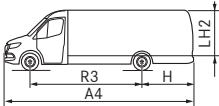
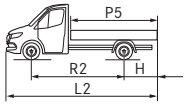
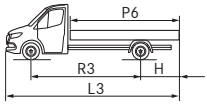
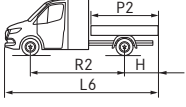
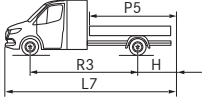
Weight variant		3.0 t (IT3) 3.0t ¹¹⁾ /3.025t ^{6) 11)} / 3.19 t		3.5 t (IT4) 3.2 t/3.49 t ^{2) 11)} /3.5 t ²⁾ /3.55 t ^{2) 6)}	
Rear axle		with single tires		with single tires	
		R2/A2	R2/A2	R3/A3	R3/A4
FKA	LH1	907623 ²²⁾  MU1,3,5	907633 ²⁾  MU1,3,5,6		
	LH2	907623 ²²⁾  MU1,3,5	907633 ²⁾  MU1,3,5,6	907635 ²⁾  MU1,3,5,6	907637 ²⁾  MU1,3,5,6
FKB	LH1	907,723  MU1,3,5	907733 ²⁾  MU1,3,5,6		
	LH2	907,723  MU1,3,5	907733 ²⁾  MU1,3,5,6	907,735  MU1,3,5,6	
FHS			907133 ²⁾  MU1,3,5,6	907135 ²⁾  MU1,3,5,6	
FHL		907223 ²⁰⁾  MU1,3,5	907233 ²⁾  MU1,3,5,6	907235 ²⁾  MU1,3,5,6	

Legend (→ page 52)

3 Planning of bodies

Weight variant		4.0 t (IT9) 3.88 t/4.0 t/4.1 t/4.25 t		
Rear axle		with single tires		
		R2/A2	R3/A3	R3/A4
FKA	LH1	907643 ²⁾  MU1,3,5,6		
	LH2	907643 ²⁾  MU1,3,5,6	907645 ²⁾  MU1,3,5,6	907647 ²⁾  MU1,3,5,6
FKB	LH1	907.743 ²⁾ 10) 23)  MU1,3,5,6		
	LH2	907.743 ²⁾ 10) 23)  MU1,3,5,6	907.745 ²⁾ 10) 23)  MU1,3,5,6	
FHS		907143 ²⁾  MU1,3,5,6	907145 ²⁾  MU1,3,5,6	
FHL		907243 ²⁾  MU1,3,5,6	907245 ²⁾  MU1,3,5,6	

Legend (→ page 52)

Weight variant		5.0t ⁷⁾ (IT5)		
		3.5t/4.25t/4.49t ¹¹⁾ /4.53t ^{11) 12)} /4.6t/5.0t/5.03t ¹⁵⁾ /5.2t ²⁴⁾ /5.38t ²⁴⁾ /5.5t ⁸⁾		
Rear axle		with twin tires		
		R2/A2	R3/A3	R3/A4
FKA	LH1	907653 ¹⁾		
				
		MU1,3,5,6		
	LH2	907.653 ^{1) 3)}	907.655 ^{1) 3)}	907.657 ^{1) 3)}
				
		MU1,3,5,6	MU1,3,5,6	MU1,3,5,6
FHS		907.153 ^{1) 3)}	907.155 ^{1) 3)}	
				
		MU1,3,5,6	MU1,3,5,6	
FHL		907253 ¹⁾	907255 ¹⁾	
				
		MU1,3,5,6	MU1,3,5,6	

Wheelbase		Load compart- ment height		Vehicle length, rear overhang (front overhang 1021 mm)										Platform length, internal			
[mm]		[mm] H201		[mm]										[mm]			
				FKA/FKB				FHS/FHL with platform									
								IT3/4/9			IT5						
R2	3665	LH1	1729 (1719) ⁹⁾	A2	5932	Rear	1246	L2	6046	Rear	1360	L2	6246	Rear	1560	P2	2708
R3	4325	LH2	2020 (2010) ⁹⁾	A3	6967		1621	L3	6946		1600	L3	6946		1600	P4	3408
				A4	7367		2021	L6	6156		1470	L6	6156		1470	P5	3608
								L7	6856		1510	L7	7056		1710	P6	4308

Legend (→ page 52)

3 Planning of bodies

Model designation ⁵⁾ (code)	Permissible gross mass [t]	Tonnage code
3.0 t (IT3)	3.0	X3G ¹¹⁾
	3.025	XA1 ^{6),11),14)}
	3.19	XB5
3.5 t (IT4)	3.2	X13
	3.49	X19 ^{2,11)}
	3.5	XG7 ²⁾
	3.55	XG4 ^{2),6)}
4.0 t (IT9)	3.88	XL8
	4.0	XH1
	4.1	XL2
	4.25	XG5
5.0 t (IT5)	3.5 (load capacity reduction)	X16
	4.25	XG5
	4.49	XG9 ¹¹⁾
	4.531	XB3 ^{11),12)}
	4.6	XA4
	5.0	X5G
	5.03	XL6 ¹⁵⁾
	5.2	XH2 ²⁴⁾
	5.38	XH3 ²⁴⁾
	5.5	XL4 ⁸⁾

- 1) Also as all-wheel drive, A4M with MU6 and G43
- 2) Also as all-wheel drive, A4M with MU6 and G43; only with permissible gross mass XL4/XH2/XH3
- 3) Also with ultrawide-base tires (R9A) in combination with MU3, MU5
- 5) Depending on model designation and equipment
- 6) Not for FKB
- 7) IT5 variants only in combination with Euro VI-D (MQ0, NEDC) and Euro VI-E (MR0, WLTP)
- 8) As 4×4 series; not as ultrawide-base tire; limited to maximum of 100 km/h; for FHS in combination with X2R, 130 km/h also possible
- 9) Different values for A2
- 10) Not free, but available only for authorities and only as N2 (only in combination with ZD0+Z2N+Z2E), not for RoW
- 11) Not in combination with Z2F (vehicles outside of EU/EFTA with CoC)
- 12) Only for Mexico
- 14) For rear wheel drive, not in combination with GCC countries
- 15) Only for Jordan and Angola in combination with Euro III
- 20) For Euro VI, not in combination with Z2F (vehicles outside of EU/EFTA with CoC) and not in combination with Z2E (EU/EFTA national version)
- 22) Only for RoW (Z2F)
- 23) Only for EU (Z2E)
- 24) Only for Z2E/Z2F with CoC complete (XC9), axle loads such as 5.5t (XL4)

3.3 Selecting the basic vehicle

General information

The operating conditions of the subsequent complete vehicle are crucial to the selection of a suitable basic vehicle when planning work on the vehicle body. Please note with regard to this:

- Requirement-based basic vehicle design
- Body variant
- Series equipment and special equipment

For better orientation, also use the identification plate, model designation, and the Vehicle Identification Number (VIN) when planning; See 3.6 Vehicle identification data (→ Page 61).

- i** On the Mercedes-Benz website, you can assemble your vehicle in the Configurator and view the available items of special equipment:

<https://www.mercedes-benz.com>



If this link does not automatically redirect you to the country-specific website for your region, or if you have any questions about vehicle equipment, contact your Mercedes-Benz Sales Partner or Service Partner.

In order to ensure the best possible use and optimal operation of the vehicle, it is essential to choose the basic vehicle carefully in accordance with the intended use.

To this end, please take particular note of the following points during the planning phase and adapt them to the application:

- Wheelbase
- Engine/transmission
- Permissible axle loads
- Permissible gross mass
- Vehicle center of gravity
- Series equipment and special equipment
- Compatibility of the driver assistance systems

! NOTE

When selecting the basic vehicle, take into account the remaining payload after the conversion to ensure it is adequate for the planned typical applications for the body.

! NOTE

Before carrying out any body mounting or conversion work, check the delivered basic vehicle for compliance with the necessary requirements.

! NOTE

Along with a user-friendly and maintenance-friendly design, the careful choice of materials and compliance with anticorrosion protection measures (→ page 102) are also of great importance when planning bodies.

Vehicles with raised center of gravity position

! NOTE

For vehicles that, following modification or body mounting work, have a center of gravity height greater than 1000 mm up to a maximum of 1300 mm, including when carrying a typical load for the body: When ordering the vehicle, it is essential to factor in special equipment designed specifically for this purpose with code B01; See 4.1.2 Maximum permissible position of the vehicle center of gravity (→ Page 78) and 4.2.1 Suspension (→ page 80).

! NOTE

Bodybuilder manufacturers that intend to assemble a camper van on the Sprinter must order code X2R "Mandatory for camper van conversion by bodybuilder manufacturer".

When code X2R is ordered, code B01 "Vehicle version for high load" is added automatically.

Further information on the basic vehicles and body variants on offer can be found in 3.2 Model overview (→ page 49) or obtained from the Technical Consultancy (→ page 18). Further information, e.g. the reference mass ruling for the Sprinter, speed limiter for EU registration of Class N2 vehicles (commercial vehicles > 3500 kg), can be obtained from the bodybuilder manufacturer information bulletins in the Bodybuilder Portal (→ page 24) or from the Technical Consultancy (→ page 18).

Information on Regulation (EU) 2021/535

Regulation (EU) 2021/535 must be observed. This specifies an even distribution of the payload, meaning that the distribution of the axle loads changes in relation to the previous regulation with the same payload.

Ex factory vehicles are delivered with the relevant measures taken (e.g. amended specifications for the maximum permissible axle loads and, depending on the vehicle variant, with the special equipment for a reinforced front axle code A50).

For vehicles that have body-manufacturer-specific installations, bodies or modifications, the bodybuilder manufacturer bears sole responsibility for compliance with the legal and vehicle-related specifications.

Bodybuilder Manufacturer Information Bulletin 01/2022 provided information about amended masses and dimensions in accordance with Regulation (EU) 2021/535 and the resulting restrictions on the availability of certain variants.

Bodybuilder Manufacturer Information Bulletin 29/2022 provides information about new weight variants and the possibilities for subsequently increasing the load capacity on model series 907 in connection with Regulation (EU) 2021/535:

Prohibited weight variant	Alternative weight variant
5500 kg (code XL4)	5200 kg (code XH2) 5380 kg (code XH3)
4100 kg (code XL2)	3880 kg (code XL8)

Details on the availability of these weight variants depending on the equipment installed can be found in this Bodybuilder Manufacturer Information Bulletin.

As part of a change to the registration type (e.g. N2 to M1 for special purposes), a type approval in the second stage or an individual inspection, it is technically permissible to increase the load capacity for the weight variants 5200 kg and 5380 kg to 5500 kg; the weight variant 3880 kg can be increased to 4100 kg accordingly.

If one of the previously mentioned load uprating options is required for your body solution, contact the Technical Consultancy (→ page 18).

For certain versions of closed model designations with the weight variants 4.1 t, 5.0 t and 5.5 t that have had to be blocked ex factory due to Regulation (EU) 2021/535, the option exists – depending on the equipment and explicitly only in combination with a body or modification of the vehicle by the bodybuilder manufacturer – to obtain a type approval in the second stage.

To this end, the special equipment "Mandatory curb weights in CoC document by bodybuilder manufacturer" (code Z3P) is available, with which the basic vehicle is delivered ex factory with incomplete CoC documents without the curb weight being entered. For the type approval in the second stage, the bodybuilder manufacturer must then determine the curb weight of the vehicle including the complete body or modification and present this to the registration authority for entry in the registration documents.

 Observe the information in Bodybuilder Manufacturer Information Bulletin 12/2023, which can be found in the Bodybuilder Portal.

If you have any questions, please contact the Technical Consultancy team (→ page 18).

It is recommended to coordinate the approval and registration requirements of a body or modification with the Technical Service and/or with the relevant type-approval authority during the planning phase.

It is important to ensure that Regulation (EU) 2021/535 Annex XIII and Regulation 1230/2012/EC continue to be complied with. This can be done by means of an acceptance inspection by the Technical Service at the bodybuilder manufacturer.

Information on engine cooling for a permissible gross combination mass greater than 5.5 t

NOTE

For vehicles with engine OM654 and a planned permissible gross combination mass greater than 5.5 t, a pre-installation for installing a trailer coupling with increased towing capacity is essential to ensure adequate engine cooling.

For this purpose, one of the following special equipment options must already be selected when ordering the vehicle in accordance with the weight requirement:

- Code QA5: Pre-installation of trailer coupling for trailer load up to 2800 kg
- Code QA3: Pre-installation of trailer coupling for trailer load up to 3500 kg

See 6.6.7 Trailer coupling (→ page 193).

In the case of vehicles without this special equipment, a subsequent increase in the permissible gross combination mass is not possible.

Information on changing the registration type from N2 to M2/M3

! NOTE

Note that when converting a panel van with N2 registration to vehicles of class M2 or M3, the code Z0X or Z0Y is mandatory in order to comply with legal requirements for external noise.

This must already be taken into account in the planning phase when the vehicle is ordered.

To this end, also observe Bodybuilder Manufacturer Information Bulletin 06/2023.

If you have any questions, please contact the Technical Consultancy team (→ page 18).

Vehicle configurations with "Mandatory assembly with Letter of Compatibility" (code X88)

Depending on vehicle configurations with the following codes, there may be an obligation for bodybuilder manufacturers to request a Letter of Compatibility.

These codes are linked with code X88 "Mandatory assembly with Letter of Compatibility" so that code X88 must be confirmed when the vehicle is ordered.

Code	Equipment
SV4	Airbag pre-installation for fully integrated vehicles with crash certificate
E3S	Pre-installation for seat occupancy recognition for driver's seat
J1E	Pre-installation for seat belt warning system for front passenger seat
JZ8	Omission of Blind Spot Assist
F50	Cab base vehicle/"cowl vehicle"
X5T	Mandatory 3-axle conversion
JA0	Omission of Crosswind Assist

This list represents the current status at the point in time of the publication of this edition of the Body/Equipment Mounting Directive. An expansion to include additional codes is possible due to modifications to the product range or due to legal requirements.

If you have any questions, please contact the Technical Consultancy team (→ page 18).

For information on applying for a Letter of Compatibility, See 2.1.3 Applying for Letters of Compatibility (→ Page 20).

i On the Mercedes-Benz website, you can assemble your vehicle in the Configurator and view the available items of special equipment:

<https://www.mercedes-benz.com>



If this link does not automatically redirect you to the country-specific website for your region, contact your Mercedes-Benz Sales Partner or Service Partner.

3.4 Vehicle modifications

Before starting body mounting work, the bodybuilder manufacturer must check that:

- the basic vehicle is suitable for the planned body,
- the vehicle and equipment remain suitable for the operating conditions even after the body mounting work has been carried out.

To aid in the planning of bodies, 2D drawings (offer drawings), product information and technical data can be requested from the Technical Consultancy, or this information can be retrieved through the communication system (→ page 24).

Furthermore, you should also note the special equipment that is available from the factory (→ page 74).

As supplied ex factory, all vehicles comply with EU and EC regulations and directives, UNECE regulations, and country-specific specifications (except for some vehicles for non-European countries).

! NOTE

After the modifications have been carried out by the bodybuilder manufacturer, the bodybuilder manufacturer must independently ensure that the vehicles still comply with all legal specifications in accordance with EU and EC regulations and directives, UNECE regulations and country-specific regulations, and/or the national laws and regulations applicable to the location in which the vehicle is to be used (including specifications for lighting systems, pedestrian protection, passive and active safety, etc.).

Inquire with the Technical Service for Vehicle Registrations with regard to the legal specifications relevant for your body or modification.

Observe additional country-specific regulations, such as accident prevention regulations or general machinery directives See 2.6 Accident prevention (→ Page 43).

It is recommended to coordinate the approval and registration requirements of a body or modification with the Technical Service and/or with the relevant type-approval authority during the planning phase.

! NOTE

After commissioning and before the vehicle is placed on the market, it must always be ensured that all existing vehicle and driving assistance systems are functioning flawlessly. For this, contact a technical service that has been designated by the responsible registration authority and that possesses the necessary knowledge and testing equipment.

For the disassembly and subsequent reinstallation of series production parts on the basic vehicle when performing modifications and body mounting work, the information and specifications in the Workshop Information System must be observed(→ page 26).

! NOTE

Adequate clearance must be maintained in order to ensure the function and operational safety of major assemblies.

⚠ WARNING

Any modifications to the steering and the brake system may result in these systems malfunctioning and ultimately failing. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Do not carry out any modifications to the steering or brake system!

Vehicle inspection

The bodybuilder manufacturer must inform the officially recognized approval official or inspector of any modifications to the basic vehicle. Acceptance and inspection organizations will decide on compliance with legislation and regulations after any modifications have been made to the basic vehicle, and therefore on the eligibility for registration of the complete vehicle.

! NOTE

Observe all national laws, directives, and registration regulations!

Acceptance inspections conducted by official test facilities or official approvals do not inevitably guarantee compatibility with all of the functions and systems in the basic vehicle.

If you have any questions with regard to ensuring compatibility, contact the Technical Consultancy (→ page 18).

3.5 Dimensions and weights

Modifications to the vehicle width, vehicle height or vehicle length must not exceed the limit values specified in the current version of the Body/Equipment Mounting Directive.

Take dimensions and weights from the 2D drawings (offer drawings) and technical data in the Mercedes-Benz Bodybuilder Portal (→ page 24), the ADVANTAGES product information system (→ page 29), and the technical limit values (→ page 78). These are based on a vehicle that is fitted with standard equipment. Items of special equipment are not taken into consideration.

Observe weight tolerances of +5% in production (according to DIN 70020 in Germany).

! NOTE

The binding weight specifications for the vehicle as shown in the vehicle documents and on the identification plate must be observed, See 3.6 Vehicle identification data (→ Page 61).

The permissible axle loads and the maximum permissible gross mass must not be exceeded.

Further information about permissible axle loads and other weight specifications (e.g. permissible gross mass or curb weight) can be obtained from your Mercedes-Benz Sales Partner or from the Technical Consultancy (→ page 18). You can also obtain information about weight modifications through this channel.

- ▶ Before starting any body mounting work, determine the actual curb weight and the corresponding axle loads at curb weight by weighing – see notes on weighing in 9.1.1 Determination of the center of gravity in the x-direction (→ page 391) – or using the Vehicle Identification Number and the curb weight of your vehicle ex factory and its load distribution via your Mercedes-Benz partner, and then document this information.
- ▶ Following completion of all body mounting work, measure the actual curb weight again by weighing; See 9.1.1 Determination of the center of gravity in the x-direction (→ Page 391). At the same time, determine the corresponding axle loads by weighing. The three variables of the curb weight, the front axle load and the rear axle load – all in the completed state – must all be documented again.
- ▶ In the case of a permanent axle load increase of 250 kg or more at the front axle on completion of all body mounting work compared to the actual basic vehicle delivered to the bodybuilder manufacturer (curb weight in standard configuration), the suspension must be measured and the correct toe and camber values must be set at the front axle.

These values can be obtained from your Mercedes-Benz Service Partner or directly from the Workshop Information System documents regarding wheel alignment (→ page 26).

Implement the information on the curb weight and corresponding axle loads before and after body mounting work; see "Checking the wheel alignment" in 3.11.3 Work before handing over the modified vehicle (→ page 73) as well as in 3.12 Special equipment (→ page 74), 4.2.9 Wheel alignment (→ page 84) and 6.1.1 General information on the suspension (→ page 106).

WARNING

The vehicle's tire load capacity must not be exceeded by overloading the vehicle beyond its specific gross vehicle mass. The tires could otherwise overheat and suffer damage. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

The braking distance may increase considerably when the vehicle is overloaded.

The permissible weights can be found in the vehicle identification data on the vehicle (→ page 61) and in the "Technical details" section (→ page 398).

WARNING

If the permissible axle loads or gross mass are exceeded on vehicles equipped with the ESP® system, this system may no longer function correctly. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Observe the permissible axle loads and gross mass.

For further information on the operating principle of the ESP®, See 8.9.1 Electronic Stability Program (ESP®) (→ Page 308).

Take into account here the permissible number of vehicle occupants and a sufficient margin for the payload.

Take the weight of special equipment into consideration when making calculations.

NOTE

Observe all national laws, directives, and registration regulations.

3.6 Vehicle identification data

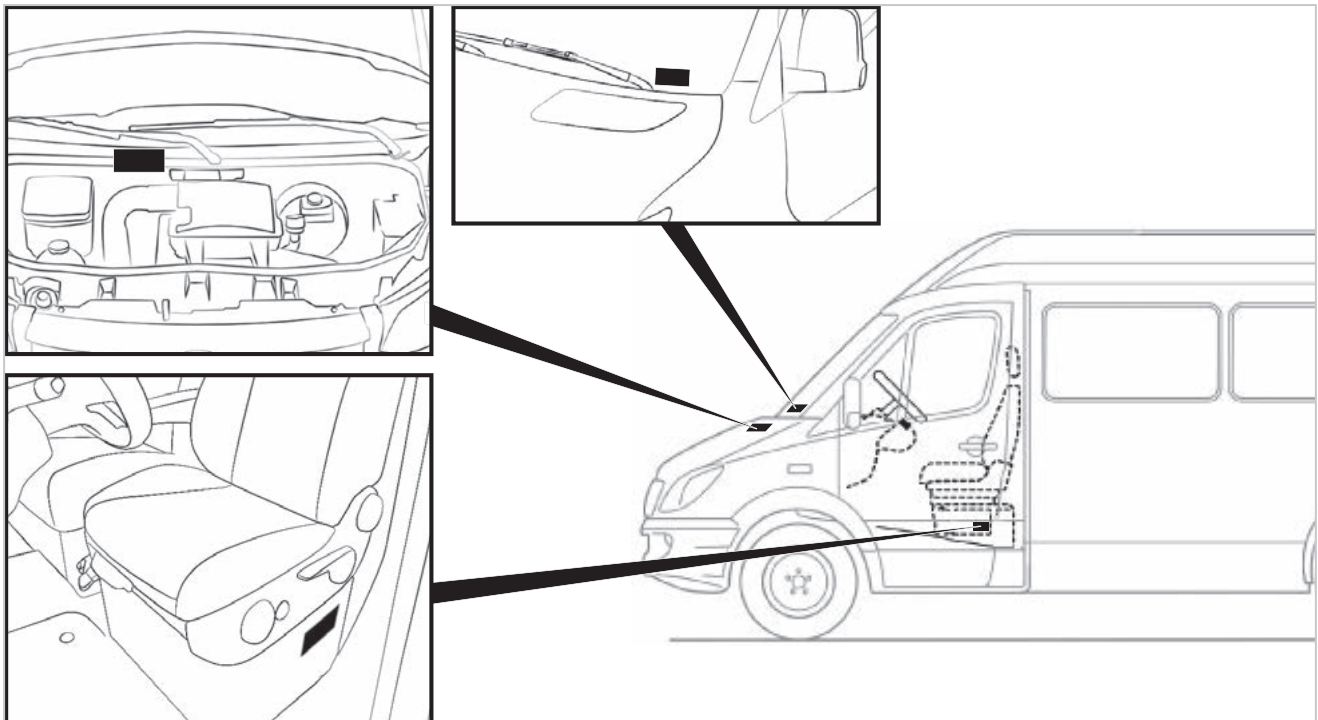
! NOTE

The vehicle identification number (VIN) and the identification plate of the vehicle must neither be changed nor be fitted to a different point on the vehicle.

The vehicle identification number is embossed in the belt rail in the engine compartment.

For certain countries, the externally visible vehicle identification number is attached as a plate to the lower end of the windshield.

The identification plate bearing the vehicle identification number and the permissible weights is located at the side on the seat box for the driver's seat.



Position of vehicle identification data

3.6.1 QR code rescue sticker



Attachment (example) of QR code rescue sticker at left B-pillar (driver's side)

All vehicles are supplied with a QR code rescue sticker from the plants at Düsseldorf and Ludwigsfelde. The rescue stickers in the Sprinter (except those with code F50) are affixed to the left and right B-pillars.

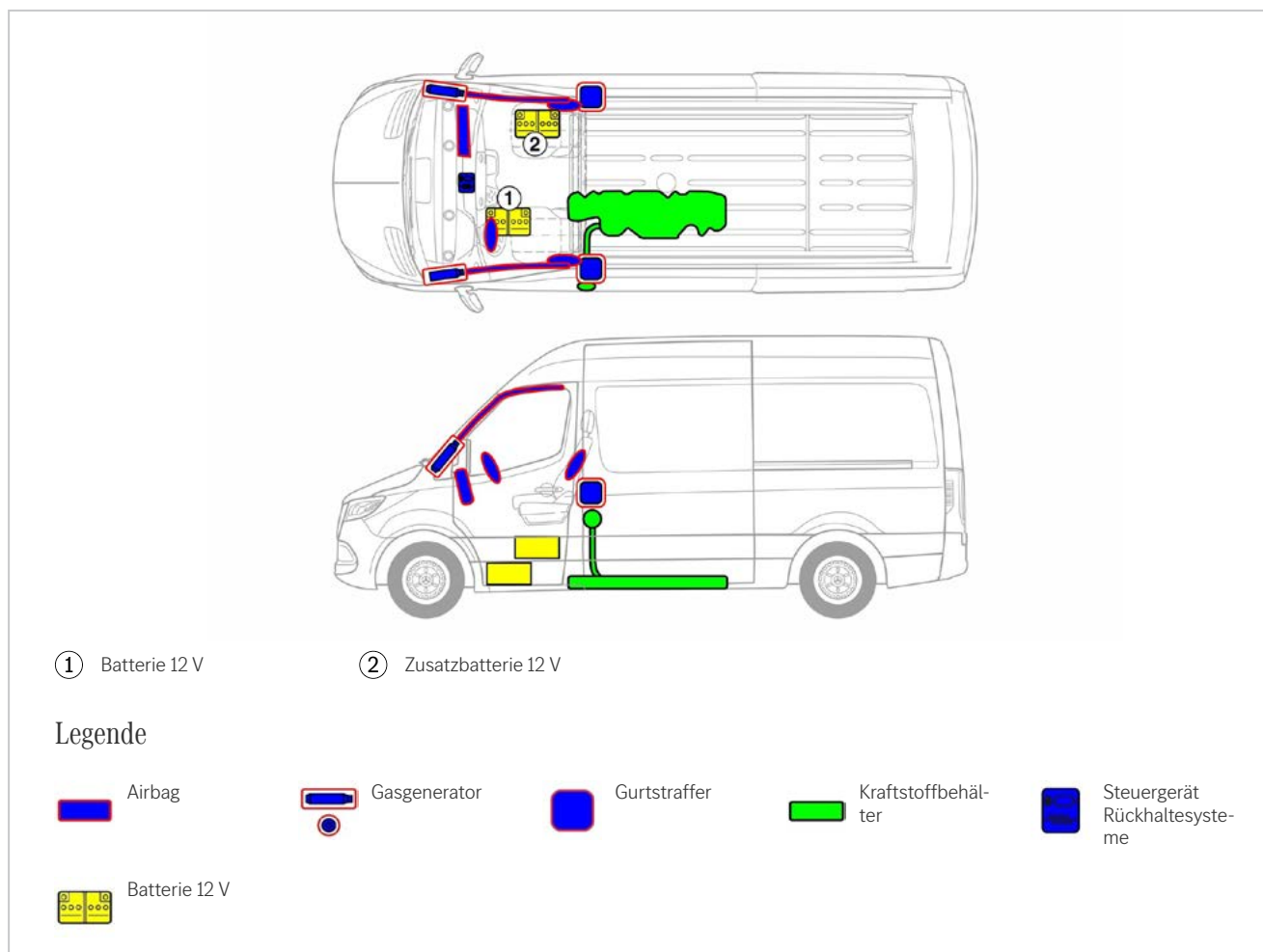
For vehicles with code F50, the rescue stickers are placed in the wallet with the vehicle documents.

When the QR code on the sticker is scanned using a smartphone or tablet and there is an internet connection present, the rescue card or the rescue data sheets for the vehicle is/are displayed. Rescue crews therefore immediately see where airbags, the battery, fuel tanks, high-voltage lines (above 30 V AC, above 60 V DC), pressure cylinders and other component parts critical for the rescue are located, enabling them to use rescue shears with low risk, for example.

The rescue card corresponds to the vehicle in the state that it was in when supplied by Mercedes-Benz.

If component parts that are visible on the rescue card (see following illustration) – where changes to these parts (in accordance with the specifications in this Body/Equipment Mounting Directive) are permissible – are removed, modified or added in any way whatsoever, then the rescue labels provided at the factory are rendered unusable.

In this case, the bodybuilder manufacturer must produce a new rescue card that illustrates this information for the modified vehicle as per ISO17840-1 to 4. The rescue cards can be made available either on the Internet (where they are accessed by scanning a QR code) or provided in the vehicle in printed form, preferably behind the driver's sun visor.



Example rescue card for panel van model series 907 (analogous for open model designations)

If the QR code is destroyed in the course of your body mounting work, but the rescue-relevant components remain unchanged, you must obtain a replacement via your Mercedes-Benz Service Partner and affix it to the previously mentioned positions on the vehicle. Alternatively, the same partner can provide you with a printed version of the vehicle's rescue card.

- ① Further information on the rescue sticker in Mercedes-Benz vehicles is available from:

<http://rk.mb-qr.com/en>



- ① Rescue cards for the various vehicle versions and model series can be viewed at:

<http://rk.mb-qr.com/en/#rescue-card-selector>



3.7 Vehicle stability

For acceptance of the vehicle with the body, calculated proof of the height of the center of gravity of the laden vehicle must be submitted in accordance with UN R 13 or in accordance with UN R 13H for Class M1 vehicles (cars).

Refer to the explanations on the center of gravity position in chapter 4 Technical limit values for planning (→ page 78).

In the case of bodies for cargoes with unfavorable centers of gravity (e.g. rear loads, high loads and side loads), Mercedes-Benz AG will make no statements concerning

- handling characteristics,
- braking characteristics,
- steering characteristics, and
- ESP® control response,

as these aspects are substantially affected by body mounting work, and can therefore only be assessed by the bodybuilder manufacturer.

! NOTE

On vehicles equipped with ESP®, where body mounting work causes an extreme displacement of the centers of gravity, deactivate the ESP® while observing and complying with all country-specific registration requirements.

If ESP® has been deactivated, the driver will then have to adapt their style of driving accordingly (reducing cornering speed, avoiding sudden steering wheel movements, etc.). When driving dynamics become critical, the vehicle behaves like a vehicle without ESP®. The permissible axle loads, gross vehicle masses, and center of gravity locations must be complied with.

If you have any questions or require further information, contact the Technical Consultancy (→ page 18).

The technically permissible wheel and axle loads and the permissible gross masses of the vehicle must not be exceeded due to the bodies or loading in the ready-to-drive condition.

! WARNING

If the permissible axle loads are exceeded on vehicles equipped with the ESP® system, this system may no longer function correctly. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Observe the permissible axle loads.

Following modifications to the vehicle that could result in changes to the longitudinal and lateral inclination (e.g. installation of heavy parts), the sensor cluster must be recalibrated in order to ensure that the ESP® functions correctly.

It is recommended to have this calibration work carried out by a qualified specialist workshop.

For further information on the operating principle of the ESP®, See 8.9.1 Electronic Stability Program (ESP®) (→ Page 308).

More information on permissible weights is available in the vehicle identification data on the vehicle (→ page 61).

3.8 Tires

The bodybuilder manufacturer must ensure that

- there is always sufficient space between the tire and the fender or wheel well, even when snow or anti-skid chains are fitted and the suspension is fully compressed (including allowing for the degree of axle articulation) (→ page 127), and that the relevant information in the 2D drawings (offer drawings) is observed,
- and that only tire sizes approved by Mercedes-Benz are used, sSee 4.2.3 Approved tire sizes (→ Page 83).

In particular, comply with country-specific regulations concerning the approval of tires. These regulations may define a specific type of tire for your vehicle or may forbid the use of certain tire types that are approved in other countries.

WARNING

Exceeding the specified tire load capacity or the permissible speed rating can lead to tire damage or tire failure. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Therefore, only fit tires of a type and size approved by Mercedes-Benz for your vehicle model and observe the tire load capacity required for your vehicle and the speed rating.

If you have other wheels (not approved by Mercedes-Benz) fitted:

- The wheel brakes or suspension system components could be damaged,
- Wheel and tire clearance is no longer guaranteed,
- This can cause the wheel brakes or components of the suspension system to no longer function correctly,
- Driving assistance systems may be subject to faults, delays, or may not even react at all.

 Observe the additional information and specifications See 4.2.3 Approved tire sizes (→ Page 83).

The determined speed information of the vehicle is displayed in the instrument cluster and is important for controlling the driving assistance systems. The display accuracy of the speedometer and the odometer is governed by law. The determination of the speed information depends on the tire size and the rolling circumference of the wheels and thus on their rim diameter. The rim diameter is always given in inches.

Therefore, the control units of the vehicle can be coded for the following wheel size groups:

Wheel size group	Tires model series 907
Wheel size group 1	195/75 R16C 205/75 R16C 235/65 R16C 235/60 R17C
Wheel size group 2	225/75 R16C
Wheel size group 3	225/75 R16C FA 285/65 R16C RA or 205/75 R16C FA 285/65 R16C RA
Wheel size group 4	245/65 R17XL

-  It is recommended that you stay within one wheel size group when changing tires.
- This avoids the need to recode control units.
 - If you change the wheel size, e.g. during a wheel change for winter operation, check their assignment to the wheel size group. If the assignment to the wheel size group changes, you must have the control units recoded at a qualified specialist workshop.
 - The display accuracy of the speedometer and the odometer will otherwise be outside the legally prescribed tolerance. It can also impermissibly deviate downward, i.e. the current vehicle speed is then higher than the speed indicated by the speedometer.
 - Driving assistance systems may be impaired in their function in the event of a deviation outside the tolerance range or may detect a fault and shut down.

3.9 Threaded connections and weld joints

3.9.1 Threaded connections

For the threaded connections between the body and the basic vehicle or the vehicle frame, the use of flanged bolts and flanged nuts is recommended.

If it is necessary to replace standard screws/nuts, it is only permitted to use screws/nuts:

- of the same diameter
- of the same strength
- of the same screw standard or screw type
- of the same surface coating (anticorrosion protection, friction factor)
- with the same pitch

The use of Mercedes-Benz standard parts is recommended.

WARNING

Do not modify any screw connections that are relevant to safety, e.g. that are required for wheel control, steering or braking functions. They may otherwise no longer function correctly.

This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Reassembly must be performed in accordance with Mercedes-Benz service instructions and using suitable standard parts. The use of Mercedes-Benz genuine parts is recommended.

- Apply VDI guideline 2862 to all assembly work.
- Do not reduce the screw grip clearance, make changes to a stretch shank, or use screws with a shorter thread clearance.
- Observe the settling properties of threaded connections.
- When re-tightening threaded connections on the overall vehicle, check whether a torque specification from Mercedes-Benz has been recorded and needs to be complied with by the bodybuilder manufacturer. Further information on this is available in the Workshop Information System (→ page 26)) or from your Mercedes-Benz Service Partner.
- The use of Mercedes-Benz tightening torques requires the screw connection to have a total friction factor in the range $\mu_{\text{tot}} = 0.08-0.14$.
- If screws are tightened to the required torque and tightening angle by Mercedes-Benz, a constructive modification is not possible.
- Use the Workshop Information System (→ page 26) to determine whether screws and nuts of suspension components must only be tightened when the vehicle is in ready-to-drive condition.
- Additionally clamped component parts of the body must exhibit an identical or higher strength than the previous clamped assembly.

 Information is available from any Mercedes-Benz Service Partner.

WARNING

Screws or nuts with locking splines, microencapsulated screws and self-locking nuts must always be replaced after a single use. Before new micro-encapsulated screws are screwed in, the mating thread must be recut or the nuts replaced to remove all residual screw locking compound. Finally, the recut through-tapped or blind-tapped holes must be blown out with air, because any adhesive residue in the thread would prevent correct tightening of the screws.

If these instructions are not observed, bending forces could act on the screw due to the lack of pretension and cause the screw to break. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

WARNING

There is a risk of injury when micro-encapsulated screws are loosened because of the sudden loosening of the screws. For this reason, ensure you have sufficient freedom of movement when loosening micro-encapsulated screws.

 Further information on special screw connections is available in the Workshop Information System (→ page 26)) or from the Technical Consultancy (→ page 18).

NOTE

Observe specifications for areas which may not be drilled on for threaded connections, in 6.2.1 General information on the body-in-white (→ page 112).

3.9.2 Weld joints

General information

In order to maintain the high standard of welding demanded by Mercedes-Benz, welding work must only be carried out by appropriately qualified welders.

For high-quality weld seams, we recommend that you:

- Clean the area to be welded thoroughly
- Make several short welding beads rather than one long bead
- Make symmetrical beads to limit shrinkage
- Avoid more than three welds at any one point
- Avoid welding in strain-hardened zones

! NOTE

Disconnect the battery before performing welding work.

Protect airbags, the airbag control unit, airbag sensors and components of the seat belts against weld spatters and remove them if necessary.

! NOTE

Observe specifications for areas in which welding is not permitted, in 6.2.1 General information on the body-in-white (→ page 112).

In addition, observe the information and specifications on preventing damage in 5.2 Welding work (→ page 101).

Selecting welding procedures

The mechanical properties of weld seams depend on selecting an adequate welding procedure and on the geometry of the elements to be joined.

If overlapping sheet metal is to be welded, the choice of welding method will depend on the access to the sides of the workpiece:

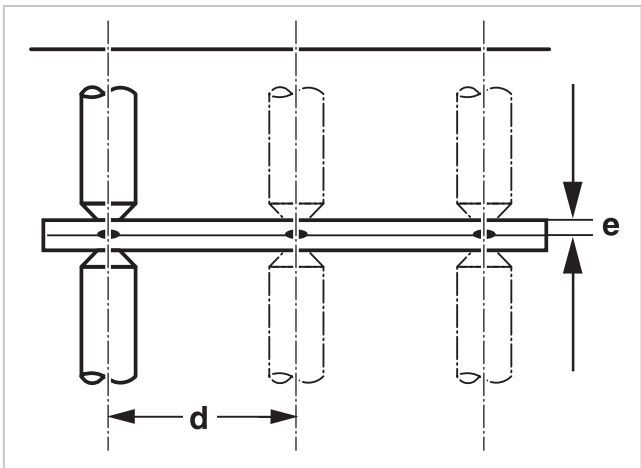
Access to side	1	Shielding gas plug welding
	2	Resistance spot welding

Resistance spot welding

Resistance spot welding is used for welding overlapping parts which are accessible from both sides. Avoid any spot welding of more than two sheet layers.

Distance between spot welds

To avoid shunt effects, the specified distances between the spot welds must be maintained ($d = 10e + 10 \text{ mm}$).

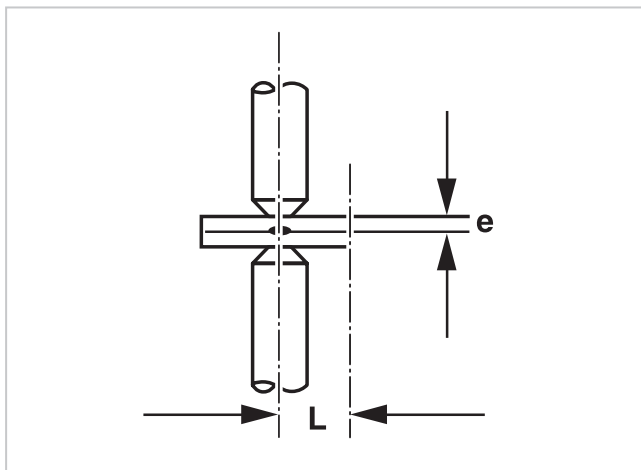


Ratio of sheet thickness to distance between spot welds

- d Distance between spot welds
- e Sheet thickness

Distance from sheet edge

To avoid melting core damage, the specified distances to the sheet edge must be maintained ($L = 3e + 2 \text{ mm}$).



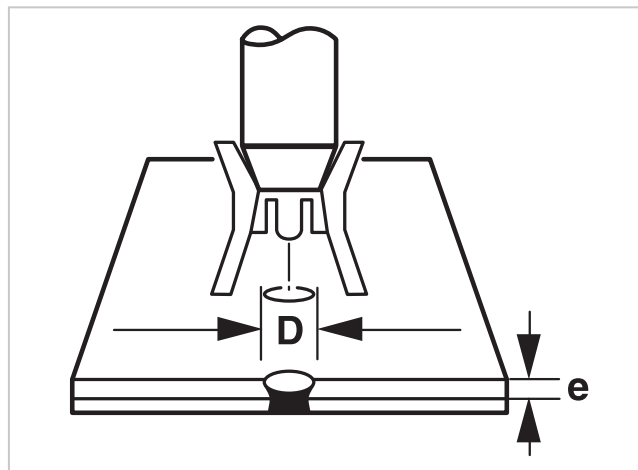
Ratio of sheet thickness to distance from the edge

- e Sheet thickness
- L Distance from sheet edge

Shielding gas plug welding

If overlapping sheets are only accessible from one side, use either shielding gas plug welding or tack welding for the weld joint.

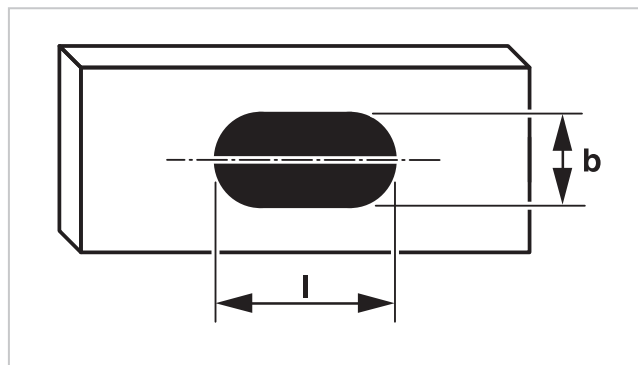
If the joint is produced by stamping or drilling followed by plug welding, the drilled area must be deburred before welding.



Ratio of sheet thickness to plug hole diameter

D = Plug hole diameter [mm]	4.5	5	5.5	6	6.5	7.0
e = Sheet thickness [mm]	0.6	0.7	1.0	1.25	1.5	2.0

The mechanical quality can be additionally improved by the use of "elongated holes" ($l = 2b$).

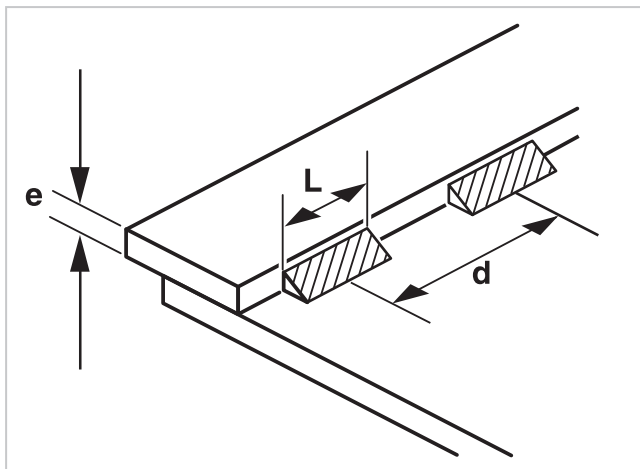


Ratio of width to length of elongated holes

- b Width of elongated hole
- l Length of elongated hole

Tack welding

In case of sheet thicknesses > 2 mm, overlapping sheets may also be joined by tack welding ($30 \text{ mm} < L < 40e$; $d > 2L$).



Dimensions for tack welding

- d Distance, tack weld
e Sheet thickness
L Length of tack weld

Anticorrosion protection after welding

After completion of all welding work on the vehicle, comply with the specified anti-corrosion protection measures (→ page 102).

3.10 Soundproofing

If modifications are carried out on any parts whose operation produces noise, e.g.:

- Engine
- Exhaust system
- Underfloor paneling
- Intake air system
- Tires, etc.

Carry out noise measurements.

! NOTE

Observe all national laws, directives, and registration regulations.

Observe UN R 51 in the EU.

Soundproofing parts fitted as standard must not be removed or modified.

The level of interior noise must not be adversely affected.

! NOTE

For all modifications to the vehicle, the external noise of the vehicle specified in UN R 51 must be complied with.

! NOTE

To ensure that modifications do not change the vehicle's sound levels, minimize the interior sound levels when planning bodies (→ page 162).

Observe the information and specifications on the underfloor paneling as a soundproofing measure in 6.2.13 Underfloor paneling (→ page 134).

In the case of a planned conversion of a panel van with N2 registration to a vehicle in classes M2 or M3, observe the note in 3.3 Selecting the basic vehicle (→ page 53).

3.11 Maintenance and repair

Maintenance and repair of the vehicle must not be made unnecessarily difficult by the body. Observe the Operator's Manual in this regard.

- Maintenance points and major assemblies must be easily accessible.
- Stowage boxes must be fitted with maintenance flaps or removable rear panels.
- To facilitate maintenance and repair work, ensure access to the fastening points of series production parts on the basic vehicle. This also affects parts on the door mechanism (e.g. hinges on hinged doors, guide rails/carriages of load compartment sliding door, etc.).

i Any additional work that arises due to the body in terms of warranty-related work, maintenance or repair work will be assessed in each instance by Mercedes-Benz AG, who may not accept liability for this. If you have any questions with regard to this, contact your Mercedes-Benz Service Partner.

- The battery box must be sufficiently ventilated, with provision for air to enter and exit.
- Check the condition and capacity of batteries and service them in accordance with the manufacturer's specifications (→ page 72).

! NOTE

In the event of extended periods of non-operational time, damage may occur to the on-board electrical system and, depending on the equipment, to the auxiliary batteries as well. This can be avoided by disconnecting the batteries and storing them in an appropriate manner (→ page 72) or by activating the deep discharge protection (Standby Mode) (→ page 285).

If the vehicle needs to be started using an external power supply, the jump-start connection point in the engine compartment must be used.

i Information and specifications on how to use the jump-start connection point correctly can be found in the Operator's Manual for your vehicle.

! NOTE

The auxiliary battery (code E2I, E2M) or other auxiliary batteries connected to the on-board electrical system via the battery cutoff relay must not be used to start the vehicle via an external power supply, as this could otherwise result in damage to the vehicle (→ page 269).

The following must be observed by the bodybuilder manufacturer before delivery of the vehicle:

- Check the headlamp adjustment or have this checked at a qualified specialist workshop. It is recommended to employ the services of a Mercedes-Benz Service Partner for this purpose.
- Retighten the wheel nuts/wheel bolts to the specified tightening torque.

It is recommended to adapt the scopes of maintenance work on the body to the vehicle using the Mercedes-Benz maintenance systems. This applies both to the scope and type of maintenance work, and for determining the service due dates based on time elapsed or mileage.

The Operator's Manual and maintenance procedures for the body mounting work carried out, along with any additional major assemblies installed, must be included in the vehicle by the bodybuilder manufacturer and they must be in the language of the country in which the vehicle is to be used.

3.11.1 Storing the vehicle

Storage in an enclosed space

- ▶ Clean the overall vehicle.
- ▶ Check the oil and coolant levels.
- ▶ Inflate the tires to 0.5 bar above the specified tire pressure.
- ▶ Release the parking brake and position the wedges.
- ▶ Observe 3.11.2 Maintenance and storage of batteries (→ page 72).

Storing the vehicle in the open (less than 1 month)

- ▶ Carry out the same steps as for storing in enclosed spaces.
- ▶ Close all air inlets and set the heating system to "Off".
- ▶ Observe 3.11.2 Maintenance and storage of batteries (→ page 72).

Storing the vehicle in the open (longer than 1 month)

- ▶ Carry out the same steps as for storing in enclosed spaces.
- ▶ Fold windshield wipers away from windshield.
- ▶ Close all air inlets and set the heating system to "Off".
- ▶ Observe 3.11.2 Maintenance and storage of batteries (→ page 72).

Maintenance work on the stored vehicle (in storage for more than 1 month)

- ▶ Check the oil level every month.
- ▶ Check the coolant every month.
- ▶ Check the tire pressure every month.

Removing the vehicle from storage

- ▶ Check the fluid levels in the vehicle.
- ▶ Observe manufacturer-specific information on operating fluids (e.g. maximum useful life, seasonal use, etc.) and, if necessary, replace the operating fluids.
- ▶ Set the tire pressure as specified.
- ▶ Check the state of charge of the on-board electrical system battery – and, depending on the equipment, the auxiliary batteries – and install the batteries.
- ▶ Clean the overall vehicle.

3.11.2 Maintenance and storage of batteries

For longer non-operational periods or periods of storage, ensure that the 12 V batteries (the on-board electrical system battery and, depending on the equipment, the auxiliary batteries – hereinafter simply referred to as the "batteries") are always in a charged state (more than 80% – corresponds to an open circuit voltage of approx. 12.55 V).

- ▶ For non-operational periods up to 4 weeks, disconnect the batteries at the beginning of the non-operational period, activate the standby mode (→ page 285), or isolate the battery main switch in vehicles with code E30.
- ▶ For non-operational periods exceeding 4 weeks, remove the batteries and store them in a dry place at temperatures of between 0 °C and 30 °C. Charge the batteries before putting them into storage. The residual charging current should be less than 5 A and the voltage higher than 13.5 V. Recharge the batteries every 4 weeks if installed in the vehicle, or every 6 months if removed from the vehicle, until the residual charging current is less than 5 A.
- ▶ Store the batteries in an upright position.
- ▶ The battery voltage must be kept above 12.55 V at all times.
- ▶ If the battery voltage drops below 12.55 V, but not below 12.1 V, the battery must be recharged.

! NOTE

If the open circuit voltage (under no load for 4 hours when removed) of the batteries drops below 11 V, they will be damaged and will have to be replaced.

- ① Observe further information and specifications on charging the batteries in 8.3 Battery (→ page 267) and in the Operator's Manual.

! NOTE

Under no circumstances must the batteries be charged by operating the vehicle in engine idle. Otherwise, this could cause irreparable damage to the components of the exhaust system.

3.11.3 Work before handing over the modified vehicle

The work and modifications carried out must be confirmed by the bodybuilder manufacturer by means of an entry in the maintenance booklet.

Checking the overall vehicle

- ▶ Check that the vehicle is in perfect condition.
- ▶ Rectify any damage.

Checking the brake system

- ▶ The brake fluid must be renewed regularly in accordance with Mercedes-Benz specifications, but at least every 2 years.
- ▶ If it is not known how long a vehicle equipped with a hydraulic brake system has been non-operational, the brake fluid must be renewed.
- ▶ Check electric and hydraulic lines for damage of any kind and replace these if necessary.

Checking the batteries

- ▶ Check and, if necessary, correct the state of charge of the on-board electrical system battery – and, depending on the equipment, the auxiliary batteries – before handing over the vehicle.

Checking the tires

- ▶ Before handing over the vehicle, check that the tires are inflated to the specified air pressure and check them for any damage.
- ▶ Replace any damaged tires.

Checking the wheel alignment

- ▶ It is recommended to have the suspension geometry (toe adjustment, etc.) checked if modifications have been made by body workshops.

! NOTE

Always comply with the information under 3.5 Dimensions and weights (→ page 59) on the curb weight and corresponding axle loads before and after body mounting work, 4.2.9 Wheel alignment (→ page 84) and 6.1.1 General information on the suspension (→ page 106).

Further information can be found in the Mercedes-Benz Workshop Information System (→ page 26).

Deactivating transport mode

To avoid excessive discharge of the battery during longer non-operational times, the electrical system of the Sprinter is in transport mode by default ex factory; this is only deactivated just before the vehicles are delivered.

In transport mode, the speed is limited to 40 km/h and not all the electrical and electronic systems are active. This results in lower electrical consumption and reduces load on the battery.

For bodybuilder manufacturers where conditions dictate that the vehicles are non-operational for long periods of time, transport mode is not deactivated at the Mercedes-Benz partner, but rather at the bodybuilder manufacturer at the end of the non-operational time.

It is possible to maneuver vehicles in transport mode. However, note that transport mode must always be deactivated before the vehicle is put into service on public roads.

To deactivate transport mode with combinations of control actions before the vehicle is put into service:

- ▶ Press the starter button once to switch on the ignition (terminal 15r). The hazard warning light system must be switched off.
- ▶ Switch on the left turn signal indicator for at least 2 blinking cycles.
- ▶ Switch on the right turn signal indicator for at least 2 blinking cycles.
- ▶ Switch off the turn signal indicator.
- ▶ Switch on the hazard warning light system for at least 2 blinking cycles.
- ▶ Switch off the hazard warning light system.
- ▶ Switch on the left turn signal indicator for at least 2 blinking cycles.
- ▶ Switch on the right turn signal indicator for at least 2 blinking cycles.
- ▶ Switch off the turn signal indicator.
- ▶ Press the starter button a second time (terminal 15) and check that the message "Transport Mode" is no longer displayed in the instrument cluster.

3.12 Special equipment

⚠ WARNING

The use of unapproved parts, major assemblies, conversion parts, or accessory parts can have detrimental effects on the safety of the vehicle.

Before starting work on the vehicle, please read the chapters of the vehicle's Operator's Manual relevant to assembly and the Operator's Manual and installation instructions from the manufacturers of accessories and special equipment. You could otherwise fail to recognize dangers that could put yourself or others at risk.

It is recommended to use the special equipment offered as codes by Mercedes-Benz to optimally adapt the vehicle to the body. You can obtain information on this from your Mercedes-Benz Service Partner or from the Technical Consultancy (→ page 18).

- i** On the Mercedes-Benz website, you can assemble your vehicle in the Configurator and view the available items of special equipment:

<http://www.mercedes-benz.com>



If this link does not automatically redirect you to the country-specific website for your region, contact your Mercedes-Benz Sales Partner or Service Partner.

Special equipment (e.g. reinforced springs, frame reinforcements, additional fuel tanks, stabilizers, etc.) or retrofitted equipment increases the curb weight of the vehicle.

- ▶ Determine the actual vehicle mass and axle loads by weighing before mounting.
- ▶ After completion of all body mounting work, determine the actual curb weight and the corresponding axle loads again by weighing. Comply with 9.1.1 Determination of the center of gravity in the x-direction (→ page 391), for example, to do so.
- ▶ Document the determined values.
- ▶ The specifications in the following chapters must be complied with or implemented using these values:
 - Chapter 3.5 Dimensions and weights (→ page 59) regarding the curb weight and the associated axle loads before and after the body mounting work,
 - Checking wheel alignment as per 3.11.3 Work before handing over the modified vehicle (→ page 73),
 - 4.2.9 Wheel alignment (→ page 84),
 - 6.1.1 General information on the suspension (→ page 106).

Not all auxiliary equipment can be installed in every vehicle without problems. This applies particularly to retrofitting.

- i** If you have any questions, please contact the Technical Consultancy team (→ page 18).

3.13 Adhesive films on the vehicle exterior

The specifications that follow are presented on the basis of the panel van (FKA) closed model designation. They also apply to the corresponding surfaces of the Tourer (FKB) closed model designation and the open model designation of chassis with cab or crewcab (FHS/FHL). For bodybuilder-manufacturer-specific bodies, these specifications must be applied accordingly.

- ❗ Illustrations of the open model designation with platform ex factory (code P02) can be found in the identically named chapter of the Body/Equipment Mounting Directive for the Sprinter model series 910.

With regard to adhesive films, note the following:

- Adhesive films may only be applied to the exterior surfaces of the vehicle that are not labeled as impermissible areas in the schematic illustrations below (FKA panel van by way of example).
- In areas with sensors, the attachment of adhesive film is generally not permitted; also refer to the information on Blind Spot Assist (code JA7), Side-guard Assist (code JT7) and the pre-installation for Blind Spot Assist (code J1V) from (→ page 318), as well as the information on the parking package (codes JB6/JB7) and the Moving-off information Assist (code JF7) from (→ page 344).
- Make sure to also observe the previously mentioned specification when applying special protective films (e.g. improved splinter protection for special-purpose vehicles or vehicles with infrared-absorbing sun protection films). The attachment of adhesive films in the black printing area with the multi-purpose camera cutout and the light/rain sensor (code LA2/JF1) is not permissible either in such use cases (see following illustration).
- All national legal requirements of any kind pertaining to the application of films or similar materials on vehicles must be observed and complied with. Observe, in particular, the regulations about masking off the windshield, window systems, and lighting systems.
- The notes and specifications from the bodybuilder manufacturer's "Small Conversions" guide and the "Design Quality for Vans" guide must be observed. Both can be found in the Bodybuilder Portal:

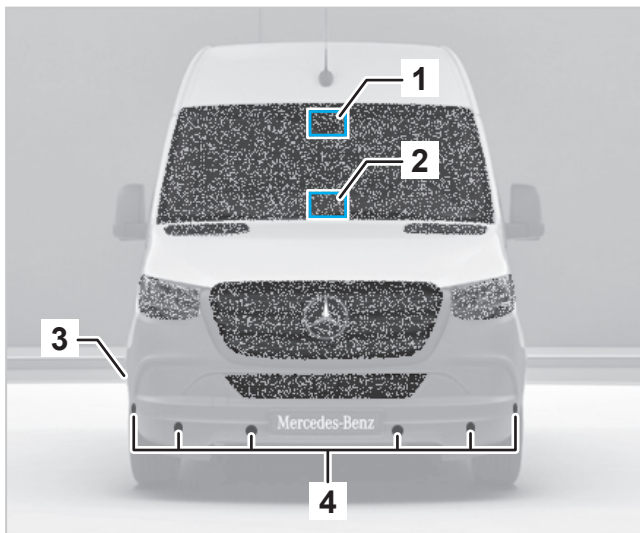
<https://bb-portal.mercedes-benz-vans.com>



⚠ WARNING

Depending on the type and thickness, paints or films can cause damping of radar waves. This could lead to malfunctions or system failure. This could cause the driver to lose control of the vehicle.

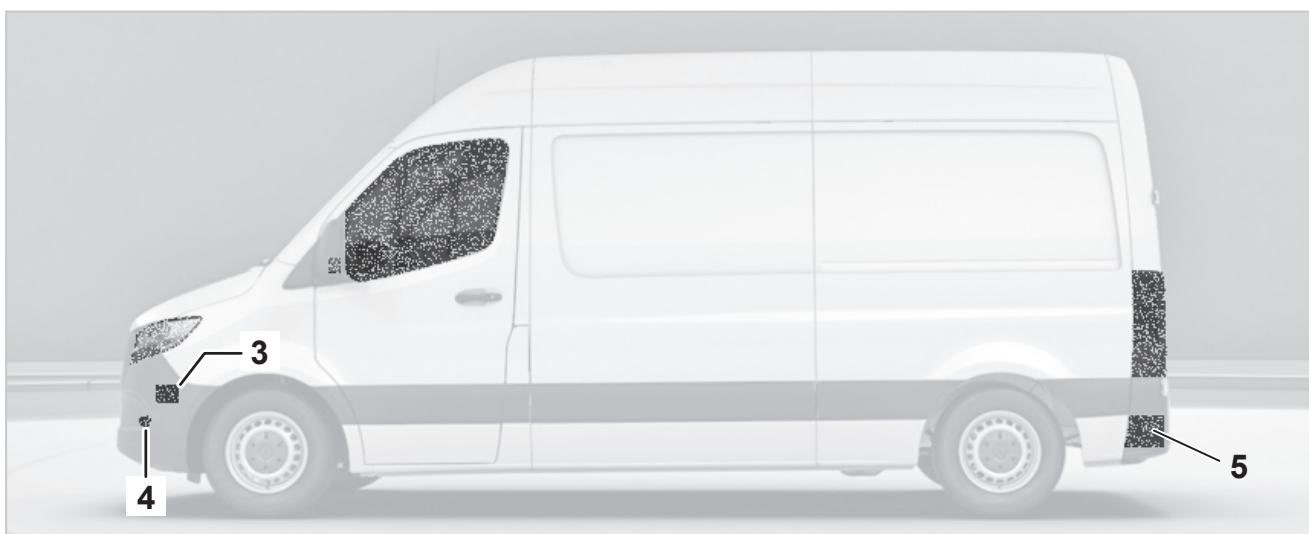
There is risk of an accident and danger to life and limb!



Vehicle front



Vehicle rear



Vehicle side

Speckled areas: Areas where adhesive applications on the vehicle are not permissible (example of FKA panel van, schematic)

- 1 Impermissible area in black printing with cutout for light/rain sensor (code LA2/JF1)
- 2 Impermissible area in black printing with cutout for multi-purpose camera
- 3 Impermissible area on **front passenger side** bumper with Sideguard Assist (code JT7); in side view, this is shown on the **driver's side**
- 4 Impermissible areas on the bumper with parking package (codes JB6/JB7) and Moving-off information Assist (code JF7)
- 5 Impermissible areas on the left/right side in the rear bumper with Blind Spot Assist (code JA7)

3.14 Vehicle markings with warning signs

For the positioning, fastening and attachment of additional parts required by law or authorities – such as warning signs for marking special-purpose vehicles (e.g. transport of dangerous goods) – the relevant information and specifications in the present Body/Equipment Mounting Directive and in the operating instructions must be observed, depending on the equipment, in addition to the legal requirements (e.g. driver's visibility or lighting equipment).

In particular, the specifications in the following chapters must be observed with regard to vehicle markings on the front of the vehicle:

- 4.4.3 Engine cooling (→ page 92)
- 8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC (→ page 314)
- 8.9.8 Parking packages/Moving-off information Assist (→ page 344)

! NOTE

After attaching the additional parts and before the vehicle is placed on the market, it must always be ensured that all existing vehicle and driving assistance systems are functioning flawlessly. For this, contact a technical service that has been designated by the responsible registration authority and that possesses the necessary knowledge and testing equipment.

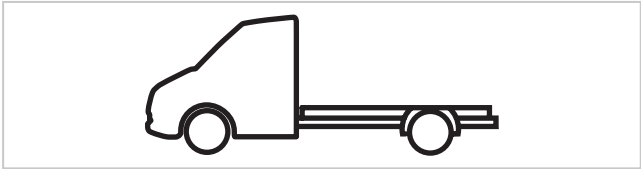
! NOTE

Observe all national laws, directives, and registration regulations.

4.1 Limit values for the basic vehicle

 **NOTE**

This chapter contains the technical limit values of the basic vehicle that are most important for planning work. Also observe the other sections of the current version of the Body/Equipment Mounting Directive.



4.1.1 Steerability

In all load states, the front axle load must represent at least the following proportion of the permissible gross mass of the vehicle:

General	Up to 4.2 t: at least 35% of the actual gross vehicle mass
	Up to 5 t: at least 30% of the actual gross vehicle mass

Comply with the permissible axle loads in all load situations.

The Sprinter is equipped with electric power steering.

 **WARNING**

Do not make modifications to the steering. This includes any change to and/or restriction of the steering angle.

Modifications to the steering can cause the electric power steering to fail.

There is risk of an accident and danger to life and limb!

4.1.2 Maximum permissible position of the vehicle center of gravity

x-direction	Resulting from the limit value for front axle load, See 4.1.1 Steerability (→ Page 78)
y-direction	On a laden vehicle, the wheel load (½ the axle load) is only permitted to be exceeded by 4%.
z-direction	Maximum 1000 mm ¹⁾ With code B01: 1000 to maximum 1300 mm ¹⁾

1) Height of vehicle center of gravity above roadway

Maximum permissible vehicle center of gravity height

The overall height of the center of gravity of a vehicle has a significant influence on its braking and handling characteristics. When designing bodies and attachments, care must therefore be taken to ensure that the center of gravity of the basic vehicle and of the laden vehicle is kept as low as possible.

The height of the overall center of gravity must be a maximum of 1300 mm above the roadway, since the Electronic Stability Program (ESP®) is only designed to maintain its basic functionality up to this point.

Irrespective of this, the tendency of vehicles to tip over rises as the vehicle center of gravity height increases, due to the laws of physics. The ESP® in the Sprinter is optimized to reduce the vehicle's tendency to tip over at overall vehicle center of gravity heights up to approx. 1000 mm.

 **NOTE**

On vehicles with a vehicle center of gravity height above 1000 mm up to a maximum of 1300 mm above the roadway, it is essential to ensure that the vehicles are equipped with the necessary suspension and ESP® variant of the special equipment "Vehicle design for high load" (code B01).

❗ Further information and specifications regarding the special equipment code B01 can be found in 4.2.1 Suspension (→ page 80).

❗ To determine the vehicle center of gravity height, observe the procedure in 9.1.2 Determination of the vehicle center of gravity in the z-direction (→ page 393).

Make sure that the correct variant is selected when ordering the vehicle. Ensure that the vehicle center of gravity points are determined and that the specifications are observed in 4.2 Limit values for the suspension (→ page 80).

❗ If you have any questions, please contact the Technical Consultancy team (→ page 18).

4.1.3 Permissible vehicle dimensions

Maximum permissible vehicle width

According to EU Directives
97/27/EC, 96/53/EC, Regulation (EU) 1230/2012,
and 2021/535 Annex XIII

General	2550 mm
---------	---------

The following table contains information on the vehicle width, which is contained in the approval for model series 907 and which must be observed for registration:

Body/vehicle width¹⁾ depending on outside mirror

Code	Equipment	Body width
–	Standard mirrors	≤ 2100 mm
FT1	Large mirror head	2100–2190 mm
FS1	Extended mirror bracket	2190–2300 mm
FS2	Extra-long mirror bracket	2300–2488 mm

1) In consequence of the current UN R 46 on outside mirrors

Additional information:

- With code FT1 – Large mirror head, the following codes are added automatically:
 - Code FU5 – Outside mirrors without turn signal indicators
 - Code L44 – Front side turn signal indicators
- Vehicles with a body width of 2100 mm or more are only available without the following code:
 - Code F64 – Electric folding outside mirrors

Maximum permissible vehicle height

According to EU Directives
97/27/EC, 96/53/EC, Regulation (EU) 1230/2012,
and 2021/535 Annex XIII

General	4000 mm
---------	---------

Vehicle length

According to EU Directives
97/27/EC, 96/53/EC, Regulation (EU) 1230/2012,
and 2021/535 Annex XIII

The different permitted vehicle length values, depending on the registration type, must be taken from the legal regulations.

4.2 Limit values for the suspension

4.2.1 Suspension

! NOTE

To ensure the correct selection and design of the suspension depending on the body or modification and its impact on the basic vehicle in terms of weight, center of gravity, etc., make sure to observe the following information and specifications as well as the detailed description in the Sprinter/eSprinter suspension design guide.

- i** You can find the suspension design guide in the Bodybuilder Portal after selecting the vehicle model in the "Technical Data" area under "Engineering and Information (Vans)".

Use of the suspension design

In cooperation with bodybuilder manufacturers, tailor-made solutions can be implemented for customer-specific requirements on the basis of the Sprinter. In order to achieve an optimal suspension design for the wide range of potential bodies, additional suspension configurations are available for customization purposes alongside the basic suspension. These are combined in different suspension packages:

Code	Suspension package
CB1	Suspension for comfort and sensitive loads
CB2	Comfort suspension with air suspension
CB7	Stabilization stage I
CB8	Stabilization stage II
CB4	Stabilization stage III

In addition, a particular special equipment code "Vehicle design for high load" (code B01) is available for vehicles used in high-load applications. This contains a package of suspension, tires, ESP® parameters, and driver assistance system parameters that have been tailored and developed specifically for bodybuilder manufacturer vehicles with a vehicle center of gravity more than 1000 mm up to a maximum of 1300 mm above the roadway.

! NOTE

The specifications on maximum center of gravity height and the special equipment code B01 in 4.1.2 Maximum permissible position of the vehicle center of gravity (→ page 78) must be observed.

When ordering code B01 ex works, code CB8 (stabilization level II) is automatically added. With code B01, code CB4 (stabilization level III) is also optionally available.

For the following vehicle configurations, code B01 can only be ordered in combination with code CB4:

- Permissible gross mass 5.2 t (code XH2), 5.38 t (code XH3) and 5.5 t (code XL4) and wheelbase R2 (3665 mm, code IR4)
- Permissible gross mass 5.0 t, wheelbase R2 (3665 mm, code IR4) and reinforced front axle (code A50)

! NOTE

For model designations 907.153 and 907.253 (FHS and FHL, 5.5 t, body length A2) with all-wheel drive (code A4M), additional measures are required for applications involving high loads and a vehicle center of gravity positioned over 1000 mm above the road-way.

When planning bodies for high loads, check whether code B01 "Vehicle version for high load" is provided ex factory with the selected vehicle specification. If not, contact the Technical Consultancy during the planning phase (→ page 18).

If code B01 is not available, a Letter of Compatibility is generally required for placing the vehicle on the market (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Clear recommendations for selecting the right suspension package according to body type and model designation follow. If the modification/body required by the customer is not listed in the following, make your decision based upon the vehicle characteristics listed and select the variant that comes as close as possible to the body in question; alternatively, contact the Technical Consultancy for further advice on how to proceed (→ page 18).

Special equipment available ex factory (model series 907)

Code	Equipment
A50	Reinforced front axle
B01	Vehicle version for high load with center of gravity above 1000 mm up to a maximum of 1300 mm
CW2¹⁾	Omission of vehicle lowering (only for model series 907 rear wheel drive 3.0/3.5 t, not available for all-wheel drive)
C47	Reinforced shock absorbers
C83	GRP rear axle springs
CB1	Suspension for comfort and sensitive loads
CB2	Comfort suspension with air suspension (model series 907 only)
CB7	Stabilization stage I – Enhanced roll stabilization with augmented body damping in comparison to basic suspension
CB8	Stabilization stage II – Enhanced roll stabilization with augmented body damping in comparison to CB7
CB4	Stabilization stage III – Maximum roll stabilization with augmented body damping in comparison to CB8
CE8	Body raised at rear axle for special applications
CT1	Vibration absorbers at the rear axle (model series 907 only)

- 1) Note that through the addition of code CW2, the body level is raised by approx. 30 mm. On closed model designations of weight variant IT4 and body lengths A3 and A4 (model designations 907.635, 907.637, 907.735), code CW2 is added as standard ex factory. It is not possible to deselect code CW2 for these model designations.

Content of the suspension packages

Code	Equipment	Benefit
CB1	Suspension for comfort and sensitive loads	Enhanced comfort due to adapted damper characteristics and the use of a progressive spring characteristic
CB2	Comfort suspension with air suspension	Raisable and lowerable for loading and unloading. Increased driving comfort through a spring characteristic that adapts to the load condition (soft with a low load); enables the same driving level regardless of the load condition.
CB7	Stabilization stage I	Increased roll stabilization and body damping compared to the basic suspension; CB7 compensates for strong roll behavior through increased loading/expansion centers of gravity. Depending on the tonnage, torsion bars are automatically added for the front and rear axles, or torsion bars with an increased diameter are used. Furthermore, dampers with an adapted characteristic are installed.
CB8	Stabilization stage II	Additional roll stabilization and body damping compared to stabilization stage I; compulsory addition of increased torsion bar diameters at the front and rear axles and an adapted damper characteristic; in addition, rear axle springs with increased stiffness are used.
CB4	Stabilization stage III	Suspension with maximum roll stabilization for bodybuilder manufacturer vehicles, CB4 includes the stabilization stage II (CB8) with adapted springs/dampers, stabilizer characteristics and the use of a progressive spring characteristic.

4.2.2 Permissible axle loads

WARNING

If the permissible axle loads are exceeded on vehicles equipped with the ESP® system, this system may no longer function correctly. Furthermore, any overloading may result in damage to the suspension and structural parts. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

The permissible axle loads as per the identification plate and vehicle documents must be complied with.

NOTE

Observe the permissible axle loads for the vehicle as listed in the vehicle documents and on the identification plate, See 3.5 Dimensions and weights (→ Page 59) and 3.6 Vehicle identification data (→ page 61).

-  Further information on the permissible axle loads – depending on the vehicle model designation and equipment – is available from your Mercedes-Benz Sales Partner or from the Technical Consultancy (→ page 18).

Permissible axle loads

Permissible gross mass [t] (code)	Front axle ¹⁾ [kg]	Rear axle ¹⁾ [kg]
3.0 (IT3)	1650/1740 ²⁾	1800
3.5 (IT4)	1650/1860 ³⁾	2250
4.0 (IT9)	1860/2000 ³⁾	2430
5.0 (IT5)	1850/2000 ³⁾	3500
5.5 (XL4)	2100 ³⁾	3600

1) Depending on model designation and equipment

2) For Tourer (FKB) model designation with front axle with increased load capacity stage 1 (code A49)

3) With front axle with increased load capacity (code A50) or with all-wheel drive (code A4M; not for code XL4)

4.2.3 Approved tire sizes

! NOTE

The information and specifications in 3.8 Tires (→ page 65) must absolutely be observed.

! NOTE

The use of appropriate tire sizes only applies in the context of the approved and intended tread designs as well as weight and speed ratings. An overview of the permissible wheel-tire combinations as well as changes and current tire values can be found at:

<https://www.mercedes-benz.de/vans>



Here, you can also find the information "Genuine parts, accessories & tires", "Tires and wheel and tire assemblies" and "Wheel and tire combinations" to download for "Sprinter model series 907/910".

- i** If this link does not automatically redirect you to the country-specific website for your region, or if you have any questions regarding the specifications for the planned usage market for your vehicles, contact your Mercedes-Benz Sales Partner or Service Partner.

! NOTE

Observe all national laws, directives and regulations.

4.2.4 Turning circle diameter

See:

- European Union: 97/27/EC, Regulation (EU) 1230/2012 and 2021/535 Annex XIII depending on the vehicle class and model
- European Union: 96/53/EC

Wheelbase [mm]	Turning circle diameter [m]		
	4×2 (3.0–4.1 t)	4×2 (5.0–5.5 t)	4×4 (3.5–5.0 t)
BR 907			
3665	13.4	13.2	14.0
4325	15.3	15.2	16.1

4.2.5 Modifications to the axles

Do not make modifications to the suspension or axles (→ page 106).

4.2.6 Modifications to the steering

The Sprinter is equipped with electric power steering.

⚠ WARNING

Do not make modifications to the steering. This includes any change to and/or restriction of the steering angle.

Modifications to the steering can cause the electric power steering to fail.

There is risk of an accident and danger to life and limb!

4.2.7 Modifications to the brake system

Do not make modifications to the brake system.

Do not make modifications to the disk brake air inflow and air outflow (→ page 108).

Under Section 41 Para. 14 StVZO (German vehicle licensing regulations), all vehicles with a permissible gross mass above 4 t must be equipped with chocks.

4.2.8 Modifications to springs, spring suspension, and dampers

Modifications to springs and dampers can only be made if they are matched at the front and rear. Only the combinations provided for ex factory must be used.

Further information is available from the Technical Consulting team (→ page 18). A Letter of Compatibility may also be required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

On no account should springs and dampers be used if they do not correspond to the characteristics of standard parts or parts obtainable as special equipment. The use of Mercedes-Benz standard parts is recommended.

Do not make modifications to the spring suspension (→ page 106).

4.2.9 Wheel alignment

Do not make modifications to the steering geometry, in particular wheel alignment settings (→ page 106).

If the body or any permanent equipment that belongs to the vehicle leads to an increased curb weight, or if the vehicle is mostly used in a fully loaded state, refer to the notes in 6.1.1 General information on the suspension (→ page 106). This could apply to, for example, recovery vehicles, fire-fighting vehicles, ambulances, semitrailer trucks, workshop vehicles, or camper vans.

4.3 Limit values for the body-in-white

4.3.1 Modifications to the body-in-white

See also 6.2 Body-in-white and body (→ page 112).

- Modifications must not be made to the front-end assembly structure, including longitudinal members and crossmembers, from the front of the vehicle up to and including the B-pillar.
- Do not make modifications to the rear portal including the roof area.
- In the event of modifications to the load-bearing structure, the total equivalent rigidity of the structure fitted by the bodybuilder manufacturer must at least equate to that of the series production vehicle.
- In the case of bodies or installations that are fitted to the body-in-white by means of a frictional connection, proof of structural durability must be submitted upon request.
- Sufficient clearance for fuel filler necks, fuel tank lines and fuel lines must be maintained.
- If modifications are made to the side wall on the panel van or the crewbus, the rigidity of the modified body must be equal to that of the basic vehicle.
- If bodies are mounted on basic vehicle cabs, protection of the fuel system may be necessary depending on the body type, See 6.3.1 Fuel system (diesel) (→ Page 136).

NOTE

Observe specifications for areas in which drilling and welding are not permitted, in 6.2.1 General information on the body-in-white (→ page 112).

4.3.2 Limit values for the vehicle frame

If the frame is extended, the material of the extension element must have the same quality and dimensions as the standard chassis frame, See 7.1.1 Material quality, general (→ Page 201).

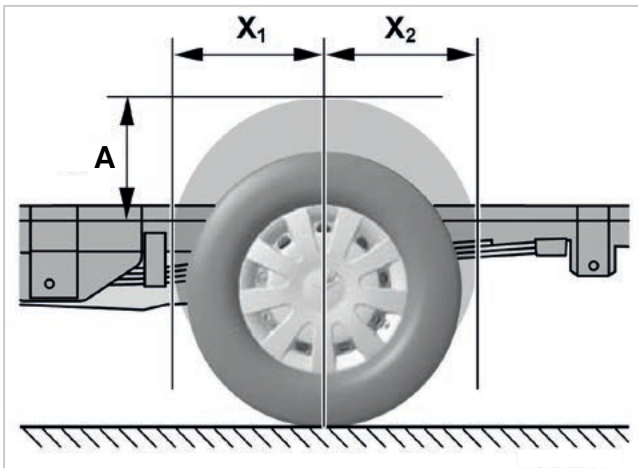
4.3.3 Lower wheel well on rear axle for closed model designations

On closed model designations, the wheel wells are designed for the wheel/tire combinations and the jounce travel (dependent on the vehicle mass and vehicle dynamics).

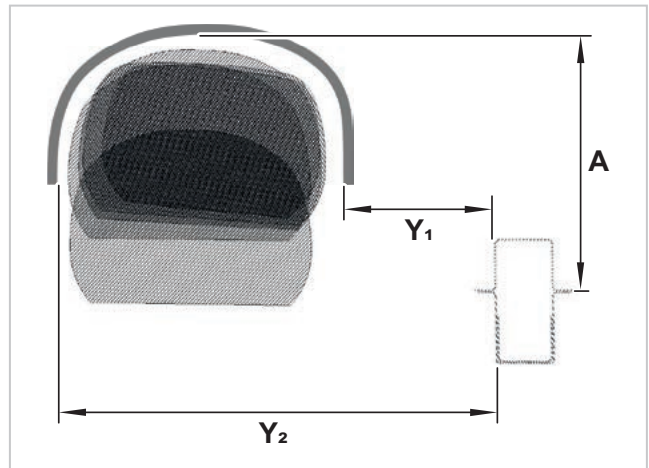
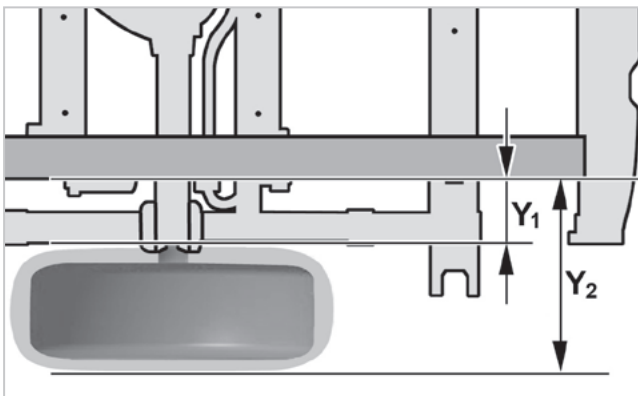
For this reason, modifications to the wheel wells are generally prohibited for closed model designations.

In exceptional cases, a change can be approved provided that this has been coordinated with the Technical Consultancy in advance (→ page 18). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

4.3.4 Lower wheel well on rear axle for open model designations

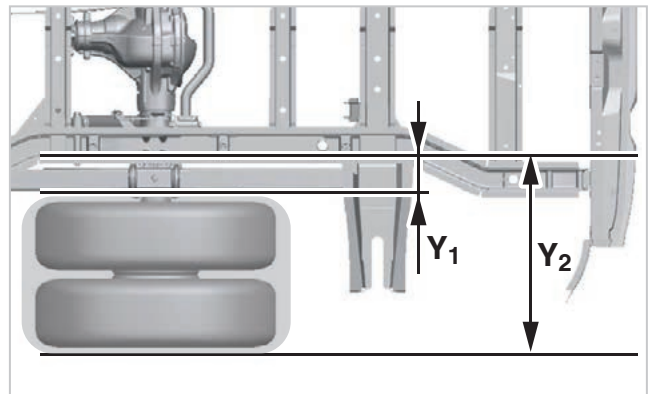


Limit values for chassis wheel wells



Minimum clearance A of the wheel well to the flange between the upper and lower flanges of the longitudinal frame member

The diagram below shows the reference edges for the dimensions Y_1 and Y_2 on vehicles with a tapering frame section (twin tires).



5 t vehicle frame with taper

Limit values/minimum distance between wheel well/fender and vehicle frame

Permissible gross mass [t]	Tires	Dimensions [mm]				
		X ₁	X ₂	Y ₁	Y ₂ ²⁾	A ³⁾
3.0	205/75 R16 C	410	410	195	520	260
3.5	235/65 R16 C	410	410	195	520	260
4.6	285/65 R16 C	445	445	245	635	260
4.6–5.0 ¹⁾	2 × 195/75 R16 C	405	405	120	630	240
	2 × 205/75 R16 C	410	410	115	635	250
3.5 (all-wheel drive)	225/75 R16 C	430	430	195	510	200
5.0 (all-wheel drive) ¹⁾	2 × 205/75 R16 C	410	410	115	638	190
5.2–5.5 ¹⁾	2 × 205/75 R16 C	410	410	115	635	250

1) On vehicles with twin tires, the inside of the inner wheel has been used for Y₁ and the outside of the outer wheel for Y₂.

2) With maximum wheel arch liner to wheel center

3) Minimum distance from frame flange to wheel well contour

The minimum distance A of the wheel well is measured from the flange between the upper and lower flanges of the longitudinal frame member up to the lowest point of the wheel well contour.

The dimensions Y₁ and Y₂ are measured from the outer surface at the longitudinal frame member.

Observe the following for dimension A with the following special equipment:

- Omission of vehicle lowering – code CW2: Dimension A is unaffected.
- Raised body for special applications – code CE8: Dimension A can be reduced by 35 mm.

 Observe the additional specifications for modifications to the wheel wells in 6.2.8 Fenders and wheel wells (→ page 127).

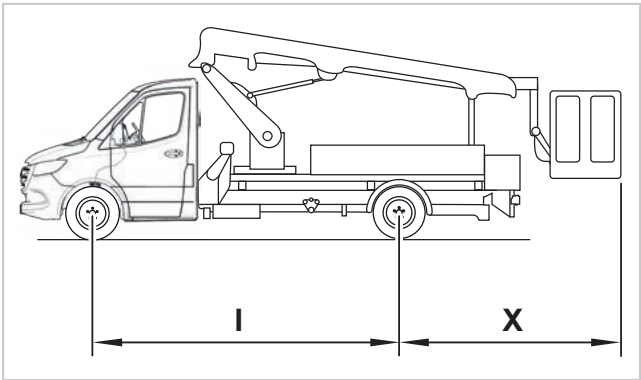
Additional tire information See 4.2.3 Approved tire sizes (→ Page 83).

4.3.5 Overhang

The maximum overhang without exceeding the permissible axle loads and centers of gravity is:

Maximum overhang	
Wheelbase I [mm]	Overhang X [mm]
3665	1850
4325	2200 ¹⁾ /2421 ²⁾

- 1) Overhang up to 2200 mm:
- Driving operation with trailer load is permissible.
 - Observe the following specifications on the permissible trailer loads and code O19 if the trailer coupling point has been relocated (→ page 89).
- 2) Overhang over 2200 up to max. 2421 mm:
- Driving operation is only permissible without trailer load.
 - Ensure that the Trailer Stability Assist (TSA) function with O-code O19 is deactivated and "coded out". In the case of camper vans (with code X2R) based on FHS/FHL or wheelbase extensions in combination with code O03, TSA is automatically deactivated by the system.
 - Observe the further information and specifications on the overhang extension in 6.2.4 Overhang extension (→ page 119) and on the Crosswind Assist in 8.9.2 Crosswind Assist (CWA) (→ page 309).



Maximum overhang using the example of an elevating work platform

- I Wheelbase
X Overhang

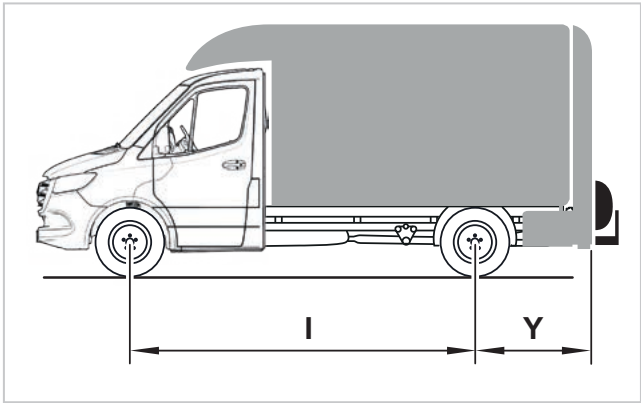
Special considerations for camper vans

The following table for the maximum possible overhang on camper vans applies only in combination with code X2R (Mandatory for camper van conversion by bodybuilder manufacturer).

The reference dimension for the maximum possible overhang ranges from the wheel center on the rear axle to the rear panel of the camper van body. Protruding component parts such as trailer couplings or bumpers are not taken into account.

Maximum overhang for code X2R	
Wheelbase I [mm]	Overhang Y [mm]
3665 (code IR4)	2200
4325 (code IR6)	2600

Camper vans are permitted to have a maximum overhang of 60% of the wheelbase, up to a maximum of 2600 mm.



Maximum overhang for camper vans up to the rear panel without protruding component parts (schematic)

- I Wheelbase
Y Overhang

- i** Observe the additional information and specifications on the overhang extension in 6.2.4 Overhang extension (→ page 119), on the Crosswind Assist in 8.9.2 Crosswind Assist (CWA) (→ page 309),

and on the Blind Spot Assist/Sideguard Assist in 8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist (→ page 318).

Permissible trailer loads when relocating the trailer coupling point

Permissible gross mass [t]	Wheelbase [mm]	Relocation of trailer coupling point from standard configuration				
		0–200 mm [kg]	201–500 mm [kg]	501–600 mm [kg]	601–700 mm [kg]	> 700 mm [kg]
3.5	3665	3500	3000 ¹⁾	2500 ¹⁾	2000 ¹⁾	¹⁾ + ²⁾
4.1	4325	3500	3000 ¹⁾	2500 ¹⁾	2000 ¹⁾	¹⁾ + ²⁾
5.0	3665	3500	3500 ¹⁾	3000 ¹⁾	2500 ¹⁾	¹⁾ + ²⁾
	4325	3500	3500 ¹⁾	3000 ¹⁾	2500 ¹⁾	¹⁾ + ²⁾

- 1) If the trailer coupling point is relocated by over 200 mm in relation to the standard configuration, the Trailer Stability Assist (TSA) must be deactivated through the use of code O19.
In camper vans (with code X2R) based on FHS/FHL, the TSA is automatically deactivated for system-related reasons.
- 2) If the trailer coupling point is relocated by over 700 mm in relation to the standard configuration, contact the Technical Consultancy (→ page 18).

4.3.6 Attachment to the frame

- Implement the attachment to the frame as per 6.2.2 Attachments on the frame (→ page 117) and 7.1.4 Attachment to the frame (→ page 204).
- At minimum, all the factory-fitted body consoles must be used to fasten bodies to the vehicle frame (→ page 204). These permissible body mounting points are located only on the longitudinal frame members and may be supplemented as required by additional mounting supports on the longitudinal frame members.
- The body must be secured using two screws for each mounting support.

4.3.7 Wheelbase modifications

! NOTE

With the new generation of driving assistance systems and the legal specifications that apply to this, a modification to the wheelbase for closed and open model designations for BR 907 is not permitted.

If a wheelbase modification is required for a special body or modification, contact the Technical Consultancy during the planning phase (→ page 18). In any event, note the obligation to check compatibility (→ page 19), which can be applied for via the Bodybuilder Portal (→ page 20).

4.3.8 Roof and roof load

Maximum roof load		
Panel van [kg] LH1	High roof panel van [kg] LH2	Cab Crewcab [kg]
300	150	100

The information in the table applies to dynamic driving situations.

For panel vans with a standard roof (code LH1) and high roof (code LH2), the maximum permissible static roof load is 300 kg, assuming that the vehicle is parked on level, horizontal ground.

The information in the table applies with use of the mounting rails available ex factory for roof luggage racks (mounting rails, code D13) in combination with a carrier system designed for the roof load and with an even load distribution across the entire roof area.

Always make allowances here for the dead weight of any carrier systems, and subtract it from the maximum permissible roof load.

Roof bows or structural parts must not be removed or damaged without being replaced.

The connection between the roof bow and the side wall must be of sufficient bending resistance (→ page 123).

Wheelbase [mm]	Quantity required
3665	≥ 5 bows
4325	≥ 6 bows

Roof bows	Position
1	Behind the front doors (B-pillar)
2	At the center of the sliding door to the load compartment (between the B and C-pillars)
3	At the center of the vehicle behind the sliding door to the load compartment (C-pillar)
4–6	Between the C-pillar and the rear area of the vehicle (D-pillar)

Roof height increase [mm]	Moment of inertia I_x per roof bow [mm ⁴]
≤ 250	≥ 40000
≤ 400	≥ 65000
≤ 550	≥ 86000

For modifications to the projected side surface area, refer to 8.9.2 Crosswind Assist (CWA) (→ page 309).

- ⓘ Observe the additional specifications on modifications to the roof in 6.2.11 Panel van and crewbus roof (→ page 128) and on attachments to the roofs in 6.6.3 Roof luggage racks (→ page 178).

4.4 Limit values for engine peripherals and drivetrain

- i** Vehicles with engines OM651 and OM642 are no longer available ex factory. For this reason, this information is no longer included in the present Body/Equipment Mounting Directive. If required, you can find corresponding information and specifications in previous Body/Equipment Mounting Directives in which the engines are still included. These are still available in the Body/Equipment Mounting Directives Archive in the Bodybuilder Portal.

4.4.1 Fuel system

For modifications to the fuel system, observe the specifications in 6.3.1 Fuel system (diesel) (→ page 136).

4.4.2 Modifications to the engine and drivetrain

Modifications to the engine timing and performance enhancement

! NOTE

Any interference in the engine timing by customers and bodybuilder manufacturers is not permissible.

Manipulation of or modification to the engine timing causes changes in the certified engine data and emission values, and thus results in the immediate invalidation of the operating permit.

Modifications to the engine and drivetrain

- Do not make modifications to the intake air system, See 6.3.5 Engine air intake (→ Page 147).
- As a general rule, do not make modifications to the propeller shafts, See 6.3.7 Propeller shafts (→ Page 148). If modifications (e.g. different lengths) are necessary, have these carried out by a company qualified in propeller shaft engineering. This company shall be responsible for the effects on the vehicle (e.g. NVH behavior, durability, package). When modifications are made, the fastening elements must not be reused.
- It is not possible to retrofit any engine speed regulation equipment, other than that which is available as special equipment, See 6.3.8 Engine speed regulation (→ Page 150).
- Do not make modifications to the exhaust system, especially in the vicinity of exhaust gas aftertreatment components (diesel particulate filter, catalytic converter, lambda sensor etc.), See 6.3.3 Exhaust system (→ Page 141).

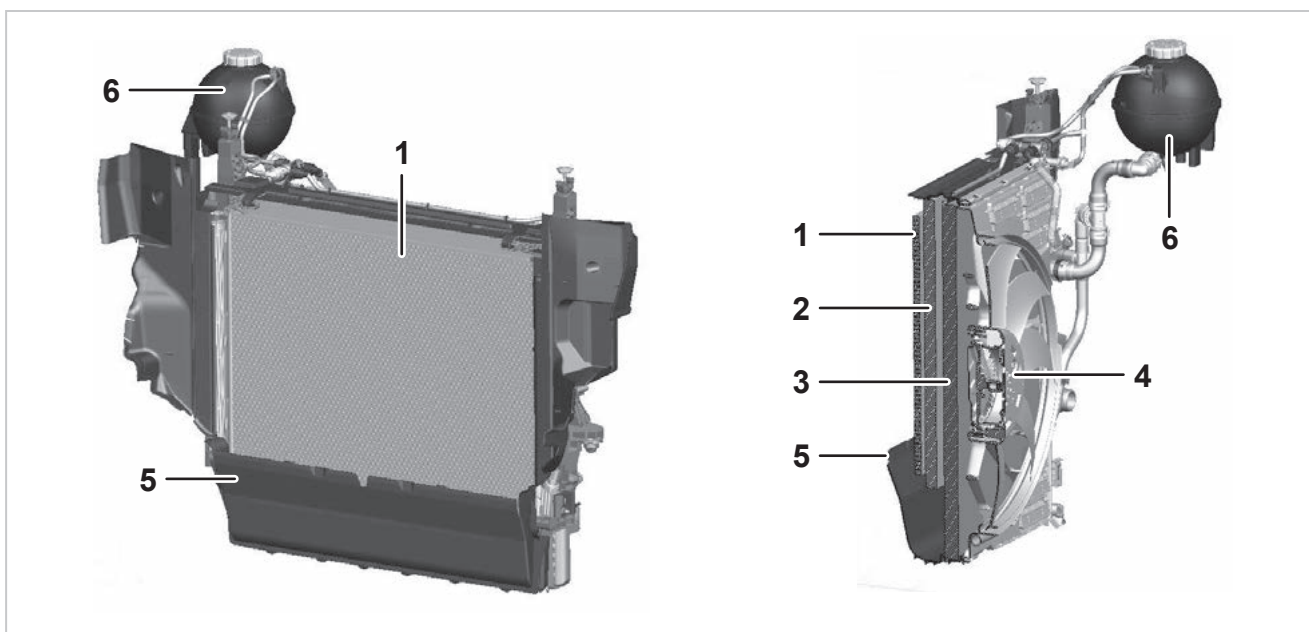
4.4.3 Engine cooling

The cooling module for the engine OM654 is shown in the following figure.

The cooler components of the refrigerant condenser, low-temperature cooler (LT cooler), high-temperature cooler (HT cooler) and electric E suction-type fan are arranged in succession. The new water-cooled charge air cooler and the transmission oil cooler (with automatic transmission) are positioned on the engine and the LT cooler supplies them.

To prevent any return flow of heated recirculated air

from the engine compartment, the cooling module is sealed off all the way round with air recirculation partitions or air ducting (for details, see the following description).



Cooling module for engine OM654

- 1 Refrigerant condenser
- 2 LT cooler
- 3 HT cooler
- 4 E-fan
- 5 Air recirculation partitions (example, installed all the way round)
- 6 Coolant expansion reservoir

Do not make modifications to the cooling system (cooler components, air ducting, hoses, and other detachable parts that are part of the cooling system).

Modifications to the front of the vehicle can fundamentally be made under the following boundary conditions.

The free cross-sectional area of the cooling air intake opening as the sum of the top front grille and bottom bumper grate is defined by the technical design; see figure "Cooling air intake openings at front of vehicle" (→ page 93). For this, it is essential to comply with the following limit values:

- Minimum: 18.2 dm²
- Maximum: 26.4 dm²

! NOTE

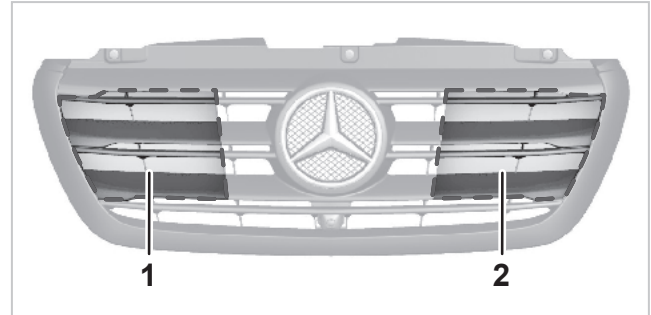
As of emissions class Euro-6d with RDE-2 (certified according to Regulation (EU) 2017/1151), the free cross-sectional areas of the coolant air intake openings are now certification-relevant.

Following any modifications to the basic vehicle, these must not be outside the limit values under any circumstances, as otherwise the emission certification issued by Mercedes-Benz AG will no longer be applicable; See 2.2.8 Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP) (→ Page 29).

If any modifications are made to the front end, the bodybuilder manufacturer must ensure compliance with the certified limit values and the vehicle's associated registration requirements.

Mercedes-Benz genuine van vehicle front

For vehicles with a Mercedes-Benz genuine vehicle front, detachable parts (horn, front flasher or similar) can be mounted in areas 1 and 2 of the front grille:



Front grille of Mercedes-Benz genuine van vehicle front

1 / 2 Areas in which detachable parts can be mounted (schematic)

If detachable parts are to be placed outside areas 1 and 2 in the front grille, verify the resulting cooling output using the test cycle specified at the end of this chapter and submit corresponding evidence of this upon request. In addition, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

If the free cooling air intake openings are reduced in size, the cooling output of the air conditioning system is lowered and – depending on the given operating status – the duration and intensity of fan operation is increased.

Do not cover the lower opening area in the bumper grate (see figure).



Cooling air intake openings at front of vehicle

- 1 Front grille
- 2 Bumper grille

Body-manufacturer-specific vehicle front

For vehicle bodies on which the bodybuilder manufacturer does not use the genuine Mercedes-Benz Van vehicle front provided ex factory, but instead integrates a bodybuilder-manufacturer-specific vehicle front (e.g. on fully integrated camper vans, buses, etc. based on a cab base vehicle, code F50), verify the cooling output of the new vehicle front using the test cycle specified at the end of this chapter and submit corresponding evidence of this upon request. In addition, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The bodybuilder manufacturer must ensure that the previously mentioned certification-relevant limit values for the free cross-section area of the cooling air intake opening and the associated approval requirements for the basic vehicle are complied with.

To this end, when calculating the free cross-section area of a bodybuilder-manufacturer-specific vehicle front, take into account the open air intake area projected in the direction of travel within the effective radiator area at 100% and outside of this at 50%.

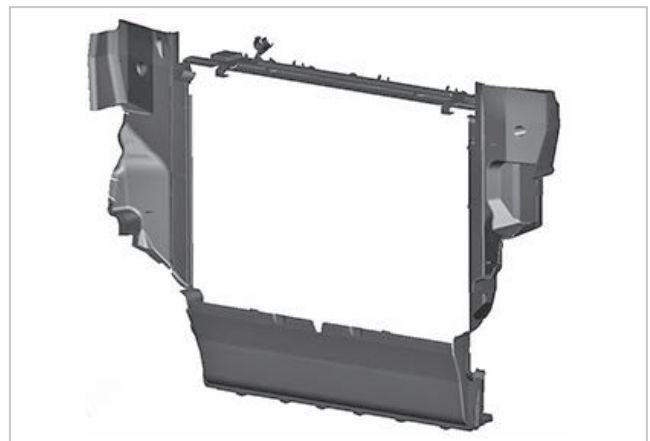
If the free cooling air intake openings are reduced in size, the cooling output of the air conditioning system is lowered and – depending on the given operating status – the duration and intensity of fan operation is increased.

Also observe the following:

- The cooling air intake openings on the vehicle front must be mounted in the immediate airflow surface of the cooler components.
- Additionally, position and design the cooling air intake openings directly in front of the cooling module at the vehicle front so that the cooling air intake is as effective as that of the genuine Mercedes-Benz Van standard component part. The cooler outlet air temperature must not exceed the limit value during the test cycle specified in this chapter.

The complete cooling module must be sealed off against any heated recirculated air from the engine compartment resulting from return flow. In other words, the standard air ducting installed on the cooling module (see figure) toward the body-manufacturer-specific vehicle front must be hermetically sealed off by correspondingly shaped component parts. Design all air ducting so that the engine compartment is completely sealed off between the new vehicle front and the cooling module. The temperature resistance of the materials used must enable component part surfaces to be heated up to a temperature of at least 100 °C.

The bodybuilder manufacturer alone shall bear full responsibility for the design of the air ducting component parts, including material selection, its professional implementation in the vehicle and testing under customer service conditions.



Air ducting on cooling module for engine OM654

Test cycle for thermal validation of cooling air intake in engine OM654

Before starting testing, comply with the following boundary conditions using the corresponding measurement technology and submit corresponding evidence of this upon request:

- The vehicle has been pre-conditioned and is at operating temperature, i.e.
 - Coolant temperature 90 °C
 - Engine oil 95 °C
 - Transmission oil between 90 and 100 °C
- Outside air temperature 30 °C
- Interior pre-conditioned to outside temperature through opening of windows, which then remain open during the tests.
- Air conditioning system with lowest specified value temperature (cooling) is activated and, on a manual air conditioning system, the maximum blower setting is set: These settings are not to be altered during the tests.

The load request and driving profile during the test are as follows:

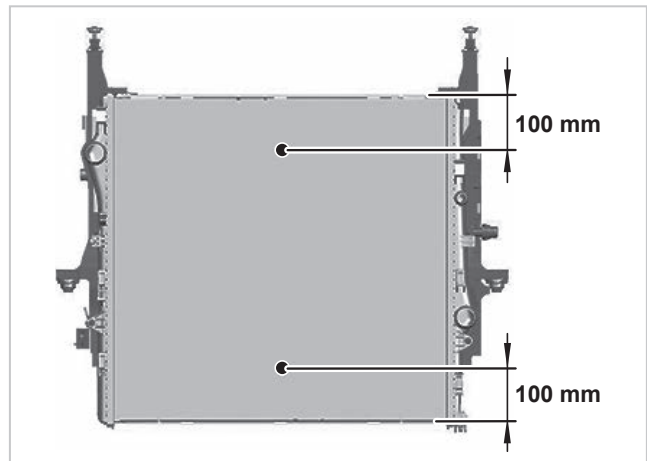
- Permissible gross combination mass (permissible gross mass + maximum trailer load)
- Uphill driving with 10% gradient
- Constant speed of 35 km/h in 2nd gear over a distance of 15 km

The test criterion for sufficient cooling efficiency is as follows:

- The cooler outlet air temperature downstream of the HT cooler must not exceed 110 °C during the entire driving period. Determine this through corresponding temperature measurements and submit corresponding evidence of this upon request.

Observe the following when positioning the measuring points for measuring the cooler outlet air temperature at the coolant radiator:

- Apply the temperature sensors so that they are each located in the middle of the HT cooler at a vertical distance of 100 mm from the upper and lower edge (see figure). The measuring beads of the temperature sensors must be located at a horizontal distance of 10 mm to the rear cooler surface.



Positioning of temperature sensors

To apply for a Letter of Compatibility, prepare and submit a test report containing the following information:

- Description of measurement technology and test setup for temperature measurement,
- Description of preconditioning process and test execution,
- Results in measurement value tables and graphs.

4.5 Limit values for the interior

NOTE

The information and specifications in 6.4.1 General information (→ page 151), 6.4.2 Safety equipment (→ page 152) and 6.4.3 Seats (→ page 160) must absolutely be observed.

4.5.1 Modifications in the area of the restraint systems

WARNING

Modifications to or work incorrectly carried out on restraint systems (seat belt and seat belt anchoring points, seat belt tensioners or airbags) or their sensor systems or wiring could cause the restraint systems to stop functioning correctly. This means, for example, that airbags or seat belt tensioners may be activated inadvertently, or may fail in the event of an accident even if the rate of deceleration exceeds the deployment threshold.

There is danger to life and limb!

Do not make modifications to restraint systems.

- Do not make modifications to airbag components and sensors or in the vicinity of them.
- Do not make subsequent modifications to the headliner or its fastenings if the vehicle is equipped with window airbags.
- Design the vehicle interior in such a way that airbags can fully deploy without impediment.
- Do not make modifications in the vicinity of the airbag control unit.

4.5.2 Modifications to seats

WARNING

Changes to the driver's and front passenger seats, the rear seats, or the mountings of the seats installed ex factory are only permitted if the requirements of the legal specifications applicable at the time that the vehicle is being placed on the market are fulfilled and corresponding evidence of this is submitted upon request:

- UN R 14 (seat and seat belt anchorages)
- UN R 16 (seat belts and restraint systems)
- UN R 17 (seats, seat anchorages and head restraints)

These requirements must be complied with even for seats with two- or three-point seat belts specific to the bodybuilder manufacturer that deviate from the series standard ex factory.

In the event of an accident, the seats could otherwise be torn or ripped from their anchorages. Passengers cannot be protected as intended.

There is danger to life and limb!

4.6 Limit values for electrics and electronics

Observe the information and specifications in 8 Electrics and electronics (→ page 265).

4.6.1 Vehicle position lamps and side marker lamps

Vehicle position lamps are required by law on all vehicles with total widths of over 2.10 m in accordance with UN R 48.

Side marker lamps are required on all vehicles with an overall length of over 6 m as specified by UN R 48.

4.6.2 Retrofitting of electrical devices

All electrical devices installed must have been tested in accordance with UN R 10 and must bear the CE marking and/or another comparable inspection marking depending on the country-specific regulations; See 8.2 Electromagnetic compatibility (EMC) (→ Page 266).

4.6.3 Mobile communication systems

The maximum transmission output (PEAK) at the base of the antenna must not exceed the values resulting from the country-specific laws regarding the maximum legitimized transmission output.

4.6.4 CAN bus

Do not make modifications to the CAN bus or the components connected to it.

The parameterizable special module (PSM, code ED5) can be used to access individual types of data available on the CAN bus (→ page 372).

4.7 Limit values for additional assemblies

Observe the following when retrofitting additional assemblies (e.g. additional refrigerant compressors, pumps, etc.):

- The operation of vehicle components must not be adversely affected.
- The freedom of movement of vehicle parts must be guaranteed in all driving situations.

4.8 Limit values for attachments

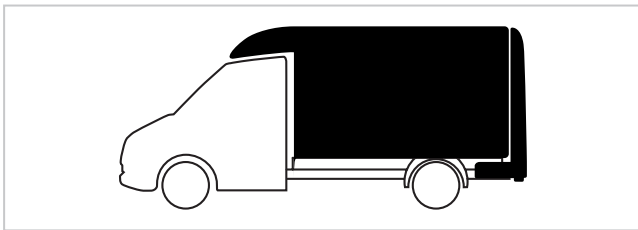
- According to UN R 73, a side underride guard is stipulated for vehicles in excess of a gross vehicle mass of 3.5 t.
- An underride guard is necessary when:
 - the distance between the rear of the vehicle and the rear axle is more than 1000 mm.
 - with the vehicle in its unladen condition, the distance between the roadway and the chassis or main parts of the body is greater than 550 mm across the entire width of the vehicle.
- The maximum permissible lifting load of a liftgate is 500 kg on a panel van and 1000 kg on chassis. A means of attachment as per 6.6.6 Liftgate (cargo liftgate) (→ page 190) is required.

Attachments on the rear frame section

On open and closed model designations from model series 907, the vehicle comes as standard with reinforcements on the vehicle frame for mounting attachments in the rear longitudinal frame member on the left or right.

If, due to a higher payload, these reinforcements are undesired, they can be deselected with code QW1 on closed model designations, depending on the equipment installed on the vehicle. Code QW1 is not available for open model designations.

4.9 Limit values for the body



Body

See also chapter 7 Design of bodies (→ page 200).

! NOTE

The standard fuel filler cap must not be removed or covered with any "blocking" parts, See 6.2.6 Modifications to the cab (→ Page 122).

! NOTE

The minimum distance between the cab and a separate body must be 50 mm.

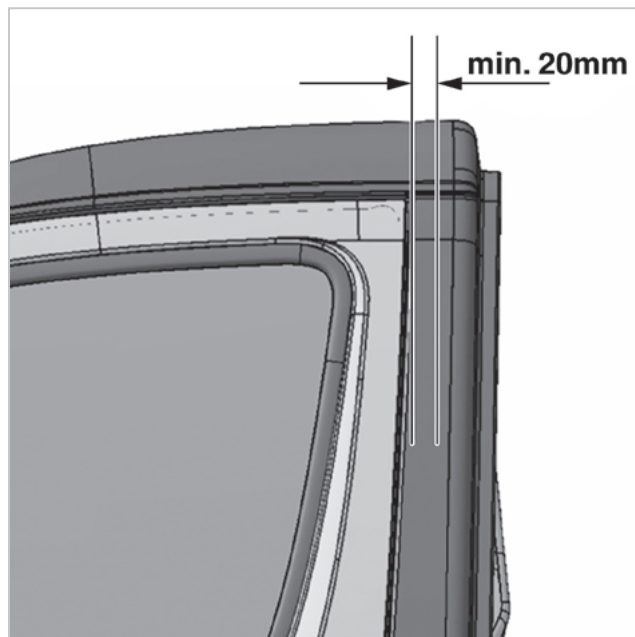
! NOTE

The minimum distance between the rear edge of the door and an integrated body must be 20 mm. Otherwise, the rear edge of the door may come into contact with the body in the event of an accident, and in extreme cases, the door may become jammed.

! WARNING

Do not attach side bars to the outside of the underbody. In the event of an accident (e.g. a side crash), these could be bent upward and block the door. This could mean that the rescue services are not able to open the door and could make recovery of the occupants more difficult.

There is danger to life and limb!



Minimum distance between rear edge of door and integrated body

General information

To simplify the quality acceptance procedure for the body scope, observe the notes in Bodybuilder Manufacturer Information Bulletin 29/2013, which can be viewed in the Bodybuilder Portal (→ page 24).

<https://bb-portal.mercedes-benz-vans.com>



- i You can access this older bodybuilder manufacturer information after registering and logging in to the Bodybuilder Portal in the "Technology and Information (Vans)" section. There, under "Archive model series", select the "Sprinter (2013 model) 906 model series". Clicking on the vehicle image takes you to "Model Selection"/"Technical Data"/"-Further Information". The last category contains the "Bodybuilder Manufacturer Information" button. Click on this, then select the Bodybuilder Manufacturer Information Bulletin, Edition 29/2013.

⚠ WARNING

Tampering with or unauthorized installations in vehicle systems, safety-relevant component parts and driving assistance systems can impair the functioning of these systems. This may lead to the failure or malfunctioning of components or safety-relevant component parts.

There is risk of an accident and danger to life and limb!

Always have work on vehicle systems, safety-relevant component parts and driving assistance systems performed at a qualified workshop.

- i Furthermore, tampering with the vehicle, safety or driving assistance systems and safety-relevant component parts can invalidate the warranty or the operating permit.

5.1 Brake hoses, cables, and lines

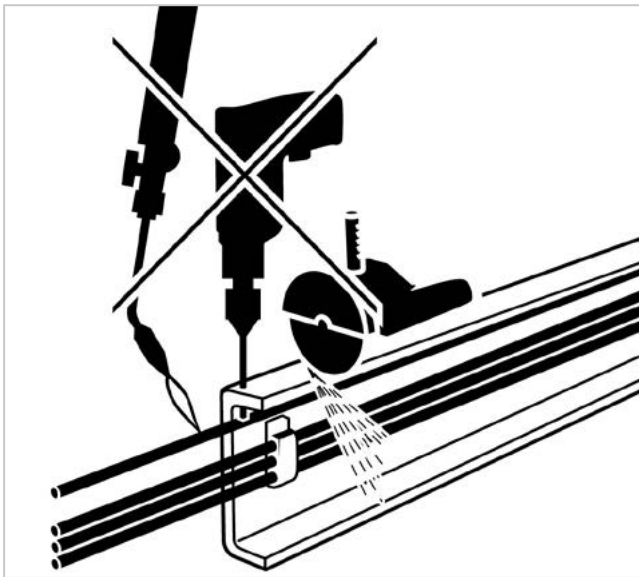
! NOTE

Observe accident prevention regulations when working on the vehicle.

! NOTE

Observe all national laws, directives and regulations.

Cover plastic lines and brake hoses before carrying out any welding, drilling, and grinding work or before working with cutting disks. If necessary, the plastic lines and brake hoses should be removed.



After installing compressed-air lines or hydraulic lines, check/test each of the systems for chafe marks, pressure loss, and leak tightness.

No other lines may be attached to brake hoses and brake lines.

No other lines may be attached to brake cables, other than those installed as standard.

Lines must be protected from heat by means of insulation.

! WARNING

Work carried out incorrectly on the brake hoses or cables may impair their function. This may lead to the failure of components or safety-relevant component parts.

There is risk of an accident and danger to life and limb!

Have all work on brake hoses and cables carried out at a qualified specialist workshop.

5.2 Welding work

⚠ WARNING

Welding in the vicinity of the restraint systems (airbags, airbag sensors, airbag control unit, or components of the seat belts) can cause these systems to no longer function as intended.

If restraint systems no longer function correctly, the vehicle occupants may be unprotected in the event of an accident.

There is danger to life and limb!

Therefore, do not weld in the area of the restraint systems.

⚠ WARNING

Welding work that is not performed correctly could lead to failure of component parts relevant to safety. It would then not be possible to rule out the risk of an accident.

There is danger to life and limb!

Have welding work carried out by a qualified specialist workshop.

⚠ WARNING

Improper welding work can result in a fire.

There is danger to life and limb!

Always comply with the legal requirements when performing welding work.

For welding work, it is essential to observe the following:

- Welding work may only be performed by trained personnel.
- Before performing welding work, remove any components that may contain flammable or explosive gases, e.g. the fuel system (→ page 136), or protect them from sparking with a fire-resistant covering.
- Before conducting any welding work in the area of the restraint systems, the component parts must be removed for the duration of the work. Important information about handling, transporting and storing airbag units is available in 6.4 Interior (→ page 151).
- Before welding, cover springs and air suspension bellows to protect them from weld pearls. Do not touch springs with welding electrodes or electrode arms.
- Welding work is not permitted on major assemblies such as the engine, transmission, axles.
- Remove and cover up the positive and negative clamps of the batteries.
- Connect the ground clamp of the welder directly to the part to be welded. Do not connect the ground clamp to major assemblies such as the engine, transmission, or axles.
- Do not touch electronic component part housings (e.g. control units) and electric lines with the welding electrode or the ground clamp of the welder.

- Weld only with electrodes connected to the positive terminal of a direct current source. Always weld from bottom to top.
 - The maximum current may be 40 A per mm of electrode diameter.
 - Use only completely dry basic-coated electrodes (diameter 2.5 mm).
 - Gas-shielded welding is permissible.
 - Only use welding wires with a thickness of between 1 and 1.2 mm.
 - The yield and tensile strength of the welding material must be at least equal to that of the material to be welded.
 - Plug welding is only permissible in the vertical webs of the longitudinal frame member.
 - Weld seams must be ground down and reinforced with angle sections to prevent notching from fusion penetration.
 - Avoid weld seams in bending radii.
 - There must be at least 15 mm between the weld seams and the outer edges.
- i** Further information on welding can be found in 3.9 Threaded connections and weld joints (→ page 66) and 6.2 Body-in-white and body (→ page 112), as well as in the Workshop Information System (→ page 26).

5.3 Anticorrosion protection measures

Surface and anti-corrosion protection measures must be carried out on the affected areas after body work on the vehicle.

i NOTE

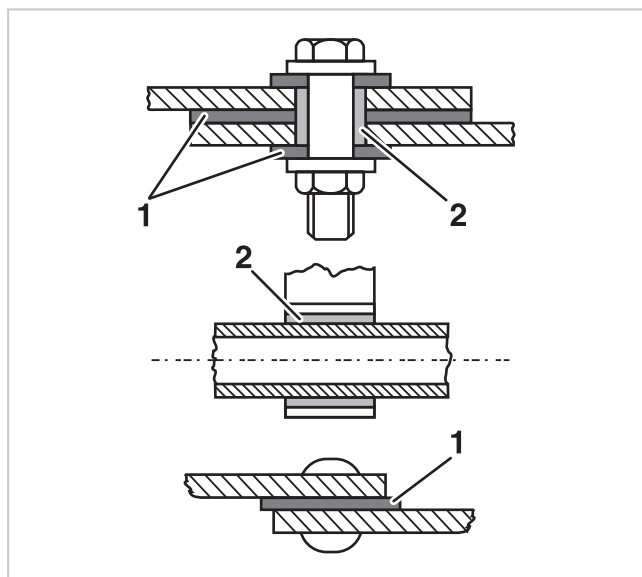
Only use preservation agents tested and approved by Mercedes-Benz AG for any anticorrosion protection measures performed.

Planning measures

Anticorrosion protection measures should be included in the planning and design stages by selecting suitable materials and designing component parts accordingly.

- i** A galvanic connection occurs if two different metals are brought into contact with each other through an electrolyte (e.g. air humidity). This causes electrochemical corrosion and the less noble of the two metals is damaged. The further apart the metals concerned are in the electrochemical potential series, the more intense the electrochemical corrosion becomes.
- For this reason, electrochemical corrosion must be prevented by insulation or by treating the component parts accordingly, or it can be minimized by selecting suitable materials.

Preventing contact corrosion by means of electrical insulation



Preventing contact corrosion

- 1 Insulating washer
- 2 Insulating sleeve

Contact corrosion can be prevented by using electrical insulation such as washers, sleeves or bushings.

Avoid any welding on inaccessible cavities.

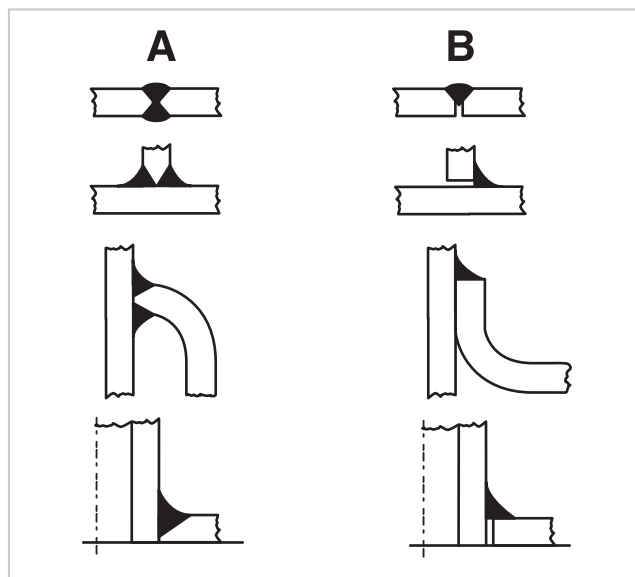
Measures implemented through the design of component parts

Corrosion can be prevented by design measures, in particular the design of connections between identical or different materials.

There is a risk of dirt or moisture accumulating in corners, edges, creases, and folds.

Design measures for counteracting corrosion can be implemented using inclined surfaces, drains, and by avoiding gaps in the joints between components.

Gaps inherent in the design of weld joints and how to avoid them



Examples of types of weld joints

- A Good (through-welded)
- B Bad (gap)

Measures implemented through coatings

The vehicle is protected against corrosion by applying protective coatings (e.g. electroplating, painting, or zinc coating applied by flame) (→ page 104).

After all work on the vehicle

- ▶ Remove drill chips.
- ▶ Deburr edges.
- ▶ Remove any burnt paint and thoroughly prepare surfaces for painting.
- ▶ Prime and paint all unprotected parts.
- ▶ Preserve cavities with wax.
- ▶ Implement anticorrosion protection measures at the underbody and frame section.

5.4 Painting and preservation work

NOTE

For drying the paint, the object temperature must be a maximum of 60 °C and the drying time must be a maximum of 30 min. At higher temperatures, the control units or other components may be damaged.

Painting or preservation damaged by the bodybuilder manufacturer must be repaired by the bodybuilder manufacturer.

Please note with regard to this:

- Adhere to the Mercedes-Benz AG quality standards for initial painting and refinishing.
- Use only materials tested and approved by Mercedes-Benz AG or equally suitable materials for performing any painting or preservation work.
- Comply with the layer thicknesses set at the factory for the individual paint coats.
- Paintwork compatibility must be guaranteed when over-coating.

 You can obtain information on the paint materials and coat thicknesses used at the factory and Mercedes-Benz paint numbers from any Mercedes-Benz Service Partner.

Cover up the following areas, among others, before carrying out any painting work:

- Disk brakes
- Brake hoses
- Pressure transformer unit for parking brake
- Contact areas between disk wheels and wheel hubs
- Contact surfaces of wheel nuts/wheel bolts
- Brake fluid reservoir
- Breathers on transmission, axles, etc.
- Sealing surfaces
- Windows
- Door locks
- Door checks in the hinges of the rear-end door
- Door checks and opening limiters in the center guide rails
- Contact surfaces on the guide rails of sliding doors
- Moving parts of the sliding door carriage
- Airbags and seat belts
- Sensors of Active Brake Assist/Active Distance Assist (→ page 314)
- Sensors of Blind Spot Assist/Rear Cross Traffic Alert/exit warning function/Sideguard Assist (→ page 318)
- Sensors of parking packages/Moving-off information Assist (→ page 344)

⚠ WARNING

Depending on the type and thickness, paints or films can cause damping of radar waves. This could lead to malfunctions or system failure. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

In repair cases, remove the existing coat of paint in such a way that the plastic cover is not damaged and the material thickness of the cover is not significantly reduced.

! NOTE

For the subsequent painting of sensors of the driving assistance systems, it is essential to observe specific specifications and limit values in the chapters relating to the affected sensors.

- 8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC (→ page 314)
- 8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist (→ page 318)
- 8.9.8 Parking packages/Moving-off information Assist (→ page 344)

5.5 Storing and handing over the vehicle**Storage**

To prevent any damage while vehicles are in storage, it is recommended that they are serviced and stored in accordance with the manufacturer's specifications (→ page 71).

Handover

To prevent damage to the vehicle or to repair any existing damage, it is recommended to have the vehicle subjected to a full function check and a complete visual inspection before it is handed over.

6.1 Suspension

! NOTE

Make sure to observe the information and specifications on suspension design in 4.2.1 Suspension (→ page 80).

6.1.1 General information on the suspension

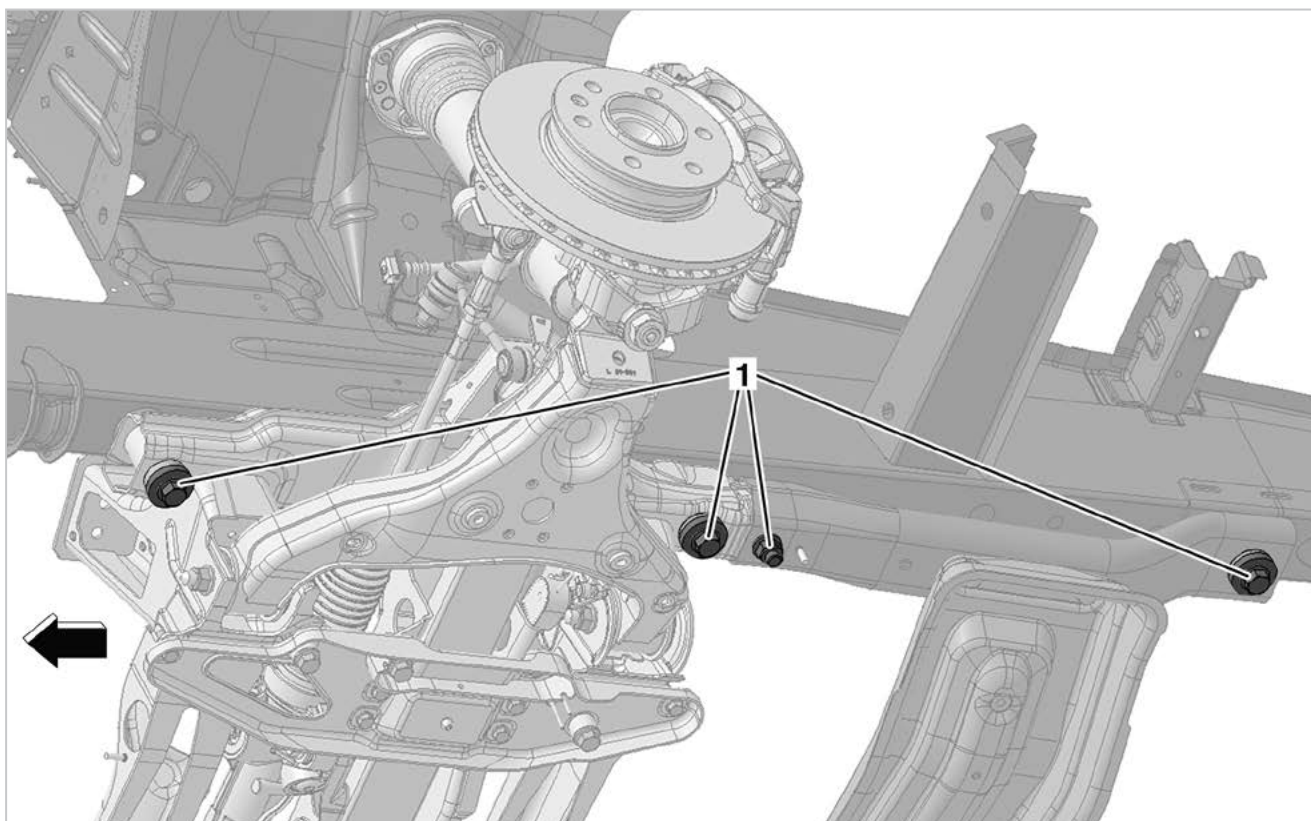
No additional detachable parts may be mounted to the threaded connection points on the front axle.

⚠ WARNING

Changes to suspension components can lead to unstable driving behavior. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Therefore, do not make any modifications to suspension parts and do not install any additional components.



Front axle

1 Bolting points on the front axle

➡ Direction of travel

In the front axle area, observe the following:

- Comply with the specifications for the steering in 4.2.6 Modifications to the steering (→ page 83).
- Do not make modifications to wheel alignment variables.
- The front axle must not be changed in any way or used to mount additional assemblies and other modifications.
- Rigid rear axle: Refrain from making changes.
- Brakes: Refrain from making changes.
- Devices, sensors, line installation for ESP®/ABS: Refrain from making changes.
- Use new screws when installing the front axle. Tighten all screws and threaded connections in accordance with the Mercedes-Benz tightening instructions.
- Observe VDI guideline 2862, in particular the section entitled "Threaded connections with special safety relevance", for all installations.
- Any shortening of the free clamping lengths, conversion to a stretch shank, or the use of screws with a shorter free thread is not permissible.
- Observe the settling properties of threaded connections.

 Information is available from any Mercedes-Benz Service Partner.

Additionally clamped component parts must exhibit an identical or higher strength than the previous clamped assembly.

The use of Mercedes-Benz tightening torques assumes coefficients of friction for the screws/bolts in the tolerance range of $\mu = 0.08-0.14$.

The use of Mercedes-Benz standard parts is recommended.

 Note the two different suspension variants for 3.5 t model designations, with which two different loading heights can be achieved. For planned load capacity increases on these vehicles, contact the Technical Consultancy (→ page 18).

NOTE

In order to prevent increased tire wear, a change in the standard wheel alignment values (camber, toe-in) at the front axle can be economical in the case of vehicles with increased curb weight due to body or equipment.

Comply with the specifications in 3.5 Dimensions and weights (→ page 59) on the curb weight and associated axle loads before and after body mounting work, "Checking the wheel alignment" in 3.11.3 Work before handing over the modified vehicle (→ page 73), 3.12 Special equipment (→ page 74), and in 4.2.9 Wheel alignment (→ page 84).

Examples of this are recovery vehicles, fire-fighting vehicles, ambulances, rescue vehicles, semitrailer trucks, workshop vehicles or camper vans. For a planned change in the standard wheel alignment values, consulting a Mercedes-Benz partner is recommended. Based on the specific axle load conditions of the complete vehicle, they can then determine the ideal camber and toe-in values, based on the workshop documents stored in the Workshop Information System (→ page 26).

The same applies to vehicles that are frequently fully laden, as it can be assumed that the maximum permissible axle loads are frequently marginally utilized here as well.

6.1.2 Springs, dampers, and stabilizers

General information

Multiple suspension variants are available to you ex factory. A suitable suspension variant must be selected depending on the planned body; See 4.2 Limit values for the suspension (→ Page 80).

Modifications to springs, dampers and stabilizers can only be made in the combinations specified by Mercedes-Benz on the front and rear axle. A Letter of Compatibility is not required in this case. Any modifications beyond this scope must be evaluated by the Technical Consultancy in each individual case.

For more information on this, contact the Technical Consultancy (→ page 18). For this, a Letter of Compatibility may be required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

- The use of Mercedes-Benz genuine parts is recommended.
- When performing installation work, avoid damaging the surfaces and anti-corrosion protection of the spring leaves.
- Before carrying out welding work, springs must be covered to protect them against weld pearls.
- Do not touch springs with welding electrodes or locking pliers.
- For modifications to springs, dampers and stabilizers as part of the Mercedes-Benz AG approved combinations on the front and rear axle on vehicles with front radar sensor, code BA3/ET4 – See 8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC (→ Page 314) – where applicable, the installation height of the front radar sensor must be compared against and adapted to the value entered there, depending on the type of modification involved. A service calibration of the front radar sensor must also be performed, as this modification also causes the driving axis to change.

⚠ WARNING

Do not use springs and dampers if they do not correspond to the characteristics of series production parts or parts obtainable as special equipment. Otherwise, if the vehicle is fitted with ESP®, this system may no longer work correctly and could ultimately fail. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

6.1.3 Brake system

⚠ WARNING

Work carried out incorrectly on the brake lines or brake hoses may impair their function. This may lead to the failure of components or safety-relevant component parts. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Work on brake lines or brake hoses must only be carried out by a qualified specialist workshop.

After completion of the work, check that the brake system is working correctly. For this, it is recommended that the approval is carried by a technical inspection authority.

If the routing has to be altered, then avoid routing across sharp edges and through narrow gaps or near moving parts.

Hydraulic brake system

- ▶ Replace hydraulic brake lines without additional connecting points using Mercedes-Benz-approved double-walled wrapped tubes of 4.75 mm × 0.7 mm or fully assembled brake lines with connecting parts (genuine Mercedes-Benz replacement parts).
- ▶ Brake lines between the master brake cylinder and the hydraulic unit must not be modified.
- ▶ The bending radius must be >18 mm.
- ▶ Lines must only be shaped in a bending device. The cross-section must not be reduced.
- ▶ Attach nuts (part no. 000 997 66 34) or retaining screws (genuine Mercedes-Benz replacement parts) to the ends of lines and make a flange (F DIN 74234).
- ▶ The inside of the lines must be cleaned before installation.
- ▶ The brake fluid must be renewed regularly in accordance with the Mercedes-Benz specifications, but at least every two years. If necessary, renew the brake fluid before delivery to the customer.
- ▶ If the change interval of the brake fluid has been exceeded or if the age of the brake fluid is unknown, the brake fluid must be replaced.
- ▶ When routing between two component parts that move in relation to each other, use a high-pressure-resistant flexible line (brake hose, Mercedes-Benz genuine brake components if possible). Make sure that the flexible lines are not stretched in any operating statuses and that they cannot chafe against other component parts.

Line installation

WARNING

A distance sufficient for the vehicle configuration in all operating conditions must be maintained between brake lines and heat sources, sharp-edged and moving parts. Otherwise, the brake system function could be impaired or the brake system could suffer total failure as a result of bubbles forming in the brake fluid or from chafe marks in the brake lines.

This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Work on brake lines must only be carried out by an authorized specialist workshop.

It is recommended to use genuine Mercedes-Benz brake line holders for the attachment of the brake lines.

The distance from holder to holder must not exceed a maximum of 500 mm.

- ▶ Route brake lines without any kinks.

Routing lines along the brake hoses and brake lines

No other lines may be attached to brake hoses and brake lines.

Brake cable for the parking brake and modifying the length of the brake cable

- ▶ If a new brake cable is required for the parking brake, the new length of the cable must be determined and a suitable new cable must be used.
- ▶ The brake cable holders are moment-optimized; modifications are therefore not permissible.
- ▶ Route brake cables without any kinks.
- ▶ Do not make any angle changes at the end pieces of the brake cable conduit.
- ▶ No other lines may be attached to brake cables, other than those installed as standard.

i If you have any questions about the standard brake cable, contact the Technical Consultancy (→ page 18).

Disk brakes

Cooling must not be impaired by attaching spoilers below the bumper, additional wheel trims, or splash shields, etc.

For fully integrated bodies (7.14 Bodies on chassis with cab base (code F28, F50) (→ page 227)), make sure that the ventilation in the wheel wells for cooling the brakes functions in the same way as on the corresponding panel van model series 907.

⚠ WARNING

On no account should modifications be made to the air inflow and air outflow of the brake system. Any modifications to the steering and the brake system may result in these systems malfunctioning and ultimately failing. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Therefore, make sure there is always a sufficient flow of cooling air present.

⚠ WARNING

Any modifications to brake components may result in these systems not functioning correctly and ultimately failing. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

Modifications to brake components are not permissible.

Additional brakes (retarders)

i Information on this can be found in 6.5.5 Auxiliary brakes (retarders) (→ page 176) and in the description of the electrical interface in 8.12 Pre-installation for retarder (→ page 377).

6.1.4 Air suspension

For more information on retrofitting air suspension systems, contact the Technical Consultancy (→ page 18).

⚠ WARNING

Do not use springs and dampers if they do not correspond to the characteristics of series production parts, certified components, or parts obtainable as special equipment. This applies in particular to the retrofitting of air suspension systems. Otherwise, if the vehicle is fitted with ESP®, this system may no longer work correctly and could ultimately fail. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

⚠ WARNING

If attachments are fitted to the front part of the frame, this modified crash structure may cause the airbag units to function incorrectly on vehicles with this equipment.

Retrofitting of air suspension systems at the front axle must therefore only be carried out following coordination with the Technical Consultancy (→ page 18).

6.1.5 Wheels and tires**! NOTE**

The information and specifications in 3.8 Tires (→ page 65) and 4.2.3 Approved tire sizes (→ page 83) must absolutely be observed.

- i** Further information about wheels and tires is available from any Mercedes-Benz Service Partner.

6.1.6 Spare wheel

As standard, the Sprinter – BR 907 is not equipped with a spare wheel or a flat tire repair system such as the TIREFIT kit. A spare wheel can be ordered as special equipment at extra cost under code R87. Country-specific equipment or special equipment may include a spare wheel. When a vehicle is equipped with a spare wheel, the spare wheel itself must be fitted with a functioning sensor and a sensor battery that is neither discharged nor too old for the tire pressure loss warning system (code RY2).

The position under the frame in the rear area can be selected for the attachment of the spare wheel (code R65); see your vehicle operator's manual.

For incomplete chassis, See 7.14 Bodies on chassis with cab base (code F28, F50) (→ Page 227), a provisional attachment of the spare wheel on the longitudinal frame member is available for vehicle delivery (code R60).

! NOTE

The temporary mounting is not permissible for operating the vehicle on public roads. Therefore, with this variant, the bodybuilder manufacturer is responsible for mounting the spare wheel at a suitable, alternative position before the vehicle is brought into service. The bodybuilder manufacturer is responsible for validating the road safety in all vehicle driving conditions.

Observe the following in this respect:

- Attachment is only permitted on frame or structural components with sufficient strength and stiffness
- For attachment systems in the rear area or on the roof, observe the maximum permissible loads in 6.2 Body-in-white and body (→ page 112).
- The spare wheel must be easily accessible and it must be easy to use the attachment mechanism.
- Secure the spare wheel two-fold against falling off.

! NOTE

Observe all national laws, directives, and registration regulations.

6.2 Body-in-white and body

6.2.1 General information on the body-in-white

Modifications to the body must not have a negative effect on the function or strength of vehicle major assemblies or control elements or on the strength of load-bearing parts.

When converting vehicles and mounting bodies, it is not permissible to make modifications that affect the function or freedom of movement of chassis parts (e.g. during maintenance and inspection work) or the access to these parts.

Observe the following in this respect:

- The tire pressure loss warning system (Tire Pressure Management System, TPMS) may malfunction if modifications are made in the immediate vicinity of the antennas and wheels, See 8.9.7 Tire pressure loss warning system (→ Page 342).
- Modifications must not be made to the front-end assembly structure, including longitudinal members and crossmembers, from the front through to one meter behind the B-pillar, as this can otherwise impair the strength necessary for passive safety; See 4.3.1 Modifications to the body-in-white (→ Page 85).
- The clearance for the fuel filler neck, fuel tank, and fuel lines must be maintained (→ page 136).
- Do not change the rigidity in the area of the filler necks.
- Avoid any sharp-edged corners.
- It is not permissible to drill holes in or perform welding work on the A-pillar or B-pillar.
- It is not permissible to make cuts in the C or D-pillar, including the associated roof bows.
- If the roof paneling and roof bows are removed and if it is not possible to install a continuous sectional frame, additional roof bows must be fitted. The all-round concept of the side, front and rear roof structure must be maintained. In doing so, a sufficiently high equivalent rigidity must be guaranteed, See 6.2.12 Cutting the cab roof and B-pillar roof bow (→ Page 132).

! NOTE

The longitudinal frame members are critical structural components for passive safety.

If work is performed in these areas, the structure necessary for passive safety can be disrupted.

Holes in the longitudinal frame member are the result of the production process and are not suitable for fastening bodies; otherwise, there is a risk of damage to the frame.

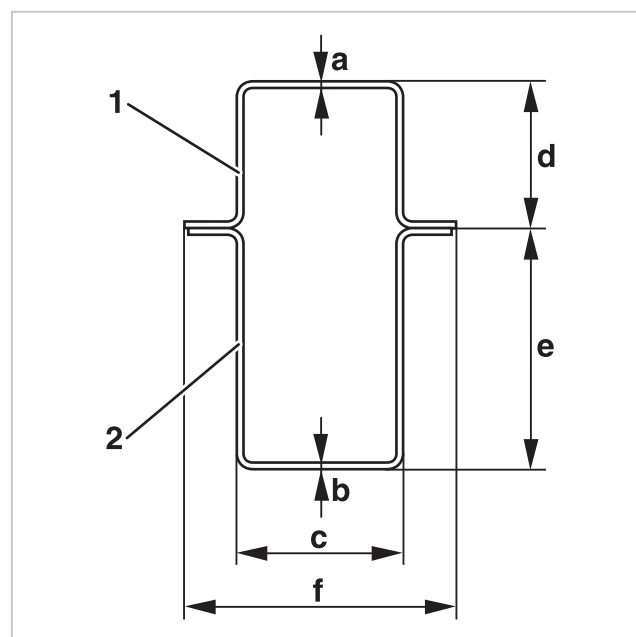
- The following applies to panel vans: Modifications are only permitted to be made to the rear portal, including the roof area, in exceptional circumstances. For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- The attachment of additional equipment to the longitudinal members and crossmembers must be implemented using mounting consoles (plug welding). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- Check the trailer connections for proper operation.
- If a trailer coupling is installed, the necessary reinforcements must be present (→ page 193).
- If bodies are mounted on basic vehicle "chassis", protection of the fuel system is necessary; See 6.3.1 Fuel system (diesel) (→ Page 136).
- If a vehicle is to be converted to a low-floor vehicle, a Letter of Compatibility is required for this (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20). Also observe the paragraph on the Euro VI exhaust system for low-floor vehicles (→ page 143).

NOTE

The factory-installed wooden or plastic floors, including the integrated tie-down eyes as well as the partition, represent a certified as-built configuration in the panel van.

- If these components are removed, ensure that an equivalent substitute is refitted for load securing as per the international requirements in accordance with ISO 27956.

Section dimensions of longitudinal frame members



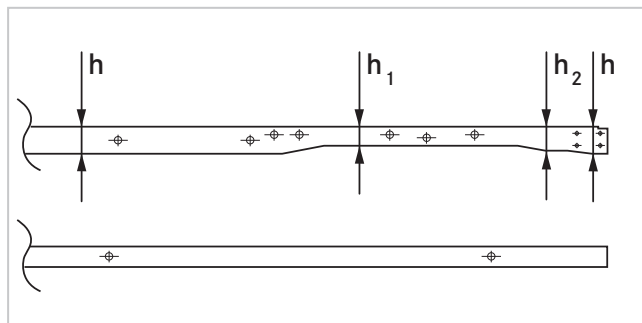
Dimensioning of upper and lower flange

- 1 Upper flange
2 Lower flange

Permissible gross mass [t]	Dimensions [mm]					
	a	b	c	d	e	f
3.5 Chassis	2	2	70	61	119 84 ¹⁾	118
5 Chassis	3	3	70	80	120 100 ¹⁾	126
3.5 Panel van/ crewbus		1.5	70	-	120 85 ¹⁾	93
5 Panel van/ crewbus		3	70	-	120 100 ¹⁾	118

1) In the area of the rear axle

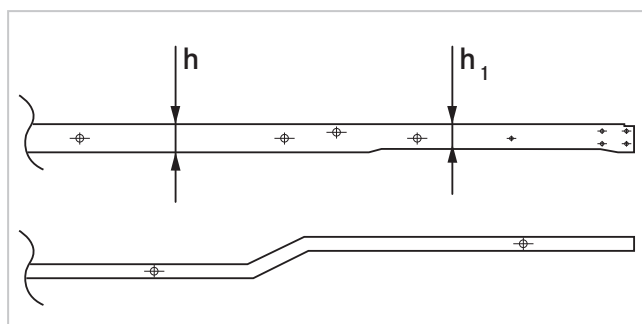
3.5 t longitudinal frame member



Dimensions of the lower flange of the longitudinal frame member

Dimensions [mm]	
h	120
h₁	85
h₂	110

5 t longitudinal frame member



Dimensions of the lower flange of the longitudinal frame member

Dimensions [mm]	
h	120
h₁	100

Welding work on the body-in-white

Welding work must only be performed by trained personnel.

NOTE

Areas where welding work is not permitted:

- On the A-pillar and B-pillar
- In the upper or lower flange of the longitudinal frame member (horizontal surfaces)
- In bending radii
- On major assemblies such as the engine, transmission, axles, etc.
- In the area of the airbags, airbag sensors, airbag control unit, seat belts
- In the area of the driver assistance systems, e.g. the front and rear radars

Only plug welding is permitted on the vertical surfaces of longitudinal frame members for material thicknesses of up to 2 mm. Tack welding can also be used for material thicknesses over 2 mm.

i Further information on the welding process can be found in 3.9.2 Weld joints (→ page 68)

i Further information on welding can be found in 3 Planning of bodies (→ page 47) and 5 Damage prevention (→ page 99), as well as in the Workshop Information System (→ page 26).

WARNING

Welding in the vicinity of the restraint systems (airbag or belts) can cause these systems to no longer function correctly.

If restraint systems no longer function correctly, the vehicle occupants may be unprotected in the event of an accident.

There is danger to life and limb!

Therefore, never weld in the area of the restraint systems.

Drilling work on the body-in-white

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Areas that must not be drilled:

- On the A-pillar and B-pillar
Exception: See 7.17 Semi-integrated bodies and optional mounting of free-standing bodies (→ Page 259)
- In the upper or lower flange of the longitudinal frame member (horizontal surfaces)
- Areas with a load-bearing function for the front and rear axle or parts fastened to the frame
- At load application points (e.g. spring brackets, holders etc.)
- In the area of the airbags, airbag sensors, airbag control unit, seat belts
- In the area of the driver assistance systems, e.g. the front and rear radars

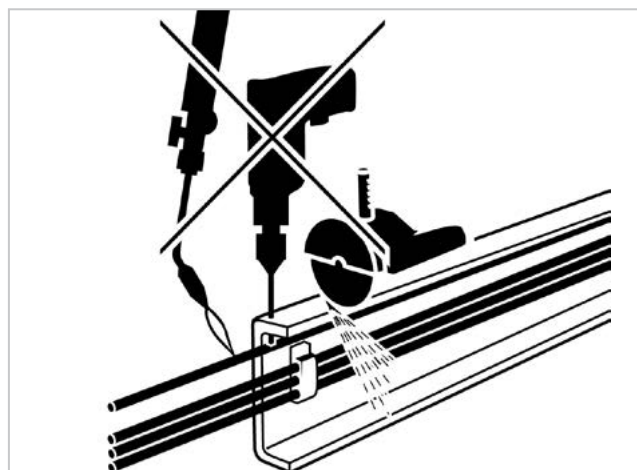
! WARNING

Drilling in the vicinity of the restraint systems (airbag or belts) can cause these systems to no longer function correctly.

If restraint systems no longer function correctly, the vehicle occupants may be unprotected in the event of an accident.

There is danger to life and limb!

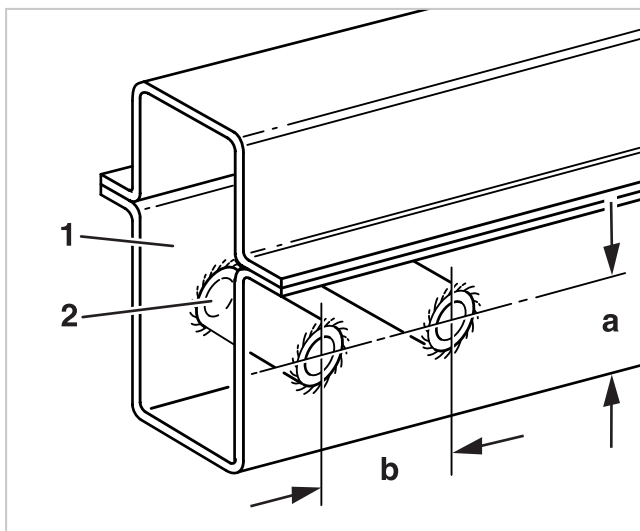
Therefore, never drill in the area of the restraint systems.

**! NOTE**

Holes in the longitudinal frame members and cross-members are the result of the production process and are not suitable for body mounting work.

Never use the holes produced during the production process, as otherwise frame parts may be damaged.

Drilling in the longitudinal frame member web is permissible in accordance with the figure and if spacer bushings are used. The steel spacer sleeves must have adequate strength to absorb the screwdriving forces and be welded to the longitudinal frame member.



Drilling in the longitudinal frame member at the rear frame end (schematic)

- 1 Vehicle frame
- 2 Spacer bushings
- a Distance of at least 20% of the frame height
- b Distance between bores at least 50 mm

On open model designations (chassis without or with platform ex factory, cab base vehicles code F50/F28), it is only permissible to introduce bores in the longitudinal frame members (in compliance with the previously mentioned specifications) in the areas shown below. When doing so, a sufficient distance of at least 20 mm must be maintained from existing ex-factory holes, reinforcements, the connecting flanges for the component parts, and in areas where other components are attached, etc. (list not exhaustive).

On closed model designations (panel vans or Tourers), bores in the above-mentioned areas are only permitted if the specifications are observed. For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).



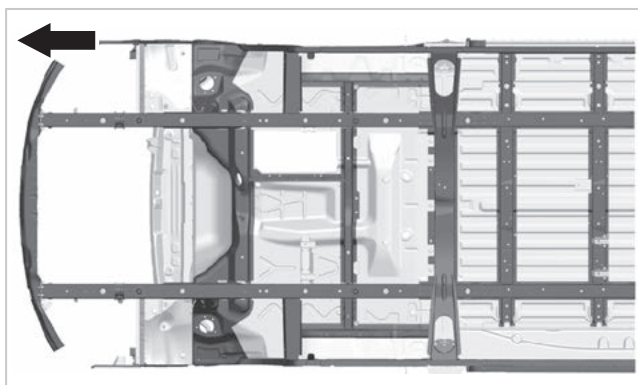
Marking: permissible areas for bores in the longitudinal member web on open model designations with body length A2 (schematic)

- ▶ After drilling, deburr and ream all bores.
- ▶ Remove chips from the frame and treat the bores with cavity preservation.
- ▶ Restore the anticorrosion protection in accordance with Mercedes-Benz guidelines.

6.2.2 Attachments on the frame

Attachments on the front frame section

Do not make interventions (holes, cuts, etc.) in the longitudinal member and crossmember structure from the front up to one meter behind the B-pillar (see illustration) and the attachment of major assemblies, brackets, component parts, etc. in the area of the front-end assembly and front axle, as this can impair the structure required for passive safety.



Longitudinal and crossmember structure in front-end assembly area (view from below)

➔ Direction of travel

⚠ WARNING

If attachments are mounted on the front frame section, the function of the forward crash structure and the airbag units may be impaired.

Attachments are therefore only permitted to be mounted on the front section of the frame following coordination with the Technical Consultancy.

❗ NOTE

The modifications must not hinder possible repair work on the series production vehicle.

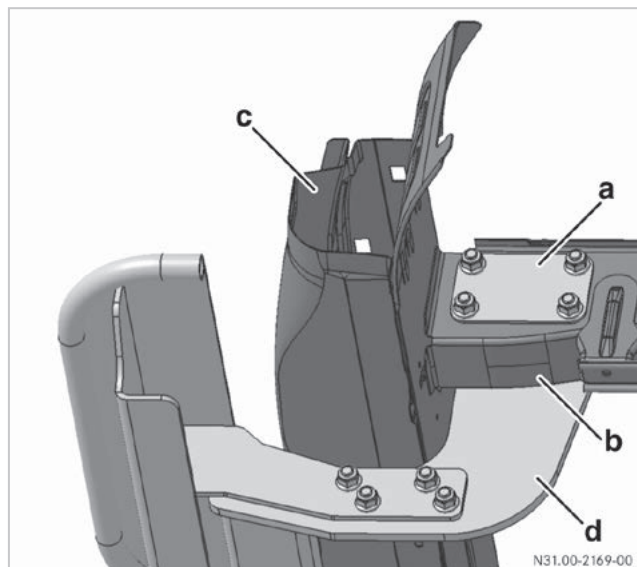
Attachments on the rear frame section

On open and closed model designations from model series 907, the vehicle comes as standard with reinforcements on the vehicle frame for mounting attachments in the rear longitudinal frame member on the left or right. If, due to a higher payload, these reinforcements are undesired, they can be deselected with code QW1 on closed model designations, depending on the equipment installed on the vehicle. Code QW1 is not available for open model designations.

If the vehicle does not have this special equipment, it is not permissible to install a trailer coupling, for example (see warning).

The fastening of additional assemblies or attachments to the rear frame section must be analogous to the fastening of the trailer coupling available as special equipment.

In order to cope with larger forces and moments, provide additional bracing on the frame end crossmember to handle moment loads.



End crossmember on longitudinal frame member (right)

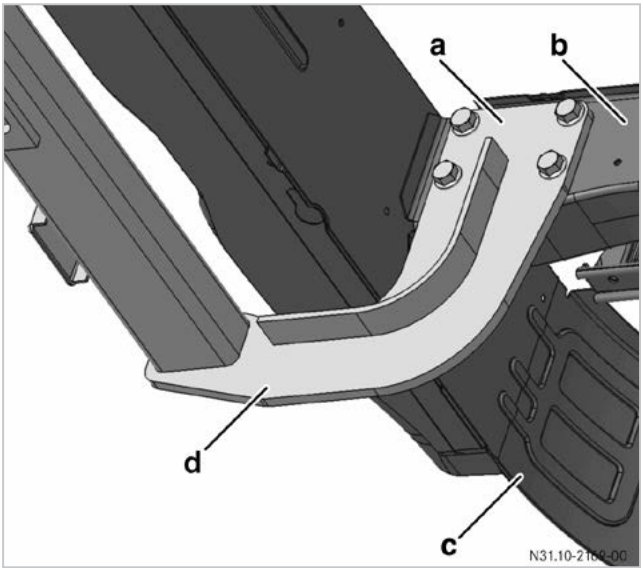
- a Attachment of assembly stand at longitudinal frame member
- b Lower flange of longitudinal frame member
- c Frame end crossmember
- d Assembly stand for trailer coupling

 **WARNING**

If the retrofitting of a trailer hitch is not permitted and, despite this, a trailer hitch or other component parts are installed, the longitudinal frame member will be weakened and could break. If this happens, the trailer can detach from the vehicle.

There is risk of an accident and danger to life and limb!

Only retrofit a trailer hitch if this is permitted.



Inside view of end crossmember on right longitudinal frame member

- a Attachment of assembly stand at longitudinal frame member
- b Lower flange of longitudinal frame member
- c Frame end crossmember
- d Assembly stand for trailer coupling

For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Body-builder Portal (→ page 20).

Further information on the hole patterns of the different trailer coupling variants is available under 10.2 Trailer coupling hole patterns (→ page 403).

Attachment using mounting supports

As a minimum, use all the factory-installed mounting supports for attaching bodies to the vehicle frame. The bodies must rest on the mounting supports and may also rest on the longitudinal frame members. It is recommended that the body should not rest on the end crossmember. You can find further information in 7.1.4 Attachment to the frame (→ page 204).

6.2.3 Material for the vehicle frame

If the frame is extended, the material of the extension element must have the same quality and dimensions as the standard chassis (only permissible in the substructure area).

 **NOTE**

Lengthening in the area of the front-end assembly is not permissible (impact load path)!

Material	Yield strength [N/mm ²]	Tensile strength [N/mm ²]
Steel	235–340	240–510

6.2.4 Overhang extension

Modifications to the overhang are possible, but must always take the permissible axle loads and the minimum front axle load into account.

On vehicles with a closed body (crewbus or panel van), an overhang extension is only permissible following coordination with the Technical Consultancy (→ page 18).

In open model designations, a frame extension in compliance with the following specifications is possible:

- Observe the restrictions and specifications for vehicles with Blind Spot Assist/Sideguard Assist in 8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist (→ page 318) and reversing camera in 8.9.9 Reversing camera (→ page 347).
 - Implement a frame extension with reinforcements in accordance with the following schematic illustrations.
 - A screw connection (in the X-direction) of a frame extension on the screw-on brackets at the frame end is not permitted.
 - It is recommended to select the following special equipment for bodies with frame extension:
 - Frame end without screw-on brackets code Q74
 - Omission of end crossmember code Q72
- ▶ For a frame extension of over 350 mm, an additional crossmember must be installed.
- ▶ Any additional crossmembers must have the same functionality as standard crossmembers.
- ▶ Standard mounting supports must be used at the end of the frame.
- ▶ The interval between the mounting supports must not exceed 500 mm.
- ▶ If the frame overhang is extended, the function of Trailer Stability Assist (TSA) and the permissible trailer load specified in the vehicle registration document must be checked and, if necessary, be reduced or even omitted; See 4.3.5 Overhang (→ Page 88).
- ▶ Reinforce the frame overhang.

▶ If the overhang is extended, enter the new overhang dimension and the distance of the rear radar from the rear axle in both rear radars (BSM) by new SCN coding with O-codes.
On no account must the original positions of the radar sensors in the bumpers be changed. This adjustment must be performed by a Mercedes-Benz Service Partner. For further information See 8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist (→ Page 318).

▶ Following the modification, the bodybuilder manufacturer must determine the vehicle's centers of gravity as per 9.1 Vehicle center of gravity (→ page 391).

- Comply with the technically permissible axle loads.
- Comply with the minimum front axle load under all load conditions (→ page 78).

❗ For more information, contact the Technical Consultancy (→ page 18).

Maximum overhang

If the limit values given in 4.3.5 Overhang (→ page 88) for the overhang and the technically permissible maximum rear axle load are complied with, the original trailer load will still apply and ESP® operation will not be influenced if the trailer coupling point is not relocated. If the trailer coupling point is relocated, observe the additional information in chapter 4.3.5 Overhang (→ page 88).

If the underride guard needs to be repositioned due to the overhang extension, the attachment must be the same as that of the original vehicle; See 6.6.8 Underride guard (→ Page 198).

Observe the country-specific legal regulations pertaining to the underride guard.

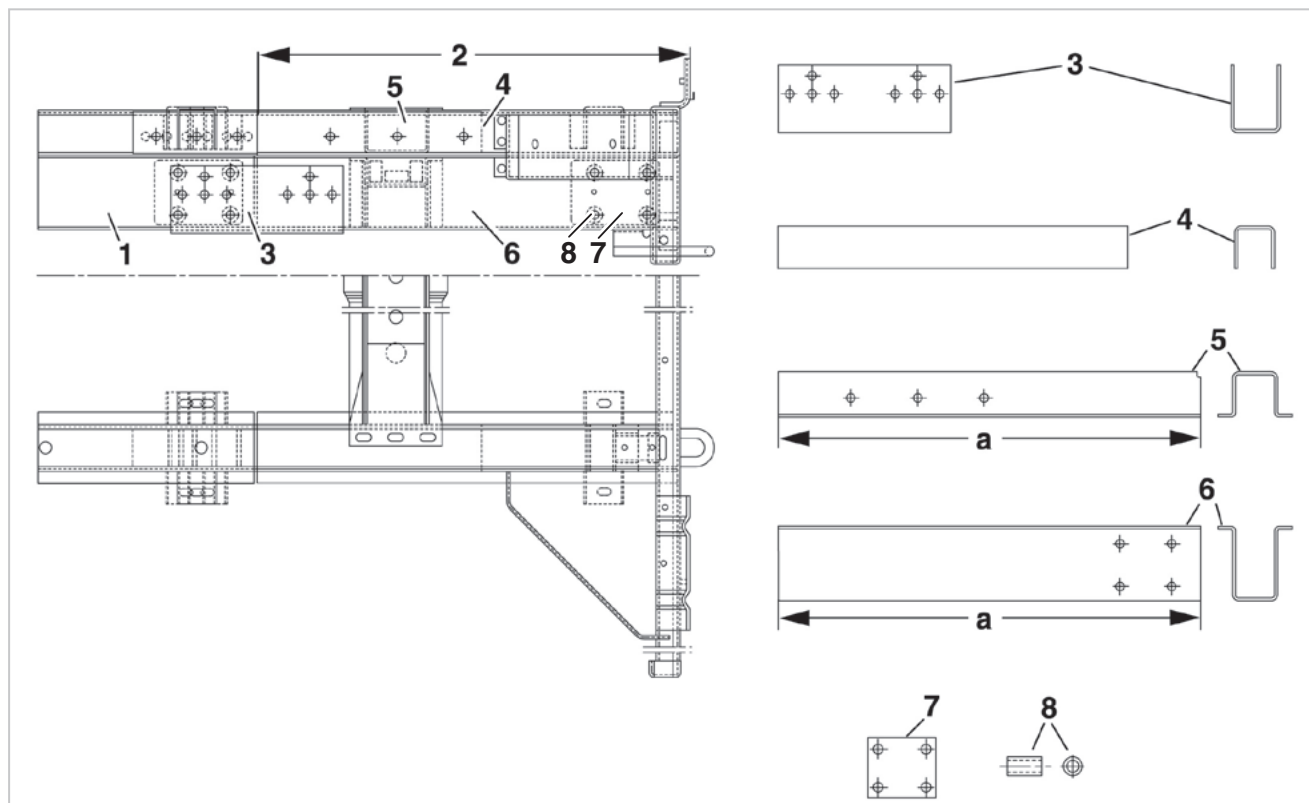
Observe the information and specifications for the commissioning of control units after completion of the body mounting work, before the vehicle is placed on the market, See 8.19 Commissioning of control units (→ Page 387).

6 Modifications to the basic vehicle

For the design of the frame extension in the case of an overhang extension, see the following illustrations.

Check the effects on the permissible trailer load and the Trailer Stability Assist (TSA) function; See 4.3.5 Overhang (→ Page 88).

Vehicles with 3.0 t and 3.5 t



Frame extension with overhang extension (schematic)

- 1 Longitudinal frame member
- 2 Frame extension
- 3 Outer reinforcement
- 4 Inner reinforcement
- 5 Body support extension
(wall thickness 3.5 t: 2 mm)
- 6 Vehicle frame extension
(wall thickness 3.5 t: 2 mm)
- 7 Reinforcement plate min. 2 mm
- 8 Spacer bushing, pipe 24 × 4
M steel or St 35 NBK
- a Dimension defined by bodybuilder manufacturer

! NOTE

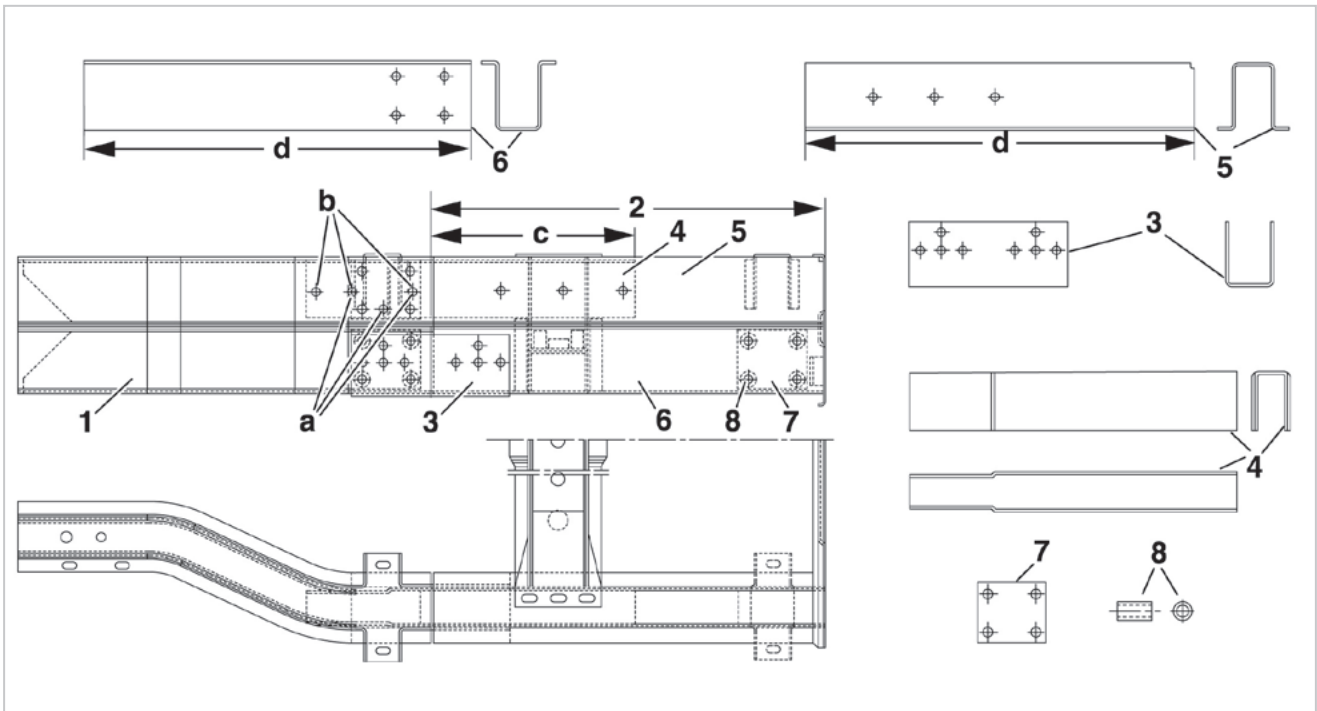
After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

! NOTE

Observe the country-specific regulations and directives.

A longitudinal frame member reinforcement (see 7 and 8) is required for the assembly of attachments (e.g. trailer coupling, underride guard, or cargo liftgate).

Vehicles with 4.6 t and 5.0 t



Frame extension with overhang extension (schematic)

- | | | | |
|---|---|---|---|
| 1 | Longitudinal frame member | a | Bore holes, 3665 mm wheelbase |
| 2 | Frame extension | b | Bore holes, 4325 mm wheelbase |
| 3 | Outer reinforcement | c | 350 mm (3665 mm wheelbase),
300 mm (4325 mm wheelbase) |
| 4 | Inner reinforcement
(wall thickness 5 t: 3 mm) | d | Dimension defined by bodybuilder manufacturer |
| 5 | Body support extension | | |
| 6 | Vehicle frame extension
(wall thickness 5 t: 3 mm) | | |
| 7 | Reinforcement plate min. 2 mm | | |
| 8 | Spacer bushing, pipe 24 × 4
M steel or St 35 NBK | | |

! NOTE

Observe the country-specific regulations and directives.

A longitudinal frame member reinforcement (see 7 and 8) is required for the assembly of attachments (e.g. trailer coupling, underride guard, or cargo liftgate).

6.2.5 Wheelbase modifications for open and closed model designations

-  Observe the information in 4.3.7 Wheelbase modifications (→ page 89).

6.2.6 Modifications to the cab

For all modifications to the cab, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20). Rigidly installed equipment or conversions must satisfy the requirements of legislation relating to head-on impact as specified in UN R 21 and FMVSS 201.

WARNING

Modifications to the cab must not impair the function of any component parts relevant to safety (e.g. airbag units, sensors, pedals, gearshift lever, lines, or others). This may lead to the failure of components or safety-relevant component parts.

There is risk of an accident and danger to life and limb!

WARNING

If aftermarket modifications are made to the headliner or the roof paneling between the A-pillar and the B-pillar, the deployment of the window airbag may no longer function correctly.

If window airbags no longer function correctly, the vehicle occupants may receive inadequate protection in the event of an accident.

There is danger to life and limb!

Do not make subsequent modifications to the headliner or the roof paneling between the A-pillar and the B-pillar if the vehicle is equipped with window airbags.

NOTE

If the fuel filler cap is removed or parts are attached to the fuel filler cap, blocking may occur in the event of an accident. As a result, the area intended for the protection of the tank filler neck in the B-pillar may not function properly under certain loads (e.g. in the event of a crash).

Therefore, do not cover the cap with trim parts, and never mount "blocking" parts on the B-pillar.

The strength and rigidity of the cab structure must not be impaired.

The intake of air into the engine must not be hindered.

Modifications to the cab will cause a change in the vehicle center of gravity. Maintain the permissible center of gravity limits and axle loads.

NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Changes to cab roof, general

Modifications to the cab roof (e.g. lowering the roof height) may only be undertaken following coordination with the Technical Consultancy (→ page 18) or as per 6.2.12 Cutting the cab roof and B-pillar roof bow (→ page 132).

Plastic roofs are suitable for the installation of roof hatches only to a limited extent.

The roof load-bearing capacity is limited (→ page 90).

! NOTE

Roof bows or supporting parts must not be removed or modified without being replaced.

i Information on over-cab attachments and air deflectors is also available in the chapter "Attachments" (→ page 177).

Maintain the permissible center of gravity limits and axle loads.

Modifying the cab rear panel

If it is necessary to cut through the cab rear panel, it is possible to do this in connection with a continuous surrounding frame. The equivalent rigidity of the frame must be at least equal to the original rigidity.

Partitions may be totally or partially removed. Also note 7.4 Modifications to the panel van (→ page 212).

Comply with the statutory standards (ISO, etc.) and the relevant licensing regulations.

6.2.7 Side wall, windows, doors, and flaps

Side wall

If modifications are made to the side wall on the panel van or the crewbus, the rigidity of the modified body must be equal to that of the basic vehicle.

The front of the roof frame, the roof bow at the B-pillar, and the A-pillar and the B-pillar must be retained and their function must not be impaired in any way.

For modifications to these structural components, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

For modifications to the projected side surface area, refer to 8.9.2 Crosswind Assist (CWA) (→ page 309).

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Windows

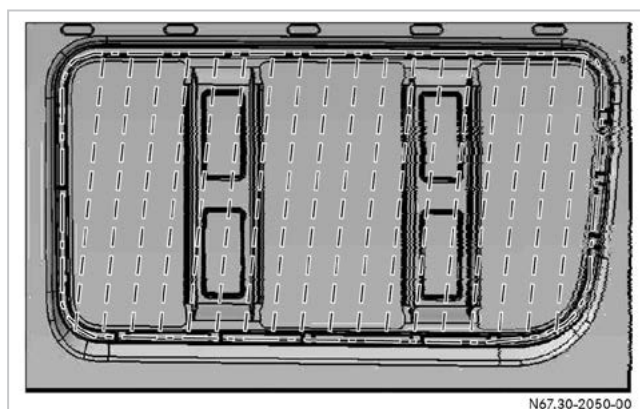
- ▶ Windows must be inserted with a stable frame.
- ▶ Connect this frame with a frictional connection to other body elements.
- ▶ If modifications need to be carried out to the supporting structure of the basic vehicle (pillars, reinforcements, attachment of roof bows) in order to retrofit windows (panoramic glazing), the rigidity of the modified body must be equal to that of the basic vehicle.

If the bodybuilder manufacturer intends to install its own windows, code W94 "No windows in bus version" is available from the factory.

- ❗ You can find further information on modifications to the side wall in 6.6.4 Shelves and load rails in the load compartment (→ page 178).

! NOTE

Do not install panoramic glazing on a panel van basic vehicle by cutting out the stamped window shape without ensuring equivalent rigidity. Otherwise there is a risk of damage to the side wall.



Example of the side wall body-in-white structure and window surface at the panel van or crewbus

- ❗ For conversions with subsequent cutting out of the stamped window shape with stiffening measures, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- ❗ A design proposal with reinforcement measures can be provided by the Technical Consultancy (→ page 18). With regard to this, also observe 7.4 Modifications to the panel van (→ page 212).

If the bodybuilder manufacturer intends to install its own windows in the rear-end doors, note the following in combination with code W78 "Windshield wiper on rear-end door":

- To guarantee proper functioning of the rear window wipers, the geometry of the windows supplied by the bodybuilder manufacturer must match that of the windows available as standard equipment.
- The wiper rubbers of the rear window wipers must make contact with the windows across the entire swept area.
- The rear windows must be 3 mm thick.
- The rear windows must not protrude over the door paneling.

Doors and flaps

⚠ WARNING

Modifications to

- door kinematics
- guidance systems (rails, sliding carriages etc.)
- locking systems (closing assists, anti-entrapment strips, locks, arresters etc.)
- end stop systems (buffers etc.)
- body-in-white structure

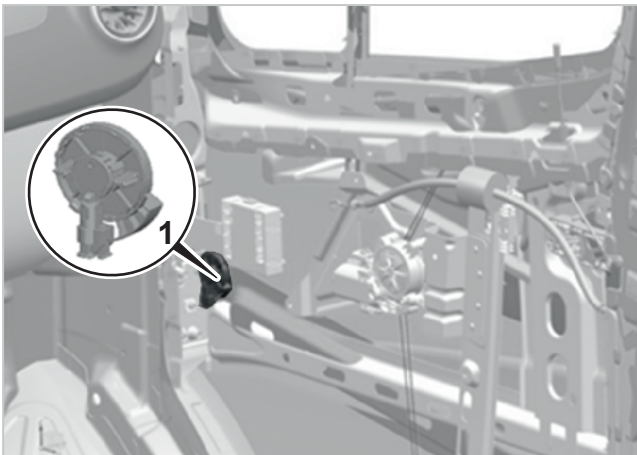
may result e.g. in the door opening unintentionally.

There is risk of an accident and danger to life and limb!

Do not make any modifications to these systems.

If modifications need to be carried out to the supporting structure of the basic vehicle (crossmembers, frames, pillars, reinforcements, attachment of roof bows) in order to retrofit doors, the rigidity of the modified body must be equal to that of the basic vehicle.

The trigger sensor of the occupant protection systems is located in the door body on vehicles with window airbags or thorax/pelvis side airbags. Do not make modifications to the door body (see illustration).



Door body with sensor system

- 1 Pressure sensor
(trigger sensor of the occupant protection systems)

Seats in the living and passenger compartment must be directly accessible from the outside by a door or from the cab.

It must be possible to open locked doors quickly and easily from the inside.

The doors must open wide enough and the door entrances must be shaped in such a way as to enable persons to get in and out of the vehicle safely and comfortably.

The maximum permitted height of the bottom step above the roadway is 400 mm.

Installations must ensure sufficient clearance from the inside handles in all positions (anti-pinch protection).

Take into account the new trim parts as of model year 2024 for the additional measure "catch bearing" (front catch) of the load compartment sliding door in the center area of the B-pillar when installing or modifying, see information on the partition in 7.4 Modifications to the panel van (→ page 212).

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

! NOTE

For all vehicle door designs (front doors, sliding doors, rear-end doors), do not make modifications to the locking system, the immediate area around the door, or the area of the pillars/crossmembers.

The attachment of additional mechanical locks (e.g. slam locks) is not permitted.

Sliding door to load compartment (LST)

The system for the load compartment sliding door (LST) has been designed for a maximum door weight of 65 kg. Modifications must not cause this weight to be exceeded.

The following table shows the ex factory weight of the load compartment sliding door for each vehicle model and wheelbase. The delta corresponds to the maximum permissible additional weight on the sliding door to the load compartment.

Maximum permissible additional weight LST

(total weight of LST - basic equipment = additional weight)

Roof height	Wheelbase as of 3665 mm Body length as of A2 [kg]
LH1	65 - 49 = 16
LH2	65 - 54 = 11

For the special equipment items with the following codes, the respective weights must be added to the basic equipment (see table).

If equipment needs to be removed, take it into account in the weight analysis.

- Electric closing assist (code T50/T51): 3 kg
- Electric operation (code T55/T56): 6 kg (including code T50/T51)
- Window at front of sliding door (code W17):
As of body length A2 = 4 kg
- Laminated glass (code W17+F43):
As of body length A2 = 7 kg
- Sliding window in side wall (code W18):
As of body length A2 = 6.5 kg
- Luxury door trim (code V21): 2 kg
(only for RWD A2LH2 RE)
- Speakers (code EL9): 2 kg

Rear-end doors (HDT)

The system for the rear-end doors has been designed for a maximum weight of 38 kg per rear-end door, and this must not be exceeded due to modifications.

The following table shows the ex factory weight of the rear-end doors for each vehicle model and roof height. The delta weight, including special equipment, corresponds to the maximum permissible additional weight on the rear-end door.

Maximum permissible additional weight HDT

(total weight of HDT - basic equipment = additional weight)

Roof height	Rear-end door	
	Left [kg]	Right [kg]
LH1	38 - 27.5 = 10.5	38 - 26 = 12
LH2	38 - 30 = 8	38 - 28 = 10

For the special equipment items with the following codes, the respective weights must be added to the basic equipment (see table).

If equipment needs to be removed, take it into account in the weight analysis.

- Window (code W61/W78): 2 kg
- Luxury interior trim (code V21): 1.5 kg
- Windshield washer system (code W78): 1 kg

Furthermore, take care to ensure an extensive connection and CoG distribution at the doors.

The bodybuilder manufacturer alone is responsible for the conversions and modifications it makes to the vehicle.

! NOTE

When selecting a carrier system for bicycles or other transported items on the rear end, observe the specified restrictions on additional masses on the rear-end doors. Take into account the mass of the carrier system including the transported items, e.g. bicycles.

If the additional mass restriction cannot be complied with, it is recommended to use a suitable carrier system supported by or attached to the ball head of the trailer coupling.

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

6.2.8 Fenders and wheel wells

Ensure that there is sufficient space between the tire and the fender or wheel well even with snow chains fitted and the suspension completely compressed (allowing for axle twist). Take into account the dimensions in the 2D drawings (offer drawings).

! WARNING

If modifications to seat attachments on the wheel well or to subsequently lowered wheel wells are carried out, this can cause damage to the vehicle (e.g. wheel well and tires).

There is risk of an accident and danger to life and limb!

Do not mount seats on the wheel well.

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Wheel wells (side wall and substructure scope)

It is possible to lower the wheel wells as long as the following preconditions and limit values are complied with:

- No component parts or sharp edges (e.g. folded seams or edges) must protrude into the wheel well.
- The maximum permitted lowering dimension must not be exceeded by any component part in the wheel well.
- Unrestricted use of snow chains is not possible: The entry "Use of snow chains subject to limitations" must be entered in the vehicle documents.

! NOTE

For modifications to the wheel wells, observe the information and specifications in chapters 4.3.3 Lower wheel well on rear axle for closed model designations (→ page 85) and 4.3.4 Lower wheel well on rear axle for open model designations (→ page 86).

6.2.9 Frame end crossmember

If special bodies are mounted, the frame end crossmember (code Q72) can be omitted ex factory; See 6.2.4 Overhang extension (→ Page 119).

i Further information is available in 6.6.8 Underride guard (→ page 198).

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

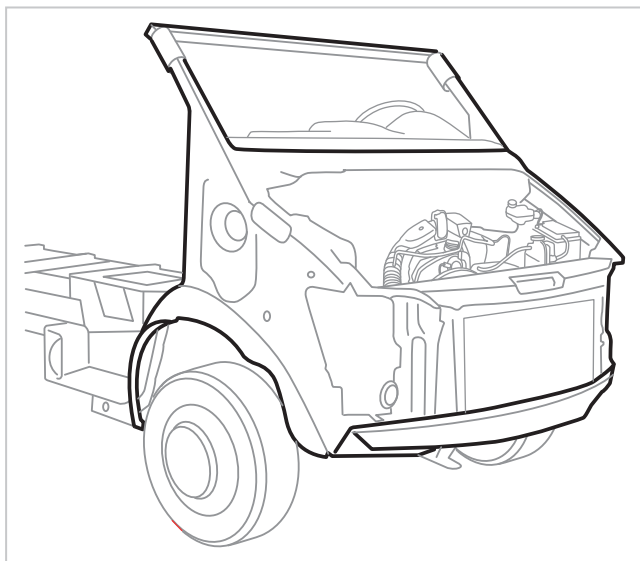
! NOTE

Observe all national laws, directives, and registration regulations.

Do not reduce the width of the wheel wells.

6.2.10 Cowl

The cowl (chassis cab base) offers bodybuilder manufacturers a basis for producing fully integrated bodies (e.g. camper vans) or special-purpose bodies. It is available ex factory as code F50 "Cab base".



Cowl chassis (schematic)

Observe the specifications on setting up cowl chassis in 7.14 Bodies on chassis with cab base (code F28, F50) (→ page 227).

Assembly with a Letter of Compatibility is mandatory (code X88), See 3.3 Selecting the basic vehicle (→ Page 53).

! NOTE

On F50 vehicles, the jack support point on the body-in-white is not to be used before and during the body mounting phase, because the structure does not have the required stability.

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

6.2.11 Panel van and crewbus roof

If modifications are made to the roof structure of panel vans or crewbuses, observe the following points:

- If the roof paneling and roof bows are removed and if no continuous sectional frame is possible, additional roof bows must be fitted. The overall design must be retained, and sufficient rigidity of the modified parts must be guaranteed.
- In order to prevent functional impairments, observe the body limits for camera-based assistance systems; See 8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist (→ Page 339) and 8.9.6 Automatic driving lights/rain sensor (→ page 340).
- For modifications to the projected side surface area, refer to 8.9.2 Crosswind Assist (CWA) (→ page 309).
- Observe the limit values for the suspension (→ page 80).

! NOTE

The rigidity of the new roof structure must be equal to that of the original standard roof.

Do not make modifications to the rear portal including the roof area.

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Attachment to the roof

Fastenings similar to roof luggage racks can be used as an option for retrofitting attachments (→ page 178).

For attachments at the roof paneling, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20). Rotating beacons and work lamps are excluded from this requirement.

For attachments at roof bows, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Cutouts in the roof

Cutouts in the roof are permissible when the following specifications are observed:

- Maintain a distance of at least 20 mm to component parts of the roof structure.
- The roof structure may only be modified in accordance with the previously mentioned specifications.
- The corners of the roof cutout must have a radius of at least 75 mm.
- Where plastic roofs are used, pay particular attention to the properties of the material, as this may be susceptible to cracking.
- Deburr all edges and seal using anticorrosion protection.
- For designs that deviate from this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- For cutouts in the cab roof, observe the specifications in 6.2.12 Cutting the cab roof and B-pillar roof bow (→ page 132).

⚠ WARNING

If aftermarket modifications are made to the headliner or the roof paneling between the A-pillar and the B-pillar, the deployment of the window airbag may no longer function correctly.

If window airbags no longer function correctly, the vehicle occupants may receive inadequate protection in the event of an accident.

There is danger to life and limb!

Do not make subsequent modifications to the headliner or the roof paneling between the A-pillar and the B-pillar if the vehicle is equipped with window airbags.

Roof height increase

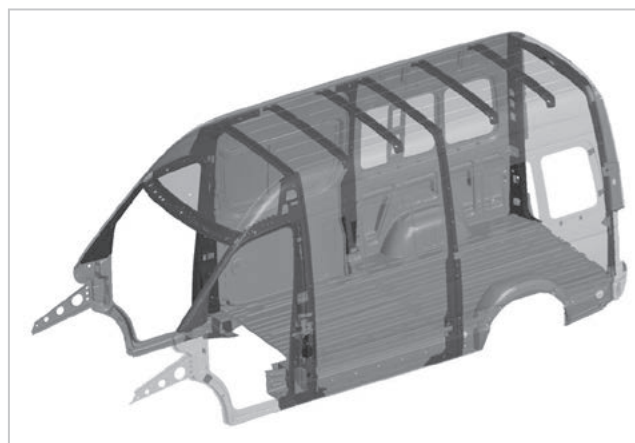
Modifications are only permitted to be made to the rear portal, including the roof area, in exceptional circumstances. For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The rigidity of the new roof structure must be equal to that of the original standard roof.

Number of roof bows

Wheelbase	Quantity required
3665 mm	≥ 5 bows
4325 mm	≥ 6 bows

Location of roof bows



Panel van roof bows

! NOTE

The bows must be secured to the side walls in such a way that a non-positive connection is guaranteed (bend-resistant connection of bow and roof frame).

Reinforce the roof bows accordingly in the event of any increase in the height of the roof.

6 Modifications to the basic vehicle

The minimum moment of inertia required I_x per roof bow is available in the table below:

Increase in roof height [mm]	Moment of inertia I_x per bow [mm ⁴]
250	≥ 40000
400	≥ 65000
550	≥ 86000

If the roof height is either reduced or not modified, adhere to a minimum required moment of inertia per bow of:

$$I_x = 33,000 \text{ mm}^4$$

The maximum roof load of the high panel roof is 150 kg if the load is evenly distributed over the entire roof surface; see also 6.6.3 Roof luggage racks

(→ page 178).

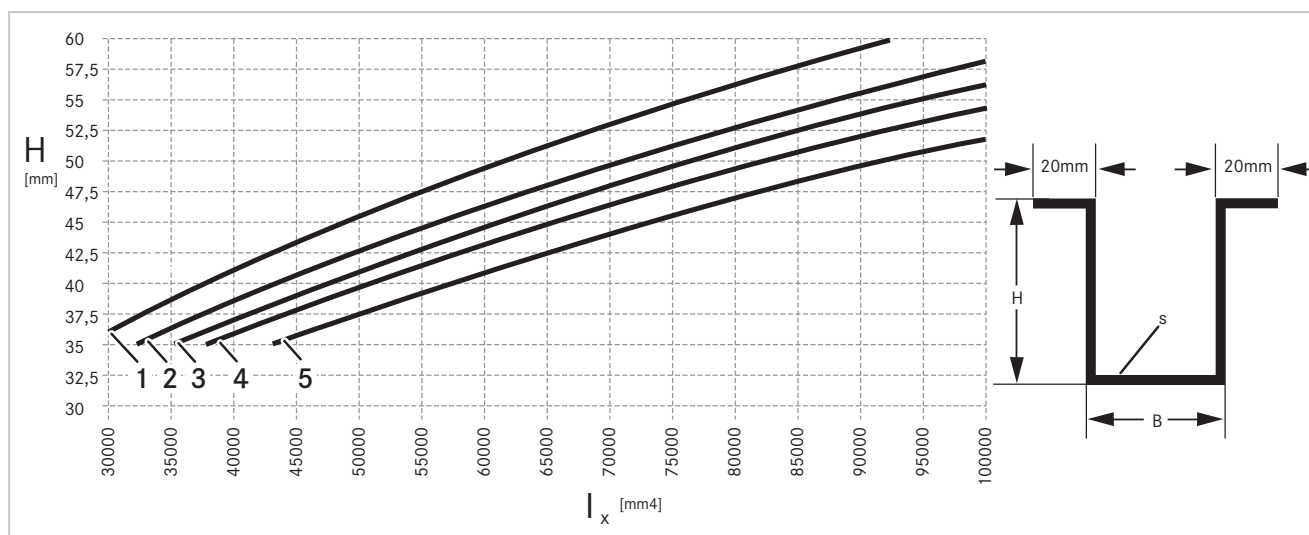
⚠ WARNING

If the maximum permissible center of gravity is exceeded on vehicles with ESP®, this system may no longer work correctly and could ultimately fail. This could cause the driver to lose control of the vehicle.

There is danger to life and limb!

Make sure that the permissible height for the vehicle center of gravity is not exceeded.

Observe 8.9.2 Crosswind Assist (CWA) (→ page 309) for modifications to the projected side surface area (increase in the height of the roof).



Required moments of inertia for roof bows with 20 mm flange with the roof paneling

- 1 $B: 50 \times s: 0.8$
- 2 $B: 40 \times s: 1.0$
- 3 $B: 50 \times s: 1.0$
- 4 $B: 60 \times s: 1.0$
- 5 $B: 50 \times s: 1.2$

Retrofitting of roof hatch and lifting roof

⚠ WARNING

If aftermarket modifications are made to the headliner or the roof paneling between the A-pillar and the B-pillar, the deployment of the window airbag and the thorax/pelvis side airbag may no longer function correctly.

If window airbags and thorax/pelvis side airbags no longer function correctly, the occupants may be insufficiently protected in the event of an accident.

There is danger to life and limb!

Do not make subsequent modifications to the headliner or the roof paneling between the A-pillar and the B-pillar if the vehicle is equipped with window airbags.

! NOTE

Roof bows or structural parts must not be removed or damaged without being replaced (→ page 132).

Maximum roof loads

Panel van [kg] LH1	High-roof panel van [kg] LH2	Cab crewcab [kg]
300	150	100

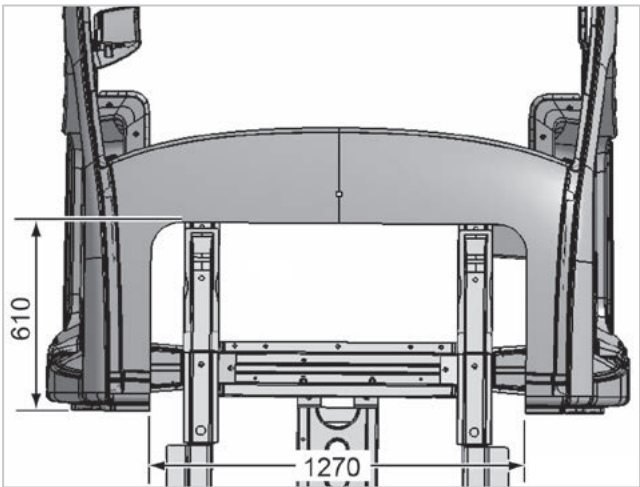
The limit value of the vehicle's maximum center of gravity must not be exceeded.

When installing roof hatches and lifting roofs, observe the following:

- Observe the specifications on cutouts in the roof and on modifications to the roof structure.
- Ideally position roof hatches so that they are centered (symmetric to the vehicle longitudinal axis).
- Comply with the limit values for the roof load-bearing capacity in the following table.
- Plastic roofs are suitable for the installation of roof hatches only to a limited extent. Glue the roof hatch over the entire surface in the area of the overlap to stiffen the roof section. The component weight must not exceed 8 kg. It is recommended to use a suitable solution from the camper van segment.
- If a lifting roof is fitted, at least $\frac{2}{3}$ of the original roof area must be retained.
- Provide a seal between the installation and roof surface to protect against water and dust ingress.
- In the event of deviations from this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

6.2.12 Cutting the cab roof and B-pillar roof bow

For semi-integrated bodies, e.g. camper vans or integral box bodies, the cab roof including B-pillar roof bows can be cut out in the indicated area (see illustration) where necessary:



Permissible roof cut

NOTE

When cutting the B-pillar roof bow, it is essential to ensure equivalent rigidity in one of the ways listed below.

In the event of deviations from the specifications of these variants or for alternative methods of ensuring equivalent rigidities developed by the bodybuilder manufacturer, a detailed evaluation by the Technical Consultancy (→ page 18) is required. A Letter of Compatibility is also required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Equivalent rigidity when cutting B-pillar roof bow

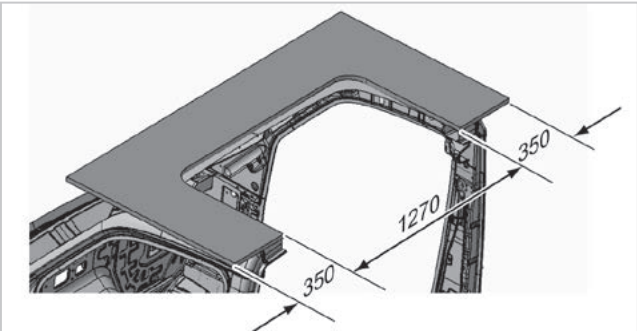
The following variants have fundamentally been approved as the basic concept. The bodybuilder manufacturer is responsible for the entire body concept. If you have any questions regarding these variants, contact the Technical Consultancy during the planning phase (→ page 18).

Variant 1: Sandwich construction and wooden board

When the B-pillar roof bow is cut, the equivalent rigidity requirements can be met by means of a wooden board or sandwich construction bonded to the basic vehicle over its entire surface (e.g. with Sikaflex 221). The arched roof contour must be adapted to form a non-positive fit with the sandwich construction or wooden board by means of an auxiliary construction.

Required bending resistance of sandwich construction and wooden board

y-axis	$EI_2 = 7 \times 10^8 \text{ N/mm}^2$
z-axis	$EI_1 = 2 \times 10^{11} \text{ N/mm}^2$



Simulating structure (sandwich construction/wooden board) bonded to cut roof structure over entire surface

Material characteristics

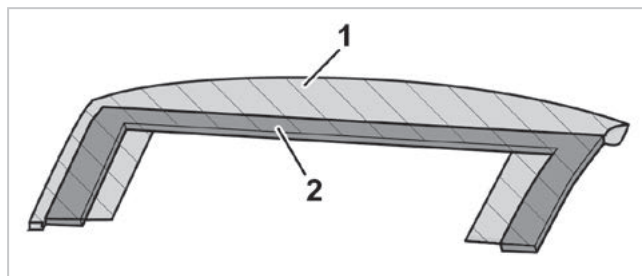
Material	Sandwich construction	Wooden board
Body	2.0 mm GRP 26.0 mm foam 2.0 mm GRP	20.0 mm wood
E [N/mm ²]	$E_{\text{GRP}} = 12000$ $E_{\text{foam}} = 80$	$E_{\text{wood}} = 3000$

Variant 2: Welded structure under cab roof

When the B-pillar roof bow is cut, the equivalent rigidity requirements can be met by means of a welded structure installed in the basic vehicle under the cut cab roof.

Required bending resistance for welded structures

y-axis	$EI_2 = 8.35 \times 10^9 \text{ N/mm}^2$
z-axis	$EI_1 = 2.36 \times 10^{11} \text{ N/mm}^2$
4325 mm	≥ 6 bows

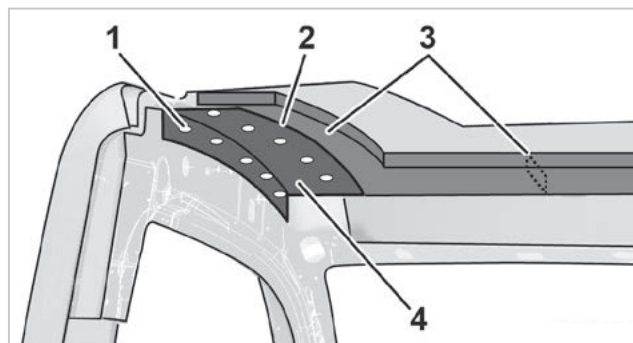


Welded structure

- 1 Roof paneling
- 2 Rectangular profile

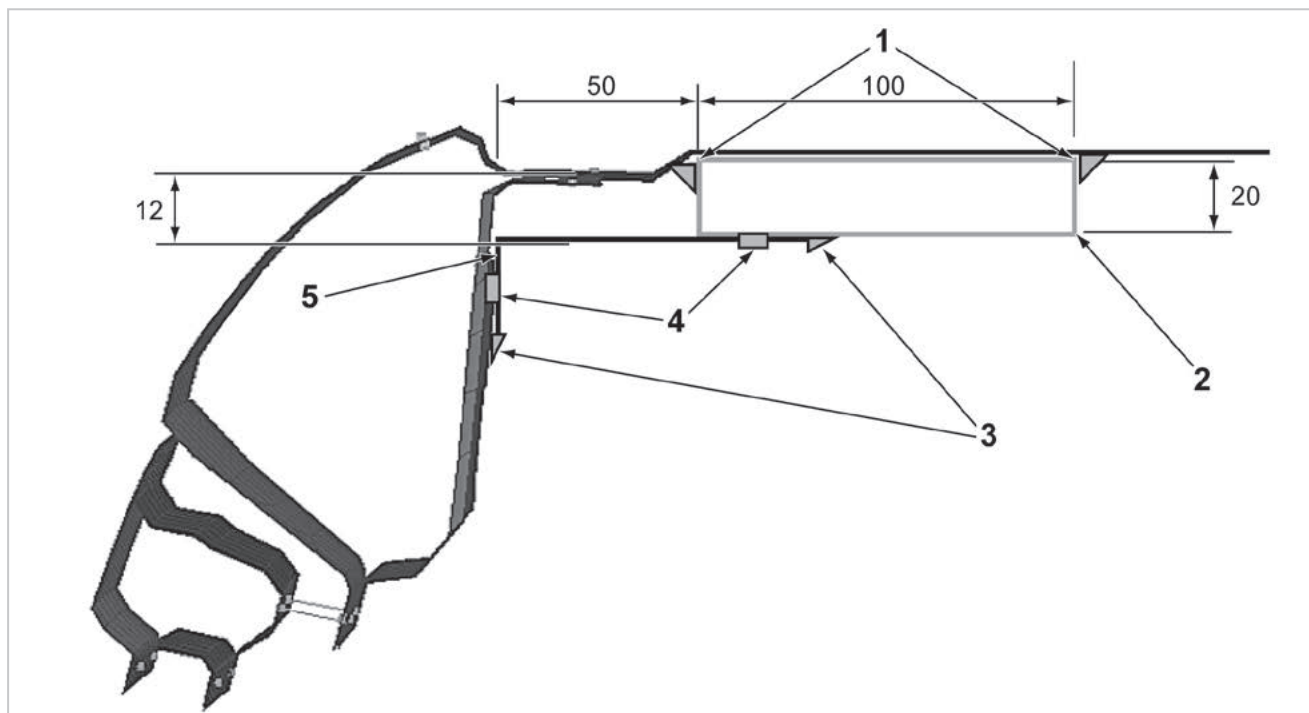
Material characteristics of subframe

Material	Steel
Height [mm]	20
Width [mm]	100
Wall thickness [mm]	1.5
E [N/mm ²]	210000



Welded structure

- 1 Plug weld
- 2 Overlap seam
- 3 Rectangular profile
- 4 Striker plate



Cross-section of welded structure

- 1 Fillet weld
- 2 Rectangular profile (2 mm wall thickness)
- 3 Overlap seam
- 4 Plug weld
- 5 Striker plate

6.2.13 Underfloor paneling

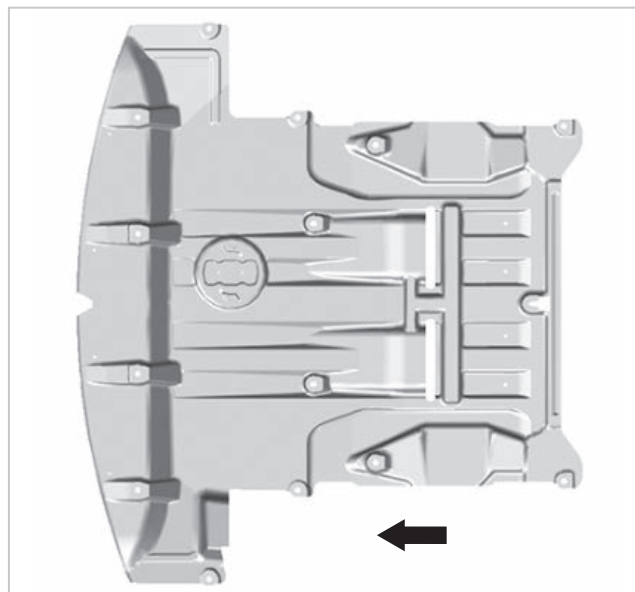
In the Sprinter, different types of underfloor paneling are available for soundproofing. Depending on the registration type and the country-specific laws and regulations, these are already available in the standard scope as single-piece or three-piece noise capsule. The one-piece noise capsule is available as special equipment with Code XM1.

On vehicles with M1 /N1 registration and OM654 engine, the front, one-piece underfloor paneling is fitted ex works.

A larger, three-part underfloor paneling is only fitted ex works with N2 registration and engine OM654, automatic transmission code G43 and in combination with axle ratio code AR5 for countries with statutory requirements.

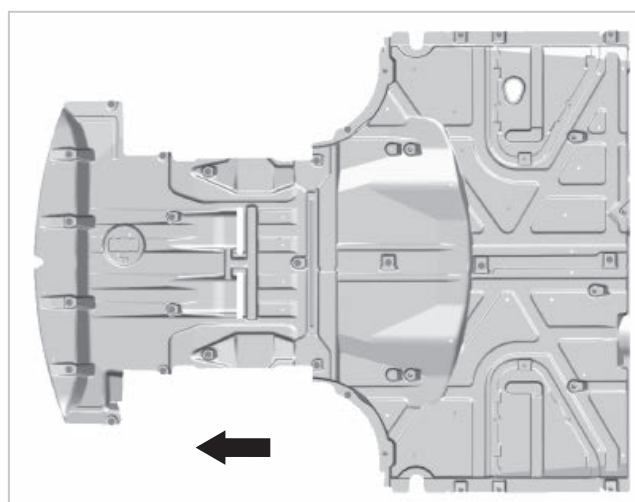
Further adjustments to the soundproofing with regard to different types of registration due to new legal requirements are currently under review. If you have any questions, please contact the Technical Consultancy team (→ page 18).

If the vehicle is ordered with the "Transmission power take-off" special equipment (code N05/07) and the associated exhaust system variants, for thermal reasons, no underfloor paneling will be installed; See 6.3.3 Exhaust system (→ Page 141).



One-piece noise capsule at front under engine

➡ Direction of travel



Three-piece noise capsule under engine and transmission and behind front axle

➡ Direction of travel

! NOTE

The underfloor paneling installed ex works represents a certified vehicle status. For this reason, no modifications shall be made to these component parts.

Bodies or modifications that result in a change to the underfloor paneling can cause the type approval of the basic vehicle to cease to be valid with regard to external noises according to UN-R51.

During the planning phase, coordinate with the Technical Service and/or the relevant type-approval authority whether and how the certified limits for the basic vehicle can be complied with.

Additional soundproofing measures for compliance with the legal specifications are available in the vehicle at different additional locations depending on the registration type and the country-specific laws and regulations.

It is recommended to not modify or replace the component parts installed ex factory.

If the registration type will be changed as a result of the body or conversion, additional or modified soundproofing measures may be required. The bodybuilder manufacturer has the overall responsibility in this respect for compliance with the country-specific laws and regulations (e.g. UN R 51); See 3.10 Soundproofing (→ Page 70).

When submitting the application for a Letter of Compatibility, provide proof of compliance with legal requirements upon request.

! NOTE

In the event of changes to the underfloor paneling or the installation of an underride guard required for body mounting or conversion work on special-purpose vehicles, contact the Technical Consultancy during the planning phase (→ page 18). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

6.3 Engine peripherals and drivetrain

NOTE

Maintenance and repair of the vehicle must not be hindered by the body (→ page 71).

-  Vehicles with engines OM651 and OM642 are no longer available ex factory. For this reason, this information is no longer included in the present Body/Equipment Mounting Directive. If required, you can find corresponding information and specifications in previous Body/Equipment Mounting Directives in which the engines are still included. These are still available in the Body/Equipment Mounting Directives Archive in the Body-builder Portal.

Modifications to the engine timing and performance enhancement

NOTE

Any interference in the engine timing by customers and bodybuilder manufacturers is not permissible.

Manipulation of or modification to the engine timing causes changes in the certified engine data and emission values, and thus results in the immediate invalidation of the operating permit.

6.3.1 Fuel system (diesel)

General

Modifications to the fuel system can only be carried out in coordination with the Technical Consultancy (→ page 18).

WARNING

Fuels are highly flammable.

There is a risk of fire and explosion if fuels are improperly handled!

Avoid all fire, open flames, smoking, and sparks.

NOTE

Non-approved modifications to the fuel system (fuel tank, lines, etc.) may lead to impaired performance and the engine running in limp-home mode.

Note the following when working on the fuel system:

- Do not attach heat-conducting components or components that restrict installation space.
- Modifications to the fuel pump, fuel line length, and fuel line routing are not permitted. Any changes to these components that are matched to each other may impair engine operation.
- Modifications and attachments (e.g. additional eye-lets) are not permitted in the vicinity of the fuel filler neck in the area of the B-pillar.
- If bodies are mounted on basic vehicle cabs, protection of the fuel system is necessary if the fill level sensor is not protected by the body. The delivery module is protected by the body ex factory.
- When installing fuel-fired auxiliary heating, observe the specifications in 6.5.2 Auxiliary heating (→ page 167).
- Observe the type-approval procedures when connecting the fuel supply to auxiliary heating systems.

Protection of the fuel system

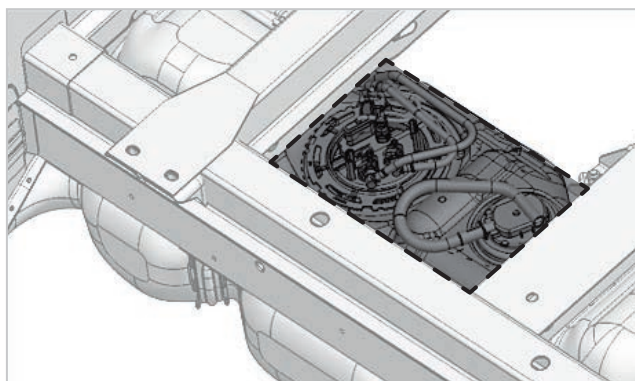
! NOTE

As a general rule, do not place any load on the fuel tank, e.g. by using it as a step.

In the case of bodies based on an open model designation, it is necessary to protect the system components on the top of the fuel tank (fill level sensor and vent valve) from falling cargo or objects as shown in the following illustration.

Without such protection in the area illustrated, these system components and the fuel tank may become damaged, rendering the vehicle unusable.

The planning and implementation of the protection, taking into account the specific vehicle attachment, is the sole responsibility of the bodybuilder manufacturer.



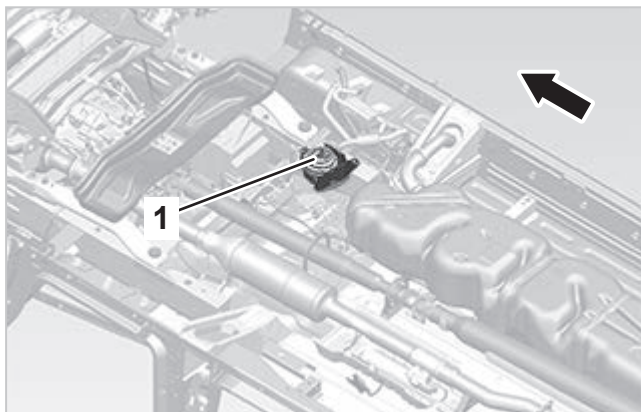
Protection of the fuel system (schematic)

Fuel filter

On engine OM654, the fuel filter is located in the substructure, in the direction of travel upstream of the fuel tank (see illustration). The position differs for the vehicles with rear wheel drive and with all-wheel drive in the X-direction.

Details on the position and component part geometry can be found in the 3D design data. For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).

No modifications may be made to the fuel filter, its position and mounting, and the related detachable parts. Ensure that access to the fuel filter for servicing and repair purposes is provided and that it is not restricted by body-manufacturer-specific detachable parts.



Position of fuel filter on OM654 engine and rear wheel drive (schematic)

- 1 Fuel filter for OM654
➔ Direction of travel

! NOTE

Impermissible modifications to the fuel filter and any access hindrance may lead to functional impairments or to component failure, and consequently to the engine running in limp-home mode.

Fuel-powered booster heater

Observe the following if fuel-powered booster heaters are retrofitted:

- No sharp edges permitted
- The fuel tank must not be subjected to load in the event of a crash. Fit deflectors if necessary.
- Fuel lines must be designed safely.
- Exhaust gases must not be directed into the vehicle interior.

Observe the type-approval procedures when connecting the fuel supply to auxiliary heating systems.

Connection of booster heater (code KL1)

Code KL1 is available as special equipment and consists of the following component parts:

- 1x transport protection with connector coupling
- 1x auxiliary heating line

! NOTE**Operation of additional external consumers**

When the fuel level is in the reserve range, do not tap fuel from externally operated consumers via the auxiliary heating line.

! NOTE**Fuel display (code J51)**

Do not make modifications to the fuel tank. If the bodybuilder manufacturer installs a tank of a different size, the fill level will no longer be displayed correctly on the instrument cluster. Modifications to the tank system are only permissible in consultation with the vehicle manufacturer Mercedes-Benz AG.

When using externally operated consumers via the auxiliary heating line in vehicles with code J51 "Fuel display, adapted for additional fuel tapping", the indicated range remaining varies from the series production configuration.

**ENVIRONMENTAL INFORMATION**

Modifications carried out incorrectly to the fuel system may have a detrimental effect on the environment.

6.3.2 SCR exhaust gas aftertreatment system

The SCR (Selective Catalytic Reduction) exhaust gas aftertreatment system is the most important element for complying with the Euro 6/VI emissions regulations. The SCR exhaust gas aftertreatment system reduces nitrogen oxide (NOx) through exhaust gas aftertreatment.

! NOTE

Modifying the components of the SCR exhaust gas aftertreatment system or changing their installation positions is not permitted. This includes the AdBlue® tank, the AdBlue® lines, and all other exhaust gas aftertreatment system components.

Apart from electronic control units and sensors, the exhaust gas aftertreatment system requires a catalytic converter and the AdBlue® additive.

AdBlue®

AdBlue® is a solution consisting of urea (32.5%) and water. AdBlue® is a noncombustible, nontoxic, colorless, odorless, and water-soluble liquid.

AdBlue® has a limited shelf life, which is affected by ambient temperature and air humidity. Take this into account when filling it from containers.

Information on handling AdBlue® and its material characteristics is available in ISO 22241.

Observe country-specific laws and regulations for proper disposal.

If AdBlue® comes into contact with painted surfaces or aluminum, the affected area must immediately be cleaned with water.

Any AdBlue® removed from the AdBlue® tank must not be reused. As the purity of this AdBlue® cannot be guaranteed, it must be properly disposed of.

i For more information, observe the vehicle's Operator's Manual.

AdBlue® tank

The AdBlue® tank system is part of the SCR exhaust gas aftertreatment system. It has been designed for all common AdBlue® containers and delivery nozzles.

The tank flange module with pumps, heating and sensors, which is part of the exhaust gas aftertreatment system, is welded together with the AdBlue® tank. The AdBlue® tank is located in the front-end assembly on the right side in front of the front axle, and is connected to the longitudinal frame member and the inner part of the A-pillar.

On the series production vehicle, the component parts of the AdBlue® tank and tank flange module, as well as the electrical and hydraulic connections, are protected against damage by the front apron, e.g. when driving over low curbs or other obstacles.

If the factory-installed front apron is omitted, take measures to protect the component parts on the underside of the SCR exhaust gas aftertreatment system.

The filler neck is located in an opening on the front of the front module carrier under the hood.

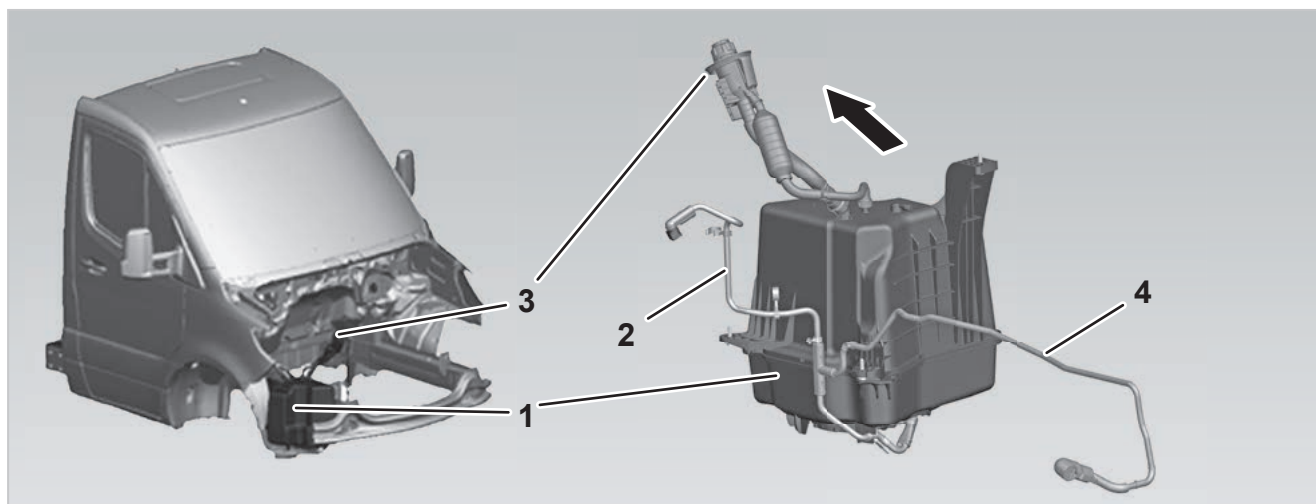
For modifications to the tank flange module, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The AdBlue® line connects the AdBlue® tank to the injection valves for metering. The line is electrically heated and designed for this system according to the line rigidity.

For modifications to the heated AdBlue® line, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The AdBlue® tank for OM654 is located at the same position in the engine compartment as in the previous engines.

The injection of AdBlue® into the exhaust system nevertheless takes place at two positions in the vicinity of the engine and in the area of the substructure.



SCR system for engine OM654

- 1 AdBlue® tank
 - 2 AdBlue® line for metering near engine
 - 3 AdBlue® filler neck
 - 4 AdBlue® line for metering in substructure area
- ➡ Direction of travel

6.3.3 Exhaust system

It is recommended to refrain from modifying the exhaust system of Sprinter vehicles. As the bodybuilder manufacturer, you should plan your scopes according to the available special equipment of exhaust system variants.

If a vehicle's exhaust system is replaced, the use of Mercedes-Benz genuine parts is recommended. When selecting a different exhaust system as special equipment, ensure that it is available for the same engine variant and emissions class from the Mercedes-Benz AG factory. This guarantees that the planned special exhaust system has been certified in this vehicle configuration. Further information on this can be obtained from your Mercedes-Benz Service Partner.

! NOTE

Observe all national laws, directives, and registration regulations.

General

! NOTE

It is important to note that modifications to the geometry of the exhaust system and its piping are only permissible under the boundary conditions described below.

The specifications on the area of the exhaust system in which modifications are permissible must be observed; see section "Exhaust system with SCR exhaust gas aftertreatment system" (→ page 145).

As part of such modifications through measurements, ensure that the exhaust gas back pressure (before/after) has not increased in comparison to the unmodified exhaust system. The bodybuilder manufacturer must be able to provide evidence of these measurements to Mercedes-Benz AG on request.

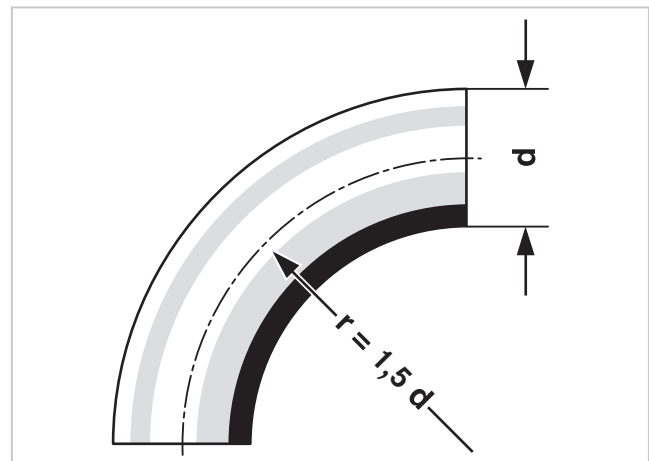
All modifications above and beyond these constitute an alteration of a certified state in terms of emissions and noise levels.

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Exhaust system geometry

Observe the following dimensions and specifications when carrying out modifications to the exhaust system geometry:



Example of a pipe bend

- Maximum pipe bend 90°
- Avoid the use of additional pipe bends
- Bending radii $> 1.5 d$

Any modification to the exhaust system in the permissible area must be designed in such a way that it is possible for the material to expand due to heating during operation. The attachment of the exhaust system and surrounding component parts must also be taken into account for this.

Minimum clearances to lines and detachable parts

When the vehicle is in operation, the temperature between the exhaust system (diesel particulate filter, catalytic converter or main muffler) and the floor panel may rise above 80 °C depending on the location. During regeneration, the diesel particulate filter can reach a surface temperature in excess of 500 °C. For this reason, shielding or insulation must be mounted on the substructure to reduce the effects of heat dissipation.

The following minimum clearances apply for fuel lines, plastic lines, electrical cables, and spare wheels, and for all component parts made of non-heat-resistant materials:

- 200 mm for exhaust systems without shielding
- 80 mm in areas with sheet metal shielding
- 40 mm in areas with sheet metal shielding and additional insulation

⚠ WARNING

If the minimum clearances to fuel lines, plastic lines, and electrical cables are not maintained, severe heat can result in a fire.

There is a risk of fire and explosion!

Always comply with the specified minimum clearances.

Be aware not only of the radiant heat emitted by the exhaust system, but also the jet of exhaust emitted behind the outlet of the exhaust pipe under all operating conditions.

! NOTE

When using the power take-off, observe the information and notes in the operator's manual of the vehicle regarding regeneration of the diesel particulate filter in the exhaust system.

For model series 907 with engine OM654, the option of manual regeneration at standstill (code K1M) is available as special equipment.

Additional shielding is required:

- Near control panels
- Near major assemblies, attachments and installations made of non-heat-resistant material

Exhaust gas emission

Particularly when designing access points and doors, make sure that exhaust flowing out of the exhaust tailpipe cannot enter the passenger compartment.

Ex factory exhaust system variants

The following exhaust system versions are available ex factory as special equipment:

Code	Description
K60	Exhaust, straight to rear
K61	Pre-installation for tailpipe for bodybuilder manufacturers
K63	Exhaust, to the side behind the rear axle
KA3	Exhaust, to the side in front of the rear axle
KA6	Exhaust system for power take-offs
KN6	Exhaust system for low-floor applications
K1M	Adjustment of tailpipe for manual regeneration when stationary

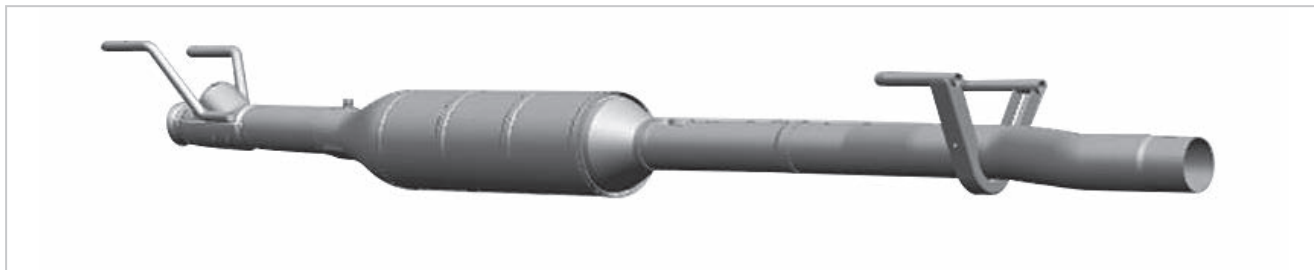
- i** Further information on special equipment is available from your Mercedes-Benz Service Partner, from the Technical Consultancy (→ page 18), or under 3.12 Special equipment (→ page 74).

Pre-installation for tailpipe for bodybuilder manufacturers

The pre-installation for tailpipe for bodybuilder manufacturers, code K61, is available as special equipment for model designations for panel vans and chassis in combination with CoC documents incomplete, code XC1, or omission of CoC documents, code XW2, exclusively for camper van manufacturers (code X2R).

This special equipment consists of an incomplete and shortened variant of the exhaust system and enables individual routing of the exhaust system in the area of the exhaust tailpipe for camper van conversions.

To obtain the operating permit, the bodybuilder manufacturer must observe the country-specific laws and regulations (e.g. UNECE regulations) during the completion of the exhaust system.



Pre-installation for tailpipe with special equipment code K61 (schematic)

Exhaust system special equipment for low-floor applications (code KN6)

For the Sprinter model series 907, an exhaust system for the Euro VI emissions class is available as special equipment (code KN6) for special bodies.

This exhaust system is designed for vehicle conversions with a lowered vehicle floor for use as low-floor vehicles with Z-redirecting transmission and has its own Stage I system approval.

A minimum unladen mass of 2585 kg must be maintained for the converted vehicle, otherwise Mercedes-Benz AG's system approval for this variant will be rendered invalid.

Observe the specifications for determining the curb weight in 2.2.8 Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP) (→ page 29).

The exhaust system code KN6 is included with the vehicle ex factory, together with the small parts required for assembly.

i Information on assembly is available in the Workshop Information System (→ page 26) or from your Mercedes-Benz Service Partner.

As a result, no exhaust system with an SCR catalytic converter is installed ex factory on the vehicle. This means that a fault code is stored in the engine control unit; an emergency operation mode is also activated whereby the engine output is throttled by 70% and the injection of AdBlue® is prevented.

After mounting the KN6 exhaust system, XENTRY Diagnosis –See 2.2.4 Diagnosis (→ Page 26) – must be used to delete this fault code in the engine control unit and thus deactivate this emergency operation mode.

For modifications to this exhaust system, including all its components, make sure to observe the following specifications for the exhaust system with SCR exhaust gas aftertreatment system. An illustration of this exhaust system is also included.

The bodybuilder manufacturer is required to obtain a separate European Type Approval for the low-floor exhaust system for its finished vehicle.

Special equipment – Exhaust system for power take-offs (code KA6) and Manual regeneration at standstill (code K1M)

The special equipment Exhaust system for power take-offs (code KA6) has different geometrical routing compared to the standard equipment due to the attachment of major assemblies to the transmission power take-off (code N05/N07).

In addition, for the possibility of regeneration of the diesel particulate filter at a standstill during operation of auxiliary assemblies, a special equipment option (code K1M) is available which is equipped with a special diffuser at the tailpipe which reduces the exhaust gas temperatures at the pipe outlet during the regeneration phase.

Available special equipment for vehicles with OM654 and code N05/N07

Code	Wheelbase 2 (Code IR4)	Wheelbase 3 (Code IR6)	Use (example)
K1M Without KA6	Available only with code X2X/ X3X ¹⁾	Available	Operation of a hydraulic pump
K1M With KA6	–	Available	Operation of a generator

1) Special equipment for elevating work platform with vehicle center of gravity height
Code X2X: up to 1000 mm
Code X3X: over 1000 mm
See Body/Equipment Mounting Directive Model Series 907 Chapter 7.13 or Model Series 910 Chapter 7.12.

Note that the special equipment with code K1M requires the power take-off to be actuated via the PSM (code ED5), which is ensured by automatic addition of an electronic engine speed control system (constant-rpm control code M53 or variable electronic rpm governor code MT4).

Body-manufacturer-specific modifications with direct actuation of the power take-off are therefore not permissible.

! NOTE

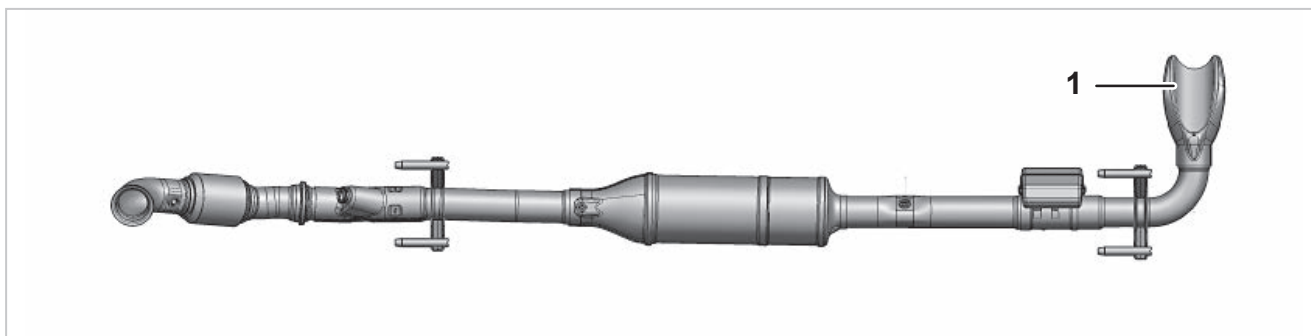
A modification of the exhaust systems in combination with the special equipment code K1M is not permitted.

Also note that vehicles with special equipment code K1M do not have underfloor paneling (code XM1) installed for thermal reasons.

! NOTE

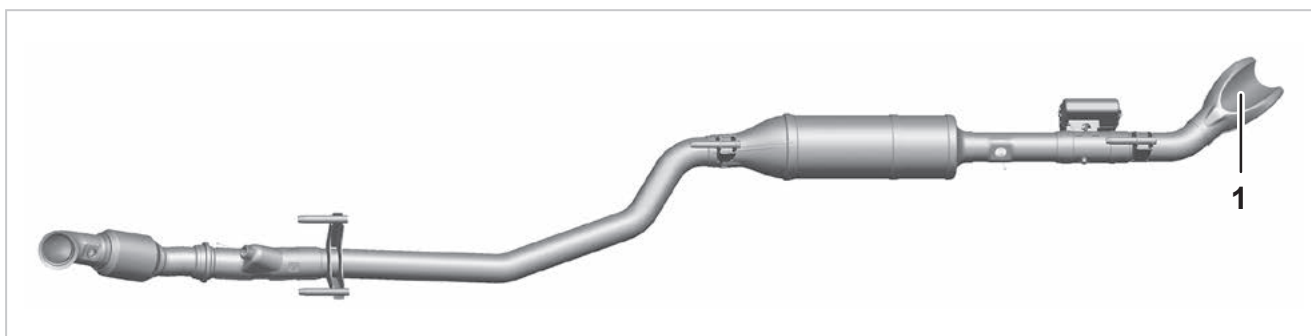
Retrofitting of bodybuilder-manufacturer-specific parts that cause a thermal insulating effect in the area of the exhaust system similar to underfloor paneling (e.g. retractable step or similar) is not permissible.

i Further information and specifications on the regeneration of the diesel particulate filter can be found in the operator's manual for your vehicle.



Exhaust system with special equipment code K1M without KA6

1 Diffuser



Exhaust system with special equipment code K1M with KA6

1 Diffuser

Exhaust system without SCR exhaust gas aftertreatment system

The length and installation position of the flexible metal hose between the exhaust manifold and the exhaust pipe must not be changed.

The free cross-section of the exhaust pipe behind the main muffler must not be reduced.

⚠ WARNING

The lengths and routings, e.g. between the diesel particulate filter and the main muffler, are optimized with regard to temperature characteristics. Modifications could lead to higher or extreme temperatures in the exhaust system and surrounding component parts (propeller shafts, fuel tank, floor panel, etc.).

There is a risk of fire!

Modifications to the exhaust system are not permissible up to the point of the main muffler.

Exhaust system with SCR exhaust gas aftertreatment system

A modification to the exhaust system with SCR exhaust gas aftertreatment system is only permissible on vehicles with engine OM654 as of a clearance of 100 mm behind the throttle flap (measured from the rear welding edge).

! NOTE

A modification of the exhaust systems in combination with the special equipment for manual regeneration when stationary (code K1M) is not permitted as a whole.

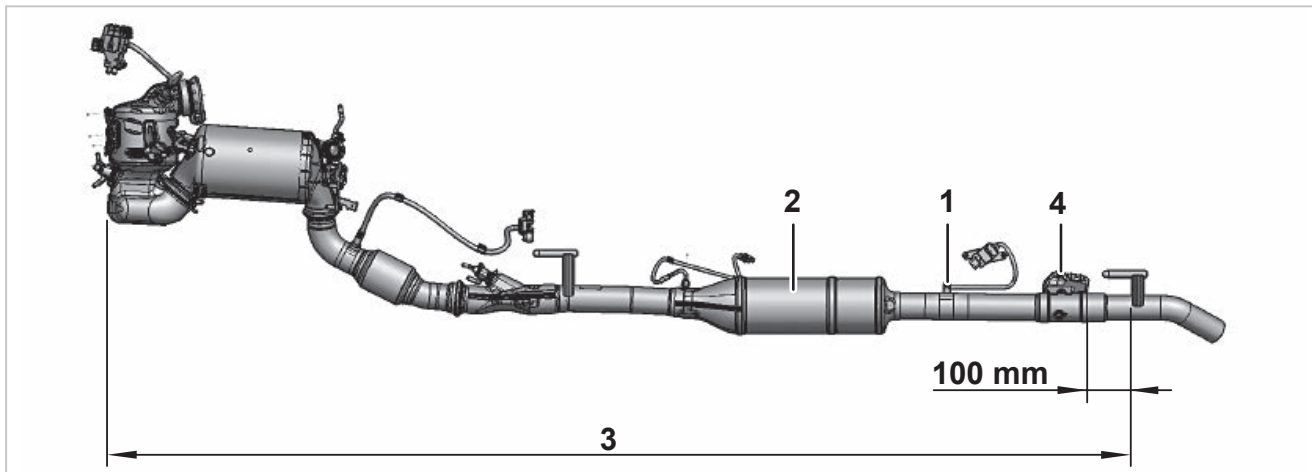
The positions and installation positions of the sensors and the throttle flap, as well as the spatial geometry of the overall system or other components, must not be modified, as this is part of the vehicle registration criteria.

6 Modifications to the basic vehicle

It is the bodybuilder manufacturer's sole responsibility to guarantee adequate clearance between the exhaust system to all the components relative to its body. Observe the previously mentioned minimum distances to the exhaust system components in this respect.

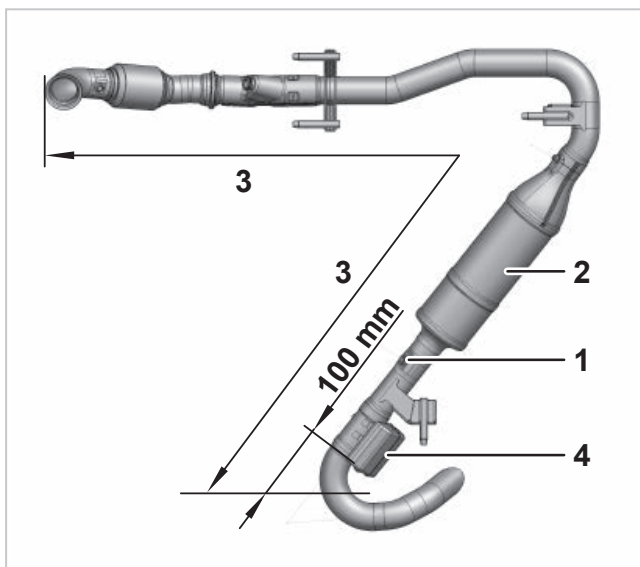
! NOTE

After any components carrying AdBlue® have been opened, they must then be hermetically sealed against the ambient air. Otherwise, AdBlue® can crystallize and clog the system components.



Exhaust system for OM654 (schematic)

- 1 NOx sensor
- 2 SCR catalytic converter
- 3 Area in which modifications are not permitted
- 4 Throttle flap



Exhaust system for OM654, special equipment KN6 (schematic)

- 1 NOx sensor
- 2 SCR catalytic converter
- 3 Area in which modifications are not permitted
- 4 Throttle flap

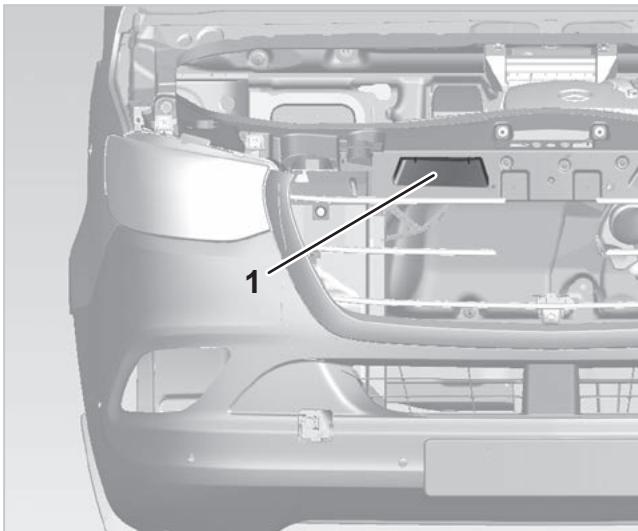
6.3.4 Engine cooling

! NOTE

The information and specifications for cooling engine OM654 in 4.4.3 Engine cooling (→ page 92) must be observed.

6.3.5 Engine air intake

In engine OM654, the engine air intake is integrated into the area of the upper crossmember of the new front module.



Intake air system for OM654

- 1 Opening of engine air intake

! NOTE

The engine air intake constitutes a certification-relevant state and must therefore remain unchanged. Any modifications to the intake system or its surroundings are not permissible, as otherwise the emissions certification will no longer be valid.

If attachments or conversions are implemented in the area of the vehicle front, the bodybuilder manufacturer bears sole responsibility for adhering to the legal certification requirement.

The air filter is secured by two rubber-mounted holders in the front module.

The design of the attachment of the air filter must be retained in the event of any modification to the front module.

Warm air

The intake of warm air will lead to a loss of engine power.

A bulkhead between the engine compartment, the vehicle interior and the intake point is therefore essential.

The intake air temperature should not exceed the outside temperature by more than 10 °C.

Water

When being drained off, water, splash water, or car wash water must not flow directly past the intake point from the body.

Make sure that water cannot reach the intake points through any fresh air inlets.

The flow velocity at the intake points must not be increased by modifications to the opening of the intake points.

Dust and dirt

Increased dust intake will lead to shorter maintenance intervals for the air filter.

6.3.6 Clearance for major assemblies

To guarantee proper functioning and operational safety of the major assemblies, sufficient clearance must be maintained (in particular to electrical, brake and fuel lines).

Take into account the dimensions in the 2D drawings (offer drawings) (→ page 24).

The distance between the cab and the body must be at least 50 mm; See 7.1.4 Attachment to the frame (→ Page 204).

6.3.7 Propeller shafts

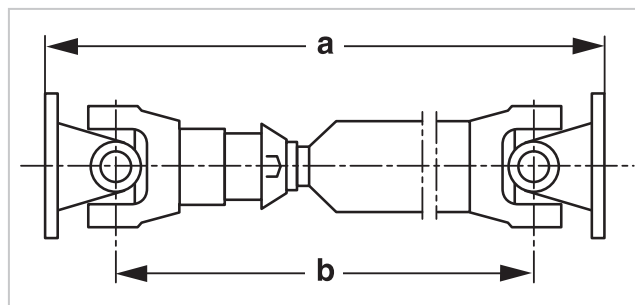
The correct layout of the propeller shaft drive prevents noise and the development of vibrations. The use of Mercedes-Benz genuine parts is recommended.

! NOTE

If the vehicle wheelbase is modified, the length of the propeller shafts must be adapted to the vehicle. The modifications must be carried out by a company qualified in propeller shaft engineering.

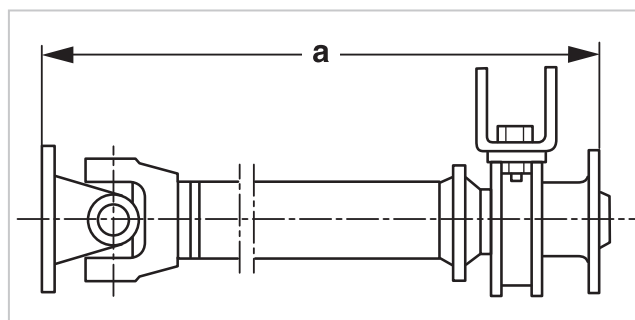
Design the propeller shaft intermediate bearings to be appropriately rigid. Also design them in such a way that no vibrations are transmitted to the vehicle structure.

When modifications are made, the fastening elements must not be reused.



Propeller shaft

- a Operating length
- b Permissible shaft length



Intermediate shaft

- a Operating length

If modifications are made, the propeller shaft location and length must be the same as a comparable series production vehicle (same model and identical or similar wheelbase).

The diameter, wall thickness and material properties of the propeller shaft tube must be the same as for a standard propeller shaft.

When extending a propeller shaft tube, check the dimensions and adjust them if necessary (increase in the diameter and wall thickness).

The check straps fitted to the substructure are for passive safety and protect the fuel tank in the event of an impact. Modification to the check straps is not permissible.

Working angles

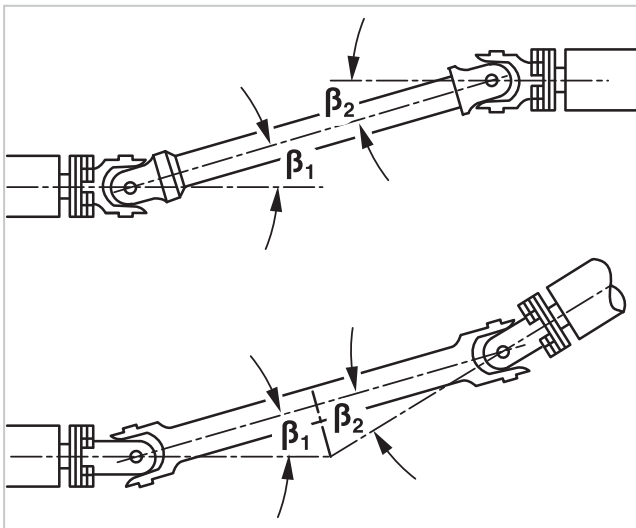
If necessary, use multiple propeller shafts with intermediate bearings.

The working angles must be identical at both joints ($\beta_1 = \beta_2$). Generally, the working angles must not exceed 4.5° , and in exceptional cases must be no greater than 6° ; proper functioning of the universal joints must also be assured. The angular difference should not exceed 1° .

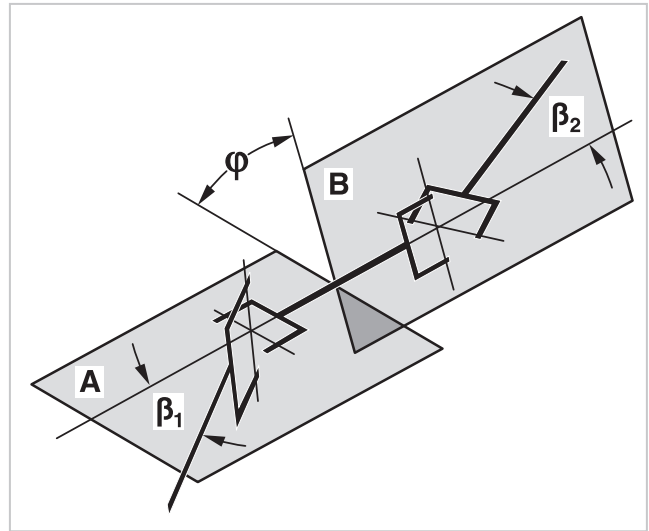
! NOTE

Working angles greater than 6° and angle errors ($\beta_1 \neq \beta_2$) lead to vibrations on the drivetrain. They shorten the service life of major assemblies and may cause damage.

In addition, complaints regarding noise and vibrations that are transferred to the cab and the body are to be expected.



Types of angle



Working angle $\beta_1 = \beta_2$

- Angle in one plane (two-dimensional offset): "W" or "Z" layout
- Angles in two planes (three-dimensional offset): The power input and power output shafts intersect in different planes (combined W and Z-offset).

In order to compensate for any irregularities, the inner joint forks can be offset. Have this fork offset designed and validated by a company qualified in propeller shaft engineering.

Balance the propeller shafts before installation.

Do not make modifications that are outside the limit values. For potential exceptions, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20). In such cases, submit drawings containing the planned propeller shaft modifications with precise dimensions (shaft length, universal joint size, material data, balance values and working angles).

6.3.8 Engine speed regulation

The engine must be able to run at a constant or variably adjustable speed in order to drive auxiliary assemblies (e.g. pumps, compressors, etc.).

The special equipment "Constant-rpm control" (code M53) and "Electronic rpm governor, variable" (code MT4) is available for some engine combinations.

When ordering this special equipment, the parameterizable special module PSM (code ED5) that is required for controlling the engine speed is automatically added. See 8.10 Parameterizable special module (PSM) (→ Page 372).

With code M53, the constant-rpm control is activated via a switch and is preset ex factory for a minimum engine speed of 1100 rpm.

On vehicles with an engine-side power take-off ex factory (code N05/N07), the minimum engine speed is 1200 rpm; See 6.5.3 Power take-offs (→ Page 169).

Through a parameterization of the PSM, three additional constant engine speeds can be determined and set via a control system of the power take-offs with the PSM during use.

With code MT4, following activation via a switch, a variable speed regulation via the rpm range of 1100 to 3800 rpm is available that can be set with a rocker switch "±" or via the control system of the power take-off with the PSM.

Depending on the alternator type and the electrical consumer power, these special equipment options may not be suitable if frequency stability is required.

! NOTE

Retrofit bodybuilder-manufacturer-specific solutions for regulating the engine speed are only possible with the "parameterizable special module" (PSM) special equipment (apart from those retrofit solutions available as special equipment code M53 ex factory). This special equipment makes it possible to control the working speed regulation externally (→ page 372). There is otherwise a risk of malfunctions and the engine could enter limp-home mode.

i If you have any questions or require further information regarding this special equipment, contact the Technical Consultancy (→ page 18).

6.4 Interior

6.4.1 General information

! NOTE

All legal requirements detailed in this section relate to Germany.

Observe all national laws, directives, and registration regulations.

! NOTE

It is essential to observe the information, notes, and warnings regarding safety equipment and the pyro-technical components, See 6.4.2 Safety equipment (→ Page 152).

When redesigning the interior through installation, conversion, or removal, the bodybuilder manufacturer shall be solely responsible for complying with all relevant legal requirements.

! NOTE

Modifications in the cockpit area and above the belt rail line must fulfill the criteria of the head impact tests in accordance with UN R 21 as well as FMVSS 201.

In the vicinity of the airbag deployment areas, See 6.4.2 Safety equipment (→ Page 152), modifications to the surface of the cockpit, interior trims and seats (wooden trims, additional installed parts, mobile phone cradle, bottle holder or similar) are not permitted.

As a general principle, nothing may be installed on moving parts (e.g. storage compartment lid) in the cockpit area.

i You can find further information on camper van conversions in 7.15 Camper vans (→ page 250).

! WARNING

If paint or surface treatment is applied to the instrument panel, the window airbags, the steering wheel impact absorber and the airbag tear seams, chemical reactions can occur on the treated surfaces. This could weaken or damage the materials meaning that the restraint systems no longer function correctly.

There is danger to life and limb!

Painting or surface treatment is not permissible on the instrument panel, window airbags, steering wheel impact absorber or airbag tear seams.

In the event of modifications in the interior, the permissible vehicle center of gravity as well as the axle loads must not be exceeded.

In general, observe the following for installations or modifications in the interior:

- For the modification of vehicles in the Federal Republic of Germany, appropriate information sheets can be requested from the relevant technical inspection authorities for motor vehicles (e.g. TÜV (Technical Inspection Association), DEKRA).
- Fittings must be made of flame-retardant material with soft edges and surfaces and must be securely mounted.
- Unimpeded access to the seats must be ensured. There must not be any protruding parts, edges, or corners that could cause injury in the area of the seats.
- Note the area at the lock of the load compartment sliding door to the B-pillar modified as of model year 2024 on vehicles with rear seats or with preparation for retrofitting of rear seats (special equipment code V40 or T01; See 7.3.1 Retrofitting seats (→ Page 209).
A bodybuilder-manufacturer-specific installation or conversion (e.g. partition, trim or similar) must be adapted accordingly to ensure the function of the new sliding door lock.
- To ensure accessibility to screw connections or when replacing series-production parts (for maintenance and service), it is advisable to provide sufficient openings at appropriate positions for conversions or work on the body with interior equipment specific to the bodybuilder manufacturer, e.g. in the middle of the side wall for bolting on the load compartment sliding door track or in the area of the rear portal at bottom left for replacement of the interior pollen filter.

6.4.2 Safety equipment

WARNING

Any systems or components that influence passive safety must not be modified. In particular, if alterations were to be made to the vehicle's structure or components, the safe operation of restraint systems would no longer be guaranteed.

There is danger to life and limb!

Modifications such as these could also result in the vehicle registration being revoked.

Therefore, the following modifications are not permissible:

- Modifications to seats, seat belts, and their mounts
- Modifications to the front-end assembly
- Modifications to the interior by installing parts in the vicinity of outlet openings and in the airbag deployment areas; see the illustrations in the section on airbags (→ page 156).
- Modifications to the instrument panel (cockpit), the steering wheel, or the trim, as well as their lamination with coverings or other types of trim, in particular in the areas of airbags
- Modifications to the A-pillar and B-pillar, the roof frame, and its trim
- Modifications to the doors

⚠ WARNING

Any modifications to restraint systems or their wiring could result in these restraint systems not functioning correctly. A restraint system can fail or unintentionally activate.

There is danger to life and limb!

Modifications to restraint systems or their wiring are therefore not permitted. This includes modifications to the belt mounting points and their positioning on the B-pillar and any modification of Mercedes-Benz genuine belt system components.

In the case of the following interventions in the structure of the vehicle, pay attention to the particular provisions under 6.4.3 Seats (→ page 160):

- Modifications to seats
- Installation of non-MB seats

! NOTE

Attachments with rigid connections to the front, side, and rear of the vehicle at the height of possible accident zones could modify characteristics of the vehicle's passive safety.

Airbag control unit and sensors

It is not permitted to modify the installation location, installation position and attachment of the airbag control unit and satellite sensors by comparison with the standard equipment on vehicles equipped with window airbags and thorax/pelvis side airbags. No modifications of any kind are permitted to the lines of these components.

Other vehicle components must not be attached to the airbag control unit, acceleration sensors, their lines or the attachment points. Comply with a minimum distance of 6 mm during the installation of component parts in the vicinity of these components.

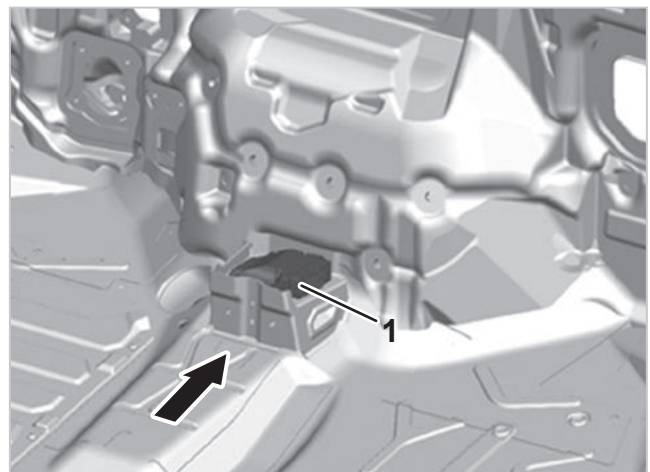
⚠ WARNING

No modifications of any kind are permitted to the components of the restraint systems (airbag control unit, frontal acceleration sensors, airbags, seat belt tensioners, etc.), as well as their lines and attachments.

Vehicle parts that generate vibrations must not be mounted in the vicinity of the airbag control unit or the sensor installation locations. Nor may modifications be made to the floor structure in the vicinity of the airbag control unit or the acceleration sensors. Otherwise, the safe operation of restraint systems is no longer guaranteed.

There is danger to life and limb!

The airbag control unit is located on the center tunnel under the center console.



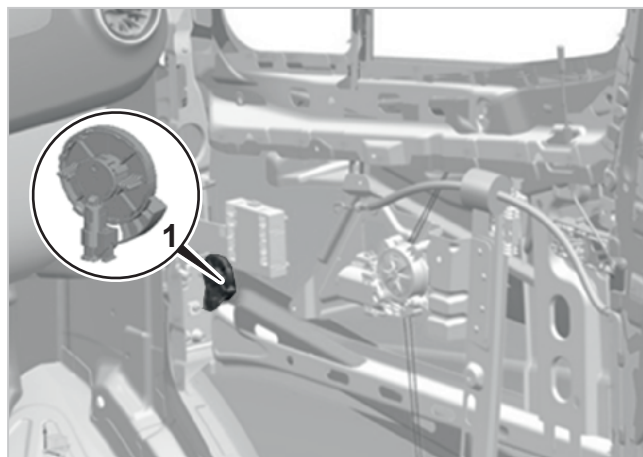
Position of the airbag control unit

- 1 Airbag control unit
➔ Direction of travel

6 Modifications to the basic vehicle

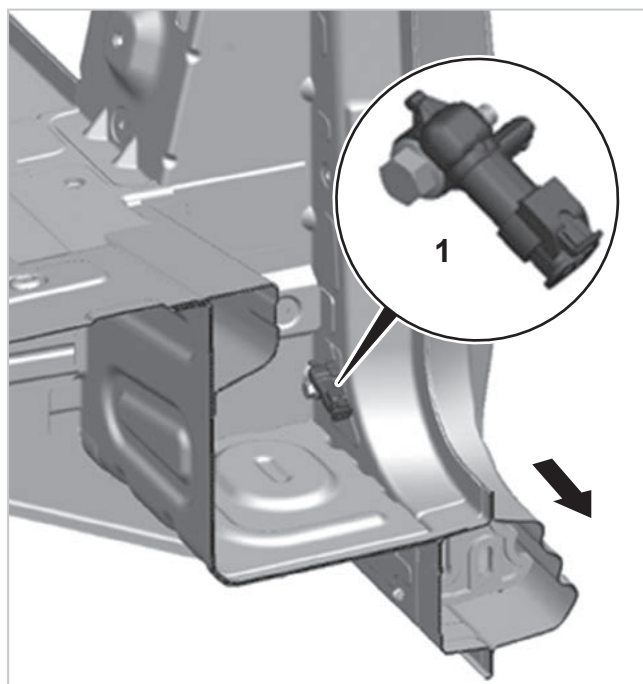
In vehicles equipped with window airbags and/or thorax/pelvis side airbags, the acceleration sensors are located toward the bottom of the B-pillar behind the door sill trim in the driver's and front passenger's entrance step modules.

The pressure sensors are located in the driver's door and/or front passenger door.



Front pressure sensor

- 1 Pressure sensor
(trigger sensor of the occupant protection systems)



Sectional view of left entrance step module at B-pillar

- 1 Acceleration sensor
(trigger sensor of the occupant protection systems)
➡ Direction of travel

⚠ WARNING

For safety reasons, airbag sensor systems, airbag control units or acceleration sensors that have been dropped onto the floor must no longer be installed. Replace these parts with the corresponding Mercedes-Benz genuine parts. Otherwise, the safe operation of these systems is no longer guaranteed. There is danger to life and limb!

Seat belts

⚠ WARNING

Parts relevant to safety such as seat belts or belt anchoring points and seat belt tensioners must not be damaged or soiled when work is carried out on the vehicle. Otherwise, these restraint systems may no longer function properly, and in the event of an accident, they would fail to provide sufficient protection. There is danger to life and limb!

! NOTE

Only the original seat belts must be fitted; otherwise, the general operating permit of the vehicle would be invalidated.

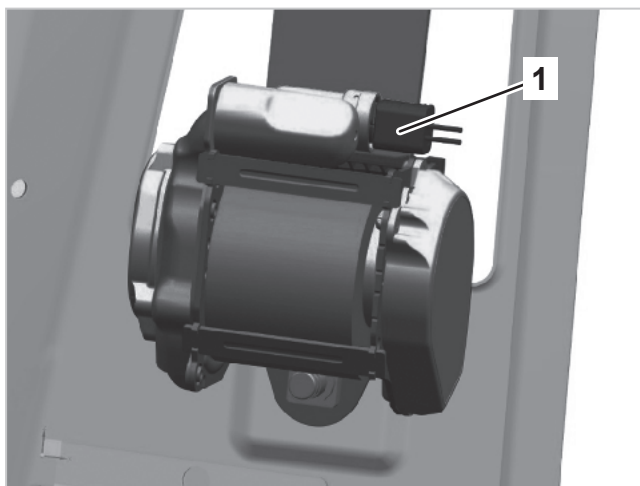
! NOTE

For seat belts, seat belt tensioners, seat belt end-fitting tensioners and anchorings for the seat belts, comply with the specifications of UN R 14, UN R 16 and UN R 17 and FMVSS or other country-specific regulations on seat belts. Verify if the previously mentioned specifications do not fully cover country-specific regulations or take precedence over these. Submit corresponding evidence of this upon request.

Vehicles with a maximum design speed in excess of 25 km/h must be equipped with seat belts.

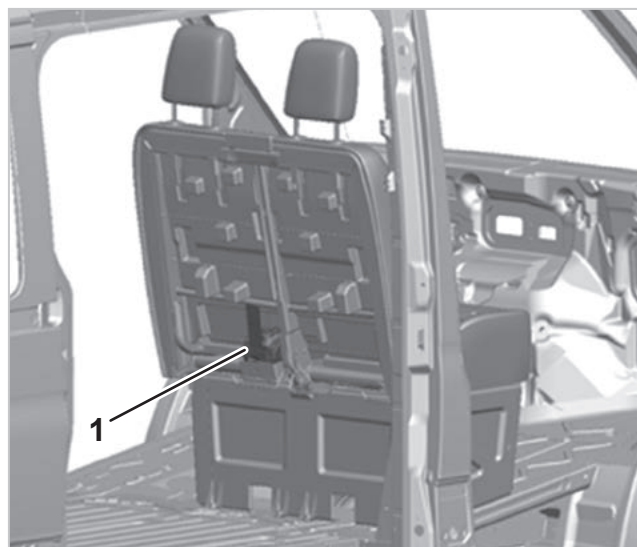
All vehicles are equipped with pyrotechnical seat belt tensioners in the retractor in the area of the front seats.

The retractors are located in the B-pillars. If a two-seater front passenger seat is present in the vehicle, an additional retractor is located in the seat backrest of the bench seat.



Retractor with pyrotechnical seat belt tensioner

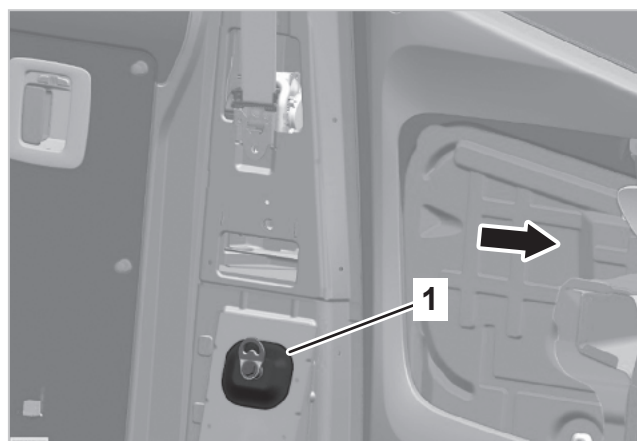
1 Connector coupling



Front passenger bench seat with retractor

1 Retractor

There is also a securing point for a seat belt end fitting at the bottom of the B-pillar, which has been tested in accordance with UN R 14, R 16 and R 17, and with a folding seat rigidly connected to the body-in-white.



Securing point for seat belt end fitting in the B-pillar

1 Seat belt end fitting

➡ Direction of travel

Airbags

All airbag units are labeled with the word "AIRBAG":

- The driver's airbag at the padded boss
- The front passenger airbag (code SA6) on the cockpit
- The window airbags (code SH9) on the A-pillar trims
- The thorax/pelvis side airbags (codes SH1 /SH2) at the sides on the backrests of the driver's seat and front passenger seat facing the vehicle exterior

An additional identifying feature is the red indicator lamp  in the instrument cluster.

The indicator lamp signals a malfunction in the restraint system, which includes functions of the seat belt warning and the ESP® sensor system.

The following illustrations show the positions and deployment areas of the driver's and front passenger airbags as well as those of the window airbag and thorax/pelvis side airbag.

The deployment areas shown are greater than the actual volume of the airbag because space is required for airbag rebound as it deploys.

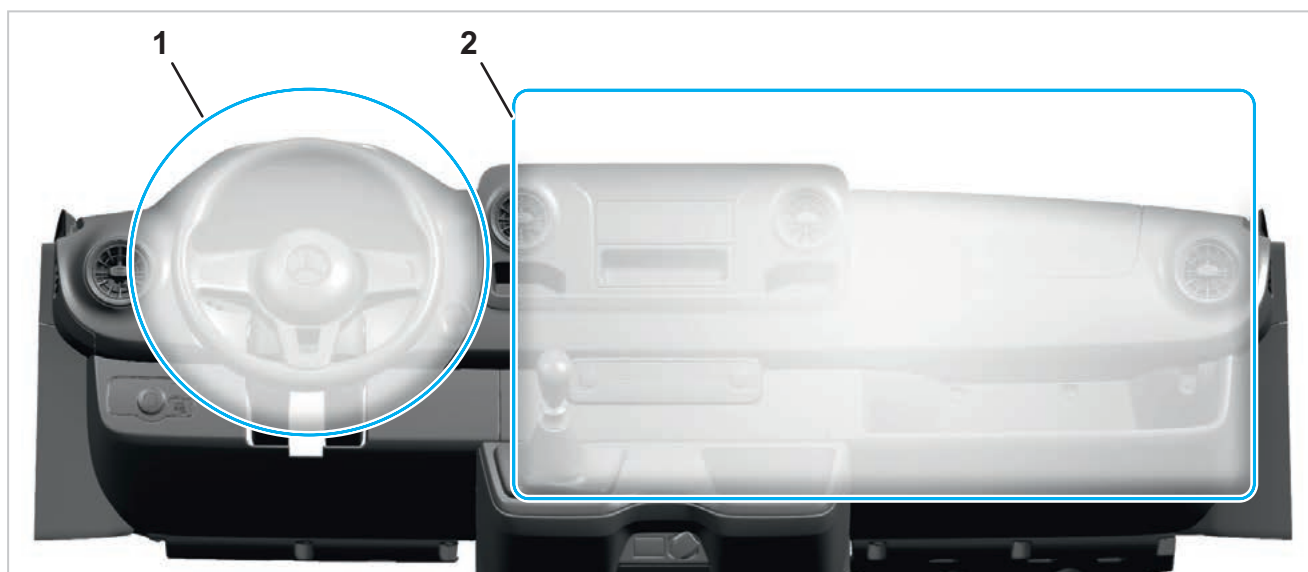
NOTE

No installations are permissible in the deployment areas of all airbags.

NOTE

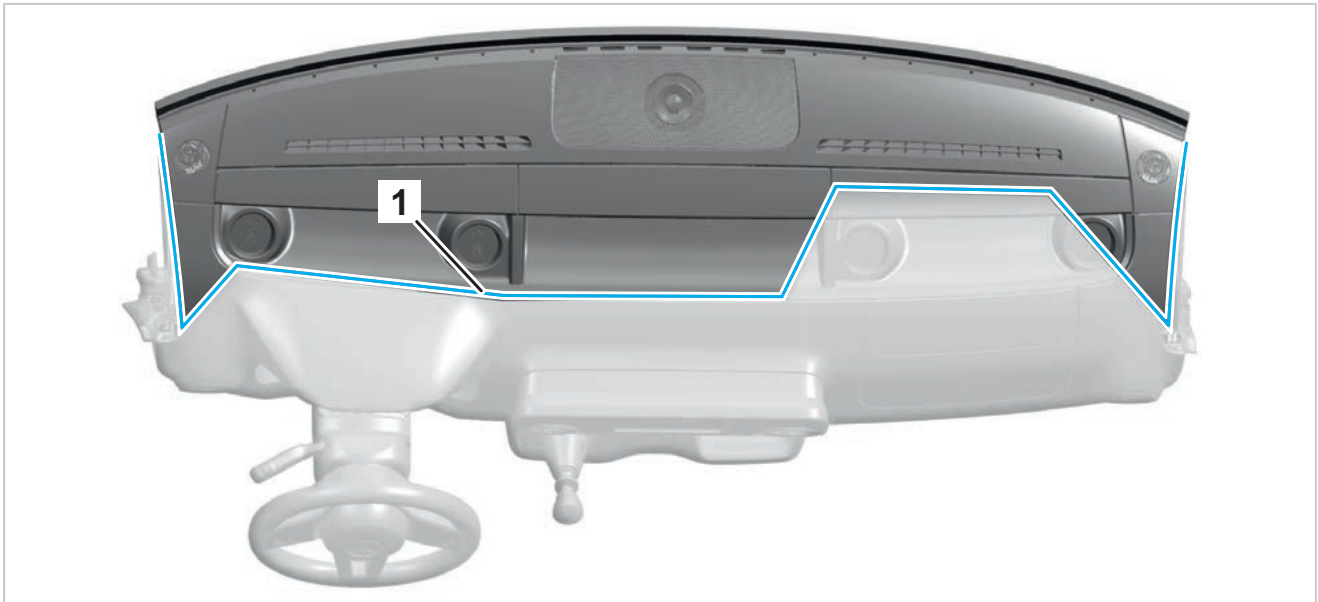
A minimum clearance of 20 mm must be maintained between installed parts and the interior trim of the A-pillar as well as the windshield.

-  If you have any questions regarding installations in the interior, contact the Technical Consultancy during the planning phase (→ page 18).



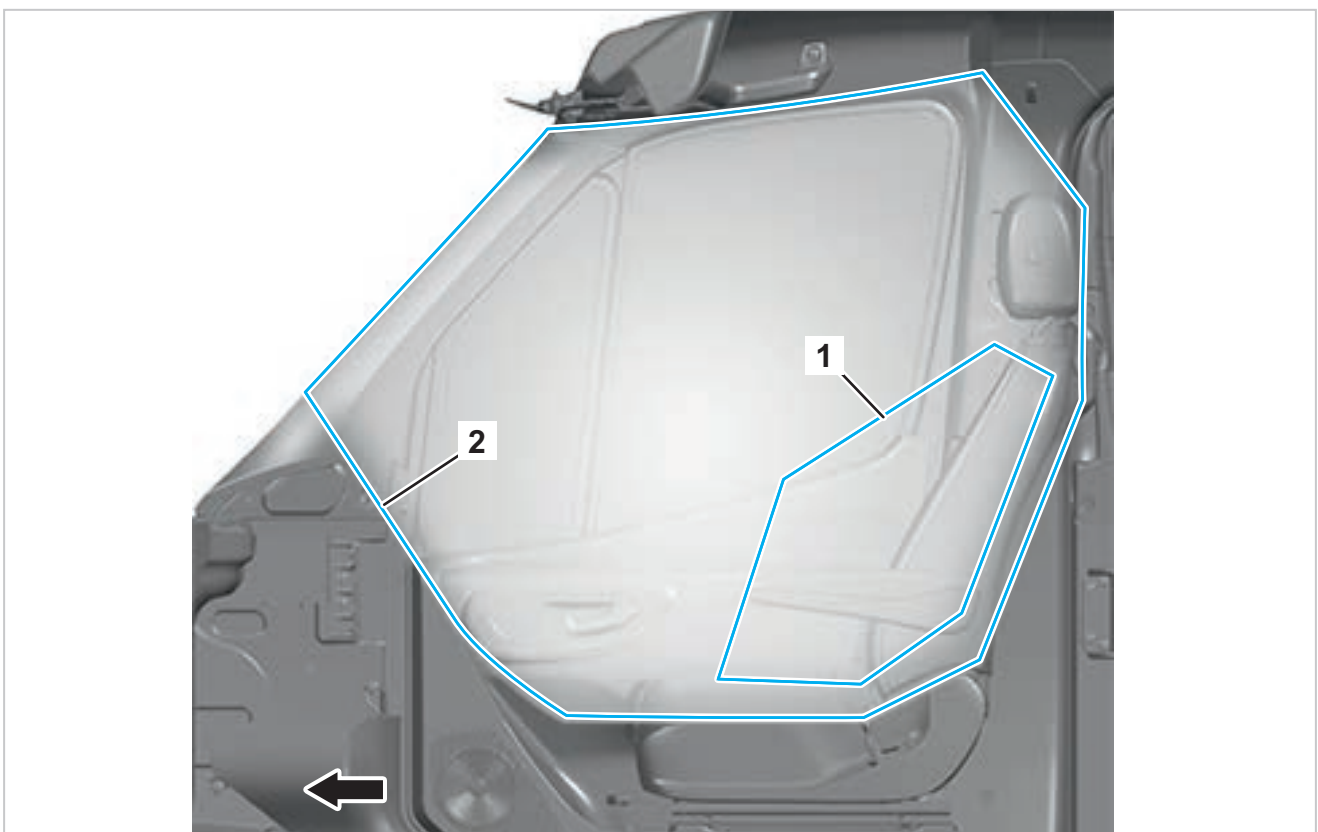
Boundary of deployment area for driver's and front passenger airbag (left-hand drive vehicle, mirror-inverted for right-hand drive vehicle, schematic)

- 1 Driver's airbag
- 2 Front passenger airbag



Boundary of deployment area for driver's and front passenger airbag (left-hand drive vehicle, mirror-inverted for right-hand drive vehicle, schematic)

- 1 Driver's and front passenger airbags



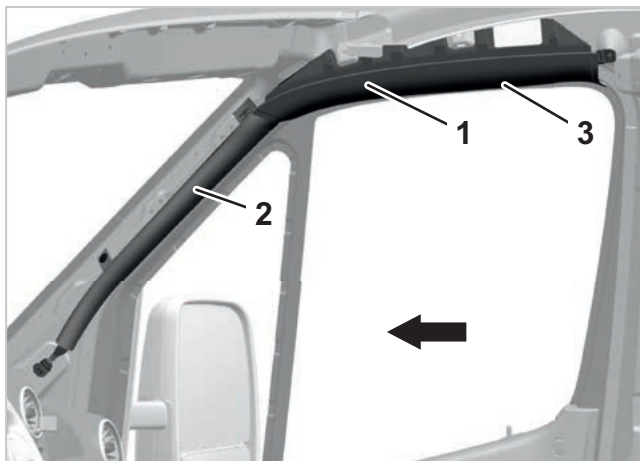
Boundary of deployment areas for side airbags (front passenger side, mirror-inverted for driver's side, schematic)

- 1 Thorax/pelvis side airbag
 2 Window airbag
 ➡ Direction of travel

⚠ WARNING

Working on the A-pillar can cause damage to the window airbag. This can result in the window airbag no longer functioning properly and it may not provide adequate protection in the event of an accident.

There is danger to life and limb!



Installation position of window airbag

- 1 Cover
- 2 Window airbag in protective covering
- 3 Gas generator (behind cover)
- ➡ Direction of travel

! NOTE

Observe the note in 7.3.1 Retrofitting seats (→ page 209) regarding the special equipment front passenger airbag removal (code SW3).

Handling pyrotechnical components

⚠ WARNING

Airbag units and seat belt tensioner units are pyrotechnical components.

There is a risk of explosion when handling pyrotechnical components!

There is danger to life and limb!

Handling (installation, removal, transportation, storage, and disposal) of airbag units and seat belt tensioner units must therefore only be carried out by personnel who are appropriately qualified. Always observe the accident prevention regulations.

The handling, transportation, and storage of pyrotechnical objects are subject to legislation concerning potentially explosive substances ("Gesetz über explosionsgefährliche Stoffe" in Germany) and must therefore be reported to the relevant trade supervisory authority.

⚠ WARNING

Removed airbag units must always be stored in such a way that the padded side faces upward. If the padded side faces downward, the airbag unit will be catapulted through the air if it is triggered accidentally.

There is danger to life and limb!

When handling pyrotechnical components of the restraint systems, it is essential to observe the following:

- Only qualified personnel may work with removed components and perform testing and assembly operations.
- Electrical testing must only be carried out in the installed state using the prescribed testing equipment; it is recommended that the testing is carried out by a Mercedes-Benz Service Partner.
- Assembly and disassembly to be conducted only when the battery is disconnected, with covered up positive terminal and disconnected test coupling/connector coupling
- After disassembling, storage must be in a suitable storeroom only, and the components concerned must only be removed just before they are to be assembled
- If work has to be interrupted, store components involved under lock and key
- Do not treat with grease, cleaning agents or similar agents
- Never expose components to temperatures of more than 100 °C.
- Replace any pyrotechnical components that have been dropped onto the floor.
- The pyrotechnical components must only be electrically tested in the installed condition with the prescribed test equipment. For safety reasons, the testing should only be carried out at a Mercedes-Benz Service Partner, or at a specialist workshop that has been specially trained to service these safety systems.

Transporting and storing pyrotechnical components

! NOTE

Do not transport any pyrotechnical components in the passenger compartment.

Transportation should always be carried out using the replacement part packaging in the vehicle trunk or load compartment and in observance of the previously mentioned warning notice.

Class T1 pyrotechnical materials must only be stored in limited quantities on premises used for commercial purposes.

Annex 6 of the appendix to the "Second Ordinance of the German Explosives Law" specifies that the following maximum storage quantities are permissible without obtaining special approval from the relevant authority, where materials are stored on premises used for commercial purposes and certain conditions are fulfilled (e.g. steel cabinet):

- General storage space: 20 kg gross
- The gross mass of the component part that has been approved in accordance with the act concerning explosives is used to calculate the actual stored mass.

Gross mass of pyrotechnical component parts

Driver's airbag	1.5 kg
Front passenger airbag	3.3 kg
Window airbag	2.1 kg
Thorax/pelvis side airbag	0.7 kg
Seat belt	1.4 kg

Disposal of pyrotechnical components

! NOTE

Have pyrotechnical components disposed of by qualified personnel. Comply with the country-specific accident prevention regulations.

Dispose of pyrotechnical components in accordance with the country-specific laws and guidelines. In Germany, pyrotechnical components must be electrically detonated by trained personnel to render them unusable prior to disposal.

- ▶ If seat belt components that have not been detonated require detonation for disposal purposes, place them into the footwell of a vehicle that has been sent for scrapping and connect them straight away using a 2-pin connector coupling.
- ▶ If the padded plates on the airbag units have not been destroyed, the airbag units must be detonated using the 2-pin connector coupling.

These safety measures are necessary because pyrotechnical materials could cause injury if activated incorrectly.

Hazards arise from disposal using cutting torches, by smelting, or if primed parts are disposed of on open fires or smoldering fires in landfill sites.

In order to ensure that no additional work arises due to these safety measures, it is recommended that you entrust the disposal of pyrotechnical substances to an external waste disposal company, which will implement the required safety measures (incl. 10 m safe distance, special ignition equipment).

When the materials are handed over, the waste disposal company must sign a declaration containing the obligation to dispose of the pyrotechnical materials in accordance with accident prevention regulations. With agreements of this kind, it must be ensured that it is not possible to extract pyrotechnical materials after disposal and to pass them on for repairs.

6.4.3 Seats

⚠ WARNING

Modifications to the driver's and front passenger seats, the rear seats, or the mountings of the seats installed ex factory are only permitted if the requirements of the legal specifications applicable at the time that the vehicle is being placed on the market are fulfilled and corresponding evidence of this is submitted upon request:

- UN R 14 (seat and seat belt anchorages)
- UN R 16 (seat belts and restraint systems)
- UN R 17 (seats, seat anchorages and head restraints)

These requirements must be complied with even for seats with two- or three-point seat belts specific to the bodybuilder manufacturer that deviate from the series standard ex factory.

In the event of an accident, the seats could otherwise be torn or ripped from their anchorages. Passengers cannot be protected as intended.

There is danger to life and limb!

⚠ WARNING

Do not secure seats to the wheel well under any circumstances. This also applies for subsequently lowered wheel wells. Otherwise, the vehicle could be damaged (e.g. wheel wells and tires).

There is risk of an accident and danger to life and limb!

! NOTE

Test certificates for seats on a rigid plate are not permissible.

Test certificates must be available for submission upon request in the course of product monitoring.

Only certificates issued by accredited testing institutes will be accepted.

! NOTE

For retrofitting seats other than those provided ex factory and for implementing a seat fastening system that deviates from the series standard, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

A Letter of Compatibility of this kind is also required for other installations in the area of the front passenger seat for vehicles with front passenger airbag (code SA6) and removal of front passenger seat (code S91).

The additional information and specifications in 6.4.1 General information (→ page 151), 6.4.2 Safety equipment (→ page 152) and 7.3.1 Retrofitting seats (→ page 209) must absolutely be observed.

! NOTE

Observe all national laws, directives, and registration regulations.

If seats are changed or seats other than those supplied ex factory are retrofitted, the following points must additionally be considered and complied with:

- Compliance with the original H-point of the standard seats available ex factory must be ensured for every modified or retrofitted front seat. The seat adjustment field also must not be increased, because deviations could have an impact on other points relevant for registration.
- On vehicles with a low driver/front passenger seat box (code S87/S88), the front seats available ex factory must not be attached directly; instead, the H-point to be adhered to must be achieved with the ex-factory-available swivel base for driver/front passenger (code SR8/SR9) or with the driver/front passenger suspension seat (code SB3/SB4).

i For more information on the H-point and seat adjustment field, contact the Technical Consultancy (→ page 18).

- On vehicles with omission of the front passenger seat box (code S99), a retrofit assembly of a standard seat with seat box is not possible, since in this case the body-in-white reinforcements and connection points are no longer available. The installation of a bodybuilder-manufacturer-specific seat is permissible under adherence to the specifications.
- In the event of a retrofitting of the front seats, the information and specifications in 7.14.7 Connecting point with omission of driver's and front passenger seat for pre-installation for seat belt warning system (code J1D/J1E) and seat occupancy recognition (code E3S) (→ page 237) must be observed. Commissioning of control units is required, See 8.19 Commissioning of control units (→ Page 387).
- Seats without seat belts are not permissible.
- When the seat belts and seats (including seat boxes) are reinstalled, the specified screws must be tightened to the specified torques (including tolerance) and the connection must be checked and documented according to the classification A/B/C in accordance with VDI 2862.

i You can find information about fasteners and torques in the Workshop Information System (→ page 26).

- For protection against production fluctuations and measurement tolerances, suitable procedures must be used (e.g. statistical methods such as Six Sigma). Submit evidence of protection against production fluctuations upon request.

Seat covers

⚠ WARNING

Unsuitable seat covers could hinder or even prevent the deployment of the airbags integrated in the seats. The airbags would then not protect the vehicle occupants as intended.

There is danger to life and limb!

Only use seat covers approved by Mercedes-Benz for the seat in question.

! NOTE

If standard seats with integrated thorax/pelvis side airbags are fitted with new seat covers or seat protectors (protective covers), verification must be provided that confirms that the deployment of the airbag does not deviate in any way from the series production configuration. Submit corresponding evidence of this upon request.

Seats with seat occupancy recognition must have this function verified in the event that a new seat cover in general or a protective cover with seat padding made of thick and/or rigid material is used.

If other seat covers or protective covers are used on seats without integrated airbags and without seat occupancy recognition, the bodybuilder manufacturer must ensure compliance with the legal specifications and submit evidence of this upon request (e.g. H-point seat layout, flame-retardant properties, REACH, etc.).

6.4.4 Reducing interior noise

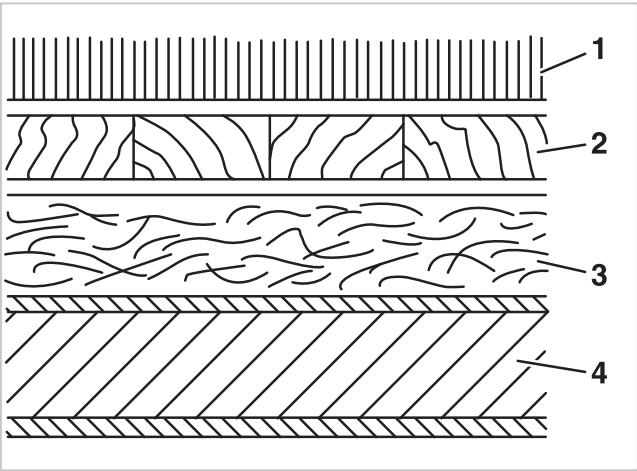
Sound insulation materials can be installed in order to reduce the noise level in the vehicle interior. These must be flame-retardant.

Floor area

A structure as shown in the illustration is advisable for insulation and noise damping. An additional covering with heavy-duty insulating foil may be provided in the area of the wheel wells.

! NOTE

Insulating foils, e.g. bituminous felt, have limited temperature resistance. Therefore, they must not be installed in the immediate vicinity of the engine or exhaust system.



- 1 Carpet (rubberized underside)
- 2 Wooden floor (12 mm plywood)
- 3 Heavy-duty insulating foil (weight per unit area 8–10 kg/m²)
- 4 Load-bearing construction

Roof and side walls

Apart from providing effective insulation, the materials used for insulation should exhibit the following properties:

- Not hygroscopic
- Not water-retaining
- Not water-absorbing
- Not water-attracting
- Water-repellent

Make arrangements for the rapid and unhindered drainage of accumulated moisture or condensation in order to avoid conditions that promote corrosion.

Keep the drain holes provided as standard. If necessary, more drain holes can be made, provided that this has been coordinated in advance with the Technical Consultancy (→ page 18).

The inside must be covered with a sound-transmitting material (perforated card, plastic, fabric cover).

WARNING

Do not make subsequent modifications to the headliner or the roof paneling between the A-pillar and the B-pillar if the vehicle is equipped with window airbags. Otherwise, the window airbag may no longer function correctly (e.g. window airbag deployment is delayed or incomplete).

There is danger to life and limb!

Seals

Openings, gaps, and slots between the engine compartment, vehicle underside, and the firewall to the vehicle interior must be carefully sealed with elastic material. Vent openings must not be fitted in the immediate vicinity of sources of noise.

Also consult the manufacturers or suppliers of sound-proofing materials. They will be able to provide suggestions on how to design optimum noise insulation specific to your modified vehicle body.

6.4.5 Ventilation

The driver cockpit and all other areas used by persons must be sufficiently ventilated.

The defrosting of the windshield and side windows must remain effective, particularly for special conversions or bodies such as camper vans or other special-purpose vehicles for which the driver's seat is incorporated without a partition to the rear passenger compartment in the open interior.

When modifying/removing the standard windshield, provide adequate defrosting/deicing and ensure that the sun sensor is reinstalled (standard with code HH4, Front automatic air conditioning). An incorrectly installed sun sensor or a different inclination of the windshield results in deviations in the climate control with code HH4 (Front automatic air conditioning).

New vehicles can be delivered ex factory with the special equipment code HH9 (Manual front air conditioning), code HH4 (Front automatic air conditioning), code HH9+HH7 (Manual front air conditioning + roof-mounted air conditioning system), code HH9/HH4+H08 (Front air conditioning + roof-mounted air conditioning system) and code HH9/HH4+HK4 (Front air conditioning + roof-mounted air conditioning system).

When retrofitting major assemblies, observe 6.5 Additional assemblies (→ page 166).

6.4.6 Attachment points for load compartment trim parts on side wall and roof

To line the load compartment in FKA panel van model designations, it is recommended to use the load compartment trim parts available from Mercedes-Benz AG as special equipment ex factory.

- You can obtain information on this in the ADVANTAGES product information system or from the Technical Consultancy (→ page 18).

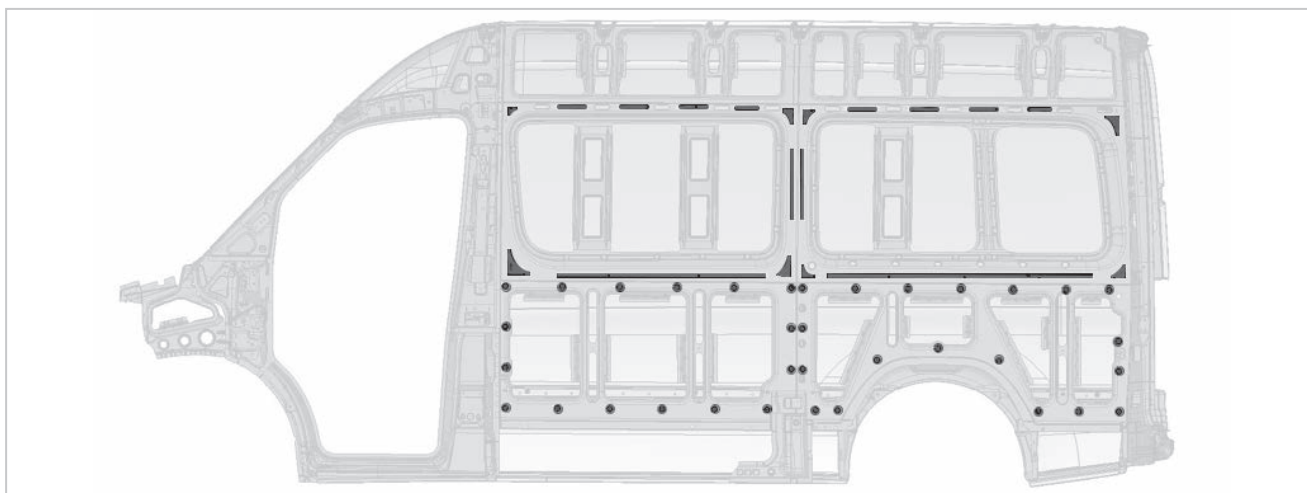
When installing load compartment trim parts specific to the bodybuilder manufacturer, ensure the following:

- Load compartment trim parts in the lower area of the side wall and on the roof are to be fastened at existing mounting holes in the body-in-white (for clips, expansion rivets, or similar fasteners) for the inner parts of the side wall and the cross bows of the roof, see following illustrations.
- In places where there are no existing mounting holes (e.g. the upper part of the side wall), additional mounting holes are permissible only in the marked areas of the side wall inner parts, see following illustrations.
- To prevent cracks from occurring when drilling extra mounting holes, a distance of at least 15 mm must be maintained from the edge of the hole to all cuts and joints, openings, other holes and panel edges of the body-in-white.
- Ensure that any bores that are subsequently introduced are executed properly, i.e. avoid chips, ensure that the drilled holes are deburred, and carry out the subsequent anticorrosion protection measures.

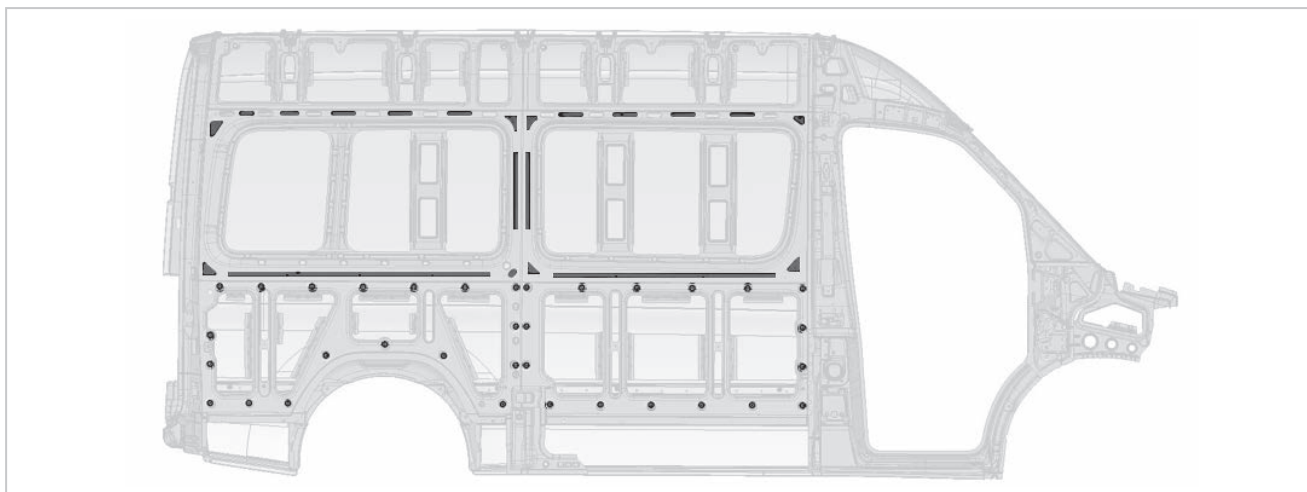
- Once the body mounting work is complete, ensure that all chips are completely removed from joints, sections, and cavities.
- Observe the specifications in this Body/Equipment Mounting Directive for the work to be performed; See 3.9 Threaded connections and weld joints (→ Page 66), "Drilling work on the body-in-white" (→ page 115), 5.3 Anticorrosion protection measures (→ page 102), 5.4 Painting and preservation work (→ page 104), and 6.2.1 General information on the body-in-white (→ page 112).
- Also observe other notes and specifications from the bodybuilder manufacturer's "Small Conversions" guideline and the "Design Quality for Vans" guideline in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>

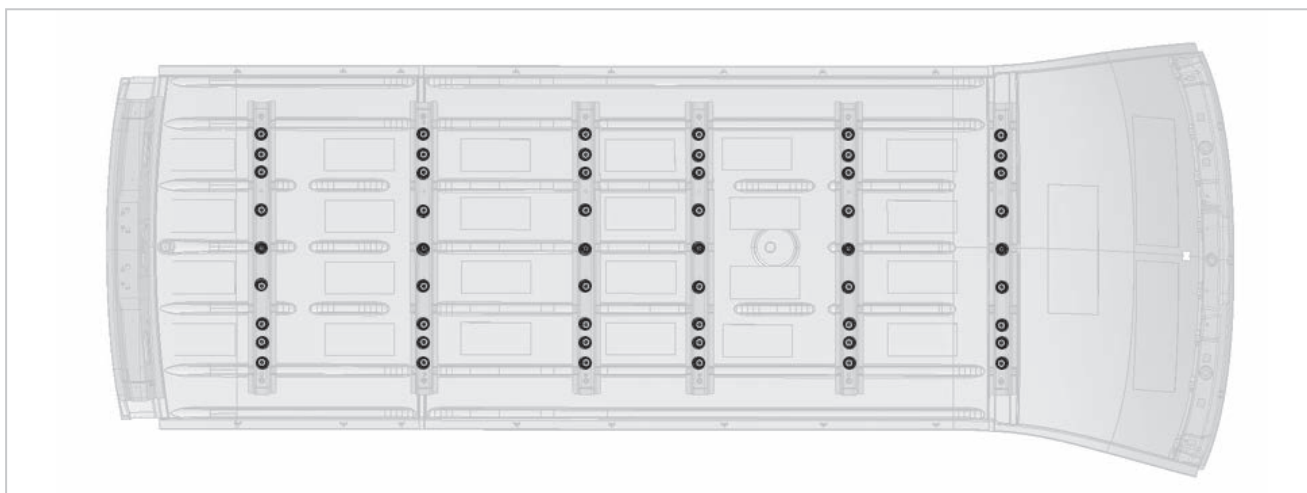




FKA panel van model designation, inside of right side wall: Permissible areas for the attachment of load compartment trim parts (dark markings, schematic)



FKA panel van model designation, inside of left side wall: Permissible areas for the attachment of load compartment trim parts (dark markings, schematic)



FKA panel van model designation, inside of roof: Permissible areas for the attachment of load compartment trim parts at the cross bows (dark markings, schematic)

6.5 Additional assemblies

When using additional assemblies, use the factory-installed power take-offs (→ page 169).

- i** Vehicles with engines OM651 and OM642 are no longer available ex factory. For this reason, this information is no longer included in the present Body/Equipment Mounting Directive.
- If required, you can find corresponding information and specifications in previous Body/Equipment Mounting Directives in which the engines are still included. These are still available in the Mounting Directives Archive in the Bodybuilder Portal.

6.5.1 Retrofitting an air conditioning system

⚠ WARNING

Prior to all mechanical and electrical work, the vehicle must be rendered free of voltage/current (disconnecting the battery is absolutely essential).

All electrical devices installed must have been tested in accordance with UN R 10 and must bear the CE marking and/or another comparable marking depending on the country-specific regulations.

The installation of additional air conditioning components (compressor, evaporator, lines, etc.) on the air conditioning systems available as special equipment ex factory is not permissible.

Air conditioning system special equipment options

- Code HH9 (Manual front air conditioning system)
 - Code HH4 (Front automatic air conditioning)
 - Code HH9+HH7 (Manual front air conditioning system + roof-mounted air conditioning system)
 - Code HH9/HH4+H08 (Front air conditioning + roof-mounted air conditioning system) and
 - Code HH9/HH4+HK4 (Front air conditioning + roof-mounted air conditioning system)
- If other air conditioning systems are to be retrofitted, observe the guidelines issued by the device manufacturer.
- Observe the following points with regard to compatibility with the basic vehicle:
- The installation of an air conditioning system must not impair vehicle parts or their function.
 - The battery must have sufficient capacity and the alternator (→ page 276) must generate sufficient power.
 - Additional fuse protection of the air conditioning system electric circuit (→ page 272).
 - Attach the refrigerant compressors using the available major assembly carriers (→ page 174).
 - The additional pulley under code N63 (maximum output 8 kW) is available ex factory as special equipment for driving refrigerant compressors (→ page 174).
 - Ensure that pipes (→ page 100) and electric lines (→ page 272) are routed correctly.
 - Access to and easy maintenance of installed major assemblies must not be impaired.
 - Supply the operator's manual and the maintenance manual for the additional assemblies upon handing over the vehicle.
 - The air supply and cooling required for the engine must not be impaired (→ page 147).
 - If compact systems are mounted on the roof (evaporator, condenser, and blower), do not exceed the permissible roof load, See 4.3.8 Roof and roof load (→ Page 90).
 - For attachments at the roof, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

! NOTE

Where refrigeration systems/components are used in combination with the refrigerant R1234yf, observe all applicable laws, specifications, and safety requirements.

In particular, ensure compliance with the safety regulations as per ISO 13043 at both the component and overall system level (FMEA, leak tightness requirements, concentration measurements, etc.).

In addition, observe the safety requirements with regard to inflammability and risk of combustibility in combination with R1234yf (maintain best-possible packaging or largest-possible distance to hot parts; if necessary, install additional devices to ensure the required safety).

If R1234yf is used, a specific information label with details of the quantity and combustion risk must be considered and installed in a visible location.

6.5.2 Auxiliary heating

The following special equipment is recommended ex factory for the auxiliary heating:

Code	Equipment
H11	Hot-air auxiliary heating
H12	Hot-water auxiliary heating (stationary heater)
HZ9	Hot-water auxiliary heating (only as booster heater)
H13	Additional heat exchanger on substructure
HH2	Electrical hot-air auxiliary heating (PTC)

Code H88 (Preparation for additional heat exchanger), available as special equipment ex factory, allows an additional heat exchanger to be connected to the existing heater circuit (including the stationary heater). Take into account the mechanical and electrical requirements, including adjustment of the coolant quantity for reliable engine operation.

Belt tensioning for engine OM654**! NOTE**

Modifications must not be made to the belt pulley diameter in the main belt drive (primary chain drive) on engine OM654.

For the belt tension of the engine-side power take-off, see the section "Adjusting the belt tension for engine power take-off OM654" (→ page 175).

Retrofitting auxiliary heating

! NOTE

When retrofitting a fuel-powered auxiliary heating system (stationary heater), the special equipment "Fill level sensor for auxiliary heating" (code KL1) and "Fuel display" (code J51) is required; See 6.3.1 Fuel system (diesel) (→ Page 136). This provides an additional connection for taking fuel on the fill level sensor.

If this instruction is ignored, the following problems can occur:

- Inadequate fuel supply to the fuel injection system
- Faulty fuel return
- Undesirable vaporization in the supply system
- Risk of damage to auxiliary heating through increased pressure in vehicle fuel system

When using the special equipment for auxiliary heating ex factory, this type of fuel connection is already present.

! NOTE

If you are planning to install an auxiliary heating system other than the special equipment that is available ex factory – which in turn will affect the cooling, hot water and oil circuits of engine OM654 – contact the Technical Consultancy during the planning phase (→ page 18).

! NOTE

The specifications for fire protection, impact performance, and accident protection must be observed when retrofitting an auxiliary heating system.

Risks of poisoning or asphyxiation due to gaseous substances must be prevented through suitable selection of the installation location and sealing measures.

Exhaust gases must not penetrate into the vehicle interior while the auxiliary heating is in operation.

The vehicle floor must be gas-tight if exhaust gases are routed out under the vehicle.

Openings in the vehicle floor provided for control elements must be sealed with sealing bellows.

The following components of the auxiliary heating must not be mounted within the cab, the passenger compartment, or load compartment:

- Exhaust pipes of the auxiliary heating
- Vents for components of the auxiliary heating system
- Fuel lines
- Fuel tank and filler neck

Route all pipes, hoses, and electric lines so that they are free of mechanical stress.

Observe the specifications with regard to the distance to the exhaust system in 6.3.3 Exhaust system (→ page 141).

i You can find further information in 3.12 Special equipment (→ page 74).

6.5.3 Power take-offs

General information

Power take-off versions available ex factory:

- Transmission power take-off
- Front engine power take-off

The power take-off version and ratio selection both depend on the power output and rotational speed of the major assembly to be driven.

Transmission power take-offs can only be engaged and disengaged when the vehicle is stationary.

The maximum transferable torques for each of the power take-offs are guide values for shock-free and vibration-free operation. These figures are based on a highly durable gearing design and a service life calculated in compliance with the German standard DIN 622. The additional inertia forces of the driven major assemblies are not taken into account.

For transmission power take-offs, the ratio chosen should ensure that a minimum engine speed of 1200 rpm at $P = 28 \text{ kW}$ or 1500 rpm at $P = 40 \text{ kW}$ (in combination with transmission oil cooling) is maintained. The power output should be within the range of the maximum engine torque.

Exposed propeller shafts, fan wheels or belt pulleys must be covered.

No belts or chain drives must be fitted to the drive shaft or drive flange of a power take-off.

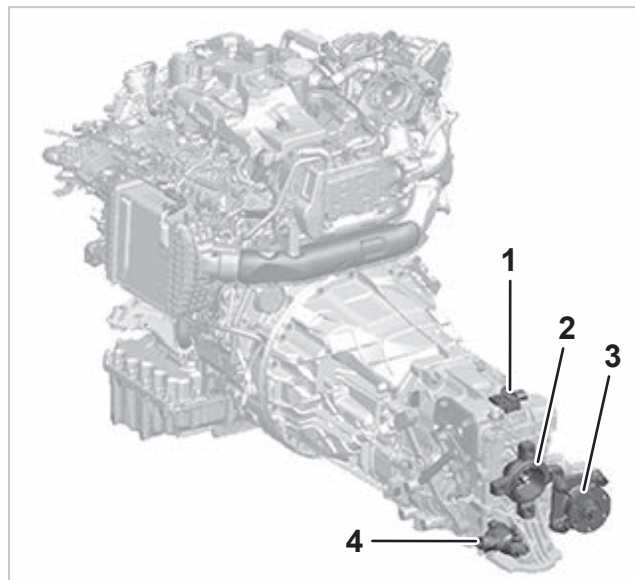
! NOTE

When using the power take-off, observe the information and notes in the operator's manual of the vehicle regarding regeneration of the diesel particulate filter in the exhaust system.

a) Transmission power take-off

The power take-off on the transmission side is available in combination with the following manual transmissions:

- 6-speed manual transmission ECO Gear (code GD8)
- 6-speed manual transmission ECO Gear High Torque (code GD9)



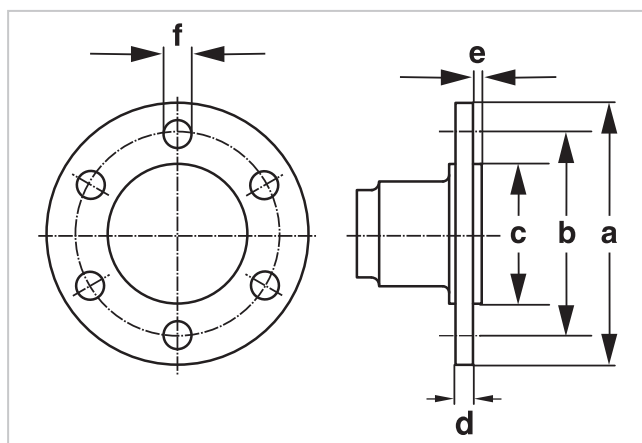
Overview of transmission power take-off (OM654 with code GD8)

- 1 Neutral-gate sensor
- 2 Transmission shaft flange
- 3 Power take-off flange
- 4 Switch cylinder for power take-off (activated by switch on instrument panel support)

The side-mounted power take-off available for the Mercedes-Benz manual transmission is available as special equipment from the factory with the following codes:

Code	Equipment
N05	without flange
N07	with flange

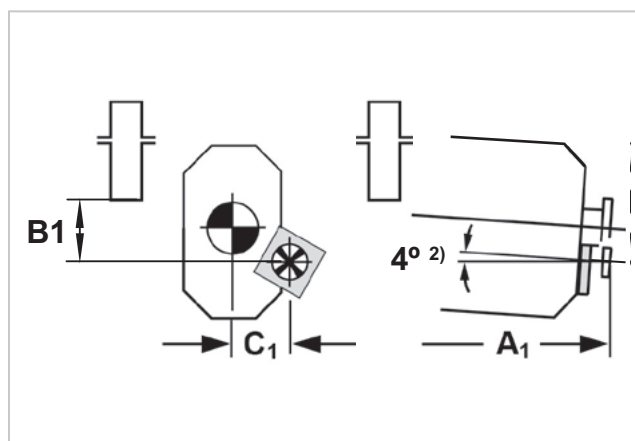
- i** Observe the information and specifications for the special equipment "Exhaust system for power take-offs" (code KA6) and "Manual regeneration of diesel particulate filter at standstill" (code K1M) in 6.3.3 Exhaust system (→ page 141).



Dimensions of coupling flange

Transmission	Code GD8/GD9
a [mm]	Ø 90
b [mm]	Ø 74.5
c [mm]	Ø 47 _{e8}
d [mm]	6
e [mm]	2.1
f [mm]	Ø 8 ^{A12}

Transmission	Code GD8/GD9
Number of holes	6



Dimensions of power take-off (values for OM654)

Power take-offs

I	OM654		OM654	
II	Code GD9		Code GD8	
III	N07	N05	N07	N05
IV	0.756	0.756	0.756	0.756
V	28/40 ¹⁾ (2526 rpm)	28/40 ¹⁾ (2526 rpm)	28/40 ¹⁾ (2526 rpm)	28/40 ¹⁾ (2526 rpm)
VI	140/200 ¹⁾	140/200 ¹⁾	140/200 ¹⁾	140/200 ¹⁾
VII	b	b	b	b
A1	679	632	608	560
B1	24	21	19	16
C1	126	126	126	126

1) With transmission oil cooler

- I Engine
- II Transmission
- III Power take-off designation (special equipment code)
- IV Gear ratio i_{PTO} ; input speed at power take-off $n_{PTO} = i_{PTO} \times N_{Engine}$
- V Continuous output at power take-off in kW at engine speed rpm
- VI Maximum transferable torque at power take-off in Nm

- VII Rotation direction viewed in direction of travel
 - a) Counterclockwise
 - b) Clockwise
- A1 Distance from rear edge of coupling flange to center of front axle measured in mm
- B1 Distance from center of coupling flange to lower edge of longitudinal frame member in mm
- C1 Distance from center of coupling flange to center of transmission flange in mm

Technical data

- ❗ The maximum weight moment at the power take-off transmission flange is 15 Nm for the direct flange-mounted PTO with code N05.

$$n_{PTO} = 0.756 \times n_{Engine} \text{ (code GD8/GD9)}$$

Max. continuous output Code GD8/GD9	28 kW at 2526 rpm (engine speed)
Max. torque	140 Nm at 1200 rpm (engine speed)

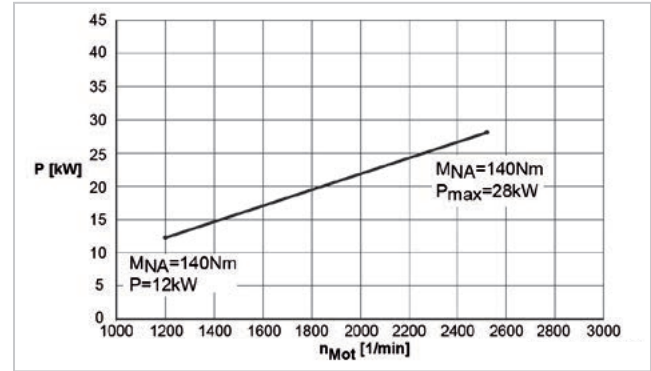
- ❗ The direction of rotation is clockwise when viewed in the direction of travel.

If transmission oil cooling is retrofitted

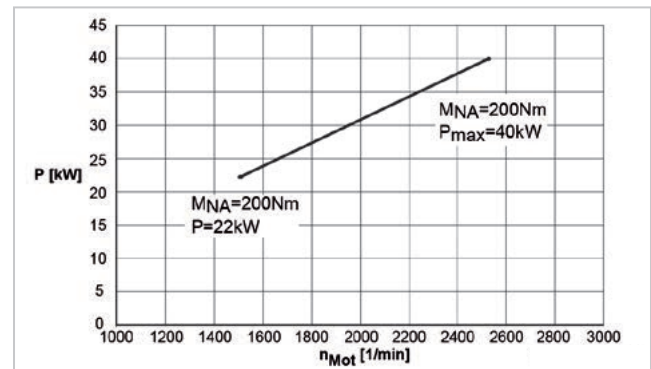
Max. continuous output Code GD8/GD9	40 kW at 2526 rpm (engine speed)
Max. torque	200 Nm at 1500 rpm (engine speed)

! NOTE

Avoid continuous excessive torque take-off, as this can otherwise lead to transmission damage.

**Code GD8/GD9 without cooling,
power take-off ratio: 0.756**

	Minimum	Maximum
P [kW]	12	28
n_{Engine} [rpm]	1200	2526
n_{PTO} [rpm]	908	1909

**Transmission code GD8/GD9 with cooling,
power take-off ratio: 0.756**

	Minimum	Maximum
P [kW]	22	40
n_{Engine} [rpm]	1500	2526
n_{PTO} [rpm]	1134	1909

n_{Engine} Engine speed
 n_{PTO} Power take-off speed

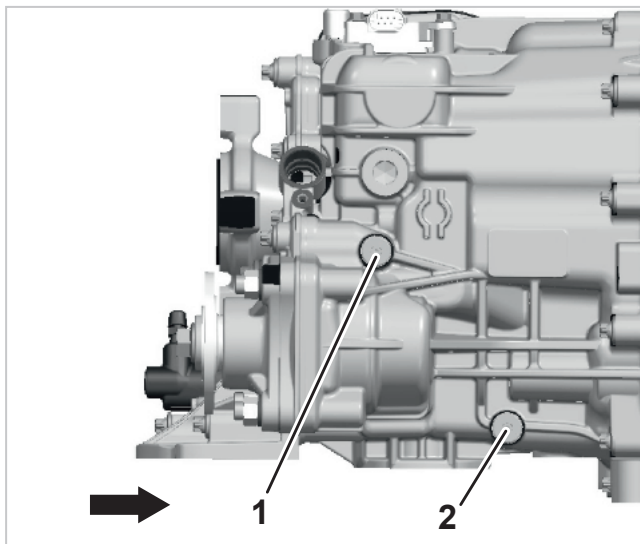
Transmission oil cooling

! NOTE

The transmission oil temperature in continuous operation (over 30 minutes) must not exceed 120 °C.

Exceeding the maximum permissible transmission oil temperature can cause damage to the transmission and must therefore be avoided by means of the following transmission oil cooling system and limitation of the power draw on the power take-off.

All transmissions of code GD8/GD9 with power take-off (code N05/N07) are equipped with connections for a separate oil cooler circuit on the right-hand side of the transmission housing; see following illustration:



Position of oil cooler connections on code GD8/GD9

- 1 Optional oil cooler connection "in" (feed)
 - 2 Optional oil cooler connection "out" (return)
- Connections with screw connection according to DIN 908-M14×1.5×12

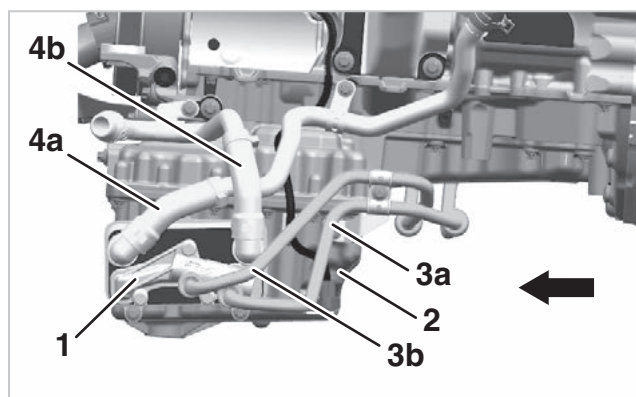
➡ Direction of travel

The special equipment pre-installation for transmission oil cooling (code GK4) is available ex works. This involves the installation of a water-cooled transmission oil cooler with the associated coolant hoses.

The transmission oil cooler on OM654 is located on the left-hand side of the oil pan on the bottom of the engine.

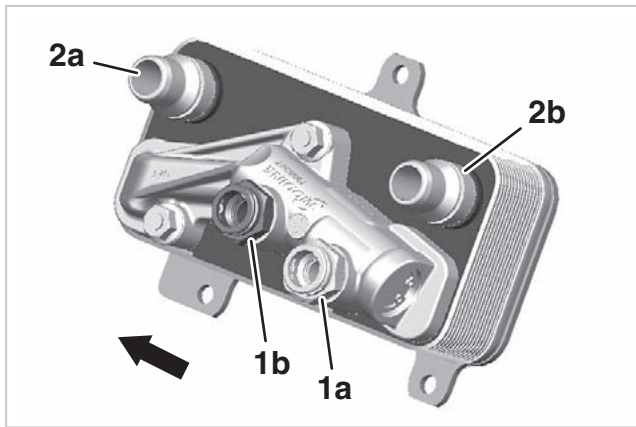
! NOTE

The coolant circuit with the factory-fitted coolant hoses and their connections to the cooler components must not be modified.



Position of oil cooler on vehicles with engine OM654 with exemplary oil line routing

- 1 Transmission oil cooler
 - 2 Oil pan of engine OM654
 - 3 Oil lines a) Feed b) Return
 - 4 Coolant hoses a) Feed b) Return
- ➡ Direction of travel



Oil cooler with connection fittings for vehicles with engine OM654

- 1 Oil line a) Feed, b) Return (Jiffy-tite 3/8" quick-release coupling)
- 2 Coolant hose a) Feed, b) Return
- ➡ Direction of travel

The oil lines between the transmission oil cooler and the transmission with power take-off, the oil pump required for circulation in the oil cooling circuit, as well as all accessory parts for line attachments, connection fittings and seals, are not included in the scope of the transmission oil cooler pre-installation (code GK4).

It is the responsibility of the bodybuilder manufacturer to professionally complete the oil cooling circuit.

For circulation of the transmission oil in the cooling circuit, it is recommended to use an oil pump with a delivery rate of 3 dm³/min.

The bodybuilder manufacturer is responsible for the appropriate design of the necessary pump capacity depending on the requirement profile, the operationally safe attachment to the vehicle, the electrical integration into the vehicle electrical system and the electronic actuation. With regard to the electrical connection, observe the specifications in 8.4 Interfaces and electric lines (→ page 271).

The oil lines between the transmission and the oil cooler must be professionally designed by the bodybuilder manufacturer in accordance with hydraulic requirements and must be fastened in such a way that they are safe to operate. The oil lines must be routed and fastened in such a way that they cannot come into contact with surrounding component parts in any operating status of the vehicle. In addition, ensure a sufficient distance and suitable heat shielding in the vicinity of component parts with high operating temperatures (e.g. exhaust system); See 6.3.3 Exhaust system (→ Page 141).

When connecting the transmission oil cooling, the additional oil volume in the oil lines, oil pump and oil cooler must be taken into account so that the oil circuit remains constant. Refill with an appropriate amount of transmission oil according to the line installation and the type of transmission oil cooling system used.

For the transmission oil cooler on OM654, provide for an additional oil volume of 0.16 dm³.

The additional oil volume required due to the oil pump and the oil lines must be determined by the bodybuilder manufacturer depending on the design.

You can find further information in 3.12 Special equipment (→ page 74).

On the PTO 2b, the maximum permissible weight moment at the transmission flange due to the intrinsic weight of a hydraulic pump is 15 Nm.

Installing propeller shafts for transmission power take-off

Observe the following when installing propeller shafts:

- Installation guidelines of the propeller shaft manufacturers
- If necessary, use multiple propeller shafts with intermediate bearings.
- The flanging surfaces must be completely flat.
- The working angles must be identical at both joints ($\beta_1 = \beta_2$). They must not be greater than 6° or less than 1° .
- Balancing plates must not be removed.
- Make sure that the marks are aligned on the propeller shafts during installation.

 More information on the layout of propeller shafts (→ page 148).

b) Front engine power take-off

Additional assemblies (e.g. a refrigerant compressor or an additional alternator) can be driven by an additional pulley on the front of the crankshaft, see also 6.5 Additional assemblies (→ page 166).

The following codes are available as the power take-off:

Code	Equipment
N62	Front engine power take-off with carrier for additional alternator
N63	Front engine power take-off with carrier for additional refrigerant compressor

These power take-offs can be obtained ex factory as special equipment.

For the power take-off with code N62, use an alternator with a freewheeling belt pulley.

Maximum transferable output

Code N62	8.5 kW
Code N63	8.0 kW

Required belt pulley for additional assemblies

- Poly-V-belt pulley with 6 grooves (a different number of grooves is not permitted)
- Outside diameter
 - Code N62: 50 mm
 - Code N63: 119 mm

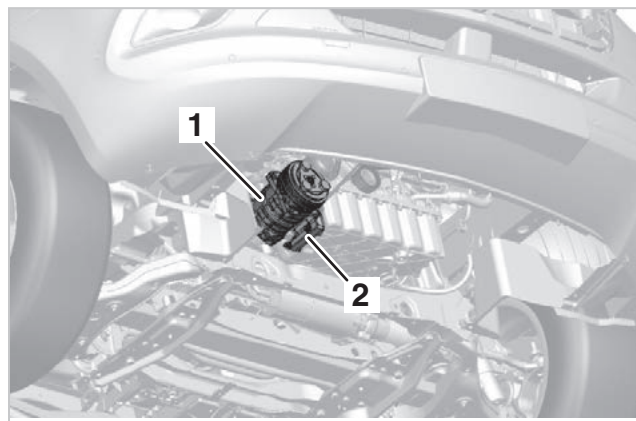
Only the following Mercedes-Benz genuine parts are permitted to be used as the belt:

OM654	Item number
Code N62	A 000 993 58 00 Stretchfit
Code N63	A 000 993 57 00 Stretchfit

The use of other belts is not permitted.

 Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

Additional assemblies can be mounted on a major assembly carrier fixed to the engine.



Additional assembly on engine-mounted major assembly carrier

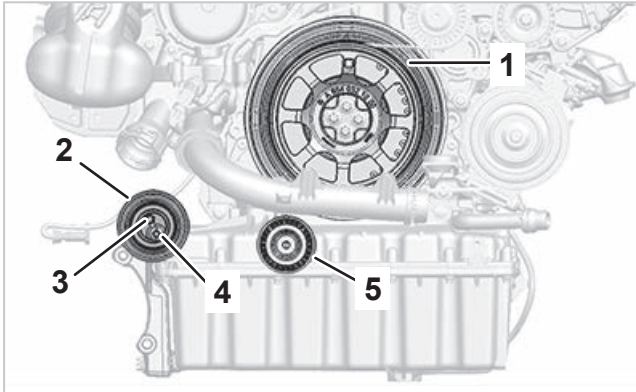
- 1 Additional refrigerant compressor
- 2 Major assembly carrier

Maximum weight of additional assemblies

Code N62	7.3 kg
Code N63	7.3 kg

Adjust belt tensioning on engine power take-off for OM654

The tension of the stretch fit belt on the engine-mounted power take-off can be adjusted using the eccentric pulley, which is known as an eccentric belt tensioner (eccentric RSP) (see following figure).



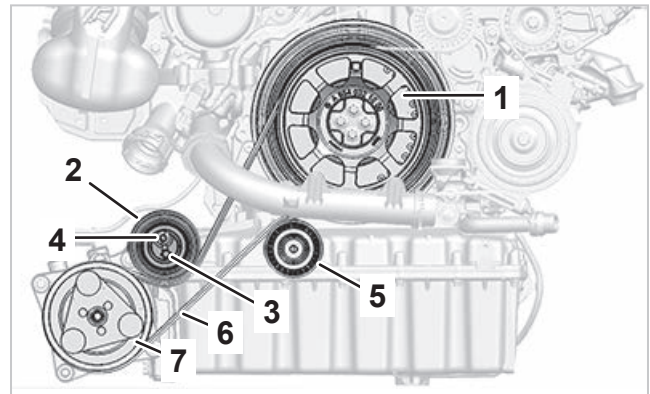
Belt tensioner on engine OM654 in as-delivered state (with code N63)

- 1 Engine-mounted belt pulley
- 2 Eccentric belt tensioner in as-delivered state
- 3 Hexalobular internal profile (T55)
- 4 External Torx bolt (E12)
- 5 Guide pulley only with power take-off code N63 (not available with code N62)

In the as-delivered state (still without belt), the eccentric belt tensioner is in a position in which the stretch fit belt can be easily placed over the guide pulley and the belt pulley of the power take-off and auxiliary assembly. In this assembly state, the hexalobular internal profile (T55) of the eccentric belt tensioner is set to an 11 o'clock position as the upper end stop, and the eccentric belt tensioner is preset at this position.

After fitting the stretch fit belt and checking if the belt is correctly positioned at all points, the eccentric belt tensioner is rotated clockwise using the hexalobular internal profile (T55) up to the lower end stop, which is after the dead center position, so that the eccentric belt tensioner then remains in this position (see following figure).

After fixing the eccentric belt tensioner in place by tightening the external Torx bolt (E12) to a torque of 30 Nm, the stretch fit belt then has the required pre-load force.



Belt tensioner in final state after assembly and tensioning of belt (with code N63)

- 1 Engine-mounted belt pulley
- 2 Eccentric belt tensioner in final state
- 3 Hexalobular internal profile (T55)
- 4 External Torx bolt (E12)
- 5 Guide pulley only with power take-off code N63 (not available with code N62)
- 6 Stretch fit belt in tensioned assembly state
- 7 Auxiliary assembly (example here of refrigerant compressor with power take-off code N63)

6.5.4 Retrofitting an alternator

The special equipment "Power take-off with code N62" is recommended for subsequently driving an additional alternator. Only alternators with a freewheel alternator pulley are permitted.

For more information on additional alternators, See 8.4.7 Alternator (→ Page 276).

6.5.5 Auxiliary brakes (retarders)

It is generally recommended to install the retarder type AF30-35 from Telma S.A. – which is available ex factory and can be ordered with code PB2 in combination with customer request codes – as this retarder has been integrated and tested by Mercedes-Benz as part of the development work for the Sprinter.

A pre-installation for the installation of the retarder is also available as special equipment: Code BR9 pre-installation for eddy current brake, See 8.12 Pre-installation for retarder (→ Page 377).

For more information on the content of the special equipment, contact the Technical Consultancy (→ page 18).

! NOTE

It is not permissible to install a different retarder (either type or manufacturer).

! NOTE

Avoid continuous excessive torque take-off, as this can otherwise lead to transmission damage.

Existing features and functions

Mechanical components:

- For the mechanical connection of the retarder to the underbody, see the illustration in 8.12 Pre-installation for retarder (→ page 377).
- Integration of the retarder in the drivetrain
- Adapted hydraulic brake line
- Thermal protection of critical component parts (fuel tanks, brake line, underbody)
- Safety concept in case of a crash
- Exhaust routing
- Warning lamp in the cockpit
- Prepared for the retrofitting of a retarder parking brake lever (multistage)

Electrics:

- Electrical connection of the retarder on the underbody
- Electrical layout and fusing of the vehicle electrics (lines/alternator/consumers)
- Coordinated CAN communication of vehicle control units and retarder control unit

See 8.12 Pre-installation for retarder (→ page 377) for further information on the electrical scope.

! NOTE

If you are using the pre-installation for retrofitting a retarder (code BR9) on 5.0 t model designations (code IT5) with wheelbase 3665 mm (code IR4), it is essential to ensure a minimum rear axle load of 2000 kg in all driving conditions after the installation of a retarder. No minimum axle load is stipulated for vehicles with comfort air suspension (code CB2).

i For more information on this special equipment, contact the Technical Consultancy (→ page 18).

6.6 Attachments

For attachments at the frame, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Always observe the permissible axle loads.

Attachments must not impair the function of vehicle parts and vehicle systems.

! NOTE

Observe all national laws, directives, and registration regulations.

Winch

Attach winches behind the cab on an assembly frame of sufficient size and strength.

! NOTE

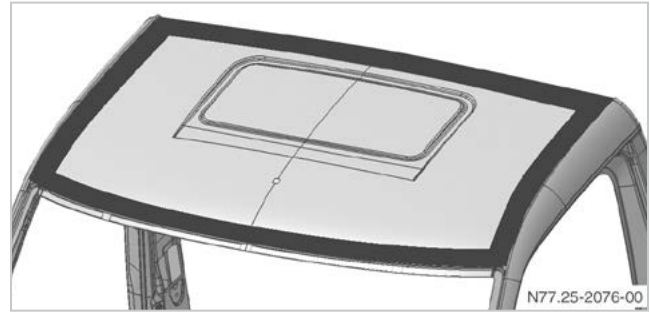
Do not attach a winch to the front section of the frame.

6.6.1 Air deflectors

Air deflectors may only be fitted onto the cab roof by applying high-strength adhesive to the whole of the area around the lateral roof frame, the front roof frame and the 1st roof bow (level with the B-pillar).

Take into account the load applied by aerodynamic drag or contact pressure. Design the attachment method in such a way that the basic vehicle is not damaged.

Take possible interference with driving assistance systems into account; See 8.9 Driving assistance systems (→ Page 307).



Area for applying adhesive when fitting air deflectors

! NOTE

Do not make additional bores for mounting in the cab roof.

For other roof attachments (e.g. air conditioning system), a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

6.6.2 Attachment above cab

- Observe the permissible vehicle center of gravity position and front axle load (→ page 78).
- Observe the specifications in 6.2.12 Cutting the cab roof and B-pillar roof bow (→ page 132).
- If the conversion causes vibrations or noise, extend the assembly frame through the cab rear panel to underneath the seat boxes and secure it. An auxiliary battery as per code E2I "Auxiliary battery for retrofitted consumers, vehicle interior" (accommodated in the front passenger seat box) is not possible with this design.
- Take possible interference with driving assistance systems into account; See 8.9 Driving assistance systems (→ Page 307).

! NOTE

Observe and adhere to the information and specifications on body limits for driving assistance systems, See 8.9 Driving assistance systems (→ Page 307).

6.6.3 Roof luggage racks

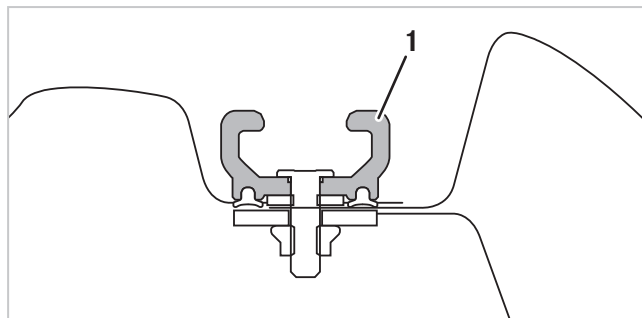
Panel vans and crewbuses

- Make sure that the load is distributed evenly across the entire roof area.
- Arrange support feet at an even distance. Apply 50 kg per pair of feet and strut as a guideline value.
- With shorter roof luggage racks, the load must be reduced proportionally.
- Observe the limit values for the maximum roof load and the minimum number of roof bows; See 4.3.8 Roof and roof load (→ Page 90).

i Further information can also be found in the ADVANTAGES product information system via the Bodybuilder Portal (→ page 29).

To make it possible to fit roof rack systems, the Sprinter – BR 907 can be equipped with mounting rails (special equipment code D13).

Note: Mounting rails (code D13) are not available ex factory for the open model designations of cab (FHS) and crewcab (FHL).



Mounting of roof luggage racks

1 Mounting rail (roof luggage rack)

Take into account the fact that large objects on the roof luggage rack may cause interference with driving assistance systems; See 8.9 Driving assistance systems (→ Page 307).

6.6.4 Shelves and load rails in the load compartment

1) Shelves in the load compartment

When installing shelves in the load compartment, observe the following:

- Shelving units must be sufficiently stable and self-supporting.
- Shelving units must rest on the crossmembers and longitudinal members of the vehicle floor and distribute the forces evenly.
- Shelving units must be attached to the load rails and tie-down eyes or to the entire contact surface on the body-in-white – in the same way as the standard rails – or to the pre-installation for shelf systems (code ZE6).
- Shelving units must be installed free of mechanical stresses. This applies to installations with the pre-installation for shelf systems (code ZE6) as well as without this pre-installation; the gap between the roof bow bracket and the longitudinal tube of the shelving must be minimized by means of shims if necessary and must not exceed 1 mm in width.

! NOTE

Shelf systems must not collapse in the event of a crash. Shelf systems must have no negative influences on the basic vehicle and its functions (in particular on passive safety).

! NOTE

Do not use attachments that apply force only to the side wall or to isolated points of the vehicle wall. Otherwise, there is a risk of damage to the side wall and the roof.

It is recommended to use the load rails available as a special equipment option for mounting and attaching shelves (code VC4 or V42); see following section 2).

- i** For further information about the side wall
See 6.2.7 Side wall, windows, doors, and flaps
(→ Page 123).

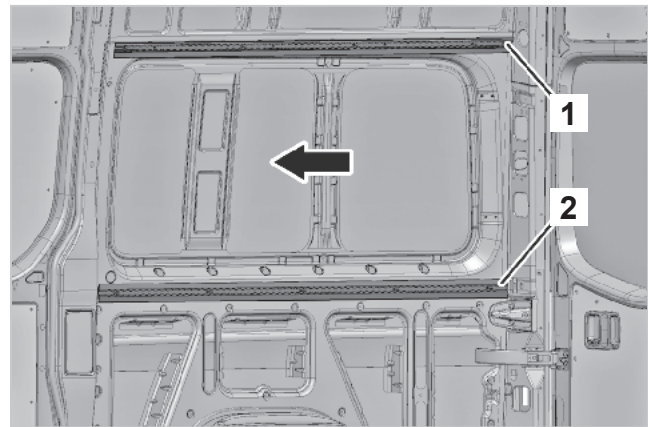
Ensure the stability of the shelf system by mounting it to the load compartment floor and supporting it at the side wall. We recommend reinforcing the shelf construction with diagonal struts.

In order to test shelf constructions with respect to their stability and installation in the load compartment, it is recommended that bodybuilder manufacturers – under their own responsibility – commission appropriate testing (e.g. based on UN R 126: crash test (frontal), sled test, tensile test) to be carried out by authorized testing institutions. Submit evidence of these validation tests upon request.

2) Load rails at the side wall ex factory

Load rails are available on the side wall ex factory in two heights as special equipment:

Code	Equipment
VC4	Load rails at the roof frame
V42	Load rails on belt rail (underneath the window stamping)



Load rails in the panel van

- 1 Load rails at the roof frame
2 Load rails at the belt rail
➡ Direction of travel

Maximum tensile forces

Code	Load rails ex factory	Permissible rated tensile force [daN]
VC4	at roof frame	150
V42	at the belt rail	250

These values only apply under the following conditions:

- The load must be standing on the floor.
- The load must be secured at two tie-down points of a load rail.
- The distance to the next load securing point on the same load rail must not exceed 1 m.

- i** Observe the operator's manual supplied with the load rails ex factory.

! NOTE

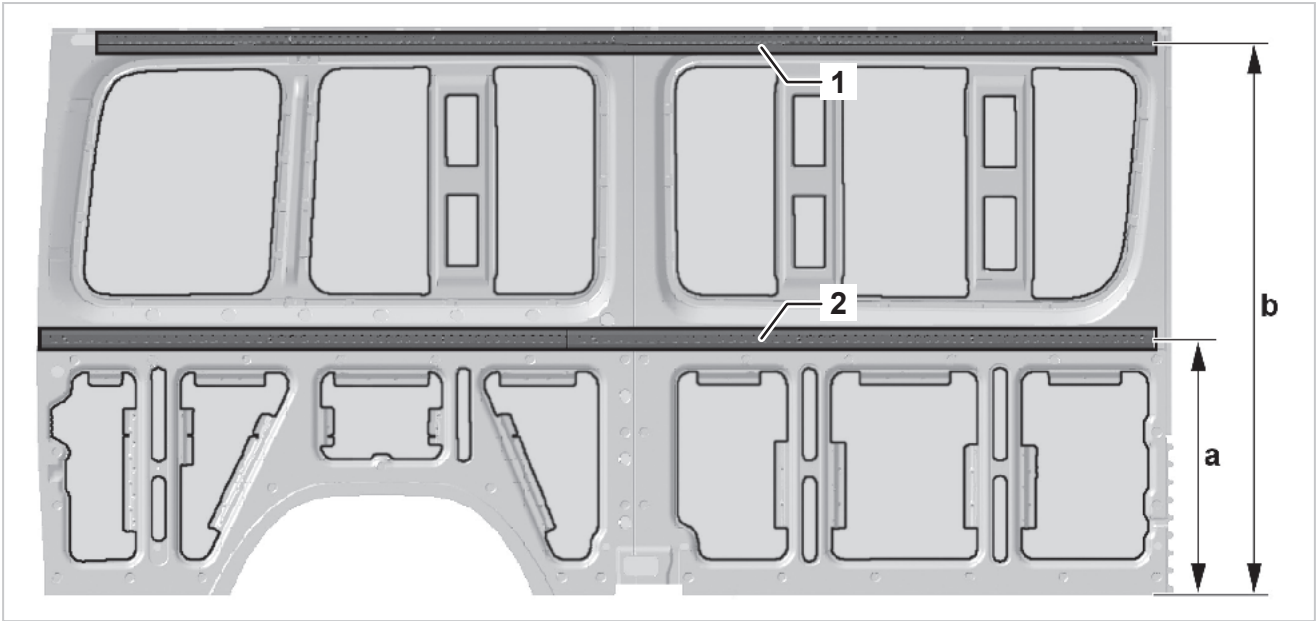
Tensioning of load securing equipment between the left and right sides of the vehicle with high tensile and compressive forces on the load rails is not permitted and must be avoided by selecting suitable equipment and methods for load securing. Otherwise, there is a risk of damage to the side wall.

3) Retrofitting load rails at the side wall

! NOTE

The subsequent installation of load rails or tie-down rails is only permitted to be carried out in the areas of the side wall designed for this purpose, analogous to the load rails ex factory.

Comply with the maximum tensile forces (see table) in all driving conditions. Otherwise, excessive tensile forces may result, causing damage to the side wall.



Position of retrofitted load rails at the side wall

- 1 Load rails at the roof frame
- 2 Load rails at the belt rail

Position of retrofitted load rails		
Center of load rail	Dimen- sion a [mm]	Dimen- sion b [mm]
Based on standard wooden floor	718	1552
Based on vehicle floor (upper crease)	729	1563

Observe the following points when retrofitting load rails to the side wall:

- Observe the information from the manufacturer of the load rails.
- Clearly indicate the maximum tensile forces (see table) in the area of the load rails (e.g. using adhesive labels) and enclose them with the operator's manual in the vehicle in suitable form.

Maximum tensile forces of riveted and glued load rails

Code	Load rails	Permissible rated tensile force [daN]
VC4	at roof frame	120
V42	at the belt rail	200

Maximum tensile forces of riveted load rails

Code	Load rails	Permissible rated tensile force [daN]
VC4	at roof frame	60
V42	at the belt rail	100

All values only apply under the following conditions:

- The load must be standing on the floor.
- The load must be secured at two tie-down points of the rail.
- The distance to the next load securing point on the same rail must not exceed 1 m.

! NOTE

Tensioning of load securing equipment between the left and right sides of the vehicle with high tensile and compressive forces on the load rails is not permitted and must be avoided by selecting suitable equipment and methods for load securing. Otherwise, there is a risk of damage to the side wall.

Recommended process adhesives**Körapur 140**

Use	Adhesive
Supplier	Kömmerring, Suppl. no. 110/75074
Part no.	A 009 989 17 71
Chemical base	1K-PU that hardens in air humidity
Use-by date	6 months, date specified on container

Köracur 110

Use	Accelerator paste
Supplier	Kömmerring, Suppl. no. 110/75074
Part no.	A 009 989 18 71
Chemical base	Water-based gel paste
Use-by date	9 months, date specified on container

Körabond HG8 1

Use	Activator
Supplier	Kömmerring, Suppl. no. 110/75074
Part no.	A 001 986 90 71
Chemical base	Silane, artificial resin
Use-by date	12 months

Requirements of rivets

When retrofitting load rails on the side walls using rivets, observe the following values:

- Cross-tension strength min. 3800 N
- Shear strength min. 3300 N
- Rivet diameter = 4.8 mm
- Head diameter = 9.3 mm
- Clamping range = 3.5 mm to 6.0 mm

Recommended process steps for adhesive bonding

- Roughen load rail
 - ▶ The entire length of the bonding surface at the rear of the mounting rail must be roughened with a wire brush. Immediately activate roughened rails.
- Activate bonding surface
 - ▶ The roughened bonding surface must be activated over the entire bonding surface using a PE bottle with felt insert. Immediately afterward, the activator that has not yet evaporated must be completely wiped off with a cleaning cloth (wipe on/wipe off procedure).
 - Flash-off time:
At least 10 minutes
 - Time until application of adhesive:
Max. 24 hours
 - Replacement cycle for felt insert:
When no longer functional
 - Replacement cycle of PE bottle:
At least once daily
- Bonding mounting rail
 - ▶ The adhesive must be applied to the entire length of the activated bonding surface as a straight, round bead. There must be a gap in the adhesive bead of approx. 10 mm at the rivet holes to prevent adhesive from being squeezed out in the visible area.

- Adhesive application
 - Bead geometry:
Round bead
 - Bead diameter:
Approx. 6 mm
 - Dwell time of adhesive in static mixer:
Max. 15 minutes
 - Time between adhesive application and joining:
Max. 15 minutes
 - Dwell time of adhesive on rail:
Max. 15 minutes
 - Time until load may be applied (curing time):
Approx. 7 days

 **NOTE**

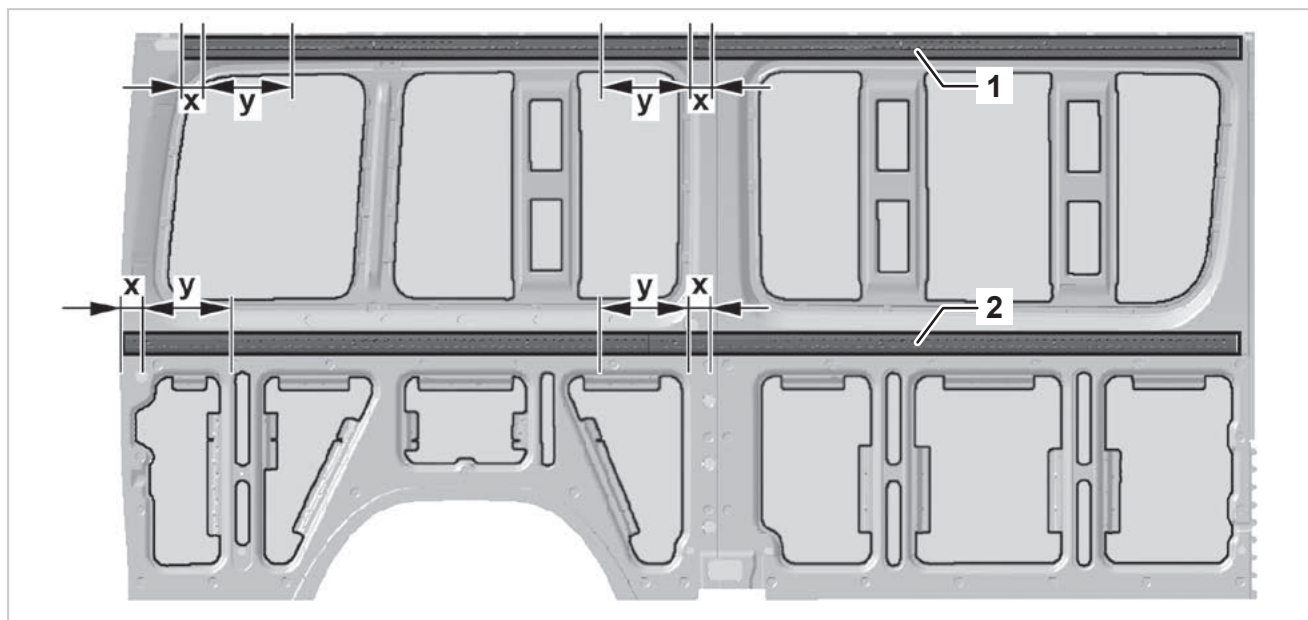
Follow the handling instructions of the adhesive manufacturer. If the specified bonding instructions are not followed, the bond will not be of the required quality.

The mounting rail with adhesive must be bonded to the vehicle interior within 15 minutes. Otherwise, excessive forces may result, causing damage to the side wall.

Rivet distance for retrofitted load rails

! NOTE

Comply with the maximum rivet distances (see table) in all driving conditions. Otherwise, excessive tensile forces may result, causing damage to the side wall.



Rivet distances for retrofitted load rails at the side wall

- 1 Load rails at the roof frame
- 2 Load rails at the belt rail

Maximum rivet distance	Dimension x [mm]	Dimension y [mm]
Riveting plus adhesive bonding	75	450
Riveting (without adhesive bonding)	25	225

4) Retrofitting load rails at the roof

! NOTE

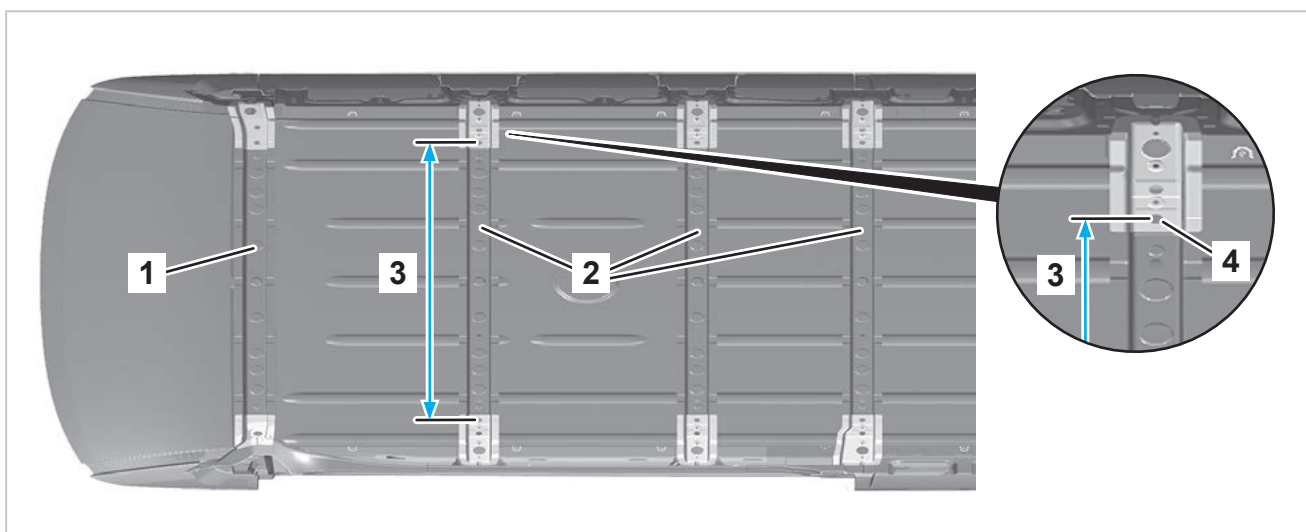
Load rails at the roof must only be used to attach load securing equipment that does not exert high tensile or compressive forces on the upper load rails when attached to the roof and floor, e.g. lock rods with locking mechanism in airline rails.

Any other use, e.g. for directly lashing loads with tension belts, is not permitted. Otherwise, excessive forces may occur and thus damage the roof structure.

When installing load rails at the roof, make sure to observe the following:

- Load rails must only be attached at the roof bows in the load compartment. Attachment at the roof structure in the area of the B-pillar and the rear portal is not permitted.

Make sure to observe the permissible installation area for this.



Permissible roof bows and permissible installation area for retrofitted load rails at the roof

- 1 Roof bow at B-pillar (not permitted)
- 2 Permissible roof bows in the load compartment
- 3 Permissible installation area
- 4 Maximum limit for permissible installation area in the center of the first bore of the side overhead consoles

- The positioning of the load rails in the load compartment floor relative to the load rails at the roof bows must be selected so that after the load securing equipment (e.g. lock rods) has been attached, a vertical alignment at a right angle (90° in all directions) to the load compartment floor is ensured.
- Observe all information from the manufacturer of the load securing equipment.
- When determining the number of attachments for load securing equipment at the load rails, observe the following load limits:

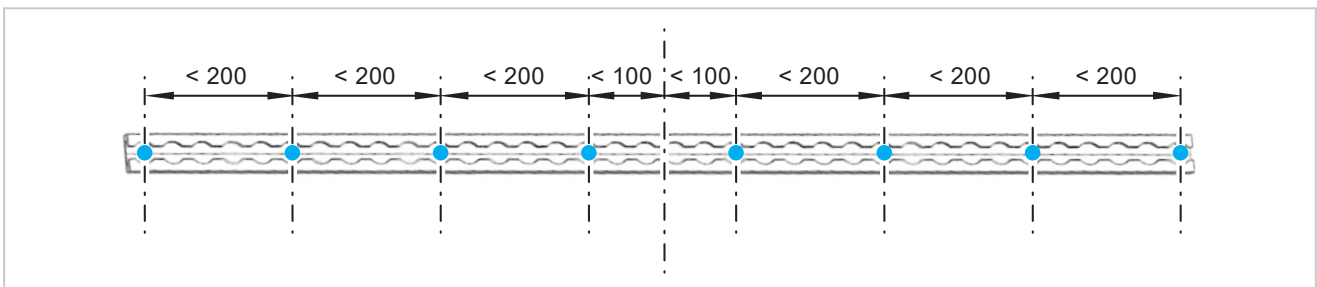
Maximum tensile forces on load rails at the roof¹⁾

	Permissible rated tensile force [daN]
Per attachment	Max. 150
In total with multiple attachments	Max. 500

- 1) Determination of the load limits as per VDA 2700 Load securing

When installing the load rail at the roof bow, make sure to observe the following:

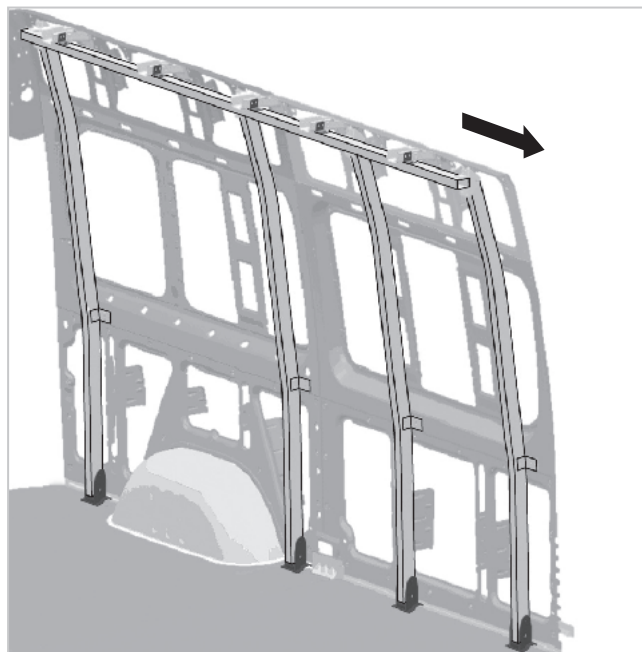
- Use at least 6 attachment points at a distance of maximum 200 mm apart, starting from the center of the load rail; see following illustration.
- Position the outer attachment points as close as possible to the ends of the load rail.
- Existing bores are permitted to be used and can be enlarged if necessary.
- Attachment material:
 - Countersunk screw M6×25 8.8 DIN 7991
 - Rivet nut M6 with flat head and shaft knurling
- Observe the installation instructions and the information on tightening torques from the manufacturers of the load rails and the attachment material.
- After installation, the load rail must rest on the roof bow without any gaps and must not protrude laterally beyond the limit of the permissible installation area.



Attachment points of the load rail at the roof
All dimensions in mm

5) Design proposal with pre-installation for shelf installation

The special equipment "Pre-installation for shelf system" (code ZE6) is available ex factory to facilitate the retrofitting of shelves. The delivery package includes angle pieces for attachment to the roof bows and mounting consoles for attachment to the load compartment floor.



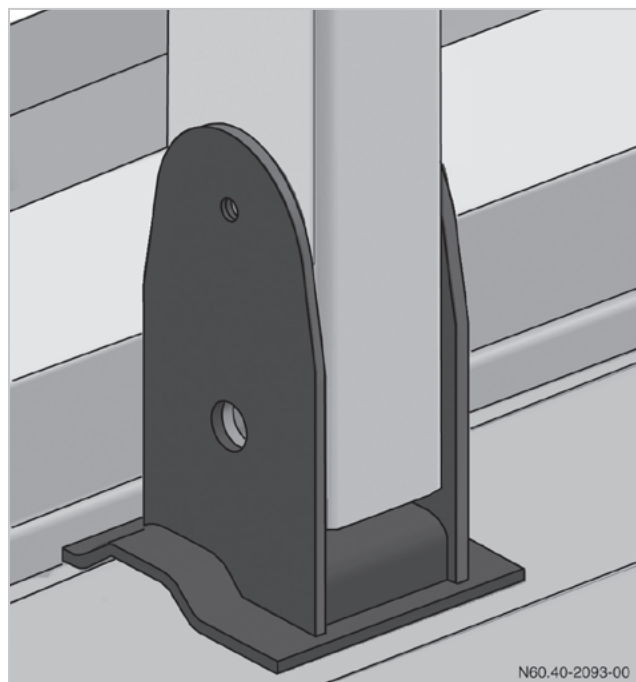
Suggested shelf attachment with ZE6

➡ Direction of travel

When installing shelves with this special equipment, observe the following:

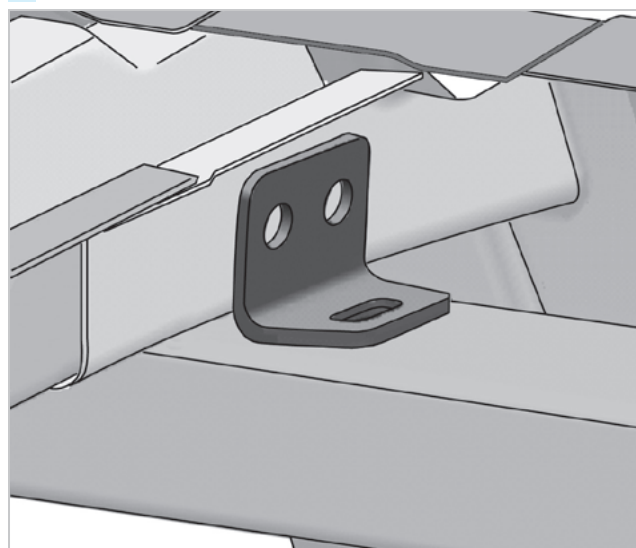
- Shelves must not be wider than 450 mm.
- The max. load-bearing capacity is 80 kg/m.
- The shelf supports must be made of steel (at least ST235JO according to DIN EN 10025) with minimum dimensions of 60 mm × 40 mm × 3 mm (length × width × thickness).

- ▶ Bolt the shelf supports to the floor with the mounting supports.

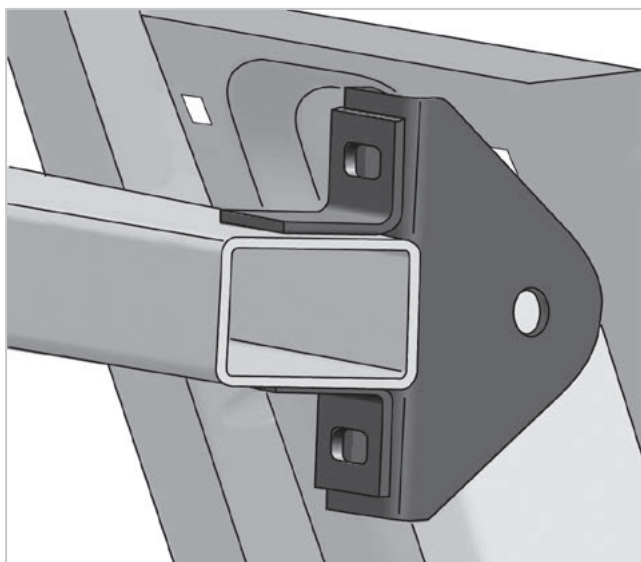


Bolting the mounting supports to the floor

- ▶ To fix the wooden floor, mount 2 angle pieces per support at the screw connection of the support and mounting console (contact surface per angle piece at least 1200 mm², dimensions 60 mm × 20 mm).
- ▶ Bolt a steel tube with a rectangular profile measuring 60 mm × 40 mm × 3 mm onto the mounting supports on the roof bows. Bolt the shelf supports to this rectangular profile at the top.

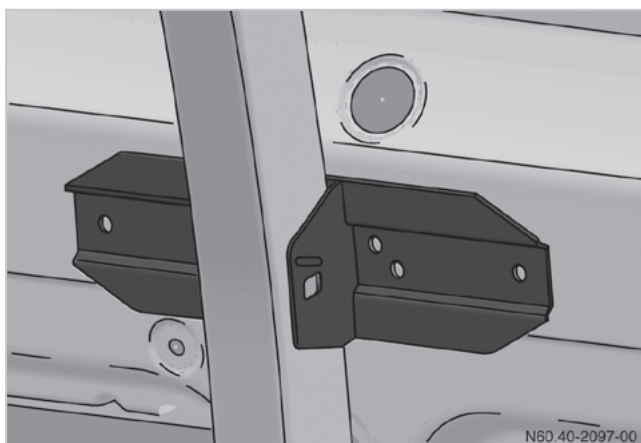


Connecting the shelf longitudinal tube to the roof bow angle piece



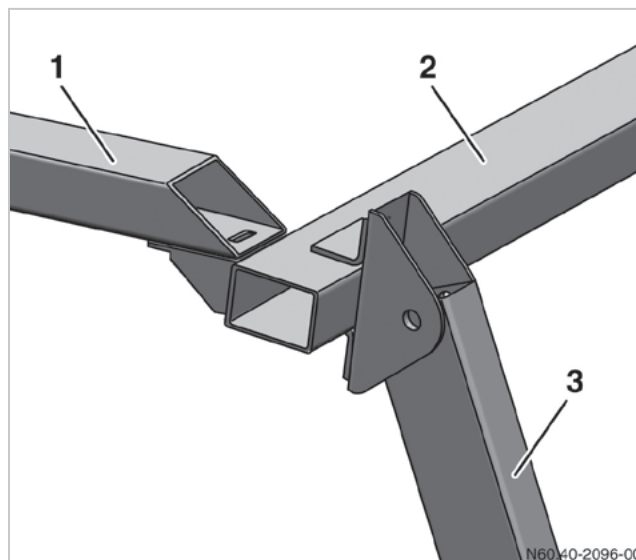
Connecting the longitudinal tube to the support

- The longitudinal tube must not be connected with the partition wall or the rear door frame.



Suggested design for mounting support on belt rail

- ▶ In addition to fixing the shelf supports to the floor or to the roof bows, fix them to the belt rail by means of a mounting support.
- ▶ The connection must be made by bonding and riveting. A minimum bonding surface area of 7000 mm² is required.



Additional connection of the longitudinal tubes

- 1 Connecting rail
- 2 Longitudinal tube
- 3 Support

- ▶ If the first or last support is more than 300 mm away from the roof bow, the longitudinal tubes must be connected together.

6.6.5 Loading crane

The size of the crane must be selected in accordance with the chassis size.

Loading cranes must be secured on an assembly frame to relieve the load acting on the frame (→ page 200).

Verify compliance with the permissible axle loads by calculating a weight balance.

The vehicle's stability must be ensured by the body-builder manufacturer. The swivel range of the crane must be limited accordingly.

Loading cranes mounted on vehicles must comply with the accident prevention regulations (UVV) and general machinery directives; See 2.6 Accident prevention (→ Page 43).

Observe country-specific statutory regulations.

Observe the installation instructions of the crane manufacturer.

-  For additional platform or dumper bodies, the dimensions of the assembly frame longitudinal members can be found in the diagram in 7.1.3 Section dimensions and dimensioning (→ page 203).

NOTE

Provide outriggers for each loading crane. The use of hydraulic outriggers is recommended for this.

The vehicle must not be raised using the outriggers, as this would damage the frame.

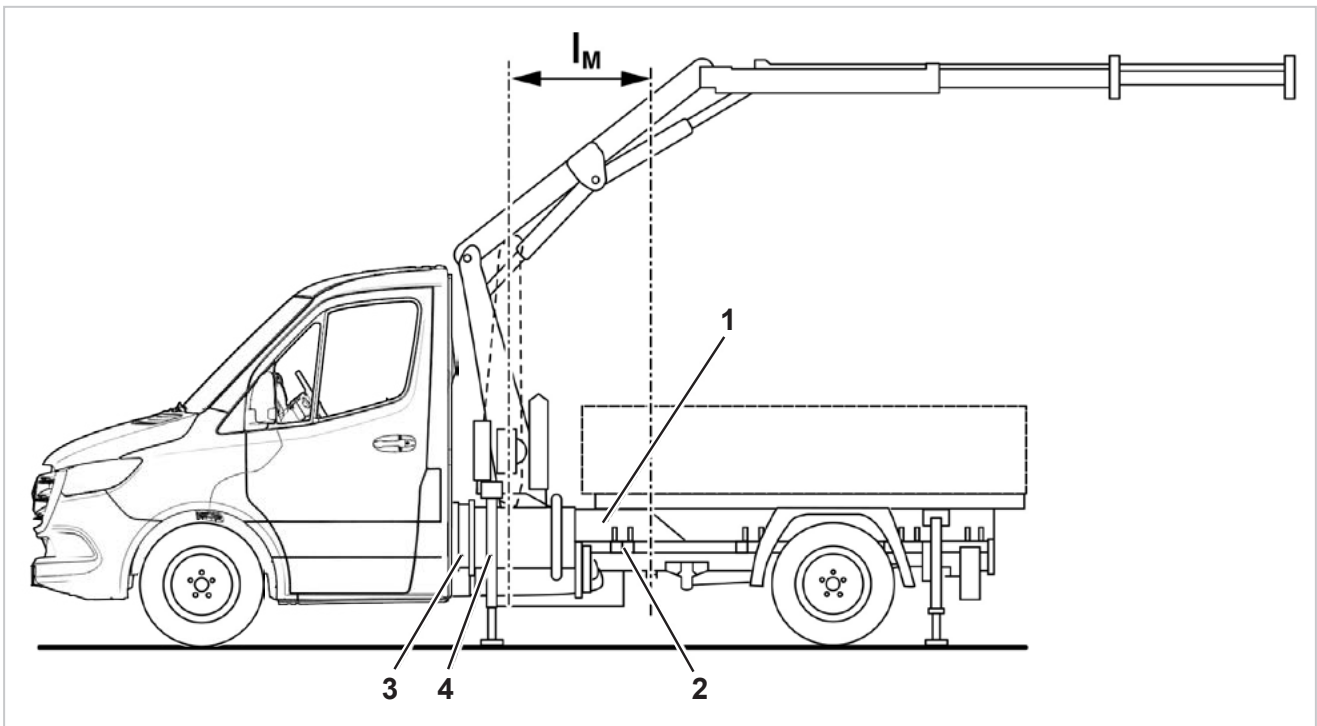
Loading crane body behind the cab

Loading cranes and outriggers must not impair the function of other major assemblies. It is recommended to implement a shear-resistant connection for connecting the assembly frame to the vehicle frame.

Assembly frame

- Maximum crane load moment ($kN \times l$): 25 kNm
- Section moduli (W_x) and material properties can be found under 7.1 Assembly frame (→ page 200).
- Section dimensions of assembly frame longitudinal members (→ page 200).
- While the crane is in operation, ensure the vehicle's stability by means of extending support feet.
 - ▶ Clearly mark outriggers that protrude beyond the vehicle when the vehicle is stationary, by means of conspicuous colors, rear reflectors and warning lights.
 - ▶ Determine the platform length according to the position and weight of the loading crane, complying with the permissible axle loads.
- If the maximum crane load moments are to be exceeded, a Letter of Compatibility is required (→ page 19); this can be requested via the Body-builder Portal (→ page 20). The crane mounting must be reinforced.
- The vehicle must only be used on flat, surfaced roads.

- A frame extension may become necessary as a result of the load distribution.
- If a stronger assembly frame is required than for the body when a loading crane is mounted behind the cab, the loading crane can be secured on a shorter assembly frame (see illustration below). The short chamfered assembly frame must have a length l_M corresponding to at least 35% of the wheelbase.
- For this attachment, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).



Loading crane body

- 1 Loading crane assembly frame
- 2 Mounting supports
- 3 Loading crane attachment
- 4 Outrigger
- l_M Length of loading crane assembly frame

Loading crane body at end of frame

 **WARNING**

Comply with the minimum front axle load (→ page 78) under all load conditions. Otherwise, adequate driving stability is no longer guaranteed.

There is risk of an accident and danger to life and limb!

- Secure the loading cranes to an assembly frame made of steel. Join the assembly frame to the vehicle frame by means of a rigid connection.
- Maximum crane load moment (kN × l): 25 kNm
- Section moduli (W_x) and material properties can be found under 7.1 Assembly frame (→ page 200).
- Section dimensions of assembly frame longitudinal members (→ page 200).
- While the crane is in operation, ensure the vehicle's stability by means of extending support feet; see the section "Assembly frame" (→ page 188).

6.6.6 Liftgate (cargo liftgate)

General information

Observe the information and specifications for driving assistance systems, in particular for the reversing camera; See 8.9.9 Reversing camera (→ Page 347).

Before installing a cargo liftgate, the available installation space must be checked by the bodybuilder manufacturer. Note the limitations in the installation space due to the installed exhaust system and tank system.

Fit the liftgate to the chassis

The special equipment "Pre-installation for liftgate" (code EV3) is recommended if you intend to retrofit a liftgate to chassis.

Among other things, this adds a more powerful alternator (code M46 or M60) and an auxiliary battery (code E2M or E2I).

In addition, include the special equipment "Power connection via terminal strip" (code EK1) in the vehicle order.

Fit the liftgate to the panel van

For retrofitting a liftgate to panel vans, the special equipment "Liftgate pre-installation" (code EV3) is not available.

If fitting a liftgate as part of the vehicle order, take into account the following codes:

Code requirements (mandatory)	
Code	Equipment
Q11	Longitudinal frame member reinforcement
EK1	Terminal strip for electric connection
E2I or E2M	Auxiliary battery in the interior Auxiliary battery in the engine compartment
Code recommendations	
Code	Equipment
M40 or M60	Alternator 14 V/200 A Alternator 14 V/250 A
W54	Double-wing rear-end door that opens as far as side wall

Prerequisites

! NOTE

If an electrohydraulic liftgate is fitted, use an alternator, a battery with higher capacity, and, in principle, an auxiliary battery.

- Within the EU, liftgates must conform to EN 1756-1.
- Liftgates must comply with the accident prevention regulations (UVV) and the general machinery directives, See 2.6 Accident prevention (→ Page 43).

! NOTE

Observe all national laws, directives, and registration regulations.

- The rear axle load must not exceed the permissible limit value, See 3.2 Model overview (→ Page 49).
- Comply with the minimum front axle load under all load conditions, See 4.1.1 Steerability (→ Page 78).
- Ensure vehicle stability in all operating statuses.
 - ▶ Use calculations to determine the load distribution. Take all special equipment into account when doing so.
 - ▶ If necessary, shorten the body length and the rear chassis overhang accordingly (chassis).
 - ▶ Hydraulic outriggers are recommended.
- Observe the legal requirements in the individual countries relating to "Underride guard" and "Lighting system" when a liftgate is mounted.

- Maximum load clearance is 600 mm, relative to the standard rear portal/standard rear crossmember
- Cuts at the end crossmember are therefore only permitted following coordination with the Technical Consultancy (→ page 18).
- Vehicle stability when loading and unloading the vehicle must be ensured by the user.

- ① Further information can also be found in the ADVANTAGES product information system via the Bodybuilder Portal (→ page 29).

! NOTE

The permissible nominal live load on the liftgate used must not be exceeded and the load clearance must be observed.

! NOTE

The standard tail lamps of the chassis must only be installed in the approved horizontal installation position as in the original condition.

For other installation positions, there are alternative tail lamps available from the manufacturer of the standard tail lamps, See 8.5.3 Tail lamps (→ Page 287).

- ① For information on the connection to ground for the liftgate/cargo liftgate with code EV3, See 8.4.12 Ground bolts (→ Page 284).
- ① For further information on electrical pre-installation for code EV3, See 8.13 Pre-installation for liftgate (→ Page 379).

Liftgate attachment

Attach the liftgate as per the section "Attachments on the rear frame section" (→ page 117).

Provide additional torque support by means of at least two screw connections fitted with spacer bushings (e.g. on the assembly frame).

- ▶ Extend the assembly frame as far forward as possible and attach it with a frictional connection to the vehicle frame.
No assembly frame is required on vehicles with a standard panel van body.

Use at least all the factory-installed mounting supports, and in particular those at the rear end of the frame, to connect the assembly frame (body and liftgate) to the vehicle frame.

Also pay attention to the need for reinforcements in the left and right rear longitudinal frame members in vehicles with a trailer coupling or the pre-installation for this, see note in 6.6.7 Trailer coupling (→ page 193).

If modifications are required to the underride guard due to the attachment of a liftgate, the strength and bending resistance of the underride guard must not be changed (→ page 198).

! NOTE

The vehicle must not be raised using the outriggers, as this would damage the frame.

Permissible lifting force of liftgate

Type	Wheelbase [mm]	Lifting force up to [kN] ¹⁾		Minimum dimension of assembly frame - longitudinal member [mm]
		Chassis	Panel van ²⁾	
211 CDI - 216 CDI	3665	5	5	None
311 CDI - 319 CDI	3665	5	5	None
	4325	5	5	
	3665	7.5	5	120×50×4
	4325	7.5	5	or factory platform (code P02) ³⁾
411 CDI - 419 CDI	3665	5	5	None
511 CDI - 519 CDI	4325	5	5	
	3665	7.5	5	80×45×3
	4325	7.5	5	or factory platform (code P02) ³⁾
	3665	10	5	120×50×4
	4325	10	5	

1) For rated load distance of 600 mm

2) Without assembly frame

3) Steel platform with aluminum dropsides

6.6.7 Trailer coupling

For trailer couplings or for their pre-installation, the following special equipment is available depending on the vehicle model designation:

Code	Equipment
E57	Electrics for trailer socket (Provides the power supply to the trailer, including lighting control)
Q10	Crossmember for trailer coupling (A special crossmember is mounted at the rear area of the vehicle with a mounting plate to attach the trailer coupling.) The design of the crossmember depends on the vehicle's tonnage.
Q11	Longitudinal frame member reinforcement (Reinforcements in the rear left/right longitudinal frame member; in model series 907, this is already standard equipment for open and closed vehicle model designations; can only be selected with code QW1 for closed vehicle model designations)
QA3	Pre-installation of trailer coupling for trailer load up to 3500 kg (On-vehicle adaptations and entries in vehicle documentation)
QA5	Pre-installation of trailer coupling for trailer load up to 2800 kg (On-vehicle adaptations and entries in vehicle documentation)
Q20	Trailer coupling (Jaw for pulling trailers with drawbar) The design of the jaw depends on the vehicle's tonnage. The permissible tongue weight is 80 kg for M1 approval/100 kg for N1/N2 approval.
Q50	Removable ball head trailer coupling (Removable trailer coupling for pulling trailers with ball head) The permissible tongue weight is 80 kg for M1 approval/100 kg for N1/N2 approval.
Q22	Fixed ball head trailer coupling (Fixed trailer coupling with installation dimension increased by 50 mm) The permissible tongue weight is 80 kg for M1 approval/100 kg for N1/N2 approval.
QA0	Trailer coupling for trailer load of 2.8 t, removable ball head (Removable trailer coupling for pulling trailers with ball head) The permissible tongue weight is 120 kg.
QA7	Trailer coupling for higher towing capacity 3.5 t Attachment of a rigid ball coupling with a maximum permissible braked trailer load of 3.5 t for the 5 t weight variant; the permissible tongue weight is 140 kg.
QA8	Trailer coupling for higher trailer load 2.8 t/3.0 t Attachment of a rigid ball coupling with a maximum permissible braked trailer load of 2.8 t for the 3.5 t weight variant, and a maximum of 3.0 t for the 5 t weight variant The permissible tongue weight is 120 kg.

! NOTE

In order to install a trailer coupling, reinforcements in the left and right rear longitudinal frame members are absolutely essential.

These are always installed as standard at the factory with code Q11 on model series 907 for closed and open models. For other ex-factory special equipment, see table (→ page 193).

If it is not planned to install a trailer coupling, the omission of this reinforcement on closed model designations can be ordered under code QW1. Code QW1 is not available for open model designations.

! NOTE

Never attach a trailer coupling to the end crossmember of the frame.

! NOTE

For the retrofitting of a trailer coupling, it is essential to take into account the special equipment "Pre-installation for trailer coupling" (code QA3 or QA5) in the vehicle configuration.

With this special equipment, required adaptations on the parts level and entries in the vehicle documentation (identification plate and CoC document) are carried out.

The use of the special equipment "Electrics for trailer socket" (code E57) is also required to ensure the wiring harness is installed in the vehicle ex factory, specifically for the trailer coupling.

After a trailer coupling has been retrofitted, the code 071 "Third-party accessory trailer hitch" must be entered in the vehicle data card. SCN coding must then be performed for the trailer coupling control unit.

SCN codings of additional control units are also required in accordance with the commissioning specifications, See 8.19 Commissioning of control units (→ Page 387).

! WARNING

If the retrofitting of a trailer coupling is impermissible and you nevertheless attach a trailer coupling or other components, the component parts affected by this attachment may fail and result in the trailer or the other components becoming detached from the vehicle.

There is risk of an accident and danger to life and limb!

Only retrofit a trailer coupling if this is permitted.

- For vehicles with engine OM654 and a planned permissible gross combination mass greater than 5.55 t, observe the note in 3.3 Selecting the basic vehicle (→ page 53) regarding code QA5/QA3.
- It is recommended to only use trailer couplings that have been approved by Mercedes-Benz or equivalently safe trailer couplings at the specifically intended mounting points on the body-in-white (rear longitudinal frame member) (→ page 403).
- Make sure that there is access to the spare wheel on trailer couplings with a non-removable ball head (on a fully laden vehicle in particular).
- The trailer coupling attachment and clearance dimensions must comply with the regulations in the countries: Those of UN R 55 in countries that apply EU/UNECE regulations, in addition to DIN 74050 in Germany.
- If there are any deviations from the German Accident Prevention Regulations (UVV), a Technical Certificate must be requested in Germany from the Berufsgenossenschaft Verkehrswirtschaft Post-Logistik Telekommunikation (BG Verkehr):

<https://www.bg-verkehr.de>



- For information on the relationships between trailer load, overhang, and Trailer Stability Assist (TSA), See 4.3.5 Overhang (→ Page 88).

Trailer coupling dimensions

The size of the trailer coupling is determined using the drawbar ratio.

$$D = g \times \frac{m_k \times m_a}{m_k + m_a} \quad (kN)$$

D = Drawbar ratio

m_k = Permissible gross mass of towing vehicle (motor vehicle) in t

m_a = Permissible gross mass of trailer in t

g = 9.81 m/s²

In order to allow the trailer to be exchanged when used in international transport, the clearance between the center of the trailer coupling and the end of the towing vehicle must be no more than 300 mm (in Germany according to DIN 74050).

Clearance dimensions for trailer coupling

Take into account the country-specific clearance dimensions: in countries that apply EU/UNECE regulations, these are as per UN R 55.

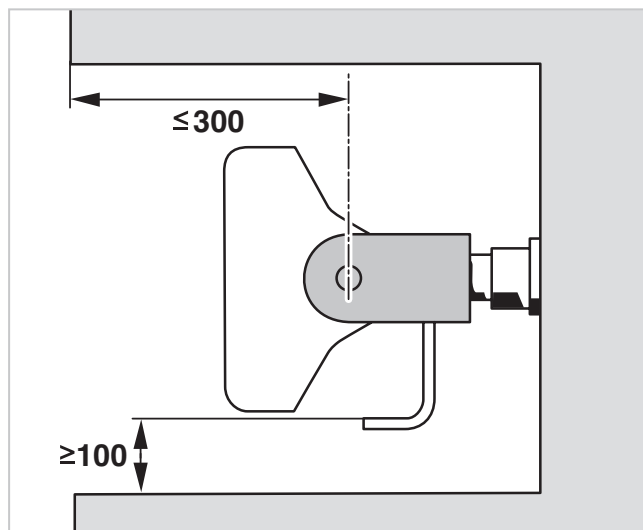
The height of the trailer coupling above the ground must be 350–420 mm when the vehicle is laden to the permissible gross mass.

Jaw

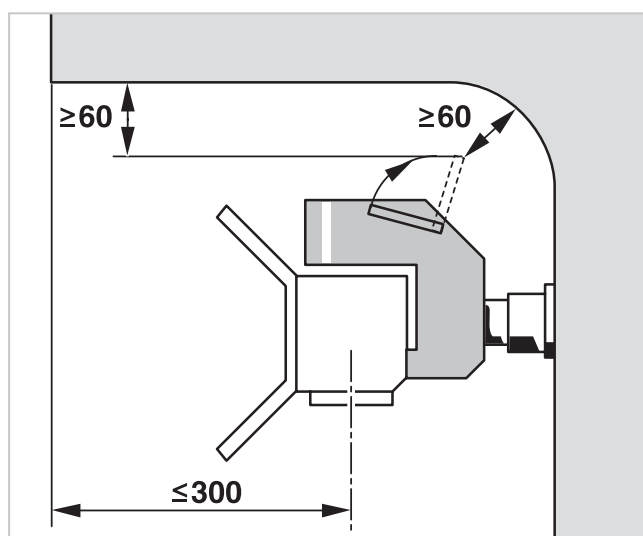
The distance from the center of the coupling bolt of the trailer coupling to the end of the body must not be more than 300 mm. Sufficient clearance must be maintained.

Reliable operation of the coupling must not be impaired.

Do not attach a jaw-type coupling to the front.



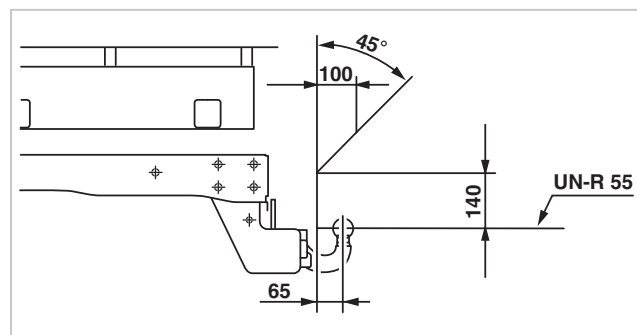
Top view of jaw-type coupling



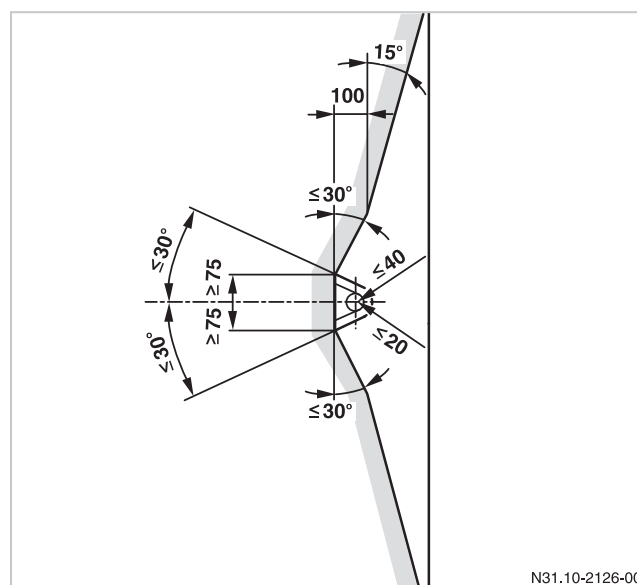
Side view of jaw-type coupling

Trailer coupling

Comply with the specified clearances.



Side view of trailer coupling



Top view of trailer coupling

⚠ WARNING

If the towing vehicle is unladen, only an unladen trailer may be towed. Otherwise, the vehicle may become unstable.

There is risk of an accident and danger to life and limb!

! NOTE

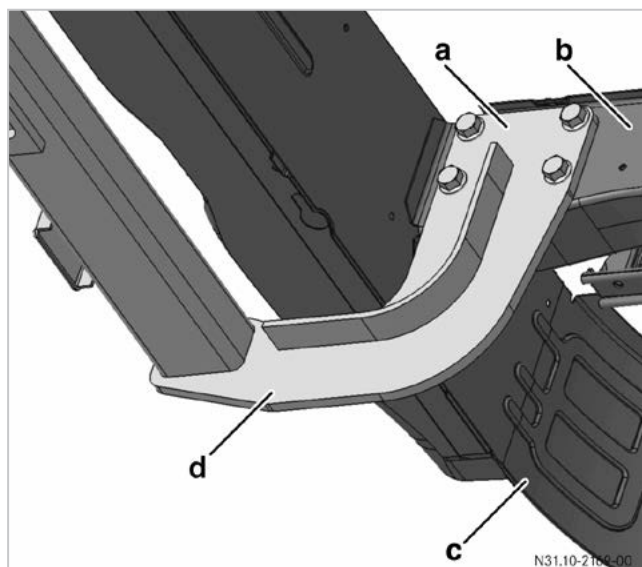
If trailer couplings have removable ball heads, supply an operator's manual in the vehicle that refers to the special features and operation of the coupling.

Trailer coupling attachment

Only fasten trailer couplings/trailer brackets to the specifically intended mounting points on the body-in-white (rear longitudinal frame member) (→ page 403).

- Do not attach these to the underride guard.
- Modifications to the underride guard must be coordinated with the Technical Service and/or with the relevant type-approval authority. The strength or the bending resistance must not be impaired.
- If the frame needs extending, fit spacer bushings to the frame for reinforcement in order to attach the trailer bracket or the end crossmember (→ page 115). This may lead to a reduction in the trailer loads and tongue weights.

Hole patterns with dimensions for attaching trailer couplings are available under 10.2 Trailer coupling hole patterns (→ page 403).



Interior view

- a Attachment of assembly stand at longitudinal frame member
- b Lower flange of longitudinal frame member
- c Frame end crossmember
- d Assembly stand for trailer coupling

Pre-installation of electrical system for trailer socket (code E57)

When special equipment code E57 is ordered, electric lines are present in the vehicle that can be used to retrofit a trailer socket.

For this purpose, a connecting point for the electrics of the trailer socket is provided at the rear of the vehicle ex works.

To activate trailer operation, a trailer control unit (see note), a trailer socket and the trailer coupling must be retrofitted to the vehicle.

! NOTE

The trailer control unit must be approved by Mercedes-Benz AG.

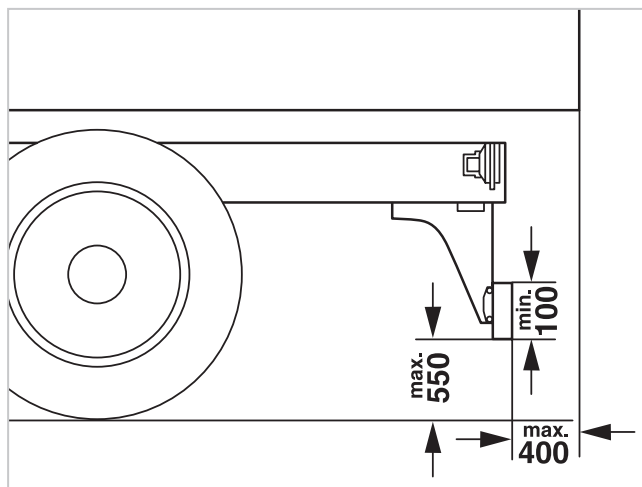
Further information on this can be obtained from your Mercedes-Benz Service Partner.

After this, enter code O71 (External accessories for trailer hitch) in the vehicle data record and carry out SCN recoding in all relevant control units in the vehicle as per the documentation in the Workshop Information System, See 8.19 Commissioning of control units (→ Page 387).

6.6.8 Underride guard

Rear underride guard

The rear underride guard fitted ex factory (except on panel vans and crewbuses) complies with UN R 58.



Side view of the underride guard location

Do not make modifications to the underride guard.

If modifications are nevertheless deemed unavoidable, these must first be coordinated with the Technical Service and/or with the relevant type-approval authority. In the countries that use UNECE regulations, UN R 58 stipulates that an underride guard must be fitted in the vehicle's rear area when, with the vehicle in its unladen condition, the distance between the roadway and the chassis or main parts of the body in the area between the insides of the tires (minus 100 mm per side) is more than 550 mm.

The bodybuilder manufacturer must clarify any potential exceptions to this specification for special purpose vehicles with the relevant national registration authority.

- ▶ The prescribed underride guard must be implemented by the bodybuilder manufacturer in accordance with UN R 58.
- ▶ Attach the underride guard as close to the end of the vehicle as possible.

Dimensions

- Distance between the roadway and the underride guard (unladen vehicle) max. 550 mm
- Width:
 - Maximum = Width of the rear axle (outer tire edges)
 - Minimum = Width of the rear axle minus 100 mm on each side; the widest axle takes precedence.
- Crossmember section height at least 100 mm
- Edge radius at least 2.5 mm

Modifications to the underride guard

If the underride guard needs to be repositioned due to the overhang extension, the attachment must be the same as that of the original vehicle.

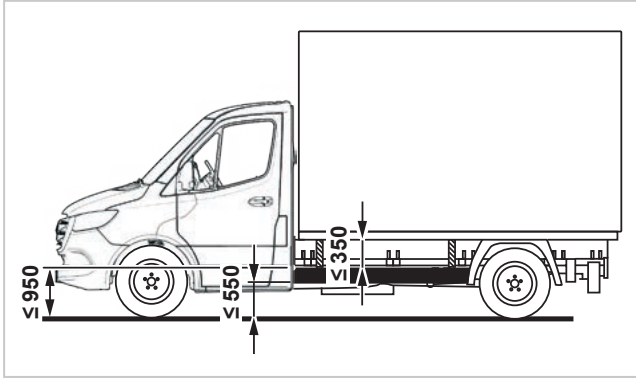
If modifications are required to the underride guard due to attachments/installations (e.g. liftgate), the strength and bending resistance of the underride guard must not be modified.

! NOTE

Observe all national laws, directives, and registration regulations.

Reinforcements are required to fasten an underride guard to the vehicle frame, See 6.2.2 Attachments on the frame (→ Page 117).

Side underride guards



Side view of the underride guard location

According to UN R 73, a side underride guard is stipulated for vehicles with a permissible gross mass in excess of 3.5 t.

The bodybuilder manufacturer must clarify any potential exceptions to this specification in relation to special purpose vehicles with the relevant national registration authority.

Component parts may be mounted within the side underride guards, e.g. battery box, compressed-air reservoir, fuel tank, lamps, reflectors, spare wheels and tool box, provided that the specified clearances are maintained.

Brake, air, or hydraulic lines and other parts must not be fastened to the side underride guard.

The function of and access to all major assemblies on the vehicle must not be impaired.

The side underride guards are secured to the platform at the factory (code C57).

For retrofitting

- The dimensions specified in the illustration must not be exceeded.
- The underride guards must be fitted in accordance with UN R 73.
- The underride guards must extend continuously from the front to the rear wherever possible.

It is permissible for adjacent parts to overlap. The overlapping edge must point to the rear or downward. The maximum permissible width of any gap between segments is 25 mm, provided that the rear part does not protrude significantly beyond the front part.

The side underride guard may consist of a continuous level surface. The outer surface must be smooth and generally flat. The side underride guard parts must be rigid and permanently fixed into place. They must be made of metal and/or another suitable material. The clearance of the outer surface of the underride guard to the outer edge of the vehicle must be a maximum of 120 mm. The edge radius must be at least 2.5 mm.

General information

! NOTE

This chapter contains information on the different designs of bodies. The specifications are based partially on existing vehicle equipment and partially on empirical values.

Note that the bodybuilder manufacturer assumes overall responsibility for the design and implementation of the bodybuilder-manufacturer-specific vehicle body.

All specifications in this Body/Equipment Mounting Directive must be observed.

7.1 Assembly frame

All bodies require a continuous assembly frame or a substructure that acts as a continuous assembly frame in order to ensure a flawless connection between the chassis and body; See 7.1.5 Assembly frame as floor assembly (→ Page 208) and 7.2 Self-supporting bodies (→ page 208).

The attachment to the frame must run along the frame using the existing mounting supports from the factory, see 7.1.2 Design (→ page 202) and 7.1.4 Attachment to the frame (→ page 204). Exception: On vehicles with a dropped frame, model designation 907.x5x (7.1.2 Design (→ page 202)), the assembly frame longitudinal members can run continuously in a straight line.

7.1.1 Material quality, general

Material quality of specified assembly frames made of steel:

- Assembly frame mounted with mounting supports (non-positive) = CR240LA or S235JRG2.
- For CR240LA or S235JRG2 steels complying with the DIN EN standard, analogous materials complying with the US SAE/ASTM J403/J412/J413 standards, the Japanese JIS G3445 standards, or the UK BS 970 standards can be used.

Material	Yield strength [N/mm ²]	Tensile strength [N/mm ²]
Steel	235–340	240–510

Minimum section modulus required for assembly frame $W_x^{1)}$ [cm³]

Version	Platform/ box body	Dump truck/elevating work platform	Loading crane
All weight variants	17 ²⁾	30	40

- 1) The minimum section modulus required for assembly frames is relevant to the material characteristics specified in the table of materials above, and each individual assembly frame longitudinal member must have this section modulus.
- 2) Up to the maximum standard wheelbase; above that +10%

- When using an assembly frame made of aluminum, for example, it must have at least the bending resistance ($E \times I$) of a steel assembly frame.

Guideline values for modulus of elasticity [N/mm²]:

- Aluminum: 70000
- Steel: 210000

Attachment of a liftgate See 6.6.6 Liftgate (cargo liftgate) (→ Page 190).

- i** Observe any differing specifications that may apply in relation to this chapter, see 7.5 Platform bodies (→ page 214) and 7.9 Dumper bodies (→ page 216).

7.1.2 Design

General information

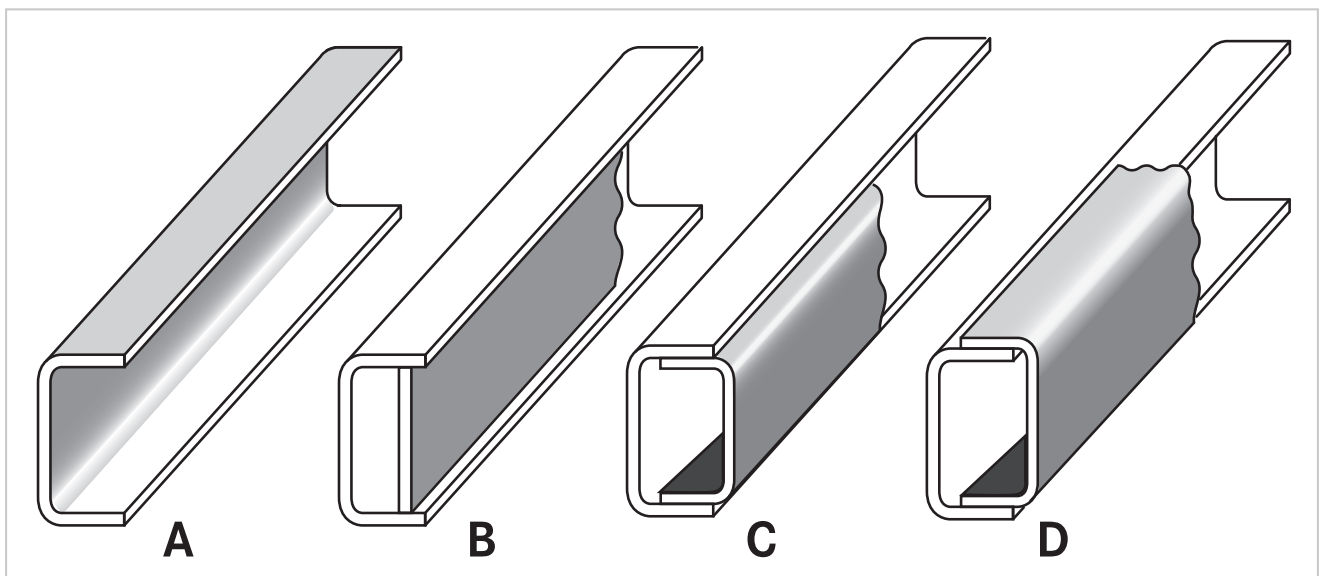
- ▶ On model designations 907.X2X, 907.X3X, and 907.X4X, arrange the assembly frame longitudinal members above the straight vehicle frame longitudinal members.
- ▶ On model designation 907.X5X, the assembly frame longitudinal members can be straight through, see note on offset frames at the end of this chapter.
- ▶ At the positions of the double body consoles (see 2 in illustration 7.1.4 Attachment to the frame (→ page 204)), the assembly frame crossmembers must be positioned above the vehicle frame crossmembers.
- ▶ For the additional assembly frame crossmembers, a location above the chassis frame crossmembers is recommended.

- ▶ Extend the assembly frame longitudinal members as far toward the front of the vehicle as possible, to reinforce the point behind the cab which is critical with regard to bending stress, thereby preventing vibration problems.
- ▶ The body must have a torsion-free attachment to the mounting supports on the longitudinal frame member.
- ▶ Position the vehicle on a flat, horizontal surface before mounting the body.

If very high assembly frame longitudinal members or low overall frame heights need to be achieved, the following non-positive connections are possible for U sections:

- Closed as a box
- Nested
- Overlapping

This increases the section modulus and torsional stability.



Frame sections

- A Open U-section
- B Closed U-section
- C Nested U-section
- D Overlapping U-section

Assembly frame with dropped frame

On vehicles with a dropped frame (permissible gross mass ≥ 4.6 t), the assembly frame longitudinal members can run continuously in a straight line.

7.1.3 Section dimensions and dimensioning

For the longitudinal members, use flanged U sections or commercially available U sections for vehicle construction (not rolled steel sections). Box sections are also permitted as longitudinal member sections.

The dimensions of the assembly frame longitudinal members are a function of the section modulus (W_x) required for the body and the chassis (→ page 201).

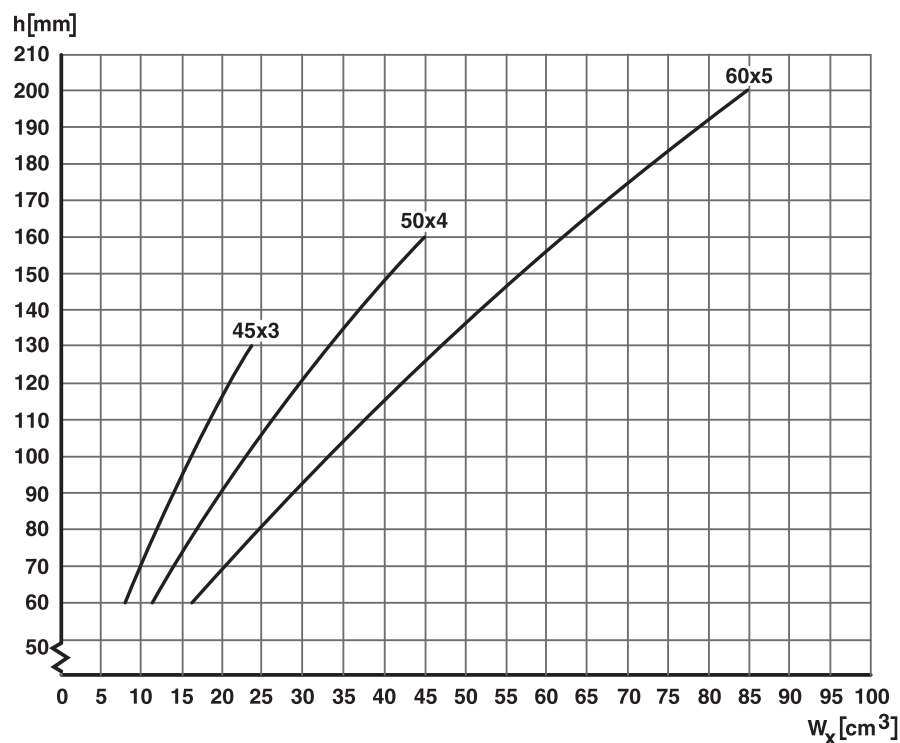
The specified section moduli and section dimensions refer to longitudinal frame members subjected to identical loads on both sides.

Please refer to the table below for the section dimensions of assembly frame longitudinal members (open section).

The assembly frame and the chassis should have approximately the same flange width.

! NOTE

If more than one body is mounted on the same chassis (e.g. platform and liftgate), the larger of the specified section moduli must be taken to determine the assembly frame.



Dimensioning of the assembly frame longitudinal member

h Section height in mm

W_x Section modulus in cm³

7.1.4 Attachment to the frame

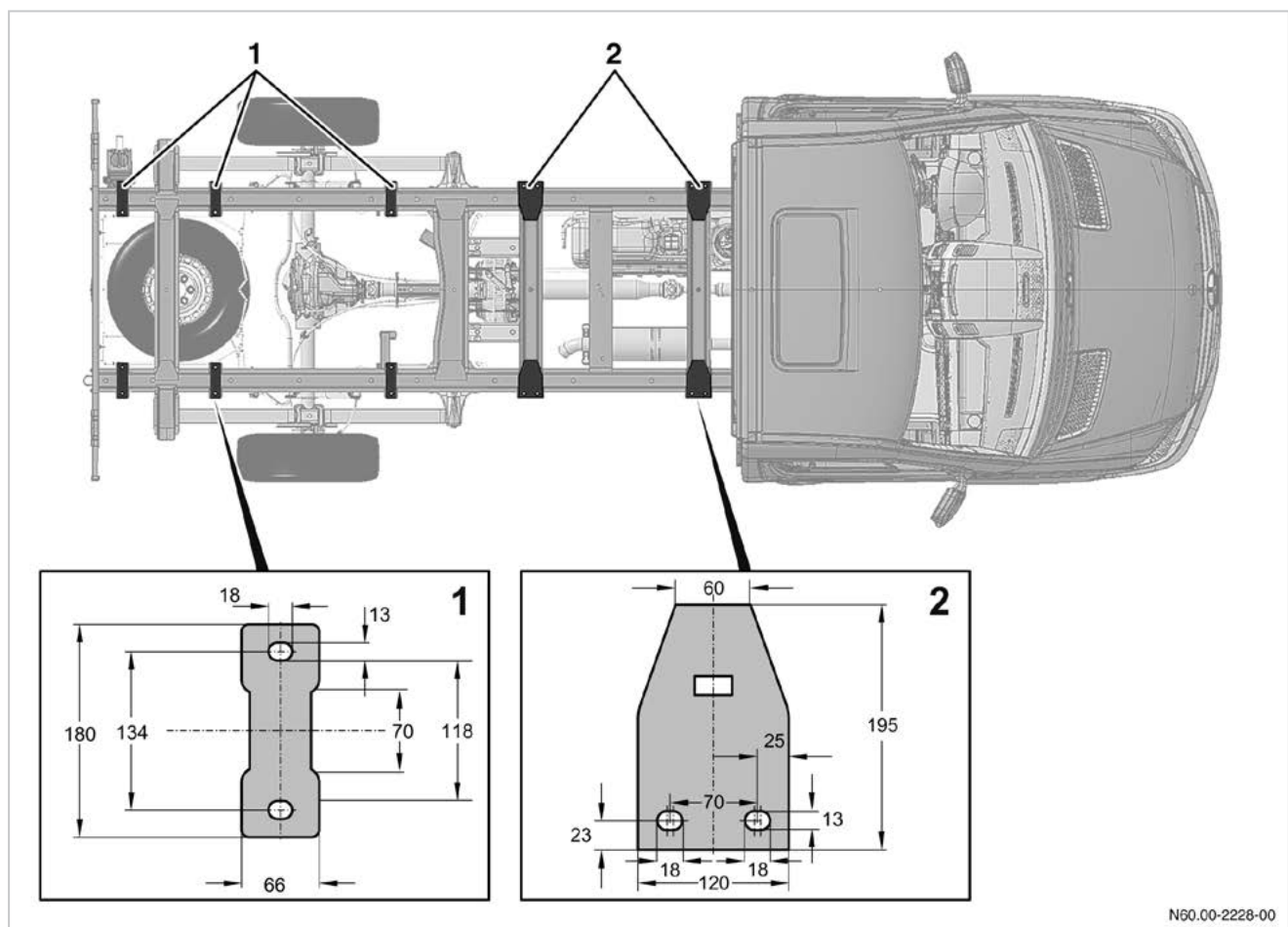
As a minimum, use all the factory-installed mounting supports for attaching bodies to the vehicle frame. The mounting supports are located on the longitudinal frame members and additional mounting supports may be fitted to the longitudinal frame members as required. Use all the mounting supports, in particular those at the rear end of the frame, to attach a liftgate (cargo liftgate) and the body.

! NOTE

The minimum distance between the body and the cab must be 50 mm.

The bodies must rest on the mounting supports and are also permitted to rest on the crossmembers. It is recommended that the body should not rest on the end crossmember. If prefabricated assembly frames are used, take into consideration the manufacturing tolerances of the vehicle frame width (maximum +6/-3 mm).

i The positions for the model-designation-dependent mounting supports can be found in the 2D drawings (offer drawings) (→ page 24).



N60.00-2228-00

Designs of attachment points on the frame

Additional mounting supports

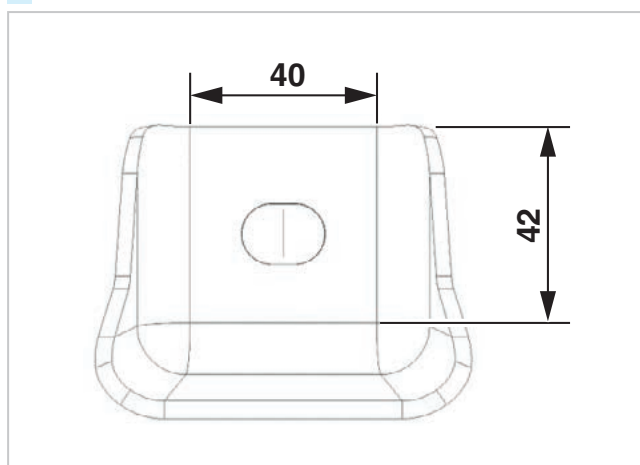
If it is necessary to fit additional body consoles, make sure that you comply with the welding specifications (→ page 101).

- Plug welding is only permissible in the vertical webs of the longitudinal frame member.
- Do not perform any welding work in bending radii.

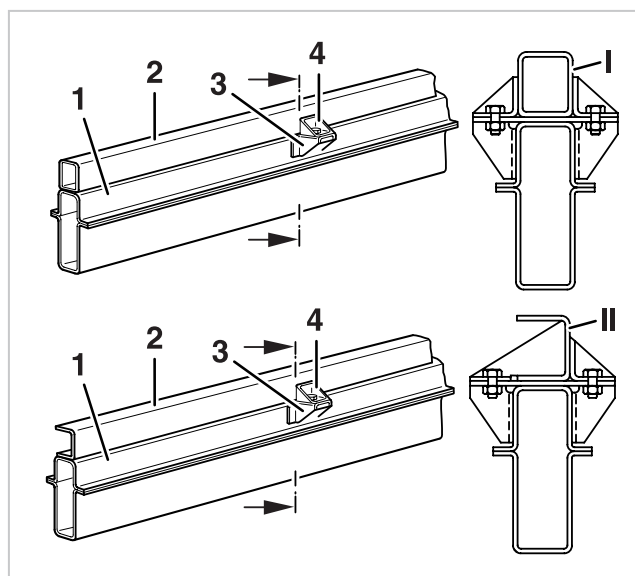
The mounting supports must be attached using two screws for each mounting support.

Attachment of the mounting supports

- ▶ For rigid bodies in particular, to ensure optimal transmission of force into the mounting supports on the vehicle (see illustration "Designs of attachment points on the frame") as well as at the longitudinal member, make sure that the mounting supports of the subframe rest against a flat support. If possible, the mounting supports of the subframe should have identical dimensions to the body consoles on the vehicle.
- ▶ Distribute the application of force at screw connections across the maximum possible surface area by using washers with a diameter of approx. 35–40 mm or square shims with a length along each edge of approx. 35–40 mm.

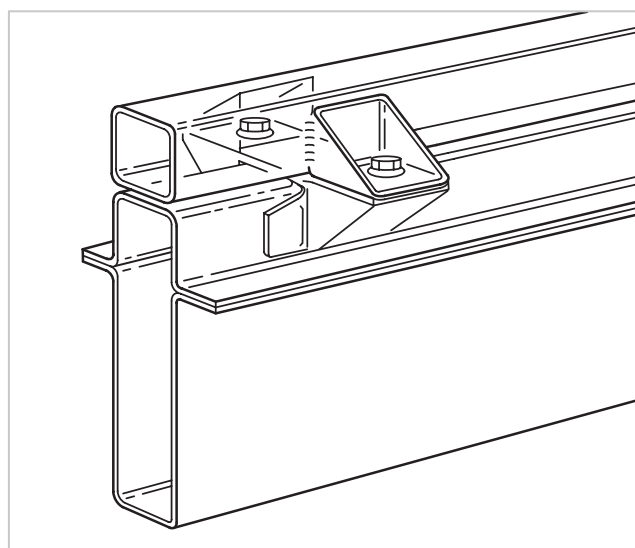


Mounting supports (dimensions in mm)

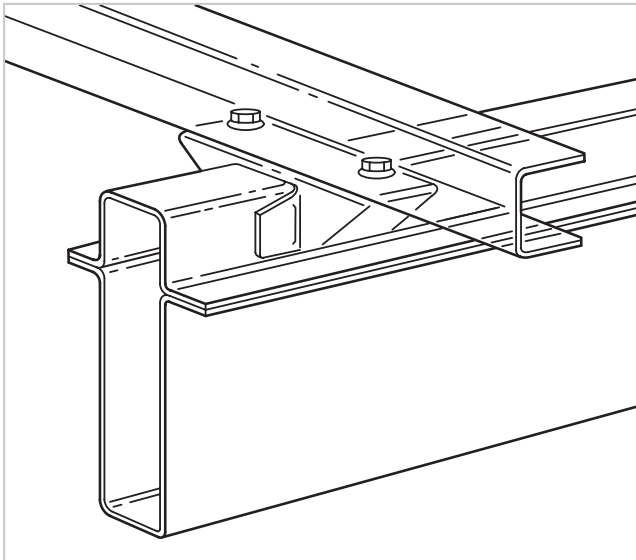


Example of a mounting support design

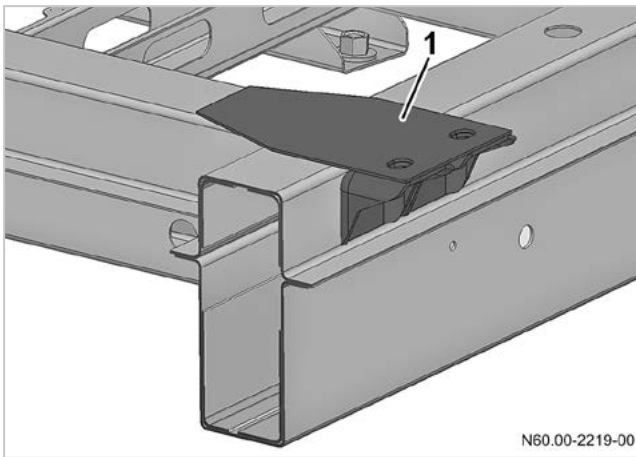
- I Box section
- II U-section
- 1 Vehicle frame
- 2 Assembly frame
- 3 Standard mounting support
- 4 Mounting support



Attachment to a longitudinal member



Attachment to a crossmember



Attachment to a crossmember

1 Mounting support

Select the number of attachments to ensure adequate transfer of all longitudinal and lateral forces.

Correct attachment is a decisive factor for:

- vehicle handling and operating safety
- the durability of the vehicle frame and the body.

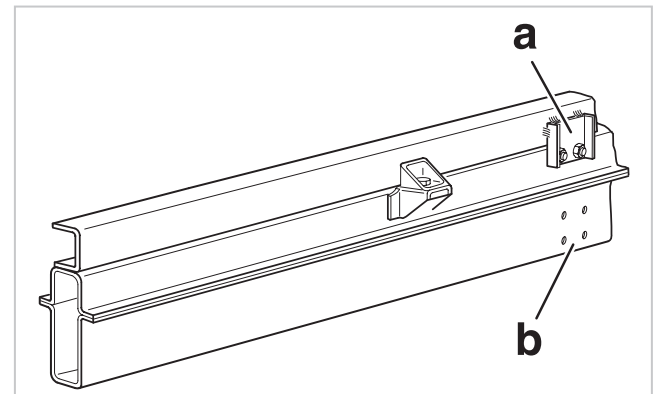
Rigid connection

i Also note 7.10 Semitrailer trucks (→ page 218).

For a rigid connection, the assembly frame longitudinal members must be secured in both longitudinal and transverse directions. This will allow movement of the assembly frame longitudinal member only under specific conditions.

The body can be secured to the sides of the upper flanges on the longitudinal frame member. Spacer bushings are required for reinforcement; these must be welded to the frame.

For rigid connections, a double support per longitudinal frame member is required; see the following illustration. The rigid connection is required e.g. for "Loading crane at frame end" and is recommended for "Loading crane behind the cab".

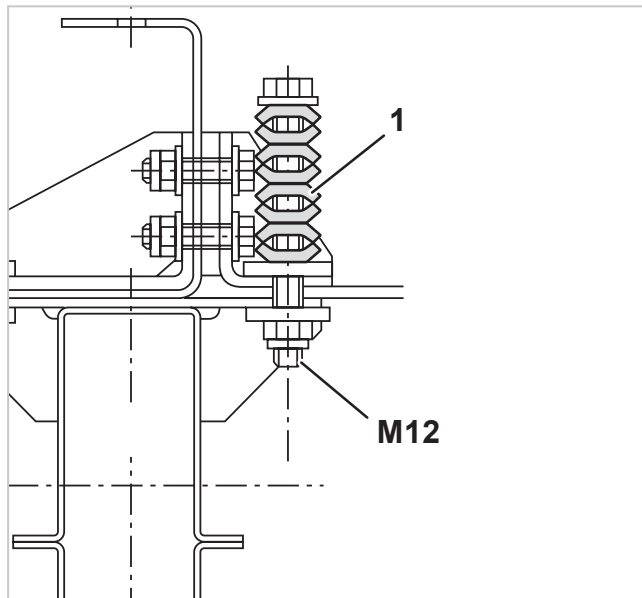


Double support (rigid connection)

- a Rigid connection at frame end
- b Standard bores at frame end

Elastic connection

In the case of extremely rigid bodies (e.g. box bodies), it is recommended to provide the front connecting points (in the area behind the cab) with elastic attachments and connections (e.g. by shimming disk springs or elastomer bearings).



Elastic connection (schematic)

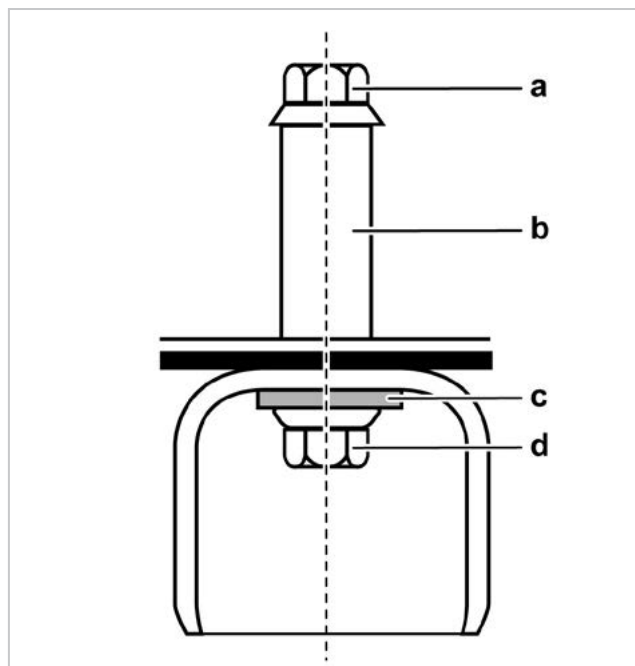
1 Disk spring



Elastomer bearings/bushings

Threaded connections locked to prevent loosening

On rigid bodies (e.g. self-supporting bodies and box bodies with assembly frame, or elevating work platforms), provide threaded connections locked to prevent loosening with spacer sleeves at the first and second mounting supports. The dimensions of the spacer sleeves must be adequate to ensure that they cannot become deformed. Also use these stretch bolt connections locked to prevent loosening for attaching bodies such as refuse presses or closed box bodies ≥ 1000 mm long behind the cab in combination with a dumper bed or platform on the vehicle frame.



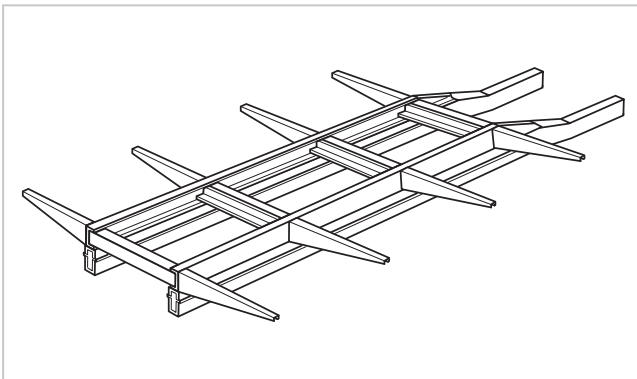
Suggested method for producing a threaded connection locked to prevent loosening

- a Screw with flange M12 $\times$ 90, strength 10.9
- b Spacer sleeve 22-13 $\times$ 50
- c Washer DIN 7349-13-ST
- d Nut with flange M12, strength 10.9

7.1.5 Assembly frame as floor assembly

An assembly frame with continuous longitudinal members is not required if the body floor assembly can take on the function of the assembly frame.

The longitudinal members can also be integrated in the body. If the assembly frame longitudinal members are intersected by the crossmembers, the connection between the longitudinal members and crossmembers must be rigid and resistant to torsion and bending.

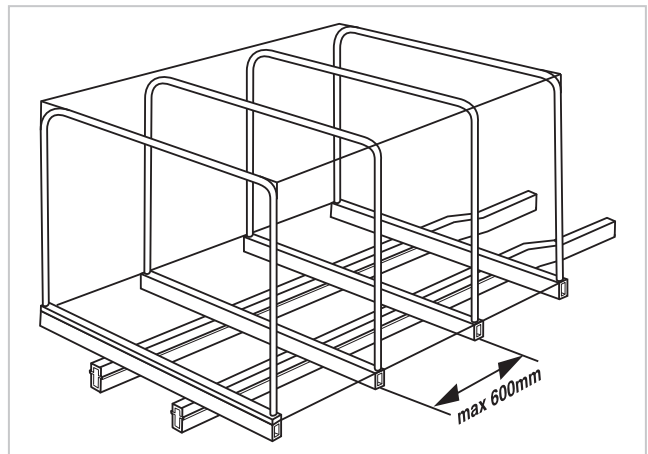


Example of a floor assembly design

7.2 Self-supporting bodies

An assembly frame with continuous longitudinal members is not required if the body's floor assembly can assume the functions of the assembly frame.

Self-supporting bodies must comply with the properties of the specified assembly frame. The body's floor assembly must have equivalent rigidity and a section modulus similar to the assembly frame. Attach the floor assembly of the self-supporting body on the vehicle frame in line with the attachment of a body frame / assembly frame using stretch bolt connections locked to prevent loosening (→ page 207).



Example of a body design

7.3 Modifications to the interior

7.3.1 Retrofitting seats

! NOTE

All seats that deviate from the standard seating must satisfy the requirements in chapters 6.4.1 General information (→ page 151), 6.4.2 Safety equipment (→ page 152), and 6.4.3 Seats (→ page 160).

⚠ WARNING

If seats are subsequently mounted on thin-walled sheet metal parts of the vehicle without reinforcement, the seats could become detached from their anchoring points in the event of an accident. Passengers cannot be protected as intended.

There is danger to life and limb!

Retrofitted seats, their seat fastenings and seat belts must therefore always be tested in accordance with legal specifications and a sufficient strength must be proven, See 6.4.3 Seats (→ Page 160).

! NOTE

Retrofitting a seat on the front passenger side or that fastens to the partition (including folding seat or another seat option) is not permissible in vehicles with registration without front passenger seat (code Z0S).

This affects vehicles with special equipment "Removal of front passenger seat" (code S91) in combination with "Removal of front passenger airbag" (code SW3), as well as the consequent "Removal of front passenger side seat belt" (code SW1) ex factory, delivered in countries where GSR II (General Safety Regulation) is valid.

! NOTE

If the vehicle is converted for transporting persons with M2 or M3 registration, in addition to the legal specifications in 6.4.3 Seats (→ page 160), UN R 80 (seats and seat anchorages in buses) must be fulfilled as per the version valid at the time the vehicle is brought into service, with corresponding evidence of this submitted upon request.

⚠ WARNING

If seats other than those fitted at the factory are installed in conjunction with seat belts available from the factory, only seat belt buckles that are compatible with the seat belt tongues of the factory-supplied seat belts may be used. Otherwise, the seat belt cannot be locked in the seat belt buckle as intended. Passengers cannot be sufficiently protected.

There is danger to life and limb!

Only use seat belt tongues that are approved for the factory-fitted seat belts.

Only the component parts of the series production supplier must be used for the installation of seat belts and seat belt buckles:

Contact details

Address	Autoliv B.V. & Co. KG Postfach 109 25333 Elmshorn, Germany
Telephone	+49 (0)41 21 797-0

Ensure that all regulations relevant to approval (e.g. seat belt buckle position) are observed when fitting seat belts and seat belt buckles other than those available from the factory.

Rear seats in the passenger compartment and cargo area

For retrofitting rear seats to panel vans in model series 907, it is recommended that the body-in-white floor assembly for crewbus vehicles be ordered from the factory as special equipment (code V40).

The number of mounting options for the crewbus floor assembly depends on the model series and the equipment or registration variants (the seat shells for the bench seats are not included in the delivery package).

! NOTE

Adherence to the legal specifications for the transport of persons in the rear passenger compartment must be independently ensured by the bodybuilder manufacturer, e.g. as per UNECE regulations, specifications as per GSR II (General Safety Regulation) and/or other country-specific specifications.

With the body-in-white floor assembly for crewbus vehicles (code V40), for camper vans (code X2R), and for special-purpose vehicles (code ZB7), additional measures necessary for this purpose are installed, among other things, on the sliding door to the load compartment.

! NOTE

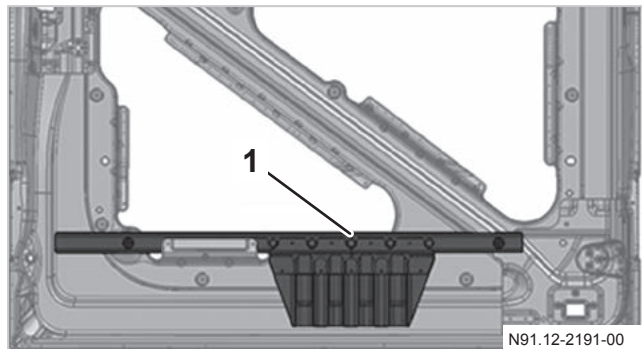
In all cases deviating from this, it is essential to order the additional measures on the sliding door to the load compartment with the special equipment "Load compartment sliding door for bodybuilder manufacturer conversion of passengers behind B-pillar" (code T01) for installation ex works.

In this regard, in countries where GSR II applies, note that for vehicles with a partition at the B-pillar ex factory (code D50, D51 and D64), the equipment code T01 cannot be ordered due to the additional measure "catch bearing". Retrofitting of seats in the rear passenger compartment is not permissible on such vehicles. Alternatively, the special equipment "Pre-installation for retrofitting a partition" (code D62) can be selected and a bodybuilder-manufacturer-specific partition can be retrofitted; See 7.4 Modifications to the panel van (→ Page 212).

i For further information on special equipment, contact your Mercedes-Benz Service Partner or the Technical Consultancy (→ page 18).

Reinforcement of sliding door to load compartment (LST) with rear seats

For the model series 907, the impact cushion reinforcement is added automatically as required for vehicles with a long sliding door to the load compartment (body length A2 and over) and vehicles with crewbus floor assembly (code V40) for countries with appropriate legislation that requires the installation of this reinforcement.



Right load compartment sliding door from outside without paneling

1 Assembly part "impact protection reinforcement"

The reinforcement is required when installing rear seats in the panel van and crewbus in the following countries:

- Mexico, USA, Canada
- GCC Arab States of the Gulf (Kuwait, Bahrain, Saudi Arabia, Qatar, United Arab Emirates, Oman)
GCC = Gulf Cooperation Council

If the bodybuilder manufacturer retrofits rear seating for these countries, ensure that the impact cushion reinforcement is installed.

In combination with the "Crewbus floor assembly" special equipment (code V40), the impact cushion reinforcement is already installed ex factory.

For retrofitting, the assembly part is available from your Mercedes-Benz Service partner under the following item numbers:

- A 910 730 47 00 Left reinforcement
- A 910 730 48 00 Right reinforcement

 Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

Use the fasteners and tightening torques documented in the Workshop Information System (→ page 26) for installation.

NOTE

Observe all national laws, directives, and registration regulations.

7.4 Modifications to the panel van

Floor assembly and side walls

On panel vans, the body and the chassis form a self-supporting unit. If body parts are modified or fitted, they must only be welded if a bonded connection is not possible.

For this reason, windows, roof hatches and vent openings must be mounted in a sturdy frame. Connect this frame with a frictional connection to other body elements.

Rear panel of cab

If an opening is made in the cab rear panel, surround it with a sectional frame.

Reinforce the remaining struts and spars by additional gusset plates and connect them to the sectional frame (e.g. by bonding).

Also note 6.2.6 Modifications to the cab (→ page 122).

Partitions

To fulfill country-specific regulations for proper load securing, vehicles with N approval for goods transportation should be equipped with a partition tested in accordance with ISO 27956. ISO 27955 applies for vehicles with M1 approval.

A vehicle configuration fitted with a partition at the factory is strongly recommended; see the following overview of available special equipment.

Special equipment for partitions

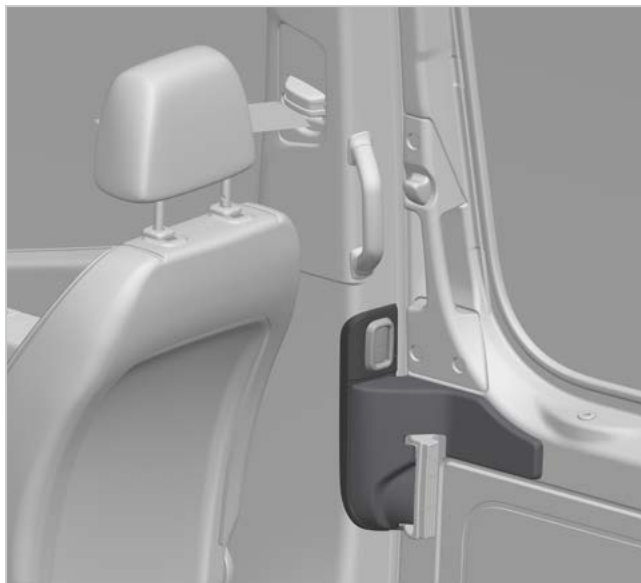
Code	Description
D50	Full-width partition wall (standard)
D51	Full-width partition with window
D56	Full-width partition wall at C-pillar
D62	Pre-installation for retrofitting a partition
D64	Partition wall with sliding door
D93	Omission of partition

For the installation of a partition specific to a body-builder manufacturer, select a vehicle configuration with the special equipment "Pre-installation for retrofitting a partition" (code D62).

Also note the new trim parts as of model year 2024 for the additional measure "catch bearing" (front catch) of the load compartment sliding door in the center area of the sliding door lock at the B-pillar; see following illustration.

A bodybuilder-manufacturer-specific partition or other parts planned for the installation must be adapted to the new component part contours in this area.

- ❗ For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).



Additional measure "catch bearing" (schematic)

For retrofits, modifications or any kind of attachments to the partition, ensure that the previously mentioned specifications are complied with and submit corresponding evidence of this upon request. It is recommended to have this documented evidence confirmed by an authorized testing organization.

If the partition is removed, the vehicle will no longer comply with the previously mentioned specifications. In this case, as a requirement for registration, at least the tie-down eyes must be installed.

If the partition/divider is modified or has parts attached to it, or if it is retrofitted or removed, the functioning of safety equipment (e.g. the deployment areas of airbags) must not be affected, See 6.4.2 Safety equipment (→ Page 152).

In these cases, the bodybuilder manufacturer must ensure that the product satisfies the requirements of a safe product; if necessary, by means of other measures as the bodybuilder manufacturer sees fit. In this respect, the ISO standard represents only the technical minimum standard.

For the attachment of seats at the partition (e.g. folding seats), observe the specifications in 7.3.1 Retrofitting seats (→ page 209).

- ❗ Further information on special equipment is available from your Mercedes-Benz Service Partner, from the Technical Consultancy (→ page 18), or under 3.12 Special equipment (→ page 74).

Roof

You can find information about modifications to the roof in 6.2.11 Panel van and crewbus roof (→ page 128).

7.5 Platform bodies

A platform with code P02 is available ex works as special equipment for chassis.

Observe the following specifications for body-manufacturer-specific platform bodies.

To ensure uniform loading of the vehicle frame, the body must be attached to the vehicle frame by means of an assembly frame (U section longitudinal members) (→ page 200).

As a minimum, use all the factory-installed mounting supports for attaching platforms to the vehicle frame.

If the standard platform is subjected to point loads (e.g. for the transportation of cable drums, coils, etc.), the substructure and the platform floor must be reinforced to support the load.

Before start of installation:

- ▶ Weigh the chassis and establish body length.

Chassis with crewcab:

- ▶ If necessary, shorten the rear frame overhang to prevent the permissible rear axle load from being exceeded and to ensure that the minimum front axle load is maintained.

- ▶ Install reflectors on the body in accordance with the legal requirements as per UN R 48 (→ page 286).

- Section moduli (W_x) and material properties can be found under 7.1 Assembly frame (→ page 200).
- For the section dimensions of the assembly frame longitudinal members, see the diagram (→ page 203).

! NOTE

Where bodies include detachable parts that move independently, ensure that there is adequate clearance between the detachable parts and the basic vehicle; otherwise, they may collide with the basic vehicle, resulting in damage.

7.6 Box bodies

To ensure uniform loading of the vehicle frame, the body must be attached to the vehicle frame by means of an assembly frame (U section longitudinal members) (→ page 200). In the case of self-supporting box bodies, its function and the equivalent rigidity of the floor assembly of the box body must be guaranteed. Also note 7.1.5 Assembly frame as floor assembly (→ page 208) and 7.2 Self-supporting bodies (→ page 208).

At the least, use all the factory-installed mounting supports for the box bodies on the vehicle frame. Implement the specifications on mounting on the vehicle frame in 4.3.6 Attachment to the frame (→ page 89), 6.2.2 Attachments on the frame (→ page 117), and 7.1.4 Attachment to the frame (→ page 204).

Attach the floor assembly (7.1.5 Assembly frame as floor assembly (→ page 208)) and a self-supporting body (7.2 Self-supporting bodies (→ page 208)) to the vehicle frame in the same way as a body assembly frame is mounted on the vehicle frame.

On self-supporting bodies or box bodies with assembly frames, provide stretch bolt connections locked to prevent loosening and spacer sleeves behind the cab at the first and second mounting supports. The dimensions of the spacer sleeves must be adequate to ensure that they cannot become deformed (→ page 207).

For section moduli (W_x) and material properties, See 7.1 Assembly frame (→ Page 200).

! NOTE

The minimum distance between the cab and a separate body must be > 50 mm.

- ① For information on integral box bodies, See 7.17 Semi-integrated bodies and optional mounting of free-standing bodies (→ Page 259).

7.7 Refrigerated vehicles and temperature-controlled vehicles

Refer to the chapters:

- 6.5.1 Retrofitting an air conditioning system (→ page 166).
- 6.5.3 Power take-offs (→ page 169).
- 6.2.11 Panel van and crewbus roof (→ page 128).
- 8.4.6 Retrofitting of electrical devices (→ page 275).
- 6.4.4 Reducing interior noise (→ page 162).

With panel vans, ensure easy access to the components of the door mechanism (e.g. guide rails and hinges) so as not to hinder possible repair work; See 3.11 Maintenance and repair (→ Page 71).

! NOTE

On panel vans, the insulation increases the weight of the doors and therefore the load on the hinges, carriages and locking systems.

Observe the specifications for the maximum permissible additional weights on the doors, See 6.2.7 Side wall, windows, doors, and flaps (→ Page 123).

These requirements and information are also applicable for vehicles used to transport loads under constant conditions within the load compartment, e.g. with a constant temperature higher than 0 °C.

! NOTE

People working inside the vehicle may inadvertently become trapped by doors closing by themselves or being closed by other persons.

The bodies of refrigerated/temperature-controlled vehicles must be designed so that any persons shut inside the vehicle can open the door outward from the inside in any situation.

It is recommended to fit the doors of refrigerated bodies with an emergency release so that they can be opened easily from the inside at any time, even when locked.

! NOTE

Where refrigeration systems/components are used in combination with the refrigerant R1234yf, observe all applicable laws, specifications, and safety requirements.

In particular, ensure compliance with the safety regulations as per ISO 13043 at both the component and overall system level (FMEA, leak tightness requirements, concentration measurements, etc.).

In addition, observe the safety requirements with regard to inflammability and risk of combustibility in combination with R1234yf (maintain best-possible packaging or largest-possible distance to hot parts; if necessary, install additional devices to ensure the required safety).

If R1234yf is used, a specific information label with details of the quantity and combustion risk must be considered and installed in a visible location.

7.8 Vehicles for transportation of technical gases

Equip vehicles for transporting technical gases that have tanks and lines for liquefied gases with adequate thermal insulation from the rest of the vehicle. This is even more important for lines that are passed through the body or the body-in-white.

To allow venting of supercooled liquefied gases, a distance of at least three meters is required between the vent opening and the vehicle. If venting is necessary, make sure that the vent opening points away from the vehicle. Catch any escaping condensation in a suitable container.

7.9 Dumper bodies

To prepare the vehicle for being equipped with a dumper body, it is recommended to order the special equipment "Pre-installation for three-way dump truck" (code P57).

This pre-installation comprises a rocker switch with indicator lamp in the cockpit and the associated wiring harness with power supply for subsequently connecting the electrohydraulic drive system for a dump truck.

i Contact your Mercedes-Benz Sales Partner for further details about this special equipment.

For vehicles with dumper bodies, observe the country-specific guidelines and laws as well as accident prevention regulations (UVV) and general machinery directives; See 2.6 Accident prevention (→ Page 43).

On vehicles with automatic transmission, hydraulic units cannot be driven by the transmission power take-off; See 6.5.3 Power take-offs (→ Page 169).

Observe the permissible axle loads.

i Also observe the section "Side underride guards" (→ page 199) and 6.6.8 Underride guard (→ page 198).

Tilt support

- ▶ Position the rear tilt support as close to the rear axle as possible on three-way dumper and rear dumper bodies.
- ▶ The folded-down dropside must not strike the end of the frame, the lighting systems or the trailer coupling.
- ▶ Provide guide brackets for the front tilt supports, so that the tilt supports are guided when the dump body is lowered.

Safety devices

- ▶ Observe country-specific directives and laws.
- ▶ Install a support (fold-out support) to prevent the dump body from dropping.
- ▶ Secure operating devices against accidental operation.
- ▶ Connect a "dump body" indicator lamp as a visual warning when the dump body is not tilted all the way back (drive position).

Tilting press

- ▶ The press carrier is attached to crossmembers in the assembly frame.
- ▶ Position the crossmembers of the assembly frame and those of the chassis above each other where possible.
- ▶ On three-way dumper bodies, the application point of the tilting press must be in front of the center of gravity of the body and the payload.

Assembly frame

If chassis are provided with dumper bodies, the assembly frame must have the correct dimensions to support the high loads to which the vehicle will be subjected.

Observe the following points:

- ▶ Attach assembly frames to mounting supports as per 7.1.4 Attachment to the frame (→ page 204).
- ▶ Make sure that the steel longitudinal members and crossmembers have the correct dimensions
- ▶ Close off the rear area of the assembly frame to form a box and, if necessary, reinforce the assembly frame by installing a diagonal cross or by taking other appropriate measures.
- ▶ As a minimum, use all the factory-installed mounting supports for attaching dumper bodies to the vehicle frame.
- ▶ Vehicles with dumper bodies can only be used under normal operating conditions. If the vehicle is intended to be used in severe operating conditions, coordination with the Technical Consultancy is recommended (→ page 18).
- ▶ For section moduli W_x and material properties, See 7.1 Assembly frame (→ Page 200).

7.10 Semitrailer trucks

Chassis may generally be converted into semitrailer truck vehicles provided that this conversion complies with national regulations and laws. The registration eligibility of a Sprinter converted to a tractor/semitrailer combination must be assured by the bodybuilder manufacturer carrying out the work according to the country-specific registration requirements.

Obtain and comply with the specific state and national vehicle registration requirements with regard to the topics "Electronic Stability Program (ESP®)" and "Tire pressure loss warning system".

A functioning electronic stability program is required by law in the EU for all classes of vehicle.

Operation of the Sprinter as a semitrailer truck with functioning Electronic Stability Program ESP® is possible under the following conditions:

1. The semitrailer used must be equipped with an anti-lock braking system (ABS) as a minimum. Combining the Sprinter with a semitrailer without ABS is not permitted.
2. The brake systems of the semitrailer truck and semitrailer, the compressed air supply, and the compressed air reservoir must be designed in accordance with UN R 13, making sure to observe Annex 10 of UN R 13 in particular.
3. When using conventional trailer control valves, **without** additional sensing of the actual wheel braking forces for autonomous brake applications, the following specifications apply to the special equipment of the driving assistance systems:
 - a) Code restrictions that are not permitted to be selected:
 - Active Brake Assist (code BA3)
 - Active Distance Assist DISTRONIC (code ET4)
 - Cruise control (code MS1)
 - b) Code requirements that must be selected:
 - Omission of Active Brake Assist (code BW6)
4. Check country-specific legal specifications on the obligation to install and use active emergency brake assists with autonomous brake applications as of the application of GSR II (General Safety Regulation).

Autonomous brake applications may be triggered, for example, by the driving assistance systems available ex factory:

 - Active Brake Assist (BA3)
 - Cruise control (MS1)
 - Active Distance Assist DISTRONIC (code ET4)
5. The use of the driving assistance systems mentioned under 4. (codes BA3, ET4 and MS1) is only possible with sensing and processing of the actual wheel braking forces (sensing of hydraulic pressure per wheel on the base model vehicle or CAN signal via PSM). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

NOTE

It is not permissible to mount a semitrailer truck body without the forwarding of autonomous brake application signals from the driving assistance systems to the semitrailer through the trailer control valve.

! NOTE

The basic vehicle is delivered with the code XC1 "CoC documents incomplete" and in this variant does not have a brake approval for operation as a semitrailer truck.

Obtaining this brake approval is the sole responsibility of the bodybuilder manufacturer.

Proof of compliance with UN R 13 (including Appendix 10) in the form of a brake approval must be provided by the bodybuilder manufacturer and its responsible inspection authority (e.g. DEKRA/TÜV) and must be submitted in the case of an individual approval or EC/small series type approval.

We reserve the right to demand corresponding evidence within the scope of product monitoring.

! NOTE

For vehicles with engine OM654 and a planned permissible gross combination mass greater than 5.55 t, a pre-installation for trailer coupling must be provided to ensure sufficient engine cooling.

For this purpose, select one of the special equipment codes QA5 or QA3 in accordance with the weight requirement when ordering the vehicle, see the following overview of code specifications and 6.6.7 Trailer coupling (→ page 193).

Alternatively, a combination of the special equipment codes IT5 + MU6 + GD9 + AR5 can be selected for the vehicle configuration, see code specifications below.

Vehicles without this special equipment cannot be retrofitted.

However, in order to achieve optimum handling characteristics, it is recommended to use an RSC system (Roll Stability Control) in the semitrailer that will automatically intervene whenever the trailer becomes unstable.

Since the RSC of the semitrailer acts autonomously, no provision is made for signal transmission from the towing vehicle. No additional electrical connections are necessary. The manufacturer of the semitrailer brake system must be consulted to ascertain whether it is possible to retrofit an RSC system on a semitrailer with ABS.

If an increase in the gross combination mass is planned, contact the Technical Consultancy (→ page 18).

More information on the ESP® is available in 8.9.1 Electronic Stability Program (ESP®) (→ page 308).

For the conversion of chassis to a semitrailer truck in conjunction with the aforementioned driving assistance systems, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The longitudinal frame members must be reinforced by an appropriate subframe or a semitrailer bracket.

Furthermore, take into account the notes in 4.3.1 Modifications to the body-in-white (→ page 85) and 4.3.6 Attachment to the frame (→ page 89).

Code requirements for conversion to a semitrailer truck

With trailer control valves between master brake cylinder and ESP® hydraulic unit:

Code requirements (mandatory)

Code	Equipment
XC1	COC documents incomplete
BW6	Omission of Active Brake Assist
CB7 or CB2	Stabilization stage I Comfort suspension with air suspension
QA5 or QA3	Pre-installation of trailer coupling for trailer load up to 2800 kg Pre-installation of trailer coupling for trailer load up to 3500 kg

Alternative code combination to QA5 or QA3 (all 4 codes must be selected)

Code	Equipment
IT5	Model designation 5.0 t
MU6	Engine OM654 140 kW
GD9	6-speed manual transmission eco-gear high-torque
AR5	Axle ratio $i = 4.727$

Code recommendations

Code	Equipment
A50	Front axle with increased load capacity
ED4	Up-rated battery (12 V/92 A)
E2I or E2M	Auxiliary battery in vehicle interior Auxiliary battery in the engine compartment
EK1	Terminal strip for auxiliary consumers
E57	Electrics for trailer socket

Code restrictions (not usable)

Code	Equipment
BA3	Active Brake Assist
ET4	Active Distance Assist DISTRONIC
JA7	Blind Spot Assist
JB6	Parking package with 360° camera
JB7	Parking package with reversing camera
MS1	Cruise control
R9A	Ultrawide-base tires (front axle 225/75 R16C, rear axle 285/65 R16 C)
R9X	Ultrawide-base tires (front axle 205/75 R16C, rear axle 285/65 R16 C)
A4M	All-wheel drive

Gross combination mass

The technically permissible gross combination mass (with regard to the drivetrain) is limited to a maximum of 7000 kg as a matter of principle.

If a technically possible gross combination mass of more than 7000 kg up to a maximum of 8750 kg is required, select the following code combination:

IT5 + MU6 + GD9 + AR5 (see above code requirements) or the collective code XP6.

Code XP6 adds both this code combination and also a trailer coupling (code QA7 or QA8); See 6.6.7 Trailer coupling (→ Page 193).

! NOTE

The maximum technically possible gross combination mass refers only to the drivetrain. A brake approval in connection with a conversion to a semitrailer truck is not available. This must be obtained by the body-builder manufacturer.

For a change in the weight of the incomplete vehicle, contact the Technical Consultancy (→ page 18).

Assembly frame for semitrailer truck

If the vehicle is used as a semitrailer truck, a steel assembly frame made of rectangular tube measuring $100 \times 60 \times 3$ mm (or $s = 4$ mm) with a section modulus W_x of at least 25 cm^3 is required; See 7.1.1 Material quality, general (→ Page 201). The assembly frame must extend rearward as far as the standard chassis end and, toward the front, as far as the first mounting support behind the cab.

The assembly frame must be mounted according to 7.1.4 Attachment to the frame (→ page 204) using, at the least, all the ex factory mounting supports.

In addition, the connection between the frame and the assembly frame must be rigid at the frame end. This should be executed at the upper flange of the longitudinal frame member (→ page 204).

Make an additional shear-resistant connection at the front end of the longitudinal frame member.

Electrical connection for the semitrailer

Connect all additional electrical consumers as per 8.4 Interfaces and electric lines (→ page 271) and 8.4.6 Retrofitting of electrical devices (→ page 275).

- The connection lines must not chafe against body components.
- The bodybuilder manufacturer must ensure freedom of movement when cornering.
- Connection lines must not get caught on the semitrailer or pull on the trailer socket.
- During driving operation without a semitrailer, the connection lines must be secured correctly.

Brake system

The semitrailer brake system must be connected to the brake system of the semitrailer truck. It is not permitted to use inertia-activated brakes in combination with a semitrailer truck of model series 907.

The bodybuilder manufacturer must guarantee the following:

- A certified hydraulic-pneumatic control valve must be installed in the vehicle brake system for the actuation of the semitrailer brake.
- The hydraulic-pneumatic control valve must be integrated into the standard brake system between the master brake cylinder and the ABS/ASR hydraulic unit, with a maximum volume absorption of the control valve of 0.6 cm³ per brake circuit.
- When using hydraulic-pneumatic control valves, observe the previously mentioned code specifications. The use of the previously mentioned driving assistance systems requires signal processing of the actual wheel braking forces (hydraulic pressure per wheel or CAN signal via PSM). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

! NOTE

The brake systems of the semitrailer truck and the semitrailer as well as the energy supply must be designed in accordance with country-specific guidelines and legislation.

The semitrailer manufacturer and the bodybuilder manufacturer are responsible for the correct functioning of the semitrailer brake.

Assembly plate and fifth-wheel coupling

The bodybuilder manufacturer must ensure that the assembly plate and fifth-wheel coupling are adequately dimensioned.

Observe the manufacturer's information and installation instructions for the assembly plate and the fifth-wheel coupling.

7.11 Recovery vehicles

Vehicles with bodies for rescue or recovery equipment must be attached with assembly frames of adequate dimensions (→ page 204).

The bodies must also be fitted with two rigid connections for each longitudinal frame member (→ page 200).

For the installation of recovery or towing vehicles, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

For the attachment of winches, also observe 6.6.8 Underride guard (→ page 198) as well as the sections "Winches" (→ page 177) and "Side underride guards" (→ page 199).

Observe country-specific regulations, as well as accident prevention regulations (UVV) and general machinery directives; See 2.6 Accident prevention (→ Page 43).

7.12 Torsionally stiff body types

Torsionally stiff bodies can be found, for example, in municipal vehicles, refuse presses, closed box bodies ≥ 1000 mm behind the cab in combination with a dumper or platform, fire department cases or road cleaning vehicles. With torsionally stiff bodies, attach the body and assembly frame in the front area of the frame via stretch bolt connections consisting of spacer sleeves and stretch bolts that are locked to prevent loosening (→ page 207). As a minimum, use all the factory-installed mounting supports.

If required, the assembly frame must be additionally reinforced at the rear by fitting a diagonal cross.

Please also observe 8.4.6 Retrofitting of electrical devices (→ page 275).

For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

7.13 Elevating work platforms

General information

To ensure an optimum suspension design for vehicles from model series 907 with elevating work platforms, the following pre-installations are available as special equipment:

- Code X2X: Pre-installation for elevating work platforms with a center of gravity height of up to 1000 mm
This pre-installation contains the suspension stability package code CB7 and can also be combined with the GRP spring code C83.
- Code X3X: Pre-installation for elevating work platforms with a center of gravity height over 1000–1300 mm
This pre-installation contains the suspension stability package for high-load applications code B01+CB4.

The specifications in 4.1.2 Maximum permissible position of the vehicle center of gravity (→ page 78) and 4.2.1 Suspension (→ page 80) must absolutely be observed.

Also note that for vehicles from model series 907 with wheelbase R2 (IR4), the special equipment "Manual regeneration of diesel particulate filter at standstill" (code K1M) is only available without the exhaust system for power take-offs (code KA6), See 6.3.3 Exhaust system (→ Page 141).

-  If you have any questions about the vehicle configuration, contact the Technical Consultancy (→ page 18).

NOTE

Where bodies include detachable parts that move independently, ensure that there is adequate clearance between the detachable parts and the basic vehicle; otherwise, they may collide with the basic vehicle, resulting in damage.

NOTE

The elevating work platform must only be used when the vehicle is fully raised on its outriggers. When the vehicle is raised, there must be no additional loads in or on the cab. There is otherwise a risk of damage to the frame. The vehicle must not be moved with the elevating work platform extended. The frame may be damaged if the vehicle is moved with the elevating work platform extended.

The bodybuilder manufacturer must implement a safety facility to prevent the vehicle from being moved with the elevating work platform extended. This may be done through the control system of the elevating work platform, for example, or in conjunction with the parameterizable special module PSM (→ page 372).

NOTE

Observe country-specific regulations, accident prevention regulations (UVV) and general machinery directives; See 2.6 Accident prevention (→ Page 43).

If chassis are equipped with elevating work platforms, the following points must be observed due to high loads:

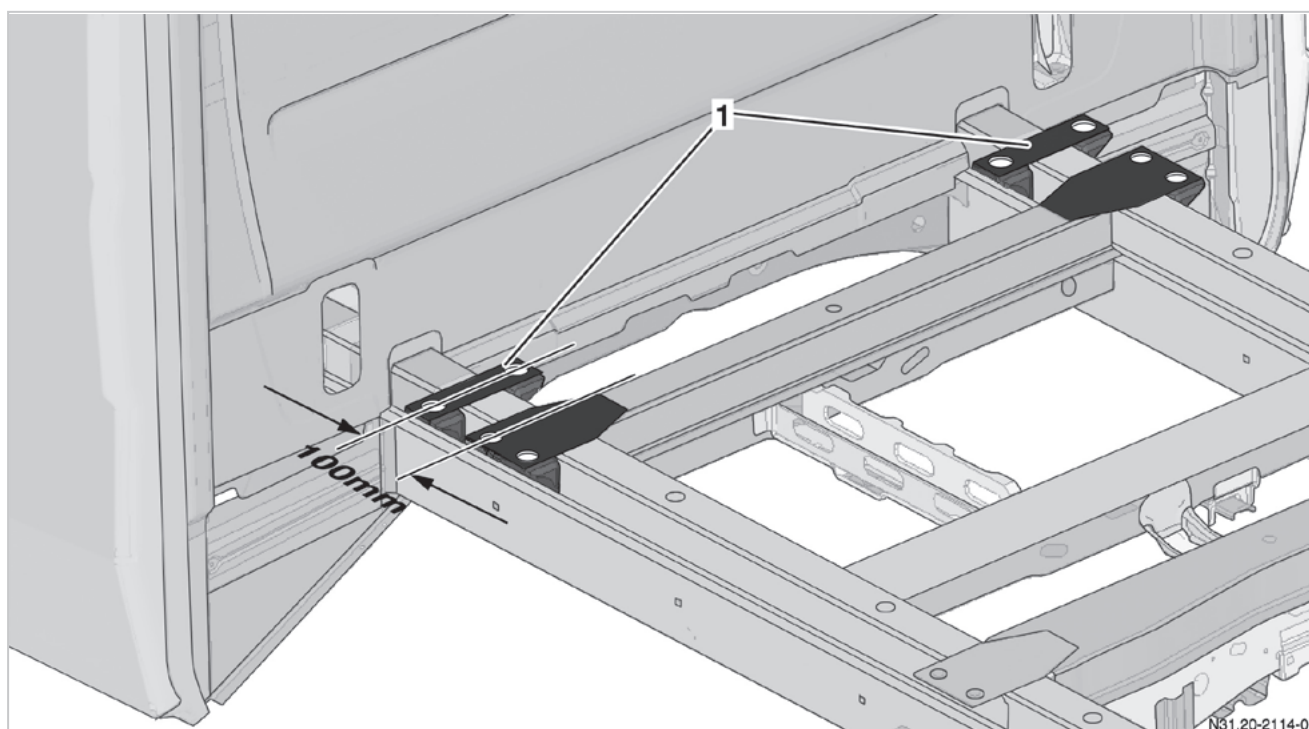
- For the retrofitting of elevating work platforms, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- The stability of the elevating work platform must be ensured by the bodybuilder manufacturer.
- The outriggers must comply with the information (→ page 188) on outriggers for loading cranes.
- The bodybuilder manufacturer must produce an additional Operator's Manual for the lifting facility and supply it with the vehicle in the national languages of the target markets. The Operator's Manual must contain the warning that no persons or loads are permitted in the cab when the vehicle is in the raised state.
- To ensure the uniform loading of the vehicle frame, the body must be attached to the vehicle frame by means of an assembly frame.
- At the least, all the factory-installed mounting supports must be fastened to the assembly frame.
- An additional mounting support must be installed for each longitudinal frame member in the area behind the cab (see example illustration).
- The attachment of the first and additional mounting consoles must be made with threaded connections locked to prevent loosening and spacer sleeves (→ page 207).
- The introduction of force in the supports must be exactly halfway between the two standard double bracket pairs behind the cab on the assembly frame. In addition, the assembly frame must be sufficiently protected against torsion in the area of force introduction by means of a crossmember.

Required additional mounting support

To ensure a uniform introduction of force in the chassis frame, one additional mounting support is required for each longitudinal frame member in the area behind the cab.

The mounting supports must be of at least the same quality as the standard material – See 6.2.3 Material for the vehicle frame (→ Page 118) – and have a wall thickness of 3 mm.

The hole spacing of the additional mounting support to the next adjacent hole in the existing mounting support must measure 100 mm.



Mounting supports

1 Additional mounting supports

When installing additional mounting supports, it is recommended to use Mercedes-Benz genuine parts.

For more information on the standard positions and dimensions of the body consoles, See 2.2.1 Body-builder Portal (→ Page 24) and 7.1.4 Attachment to the frame (→ page 204).

Assembly frame

The installation of elevating work platforms on a chassis requires an assembly frame of sufficient size.

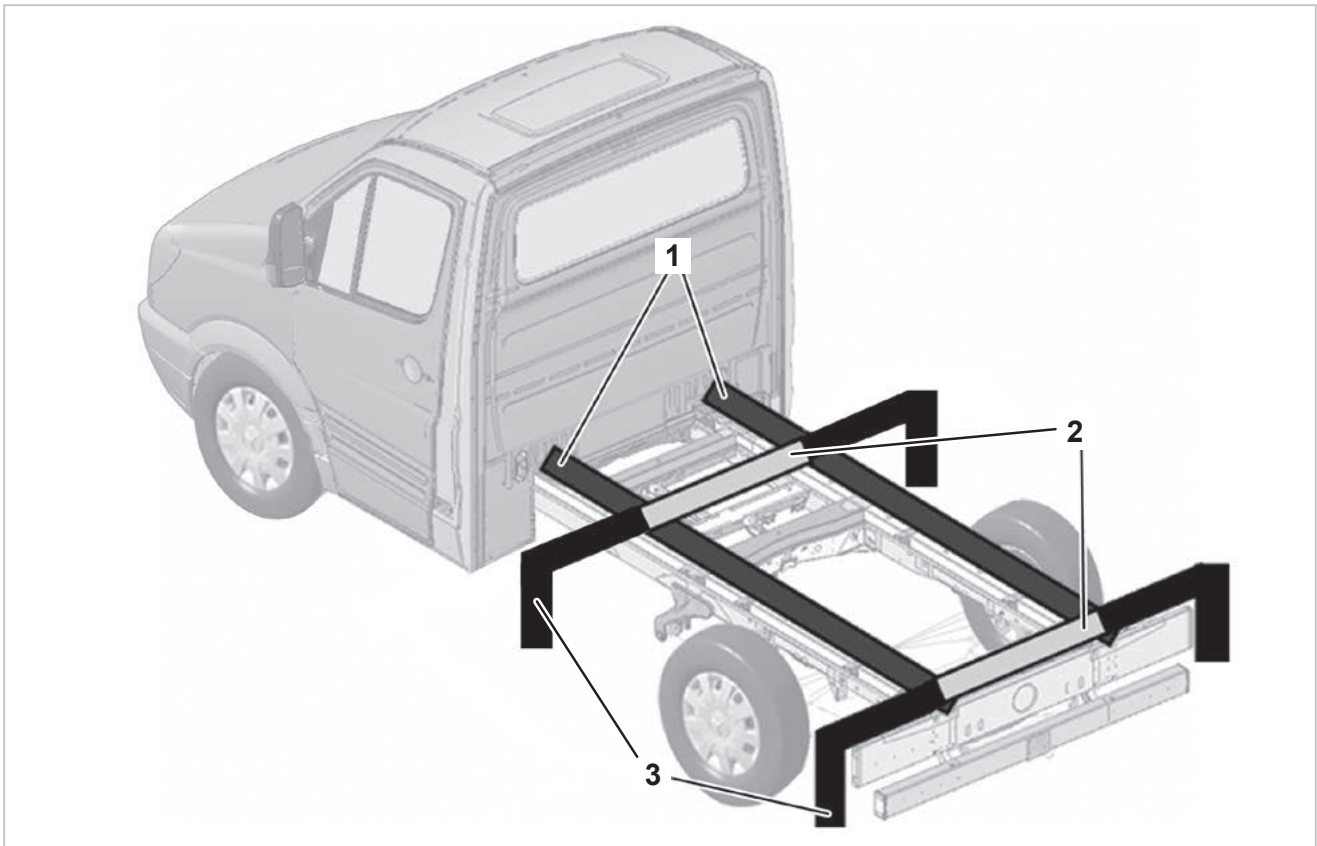
Section moduli (W_x) and material properties can be found under 7.1 Assembly frame (→ page 200).

The assembly frame must be attached in the same way to all mounting supports. The attachment of the assembly frame at the first and additional mounting consoles must be made with threaded connections locked to

prevent loosening and spacer sleeves (→ page 207).

The introduction of force into the assembly frame by the outriggers must take place exactly halfway between the two double bracket pairs installed behind the cab as standard.

In the area of the introduction of force in the assembly frame by the outriggers, a rigid crossmember (front and rear) must be installed to protect the assembly frame adequately against torsion.



Attachment of assembly frame to mounting supports

- 1 Area of additional mounting consoles
- 2 Required crossmembers for assembly frame in area of force introduction from outriggers
- 3 Outriggers

Regeneration of diesel particulate filter at standstill

With respect to the possibility of having regeneration of the diesel particulate filter at standstill (special equipment code K1M) in combination with elevating work platforms, observe the information and specifications in 6.3.3 Exhaust system (→ page 141).

7.14 Bodies on chassis with cab base (code F28, F50)

7.14.1 Cab base with cowl (code F50) and cab base with doors (code F28)

Registration eligibility of cab base vehicles (code F50, F28)

On leaving the factory, the cab base vehicles (code F50, F28) initially only have incomplete CoC documents. The bodybuilder manufacturer must ensure that the vehicle has complete type approval.

Refer to the information on requirements for the approval and registration eligibility of vehicles as well as on European Type Approval (ETA) and the EC Certificate of Conformity (CoC) (→ page 28).

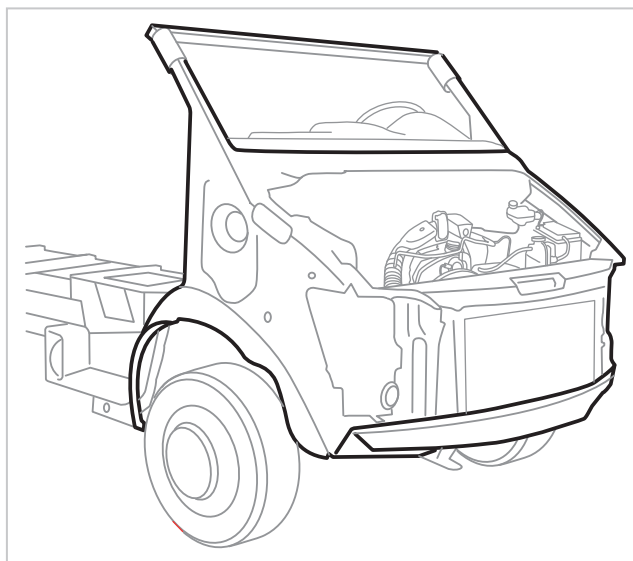
Modifications or conversions to standard vehicles (e.g. the installation of a lifting roof) may invalidate the operating permit. Comply with the approval conditions and requirements of the European type approval (ETA). In Germany, modifications to the vehicle must therefore be inspected by the relevant technical authorities in accordance with Section 19, Paragraph 2 of the StVZO (German vehicle licensing regulations).

Commissioning of control units

Observe the information and specifications for the commissioning of control units after completion of the body mounting work or conversion work, before the vehicle is placed on the market, See 8.19 Commissioning of control units (→ Page 387).

a) Cab base and cowl (code F50)

The chassis base (cowl) offers bodybuilder manufacturers a base for producing fully integrated bodies (e.g. camper vans) or special-purpose bodies. It is available from the factory under code F50 (→ page 74).



Cowl chassis (schematic)

! NOTE

For bodies on cowl chassis, observe all country-specific laws, directives and registration regulations.

Assembly with a Letter of Compatibility is mandatory (code X88), See 3.3 Selecting the basic vehicle (→ Page 53).

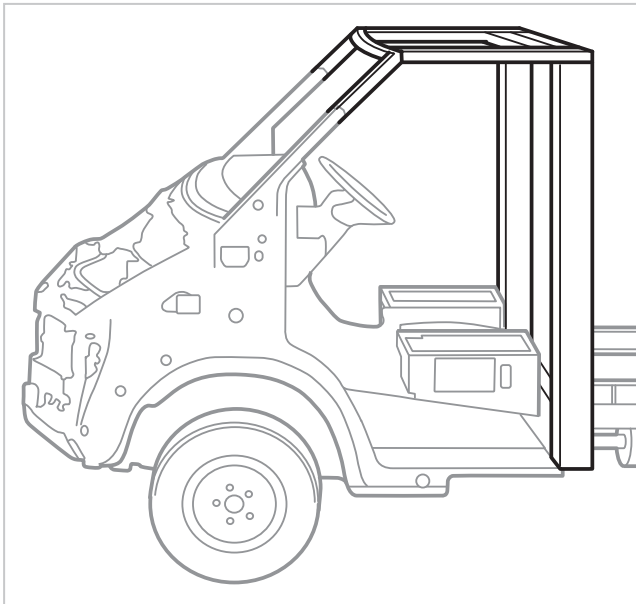
If bodies are mounted on cowl chassis, ensure that the cab structure has the same rigidity as the series production vehicle.

The front body part must be designed as a self-supporting structure through to the B-pillar.

To provide equivalent rigidity, it is strongly recommended to reproduce a new cell structure comprising

- A-pillar
- B-pillar
- Roof crossmembers
- B-pillar substructure crossmembers

that is identical with the original structure.



Cowl with cell structure (schematic)

The attachments between the crossmembers and the cab A-pillars or B-pillars must be positively connected. Make a separate, non-positive connection between the headlamp frame and the inner part of the A-pillar – do not bond this connection.

Do not connect a non-steel fender by means of a common connection to the headlamp frame and the inner part of the A-pillar.

For bodies on cowl chassis, also observe the following chapters:

- 3.11 Maintenance and repair (→ page 71)
- 6.1.3 Brake system (→ page 108)
- 6.3.2 SCR exhaust gas aftertreatment system (→ page 139)
- 6.3.4 Engine cooling (→ page 147)
- 6.3.5 Engine air intake (→ page 147)

! NOTE

After completion of all work on the vehicle, observe the anticorrosion protection measures (→ page 102).

Installation of an engine compartment flap by the bodybuilder manufacturer

If modifications are made to the engine compartment flap (installed by the bodybuilder manufacturer), make sure that the functions integrated there, such as the water separation feature for the air heating system, are not affected. If necessary, the water separation feature must be replaced by parts with the same function. All functions of the modified engine compartment flap must be equivalent to those of the series production component part.

Also observe 7.14.18 Fresh-air intake with code F50 (→ page 247) and 7.14.19 Hood contact switch with code F50 (→ page 247).

Windshield wiper system

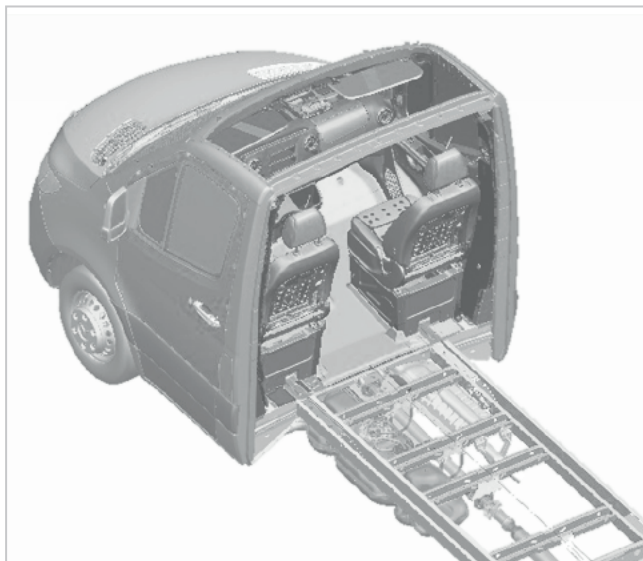
Cowl cab base vehicles (code F50) are assembled as standard without a windshield wiper system.

With the special equipment pre-installation for windshield wipers (code F45), a wiper motor with linkage to the two drive axles and with electrical connection to the vehicle electrical system is installed.

The wiper arms and wiper blades must be completed by the bodybuilder manufacturer depending on the windshield design.

b) Cab base with doors (code F28)

On vehicles with cab base and doors, the cab rear panel and cab roof are omitted ex factory. In addition, an auxiliary roof bow is fitted above the B-pillars to stiffen the cab.



Cab base with doors (code F28)

Omission and cutting of B-pillar auxiliary roof bow

If the B-pillar auxiliary roof bows are cut or omitted, reinforcement measures (→ page 132) are necessary.

! NOTE

For alternative methods of ensuring equivalent rigidity developed by the bodybuilder manufacturer, a detailed evaluation by the Technical Consultancy (→ page 18) is required.

For the implementation of the equivalent stiffness, a Letter of Compatibility is recommended (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

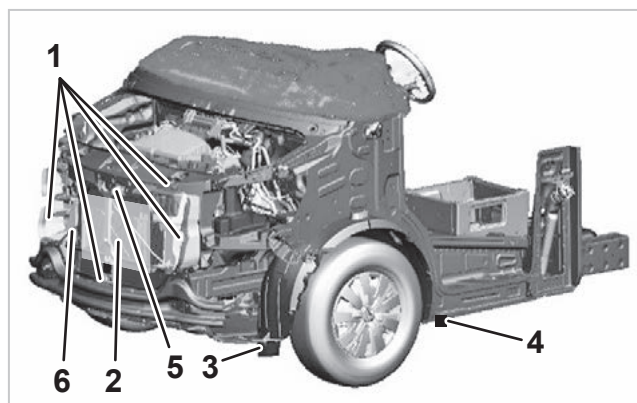
7.14.2 Mechanical equipment

The following information is relevant to all bodybuilder manufacturers who use cab base vehicles (code F28, F50) as a basis, even if the text addresses specific bodybuilder manufacturers, e.g. camper van manufacturers.

Integration in the cab base vehicle

When designing cowl cab base vehicles (code F50), take into account the following:

- Radiator opening area and radiator section sealing to ensure the required cooling output in the engine compartment
- Brake ventilation to achieve the optimum braking effect
- Position of the jack support point to ensure access to it and its strength

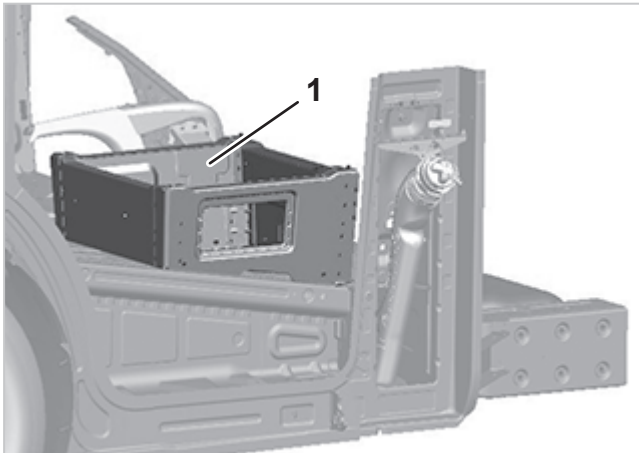


Cab base vehicle (schematic)

- 1 Radiator section sealing
- 2 Radiator opening area
- 3 Brake ventilation
- 4 Jack support
- 5 Intake air system
- 6 SCR (AdBlue® tank) including electronic component

Reinforced seat base for seat with integrated three-point seat belt (code S86)

The seat with integrated three-point seat belt is not included in the delivery package supplied by Mercedes-Benz AG.



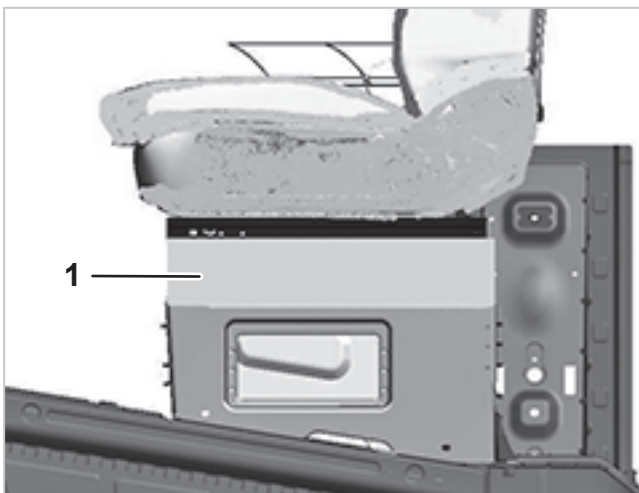
Reinforced seat box (schematic)

1 Reinforced seat base

The reinforced seat base is designed to absorb increased forces (caused by the seat with integrated three-point seat belt).

Boundary conditions

- Certification takes place in the system (seat, seat box and body) and must be performed by the bodybuilder manufacturer.
- The assumed points of load application on the seat (when designing the reinforced seat box) must be considered when selecting the seat.



Side view of reinforced seat base

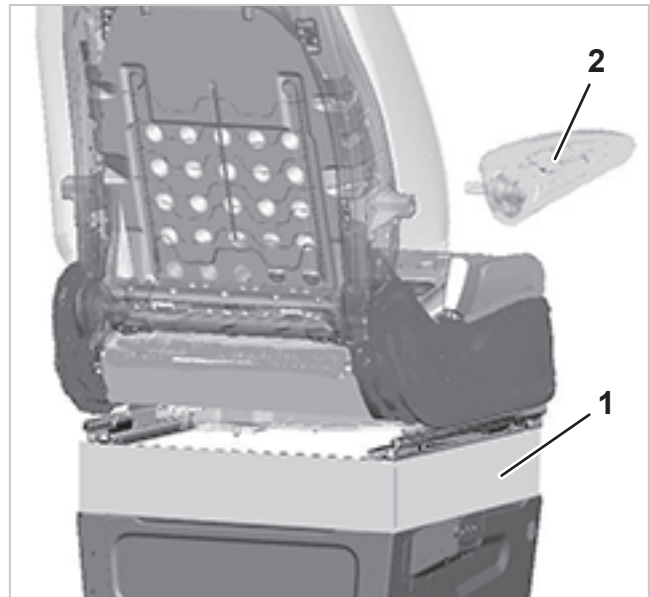
1 Swivel console

Pre-installation in seat (driver and front passenger) for mounting second armrest

The seats (driver and front passenger) of the Sprinter can be ordered by the bodybuilder manufacturer with an additional mount for a second armrest.

Boundary conditions

- The geometry of the connection element of the Sprinter armrest must be adopted.
- The second armrest is a retrofit.



Seat (schematic)

1 Swivel console
2 Armrest

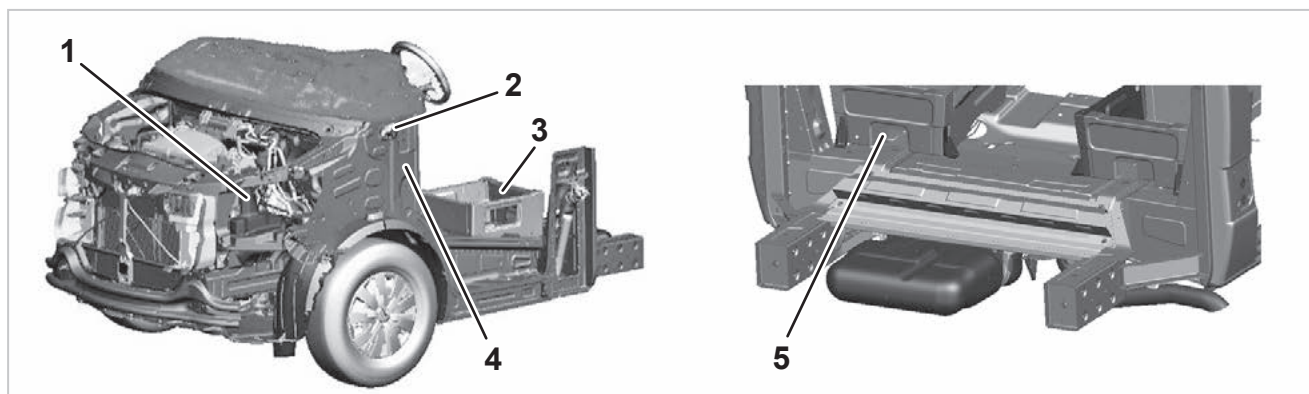
For further information on seats and seat boxes See 4.5 Limit values for the interior (→ Page 96) and 6.4 Interior (→ page 151).

7.14.3 Wiring harness connecting points

The connecting points shown below, divided into series production packages and special equipment options, mainly concern the chassis with cab base versions (code F28, F50).

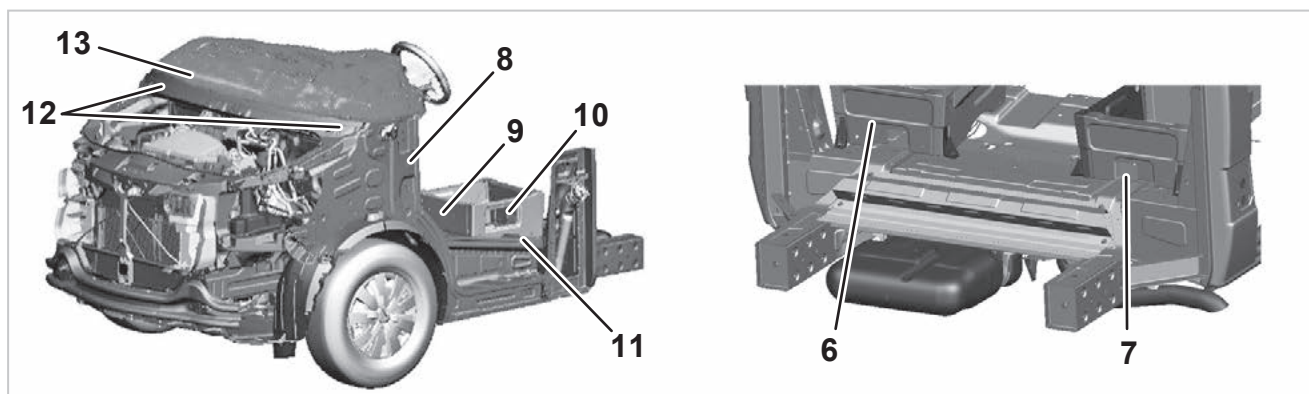
Certain connecting points are only relevant in the case of conversion to camper vans. Other connecting points are specific to the traction head. They are shown in the overview, but are not described in further detail.

Details on the specific topics of the traction head can be found in the Body/Equipment Mounting Directive for model series 910 in chapter 7.15.



Connecting points for series production packages

- 1 Wiring harness connecting point for headlamps (front passenger in mirror reverse)
- 2 Wiring harness connecting point for automatic driving lights (code LA2) and rain sensor (code JF1) with omission of windshield
- 3 Wiring harness seat connecting point for driver-side with omission of seat (front passenger in mirror reverse) for seat belt warning system and seat belt buckle pre-installation
- 4 Wiring harness door connecting point for driver and front passenger side with omission of door (front passenger in mirror reverse)
- 5 Wiring harness bodybuilder manufacturer connecting point A, B, and C = traction-head-specific connecting point (model series 910 only; for details, see the Body/Equipment Mounting Directive for model series 910, chapter 7.15)



Connecting points for special equipment options

- 6 Wiring harness of connecting point for pre-installation for Blind Spot Assist (code J1V), electrics of rear speaker (code EP7), reversing camera (code FR7), rearview camera for digital inside mirror (code F85), and parking system cameras with 360° all-around visibility (code JB1, JB2, JB3)
- 7 Antenna switch box/GPS interface wiring harness connecting point (code E4A), details (→ page 241)
- 8 Bodybuilder manufacturer interface H (code E2A); for traction-head-specific details, see Body/Equipment Mounting Directive for model series 910, chapter 7.15
- 9 Bodybuilder manufacturer interface M (code E5M); for traction-head-specific details, see Body/Equipment Mounting Directive for model series 910, chapter 7.15
- 10 Wiring harness seat connecting point for driver with omission of seat (front passenger in mirror reverse) for seat heating pre-installation
- 11 Additional fuses for camper vans (code E1R and E1Y)
- 12 Wiring harness for windshield heater pre-installation (code E2F) with omission of windshield
- 13 DSP Box wiring harness connecting point (code E3D), for details (→ page 244)

7.14.4 Automatic driving lights (code LA2) / rain sensor (code JF1)

Relevance

- On cab base vehicles (code F28) with windshield: When the vehicle leaves the factory, these systems are fully functional.
- On cowl cab base vehicles (code F50) and with omission of windshield (code F92), a kit is supplied for the bodybuilder manufacturer. For an installation description, See 7.14.5 Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1) on vehicles without windshield (→ Page 233).

Automatic driving lights function

If the brightness value drops below a certain value, the automatic driving lights function switches on the driving lights automatically. If the value measured by the light sensor increases to a certain level, the automatic driving lights switch off again.

Weather-related visibility impediments such as snow or rain are automatically taken into account by the automatic driving lights function.

The automatic driving lights function increases visibility and safety in road traffic.

Rain sensor function

When the windshield wiper lever is set to the appropriate position, the windshield wipers are activated.

The rain sensor transmits infrared light at a certain angle into the windshield, which then reflects the light differently depending on how wet it is. An electronics system uses the amount of light detected to control the wiping cycle, from a single wipe through to continuous wiping. This ensures relaxed driving even in varying levels of rainfall, by doing away with the need to manually adjust the wiping interval.

Boundary conditions and completion

If a superstructure is fitted (e.g. on alcove vehicles), this can have a negative effect on the automatic driving lights and rain sensor.

Observe the further information and specifications in chapter 7.14.5 Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1) on vehicles without windshield (→ page 233) and 8.9.6 Automatic driving lights/rain sensor (→ page 340).



Sensor at the windshield (example)

7.14.5 Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1) on vehicles without windshield

Relevance

- Cowl cab base vehicles (code F50), e.g. fully integrated camper vans
- Vehicles with special equipment "Omission of windshield" (code F92)

Automatic driving lights/rain sensor connecting point

The special equipment "Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1)" is available ex factory. For information on the assistance system, See 7.14.4 Automatic driving lights (code LA2) / rain sensor (code JF1) (→ Page 232).

The appropriate light/rain sensor incl. retaining plate and cover is supplied with the vehicle.

The bodybuilder manufacturer needs to assemble and install a wiring harness between the connecting point and the sensor.

When the vehicle is ordered with code F50 or code F92, the sensor system for the sensor cannot be installed ex factory. This system must therefore be completed by the bodybuilder manufacturer before the vehicle is brought into service. To do this, three lines must be routed from the connecting points (X130/3*1-B, tml. 30 and tml. 31) to the sensors.

For a description, See 7.14.3 Wiring harness connecting points (→ Page 231).

Boundary conditions

Adhere to the following specifications regarding the installation of the light/rain sensor.

In addition, observe the specifications regarding the length of a roof overhang, See 8.9.6 Automatic driving lights/rain sensor (→ Page 340).

NOTE

In the event of deviations from the following specifications, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

-  Contact the Technical Consultancy during the planning phase (→ page 18).

Completion

For completion, the following components are supplied with the vehicle:

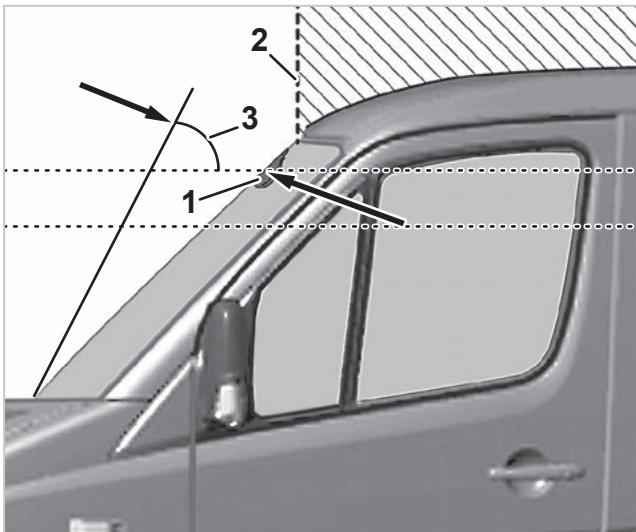
- Light/rain sensor
- Covers and mounting plates:
 - Mounting plate for light/rain sensor
 - Cover for the sensor
 - Cover M-part

Information on installation

The installation of a light/rain sensor can only be implemented using the standard sensor type, because the sensor communicates with the vehicle over the VCS LIN interface.

When installing the sensor, observe the following boundary conditions:

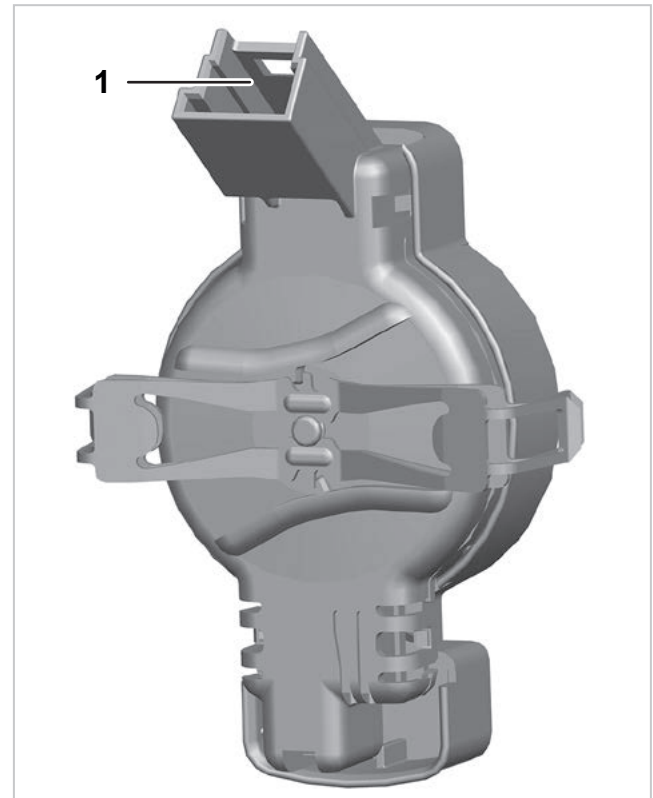
1. In order to guarantee that all functions operate without restrictions, the angle of the windshield relative to the roadway at the sensor position must be within the specified limit values between 37° and 53°.
 - If the bodybuilder manufacturer installs the sensor at a greater windshield angle up to max. 70°, the convenience functions may be impaired due to the deviation from the angle limits (e.g. the driving lights or the wipe cycle of the windshield wipers may be activated/deactivated at different times).
 - The windshield thickness at the sensor location must be between 4.5 and max. 6 mm. For larger windshield thicknesses, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).



Limit for bodies on vehicles with light/rain sensor

- 1 Light/rain sensor
- 2 Body limit – ideal state without roof overhang; for more information on roof overhangs, See 8.9.6 Automatic driving lights/rain sensor (→ Page 340).
- 3 Angle limit of the windshield

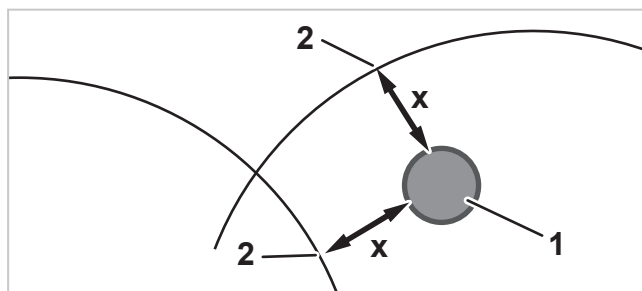
2. The horizontal and vertical curvature radius of the windshield must be ≥ 1400 mm in the area of the sensor.
3. The sensor must be positioned on the windshield in accordance with the specifications below. Deviations will result in function impairments or even failure of the sensor.
 - Position the sensor so that the electrical connection socket is facing upward toward the cab roof; see following illustration.



Vertical alignment of the light/rain sensor

- 1 Electrical connection socket in the direction of the cab roof

- To mount the sensor holder on the windshield, the distance x between the sensor position and the end of the wiper blade must be at least 35 mm; see following illustration.



Distance from sensor to end of wiper blade (schematic)

- 1 Sensor
- 2 End of wiper blade
- x Distance > 35 mm

i More information on commissioning the automatic driving lights (code LA2) and rain sensor (code JF1) can be found in the Workshop Information System (→ page 26).

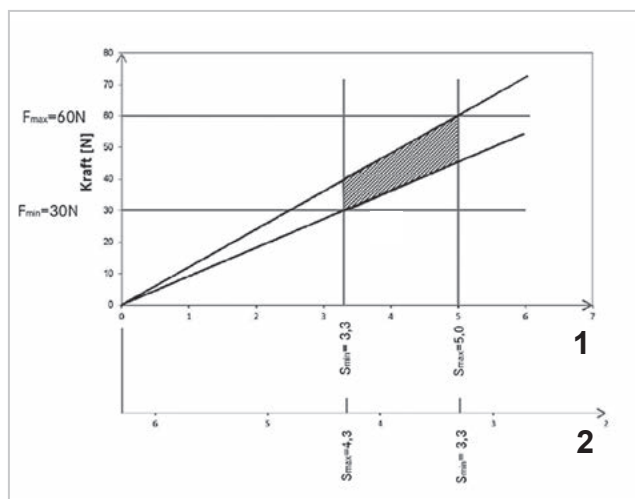
4. The transmission value of the windshield must not differ from that of the standard windshield.

- The transmission in the IR range (800–1100 nm) must not drop below 23%. If you have any questions, please contact the Technical Consultancy team (→ page 18).

Attachment of the carrier plate

If the windshield is screen-printed on the inside (layer 4), the opening for the sensor must be at least >28 mm in diameter. If the screen printing is on a different layer of the windshield, the diameter must be adjusted accordingly.

The following contact pressure of the sensors on the windshield must be guaranteed.



Spring characteristic

- 1 Travel referred to spring
- 2 Travel referred to windshield

Adhesive bonding

- ▶ The carrier plate/mounting plate must be primed or plasma-treated in the gluing area before it is stuck to the windshield.
- ▶ The holding ability of the bond between the carrier plate and the windshield must be tested with a dummy subjected to a tensile force of 80 N at a temperature of 20 °C in the opposite direction to the windshield.
- ▶ The holding ability of the bond between the carrier plate and the windshield at high temperature must be tested with a dummy subjected to 4 times the static load at 90 °C in the opposite direction to the windshield.

Commissioning and attachment of the sensors

- i** More information on installing and commissioning the automatic driving lights (code LA2) and rain sensor (code JF1) can be found in the Workshop Information System (→ page 26) and in 8.19 Commissioning of control units (→ page 387).

7.14.6 Multi-purpose camera (MPC) on vehicles without windshield

Relevance

In the case of cab base vehicles with cowl (code F50) or vehicles with the special equipment "Omission of windshield" (code F92), the MPC is enclosed for completion by the bodybuilder manufacturer.

- On cab base vehicles (code F28) with windshield ex factory, this system is supplied in a functional state. Nevertheless, calibration of the MPC is an essential requirement following completion of the mounting or conversion work.

Function

For a description, See 8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist (→ Page 339).

Boundary conditions and completion

The MPC is supplied with the vehicle for subsequent completion by the bodybuilder manufacturer, e.g. for fully integrated camper vans.

For the installation of the MPC in connection with a bodybuilder-manufacturer-specific windshield, a detailed running-in note is available with the required information and specifications.

The connecting point on the vehicle side is located on the left-hand side of the vehicle in the lower area of the A-pillar at the cockpit.

The electrical connection of the MPC to this connecting point with a wiring harness designed for this purpose (power supply and CAN line as well as electrical contacts) must be carried out by the bodybuilder manufacturer in accordance with the installation instructions.

A test and startup specification applies for the commissioning and calibration.

- Please contact the Technical Consultancy team (→ page 18) to receive the installation specification and further information on commissioning and if you have any questions.
- Details regarding the MPC and the associated components for standard vehicles (e.g. camera holders) can be found in the 3D design data (→ page 25).

 **NOTE**

For these body situations, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

- When ordering the vehicle configuration with code F50, and therefore also with code F92, code X88 "Obligation for body with letter of compatibility" is automatically assigned.

7.14.7 Connecting point with omission of driver's and front passenger seat for pre-installation for seat belt warning system (code J1D/J1E) and seat occupancy recognition (code E3S)

Relevance

- Cab base vehicles (code F28) without seats ex factory
- Cowl cab base vehicles (code F50) without seats ex factory

Assembly with a Letter of Compatibility is mandatory (code X88) (→ page 56).

! NOTE

The additional information and specifications in 6.4.1 General information (→ page 151), 6.4.2 Safety equipment (→ page 152) and 7.3.1 Retrofitting seats (→ page 209) must absolutely be observed.

Function

With code S90 (Omission of driver's seat) and code S91 (Omission of front passenger seat), it is possible to order a vehicle without driver's/front passenger seat ex factory and use bodybuilder-manufacturer-specific seats.

In this case, due to the legal requirements relating to the seat belt warning in accordance with UN R 16.07, the pre-installation for the seat belt warning system with code J1D (for driver's seat) and code J1E (for front passenger seat) will be installed ex factory in such vehicles.

In addition, the signal from the seat belt buckle is needed to trigger restraint systems such as seat belt tensioners or airbags.

The pre-installation for a seat belt warning system enables the bodybuilder manufacturer to install third-party seats with electronic seat belt buckles. The seat belt warning system for the front passenger seat additionally requires the seat occupancy recognition system, which must be connected into the pre-installation for the front passenger seat belt warning system.

Seat	Description (code)
Front passenger seat	Pre-installation for seat belt warning system for front passenger seat (code J1E): The vehicle has a connecting point with plug to which the bodybuilder manufacturer's seat belt buckle as well as a seat occupancy recognition mat are connected.
Driver's seat	Pre-installation for seat belt warning system for driver's seat (code J1D): In the vehicle, there is a connection point with a plug to which the seat belt buckle is connected by the bodybuilder manufacturer.

The connection point allows the use of bodybuilder-manufacturer-specific seats while ensuring compliance with legal requirements.

In addition, there is a pre-installation for driver's seat occupancy recognition (code E3S); this must be completed in order for the ECO automatic engine start/stop system (code MJ8), electric parking brake (code B25), and HOLD function (code BH1) systems to function.

The connecting point includes a plug to which the seat occupancy recognition mat from IEE is connected (the bodybuilder manufacturer can be given direct purchase rights for this). For this, contact your Mercedes-Benz Service or the Technical Consultancy (→ page 18).

Boundary conditions and information on installation

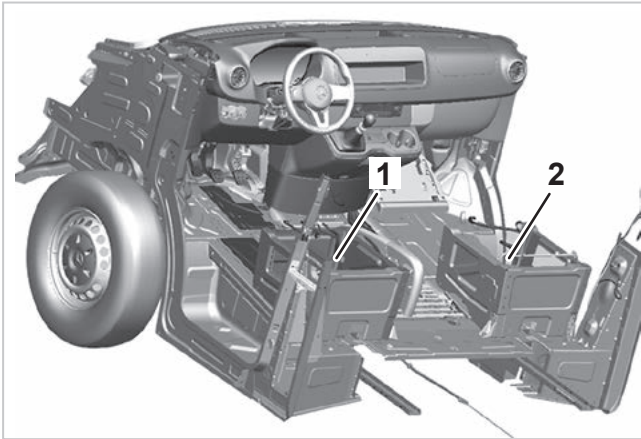
A connecting point for the seat belt buckle and for the seat occupancy recognition system is provided ex factory.

The bodybuilder manufacturer is responsible for installing the seat belt buckle and the seat occupancy recognition mat.

Usage is only permitted if proof that the seat occupancy recognition system functions correctly is provided and submitted upon request.

Completion

- Each seat belt warning system must be connected to one plug.
- More information about the seats is available in 6.4.3 Seats (→ page 160).

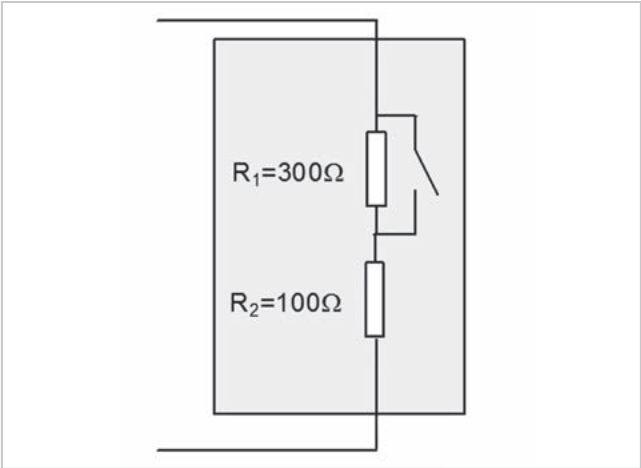


Electrical interfaces for seat belt warning system pre-installations

- 1 Driver-seat side
- 2 Front passenger seat-side

Requirements

- ▶ The bodybuilder manufacturer must install the seat belt buckle. Dimension the seat belt buckle so that the requirements for the signal are satisfied.
- Logic of driver and front passenger seat belt buckle on individual seats:
 - Switch open: unbelted
 - Switch closed: belted
- Resistances R1, R2: max. 10% tolerance

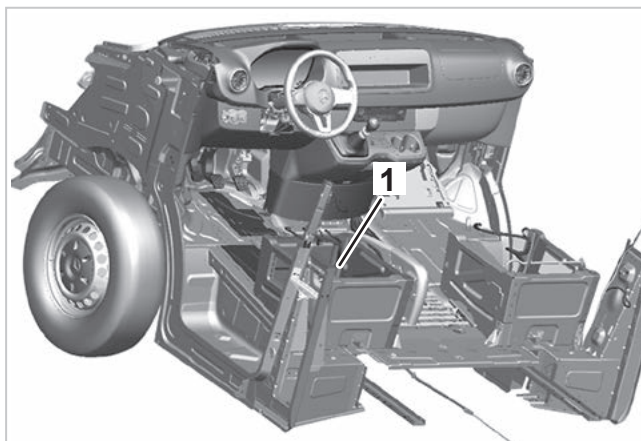


Connection scheme for seat belt buckle switch

Seat belt buckle switch				
	Min.	Typ.	Max.	Unit
Scanning current	–	12	20	mA
Repeat rate	–	50	210	ms
Pulse width	–	784	800	µs

Completion

- The seat occupancy recognition system must be connected to the plug.
- Coordinate details on completion and installation with the Technical Consultancy (→ page 18).
- For more information about the seats, See 6.4.3 Seats (→ Page 160).



Electrical interfaces

- 1 Pre-installation for seat occupancy recognition

Special equipment options that require seat occupancy recognition include:

- ECO automatic start/stop (code MJ8)
- Electric parking brake (code B25)
- HOLD function (code BH1)
- Active Distance Assist DISTRONIC (code ET4)

7.14.8 Driver's and front passenger door connecting point

Relevance

- Cab base vehicles (code F28) and all other vehicles with omission of left cab door (code FW8) and right cab door (code FW9)
- Cowl cab base vehicles (code F50)

Function

The two vehicle doors ensure reliable operation of a wide variety of functions that can or must be implemented in combination with the door control unit.

The door control unit must be connected to the body CAN and sends and/or receives necessary vehicle information via this.

The installation and commissioning of both door control units is also necessary if only a side panel without access is implemented instead of the driver's or front passenger door.

The door connecting point is an interface used for completing the following scopes:

- Door control unit and the functions implemented by the door control unit
- Speaker connection (special equipment, added via radio)

Functions that are implemented by the door control unit

- Exterior light functions: Entrance lamp, exit warning light (only with code JB6)
- Exterior mirror functions: Turn signal indicators, Blind Spot Assist indicator (only with codes JA7/JV1+OB2), heating, motors
- Actuators: Power window motor, central locking system
- Rotary tumbler door contact
- Actuation: Window lifter, seat heating, central locking system, mirror adjustment
- Front side turn signal indicators with code L44 (instead of mirror indicators; pin assignment must be observed)

Boundary conditions and information on installation

1. The door control unit must be connected and put into operation. Observe the current coding. Without the door control unit, the vehicle is not operational.
2. A door connecting point (right and left) is provided ex factory. The door control unit is supplied with the vehicle.
3. The code FX3 can be used to additionally order the door wiring harness from the door connecting point to the door control unit and to the optional special equipment.
4. The listed functions only function in combination with the door control units.
5. Further components, such as switches or electric motors, can be provided on request.
6. The door control unit and the other components must be installed and put into operation by the body-builder manufacturer.

Observe the further information and specifications in 8.19 Commissioning of control units (→ page 387).

7.14.9 Headlamp connecting point

Relevance

- Cowl cab base vehicles (code F50)

Function

The headlamp connecting point is an interface used for completing the following scopes:

- Standard packages: Headlamps and turn signal indicators
- Special equipment option: Front fog lamps (code L13)

It is possible to use own halogen or LED headlamps if the power data (table) given in 10.1 Bulb ratings of tail lamps (→ page 398) are complied with.

The provision of Mercedes-Benz series production headlamps can be requested.

The vehicle can be registered on completion of the series production packages.

Boundary conditions and completion

Power data specified for light sources by Mercedes-Benz AG must be complied with. This is necessary to ensure that the lamp failure indicator functions correctly.

For power output data, See 8.5 Lighting (→ Page 286) and 10.1 Bulb ratings of tail lamps (→ page 398).

7.14.10 Connecting point for antenna switchover box for third-party antennas (code E4A)

Relevance

- All vehicles without roof ex factory as cab base with doors (code F28) or cowl cab base (code F50)

Function

With the special equipment option "Antenna interface for radio, navigation and mobile radio (code E4A)", Mercedes-Benz AG provides an interface for connecting a third-party antenna with AM/FM and DAB. This interface is located in the driver seat frame.

An antenna switchover box is connected downstream to this connecting point ex factory. The signals from the external antenna are adapted to the Mercedes-Benz-specific signal environment using the antenna switch box.

A 12 V power supply for the third-party antenna is provided at this connecting point.

For the bodybuilder manufacturer, the connecting point enables simple connection of a third-party antenna to the Mercedes-Benz head over a defined interface.

Boundary conditions and completion

Supply the antenna switchover box with the following signals:

- FM (87–108 MHz)/AM (531–1720 kHz and 5.9–26.1 MHz)
- DAB band III (174–240 MHz)
- Impedance 50 Ω

Supply the connection of the navigation system with the following signal:

- GNSS 1575–1606 MHz
- Right hand circulation polarization
- Impedance 50 Ω
- Supply voltage 4.5–5.5 V
- Power consumption 20 mA $\pm$ 3 mA

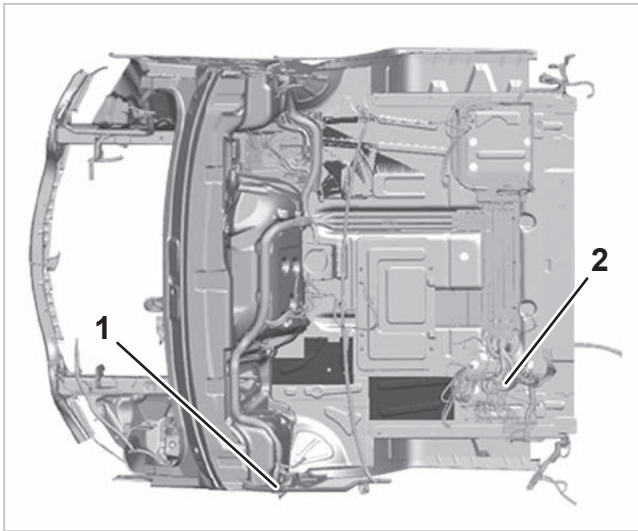
For further information on the antenna interface for bodybuilder manufacturers, See 8.17.2 Antennas (→ Page 383).

7.14.11 Pre-installation for bodybuilder manufacturer interface H (code E2A) and bodybuilder manufacturer interface M (code E5M)

Pre-installation for bodybuilder manufacturer interfaces H and M

The pre-installation for bodybuilder manufacturer interface H (tapping of bodybuilder manufacturer special signals in cockpit, code E2A) and for bodybuilder manufacturer interface M (expansion of PSM standard contact, code E5M) allows the bodybuilder manufacturer to tap the following signals in the cockpit:

- Battery network
- Tail lamps
- Instrument lighting
- Signal lines routed through from interface H. In combination with the PSM (code ED5), vehicle signals can also be supplied.



Interfaces for bodybuilder manufacturer

- 1 Bodybuilder manufacturer interface H
- 2 Bodybuilder manufacturer interface M

Function of bodybuilder manufacturer interface H

Bodybuilder manufacturer interface H (code E2A) provides an additional interface in the area of the A-pillar on the driver's side. Bodybuilder manufacturer interface H allows the following signals to be tapped more easily.

In addition, 5 signals are routed from interface H through to interface M. These signals are freely available to the bodybuilder manufacturer. They can be used, for example, to send signals from the PSM (code ED5) to interface M (code E5M).

Boundary conditions and completion

The plug can be fitted with a wiring harness from the bodybuilder manufacturer according to requirements.

The following lines/signals are provided:

- 5 signal lines
- Left and right turn signal indicators and one reversing light
- Tml. 58d dimmable instrument lighting
- Tml. 15, tml. 30T (shutoff on weak battery) and tml. 31

The maximum power values and parameters of the connections must be complied with.

Function of bodybuilder manufacturer interface M

Bodybuilder manufacturer interface M (code E5M) allows the signals from the PSM to be tapped more easily without the need for complex cable routing.

Bodybuilder manufacturer interface M provides an additional interface in the driver seat frame. The signals from the PSM are routed out from this interface. In addition, 5 signals can be routed from bodybuilder manufacturer interface M through to bodybuilder manufacturer interface H. These signals are freely available to the bodybuilder manufacturer.

Details on the following interfaces can be obtained from the Technical Consultancy (→ page 18):

- Bodybuilder manufacturer interface M - plug 1
- Bodybuilder manufacturer interface M - plug 2
- Bodybuilder manufacturer interface M - plug 3

Requirements

- The fuse ratings can be found in the supplement "Fuse assignments"; See 8.4.2 Electric lines and electrical fuses (→ Page 272).
- Access to the signals from the tail lamps is only permitted in combination with the additional turn signal module.
- Access to the reversing light signal is only permitted in combination with the additional turn signal module.
- Tml. 58d driver module requirements (battery voltage): maximum current ≤ 1 A

7.14.12 Pre-installation for windshield heater (code E2F) with omission of windshield

Relevance

With code F92 "Omission of windshield", the windshield can be omitted ex factory on cab base vehicles. This enables the bodybuilder manufacturer to install a third-party windshield.

The vehicle special equipment "Pre-installation for windshield heater" (code E2F) provides the pre-installation for the heated windshield.

Function

The control unit, the control switch, and two connecting points on the left and right in the area of the A-pillar are installed ex factory.

The bodybuilder manufacturer can install a heated windshield with integrated heating wires that complies with the relevant specification (see boundary conditions).

A switch with indicator lamp is used to switch on the windshield heater. It can only be operated when the engine is running. The heater is activated by a switch and switches off automatically after a certain time, or can be switched off manually using the rocker switch.

The window heating helps to keep the windshield free of condensation, snow and ice during winter.

Boundary conditions and completion

The heated windshield is installed by the bodybuilder manufacturer.

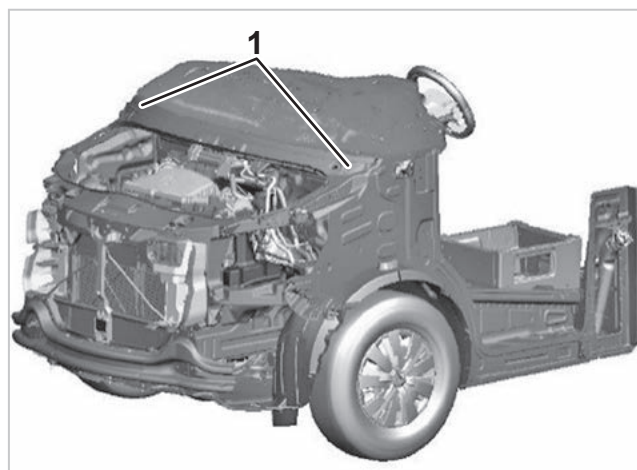
The heated windshield must be connected to the two plugs.

The maximum current of the windshield heater must not exceed 33.5 A at each connecting point.

The fuse ratings can be found in the supplement "Fuse assignments"; See 8.4.2 Electric lines and electrical fuses (→ Page 272).

Comply with the following electrical parameters for the heated windshield:

- Power 790 W $\pm$ 15%
- Maximum current per connection is 33.5 A
- U = 13.5 V



Electrical interfaces

- 1 Connecting points for heated windshield

7.14.13 Airbag pre-installation with crash certificate for fully integrated vehicles (code SV4)

Relevance

- Cab base vehicles (code F28): complete as standard equipment ex factory
- Cowl cab base vehicles (code F50): Crash certificate required, special equipment ex factory

Assembly with a Letter of Compatibility is mandatory (code X88), See 3.3 Selecting the basic vehicle (→ Page 53).

Function

As safety-relevant component parts, the driver's airbag and front passenger airbag are installed complete ex factory.

For fully integrated vehicles (code F50), it is possible to order an airbag pre-installation with crash certificate for fully integrated vehicles (code SV4) ex factory. This can only be ordered if the bodybuilder manufacturer is able to provide a crash certificate.

If work is due to be carried out on the airbag and associated components, always perform this work in accordance with the bodybuilder manufacturer directive.

Boundary conditions for semi-integrated vehicles

No special conditions, as installation is completely ex factory

Boundary conditions for fully integrated vehicles

With the airbag retrofit, fully integrated vehicles can also be equipped with the safety-relevant airbag on the driver's and front passenger side.

The airbag pre-installation can only be ordered if the bodybuilder manufacturer is able to provide a crash certificate.

For further details on the various airbags, See 4.5 Limit values for the interior (→ Page 96) and 6.4 Interior (→ page 151).

7.14.14 Pre-installation for a digital signal processor (DSP) box (code E3D)

The pre-installation for a digital signal processor box enables the bodybuilder manufacturer to install its own bodybuilder-manufacturer-specific DSP box that is connected to the Mercedes-Benz telematics and infotainment system. The DSP box allows the bodybuilder manufacturer to install custom sound systems.

Completion

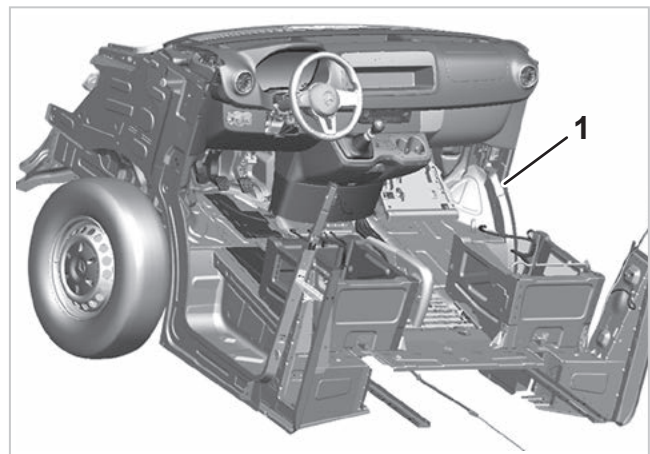
The pre-installation provides two standardized ISO plugs. These two plugs are plugged into the digital signal processor box.

- One plug for the input signals: speaker signals and supply lines
- One plug for the output signals

Coordinate further details on completion and installation with the Technical Consultancy (→ page 18).

Requirements

- The fuse ratings can be found in the supplement "Fuse assignments"; See 8.4.2 Electric lines and electrical fuses (→ Page 272).
- In combination with the Mercedes-Benz telematics and Infotainment system, the speakers must satisfy certain requirements; see 8.17.3 Speakers (→ page 385).



Electrical interface

1 Digital signal processor box

7.14.15 Pre-installation for rear speakers (code EP7)

The electrical pre-installation for rear speakers, code EP7, provides an additional electrical interface.

This interface enables the bodybuilder manufacturer to install its own rear speakers.

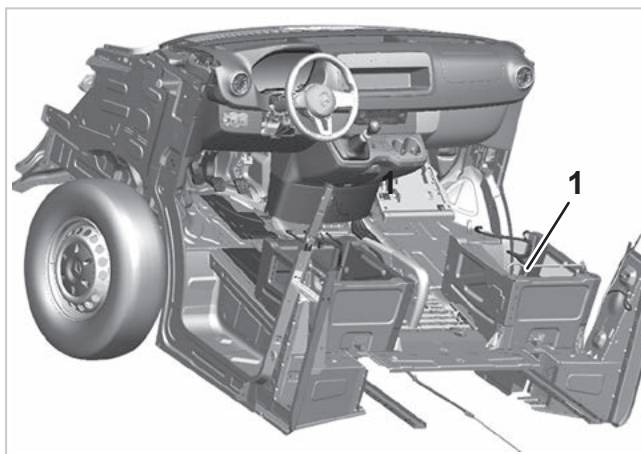
Completion

The pre-installation provides a plug that is connected to the speakers.

Coordinate further details on completion and installation with the Technical Consultancy (→ page 18).

Requirements

In combination with the Mercedes-Benz telematics and Infotainment system, the speakers must satisfy certain requirements; See 8.17.3 Speakers (→ Page 385).



Electrical interface

1 Pre-installation for rear speakers

7.14.16 Overhead control panel switch module with code F50

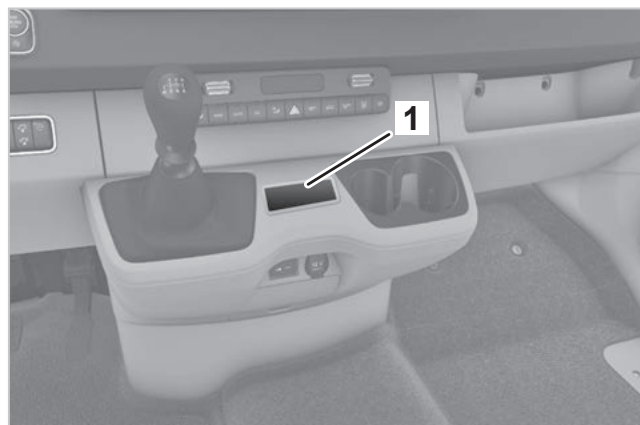
In contrast to the production version, cowl cab base vehicles (code F50) have no overhead control panel in the roof because the interior lighting is implemented via the PSM. Instead, the OCP switch module is accommodated in the covering shell of the center/shift console.

Completion

The electronics housed in the OCP switch module for cab base vehicles are required for the roof LIN and for the power supply to the light sensor on the windshield. On cab base vehicles, bodybuilder manufacturers must position and connect the sun sensor (automatic driving lights function) on the windshield according to the specifications, See 6.4.5 Ventilation (→ Page 163).

Boundary conditions

Furthermore, the OCP switch module holds the SOS pushbutton switch for the Mercedes-Benz emergency call system (code EY5) and the pushbutton switch for Mercedes-Benz roadside assistance call management (code EY6). In combination with code EY5, the microphone of the E-call emergency call system is accommodated on the left-hand side of the OCP switch module.



Center/shift console in cowl cab base vehicles (code F50)

1 OCP module with code F50

7.14.17 E-Call in cab base vehicles

The Mercedes-Benz emergency call system can help to decisively reduce the time between an accident and the arrival of emergency services at the site of the accident. The data is transferred via the communications module for Mercedes PRO in the vehicle. The emergency call is thus not dependent on whether a mobile phone is switched on, but requires a GSM connection and the ability to locate the vehicle via GPS. The emergency call can be triggered automatically by the crash sensors or manually by the driver pressing the SOS button. The emergency call will then be directed to the Mercedes-Benz emergency call number or to a local emergency call center.

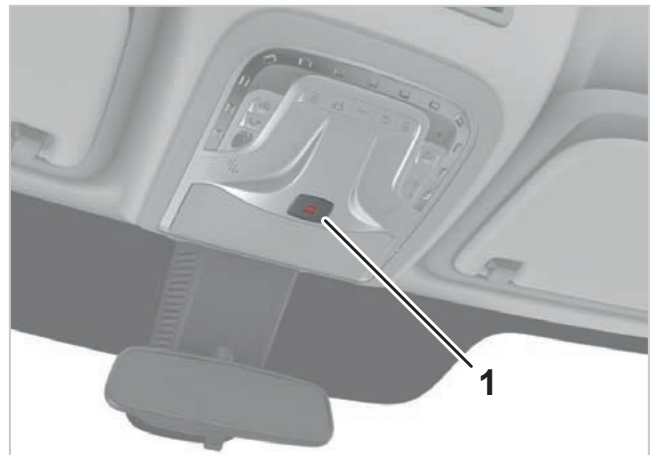


SOS pushbutton switch

Completion

On cab base vehicles without roof and rear panel (code F28), the SOS button and the associated microphone are integrated in the overhead control panel. This is supplied ex factory. For integration in the body, particular attention must be paid to the accessibility of the pushbutton switch for the driver and front passenger.

In a cowl cab base vehicle (code F50), the SOS button is located in the OCP shift module in the center of the covering shell of the center/shift console in the cockpit. The microphone for the E-call emergency call system is installed on the left-hand side of the OCP module. For a more detailed description, See 7.14.16 Overhead control panel switch module with code F50 (→ Page 245).



SOS button in overhead control panel

1 Position of SOS button

Boundary conditions

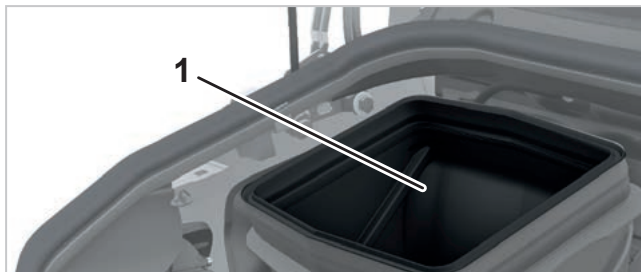
In emergency call mode, the center speaker is actuated for voice output. It is located centrally at $y = 0$ (in the center of the vehicle) at the front of the cockpit shell directly behind the windshield (F28 and standard). The bodybuilder manufacturer must take particular care to ensure that the acoustics of the center speaker are not altered by installations mounted over it, because this is a certified system. Likewise, the center speaker must not be replaced or modified in any vehicles with Mercedes-Benz emergency call system (code EY5).



Position of center speaker

7.14.18 Fresh-air intake with code F50

Cowl cab base vehicles (code F50) are not fitted with a hood ex factory. The geometry and position of the factory-installed air intake box have been adapted to the standard hood of the Sprinter in order to guarantee a fresh-air intake.



Fresh-air intake with code F50

1 Air conditioning system air intake box

Completion

Design the vehicle body so that no air is drawn in from the engine compartment (i.e. including hermetic sealing). Failure to comply with this can result in health hazards for the occupants and/or serious traffic accidents.

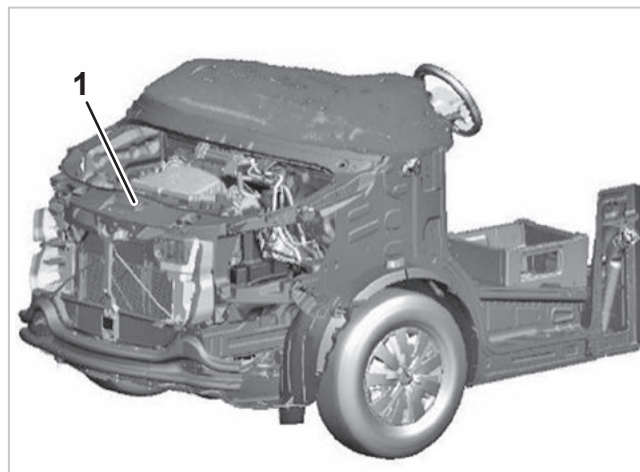
! NOTE

The pollen filters are located inside the air conditioning unit.

7.14.19 Hood contact switch with code F50

Cowl cab base vehicles (code F50) are delivered without hood, front apron, front lights, fenders, windshield, roof, A- and B-pillars, doors, rear panel as well as other parts.

A switch monitors whether the hood is closed to ensure that the ECO automatic engine start/stop system (code MJ8) and various assistance systems (including Active Distance Assist DISTRONIC code ET4) operate properly. The switch is installed and electrically connected in the hood catch in the front module carrier.

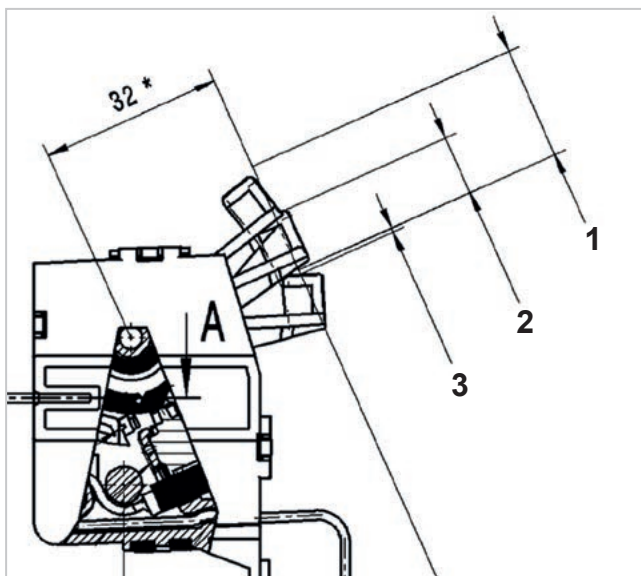


Electrical interface

1 Position of hood contact switch

Boundary conditions

The hood contact switch must not be replaced. It must be ensured that the kinematics of the replacement operating system installed by the bodybuilder manufacturer match the standard production version of the Sprinter and that the following switch points are attained and not exceeded.

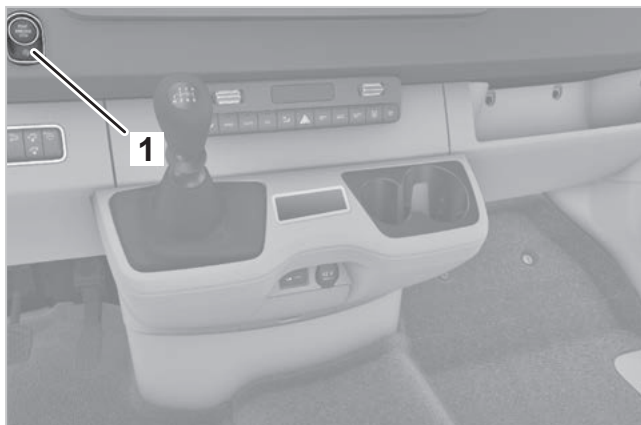


Hood contact switch switch points

- 1 Rest position 20
- 2 Switch point 10.6 ± 0.2
- 3 Permissible end position 0.8

7.14.20 Reduced Keyless-Start detection for bodybuilder manufacturer conversion with code F50

Sprinter vehicles are equipped as standard with the Keyless Start system. Drive authorization is obtained by pressing the start/stop pushbutton switch (see illustration) by means of Keyless Start.

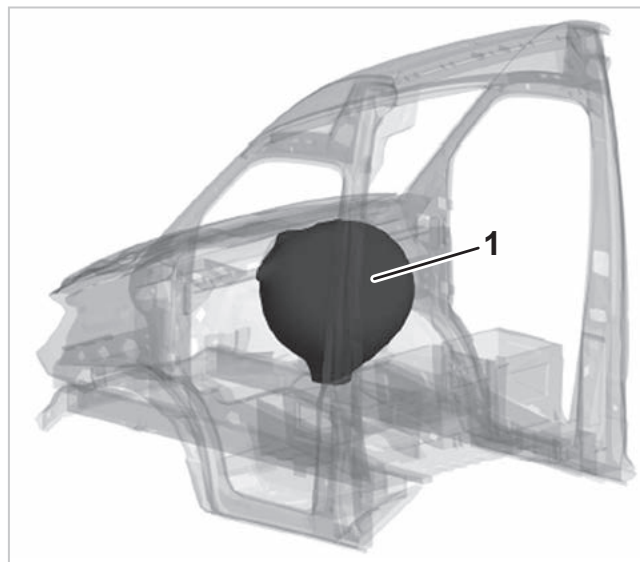


Sprinter center/shift console

- 1 Start/stop button

Boundary conditions

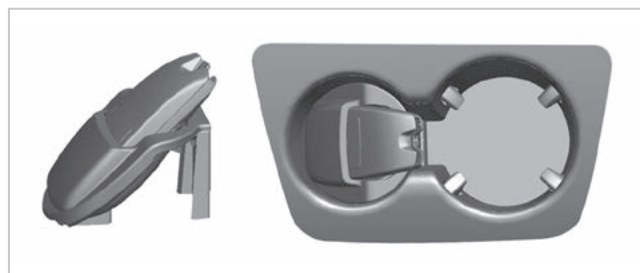
On vehicle bodies based on a cowl cab base (code F50), the non-standard body environment means that there is a danger that the Keyless Start system may not be able to make a correct and verified differentiation between the interior and the exterior. To ensure that the key is inside the vehicle during every trip, the signal range of the radio frequency antenna in the cockpit of cowl cab base vehicles (code F50) has been reduced to a smaller radius. This equipment is added automatically ex factory via code FX8, reduced Keyless-Start detection for bodybuilder manufacturer conversions, to every cowl cab base vehicle (code F50). The following illustration shows the reduced signal range of the radio antenna in the center cockpit area in which the radio signal of the Keyless Start key can be detected.



Interior area (schematic)

- 1 Detection area of key radio antenna

For trouble-free driving operation in vehicles with this reduced signal range for the radio antenna (code FX8), the key must always be positioned in the designated key holder. Designed specifically to be a plug-in key holder, it is part of the code FX8 scope and is installed at the factory in the small cup holder in the center console, see figures.



Key holder in the cup holder (schematic)

! NOTE

It is not permissible to modify the position of the key holder.

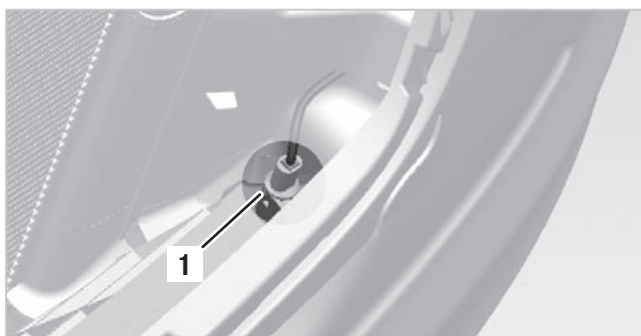


Sprinter center/shift console

1 Position of key holder

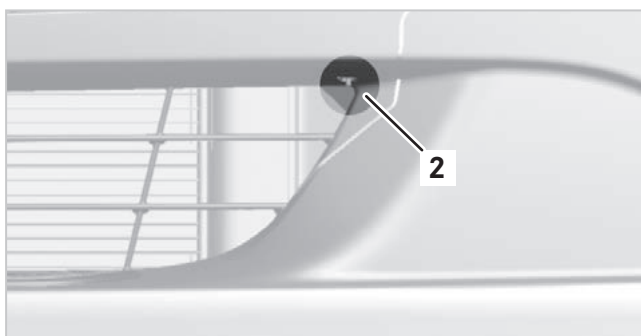
7.14.21 Temperature sensor with code F50

In the Sprinter, a temperature sensor is integrated in the front bumper on the left-hand side of the vehicle in the area of the top left corner of the air inlet grille; this has an important function for the engine timing, see figure.



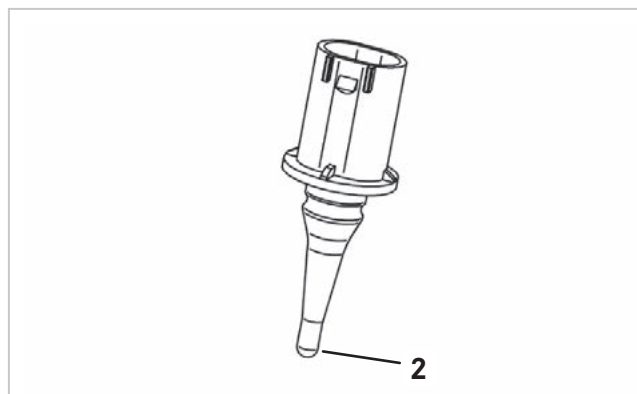
Temperature sensor position in vehicles with bumper

1 Temperature sensor in front bumper



Outer temperature sensor position

2 Sensor tip



Temperature sensor (without line)

2 Sensor tip

For all vehicles with the standard bumper, the position of the temperature sensor must not be changed and the free air flow around the sensor tip must not be restricted.

For cowl cab base vehicles (code F50) without bumper, this temperature sensor cannot be installed ex factory.

In this case, the temperature sensor with associated wiring harness is pre-fastened with a cable tie for vehicle delivery; the bodybuilder manufacturer must then position and install this.

Install the temperature sensor in a similar position to that in vehicles with bumper so that the measurement results are not influenced. Observe the following specifications in particular:

- The function of the temperature measurement must be ensured by a free flow of ambient air around the sensor tip.
- The sensor tip must not be exposed to any return flow due to heated exhaust air from the cooler or engine, as well as direct solar radiation.
- The temperature sensor must be fixed so that it is reliably installed in all driving situations, and so that the free sensor tip does not come into contact with surrounding component parts.

7.15 Camper vans

General information

Prior to conversion into a camper van, please note:

- All relevant national laws and specifications regarding the operating permit that are valid in the location of use must be complied with.
- The legal requirements for interior design and equipment for camper vans must be fulfilled.
- When installing camper-van-specific additional scopes in the area of the B-pillar, e.g. darkening systems, take into account the new trim parts as of model year 2024 for the additional measure "catch bearing" (front catch) of the load compartment sliding door in the center area of the B-pillar, See 7.4 Modifications to the panel van (→ Page 212).

! NOTE

Bodybuilder manufacturers that intend to assemble a camper van on the Sprinter must order code X2R "Mandatory for camper van conversion by bodybuilder manufacturer".

When code X2R is ordered, code B01 "Vehicle version for high load" is added automatically.

Mercedes-Benz Advanced Control (MBAC)

The MBAC is a control unit/interface module that removes the separation between the basic vehicle and the living area. It networks the components before and after the B-pillar while also enabling easy and comfortable operation. Thanks to MBAC, camper vans based on Mercedes-Benz will in future be a connected home on wheels.

The customer controls the MBAC intuitively via the MBUX multimedia system – an optional 10.25-inch display that can be positioned anywhere in the body – or with their smartphone.

MBAC only supports smartphones running on an iOS operating system.

i Further information and the complete description of the MBAC can be found in the supplemental MBAC directive in the Bodybuilder Portal.

Registration eligibility of camper vans

On leaving the factory, the cab base vehicles (F50, F28) initially only have incomplete CoC documents.

The bodybuilder manufacturer must ensure that the vehicle has complete type approval.

Refer to the information on requirements for the approval and registration eligibility of vehicles as well as on European Type Approval (ETA) and the EC Certificate of Conformity (CoC) (→ page 28).

Modifications or conversions to standard vehicles (e.g. the installation of a lifting roof) may invalidate the operating permit. The approval conditions and requirements of the European Type Approval (ETA) must be met. In Germany, modifications to the vehicle must therefore be inspected by the relevant technical authorities in accordance with Section 19, Paragraph 2 of the StVZO (German vehicle licensing regulations). After the modifications have been entered, submit the vehicle documents to the relevant vehicle licensing agency so that a new operating permit can be issued.

For the modification of vehicles in Germany, appropriate information sheets can be requested from the relevant technical inspection stations for motor vehicles (e.g. TÜV (Technical Inspection Association), DEKRA).

For the vehicle body based on code F50, assembly with a Letter of Compatibility is mandatory (code X88); See 3.3 Selecting the basic vehicle (→ Page 53).

Commissioning of control units

Observe the information and specifications for the commissioning of control units after completion of the body mounting work or conversion work, before the vehicle is placed on the market, See 8.19 Commissioning of control units (→ Page 387).

7.15.1 Body-in-white of camper vans

Boundary conditions and notes

- Ensure access to the engine, drivetrain, suspension, door mechanism (e.g. guide rails and hinges) and other components for ease of repair reasons; See 3.11 Maintenance and repair (→ Page 71).
- The standard fuel filler cap must not be removed or covered with any "blocking" parts.

! NOTE

If the fuel filler cap is removed or parts are attached to the fuel filler cap, blocking may occur in the event of an accident. Because of this, the protrusion space in the B-pillar may no longer function correctly. Do not cover the cap with trim parts, and never mount "blocking" parts on the B-pillar.

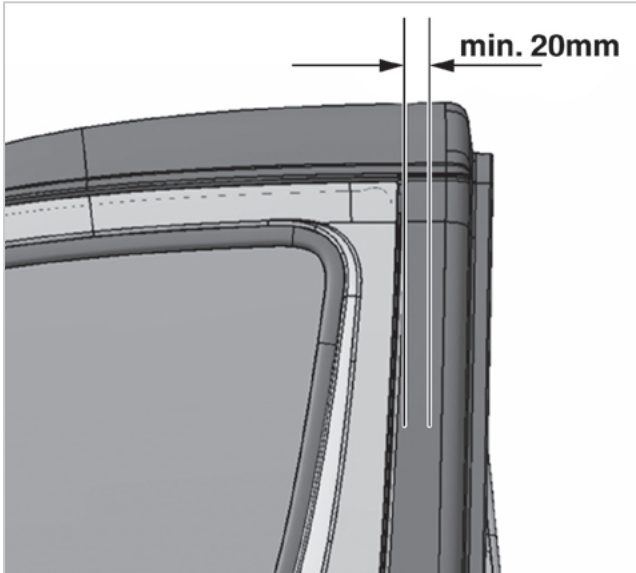
Attachment to the frame

- In the case of chassis, the attachment to the basic vehicle must use at least all of the factory-fitted mounting supports (→ page 204). These permissible body fastening points are located only on the longitudinal frame members and may be supplemented as required by additional mounting supports on the longitudinal frame members.
 - The body must be secured using two screws for each mounting support.
- i** For information on integrated camper vans, See 7.17 Semi-integrated bodies and optional mounting of free-standing bodies (→ Page 259).

! NOTE

The minimum distance between the rear edge of the door and an integrated body must be > 20 mm.

Otherwise, the rear edge of the door may come into contact with the body in the event of an accident, and in extreme cases, the door may be jammed.



Minimum distance between rear edge of door and integrated body

Pay particular attention to the following chapters:

- 3.5 Dimensions and weights (→ page 59)
- 4.2 Limit values for the suspension (→ page 80)
- 4.2.2 Permissible axle loads (→ page 82)
- 6 Modifications to the basic vehicle (→ page 106)
- 6.2.8 Fenders and wheel wells (→ page 127)
- 8 Electrics and electronics (→ page 265)
- 8.9.2 Crosswind Assist (CWA) (→ page 309)
- 8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist (→ page 339)
- 8.9.6 Automatic driving lights/rain sensor (→ page 340)

7.15.2 Suspension of camper vans

Detailed information on the suspension of camper vans is available under 4.2 Limit values for the suspension (→ page 80).

Special considerations for camper vans

Bodybuilder manufacturers that intend to assemble a camper van on the Sprinter must order code X2R "Mandatory for camper van conversion by bodybuilder manufacturer".

When code X2R is ordered, code B01 "Vehicle version for high load" is added automatically.

Make sure that the correct variant is selected when ordering the vehicle. Observe the specifications in 4.2 Limit values for the suspension (→ page 80) and the notes on the center of gravity in 9.1 Vehicle center of gravity (→ page 391).

Additional information on the basic vehicle configuration can be found in the ADVANTAGES product information system (→ page 29).

i Detailed information on cab base vehicles is available in 7.14 Bodies on chassis with cab base (code F28, F50) (→ page 227) or from the Technical Consultancy (→ page 18).

7.15.3 Electrics and electronics for camper vans

Overviews of wiring harness connecting points for series production packages and special equipment options

For more information on all connecting points, See 7.14.3 Wiring harness connecting points (→ Page 231).

Also pay particular attention to the following chapters:

- 6.5 Additional assemblies (→ page 166)
- 8 Electrics and electronics (→ page 265)
- 8.3 Battery (→ page 267)
- 8.4 Interfaces and electric lines (→ page 271)
- 8.9.2 Crosswind Assist (CWA) (→ page 309)
- 8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC (→ page 314)
- 8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist (→ page 318)
- 8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist (→ page 339)
- 8.9.6 Automatic driving lights/rain sensor (→ page 340)
- 8.9.7 Tire pressure loss warning system (→ page 342)
- 10.1 Bulb ratings of tail lamps (→ page 398)

7.15.4 Connecting point for a digital signal processor (DSP) box (code E3D)

The Digital Signal Processor (DSP box) connecting point allows the audio signals to be tapped more easily. The distribution and scaling of the speakers can be individually adapted to the surroundings. It is possible to extend an external amplifier.

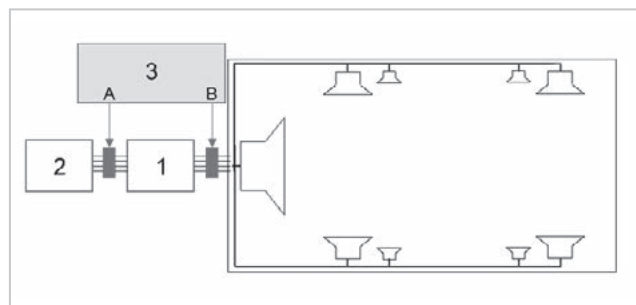
The DSP box connecting point is always installed on the right (relative to the direction of travel) and level with the A-pillar.

Boundary conditions

The rear left/right and front left/right speaker signals (coming from the head unit or radio pre-installation) as well as terminal 30T, 15 and 31 are provided ex factory in plug 1 at the A-pillar.

At plug 2, the rear left/right and front left/right speaker signals (from the DSP box) are forwarded to the speaker connecting points.

DSP box and speakers that are not installed ex factory are completed by the bodybuilder manufacturer.



Completed system (schematic)

- 1 DSP box (bodybuilder manufacturer responsible for installation)
- 2 Head unit
- 3 DSP box connecting point
- A Plug 1
- B Plug 2

7.15.5 Additional main fuses for camper vans
(codes E1R and E1Y)

Function

The main fuse box contains 4 additional fuses that are not used by the basic vehicle (code E1R and E1Y). Here, the bodybuilder manufacturer can – with the various fuse values at fuse location F9, see following table – connect defined consumers.

Three 30A output fuses are provided in the seat base, which are connected to F8 via a line, see illustration "Additional fuse for camper vans".

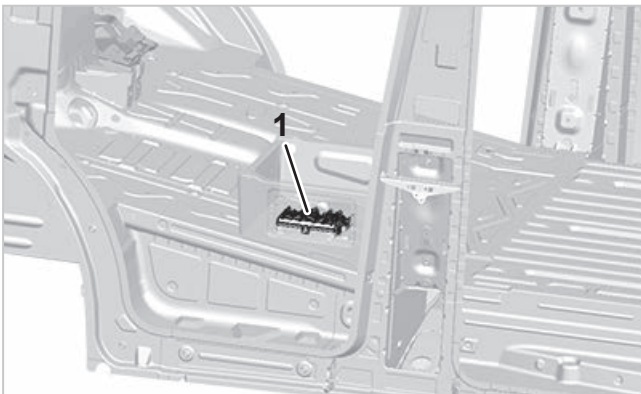
This facilitates the connection of electrical consumers to terminal 30T.

! NOTE

Do not connect any consumers to the additional main fuses (code E1R and E1Y, with connection to terminal 30T) which require a quiescent current > 1 mA when the vehicle is idle (terminal 15 or D+ inactive/engine standstill).

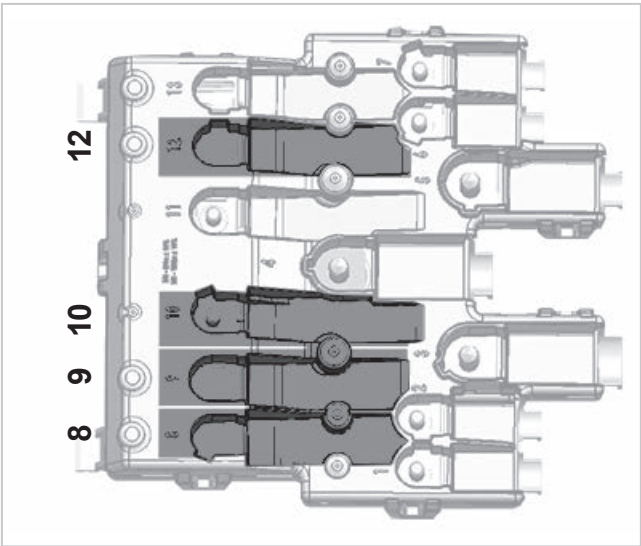
Fuse location F10 is occupied if the special equipment rear air conditioning system (code HH7, HK4 or H08) is used, and is therefore no longer available.

Fuse location F12 is occupied if the special equipment air spring (code CB2) is used, and is therefore no longer available.



Position of fuse box in model series 907 vehicle

1 Position of fuse box



Additional fuse for camper vans

Code	E1R	E1Y
Main fuse location	I _{max.} [A]	
F8	100	100
F9	150	80
F10	80	80
F12	40	40

i The fuse assignments and values that are valid for the vehicle are available in the supplement "Fuse assignment", which is part of the accompanying vehicle documents and which is available in XENTRY, See 2.2.4 Diagnosis (→ Page 26).

<https://xentry.mercedes-benz.com/home>



7.15.6 Features for camper vans

Pre-installation for switch panel (code E3J)

The Sprinter is equipped with two switch panels ex factory: A two-switch panel and a six-switch panel. The switch frames have dummy covers.

The bodybuilder manufacturer can obtain special switches via Eugen Kurz KG See 8.4.5 Operating switches (pre-installation) (→ Page 274) and deploy these, i.e. instead of installing the blind plugs and actuating via the PSM.

Advantage: By using the available switch panels and switches, bodybuilder manufacturers can implement their own functions in the Sprinter design.

-  For further information on operating switches, See 8.4.5 Operating switches (pre-installation) (→ Page 274).

Radio operating time

With the special equipment code E1X (MBUX multimedia system with longer shutoff delay), the radio operating time has been increased to 6 hours (depending on the battery charge level).

For a long radio operating time, it is recommended to fully charge the on-board electrical system battery.

For camper vans with code X2R (Mandatory for camper van conversion by bodybuilder manufacturer), this code E1X is added automatically.

NOTE

In order to use multimedia systems while stationary with an external power supply, a power feed to the on-board electrical system battery of at least 140 W is required.

Long-term startability

To ensure the starting ability even in the case of long periods of non-operational time, an extra function has been developed for the Sprinter. When the battery reaches a critical value, this function becomes active and maintains the startability of the vehicle.

Own Points of Interest in the navigation system

Bodybuilder manufacturers for camper vans have the option of integrating their own points of interest (POIs) in the Mercedes-Benz vehicle navigation system. This allows the end customer of the camper van to navigate directly to the nearest Mercedes-Benz service point or contract partner.

Pre-installation for rear speakers

A pre-installation for rear speakers (code EP7) is available to allow camper van manufacturers to connect additional speakers to the radio in their body.

For more information on this, See 7.14.15 Pre-installation for rear speakers (code EP7) (→ Page 245).

Retrofitting rear seats

For the retrofitting of seats in the rear passenger compartment, observe the information and specifications in 7.3.1 Retrofitting seats (→ page 209).

7.15.7 Airbags for camper vans

Additional airbag systems can be ordered for camper vans under the following special equipment codes:

- Driver thorax/pelvis side airbag (code SH1)
- Front passenger thorax/pelvis side airbag (code SH2)
- Window airbags for driver and front passenger (code SH9)

For the use of this special equipment in camper vans, observe the particular specifications and information in the following chapters:

- 4.5.1 Modifications in the area of the restraint systems (→ page 96)
- 5.2 Welding work (→ page 101)
- 5.4 Painting and preservation work (→ page 104)
- 6.1.4 Air suspension (→ page 110)
- 6.2.1 General information on the body-in-white (→ page 112)
- 6.2.2 Attachments on the frame (→ page 117)
- 6.4.1 General information (→ page 151)
- 6.4.2 Safety equipment (→ page 152)
- 6.4.3 Seats (→ page 160)
- 7.4 Modifications to the panel van (→ page 212)
- 7.14.7 Connecting point with omission of driver's and front passenger seat for pre-installation for seat belt warning system (code J1D/J1E) and seat occupancy recognition (code E3S) (→ page 237)
- 7.14.14 Pre-installation for a digital signal processor (DSP) box (code E3D) (→ page 244)

Additional requirements for window airbags (code SH9)

- Code SH9 can only be ordered in combination with code F28 or for closed model designations.
- Side window blackout curtains or awnings on side windows are not permitted in combination with this equipment.
- Installed windshield blackout curtains/awnings/shelves and their retaining systems must not impede the deployment areas of the airbags.
- All work on an A-pillar with window airbags is prohibited.
- Holders on the A-pillar trim are not permitted.

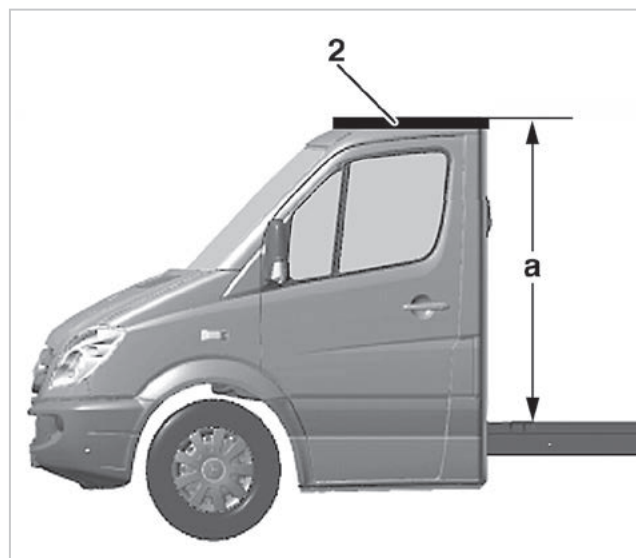
Additional requirements for thorax/pelvis side airbags (code SH1/SH2)

- Codes SH1/SH2 can only be ordered in combination with code F28 or for closed model designations.
- Installed window blackout curtains/awnings and their retaining systems on driver/front passenger side windows must not impede the deployment areas of the airbags.
- The use of armrests on both sides of the seat is not permitted in combination with this system.
- Bodybuilder-manufacturer-specific seat covers or seat protectors (protective covers) are only permitted in combination with the thorax/pelvis side airbag system with a verification that the airbag deployment does not deviate from the series production configuration, See 6.4.3 Seats (→ Page 160). Submit corresponding evidence of this upon request.
- Safe routing of the airbag ignition lines must be guaranteed by means of dedicated seat swivel consoles (no twisting/pressure points on ignition lines).

7.16 Bodies on chassis with lowered roof

A lowered roof (code FA1) is available for semi-integrated and alcove bodies on cab base vehicles with doors (code F28). Code FA1 comprises the following changes from the standard version:

- The height of the roof is reduced by approx. 70 mm.
- The vehicle is cut in the area of the roof/door portal, reinforced by means of body-in-white modifications and then painted in the area of the body-in-white modifications.
- The standard-equipment sun visors and grab handles are mounted in the same position at modified attachment points.
- The headliner can be attached at the previous attachment points, but must be trimmed and adjusted to fit the interior at the front and sides by the bodybuilder manufacturer.
- The vehicle is fitted with a temporary roof bow for transportation.
- Before the body is erected, it is necessary for the bodybuilder manufacturer to install the auxiliary roof frame supplied at the attachment points provided in order to ensure adequate equivalent rigidity. This provides a lowered attachment plane for bodies.



Location of auxiliary roof frame

- 2 Auxiliary roof frame
- a Distance between top edge of longitudinal frame member and top edge of auxiliary roof frame

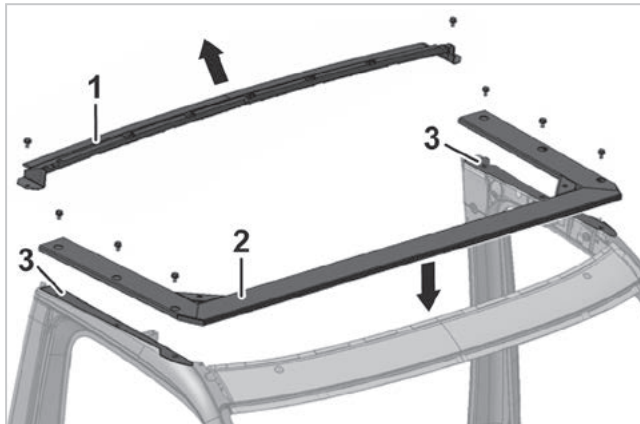
Gross combination mass [t]	Distance a [mm]
3.0–4.0	1560
5.0	1540

! NOTE

Observe and adhere to the information and specifications on body limits for driving assistance systems, See 8.9 Driving assistance systems (→ Page 307).

7.16.1 Fitting the auxiliary roof frame

- ▶ Detach the temporary roof bow (1) before mounting the auxiliary roof frame (2).
- ▶ Then mount the auxiliary roof frame (2) at the intended points using six round head bolts M10 × 20 - 10.9 (tightening torque 40 Nm ± 2 Nm).



Mounting of auxiliary roof frame

- 1 Transport roof bow
- 2 Auxiliary roof frame
- 3 Cutting area with reinforcements (body-in-white modifications)

7.16.2 Fitting the body to the auxiliary roof frame

The body can be attached to the auxiliary roof frame by:

- Screws
- Rivets
- Adhesive bonding
- Welding

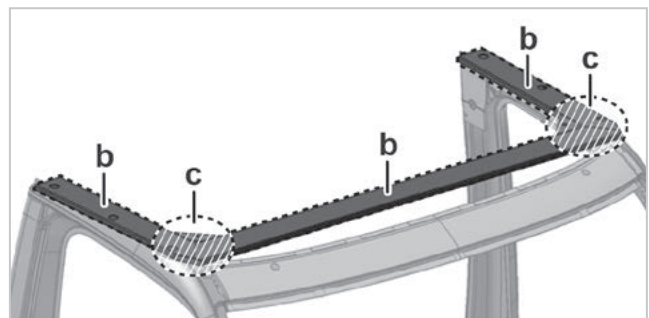
! NOTE

Holes must not be drilled in the corners of the auxiliary roof frame.

The auxiliary roof frame must not be cut.

The introduction of force to the auxiliary roof frame must occur by way of an area load (line load). Point loads must not be introduced into the auxiliary roof frame.

The load on the auxiliary roof frame while driving must not exceed 100 kg. When the vehicle is stationary, a load of 200 kg is permissible.



Attachment areas for auxiliary roof frame on vehicles with lowered roof

- b Attachment area
- c No drilling allowed

Please also observe 5.3 Anticorrosion protection measures (→ page 102). If you have any questions, please contact the Technical Consultancy team (→ page 18).

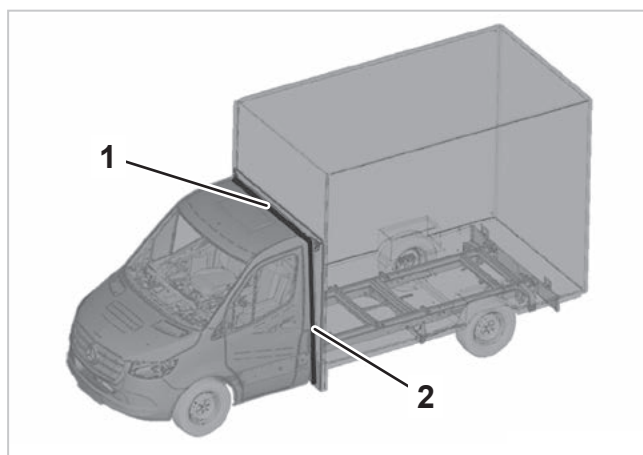
7.17 Semi-integrated bodies and optional mounting of free-standing bodies

A non-positive connection between cab and body is required on vehicles with semi-integrated bodies, e.g. semi-integrated camper vans, integral box bodies etc.

For vehicles with free-standing bodies (e.g. box bodies), it is also possible to optionally connect this to the cab. For this, the same specifications apply as for semi-integrated bodies. The body must have adequate inherent rigidity in the attachment area (avoid local compression).

The connection between the cab and the body must be continuous at the B-pillars and the roof frame (either full length or in sections). For a concrete implementation as well as other measures, contact the Technical Consultancy (→ page 18).

The descriptions apply for cabs both with and without a rear panel.



Attachment of body to B-pillar and roof using example of free-standing body (schematic)

- 1 Roof
- 2 B-pillar

Attachment of body to B-pillar (z-axis)

The body side wall must always be connected to the B-pillar. The connection between the body and the basic vehicle must be non-positive.

It must be assured that forces are transmitted between the body and the B-pillar. This can be achieved by e.g.:

Variant 1

Attachment of body to B-pillar via a stay plate with

$$t = 2 \text{ mm angled at approx. } 2 \times 45^\circ$$

The stay plate must be bonded across its entire surface area.

In addition to full-surface bonding, rivets can also be used for fastening purposes.

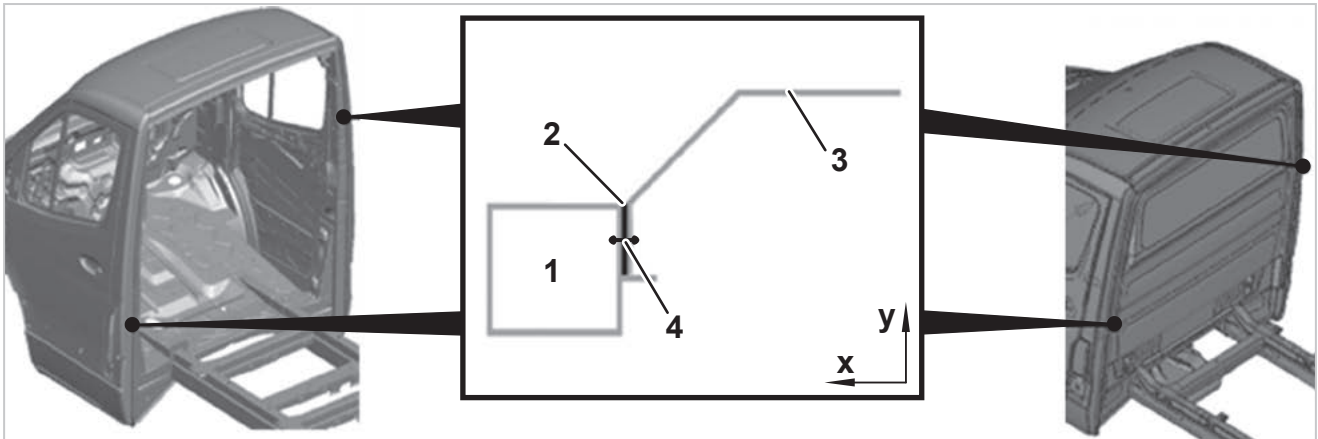
Observe the following specifications in this respect:

- A rivet connection without full-surface bonding is not permitted.
- Only use steel blind rivets.
- Maximum rivet diameter is 6 mm
- When riveting, only apply a rivet force in the longitudinal direction, not in the radial direction.
- Only position the bores and rivets on the rear of the B-pillar (opposite to the direction of travel).
- A maximum of 6 bores are permissible at regular intervals between the top of the roof frame and the bottom connection between the B-pillar and floor/frame.
- To drill the bores, use a drilling template with depth limiter set according to the sheet thickness.
- Under no circumstances must internal component parts (e.g. parts of the seat belt or left tank filler neck) be damaged by the drilling work.

7 Design of bodies

- Fully suction off all drill chips.
- When drilling holes in the vicinity of the seat belt retractor (height approx. $Z = 600$ mm in 3D design data), remove this component part, store it safely and without damage, and re-install it once drilling is complete.

 For more information and details, contact the Technical Consultancy (→ page 18).

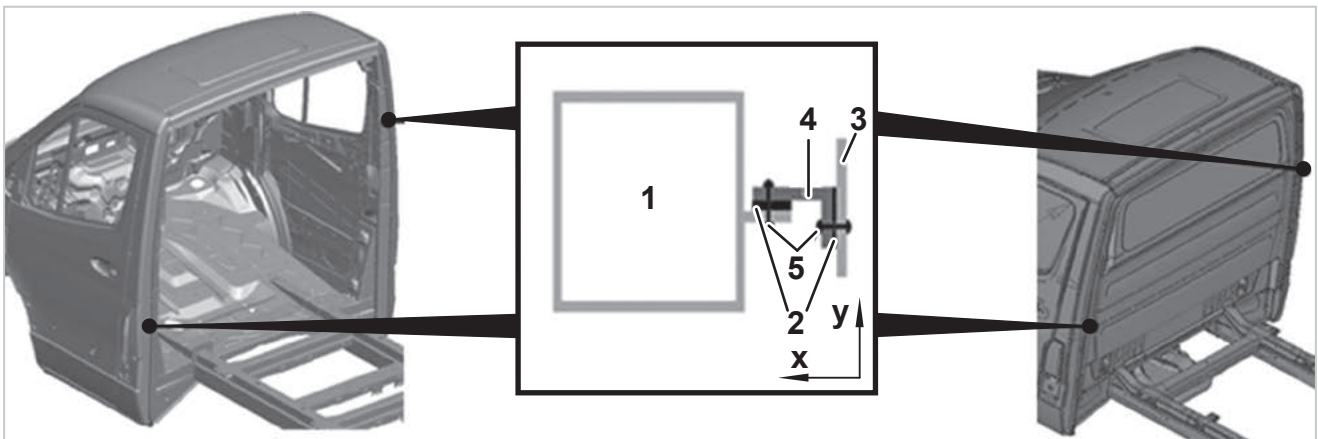


Variant 1: Attachment of body to B-pillar via stay plate (example with and without rear panel)

- 1 B-pillar
- 2 Bonding flange
- 3 Stay plate
- 4 Rivet (optional extra)

Variant 2

Attachment of body to welding flange of B-pillar with angle pieces



Variant 2: Attachment of body to B-pillar via bracket (example with and without rear panel)

- 1 B-pillar
- 2 Bonding flange
- 3 Firewall of body
- 4 Angle piece
- 5 Rivet

Attachment of body to roof frame (y-axis)

In addition to the connection between the body side wall and the basic vehicle, it is necessary to form a non-positive connection between the body and the basic vehicle in the area of the roof frame on vehicles with integral bodies. This can be achieved by e.g.:

Variant 1

Attachment of body to roof frame via a stay plate with
 $t = 2 \text{ mm}$ angled at approx. $2 \times 45^\circ$

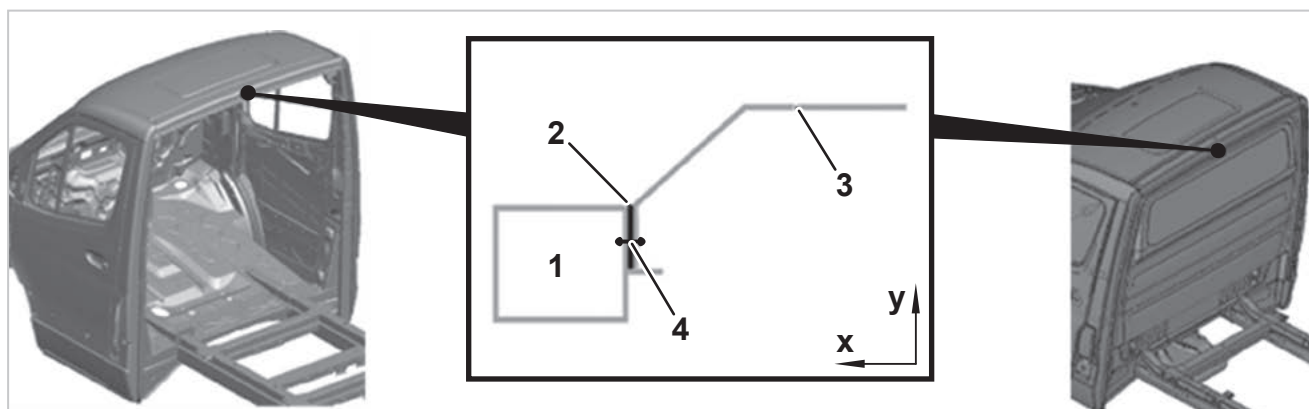
The stay plate must be bonded across its entire surface area.

In addition to full-surface bonding, rivets can also be used for fastening purposes.

Observe the following specifications in this respect:

- A rivet connection without full-surface bonding is not permitted.
- Only use steel blind rivets.
- Maximum rivet diameter is 6 mm
- When riveting, only apply a rivet force in the longitudinal direction, not in the radial direction.
- Only position the bores and rivets on the rear of the roof frame (opposite to the direction of travel).
- A maximum of 6 bores are recommended at regular intervals between the B-pillar on the left and right.
- To drill the bores, a drilling template with depth limiter is recommended.
- Fully suction off all drill chips.

 For more information and details, contact the Technical Consultancy (→ page 18).

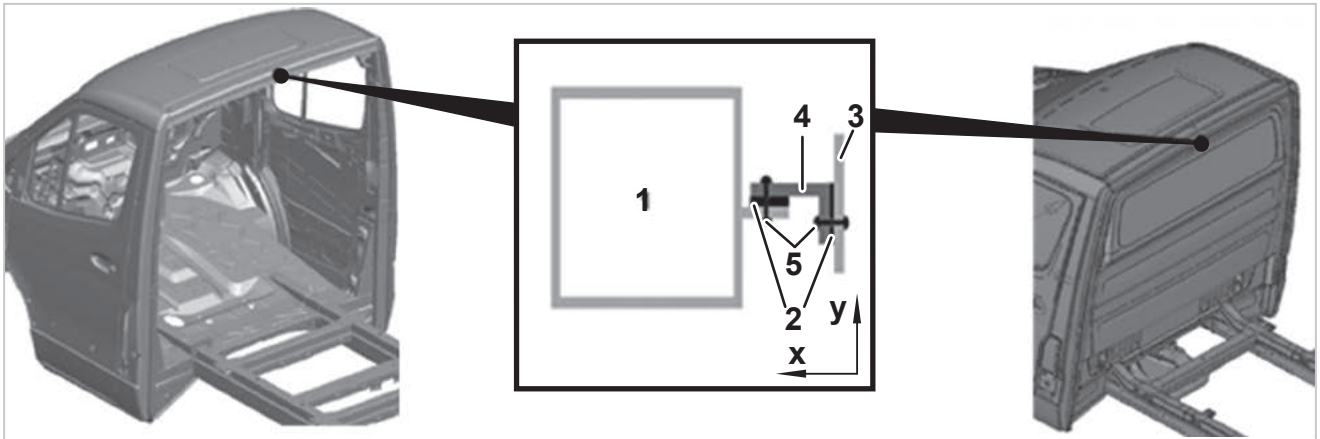


Variant 1: Attachment of body to roof frame via stay plate (example with and without rear panel)

- 1 Roof frame
- 2 Bonding flange
- 3 Stay plate
- 4 Rivet (optional extra)

Variant 2

Attachment of body to welding flange of roof frame
with angle pieces



Variant 2: Attachment of body to roof frame welding flange via bracket (example of with and without rear panel)

- 1 Roof frame
- 2 Bonding flange
- 3 Firewall of body
- 4 Angle piece
- 5 Rivet

On vehicles with a cut roof frame, the bodybuilder manufacturer must ensure force flux to the simulating structure (→ page 132). For this, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

7.18 Buses

General information

Before conversion to buses, observe the country-specific statutory regulations for the operating permit as a bus of vehicle class M2 and M3, see Regulations (EU) 2018/858 Annex II and 2007/46/EC Annex IV. In addition, observe all UN regulations and/or other legal specifications from country-specific regulations for converting to buses for the transport of persons.

Further general technical information:

- In the case of a planned conversion of a panel van with N2 registration to a vehicle in classes M2 or M3, observe the note in 3.3 Selecting the basic vehicle (→ page 53) and the laws and regulations that apply to this vehicle class.
- Ensure an equivalent simulating structure in the body-in-white (for modifications to the roof and roof bow structure).
- Ensure that the installation is free of tension.
- Agree conversion solutions with Mercedes-Benz AG (operational durability, seat modifications, etc.).
- For information on using a retarder, see 6.5.5 Auxiliary brakes (retarders) (→ page 176) and the description of the electrical interface at 8.12 Pre-installation for retarder (→ page 377).

Body-in-white

If the body-in-white is modified or cut, e.g. to lower the vehicle floor in order to gain cargo capacity, to install emergency exits or similar, the anticorrosion protection measures specified by Mercedes-Benz AG must be carried out and all specifications in the current Body/Equipment Mounting Directive regarding modifications to the body-in-white must be observed.

Pay particular attention to the following chapters:

- 4.3.1 Modifications to the body-in-white (→ page 85)
- 4.3.2 Limit values for the vehicle frame (→ page 85)
- 4.3.7 Wheelbase modifications (→ page 89)
- 4.3.8 Roof and roof load (→ page 90)
- 6.2 Body-in-white and body (→ page 112)
- 6.3.3 Exhaust system (→ page 141)

Installing own windows

If the bodybuilder manufacturer intends to install their own windows, code W94 "No windows in bus version" is available from the factory. Do not install panoramic glazing on a panel van basic vehicle by cutting out the stamped window shape (without code W78) without ensuring equivalent rigidity.

Observe the information and specifications on cutting the side wall for the installation of windows as well as the obligation to apply for a Letter of Compatibility when modifying structural components; See 6.2.7 Side wall, windows, doors, and flaps (→ Page 123).

Retrofitting an air conditioning system

See 6.5.1 Retrofitting an air conditioning system (→ page 166).

- The additional belt pulley, code N63, is available ex factory as special equipment for driving refrigerant compressors.
- All electrical equipment installed must be tested in accordance with EC Directive 72/245/EEC and must bear the EC mark.
- If other air conditioning systems are to be retrofitted, observe the guidelines issued by the device manufacturer.
- Do not impair vehicle parts or their function by installing an air conditioning system.
- If compact systems (evaporator, condenser, and blower) are mounted on the cab roof, do not exceed the permissible roof loads: 4.3.8 Roof and roof load (→ page 90).
- For attachments at the roof, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- Ensure that pipes and electric lines are routed correctly.
- More information on retrofitting air conditioning systems is available from the Technical Consultancy (→ page 18).

Interior

For all modifications to the interior, take into account:

- Interior equipment
- General bus directives

See the following chapters on the vehicle interior:

- 4.5.1 Modifications in the area of the restraint systems (→ page 96)
- 4.5.2 Modifications to seats (→ page 96)
- 6.4 Interior (→ page 151)

Doors and entrances

All modifications to doors or entrances must comply with the specifications in the following chapters:

- 8.8 Windows and doors (→ page 305)
- 6.2.7 Side wall, windows, doors, and flaps (→ page 123)
- 6.4.2 Safety equipment (→ page 152)

Further information on subsequent modifications, e.g. to the body-in-white, interior as well as doors and entrances, is available from the Technical Consultancy (→ page 18).

8.1 General information

⚠ WARNING

Modifications to electronic component parts, their software or wiring can impair their functioning and/or the functioning of other networked components. Safety-relevant systems in particular may also be affected. Because of this, they may no longer function properly and/or compromise the operational safety of the vehicle.

There is risk of an accident and danger to life and limb!

Never carry out any modifications to the wiring and electronic component parts or their software. Have all work on electrical and electronic devices carried out at a qualified specialist workshop.

! NOTE

Ensure a positive total charge balance when additional electrical consumers are installed (See 7.17 Semi-integrated bodies and optional mounting of free-standing bodies (→ Page 259) and 8.4.6 Retrofitting of electrical devices (→ page 275)).

Do not release or remove the battery clamps when the engine is running. If the vehicle is equipped with code E30 (1-pin battery main switch), the isolator must not be actuated while the engine is running. The information and specifications in the Operator's Manual for your vehicle must be adhered to.

Rapid-charge the batteries only after disconnecting them from the vehicle's electrical system. Both the positive and negative clamps must be disconnected.

! NOTE

When carrying out work on the electrical system of the basic vehicle, e.g. at the electrical interfaces in the seat box (retrofitting of electric lines and/or components), ensure that the lines are routed appropriately and that the electrical connections and fasteners are appropriately designed.

If electrical installation or modification work is not executed correctly, this may result in unnecessary fault sources and/or considerable additional work in the event of a repair; the cost of such additional work shall be borne by the bodybuilder manufacturer that executed the installation/modification work (e.g. swapped connections, damage to the lines, or fire damage as result of insufficient clearances).

- Some of the safety systems only function when the engine is running. For this reason, do not switch off the engine when the vehicle is in motion.
- Electrical and electronic components must fulfill the test requirements of ISO 16750.
- Observe the following information (→ page 267) when installing auxiliary batteries.
- Cables routed in the vicinity of exhaust systems must be insulated against high temperatures (→ page 141).
- Cables must be routed in such a way that there are no chafing points (→ page 109).
- Prior to extended periods of non-operational time, take into account 3.11.2 Maintenance and storage of batteries (→ page 72).

- As of model year 2024, in accordance with legal regulations, vehicles in markets with validity GSR-II (General Safety Regulation) shall have a pre-installation installed that allows for the retrofitting of an alcohol immobilizer as per EN 50436 in qualified specialist workshops. Upon completion, this is to be commissioned in XENTRY (O-Code 000) by the specialist workshop.
-  Contact your Mercedes-Benz Service Partner if such a measure is necessary for your target market or end customer.

 **NOTE**

The bodybuilder manufacturer is responsible for the functional safety of the body mounting work it performs in compliance with the applicable standards and regulations. Where applicable, the bodybuilder manufacturer must also ensure compliance with the international standard ISO 26262 regarding functional safety along with the international standard ISO 21434 and UN R 155 regarding cyber security; See 2.3 Product safety and product liability (→ Page 37).

Commissioning of control units

Observe the information and specifications for the commissioning of control units after completion of the body mounting work or conversion work, before the vehicle is placed on the market, See 8.19 Commissioning of control units (→ Page 387).

8.2 Electromagnetic compatibility (EMC)

Electromagnetic compatibility describes the ability of an electrical system to act neutrally in the vicinity of other systems when operating at full function. The system in question does not interfere with any of the active systems in the vicinity, nor does it suffer any interference.

Electrical interference occurs in vehicle on-board electrical systems because of the various consumers.

Mercedes-Benz tests all factory-installed electrical and electronic components for their electromagnetic compatibility in the vehicle.

Subsequent modifications may impair comfort-related aspects (e.g. radio noise) or result in malfunctions (e.g. government radio, high beam, turn signal indicators). The body of the bodybuilder manufacturer can also be influenced by EMC (e.g. by antennas).

 **WARNING**

In isolated cases, safety-critical conditions (e.g. influence on the accelerator pedal) may also occur.

There is danger to life and limb!

Therefore, observe and adhere to the following note.

! NOTE

In accordance with legislation (in Germany, Article 14 of the Electromagnetic Compatibility Law (EMVG), or other national regulations), the electromagnetic compatibility of electrical and electronic systems must be ensured when such systems are retrofitted.

It is recommended to coordinate the assessment of the EMC with a Technical Service for EMC.

For EMC relevance, the bodybuilder manufacturer must apply for certification as per UN R 10 with a Technical Service for EMC.

All electrical devices installed must have been tested in accordance with UN R 10 and must bear the CE marking and/or another comparable marking depending on the country-specific regulations.

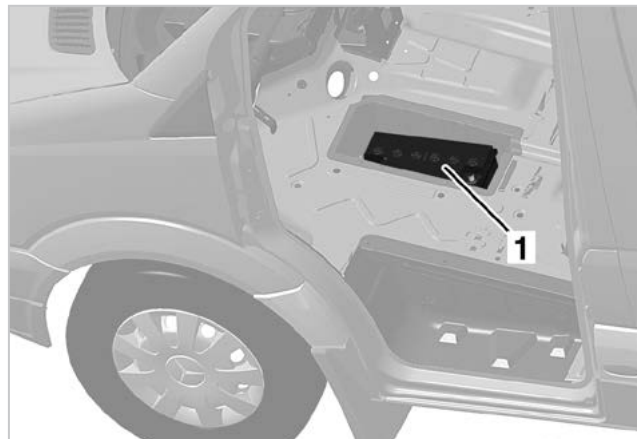
Observe the information in 8.4.6 Retrofitting of electrical devices (→ page 275), 8.6 Mobile communication systems (→ page 299), and 4.6 Limit values for electrics and electronics (→ page 97).

The following standards provide information on this:

- CISPR 12
- CISPR 25
- DIN EN 55012
- DIN EN 55025
- ISO 7637
- ISO 10605
- ISO 11451
- ISO 11452
- MBN 50284
- UN R 10

8.3 Battery**8.3.1 On-board electrical system battery**

The on-board electrical system battery is located in the floor on the left-hand side, to the front of the driver's seat.



Installation position of on-board electrical system battery

1 On-board electrical system battery

The following on-board electrical system batteries are available ex factory depending on the equipment (model AGM):

Series/option	Code	Size	Capacity [Ah]
Series	ED1	H6	70
Option	ED4	H8	92

In the event of high electricity demand and current collection for additional electrical consumers, use an auxiliary battery, See 8.3.2 Auxiliary battery (→ Page 269).

Always use an AGM battery on vehicles with ECO automatic start/stop system "plus" (code MJ8).

! NOTE

No electric lines are permitted to additionally be connected directly to the negative terminal of the on-board electrical system battery.

The positive terminal of the on-board electrical system battery must not be modified and connecting additional electrical consumers there is not permissible.

! NOTE

The vehicle systems (standard and factory-fitted special equipment) are powered by the on-board electrical system battery.

The on-board electrical management system monitors the voltage of the on-board electrical system battery and stops the function of the vehicle systems when the state of charge drops below the 40% limit, in order to preserve the startability of the vehicle. The electronic ignition lock and the tachograph are not affected by this.

Where an auxiliary battery is used as per 8.3.2 Auxiliary battery (→ page 269), the consumers connected to it are also not impaired in their function.

If one or more vehicle systems are operated while the vehicle is stationary (e.g. audio devices or interior lamps), note the state of charge and the limited capacity of the on-board electrical system battery.

External power supply in the 12 V on-board electrical system

Time-restricted charging of the on-board electrical system battery or permanent power supply through an external power supply into the on-board electrical system battery may only take place with a charger in compliance with the following specifications:

- A connection to the on-board electrical system is only permitted via terminal 30.
- The jump-start connection point in the engine compartment is recommended for charging the on-board electrical system battery.
- An external power supply is only permitted during engine standstill (terminal 15 or D+ inactive).
- The charger must be approved for AGM batteries and must also be restricted to the following values:
 - Start charging voltage maximum 14.8 V
 - Quiescent current consumption below 1 mA
- It is essential that the charging process or external power supply complies with DIN EN 50342-1, Section 5.1.
- The charging cable must be secured immediately at the connection point to the on-board electrical system with a fuse in accordance with the specifications in 8.4.2 Electric lines and electrical fuses (→ page 272).
- Any other types of external power supply, for example via solar-powered chargers (solar controllers), are not permissible for charging the on-board electrical system battery.

i Before starting the charging process, familiarize yourself with the specifications and further information in the Operator's Manual for the vehicle and charger.

8.3.2 Auxiliary battery

General

! NOTE

Capacities > 100 Ah must not be connected directly to the on-board electrical system, because this may damage the basic vehicle.

A positive total charge balance must be ensured through selection of a suitable alternator.

When consumers are powered by the auxiliary battery, ensure that the permissible total current of 80 A across the battery cutoff relay in total is not exceeded. Take into account the battery charging current with the engine running here.

If the total current is higher, even for a short time (peak), contact the Technical Consultancy (→ page 18).

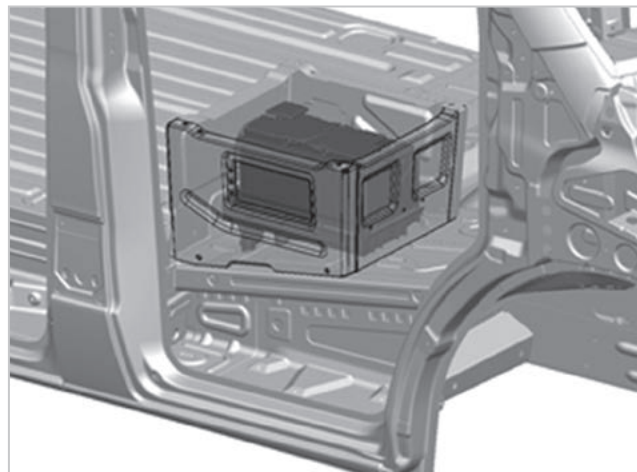
Use only the same type of battery (AGM) for the on-board electrical system battery and auxiliary battery.

It is recommended to always use the special equipment options offered by Mercedes-Benz AG ex factory if these are required for an auxiliary battery.

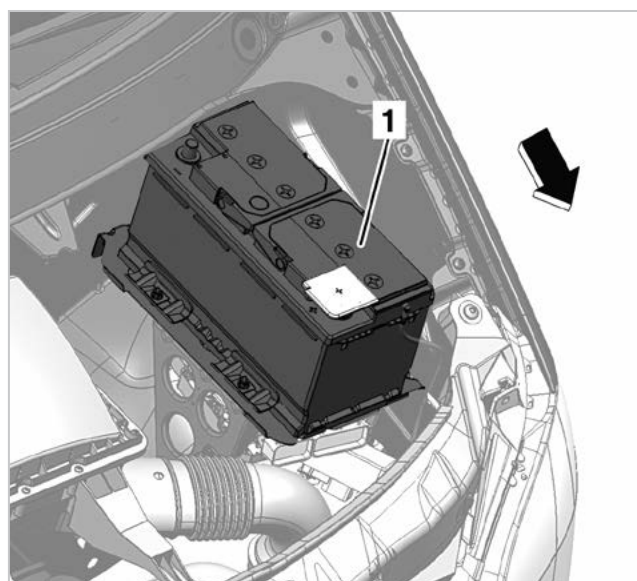
Code	Size	Capacity [Ah]	Installation location
E2I	H8	92	Front passenger seat box
E2M	H6	70	Engine compartment

The required battery cutoff relay (code E36) is installed automatically along with these auxiliary batteries.

In order to facilitate current collection from the auxiliary battery (code E2I or E2M), a current collector (fuse box) is available in the driver seat frame. This means that it is not necessary to route additional lines from the vehicle interior (bodybuilder manufacturer consumers) to the auxiliary battery in the engine compartment; See 8.4.8 Current collection (→ Page 277).



Auxiliary battery in the front passenger seat box (code E2I), example for left-hand drive vehicle



Auxiliary battery in the engine compartment (code E2M), example for left-hand drive vehicle

1 Auxiliary battery (code E2M)

➡ Direction of travel

If the vehicle needs to be started using an external power supply, the jump-start connection point in the engine compartment must be used.

- Information and specifications on how to use the jump-start connection point correctly can be found in the Operator's Manual for your vehicle.

! NOTE

The auxiliary battery (code E2I, E2M) or other auxiliary batteries connected to the on-board electrical system via the battery cutoff relay must not be used to start the vehicle via an external power supply, as this could result in damage to the vehicle.

Retrofitting auxiliary batteries

Auxiliary batteries specific to the bodybuilder manufacturer must be connected to the vehicle on-board electrical system using the following battery cutoff relay and a corresponding fuse.

! NOTE

The necessary special equipment code E36 must be taken into account when ordering the vehicle to ensure that the battery cutoff relay is installed in the factory – retrofits are not possible.

- For more information, contact the Technical Consultancy (→ page 18).

When installing an auxiliary battery, ensure that the same type of battery (AGM) is used as the on-board electrical system battery.

If the auxiliary battery is located in the passenger compartment, sufficiently dimensioned ventilation must be realized through a central vent hose leading into the atmosphere. If necessary, block the 2nd duct in the battery with a stop plug.

The auxiliary battery must only be used to power auxiliary consumers such as the stationary heater, loading aids, or electrical equipment in camper vans (refrigerator, etc.).

! NOTE

If the vehicle is already equipped with an auxiliary battery, no further auxiliary batteries may be connected in parallel without a charging current limiter. This can be realized through supplementary electronics. A maximum charging current for both auxiliary batteries of 80 A in total must be ensured by the bodybuilder manufacturer. If this is not the case, the basic vehicle may be damaged.

A positive total charge balance is to be assured through selection of a suitable alternator.

For the retrofitting of more than one auxiliary battery, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Charging the auxiliary battery by external power supply

An external power supply to charge the auxiliary battery with battery cutoff relay code E36 via a suitable charger or a solar controller is only permitted to be used during engine standstill (battery cutoff relay open, or terminal 15 or D+ inactive).

8.3.3 Battery maintenance and storage

Batteries must be regularly checked for voltage drop (self-discharge) – even when removed. With low-maintenance batteries, only the fluid level does not need to be checked.

- ❗ Further information on maintaining and storing batteries is available under 3.11.2 Maintenance and storage of batteries (→ page 72).

8.3.4 Disconnecting the on-board electrical system voltage

Whenever the vehicle is to be stored for long periods of non-operational time – See 3.11 Maintenance and repair (→ Page 71) – or if work is to be carried out on electrical systems, it is recommended that the on-board electrical system be de-energized by disconnecting the 12 V power supply at the terminals of the on-board electrical system battery and auxiliary battery or by actuating the single-terminal battery main switch (special equipment code E30).

- ❗ For this, observe the information and specifications regarding the procedure in the operator's manual of your vehicle.

8.4 Interfaces and electric lines

8.4.1 CAN bus and networking

⚠ WARNING

As all consumers are networked and internally monitored by the system, no modifications are permitted to be made to the CAN bus (e.g. interrupting, extending, or tapping). Any change to the subharness in terms of length, cross-section, or resistance can result in safety-relevant component parts failing or comfort impairments.

Internal and external vehicle diagnosis can be carried out by means of the OBD diagnostics connection (SAE 1962). Every control unit is capable of self-diagnosis and is equipped with an internal fault memory.

Communication with the control units can be established using a diagnostic device and the software developed for this unit. At Mercedes-Benz, this is the "XENTRY" diagnosis system with the associated hardware and software, the XENTRY Kit.

- ❗ You can obtain further information from your Mercedes-Benz Service Partner.

8.4.2 Electric lines and electrical fuses

If the routing has to be altered, then avoid routing across sharp edges and through narrow gaps or near moving parts.

! NOTE

All existing line routings in the vehicle must not be impaired by any modification measures in terms of properties such as current and signal transmission, insulation, EMC, component part quality, line attachment, etc. (incomplete list).

! NOTE

When carrying out work on the electrical system of the basic vehicle, e.g. at the electrical interfaces in the seat box (retrofitting of electric lines and/or components), ensure that the lines are routed appropriately and that the electrical connections and fasteners are appropriately designed.

If electrical installation or modification work is not executed correctly, this may result in unnecessary fault sources (e.g. confusing connections, damage to lines, or risk of fire as a consequence of insufficient distances) and/or considerable additional work in the event of a repair. The cost of this shall be borne by the bodybuilder manufacturer.

Only lead-free PVC-jacketed cables with an insulation operating temperature limit > 105 °C must be used. Make professional and watertight connections.

Dimension the line in accordance with the current level drawn and protect it with fuses.

The following table applies to cables with an insulation operating temperature limit of > 105 °C.

Maximum per- manent current strength [A]	Fuse rating [A]	Line cross-sec- tion [mm²]
0–4.9	5 ¹⁾	0.5
5–9.9	10 ¹⁾	1
10–18	20 ¹⁾	2.5
19–28	30 ¹⁾	4
29–35	40 ²⁾	6
36–48	50 ²⁾	10
49–69	70 ²⁾	16
70–98	100	25
99–123	125	35
124–148	150	50

- 1) Type C; DIN 72581 male spade connector
- 2) Type E; DIN 72581 male spade connector

i The fuse assignments and values that are valid for the vehicle are available in the supplement "Fuse assignment", which is part of the accompanying vehicle documents and which is available in XENTRY, See 2.2.4 Diagnosis (→ Page 26).

<https://xentry.mercedes-benz.com/home>



8.4.3 Cable extension

Use the same or a higher cable cross-section for cable extensions. The use of lines complying with LV112 is recommended for this. The protective effect of fuse elements must not be impaired.

Secure connections between electric lines and electrical components (e.g. plugs) against rocking to and fro, through the use of suitable mounting material.

Make all connections professionally and watertight in accordance with IP69k (protected for high-pressure cleaning).

Adjust lines to the ABS sensors on the rear axle to suit the specific vehicle; they are permitted to be lengthened by no more than 2.7 m. Twist the lines to each sensor with a loop length of 30 ± 5 mm.

Lengthen wiring harnesses of comfort-related and passive safety systems, the on-board electrical system and bus system

The Mercedes-Benz work instructions on the repair of cables in the Workshop Information System (→ page 26) also apply for the lengthening of cables. Work on the vehicle must always be carried out using the latest workshop aids (e.g. EPC net, WIS net, XENTRY, special tools) from the manufacturer for the vehicle in question. Observe the safety notes and safety precautions and take precautions against electrostatic discharge (ESD). Information on the safety regulations and safety precautions can be found in the brochure: "Risk factor ESD, understanding and avoiding electrostatic discharge" (order no. 6516 1310). You can request this from your Mercedes-Benz Service Partner. Outside of Germany, contact the designated partner who is responsible for your market. ESD protection is only effective when the safety regulations and safety precautions are complied with. Carry out all work steps, including the test scopes, according to the instructions in the Workshop Information System.

Which modifications are not permitted, and what must not be extended?

- Airbag control unit connector coupling
- ESP® control unit connector coupling
- Engine control unit connector coupling
- High-voltage lines
- Antenna lines
- Optical fiber cables
- Lines of bus systems, See 8.4.1 CAN bus and networking (→ Page 271)
- Lines with a cross-section $> 4 \text{ mm}^2$

More information on current work instructions regarding wiring harness extension, soldering and crimping according to Mercedes-Benz workshop specifications can be found in the Workshop Information System (→ page 26).

Refer to document AH00.19-D-1000-08A from the Workshop Information System for such information, e.g. the fact that an ESP® plug is not permitted to be "modified".

More information on wiring harness extension can be found in the following documents in the Workshop Information System:

- AR00.19-P-0100A (Wiring harness, general methods)
- AH00.19-P-1000-08A (Parts solutions and what is permitted to be and what must not be repaired/extended)

If the prescribed methods are followed and approved materials and tools are used, it is possible to lengthen lines without any loss in quality.

The following methods are approved:

- Soldering
- Raychem (use of end sleeves)
- Raychem in combination with electrical wiring harness (pigtail)
- Crimping

More information on selecting suitable working methods can also be found in document AR00.19-D-0100A in the Workshop Information System.

Soldering

Select the line connectors or solder sleeves. Solder the stripped wires according to the instructions in the Workshop Information System:

- AR00.19-P-0100-05A
- AR00.19-P-0100-09A

Crimping

Crimp the stripped wires according to the instructions in the Workshop Information System:

- AR00.19-P-0100-03A
- AH00.19-P-1000-09A

i More information can be found in the Workshop Information System (→ page 26). You can purchase access to the Workshop Information System at the following address:

<https://b2bconnect.mercedes-benz.com>



The methods focus on soldering (specifically Raychem) and crimping.

Soldering (specifically Raychem):

In the Raychem method, special end sleeves are used. The end sleeves contain the solder and are insulated and sealed. A special hot-air blower is used to heat them. Do not use a cigarette lighter!

The connections must be checked for a resistance of less than 0.5 Ω using the Fluke meter in accordance with the circuit diagram.

8.4.4 Additional electric circuits

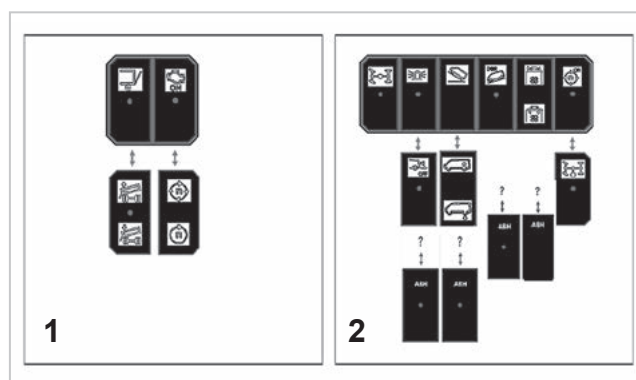
If additional electric circuits are installed, protect them against the main power circuit by fuses of adequate rating.

The lines used must be rated for the load and they must be protected against the effects of breakaway, impact, and heat.

8.4.5 Operating switches (pre-installation)

Pre-installation for switch panel

As a pre-installation for operating switches installed by the bodybuilder manufacturer, 2 switch panels are available in the area of the steering wheel. These contain 2 slots on the left of the steering wheel and 6 slots on the right of the steering wheel. Depending on the equipment installed, these may be added automatically and partially populated with switches. If they are not added automatically due to the vehicle equipment, the pre-installation for the switch panel (code E3J) can be ordered separately.



Switch panels

- 1 Van switch panel to left of steering wheel (left-hand drive vehicle)
- 2 Van switch panel to right of steering wheel (left-hand drive vehicle)

Operating switches for body functions

Four different versions of load switch are available ex factory for installation in the switch panel for body functions:

- Item number A 907 905 39 02
CO contact, single, latching, 1-pin
1 × readiness display
- Item number A 907 905 40 02
Make contact, single, momentary, 1-pin
1 × readiness display
- Item number A 907 905 41 02
CO contact, double, latching, 2×1-pin
No readiness display
- Item number A 907 905 07 09
Make contact, single, momentary, 1-pin
1 × readiness display

These switches bear no symbols and can be inscribed with laser lettering at a size of 8×8 mm.

Direct procurement for bodybuilder manufacturers:

Contact details

Address	Eugen Kurz KG Rebengasse 12 89073 Ulm, Germany
Telephone	+49 (0)731 66535
Email	info@stempel-kurz.de 

 For further information on purchasing control switches, contact the Technical Consultancy (→ page 18) or your Mercedes-Benz Service Partner.

8.4.6 Retrofitting of electrical devices

Observe the following when retrofitting additional electrical consumers:

- The bodybuilder manufacturer must ensure a positive charge balance.
- In the case of higher electrical power requirements, also use the alternators approved by Mercedes-Benz AG, See 8.4.7 Alternator (→ Page 276).
- Do not connect additional alternators to the existing on-board electrical system; the electrical systems should be galvanically isolated.
- Do not connect additional consumers to fuses already assigned.
- Do not connect additional lines (e.g. with insulation displacement terminals) to existing lines.
- Provide consumers with sufficient protection by means of additional fuses.
- All electrical consumers installed must have been tested in accordance with UN R 10 in its current version and must bear the CE marking and/or another comparable marking depending on the country-specific regulations; See 8.2 Electromagnetic compatibility (EMC) (→ Page 266).
- Any additional electrical consumers must be connected using the terminal strip for auxiliary consumers (code EK1) available ex factory as per 8.4.8 Current collection (→ page 277).

 **WARNING**

Tampering with and unauthorized installations in the electrics or on-board electronics can impair the functioning of these systems. This can lead to the failure or malfunctioning of components or component parts relevant to safety, and may result in accidents or damage to the vehicle.

There is risk of an accident and danger to life and limb!

Do not tamper with or perform any unauthorized installations in the electrics or on-board electronics.

 Furthermore, tampering with the electrics or on-board electronics can invalidate the warranty/operating permit.

8.4.7 Alternator

If additional electrical consumers are retrofitted, the increased power requirement can be met by fitting higher-rated alternators.

The following alternators are available as special equipment (SA) ex factory:

Alternator availability		
Code	I [A]	OM654
M40	200	Series
M60	250	optional
M61	280	optional

When using additional assemblies, use the factory-installed power take-offs, See 6.5.3 Power take-offs (→ Page 169).

For retrofitting alternators, it is recommended to order the special equipment code N62 available ex factory (front engine power take-off with carrier for additional alternator); See 6.5.3 Power take-offs (→ Page 169).

When retrofitting other alternators, observe:

- Only alternators with a freewheeling belt pulley are permissible.
- The installation of a generator must not impair vehicle parts and their function.
- The battery must have sufficient capacity and the alternator must generate sufficient power (→ page 276).
- Provide the alternator circuit with additional fuse protection (→ page 272).
- Dimension the line cross section in accordance with the tapped amperage (→ page 272). The higher power draw may necessitate the replacement of the starter/alternator wiring harness. The use of Mercedes-Benz genuine parts is recommended.
- Ensure that electric lines are routed correctly (→ page 272).
- There should be no impairment of the access to or easy maintenance of installed major assemblies.
- Do not impair the required air supply and cooling of the engine (→ page 147).
- Observe the guidelines of the equipment manufacturer for compatibility with the basic vehicle.
- Supply the operator's manual and the maintenance manual for the additional assemblies upon handing over the vehicle.

8.4.8 Current collection

Depending on the vehicle's as-built configuration, there are different possibilities for auxiliary consumers to collect current. Current can be collected via the terminal strip for electrical connection (special equipment EK1) or at the auxiliary battery tapping point.

- ❗ More information on tapping power for special equipment is available from the Technical Consultancy (→ page 18).

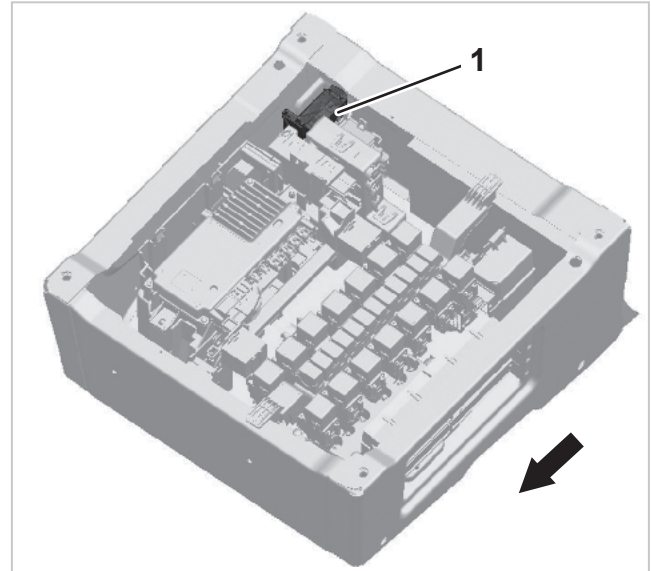
! NOTE

When installing additional electrical consumers, especially on vehicles with factory-installed special equipment that uses the auxiliary battery (fuse box in driver seat frame), the bodybuilder manufacturer must ensure a positive overall charging balance.

- ❗ More information on special equipment is available from your Mercedes-Benz Service Partner, from the Technical Consultancy (→ page 18), or in 3.12 Special equipment (→ page 74).

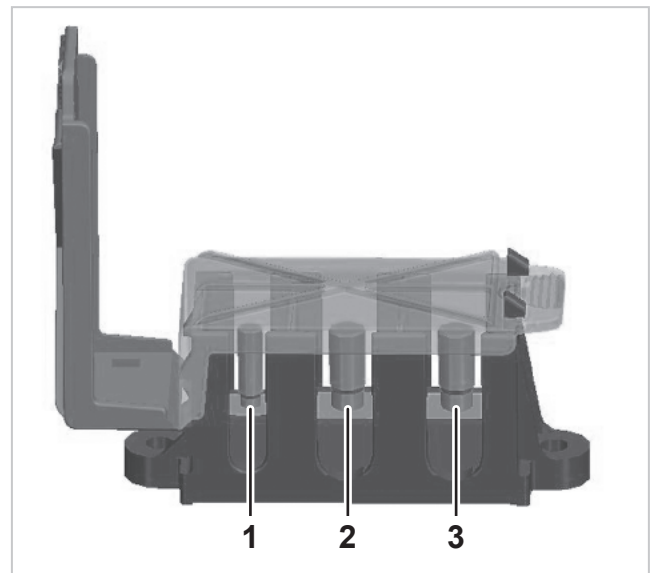
Current collection via terminal strip for electrical connection

The terminal strip (special equipment code EK1) is installed inside the driver seat frame (at the center, on the left-hand side in direction of travel) and has three connectors:



Position of terminal strip (code EK1) in driver's seat frame (left-hand drive vehicle)

- 1 Terminal strip (code EK1)
- ➔ Direction of travel



Terminal strip (code EK1)

	Terminal	Voltage/fuse rating
1	Terminal D+	12 V/10 A
2	Terminal 30T	12 V/25 A
3	Terminal 15	12 V/15 A

Tighten the electrical joint bolts in the fuse box to the following tightening torques:

Screw size	Tightening torque [Nm]
M4	3 ±0.3
M6	6 ±0.6

The terminal strip (code EK1) is supplied directly from the on-board electrical system battery, and is the supply point in the main on-board electrical system at which the bodybuilder manufacturer is permitted to connect its consumers. Other consumers can be connected via the PSM and the auxiliary battery.

! NOTE

Note that no consumers are permitted to be connected to terminal 30T of the terminal strip (code EK1) that require a quiescent current > 1 mA when the vehicle is stopped (terminal 15 or D+ inactive/engine standstill).

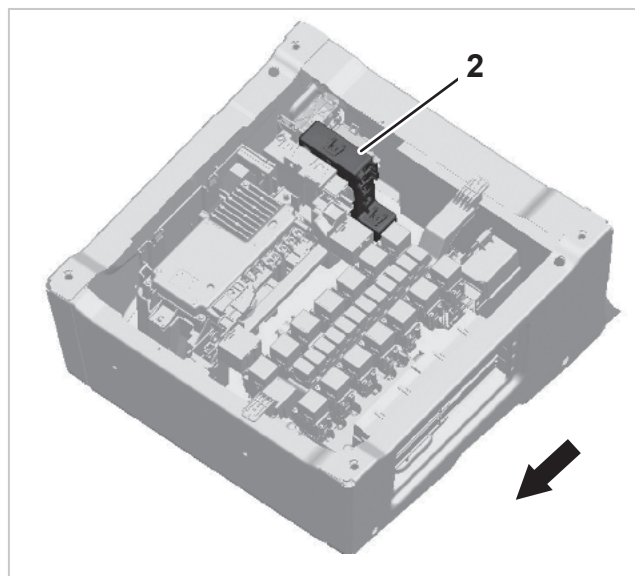
If the total quiescent current is higher, even for a short time (peak), contact the Technical Consultancy (→ page 18).

In start/stop mode, brief voltage drops (at restart) up to 7.0 V at the terminal strip (code EK1) are possible. The voltage drops are typically in the region of 9.0 V.

Current collection at auxiliary battery tapping point

In order to facilitate current collection from the auxiliary battery (code E2I or E2M), a current collector (fuse box) is available in the driver seat frame. The tapping point is the busbar in the fuse box with the direct line from the auxiliary battery (positive terminal).

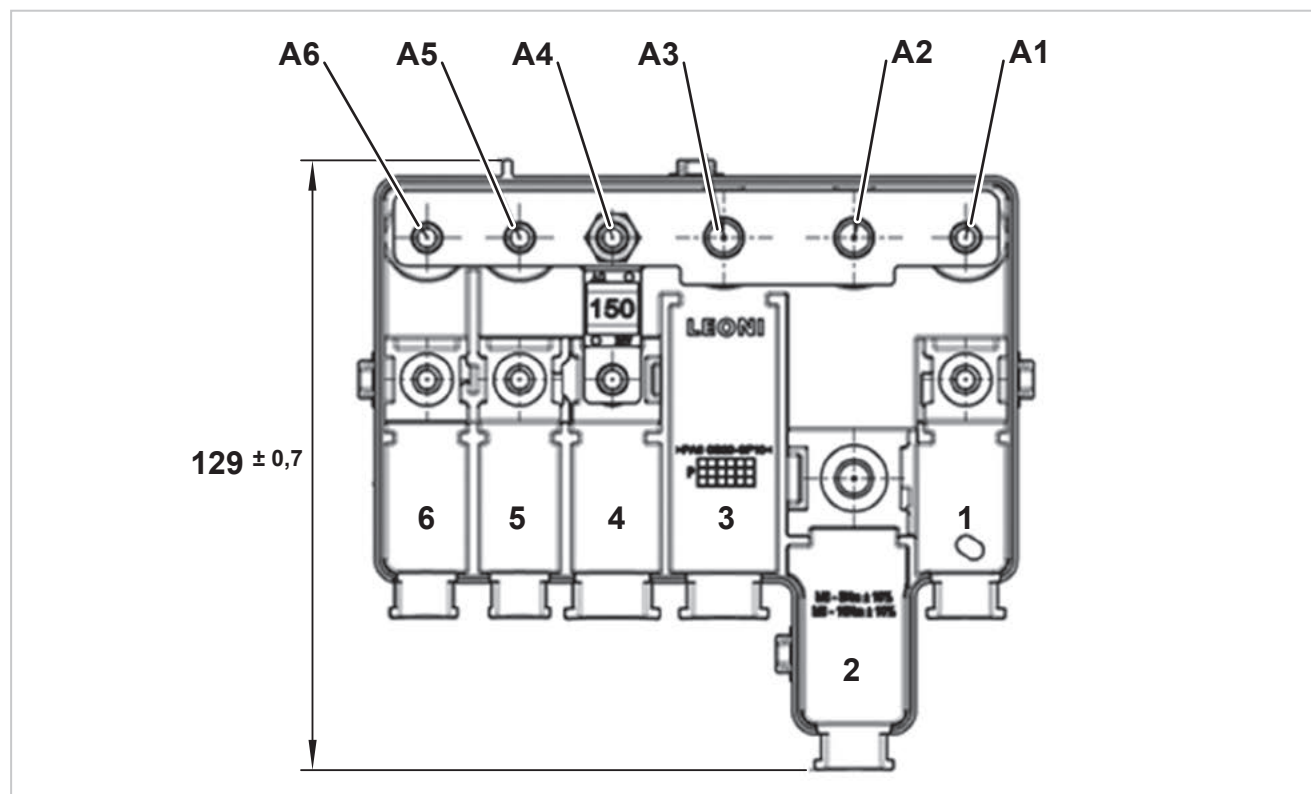
The cap must be removed to connect and install the fuse.



Position of auxiliary battery fuse box in driver seat frame (left-hand drive vehicle)

- 2 Fuse box
- ➡ Direction of travel

Fuse box



Assignment of fuse box (top view)

Con- nec- tion no.	Min./max. permissible fuse rating	Fuse type	Function	Maximum line cross-sec- tion	Cable lug	Coding	Bolt
A1	-/100 A	SF30	Retarder	25 mm ²	Complies with work spec., see drawing	2	M6
A2	-/250 A	SF51	Pre-installation: Cargo liftgate for three-way dump truck	35 mm ²	Complies with work spec., see drawing	8	M8
A3	-	-	Auxiliary battery	35 mm ²	Complies with work spec., see drawing	2	M8
A4	125 A/150 A	SF30	Battery cutoff relay	35 mm ²	N046234006005	-	M6
A5	-/80 A	SF30	Sockets, auxiliary heating	2 × 16 mm ²	Complies with work spec., see drawing	8	M6
A6	-/80 A	SF30	Sockets, auxiliary heating	2 × 16 mm ²	Complies with work spec., see drawing	7	M6

Cable lugs complying with the work specification (see table) are preferred. Torque relief on these is through the housing. If other cable lugs are used, use separate torque reliefs (A 907 545 52 00) at the M6 bolts.

Work specification for cable lugs	
M6	A 003 000 25 99
M8	A 004 002 44 99

Tighten the electrical joint bolts in the fuse box to the following tightening torques:

Screw size	Tightening torque [Nm]
M6	8 ±0.8
M8	16 ±1.6

! NOTE

If a vehicle is already equipped ex factory with a consumer that uses the fuse box in the driver's seat box, this is only available under certain conditions. The sum of the individual currents must not exceed the maximum current-carrying capacity of the fuse box of 300 A. There is otherwise a risk of overload.

Fuse

The auxiliary consumer can be connected to the auxiliary battery with a suitable fuse as per ISO 8820 SF30 (contact cavities 1, 5, and 6) or SF51 (contact cavity 2).

Preconditions for connection to the auxiliary battery include:

- Auxiliary battery fuse box ex factory in seat box
- Use of the following retaining plates (busbar side), part number M6 A 907 545 60 00 or M8 A 907 545 61 00 for torque relief
- Suitable fuses as per ISO 8820 (SF30 or SF51)

Please note with regard to this:

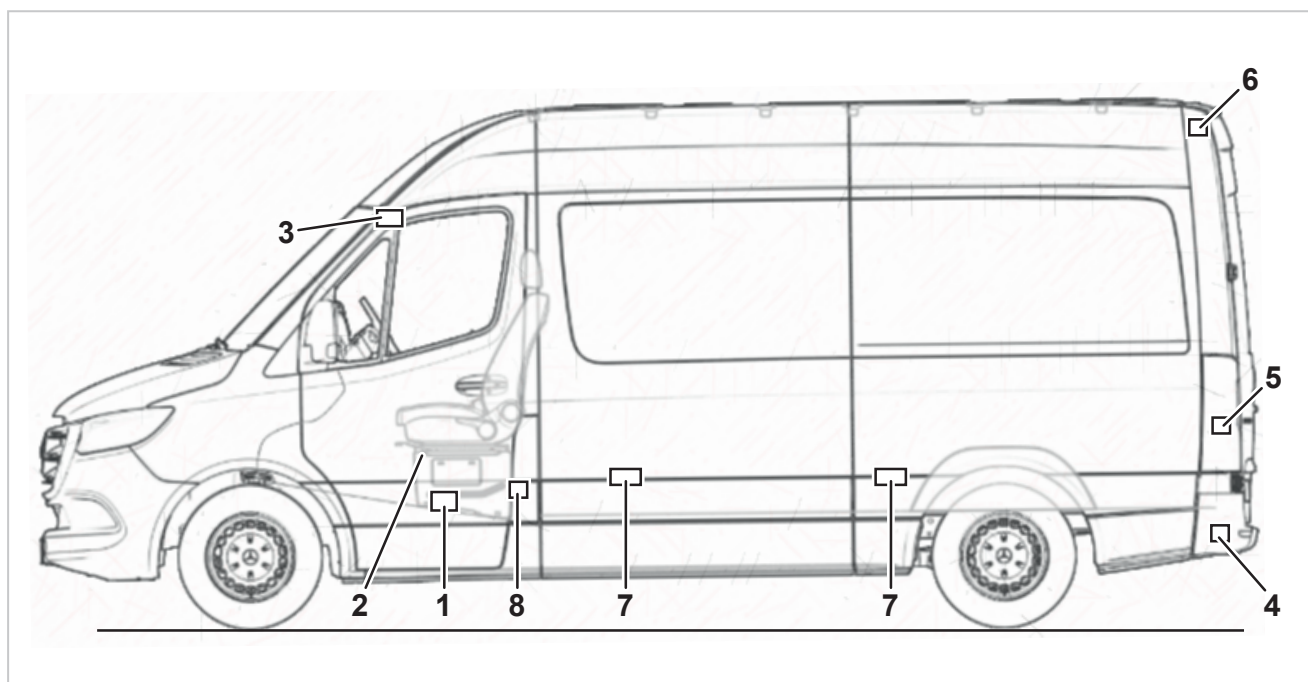
- No additional consumers may be connected to factory-installed fuses.
- If all the connections at the fuse box are fully occupied, use the positive terminal of the auxiliary battery for connecting further consumers.
- A suitable fuse box with fuse is required for these additional consumers.
- Ensure a positive total charge balance in all vehicle conditions.

On vehicles with the following special equipment, the fuse box is used at the factory and is only available for additional consumers to a limited extent (table only shows some examples):

Code	Equipment	Maximum fuse rating
BR9	Retarder pre-installation	100 A
EV3	Cargo liftgate pre-installation	250 A
P57	Three-way dump truck pre-installation	250 A

8.4.9 Interface overview

The electrical interfaces available as special equipment on the vehicle are shown in the illustration below:



Electrical interfaces on the vehicle (schematic)

	Code	Equipment
1	EK1	Terminal strip for the electrical connection on the seat box of the driver's seat
2	L72	Electrics for body interior lighting, 3-pin connecting point in left seat box in open vehicle model designations
3	LC4	Pushbutton switch in overhead control panel (OCP)
4	E57	Electrics for trailer socket
5	L76	Extended tail lamp wiring harness (2 m)
6	L77	Electrical equipment for additional turn signal lamps
7	LB2	Preparation for side marker lamps
8	E5M	Extension of PSM standard contact

Additional special equipment

Code	Equipment
BR9	Pre-installation for retarder
LV1	Pre-installation for rotating beacon
LV4	Pre-installation for front fog lamp with cornering light
LV6	Pre-installation of electrical scope for 3rd brake lamp
E2F	Pre-installation for windshield heater
FX3	Wiring harnesses for cab door
FR7	Pre-installation of electrics for ultrasonic reversing aid
FV1	Pre-installation for reversing camera with inside mirror display
JV7	Pre-installation for siren
E3J	Pre-installation for switch panel
ED8	Pre-installation of electrics for parameterizable special module
E2A	Tapping wire for bodybuilder manufacturer special signals in cockpit
E3D	Pre-installation DS box
EP7	Pre-installation of electrics for rear speakers
E4A	Antenna interface for radio, navigation and cellular telephone system
JH5	Pre-installation for communication module for digital services (UMTS)
JH6	Pre-installation for communication module for digital services (LTE)
ER0	Radio pre-installation
E7A	Pre-installation for digital radio (DAB)
EV3	Pre-installation of electrical scope for cargo liftgate
JV5	Pre-installation for accident data recorder
J1V	Pre-installation for Blind Spot Assist
L07	Clearance lamps
L90	Discontinuation of tail lamps
L91	Pre-installation for LED tail lamps
L93	Pre-installation for LED front lamps
LB2	Pre-installation for side marker lamps

NOTE

Observe the statutory regulations for installation of lighting systems, See 8.5 Lighting (→ Page 286).

For codes and specifications of the tail lamps, See 10.1 Bulb ratings of tail lamps (→ Page 398).

8.4.10 Speed signal

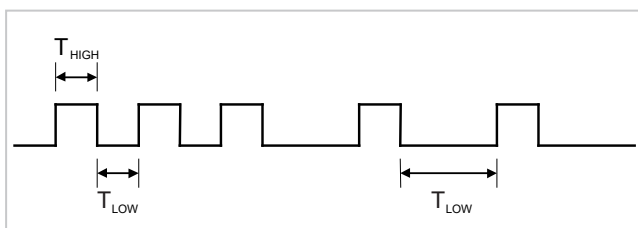
The analog vehicle speed signal in the instrument cluster has been discontinued. If required, a digital signal can be provided by the PSM via a special data entry.

! NOTE

The digital speed signal is not suitable for gauged taximeters or other measuring devices.

The digital speed signal (positive to ground) is intended for external electronics, e.g. speed-dependent volume control.

- The signal is protected against short-circuit to ground and battery voltage and is not monitored.
- The signal is output at 4 pulses per meter. The pulse duration is 4 ms.
- At 112.5 km/h, the pulse duration is the same as the pulse pause. This 1:1 ratio is maintained for higher speeds. This means that, at higher speeds, the pulse duration and the pulse pause simultaneously become shorter.



Ratio of pulse duration/pulse pause

Speed signal ($I_{max} = 20 \text{ mA}$):

$T_{HIGH} U_a \geq 8 \text{ V}$

$T_{LOW} U_a \leq 1 \text{ V}$

- The speed signal is made available on the body-builder manufacturer CAN in a format complying with SAE-J1939.
- Value range: 0–250.996 km/h
- Resolution: 1/256 km/h

8.4.11 Travel distance signal

- i As of model year 2024, a digital travel distance signal is no longer available at the PSM. If you have any questions with regard to this, contact the Technical Consultancy (→ page 18).

8.4.12 Ground bolts

For electrical connection to the ground of bodybuilder manufacturer electrical consumers, use either the following ground bolts, or the bodybuilder manufacturer should install additional ground bolts at suitable points in the body-in-white.

When doing so, always observe:

- Only the ground bolts available ex factory listed in the following may be used for the connection of further electrical consumers.

⚠ WARNING

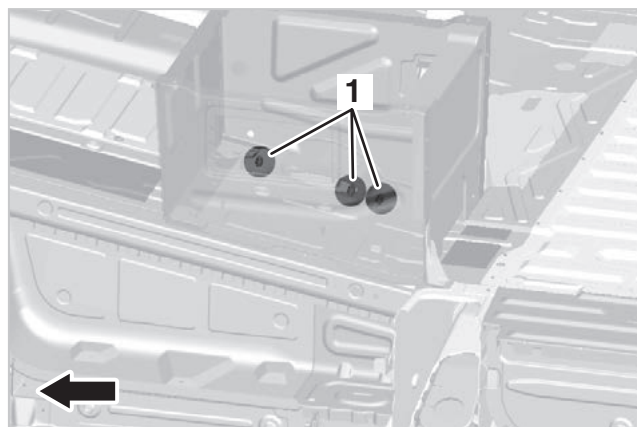
The use of any other ground bolts may lead to malfunctions of safety systems. This in turn can lead to the failure of components or safety-relevant component parts and to fault messages in the instrument cluster.

- The screw connections to ground bolts of the safety systems must not be released.
- Max. four cable lug connections may be connected to one ground bolt; these must be distributed in a uniform radial manner (e.g. 120° for three or 90° for four cable lug connections).
- The bodybuilder manufacturer must independently specify the type of cable lug connection in accordance with the electrical requirements and the ground bolt diameter.

- For the screw connection at the ground bolts, observe:
 - Torque for M6: 8 Nm $\pm$ 0.8 Nm
 - Torque for M8: 16 Nm $\pm$ 1.6 Nm
 - Type of screw connection: Class A (safety-relevant, documentation required)
 - Maintain a thread protrusion of at least three complete turns. If the thread protrusion is less, the number of cable lug connections envisaged at the affected ground bolt must be reduced.
- For any other requirements or planned deviations, contact the Technical Consultancy (→ page 18).

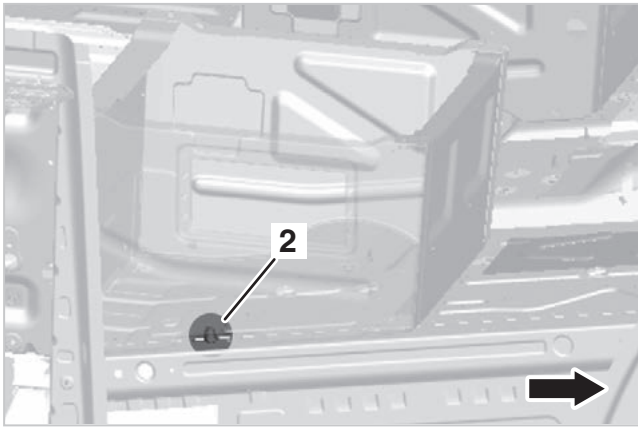
The following ground bolts are available for bodybuilder manufacturers, taking into account the previously mentioned specifications:

- 3 M6 ground bolts in driver seat box
- 1 M8 ground bolt in front passenger seat box for auxiliary battery in the vehicle interior (code E2I) and additional fuses for camper vans (codes E1R/E1Y), as well as with code VV4 (only NAFTA)
- 1 M6 ground bolt at the crossmember in front of the rear axle
- 1 M6 ground bolt for the liftgate/cargo liftgate (code EV3)



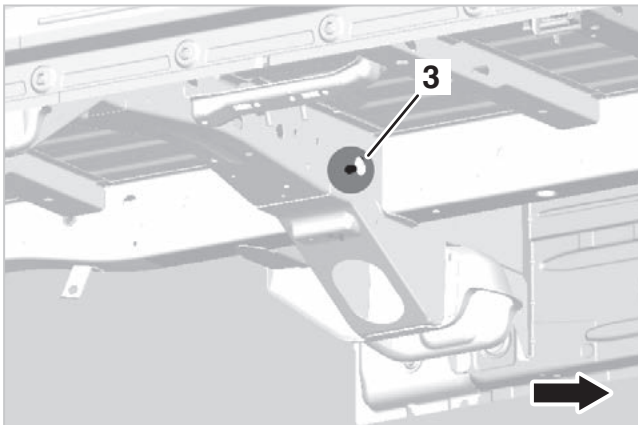
Ground bolt in driver seat box

- 1 M6 ground bolt
- ➡ Direction of travel



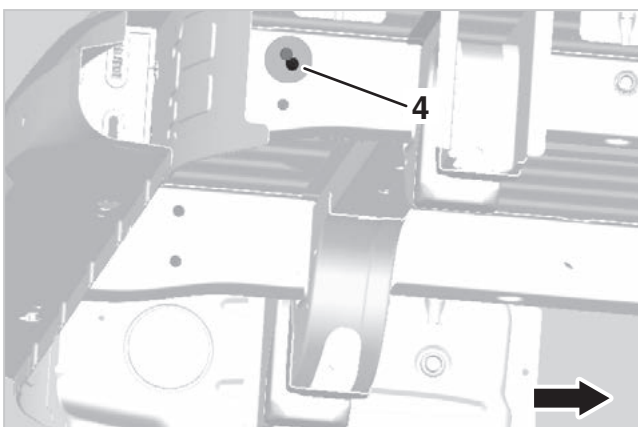
Ground bolt in front passenger seat box for codes E2I, E1R/E1Y, VV4

- 2 M8 ground bolt
 ➔ Direction of travel



Ground bolt at crossmember in front of rear axle

- 3 M6 ground bolt
 ➔ Direction of travel



Ground bolt at rear longitudinal member for code EV3

- 4 M6 ground bolt
 ➔ Direction of travel

8.4.13 Deep discharge protection (standby mode)

The deep discharge protection system can prevent impermissible deep discharging of the battery with the vehicle at a standstill (no-load phase). This function should ensure the startability of the vehicle over an extended period of time or in the event of elevated quiescent currents.

In addition, excessive quiescent currents caused by faulty components are minimized by hardware resets of the components.

The functions of the system are distributed between the vehicle locking system (the component is the electronic ignition lock (EIS)) and the on-board electrical system management system (the component is the Body Controller (BC)). The hardware component responsible is the battery disconnect switch (BTS).

Terminal 30 (BR 906) is transformed into a switchable terminal 30T (BR 907).

Another function in the deep discharge protection system is the "standby mode" function. This opens the battery disconnect switch prematurely, allowing the vehicle to be parked for long periods.

The mode can be selected in the instrument cluster (standby mode). If a head unit has been installed, it is activated there.

The standby mode allows almost all of the control units to be switched off. Subsequently, almost the only signal present at terminal 30T is the EIS or the control unit that is necessary to receive the signal of the radio remote control. The PSM and the anti-theft alarm system also continue to be supplied depending on their operating status.

8.5 Lighting

Refer to UN R 48 and the national specifications for the statutory regulations on the "Installation of lighting systems".

! NOTE

Observe the specifications and information in 10.1 Bulb ratings of tail lamps (→ page 398) for all tail lamps.

8.5.1 Headlamps

The following special equipment is available from the factory as option codes to carry out retrofitted modifications to the vehicle headlamps:

Code	Equipment	Description
L93	Pre-installation for LED headlamps	Retrofitting of other lamps and turn signal units in LED technology is possible. Actuation occurs without pulse width modulation (PWM)
L92	Omission of front lamp unit	Retrofitting of other lamp units and flashing units with conventional bulbs; actuation takes place via pulse width modulation (PWM).
LV4	Pre-installation for front fog lamp with cornering light	Retrofitting of other lamp units with conventional bulbs or in LED technology (in combination with code L93) possible; actuation takes place either with or without pulse width modulation (PWM).

! NOTE

Once body mounting work has been completed, the bodybuilder manufacturer, under its own responsibility, must adjust the headlamps or check their factory settings and correct them if they deviate from the specified values.

For the following model designations, the headlamp adjustment for the type-approval certificate must always be carried out by the bodybuilder manufacturer:

- Chassis (open model designations without platform ex factory)
- Cowl cab base vehicles (code F50)
- Cab base vehicles with cab (code F28)

The adjustment instructions with the set values for the finished vehicle in combination with the version of headlamp used are implemented by your Mercedes-Benz Service Partner. The Workshop Information System (→ page 26) also allows you to determine the currently applicable information on adjustment for your own headlamp adjustment.

! NOTE

Observe all national laws, directives, and registration regulations.

Observe the headlamp basic setting as per the identification plate.

8.5.2 Mounting additional lamps

! NOTE

Observe all national laws, directives, and registration regulations.

If moving vehicle parts cover a lighting system by more than 50% during operation, the vehicle must be safeguarded accordingly.

Reflectors at the rear of the vehicle must not be obscured by moving parts. If this is the case, additional reflectors must be attached.

An appropriate note must be attached at a point where it can easily be seen by the driver of the vehicle.

8.5.3 Tail lamps

! NOTE

Observe all national laws, directives, and registration regulations.

In the case of any bodies or conversions that could lead to the tail lamps being obscured, observe the specifications from UN R 48.

The following special equipment is available ex factory as option codes to carry out retrofitted modifications to the vehicle tail lamps:

Code	Equipment	Description/function
L76	Extended tail lamp wiring harness	The lengthened tail lamp wiring (approx. 2 m) acts as pre-installation for retrofitting tail lamps in a different location.
L77	Electrical equipment for additional turn signal lamps	On chassis with cab and crewcab, the additional line at the rear area of the vehicle is provided for additional turn signal lamps on the body.
L90¹⁾	Discontinuation of tail lamps	Retrofitting of other lamp units and flashing units in conventional bulbs possible; actuation takes place via PWM (pulse width modulation).
L91¹⁾	Pre-installation for LED tail lamps	Retrofitting of other lamp units and flashing units in LED technology possible; actuation takes place without pulse width modulation (PWM).

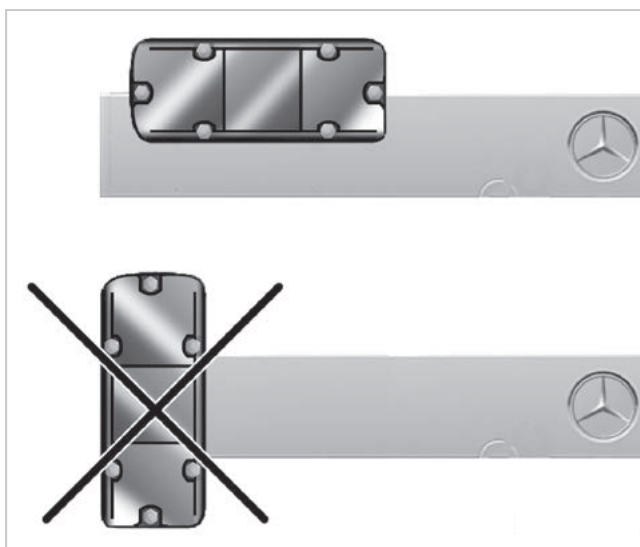
- 1) The bodybuilder manufacturer must guarantee that the loads occurring when the exterior lighting function is active consume a constant current of at least 20 mA via the respective channel. The channels of the brake light/tail light and the turn signal indicator must only be used to operate the associated functions. In other words, the parallel connection of e.g. side marker lamps to the tail light channel is not permitted.

Observe the specifications and information in 10.1 Bulb ratings of tail lamps (→ page 398) for all tail lamps.

! NOTE

On open model designations, the standard tail lamps must be installed in a horizontal position. The standard tail lamp of the open model designations is not approved for vertical installation. Avoid allowing water to enter through the ventilation holes, as this can lead to the failure of the standard tail lamps or faults with the electronic systems.

If tail lamps are to be used in any other position, the bodybuilder manufacturer must use suitable tail lamps of its own.



Standard tail lamps of the open model designations (schematic)

Tail lamps for liftgate (cargo liftgate) for open model designations

If – due to the installation of a liftgate – the use of the standard tail lamps on the vehicle rear with the license plate lighting is no longer possible due to the installation location, the company Vignal has special tail lamps available that have been approved by Mercedes-Benz for the combination with liftgates (manufacturer designation "LC5").

These tail lamps are already equipped with the appropriate electrical connectors, which facilitates their use with the standard connectors present on the wiring harness of the basic vehicle.

Depending on the installation situation, the tail lamps can be mounted horizontally or vertically. In addition, the vehicle license plate can also be illuminated independently of the tail lamps by means of separate license plate lamps that are attached in an area such as the vehicle center.

! NOTE

Observe all national laws, directives, and registration regulations.

The following type LC5 variants of the tail lamps from Vignal are available with the indicated manufacturer part numbers (or corresponding successor numbers) from specialist stores or directly via Vignal:

Variant	Vignal no.	Installation location/direction of installation	License plate lamp
1	154100	Left/horizontal	Integrated in tail lamp
2	154120	Left/horizontal	Without, with supply line for separate license plate lamp EPP98 (#196010)
3	154110	Right/horizontal	None
4	154130	Right/horizontal	Without, with supply line for separate license plate lamp EPP98 (#196010)
5	154140	Left+right/vertical	Without, with supply line for separate license plate lamp EPP98 (#196010)

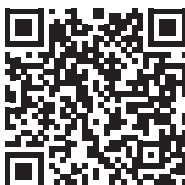
More information on the procurement sources and technical details can be found at:

<https://www.vignal-group.com>



Alternatively, send an email to:

contact@vignal-group.com



! NOTE

Depending on the assembly position, it is recommended to protect tail lamps that are installed in the direct stone chipping area of the wheels against possible damage via a separate protection device (e.g. via the attachment of a suitable protective metal sheet).

These tail lamps are equipped with bulb illuminants. Depending on the vehicle equipment, therefore have the lamp failure indicator recoded by an authorized Mercedes-Benz workshop.

A recoding of the lamp failure indicator is required if the basic vehicle contains any of the following special equipment:

- Code L22, tail lamps in partial LED technology
- Code L91, pre-installation for LED tail lamps

To avoid recoding, it is recommended to take this into account accordingly when you order the vehicle.

- ① These tail lamps can also be used for other body-manufacturer-specific bodies or conversions. For more information on this, contact the manufacturer.

Reversing light under bumper corner on closed model designations

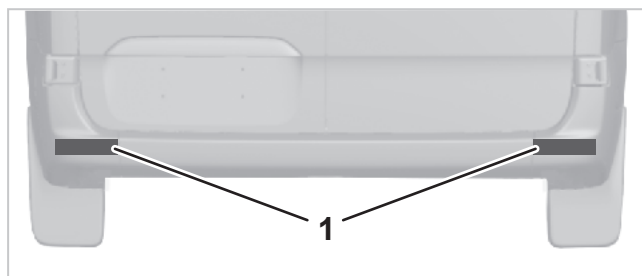
For some versions of the closed model designations 907.6xx and 907.7xx, as well as being a standard feature on the eSprinter for all model designations 907.4xx, code L21 is used to relocate the reversing light from the tail lamps to the area of the right and left corners of the rear bumper in combination with the existing reflectors.

This equipment cannot be freely selected or deselected, and the rear mud flap special equipment (code P48) is automatically added as well.

Legal requirements demand the installation and visibility of these reversing lights.

After body mounting work, it is therefore the responsibility of the bodybuilder manufacturer to continue to comply with the legal requirements.

Affected by this measure are vehicles with N1 approval and raised body (e.g. code CE8), rough road versions (code Z11 or Z12), and vehicles with all-wheel drive or electric drive (eSprinter).



Additional reversing lights for closed model designations

1 Reversing lights

 For more information on this scope, contact the Technical Consultancy (→ page 18).

8.5.4 Marker lamps

Side marker lamps (SML)

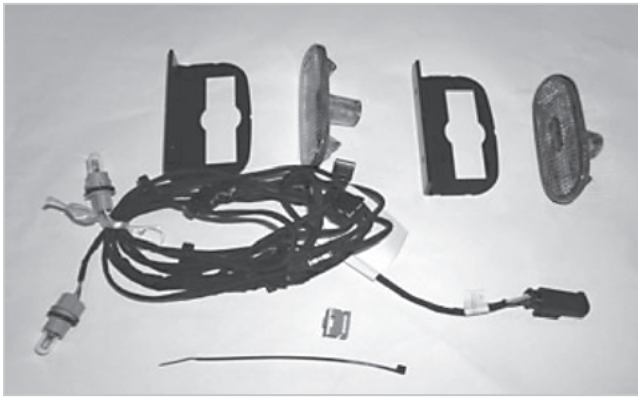
According to the regulation UN R 48, all vehicles with an overall length of over six meters must be equipped with side marker lamps.

Ex factory, the special equipment "Preparation for side marker lamps" (code LB2) is available for chassis with cab or crewcab.

In addition, side marker lamps (code LB1) are available for all model designations. With regard to this, the marker lamps and their holders are attached on chassis with cab or crewcab and without platform ex factory per vehicle side in a bag at the longitudinal frame member on the left and right for transport to the bodybuilder manufacturer.

During body mounting work, the bodybuilder manufacturer must fasten the connecting point between the main wiring harness and the adapter wiring harness for side marker lamps to the longitudinal frame member using the mounting clip A 000 991 13 70 and the cable tie A 001 997 95 90. This applies both to the side marker lamps (code LB1) and to the preparation for side marker lamps (code LB2).

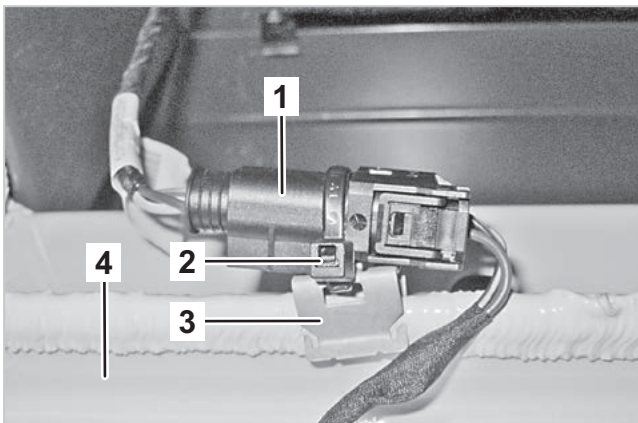
All the necessary fastening parts are included in the delivery package of code LB1 and on chassis without platform ex factory (without code P02) are also packed in the bags for the left and right sides of the vehicle. The active circuit for the side marker lamps in the body controller ex factory is also included in the delivery package.



Parts scope for the side marker lamps (code LB1) incl. fastening clip and cable ties for vehicles without platform ex factory, example for one side of the vehicle

In contrast, with code LB2, only the left and right connecting points are supplied in a bag at the rear area of the vehicle. Lamps, holders and other mounting materials for code LB2 are to be procured by the bodybuilder manufacturer and the connecting points fastened in the same way as for code LB1.

With code LB2, the lamp failure indicator for the side marker lamps is deactivated.



Correct fastening of connecting point for side marker lamps, view from below

- 1 Preparation for side marker lamps
- 2 Cable ties
- 3 Fastening clips
- 4 Longitudinal frame member

Clearance lamps and vehicle position lamps

According to the regulation UN R 48, all vehicles with a width of over 2.10 m must be equipped with clearance lamps. They are permitted to be installed for widths of 1.80 m and over.

The special equipment "Clearance lamps" (code L07) is available ex factory for special vehicle configurations.

8.5.5 Actuation of exterior lighting functions

1) Front: Halogen headlamps and pre-installation for LED headlamps

If you are not using genuine Mercedes-Benz Van standard headlamps, the relevant control unit is coded in the standard configuration ex factory by ordering code L92 or L93 (for an explanation of the codes See 10.1 Bulb ratings of tail lamps (→ Page 398)). The wiring harness is identical for both codes.

The exterior lighting functions in the headlamp are actuated discretely by the Body Controller (BCMFA2). This means that each exterior lighting function in the headlamp has a separate line.

The exception to this rule are the parking and standing lights functions. These functions are actuated via the same lines, i.e. they are implemented by the same lamps.

Fault monitoring is implemented via the signal acquisition and actuation module SAM and is always active when the relevant function is active and the corresponding illuminant is actuated. This implemented function is to be taken into account particularly when designing and using the bodybuilder manufacturer's own electrical ballasts, otherwise faults will be logged in the control unit.

The table below provides an overview of the individual lighting functions.

Explanations of the following function descriptions

- Ignition in terminal 15 position: Ignition "ON", engine start possible and/or engine running
- Ignition in terminal 15r position: Radio position, ignition "OFF", engine not running, radio operation possible and/or on-board electrical system voltage "ON"
- Ignition in terminal 15c position: Ignition "OFF", engine not running, driver's door open, on-board electrical system voltage "ON"

Exterior lighting functions in headlamps

Hardware output at body controller	Front									
	Low beams		Position lamp		Daytime running lights		High beams		Turn signal indicators	
Function	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Side										
Plug	MR1	MR2	MR2	MR1	RBA1	RBA2	MR2	MR1	MR1	MR2
Pin	20	02	01	21	17	56	04	18	01	05

Light pattern, open model										
Conventional (UNECE)										
Left parking light			X							
Right parking light				X						
Standing lights			X	X						
Low beams	X	X								
High beam/headlamp flashing							X	X		
Daytime running lights					X	X				
Turn signal indicator left/right, hazard warning light system									X	X

Low beams

Requirements for activation:

- Ignition in terminal 15 position
- Engine running.
- Rotary light switch in AUTO position
- Light sensor detects NIGHT.
- OR rotary light switch in MAN position (manual driving lights)

Standing lights and position lamp

Requirements for activation:

- Rotary light switch in STL position (standing lights); the standing lights are activated regardless of the state of the vehicle.
- OR rotary light switch in AUTO position, ignition in terminal 15r position, and light sensor detects NIGHT. If the light sensor detects DAY, no lighting function is activated.
- OR rotary light switch in MAN position and ignition in terminal 15r position

! NOTE

In the Sprinter, the standing lights are not activated when the daytime running lights or low beams are active. Only the above requirements apply.

Parking lights

Requirements for activation:

- Ignition at terminal 15c, i.e. ignition switched off and driver's door opened
- Rotary light switch in parking lamp position

Daytime running lights

Requirements for activation:

- Ignition in terminal 15 position
- Engine running.
- Rotary light switch in AUTO position
- Light sensor detects DAY.

High beam and headlamp flashing

Requirements for activation of headlamp flashing:

- Ignition in terminal 15 position

Requirements for activation of high beams:

- Low beams active

Turn signal indicators

Requirements for activation:

- Ignition in terminal 15r or terminal 15 position

2) Rear end: Tail lamps with bulbs and pre-installation for LED tail lamps

If genuine Mercedes-Benz Van standard tail lamps are not used, the relevant control unit is coded in the standard configuration ex factory by ordering code L92 or L93. The design of the wiring harness and the connecting points used vary according to the body type.

The exterior lighting functions in the tail lamp are actuated discretely by the signal acquisition and actuation module (SAM). This means that each exterior lighting function in the tail lamp has a separate line.

The exception to this rule are the parking and standing lights functions. These functions are actuated via the same lines, i.e. they are implemented by the same lamps.

On open model designations (FHS/FHL), the lines for the license plate lighting are not included, See 10.1 Bulb ratings of tail lamps (→ Page 398).

Fault monitoring is implemented via the signal acquisition and actuation module (SAM) control unit and is always active when the function is active and the corresponding illuminant is actuated. This implemented function is to be taken into account particularly when designing and using the bodybuilder manufacturer's own electrical ballasts, otherwise faults will be logged in the control unit.

Exterior lighting functions in tail lamps

Hardware output at body controller	Rear												
	Brake light		Tail lights		Turn signal indicators		Reversing light		License plate lamp		Rear fog light	Side marker lamp	
Function	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right		Left	Right
Side													
Plug	RBA1	RBA2	RBA1	RBA2	RBA1	RBA2	RBA1	RBA2	RBA1	RBA2	RBA1	RBA1	RBA2
Pin	09	50	13	60	12	61	11	62	16	57	14	15	59

Light pattern													
Left parking light			X										
Right parking light				X									
Standing lights			X	X					X ¹⁾	X ¹⁾		X ²⁾	X ²⁾
Low beams			X	X					X ¹⁾	X ¹⁾		X ²⁾	X ²⁾
Turn signal indicator left/right, hazard warning light system					X	X							
Brake light	X	X											
Reversing light							X	X					
Rear fog light											X		

1) Only FKA/FKB/FHT

2) Only FHS/FHL

Brake light

Requirements for activation:

- Ignition in terminal 15 position
- Brake pedal operated

Tail lights

Requirements for activation:

- Ignition in terminal 15 position
- Engine running.
- Rotary light switch in AUTO position
- Light sensor detects NIGHT.
- OR rotary light switch in MAN position (manual driving lights)
- OR standing lights are active.
- OR parking lights are active (left or right).

Turn signal indicators

Requirements for activation:

- Ignition in terminal 15r or terminal 15 position

Reversing light

Requirements for activation:

- Ignition in terminal 15 position
- Reverse gear is engaged.

License plate lamps

Requirements for activation:

- Low beams are active.

Rear fog light

Requirements for activation:

- Ignition in terminal 15 position
- Rear fog light activated via pushbutton switch
- Rotary light switch in MAN position
- OR rotary light switch in STL position and front fog lamps are active.
- OR rotary light switch in AUTO position and engine is running (the low beams are activated automatically).
- OR rotary light switch in AUTO position and standing lights and front fog lamps are active.

Side marker lamps

Requirements for activation:

- Standing lights or low beams are active.

See 10.1 Bulb ratings of tail lamps (→ page 398).

3) Additional exterior lighting functions

Front fog lamps

Requirements for activation:

- Ignition in terminal 15 position
- Rotary light switch in STL or MAN position
- OR rotary light switch in AUTO position
(AUTO: If the light sensor detects DAY, the standing lights are activated automatically too. If the light sensor detects NIGHT, the low beams are already active and no further automatic activations occur.)
- Front fog lamp activated via pushbutton switch

! NOTE

In the Sprinter, the cornering light function is implemented in the control unit as standard. This also applies for the pre-installation for front fog lamps with cornering light function (LV4). For these, the body-builder manufacturer may only use its own, certified front fog lamps.

Side marker lamps

Requirements for activation:

- The side marker lamps are activated as described under "Tail lights". The same conditions and notes apply.

Clearance lamp

Requirements for activation:

- The clearance lamps are activated as described under "Tail lights". The same conditions and notes apply.

8.5.6 Interior lamps

The information about the lamps and switches installed and the switching commands for the interior lamps are transmitted by the body controller via LIN to the overhead control panel.

The overhead control panel also transmits the information about whether and which pushbutton switches are actuated over LIN to the body controller.

The following special equipment for interior lighting is available ex factory:

Code	Equipment
L65	Additional LED spotlights in the load compartment of panel vans provide more light in the load compartment. One or two additional lamps are installed, depending on the wheelbase. The lamp above the rear-end door has one pushbutton switch.
L71	Motion detector, available for panel van. The interior light is activated when a valid movement is detected.
LD0	Overhead control panel with 2 reading spot lamps
LC4	Comfort overhead control panel (LED), incl. cockpit incident light
LC2	Two LED light strips/work lights in the load compartment of the panel van are always installed in conjunction with code E3I.
L1D	Overhead control panel with bus interior lighting control
LB9	Exit lamps are integrated in the door on the driver's side or front passenger side. These illuminate downward toward the ground when the door is opened. There is additional running board courtesy lighting on the sliding door (only on model series 907).
E3I	Enables the undimmed actuation of the LED light strips/work lights in the load compartment of the panel van (code LC2); this is added automatically for the vehicle configuration with code LC2.
L72	Interface for electrics for body interior lighting for open model designations

! NOTE

All interior lamps can be replaced by other body-manufacturer-specific lights. In order to ensure that the standard lamp failure indicator functions correctly, only illuminants of the same type and same output rating as standard equipment illuminants may be installed.

Lamp monitoring

The overhead control panel or the body controller monitors all outputs for open load (wire break) and short circuit.

If a lamp is not connected or is overloaded, a fault entry is stored in the fault memory of the body controller. Inform the vehicle owner and the driver of this in the accompanying documents of the body or modification. Take into account the fault entry during service when reading out the fault memory.

The interior lighting can also be actuated via the PSM, See 8.10 Parameterizable special module (PSM) (→ Page 372).

Interface for electrics for body interior lighting (code L72)

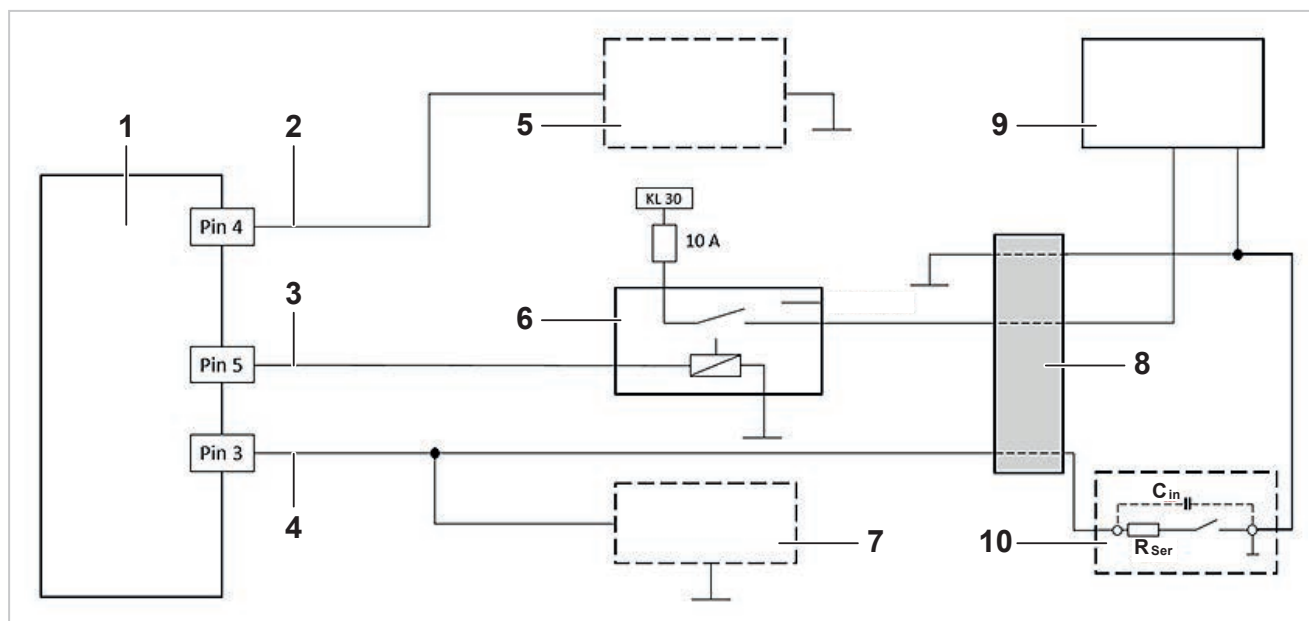
In open model designations, an electrical interface (code L72) in the driver seat box is offered to the bodybuilder manufacturer for further body and/or modification work in the rear vehicle area as special equipment.

At this interface, bodybuilder-manufacturer-specific lamps (bodybuilder manufacturer lamps) can be connected in accordance with the following specifications.

It is optionally possible to connect a pushbutton switch in the body and/or modification via which these bodybuilder manufacturer lamps can be activated and deactivated. From a functional perspective, this pushbutton switch corresponds to the rear lights pushbutton switch in the overhead control panel. If the bodybuilder manufacturer lamps are active in the body and/or modification, the symbol for the rear lights pushbutton switch lights up red in the overhead control panel and thus serves as a function confirmation.

Function:

- The bodybuilder manufacturer lamps can only be activated via a manual push of a button and do not react to automatic light functions (door opening, unlocking, etc.).
- After manual activation of the bodybuilder manufacturer lamps, these remain active for approx. 20 minutes, provided that the on-board electrical system does not detect any undervoltage (battery protection). The time is, in particular, independent of the vehicle lock status.
- Once the bodybuilder manufacturer lamps have been deactivated by the on-board electrical system due to an undervoltage, an engine start is required before the bodybuilder manufacturer lamps can be manually activated again.
- In the FHL model designation, the rear lights in the driver's cab (2nd seat row) still act like in a vehicle without code L72.
- The bodybuilder manufacturer lamps are actuated via a relay and are fused with a 10 A fuse.



Circuit diagram of code L72 interface

- 1 Overhead control panel (with 10-pin connection)
- 2 Interior lamp connection in rear passenger compartment
- 3 Relay actuation connection for special equipment code L72
- 4 Pushbutton switch connection in rear passenger compartment (input)
- 5 Interior lamp in rear passenger compartment for FHL model designation
- 6 Relay
- 7 Pushbutton switch for rear passenger compartment for FHL model designation
- 8 Interface for bodybuilder manufacturer lamps for special equipment code L72
- 9 Bodybuilder manufacturer lamps (scope of body or modification by bodybuilder manufacturer)
- 10 Pushbutton switch for bodybuilder manufacturer lamps

Parameter	Symbol	Min.	Type	Max.	Unit	Remarks
Series resistance to pushbutton switch	R_{Ser}				Ω	For an $R_{Ser} < 390 \Omega$, a reliable diagnosis of a short circuit to ground is no longer possible.
Input voltage (Normal mode)	$U_{ILL_RS_Norm}$	3.1	3.3	3.5	V	Pushbutton switch not actuated
Input voltage (Sleep mode)	$U_{ILL_RS_Sleep}$	4.1	4.3	5	V	Pushbutton switch actuated
Input capacitance	C_{in}			10	nF	
Pushbutton switch current	I_T			10	mA	$R_{Ser} = 0 \Omega$, pushbutton switch actuated
Pushbutton switch current for a stuck pushbutton switch	I_H			150	μA	$R_{Ser} = 0 \Omega$, pushbutton switch stuck
Short circuit current	I_K			10	mA	Short circuit to ground
Permissible shunt resistance	R_S	5			k Ω	Permissible shunt resistance to ground and terminal 30

8.6 Mobile communication systems

If mobile communication systems (e.g. (→ page 97) telephone, CB radio) are retrofitted, the following requirements must be fulfilled in order to avoid malfunctions developing on the vehicle at a later stage:

All items of installed electrical equipment require a type-approval certificate in accordance with UN R 10 and must bear the CE marking or another comparable marking depending on the country-specific regulations.

! NOTE

The optical fiber cable must not be kinked. The minimum bending radius is 25 mm.

8.6.1 Devices

The maximum transmission output (PEAK) at the base of the antenna must not exceed the values resulting from the country-specific laws regarding the maximum legitimized transmission output.

- Do not position mobile communication systems and holders in the deployment areas of the airbags (→ page 156).
- The equipment must be permanently installed. Mobile devices may only be operated inside the cab if they are connected to an exterior antenna that has been installed in such a manner that it is reflection-free.
- Install the transmitter separately from all other vehicle electronics.
- Protect the device against moisture and heavy physical shocks; observe the permissible operating temperature.

8.6.2 Connection and cable routing of the antenna (radio)

Observe information and installation specifications from the manufacturer.

An antenna can be installed anywhere on the roof. The maximum transmission output must not be exceeded.

- ▶ The connection must only be made at terminal 30T via an additional fuse. Disconnect the device from the electrical system before starting assistance.
- ▶ Route the lines as short as possible. The lines must be twisted and shielded (coaxial cable). Avoid chafe marks.
- ▶ Ensure that the system has good ground connections to the body (antenna and device).
- ▶ Route the antenna and connecting lines between the transmitter, receiver, and operating unit separately from the vehicle wiring harness in the vicinity of the body ground.
- ▶ Route the antenna line in such a way that it is not kinked or pinched.

Observe GGVSEB and ADR regulations.

Observe the information in 5.3 Anticorrosion protection measures (→ page 102) and 5.4 Painting and preservation work (→ page 104).

8.7 Electronic ignition lock (EIS)

8.7.1 General information

- The processes involved in the access authorization for the central locking system (CL) are verified and controlled by the signal acquisition and actuation module (SAM) and the door control unit (DCU).
- Access is authorized at the press of a button on the transmitter key.
- Drive authorization is obtained by pressing the start/stop pushbutton switch with Keyless Start. The transmitter key must be located in the area between the A and B-pillars.
- Where control units are networked, the electronic ignition starter switch sends global information such as the model series and country variant to the control units (global variant coding) on the network.

! NOTE

Drive authorization via Keyless-Start is only functional if the body areas between the A- and B-pillars, as well as the roof frame in between, are not modified; See 4.3.1 Modifications to the body-in-white (→ Page 85).

Store the vehicle keys in such a way that the engine cannot be started unintentionally. In the case of camper vans, pay special attention to the storage of the keys on account of the design features of these vehicles. This means in particular that you should not store the keys in the area between the A-pillar and B-pillar, including the roof area, see the further notes in 7.14.20 Reduced Keyless-Start detection for bodybuilder manufacturer conversion with code F50 (→ page 248).

In case of a planned modification to the previously mentioned component parts, the special equipment code FX8, See 7.14.20 Reduced Keyless-Start detection for bodybuilder manufacturer conversion with code F50 (→ Page 248), can be ordered.

For aftermarket modifications, code FX8 must be retrofitted and the electronic ignition starter switch must be coded in via SCN coding; see notes in 7.14.20 Reduced Keyless-Start detection for bodybuilder manufacturer conversion with code F50 (→ page 248). If you have any questions, please contact the Technical Consultancy team (→ page 18).

8.7.2 Central locking system and post-delivery integration of doors of bodybuilder manufacturer

General information

Bodybuilder manufacturers have the option of adapting the central locking system to the body or type of use. The following functions can be implemented via variant coding in the electronic ignition starter switch with XENTRY:

- Automatic locking when last open door is closed (post function)
- Subsequent integration of bodybuilder-manufacturer-specific doors in central locking system

An automatic locking function for speeds above 15 km/h is implemented in the vehicle as standard. More information on this function can be found in the operator's manual.

Automatic locking when last open door is closed (post function)

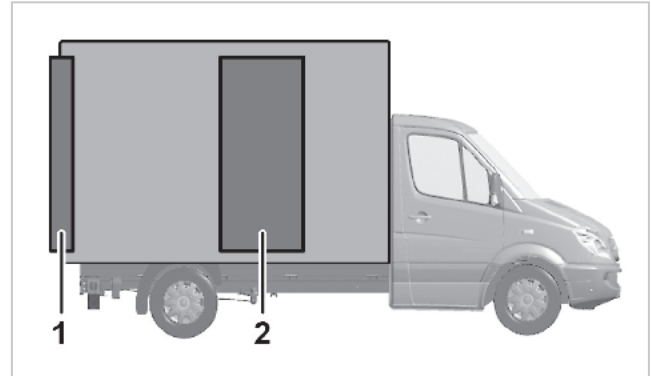
It is possible to use the O-code O22 to implement automatic locking when the last open door is closed. The function is intended to ensure that the vehicle is globally locked if the vehicle is exited, e.g. via a door in the body, and the ignition is in the radio position (terminal 15r). With ignition "on" (terminal 15), no locking of the doors is triggered.

This function requires the use of door locks with a feedback function, as the open/closed state of the door is polled and monitored. If body doors are integrated via the PSM, it is not recommended to use the O22 function.

This code can be parameterized via XENTRY, e.g. at a Mercedes-Benz Service Partner. The O code must first be entered in the vehicle data card. An SCN recoding of the electronic ignition starter switch must then be implemented.

Post-delivery integration of doors

Depending on the vehicle equipment, it is possible for bodybuilder manufacturers to integrate additional doors in the body into the central locking system of the basic vehicle. They are operated via the ignition key of the basic vehicle.



Example of bodybuilder manufacturer doors

- 1 Rear-end door
- 2 Side door

The integration of additional doors into the central locking system is only possible on chassis. Two variants are available:

- Integration of additional doors through PSM
- Integration of additional doors through SAM

Integration of additional doors through PSM

The PSM can be used to read out signal IDs (e.g. "close door", "open door") from the vehicle CAN in order to actuate additional central locking elements or relays in the body via a PSM output.

-  You can find information about parameterization options in the comprehensive "PSM function description", which is available in the Bodybuilder Portal.

<https://bb-portal.mercedes-benz-vans.com>



Advantage

Bodybuilder manufacturers can use their own door locks and components.

Disadvantage

The additional doors actuated via the PSM are not monitored for their "open" or "closed" state. The vehicle is thus not able to recognize whether all additional doors are closed and locked after a locking procedure and no indications are given on the instrument cluster. Furthermore, the vehicle locks itself automatically 40 s after unlocking if only one bodybuilder manufacturer door is opened. The reason for this is that the opening of the rotary tumbler is not detected.

Condition

The vehicle special equipment PSM (code ED5) and a special programming variant of the PSM are required. For this, contact the Technical Consultancy with regard to the PSM (→ page 18).

Integration of additional doors through SAM

Ordering the special equipment "Pre-installation for central locking system for body doors" (code EA5) allows you to integrate additional doors in the central locking system of the basic vehicle.

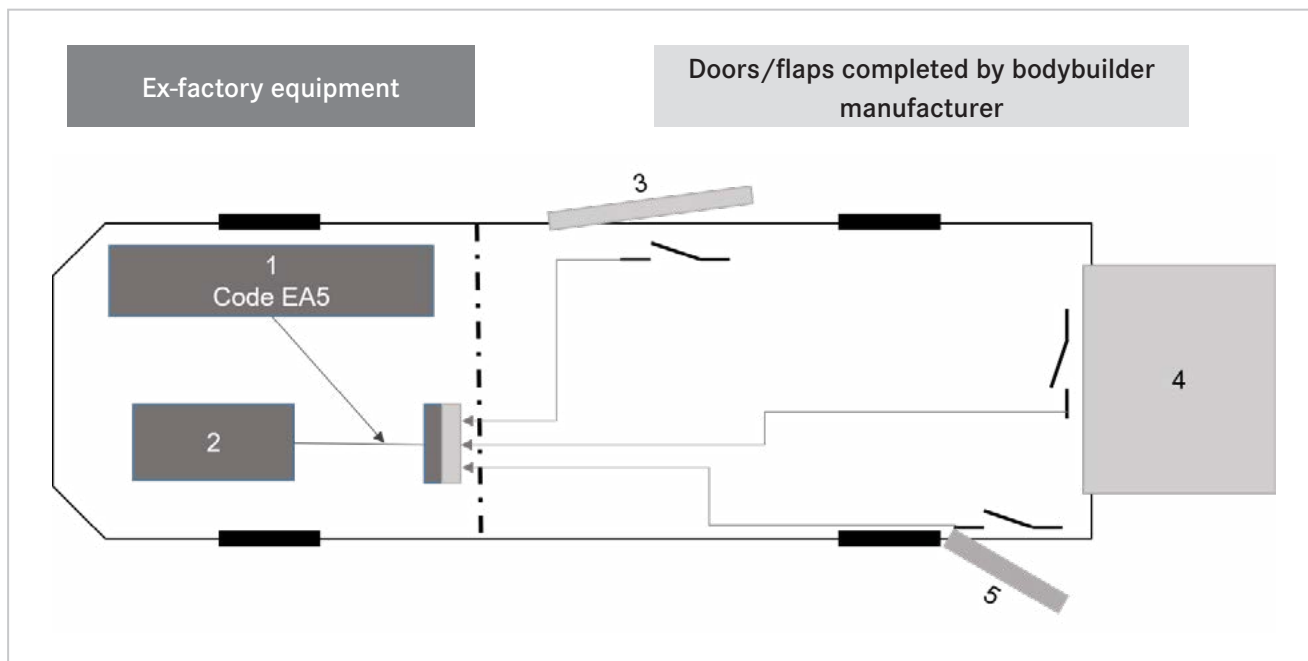
This means that a wiring harness connecting point is installed in the vehicle near the seat on the left for connecting up to three bodybuilder-manufacturer-specific doors behind the B-pillar. Activation of the door signals must be configured individually via XENTRY using O-codes. These codes can be parameterized via XENTRY, e.g. at a Mercedes-Benz Service Partner. The desired O-code must first be written in the vehicle data card, and then an SCN coding of the signal acquisition and actuation module must be carried out.

The O-codes for body doors are:

- OA5 - Parameterization for left side door present
- OA6 - Parameterization for right side door present
- O04 - Parameterization for rear-end door present

Now all the doors can be opened and closed with the vehicle key. The locks used must have an appropriate feedback function so that the central locking system in the SAM can poll the door status and the instrument cluster can display this door status correctly.

Illustration of an example body with three bodybuilder-specific doors in combination with the special equipment "Pre-installation for central locking system for body doors" (code EA5):



Top view with box or camper van body (schematic)

- 1 Electrical connecting point with code EA5
- 2 Body Controller
- 3 Door
- 4 Tailgate
- 5 Flap

Advantage

- Parameterization and installation of a PSM is not required.
- The additional doors actuated via the signal acquisition and actuation module (SAM) are monitored for their "open" or "closed" state. The vehicle is thus able to recognize whether all additional doors are closed and locked after a locking procedure and an indication is given in the instrument cluster.

Disadvantage

The door status indicators in the instrument cluster are illustrations of the panel van and have not been adapted to the body situation (e.g. box body, camper van, etc.).

In combination with the three-button key (code FY7)



Three-button key (code FY7)

With the three-button key (code FY7), under correct use of the interfaces, there is the possibility that the body door used as a tailgate can be unlocked separately. However, locking of the body doors is only possible globally, i.e. for all doors and tailgate.

Conditions

- Vehicle configuration with special equipment "Pre-installation for central locking system for body doors", code EA5
 - Use of locks with feedback function; it is recommended to use original Mercedes-Benz locks.
 - Max. three additional door locks (right door, left door, rear doors) are possible
- i More information on the integration of additional doors by the bodybuilder manufacturer (circuit diagram, components, etc.) is available from the Technical Consultancy (→ page 18).

Ambulance pre-installation

The settings required for ambulances, e.g. passive circuits for rear-end door and sliding door actuator motors, can be applied with XENTRY as follows:

- Right-hand sliding door "not present"
- Left-hand sliding door "not present"
- Rear-end door "not present"
- Common unlocking of control circuits 1 and 2
- Front passenger door "not present"

8.8 Windows and doors

8.8.1 Window lifters and window hinges

The automatic reversing function and automatic window raising can only be guaranteed with the original power window equipment.

The motor is thermally protected, i.e. the availability of the power window function may be restricted after long operating periods.

The window lifters and the window hinges can be controlled using the switches in the door operating unit. The switches in the driver's door are electronically controlled and are connected to the LIN bus. The switches in the front passenger door are voltage-coded.

All switches must only be replaced with equivalent genuine parts.

Actuation can also occur via the parameterizable special module PSM.

8.8.2 Electric sliding door to load compartment

The electrical components of the sliding door on model series 907 (only for the electric sliding door for the load compartment and particular special equipment) are connected to the on-board electrical system via a fixed electrical connection in the form of a flexible power duct (drag chain). This is located under the step for the sliding door.

The flexible power duct must be taken into consideration in the event of any modifications around the entrance. The flexible power duct can be used for the requirements of the bodybuilder manufacturer following consultation with the Technical Consultancy (→ page 18).

The system for the load compartment sliding door has been designed for a maximum weight of 65 kg per door, and this must not be exceeded due to modifications.

The ex-factory weight of the load compartment sliding door for each vehicle model and wheelbase, and additional weights for some items of special equipment, are listed in 6.2.7 Side wall, windows, doors, and flaps (→ page 123). The delta corresponds to the maximum permissible additional weight on the sliding door to the load compartment.

Add certain weights to the basic equipment depending on the special equipment installed.

If equipment needs to be removed, take it into account in the weight analysis.

Further data and information are available in 6.2.7 Side wall, windows, doors, and flaps (→ page 123).

Furthermore, observe the specifications on "Reinforcement of sliding door to load compartment (LST) with rear seats" for the countries in question; See 7.3.1 Retrofitting seats (→ Page 209).

WARNING

Modifications to

- door kinematics,
- guidance systems (rails, sliding carriages etc.),
- locking systems (closing assists, anti-entrapment strips, locks, arresters etc.),
- end stop systems (buffers etc.) and
- extensive modification of the body-in-white structure

may result e.g. in the door opening unintentionally.

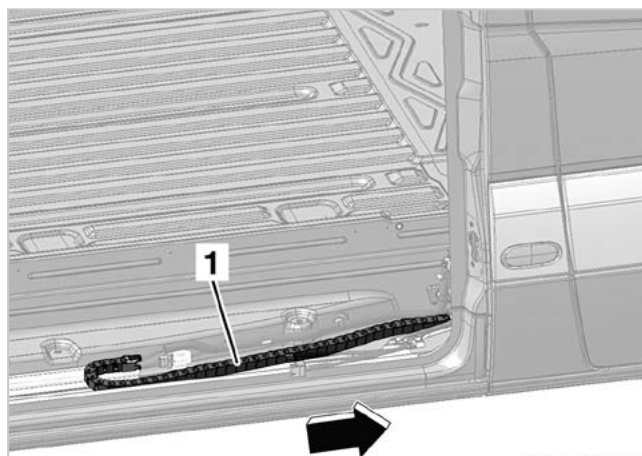
There is risk of an accident and danger to life and limb!

Do not make any modifications to these systems. Furthermore, take care to ensure an extensive connection and center of gravity distribution at the doors.

- i** The bodybuilder manufacturer alone is responsible for the conversions and modifications it makes to the vehicle.

! NOTE

Correct operation of the integrated automatic reversing function (anti-entrapment strip and path/time monitoring) must be ensured in the event of any modifications in this area, e.g. the window installation.



Sliding door with flexible power duct

- 1 Flexible power duct (drag chain)
 ➔ Direction of travel

8.8.3 Windshield wipers

The use of genuine Mercedes-Benz wiper motors is recommended.

If necessary, a second wiper motor can be connected via a relief relay ($R_i > 80 \Omega$).

The wiper motor must be connected to the signal acquisition and actuation module SAM by means of a readback line. If there is no readback line, a fault message will be stored in the fault memory of the SAM.

8.8.4 Outside mirror

The output of the mirror heater (12 V/20 W) is monitored by the door control unit. The mirror heater (if a genuine part) is deactivated if a fault entry is stored.

Design the mirror heater in line with the genuine part.

The door control unit must be modified by coding if different mirrors (without heating) are used.

The mirrors are adjusted via the operating switch in the driver's door and the adjustments are transmitted via the LIN bus to the door control unit. The door control unit in turn actuates the actuator motors.

Design the mirror adjustment drives in line with the genuine drives.

Observe the design of the outside mirror with or without Blind Spot Assist (→ page 318).

! NOTE

The bodybuilder manufacturer assumes sole responsibility for ensuring compliance with the legal requirements for outside mirrors (fields of view, length of mirror base dependent on the maximum vehicle width, etc.).

- i** The outside mirror variants installed at the factory and available as special equipment, including for retrofitting on body-manufacturer-specific bodies, can be found in 4.1.3 Permissible vehicle dimensions (→ page 79).

8.8.5 Windshield heater and rear window heater

The original heaters can be replaced with heaters with the same power rating:

- Windshield heater:
 $P = \max. 790 \text{ W at } 13 \text{ V}$
- Rear window heater:
 $P = 2 \times 151 \text{ W} \pm 15 \text{ W at } 13.5 \text{ V}$

If higher heating outputs are required, modify the relays, lines, and fuses accordingly.

If the vehicle has the battery management function, it may be necessary to adapt this function as well.

8.9 Driving assistance systems

⚠ WARNING

Driving safety and driving assistance systems are merely aids.

If the driver fails to adapt their driving style or if the driver is inattentive, the driving safety systems and driving assistance systems can neither reduce the risk of an accident nor override the laws of physics. The driver is responsible for driving safely and in keeping with traffic regulations. Their driving style must always be adapted to suit the current road, traffic, and weather conditions.

! NOTE

Always observe the information and specifications in the Operator's Manual of your vehicle. Make yourself familiar with the functions of the driving assistance systems.

The driving safety systems only work as effectively as possible when there is adequate contact between the tires and the roadway. For this, observe in particular the information on tires and recommended tire tread depths under 3.8 Tires (→ page 65), "Checking the tires" (→ page 73), 4.2 Limit values for the suspension (→ page 80), 4.2.3 Approved tire sizes (→ page 83), 6.1.5 Wheels and tires (→ page 111) and 6.2.8 Fenders and wheel wells (→ page 127).

⚠ WARNING

Tampering with or unauthorized installations in vehicle systems, safety-relevant component parts and driving assistance systems can impair the functioning of these systems. This can lead to the failure or malfunctioning of components or component parts relevant to safety, and may result in accidents or damage to the vehicle.

There is risk of an accident and danger to life and limb!

Furthermore, tampering with the vehicle, safety or driving assistance systems and safety-relevant component parts can invalidate the warranty or the operating permit.

- ❗ Information on the equipment of the vehicle with regard to driving assistance systems and combinations thereof can be obtained via the online vehicle Configurator, from your Mercedes-Benz Sales representative, or from the Technical Consultancy (→ page 18).

Commissioning of control units

Before the vehicle is put into circulation, observe the information and specifications on commissioning of control units following completion of the body mounting work; See 8.19 Commissioning of control units (→ Page 387).

! NOTE

After commissioning and before the vehicle is placed on the market, it must always be ensured that all existing vehicle and driving assistance systems are functioning flawlessly. For this, contact a technical service that has been designated by the responsible registration authority and that possesses the necessary knowledge and testing equipment.

8.9.1 Electronic Stability Program (ESP®)

ESP® is a vehicle dynamics control system that actively regulates both the longitudinal and lateral dynamics of the vehicle.

Greater driving stability is provided by ESP® with an extended sensor system that constantly compares the current actual vehicle direction with the driver's desired direction of movement.

ESP® can improve vehicle stability in all driving situations, e.g. when accelerating, braking and coasting, when driving in a straight line and cornering.

Together with the signals of other sensors, a computer monitors that the direction specified by the driver is being maintained.

If the vehicle deviates from the correct path (over-steering or under-steering), the system produces a stabilizing counteraction by applying the brakes on individual wheels.

An additional ESP® variant can be ordered with the vehicle version for high load (code B01) separately for specific applications.

Detailed information See 4.2.1 Suspension
(→ Page 80)

A functioning electronic stability program is required by law in the EU for all classes of vehicle.

This affects all countries in which EU legislation or the Directive (EU) 2018/858 applies. The EU regulation also contains exceptions for some special-purpose vehicles.

For vehicles where exceptions due to body mounting work apply, the ESP® can be deactivated under certain conditions. More information is available from the Technical Consultancy (→ page 18).

In the event of deviations from the specifications for ESP®, a Letter of Compatibility is required (→ page 19); this can be requested via the Body-builder Portal (→ page 20).

! NOTE

The special equipment code B01 is prescribed for bodybuilder manufacturer vehicles with a high center of gravity in accordance with the specifications in 4.1.2 Maximum permissible position of the vehicle center of gravity (→ page 78).

⚠ WARNING

Modifications to vehicles with ESP® may cause this system to stop functioning correctly, and this may lead to system shutdowns and incorrect control interventions. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

None of the following modifications may be made to vehicles equipped with ESP®:

- Modifications to the permissible gross mass
- Wheelbase modifications
- Modifications to the sensor system (steering angle, yaw rate, rotational speed sensors)
- Changes to the vibration characteristics at the installation location of the yaw rate sensor by modifications to the body
- Changes to the position of ESP® components
- Modifications to the suspension
- Modifications to wheels and tires
- Modifications to the engine
- Modifications to the steering
- Modifications to the brake system
- Changes to the installation position of the hydraulic unit, its holder and its mounting on the basic vehicle
- Fastening of any vibration-generating devices in the vicinity of the ESP® control unit

8.9.2 Crosswind Assist (CWA)

Crosswind Assist (code JA8) detects the influence of crosswind (track offset) through the installed ESP® components and can counteract it by applying the brakes on individual wheels with the help of the ESP®.

For the vehicle variants of model series 907, Crosswind Assist (code JA8) is added automatically as standard in combination with ESP®.

! NOTE

All body mounting work on the vehicle that changes the projected side surface area must be within the tolerance range of the Crosswind Assist test envelope that differs in its dimensions depending on the model designation and body type; see the following pages.

Measures that affect the projected side surface area include, for example, changes to the overhang, See 6.2.4 Overhang extension (→ Page 119), or the retrofitting of high roof attachments, See 4.3.8 Roof and roof load (→ Page 90).

⚠ WARNING

Increasing the projected side surface area of the vehicle within the limits of the CWA test envelope can reduce the effect of Crosswind Assist due to the changes in the physics of driving.

There is risk of an accident and danger to life and limb!

! NOTE

Crosswind Assist can be deactivated under the following conditions:

- When the option "Omission of Crosswind Assist" with code JA0 is ordered
- For post-delivery deactivation of Crosswind Assist, code JA8, this code must be deleted from the data card and the code JA0 added.
- Finally, the ESP® must be recoded. This work must be performed at a qualified specialist workshop.

i For more information, contact the Technical Consultancy (→ page 18).

! NOTE

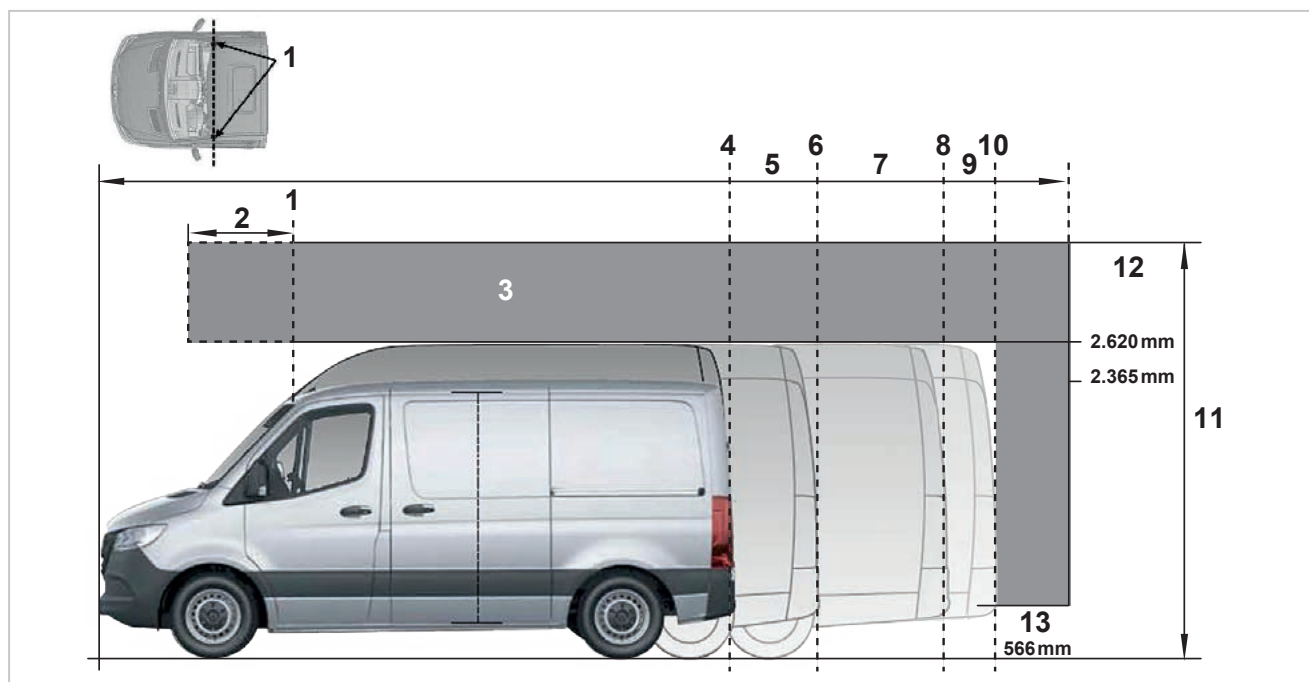
The Electronic Stability Program (ESP® 9.3) is a prerequisite for the Crosswind Assist function. In the case of conversions that necessitate the post-production deactivation of the ESP® 9.3, Crosswind Assist must be deactivated; see previous note, and 7.10 Semitrailer trucks (→ page 218).

For vehicle body mounting work, the respective country-specific registration regulations in terms of geometric dimensions and the mandatory ESP® must be observed and complied with.

Bodies with more than 2 axles (e.g. 6×2) do not have the approval of Mercedes-Benz AG for the Crosswind Assist function.

The approved geometric limits and the maximum permissible body dimensions of the body of the bodybuilder manufacturer on the model series 907 are defined and can be implemented by the CWA test envelope shown on the diagram in accordance with the controlled range of the Crosswind Assist (CWA) system.

The limits are illustrated in the diagrams below. The test envelopes shown (vehicle plus marked body area) depict the maximum permissible external dimensions of the vehicle.



Test envelope for Crosswind Assist on panel van, bus, crewbus (schematic)

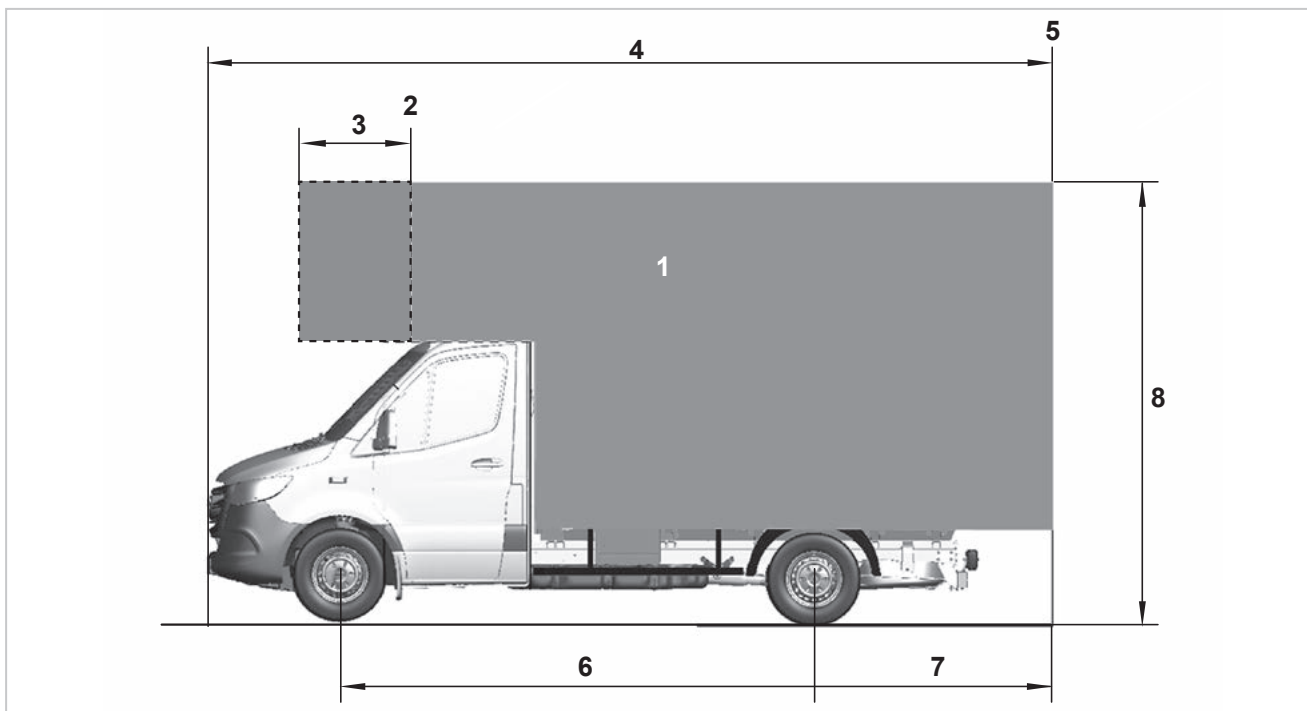
- | | |
|--|-----------------------------------|
| 1 Outside edge of windshield | 7 Maximum vehicle length for A2 |
| 2 Max. overhang over windshield | 8 Body length A3 |
| 3 CWA test envelope maximum dimensions | 9 Maximum vehicle length for A3 |
| 4 Body length A1
(no longer available as of model year 2024) | 10 Body length A4 |
| 5 Maximum vehicle length for A1
(no longer available as of model year 2024) | 11 Maximum vehicle height |
| 6 Body length A2 | 12 Vehicle heights |
| | 13 Minimum height of loading sill |

Dimensions of CWA test envelope for panel van, bus, crewbus model series 907

Tonnage		3.0 t		3.5 t		4 t			5 t		
Body length/length class		A2	A2	A3	A4	A2	A3	A4	A2	A3	A4
Length	[mm]	5932	5932	6967	7367	5932	6967	7367	5932	6967	7367
Wheelbase	[mm]	3665	3665	4325	4325	3665	4325	4325	3665	4325	4325
Maximum overhang ¹⁾	[mm]	1850	1850	2200	2200	1850	2200	2200	1850	2200	2200
Maximum vehicle length within CWA test envelope	[mm]	6536	6536	7546	7546	6536	7546	7546	6536	7546	7546
Maximum vehicle length within CWA test envelope for camper vans	[mm]	6885	6885	7941	7941	6885	7941	7941	6885	7941	7941
Maximum vehicle length within CWA test envelope	[mm]	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000
Maximum overhang of body over windshield (e.g. alcoves) ²⁾	[mm]	1150	1150	1150	1150	1150	1150	1150	1150	1150	1150

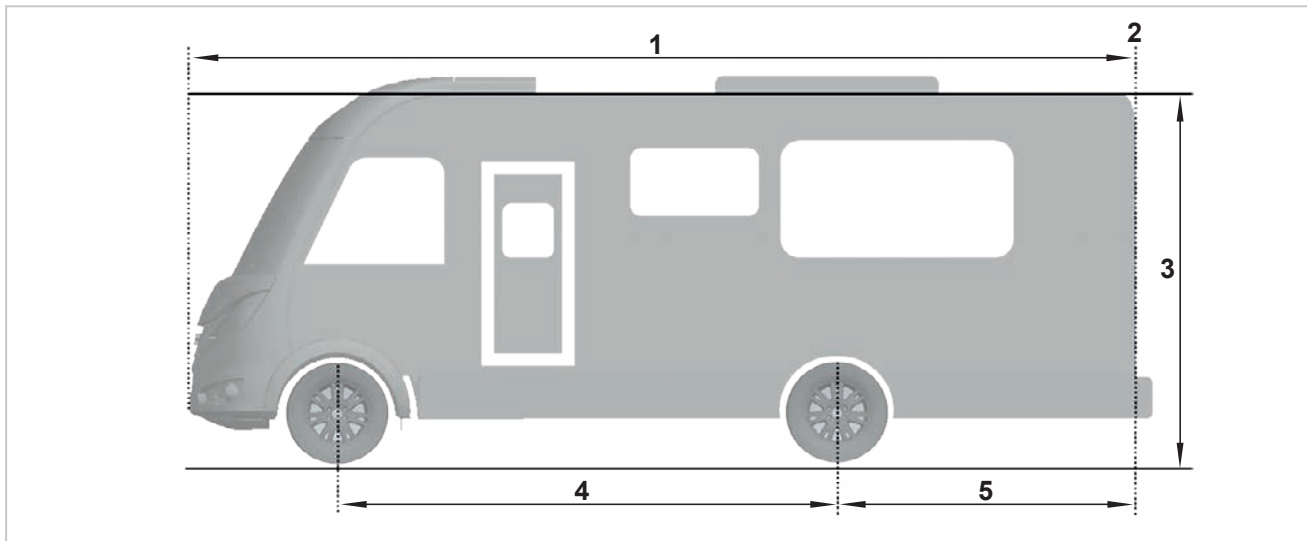
1) See 4.3.5 Overhang (→ page 88)

2) See 8.9.6 Automatic driving lights/rain sensor (→ page 340)



Test envelope for Crosswind Assist for chassis with platform vehicles, box body vehicles and semi-integrated camper vans (schematic)

- | | |
|---|--------------------------------|
| 1 CWA test envelope maximum dimensions | 5 Outside edge of vehicle body |
| 2 Outside edge of windshield | 6 Wheelbase |
| 3 Max. overhang over windshield | 7 Maximum overhang |
| 4 Maximum vehicle length within CWA test envelope | 8 Maximum vehicle height |



Test envelope for Crosswind Assist for chassis with fully integrated camper vans (schematic)

- | | |
|---|--------------------|
| 1 Maximum vehicle length within CWA test envelope | 4 Wheelbase |
| 2 Outside edge of vehicle body | 5 Maximum overhang |
| 3 Maximum vehicle height | |

Dimensions of CWA test envelope for chassis model series 907 platform vehicles, box body vehicles, camper vans, etc.

Tonnage		3.0 t	3.5 t		4 t		5 t	
Body length/length class		A2	A2	A3	A2	A3	A2	A3
Length	[mm]	6046	6046	6946	6046	6946	6246	6946
Wheelbase	[mm]	3665	3665	4325	3665	4325	3665	4325
Maximum overhang ¹⁾	[mm]	1850	1850	2200	1850	2200	1850	2200
Maximum vehicle length within CWA test envelope	[mm]	6536	6536	7546	6536	7546	6536	7546
Maximum vehicle length within CWA test envelope for camper vans	[mm]	6885	6885	7941	6885	7941	6885	7941
Maximum vehicle length within CWA test envelope	[mm]	4000	4000	4000	4000	4000	4000	4000
Maximum overhang of body over windshield (e.g. alcoves) ²⁾	[mm]	1150	1150	1150	1150	1150	1150	1150

1) See 4.3.5 Overhang (→ page 88)

2) See 8.9.6 Automatic driving lights/rain sensor (→ page 340)

Comply with the generally applicable boundary conditions:

- For the Crosswind Assist function to be used or enabled, the body width of the vehicle without outside mirrors (symmetrical in the longitudinal direction) must be no more than max. 2500 mm.
- The body/vehicle width limit values relevant for registration in terms of the installation of headlamps and outside mirrors on the vehicle are the responsibility of the bodybuilder manufacturer and must be guaranteed by them; See 4.1.3 Permissible vehicle dimensions (→ Page 79).
- The maximum body width must be less than or equal to the distance between the outside edges of the long-arm outside mirrors on the right and left.
- These test envelope guidelines do not apply to bodies on which the outer surface exhibits substantial aerodynamic differences, particularly at the front as far as the B-pillar (special bodies).
- Exceptions to this exist for fully integrated camper van bodies. Such vehicles must be assessed individually (if necessary by a practical test drive) by the department for vehicle dynamics of Mercedes-Benz AG at the bodybuilder manufacturer's expense.
- The (additional) body area must be within the marked ranges and comply with the maximum values specified in the vehicle tables.

8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC

The Active Brake Assist (code BA3) is legally prescribed depending on the country and can assist the driver in avoiding collisions with vehicles traveling or crossing ahead as well as pedestrians, and also reduce the severity of consequences from accidents.

The Active Distance Assist DISTRONIC (code ET4) automatically maintains distance from vehicles traveling in front and can therefore relieve the burden on the driver when driving on highways or in frequently stopping and starting traffic, for example.

! NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

For the purposes of these driving assistance systems, a radar sensor is installed behind the air inlet grille in the step plate of the bumper. This front radar works together with the signals from the ultrasonic sensors in the bumper and from the multi purpose camera (MPC, 10.2.1 Installation dimensions, version 1 (→ page 404)) in the lower area of the window to identify potentially dangerous situations.

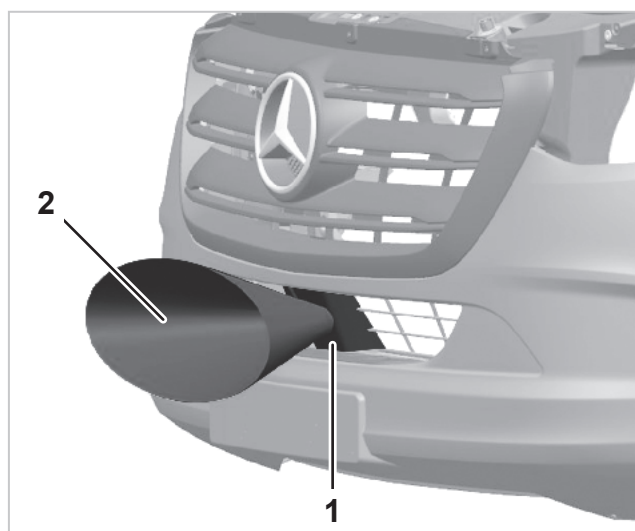
! NOTE

After changes to the vehicle width or overhang, enter the corresponding O-codes in the vehicle documentation in XENTRY, carry out SCN coding of the control unit and, in the case of changes to the vehicle position due to a change in vehicle weight, document the height of the radar sensor above the roadway.

Observe the following note on changing the inclination of the vehicle in the longitudinal direction.

i For further information See 8.19 Commissioning of control units (→ Page 387).

i Details of the front radar can be found in the 3D design data (→ page 25).



Position of front radar sensor

- 1 Position of sensor
- 2 Signal funnel (schematic), for details see design data

⚠ WARNING

Any modifications to the front radar, its attachment to the vehicle and surrounding area, as well as any bodies, installations or conversions that affect the functionality of the front radar, are impermissible.

Otherwise, driving assistance systems may no longer work as intended.

There is danger to life and limb!

To ensure that the functioning of the driving assistance systems is not affected, the following changes are not permissible:

- Changing the position of the radar sensor
- Replacing the original holder with a different one
- Removing the original front bumper or the original cover
- Replacing the original front bumper or the original cover with others
- Installing detachable parts that protrude into the signal funnel
- Installing metallic detachable parts that could shade or affect the area around the sensor

! NOTE

The area in front of the sensor (viewed in the direction of travel) also includes parts that are located beside, above, or below the sensor, but that protrude further forward than the front edge of the radar itself.

- Metal or metal-plated pipes (see note) or rounded metal or metalized surfaces located in front of or beside the radar sensor when viewed in the direction of travel; exception: When these are covered with an insulating or radar-wave-absorbing plastic or paint in the direction of the radar sensor.
- Parallel metal surfaces (see note) on both sides of the radar sensor; exception: When these are covered with an insulating or radar-wave-absorbing plastic or paint in the direction of the radar sensor.
- No metal or metalized surfaces perpendicular to each other (see note) beside, above and below the radar sensor, where the opening of the 90° angle is "visible" to the radar sensor; exception: When these are covered with an insulating or radar-wave-absorbing plastic or paint in the direction of the radar sensor.
- Additional paintwork on the cover in front of the radar sensor; observe warning!
- Application of adhesive film (wrapping) in the vicinity of the radar sensors; See 3.13 Adhesive films on the vehicle exterior (→ Page 75).
- Application of filler on the cover in front of the radar sensor; observe warning!

⚠ WARNING

Depending on the type and thickness, paints or films can cause damping of radar waves. This could lead to malfunctions or system failure. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

The area of or around the sensor must not be painted or covered with a film.

In repair cases, remove the existing coat of paint in such a way that the plastic cover is not damaged and the material thickness of the cover is not significantly reduced. The new paint coat structure must not exceed the following limits:

- A maximum of 2 color coats with a thickness of 15 µm plus 2 clearcoats may be applied.
- With metallic silver, only one coat with a thickness of 15 µm plus one coat in white 15 µm thick are permissible.
- The material thickness must be homogeneous in order to prevent distortion of the radar waves.
- Specified thickness of base material for unpainted cover: 2.42 mm ± 0.1 mm
- Specified thickness of base material for painted cover (measured without paint): 2.42 mm + 0.1 mm/- 0.2 mm

 For more information about the painting, See 5.4 Painting and preservation work (→ Page 104).

Modifications to the suspension (e.g. number of axles) not covered by the Body/Equipment Mounting Directive; See 6.1 Suspension (→ Page 106).

NOTE

Active Brake Assist is a certified system. Modifications of any kind made to the individual components, the system, or the basic vehicle (including the wheel-base, track width etc.) that are not explicitly covered by this Body/Equipment Mounting Directive constitute an alteration of the certified state.

An Active Brake Assist system is legally prescribed in the EU. The vehicle will no longer be eligible for registration if this function is deactivated. Refer to the EU legislation for exceptions.

NOTE

The Active Distance Assist DISTRONIC requires the installation of a functioning driver seat occupancy recognition system that has been approved by Mercedes-Benz AG. This can be done by ordering either the standard equipment "Driver's seat occupancy recognition" (code SK2) or the "Pre-installation for driver's seat occupancy recognition" (code E3S).

For further information, See 4.5.2 Modifications to seats (→ Page 96), 6.4.1 General information (→ page 151) and 7.14.7 Connecting point with omission of driver's and front passenger seat for pre-installation for seat belt warning system (code J1D/J1E) and seat occupancy recognition (code E3S) (→ page 237).

NOTE

For special body situations, e.g. fully integrated camper vans on the basis of cowl cab base vehicles (code F50) or government vehicles (list not exhaustive) with deviations from the previously mentioned specifications, a Letter of Compatibility is required (→ page 19); this can be requested via the Body-builder Portal (→ page 20).

 In such cases, contact the Technical Consultancy during the planning phase (→ page 18).

! NOTE

If a body installed by the bodybuilder manufacturer permanently changes the inclination of the vehicle in the longitudinal direction by more than 1.5°, a service calibration of the front radar system must be performed. If this is not done, the intermittent or permanent failure of Active Brake Assist and Active Distance Assist can be expected in subsequent driving operation.

Observe the following when determining the change in angle:

- For the angle measurement, either the vehicle must be standing on a leveled surface or the angle measurement should always be taken at the same measuring station.
- The spirit level must have an accuracy of at least 0.1°.
- The spirit level must output or display the angles digitally.
- The angle measurement before and after the conversion of the vehicle must always be taken at the same point on the chassis. Therefore, select a point that will remain accessible for the angle measurement after the conversion work. Measuring at different points on the chassis or at points on the body (indirect angle measurement) is not permissible.

! NOTE

After any damage to the front of the vehicle, have the setting and function of the radar sensor checked and recalibrated at a qualified specialist workshop. This also applies to mild collisions at low speeds, where no damage to the front of the vehicle is visible.

Adjust the parameterization of the front radar after the following changes:

- Springs: Observe 4.2.8 Modifications to springs, spring suspension, and dampers (→ page 84).
- Tire size: Observe 4.1.3 Permissible vehicle dimensions (→ page 79).
- Track width: Only in combination with a Letter of Compatibility (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).
- Vehicle width (without outside mirrors/detachable parts)

8.9.4 Blind Spot Assist/Rear Cross Traffic Alert/Exit warning function/Sideguard Assist

Available special equipment

Code	Equipment ¹⁾
JA7	Blind Spot Assist, Rear Cross Traffic Alert (RCTA) and exit warning
JT7	Sideguard Assist
J1V	Blind Spot Assist/exit warning pre-installation (only for open model designations)

- 1) The description is intended as an explanation and is not the same as the code designation.
-  Blind Spot Assist is also known as Blind Spot Monitoring (BSM) and, in combination with Sideguard Assist, as the Blind Spot Information System (BSIS).

 NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

Blind Spot Assist can warn the driver of vehicles on the adjacent lane when changing lanes.

The Rear Cross Traffic Alert function can warn the driver of traffic crossing behind the vehicle while reversing, e.g. when moving out of a parking space.

The Rear Cross Traffic Alert function is generally not available for open model designations.

The exit warning system can warn of traffic passing at the side when the driver's door or front passenger door are opened; on vehicles with a crewcab, it can also warn of traffic passing behind when the rear doors are opened.

To ensure that the exit warning system functions properly, the door control signals must be available via the CAN bus. For this purpose, commissioning with coding of the door control units must always be carried out for code J1V and generally for changes to the doors or for bodybuilder-manufacturer-specific doors; See 8.19 Commissioning of control units (→ Page 387).

 **WARNING**

The outer contours of the vehicle body, detachable parts that protrude from the outer surface of the body or retrofitted vehicle equipment must not lead to any restriction of the detection ranges or penetration of the signal funnels of the radar sensors, as otherwise the complete function of the driving assistance system is no longer ensured.

There is risk of an accident and danger to life and limb!

Ensure that there are no restrictions to the function and detection ranges during planned body or modification work, when positioning and retrofitting the radar sensors and during driving operation.

 **NOTE**

In special cases where the detection range of the radar sensors may be impaired by legally required detachable parts, please contact the Technical Consultancy team (→ page 18) to discuss the type, size and position of the required detachable parts and their compatibility with the driver assistance system.

NOTE

Bodybuilder manufacturer-specific operator's manuals must contain appropriate warnings and information for the vehicle user, which at least refer to:

- Possible danger from retrofitted detachable parts or vehicle equipment that protrude into the detection range of the radar sensors, e.g. bicycle rack at the rear of the vehicle
- Possible impairment of the function of the driving assistance system if the radar sensors and their surroundings on the vehicle are damaged
- Obligation to have the function of the driving assistance system checked by a qualified specialist workshop if damage is detected and after accidents affecting the radar sensors or their surroundings on the vehicle; it is recommended that you have this check carried out by your Mercedes-Benz Service Partner.
- The specifications for covering the radar sensors or subsequent painting

1) Equipment ex factory with code JA7 and JT7

As of model year 2024, a new system generation of radar sensors for Blind Spot Assist (code JA7) is installed on a country-dependent basis as standard equipment:

- On closed model designations with M1 /N1 registration
- On open model designations with platform ex works (code P02) with N1 approval
- For closed and open model designations with platform ex works (code P02) with N2 registration in combination with Sideguard Assist (code JT7)

In addition, the code JA7 is also available as special equipment for open model designations without a platform ex factory. These chassis vehicles with incomplete CoC documents (code XC1) must be equipped with the rear underride guard (code C71) on which the radar sensors are mounted. Always observe the specifications for chassis vehicles (→ page 321).

 Contact your sales partner at Mercedes-Benz AG to inquire about the availability of the special equipment options.

For special cases in which there is no country-specific legal obligation to use the previously mentioned standard equipment after a conversion by a bodybuilder manufacturer, it is possible to omit these when ordering a vehicle with omission codes:

Equipment code	Omission code	Equipment
JA7	JZ8	Omission of Blind Spot Assist
JT7	JZ6	Omission of Sideguard Assist

Possible examples of such cases are vehicle conversions with N2 registration to camper vans with code X2R or vehicles for passenger transport with code X2P, as well as to buses with subsequent change of registration to classes M2 or M3 by the bodybuilder manufacturer.

In the case of planned conversions as per 6.2.4 Overhang extension (→ page 119), the use of code JA7 is generally not permitted, and code JT7 is therefore also not possible.

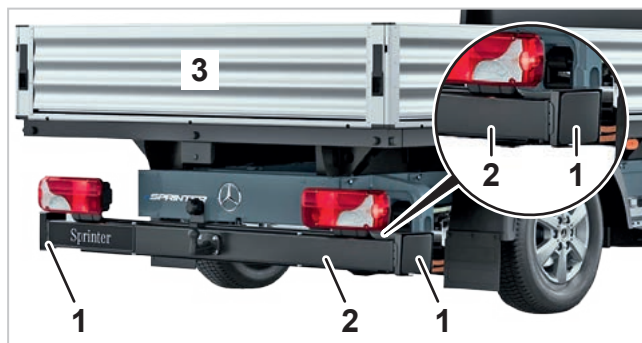
In these cases, check the country-specific legal obligation before ordering the vehicle and, if vehicle registration without these driving assistance systems is permissible, select the omission with the previously mentioned codes when ordering the vehicle.

i In any event, it is recommended to clarify the approval and registration requirements of a body or modification with the Technical Service and/or with the relevant type-approval authority during the planning phase.

If the instructions on prohibited modifications are ignored, the system may need to be deactivated. The deactivation of the system can only be carried out at a qualified specialist workshop. The Sideguard Assist can warn the driver of other road users in the restricted area of vision when maneuvering around corners in the direction of the front passenger side. In addition to Blind Spot Assist, the warning area is extended to the front of the vehicle, see the following illustration of the detection ranges.

The radar sensors of the Blind Spot Assist (code JA7) in combination with Rear Cross Traffic Alert and the exit warning system are installed on the left and right side of the vehicle under the rear bumper trim on closed model designations (panel van and Tourer).

On open model designations, the rear radar sensors for Blind Spot Assist are installed in a separate holder with cover trim, which is attached to the rear under-ride guard (code C71) on the left and right sides of the vehicle.

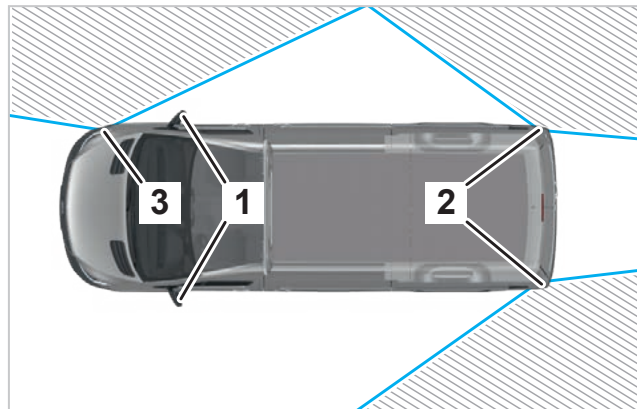


Positions of rear radar sensors (code JA7) for open model designations (example for vehicle with platform ex works)

- 1 Radar sensors, left/right
- 2 Rear under-ride guard (code C71)
- 3 Platform ex works (code P02)

For Sideguard Assist (code JT7), a radar sensor is installed under the front bumper on the front passenger side on all model designations.

The signal is conveyed to the driver by means of a light-up symbol in the outside mirrors, a warning on the instrument cluster and an acoustic warning tone. The Rear Cross Traffic Alert warning is also displayed in the head unit if the Parking Package special equipment (code JB6/JB7) is installed.



Positions and detection ranges of the radar sensors for code JA7 and JT7 (schematic), by way of example for closed model designations, same for open model designations

- 1 Display of the warning in the outside mirror
- 2 Positions of rear radar sensors (code JA7), on both sides
- 3 Position of front radar sensor (code JT7), front passenger side

To ensure the proper function of the driving assistance systems, the following changes are prohibited:

- Changing the warning lamps in the outside mirror (if no custom warning lamps are used, e.g. with omission of the doors, code FW8/FW9)
- Changing the positions of radar sensors, original holders and cover trim pieces as well as other component parts in the immediate vicinity (e.g. bumpers).
- Bodies or modifications, attachment of detachable parts or retrofitting of vehicle equipment that restrict the detection ranges or protrude into the signal funnel of the radar sensors. Observe the warning and exception (→ page 318).
Details of the signal funnel can be found in the 3D design data (→ page 25).
- Applying filler to the cover or to the corner of the bumper in front of the radar sensor; observe the warning and the following painting specifications (→ page 323).

WARNING

The cover of the radar sensors (bumper or cover trim) is made of a plastic specially designed for radar signals. For this reason, the driving assistance system may only be used with genuine parts from Mercedes-Benz. Replacement with other component parts or materials is not permissible.

There is risk of an accident and danger to life and limb!

In the event of deviations, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

The following specifications and information must also be observed when planning and carrying out body work based on chassis vehicles (open model designations without platform ex works) with special equipment code JA7:

- The planned body work must be feasible in combination with the rear underride guard (code C71) and the side-mounted radar sensors.
- Changes to the components or positions of the rear underride guard, the side radar sensors with original holders and cover trim pieces and electric lines are not permissible.
- Modifications to the chassis and vehicle frame (e.g. changing the wheelbase or rear overhang by extending the frame) are not permissible.
- Body work based on chassis vehicles (e.g. body width or overhang) is only permitted to be carried out in accordance with the specifications and information in this Body/Equipment Mounting Directive.
- The detection ranges of the radar sensors must not be restricted. It is essential to observe the warning (→ page 318).
It is recommended that compliance with this requirement be ensured as early as the design phase of the planned body work using suitable methods (e.g. design analysis with 3D design data). Details of the signal funnel can be found in the design data (→ page 25).
- The function of the radar sensors must also be ensured when the vehicle is used under typical conditions for the body. Avoid damaging the radar sensors. Damaged radar sensors must be replaced immediately and may no longer be used during driving operation.

- i** For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).
The individual item number of the radar sensor is identical for all equipment variants, but the appropriate code JA7, JT7 or J1V must be observed when selecting it.

If these requirements cannot be met, the equipment "Pre-installation for Blind Spot Assist" (code J1V) must be selected when ordering the vehicle, or – if there is no legal obligation – the omission with codes JZ8/JZ6.

Parameterization of the radar sensors that deviates from the as-delivered state is not possible with Blind Spot Assist (code JA7), which is fully installed at the factory.

Subsequent deactivation of codes JA7/JT7

If, on vehicles with codes JA7/JT7, the aforementioned modifications are not avoided and the specifications are not met (→ page 321), the driving assistance system may need to be deactivated. When deactivating the driving assistance system, it is essential to observe country-specific laws, guidelines and registration regulations that may prescribe the use of the driving assistance systems. The deactivation can only be carried out at a qualified specialist workshop and requires the following actions:

- Adaptations in the vehicle data card via VeDoc (only possible at your Mercedes-Benz Service Partner): Removal of codes JA7/JT7 and entry of codes JZ8/JZ6
- Recoding of the control units: electronic ignition starter switch, instrument cluster, head unit, door control units and radar sensors
- Replacement of the mirrors with a variant without warning display

Subsequent conversion of code JA7 to J1V

In special cases where chassis vehicles (open model designations without platform ex works) have been ordered with code JA7, but the previously mentioned specifications cannot be met, it is possible to change to the scope of pre-installation (code J1V) with additional work on the body.

To do this, the factory-fitted components of code JA7 on the vehicle must be replaced with those of code J1V and the driving assistance system must be recommissioned. You can obtain the new component parts from your Mercedes-Benz Service Partner (→ page 326).

The radar sensors are identical and can therefore continue to be used.

! NOTE

The factory-installed holders and cover trims for the radar sensors (code JA7) must not be used for the installation situation with pre-installation (code J1V).

It is therefore absolutely necessary to replace these component parts; see the information and item numbers for pre-installation in Section 2 (→ page 324).

The wiring harness available for code JA7 can be professionally extended on the side of the electrical connection to the radar sensors for each side of the vehicle using identical cables – see section 4 (→ page 333) – and the existing plug housing by a qualified electrician. The extension of the wiring harness must not exceed 2 m per vehicle side. In addition, the information and specifications in 8.4.3 Cable extension (→ page 273) and in the documents specified in the Workshop Information System must be observed.

Make sure that the existing lines are reconnected to the respective contact pins of the plug as found in the vehicle. Note the variation with or without Sideguard Assist (code JT7), see connection scheme for code J1V (→ page 334).

After the installation, the vehicle data card must be adapted via VeDoc by your Mercedes-Benz Service Partner: Removal of code JA7 and entry of code J1V. The commissioning for code J1V can then follow; see section 5 (→ page 337).

-  Further information and specifications for the pre-installation (code J1V) can be found in the following sections.
-  In this special case, it is recommended to contact the Technical Consultancy during the planning phase to coordinate further details (→ page 18).

Painting and gluing

WARNING

Paint, films or other covering materials (e.g. filler) can cause attenuation of the radar waves depending on the design and thickness. This could lead to malfunctions or system failure. This could cause the driver to lose control of the vehicle.

There is risk of an accident and danger to life and limb!

The sensor area must not be covered by other materials.

When painting in the area of the radar sensor (bumper for closed model designations or cover trim for open model designations), the following specifications must be observed.

-  For adhesive films, observe the information and specifications in 3.13 Adhesive films on the vehicle exterior (→ page 75).

In repair cases, remove the existing coat of paint in such a way that the plastic cover is not damaged and the material thickness of the cover is not significantly reduced.

The new paint coat structure must not exceed the following coat thicknesses:

Paint layer	Maximum coat thickness [µm]	
	std (single-coat, non-metallic)	metallic (two-coat)
Primer (2C)	10–15	
Base coat	None	15
Topcoat	30	15
Clearcoat	30–40	

The material thickness must be homogeneous in order to prevent distortion of the radar waves.

-  For general information about the painting, See 5.4 Painting and preservation work (→ Page 104).

NOTE

If the rear radar sensors are damaged, have the function of the driving assistance system checked at a qualified specialist workshop. This also applies to collisions at low speeds, where no damage to the front or rear area of the vehicle is visible.

NOTE

Adjust the parameterization of the rear radar sensors after the following changes:

- Rear overhang as per 6.2.4 Overhang extension (→ page 119)
- Vehicle width (without outside mirrors/detachable parts)

2) Pre-installation for Blind Spot Assist/exit warning system on open model designations (code J1V)

The pre-installation for Blind Spot Assist/exit warning system (code J1V) is available ex works for open model designations without a platform (code P02). It is automatically assigned as standard to vehicles registered in countries with GSR-II (General Safety Regulation) and CoC documents, as well as in a vehicle configuration with Sideguard Assist (code JT7).

The pre-installation (code J1V) is, in principle, not available ex works for closed model designations.

For special cases in which there is no country-specific legal obligation to use the previously mentioned standard equipment after a conversion by a bodybuilder manufacturer, it is possible to omit these when ordering a vehicle with omission codes:

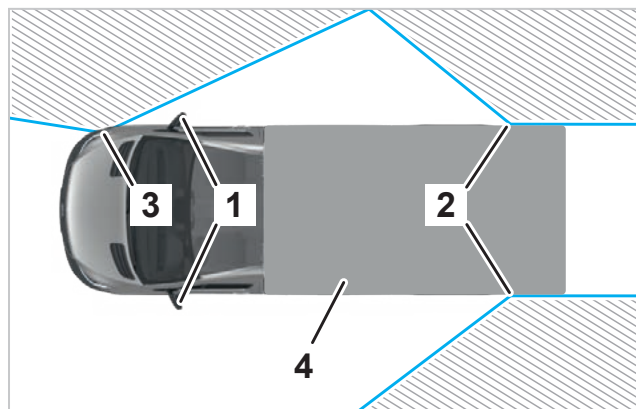
Equipment code	Omission code	Equipment
J1V	JZ9	Omission of Blind Spot Assist pre-installation
JT7	JZ6	Omission of Sideguard Assist

Possible examples can be found in section 1 for code JA7 (→ page 319).

- i In any event, it is recommended to clarify the approval and registration requirements of a body or modification with the Technical Service and/or with the relevant type-approval authority during the planning phase.
- i Contact your sales partner at Mercedes-Benz AG to inquire about the availability of the special equipment options.

From model year 2024, a new system generation of radar sensors will be used. The following new information and specifications apply.

- i If you have any questions, please contact the Technical Consultancy team (→ page 18).
- i If necessary, you can obtain information and specifications on vehicles with radar sensors prior to model year 2024 from previous Mounting Directives. These are still available in the Mounting Directives Archive in the Bodybuilder Portal.

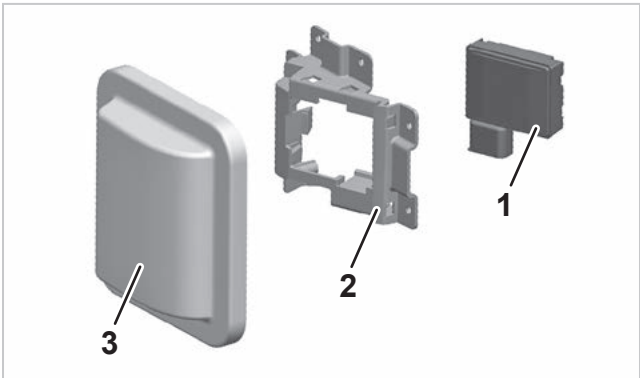


Positions and detection ranges of the radar sensors for code JT7 and J1V (schematic)

- 1 Display of the warning in the outside mirror
- 2 Rear radar sensor positions (code J1V) in the permitted installation area on the right and left side of the vehicle, see illustration in section 3 (→ page 327)
- 3 Position of front radar sensor (code JT7), front passenger side
- 4 Bodybuilder manufacturer-specific body

The delivery package of the pre-installation (code J1V) includes the component parts shown in the following illustration. They are included with the vehicle delivery or are available for replacement and conversion from your Mercedes-Benz Service Partner.

Due to a design tilt of the radar sensors of 20° to the longitudinal axis of the vehicle, see figure (→ page 330), the component parts for the left and right side of the vehicle are different.



Delivery package of the Blind Spot Assist/exit warning system pre-installation (code J1V), by way of example on left-hand side of the vehicle

- 1 Radar sensor
- 2 Holder
- 3 Cover trim

Component	Quantity	Item number	Remarks
Radar sensor CU package compl.	2	A 907 900 95 11 ¹⁾	Same component part for left and right side of vehicle Item number of individual part: A 907 901 79 04
Holder, outer left	1	A 907 884 05 00	Only for left side of vehicle
Holder, outer right	1	A 907 884 06 00	Only for right side of vehicle
Cover trim, outer right	1	A 907 885 13 01	Only for left side of vehicle
Cover trim, outer left	1	A 907 885 14 01	Only for right side of vehicle

1 New item number as of 11/2024. The radar sensor with item number A 907 900 27 11 can continue to be used.

i Mercedes-Benz item numbers are valid at the time of publication of this change information. Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

! NOTE

These components may only be retrofitted if the vehicle is equipped with pre-installation code J1V or, in special cases, is retrofitted from code JA7 to code J1V with additional work (→ page 322).

It is essential to pay attention to the different component parts for the left and right side of the vehicle during installation and electrical connection, otherwise commissioning will not be possible.

In addition, the holders and cover trims for code JA7 are different and must therefore not be used for pre-installation with code J1V – or vice versa.

- i** Details on the component parts and signal funnels (detection ranges) of the radar sensors in the respective installation position can be found in the 3D design data (→ page 25).
The individual item number of the radar sensor is identical for all equipment variants, but the appropriate code JA7, JT7 or J1V must be observed when selecting it.
- i** For special bodies and questions, please contact us by e-mail in German or English, stating the vehicle model designation:

bb-3d_data-van@mercedes-benz.com



The radar sensors must be positioned and installed in accordance with the specifications in the following sections 3 to 5 using the original holders and cover trim pieces supplied and suitable mounting on the bodybuilder manufacturer's specific body. The electrical connection must also be established and the driving assistance system put into operation.

 **NOTE**

To ensure the proper function of the driving assistance system, all specifications and information listed in section 1 (→ page 319) must also be observed for code J1V.

 **NOTE**

For vehicles with pre-installation (code J1V) as special equipment, commissioning must always be carried out, otherwise malfunctions and fault messages may occur. For this, the code X96 "Mandatory commissioning" is automatically added, See 8.19 Commissioning of control units (→ Page 387).

Commissioning must be carried out by the body-builder manufacturer before delivery to the customer, as the vehicle is only preconfigured ex works with code J1V, see section 5 (→ page 337).

When handling the radar sensors, the information and specifications for EMC must be observed, see section 4 (→ page 333).

There is no general obligation to install with a Letter of Compatibility (code X88, See 3.3 Selecting the basic vehicle (→ Page 53)). In the event of deviations from the following specifications, or for special body situations, a Letter of Compatibility is required (→ page 19); this can be requested via the Body-builder Portal (→ page 20).

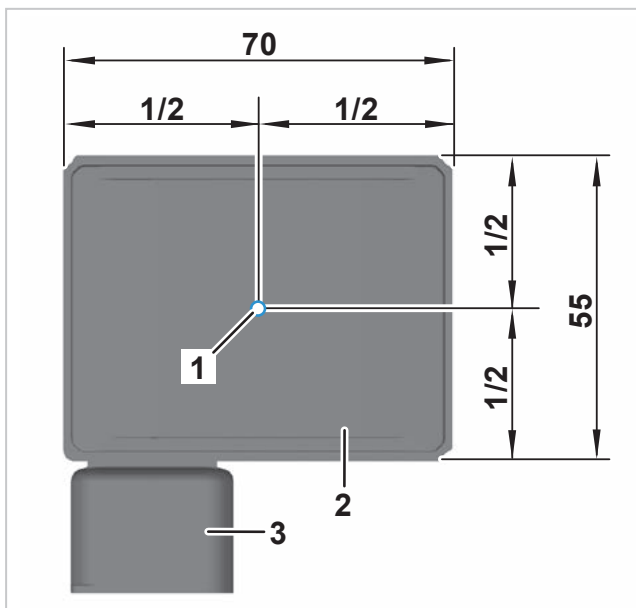
In addition, special requirements apply to all bodies in which bodybuilder manufacturer-specific outside mirrors are installed (e.g. fully integrated camper vans). In these vehicles, appropriate warning lamps must be integrated in the mirrors or at a suitable alternative location.

3) Positioning and installation of the radar sensors for open model designations (code J1V)

Radar sensors reference point

In order for the system to function correctly, the position of the radar sensor in the overall vehicle must be known.

The reference point for the dimensioning is the center of the sensor surface. This is located in the immediate vicinity of the visible injection point in the plastic housing of the sensor. To simplify matters, this injection point can be used as a reference point:



Reference point of radar sensor (not to scale)

- 1 Reference point in the visible injection point
- 2 Housing of the radar sensor
- 3 Electrical connection (socket)

i Details can be found in the 3D design data and 2D drawings.

For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).

Positioning and alignment specifications

- Note the different component parts for the left and right side of the vehicle, see table of item numbers in section 2 (→ page 325).

Details of the component part geometry can be found in the 3D design data (→ page 25).

- Always position the radar sensors so that the electrical connection (socket) points vertically downwards, see alignment specifications (→ page 330).
- In principle, installation can take place on the outside of the body or on an additional connecting part to the body.

In the case of a recessed installation position in the body, the outer surfaces of the surrounding edge of the cover trim must be flat with the outer surfaces of the directly adjacent body; see illustrations for the beam angles (→ page 330).

Also observe the specifications for the installation position in the Y-direction relative to the maximum vehicle/body width (→ page 329).

- Never install radar sensors on doors, flaps, other moving parts or thin-walled structures without sufficient rigidity. Position the sensors so that they are connected to the body at a fixed, torsionally rigid and immovable point in all operating statuses.
- If a bridge to the body or a structural part is required for correct installation of the radar sensor with the plastic holder, this bridging part must also be designed with high strength and torsional rigidity.
- It is recommended that only unmodified genuine component parts from the delivery package are used. For special bodies, however, it is permissible to reduce the component contour along predefined cutting lines by mechanical processing, see figure for mounting points (→ page 332).
Details of the dimensions can be found in the 3D design data and 2D drawings.

NOTE

The following limit values for the permissible installation position and alignment of the radar sensors in the rear area of the vehicle behind the rear axle must be observed.

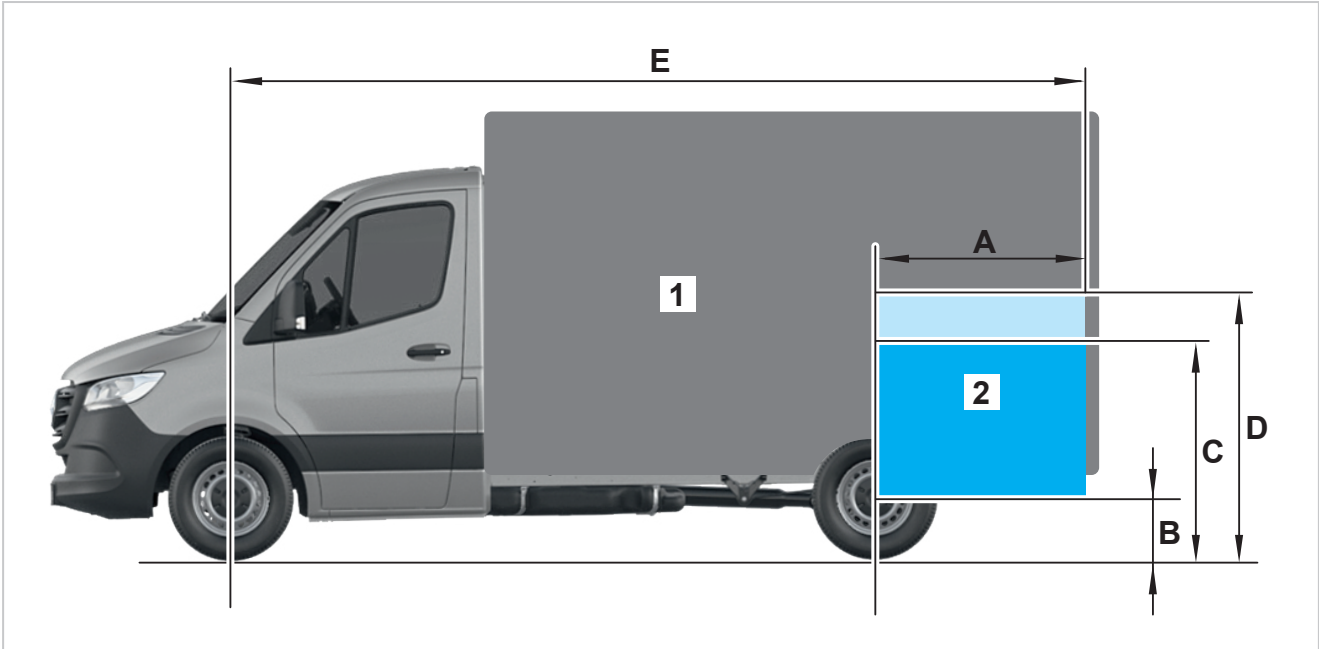
The positions of the radar sensors on the body must be documented with O-codes in XENTRY during commissioning; see section 5 (→ page 337).

In the event of deviations, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20).

Permissible range for the installation position in the X and Z directions

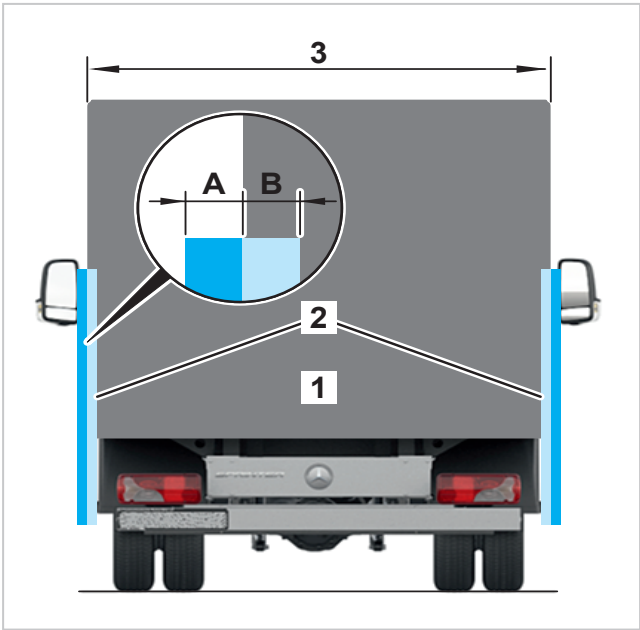
	Installation position	Limit value [mm] ¹⁾
A	Distance from the center of the rear axle ²⁾	Max. 3000
B	Height above roadway ³⁾	Min. 300
C	Height above roadway ³⁾	Max. 900
D	Height above roadway ^{3) 4)} only in exceptional cases	Max. 1200
E	Distance from center of front axle	Max. 6250

- 1) The limit values relate to the reference point of the radar sensors; see illustration (→ page 327)
- 2) For vehicles with two rear axles (three-axle or dual axle, code ZC4 or X5T), the distance in the X direction is measured from the center between the two axle centers.
- 3) The height in the Z-direction above the roadway must be measured for the curb weight of the complete vehicle with completed body.
- 4) An installation position **D** above 900 mm may only be selected in exceptional cases up to a maximum of 1200 mm if the installation position **C** up to 900 mm cannot be maintained for important reasons (e.g. design restrictions).
In such cases, it is recommended to contact the Technical Consultancy during the planning phase (→ page 18).



Permissible range for the installation position of the radar sensors in the X and Z directions (schematic)

- 1 Example of any bodybuilder manufacturer-specific body
- 2 Permissible range for the installation position behind the rear axle



Permissible range for the installation position of the radar sensors in the Y direction (schematic)

- 1 Example of any bodybuilder manufacturer-specific body
- 2 Permissible range for the installation position on the outer surfaces of the body or on a structural part
- 3 Maximum vehicle/body width

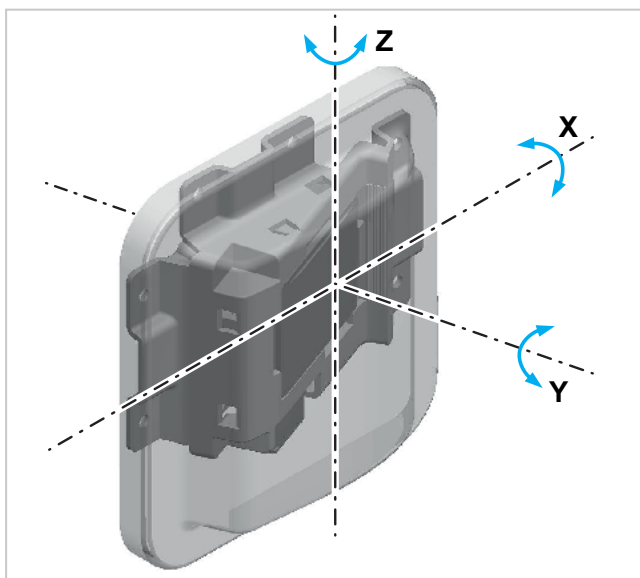
Permissible deviation of the installation position in the Y direction relative to the maximum vehicle/body width

	Installation position	Limit value [mm] ¹⁾
A	Protrusion	Max. +50
B	Recess ²⁾ only in exceptional cases	Max. -50

- 1) The limit values relate to the reference point of the radar sensors; see illustration (→ page 327)
- 2) An installation position B up to a maximum of -50 mm is only permitted to be selected in exceptional cases if the installation position A up to a maximum of +50 mm cannot be maintained for important reasons (e.g. design restrictions).
In such cases, it is recommended to contact the Technical Consultancy during the planning phase (→ page 18).

- The installation positions of the radar sensors in the Y direction depend on the maximum vehicle/body width 3 (depending on which width is the largest without outside mirrors), which is documented with O-code in XENTRY during commissioning, see section 5 (→ page 337).
- Position the radar sensors in the same Y-position as the maximum available vehicle/body width. In relation to the reference point of the radar sensor, a protrusion of up to +50 mm is recommended. Depending on the body, a deeper position down to a maximum of -50 mm can already lead to restrictions and must therefore be avoided.
- If necessary, the mounting of the radar sensor on the vehicle must be positioned further out than would be required by the body width in its directly adjacent surroundings. Observe the specifications for the detection range or signal funnel.
- The installation positions on the left and right side of the vehicle must be as mirror-inverted as possible.
- For installation positions with differences between the left and right side of the vehicle, the specifications for documenting the installation position with O-codes must be observed, see section 5 on commissioning with the available O-codes and their installation areas (→ page 337): The installation positions of the left and right radar sensor must be defined so that both positions are within the permissible installation range of common O-codes.

- The permissible limit value of the tolerance for the alignment of the radar sensors in the installation position is a maximum of $\pm 1^\circ$ in all directions; see following illustration.



Permissible tolerance of the alignment of the radar sensor in all directions max. $\pm 1^\circ$, by way of example on left side of vehicle, for details see 3D design data

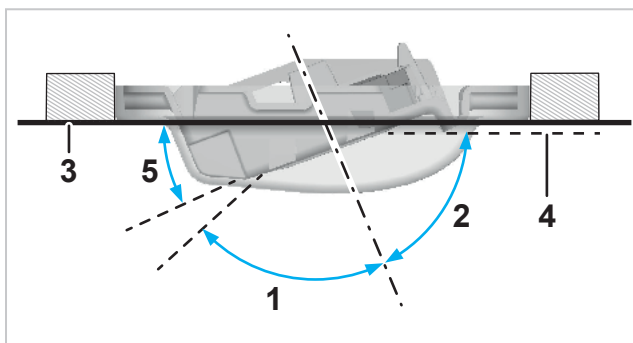
Z-axis: Alignment to the longitudinal axis of the vehicle (X-axis on the vehicle side)

X-axis: Alignment to the vertical (Z-axis on the vehicle side)

Y-axis: Orientation of the electrical connection (socket) vertically downwards (rotation around the center axis of the component part)

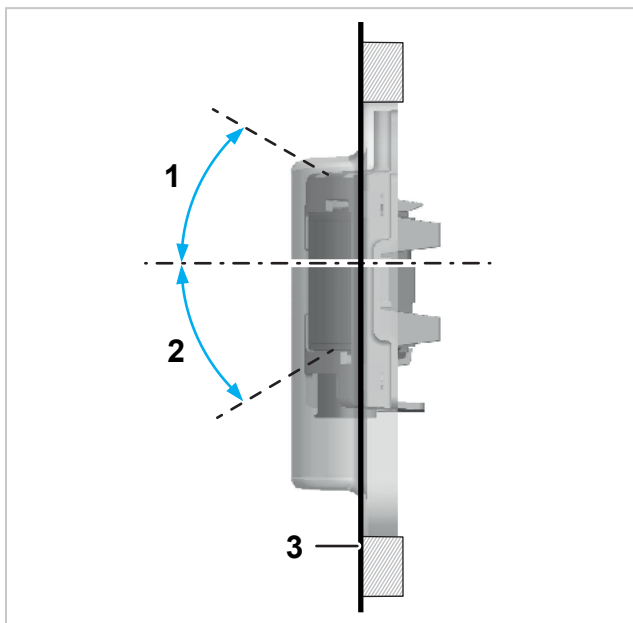
- To align the radar sensor around the Y-axis (or component part center axis) in a vehicle on a level surface, align the electrical connection (socket) of the radar sensor vertically downwards. A simulation of the entire body situation must be carried out with a complete body and typical body load or alternatively with an appropriate replacement load.
- With regard to the vehicle surroundings, observe the maximum beam angles or signal funnels of the radar sensor in all directions in the 3D design data; see following illustrations. It is essential to observe the warning (→ page 318).

- Special attention must be paid to the radar signal directed to the rear. There must be no contours or detachable parts protruding from the outer surface between the radar sensor and the rear of the vehicle that could restrict the detection range.



Horizontal beam angles (schematic), view from above, by way of example on left side of vehicle, for details see 3D design data

- 1 Beam angle 75° to the front
- 2 Beam angle 69° to the rear
- 3 External surface of body
- 4 Radar signal directed to the rear
- 5 Design tilt 20°



Vertical beam angles (schematic), view from behind, by way of example on left side of vehicle, for details see 3D design data

- 1 Beam angle 30° upwards
- 2 Beam angle 30° downwards
- 3 External surface of body

- In addition to the signal funnel of the radar sensor, note the offset surfaces in the 3D design data. This data illustrates interfaces for body contours, detachable parts or other component parts. Penetration of these interfaces must be avoided to ensure an undisturbed radar signal.
 - It is recommended to check the vehicle surroundings during the design and development of the body using the 3D design data.
Details of the signal funnel with the beam angles of the radar sensors and the offset surfaces can be found in the 3D design data.
-  For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).

Specifications for installation

WARNING

Radar sensors are very sensitive component parts, especially when exposed to external influences such as shocks or pressure loads. Sensors with visible damage to the plastic housing or sensors that have fallen from a low height onto a hard surface must no longer be used.

There is risk of an accident and danger to life and limb.

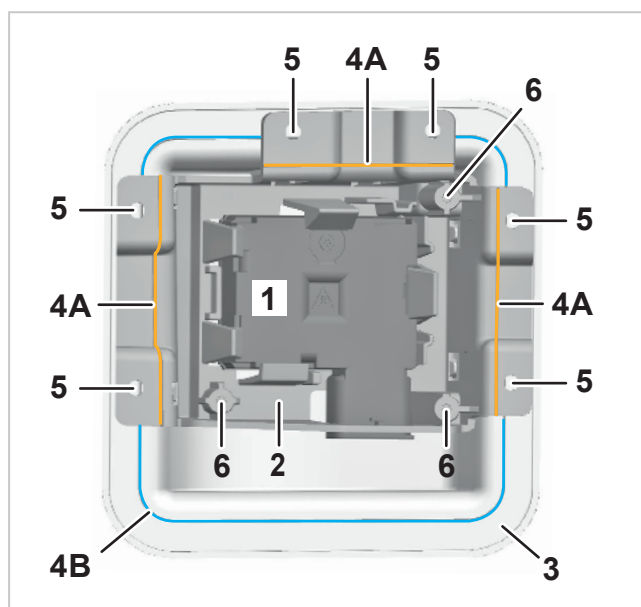
In the event of damage, replacement with new genuine Mercedes-Benz parts is absolutely essential, as otherwise the function of the driving assistance system is no longer guaranteed.

Install the sensors in the following order:

-  Snap the radar sensor into the holder using clips.
-  Establish electrical connection, see section 4 (→ page 333).
-  Screw the holder with radar sensor to the body or structural part.
-  Snap the cover trim into place using clips.
-  Seal cutouts (→ page 332).

Please note with regard to this:

- To attach the holder with the radar sensor to the body or structural part, the bodybuilder manufacturer must select suitable screws with sufficient strength and durability to suit the body situation.
- Fixing points are available for two variants, see the following illustration:
 - Original component contour: Screw connection from the outside via six screw holes **5**
 - Reduced component contour **4** for special bodies: Screw connection from the inside via all three screw bosses **6**



Fastening points for variants of the component contour (schematic), example shows holder on left side of vehicle

- 1 Radar sensor
- 2 Holder
- 3 Cover trim
- 4 Cut lines on three flanges of the holder (**4A: orange**) and all around the cover trim (**4B: blue**) to reduce the component contour
- 5 Six screw holes on flanges for screw connection from outside
- 6 Three screw bosses for screw connection from the inside with reduced component contour

- For secure fastening, we recommend using all six screw holes **5**.
A reduction in the number of screw holes is only permitted with at least one screw connection per flange. The fatigue strength of the connection to the body must be ensured.
- Observe the cutting lines visible on the component parts **4A** and **4B** for the reduction of the component contour.
- Details and dimensions of the component part variants and fixing points can be found in the 3D design data (→ page 25).
- Seal the installation additionally against water and dirt ingress from all sides (protection class IP6K) and provide anticorrosion protection between the body and cover trim, especially openings on the rear of the installation (e.g. with an additional cover cap) and between the body and covers (e.g. with a sealing seam).
- At the lower edge of the installation position on the body, provide an opening for the electric cable passage and as a condensation drain.
- The wiring between the plugs on the radar sensor and the connecting point on the vehicle must be designed and installed by the bodybuilder manufacturer, see section 4 (→ page 333).
- After installation and electrical connection, carry out commissioning with documentation of O-codes, see section 5 (→ page 337).

4) Electrical connection of the radar sensors (code J1V)

! NOTE

The workplace and personal equipment of the personnel must be equipped with ESD protection (against electrostatic discharge) when installing, electrically connecting and testing the radar sensors. This also applies for the shipping container and the warehouse areas where the sensors are packed or unpacked.

Without ESD protection, it can be assumed that the radar sensors will be damaged by electrostatic discharges. As a rule, only preliminary damage occurs at first, which only gradually expands during further operation until the sensor fails repeatedly and finally permanently.

Therefore, warranty and goodwill costs for cases of failing radar sensors in combination with code J1V in the case of non-compliance with the ESD protection specifications and a hardware failure due to this will not be borne by Mercedes-Benz AG, unless it can be demonstrated that the component part damage occurred before or during delivery to the bodybuilder manufacturer.

The installed radar sensors must be electrically connected directly to the connecting point in the vehicle by a qualified electrician using a wiring harness.

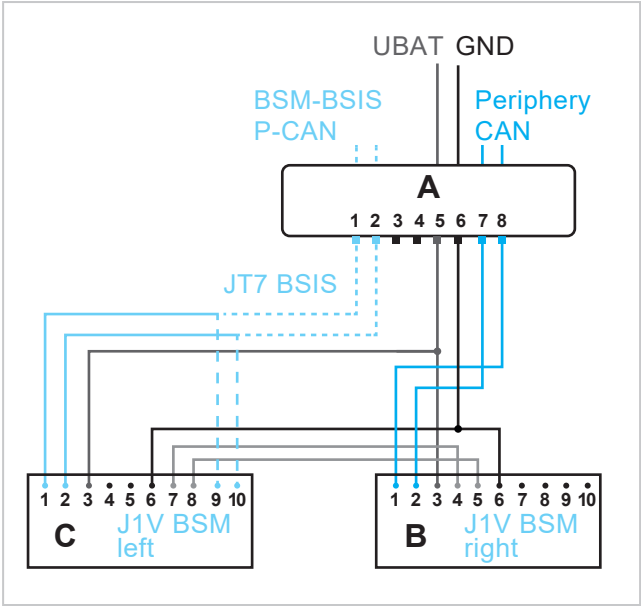
The wiring harness must satisfy the following requirements:

- The wiring harness (cable with electric couplings) made by a qualified electrician or wiring harness manufacturer must have its electrical connections and functionality checked before use (according to DS characteristic: Mandatory documentation relevant to safety). The test results must be submitted on request.
- Make sure that the connection is watertight and that the lines are routed correctly, See 8.4 Interfaces and electric lines (→ Page 271).
- The radar sensors and the lines must not be disturbed by EMC influences from other electrical components in the body.

The position of the connecting point on the vehicle for the electrical connection is located:

- on open model designations (FHS/FHL) at the rear of the vehicle in the area of the end crossmember (code Q18) or on the last frame crossmember (if the end crossmember is omitted, code Q72),

Observe the following connection scheme and pin assignments for the electrical connection of the radar sensors to the connecting point on the vehicle:



Connection scheme for radar sensors to the connecting point on the vehicle

- Contacting of pins 1 and 2 of radar sensor **C** (satellite):
- With BSIS (with code JT7): Connection dotted to pins 1 and 2 of the connecting point **A** on the vehicle
 - Without BSIS (without code JT7): Connection dashed to pins 9 and 10 on the radar sensor **C**

! NOTE

The radar sensor on the right-hand side of the vehicle **B** must be connected electrically as "Main" and the radar sensor on the left-hand side of the vehicle **C** as "Satellite", otherwise correct commissioning will not be possible.

i For BR 907, the connection schemes for the pre-installation (code J1V) and the factory-fitted equipment (code JA7) are different. If required, information on this is available in the Workshop Information System (→ page 26) or from your Mercedes-Benz Service Partner.

A Connecting point on the vehicle

Pin	Function
1	CAN-L BSIS
2	CAN-H BSIS
3-4	-
5	UBAT
6	GND
7	Periphery CAN-H
8	Periphery CAN-L

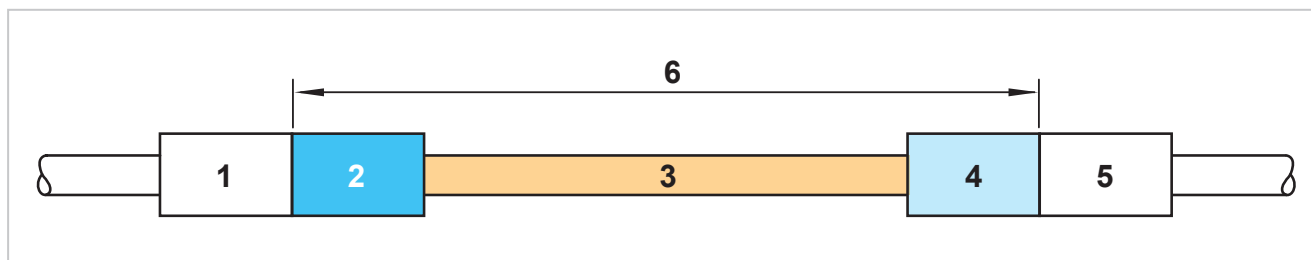
B Right radar sensor (Main)

Pin	Function
1	Periphery CAN-L
2	Periphery CAN-H
3	UBAT
4	CAN-L (to pin 7 Satellite)
5	CAN-H (to pin 8 Satellite)
6	GND
7-10	-

C Left radar sensor (Satellite)

Pin	Function
1	CAN-L (to BSIS)
2	CAN-H (to BSIS)
3	UBAT
4-5	-
6	GND
7	CAN-L (to pin 4 Main)
8	CAN-H (to pin 5 Main)
9	Termination without BSIS (to pin 1)
10	Termination without BSIS (to pin 2)

Observe the following information on the necessary parts for the production of a wiring harness:



Assignment of the line components to the following tables

For information only (white):

- 1 Vehicle-side connecting point
- 5 Direct connection to radar sensor

Required for the production of the wiring harness 6:

- 2 Connection to connecting point (blue)
- 3 Electric line (orange)
- 4 Connection to the radar sensor (light blue)

Electric line

	Type	Cross section	Manufacturer	Maximum permissible length
3	FLRY Voltage	$2 \times 0.5 \text{ mm}^2$	Commercial provider or cable manufacturer	10 m
	FLRY CAN	$2 \times 0.35 \text{ mm}^2$ Twisted with loop length 20 mm		

Observe all technical specifications and requirements in the manufacturer's data sheet.

The specified loop length of the twisted CAN lines must be observed.

Electrical contacts

	Position	Type	Type (pins)	Reference item number	Manufacturer	Manufacturer no. Coding, type
1	Vehicle-side connecting point	Voltage/ CAN	Socket (8 pins)	A 000 545 41 33	Hirschmann	805-125-551 Coding A, female
2	Connection to connecting point	Voltage/ CAN	Plug (8 pins)	A 000 545 98 33	Hirschmann	809-560-501 Coding A, male
	Voltage line	Contact	Pin contact	A 000 982 62 28	TE connectivity	1718761-3
		Seal		A 000 545 69 80		0-967067-1
	CAN line	Contact	Pin contact	A 000 982 61 28		1718759-3
		Seal		A 000 545 68 80		0-967067-2
	Seal	Blind plug		A 000 545 87 80		0-967056-1

	Position	Type	Type (pins)	Reference item number	Manufacturer	Manufacturer no. Coding, type
4 ¹⁾	Connection to the radar sensor	Voltage/ CAN	Plug (10 pins)	A 236 540 03 39	Hirschmann	812-726-501 Coding B, female
	Voltage line	Contact	Socket contact	A 008 545 63 26	TE connectivity	5-965907-1
		Seal		A 000 545 69 80		0-967067-1
	CAN line	Contact	Socket contact	A 008 545 54 26		5-963727-1
		Seal		A 000 545 68 80		0-967067-2
	Seal	Blind plug		A 000 545 87 80		0-967056-1
5	Direct connection to radar sensor	Voltage/ CAN	Socket (10 pins)	A 907 900 27 11 ²⁾	–	Coding B, male

- 1) Parts are required twice for right radar sensor **B** (Main) and left **C** (Satellite), see connection scheme (→ page 334)
- 2) Integrated in the housing of the radar sensor

 If you have any questions about the electrical connection, please contact the Technical Consultancy team (→ page 18).

 Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

5) Commissioning of the driving assistance system (code J1V)

After installation of the two radar sensors and their electrical connection to the connecting point on the vehicle, commissioning must be performed via the XENTRY Kit Mercedes-Benz diagnostic tool before the vehicle is put into circulation. In addition to activating the driving assistance system with O-code OB2 and other information on vehicle modifications (e.g. overhang, wheelbase, etc.), the exact position of the radar sensors must be reported in the Bodybuilder Portal for the respective VIN. The corresponding O-codes are thus supplied to the vehicle data card by the system.

You can obtain access for entering these values in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ After logging into the Bodybuilder Portal, select the "Order data" button.
- ▶ Select the order for your vehicle using the order number to access the specific order overview.
- ▶ Select the "Feedback data for commissioning" button.
- ▶ When commissioning for the first time, select the sub-item "Activate pre-installation". There you will find the pre-installations ordered for the vehicle and the field for entering the sensor position.

! NOTE

After the vehicle has been put into operation for the first time or after a subsequent change (e.g. to the position of the sensor), adjustment via the Bodybuilder Portal is no longer possible.

In this case, the O-codes, e.g. when changing from JA7 to J1V or for specifying the sensor position, must be adjusted manually via XENTRY in the vehicle data card and the affected control units must be re-coded according to the "Commissioning of control units" guide.

- ❗ More information on the procedure can be found in the "Commissioning of control units" guide in the Bodybuilder Portal, See 8.19 Commissioning of control units (→ Page 387).
- ❗ For information on XENTRY, See 2.2.4 Diagnosis (→ Page 26).
- ❗ It is recommended to have the commissioning performed by a Mercedes-Benz Service Partner.

Overview of the O-codes for commissioning the radar sensors with code J1V

The O-codes assigned when entering the sensor position are listed in the following tables for planning the body and determining the installation position.

In addition to the specifications and illustrations in section 3 (→ page 327), the following points must be observed:

- The O-codes for the X and Z positions must be unique and identical for both left and right radar sensors. It is therefore not possible to use two different O-codes with different installation areas for the documentation of a vehicle-specific installation situation.
Example: When planning an installation height above the roadway of 650 mm ± 50 mm with the use of O-code O8N, the positions of both radar sensors must be between 600 mm and 700 mm.

- If the input value corresponds exactly to the limit value between two O-codes, the higher installation range must be selected.

Example: For the installation height of 700 mm above the roadway, O8O (700–800 mm) must be selected, not O8N (600–700 mm).

- The following tables apply to vehicles with a fully completed body in running order without load.

! NOTE

Always ensure that the detection ranges or signal funnels of the radar sensors are not impaired, see section 3 (→ page 327).

- ① If you have any questions about the O-codes or if you cannot find the corresponding O-codes for your body situation in this overview, please contact the Technical Consultancy team (→ page 18).

a) Position of radar sensor in Z direction:

Height above roadway

O-code	Rear blind spot radar: Position Z above roadway [mm]	Permissible height [mm]
O8K	350 ± 50	300–400
O8L	450 ± 50	400–500
O8M	550 ± 50	500–600
O8N	650 ± 50	600–700
O8O	750 ± 50	700–800
O8P	850 ± 50	800–900
O8Q	950 ± 50	900–1000
O8R	1050 ± 50	1000–1100
O8S	1150 ± 50	1100–1200

b) Position of radar sensor in X direction:

Distance to center of rear axle (RA)

O-code	Rear blind spot radar: Position X relative to rear axle [mm]	Permissible distance [mm]
O7K	100 ± 100	0–200
O7L	300 ± 100	200–400
O7M	500 ± 100	400–600
O7N	700 ± 100	600–800
O7O	900 ± 100	800–1000
O7P	Additional X to RA +1000	1000–2000
O7Q	Additional X to RA +2000	2000–3000

- 1) The value ranges of the O-codes O7P or O7Q must also be selected.

Examples: 1100 mm: O7K+O7P; 2400 mm: O7M+O7P; 3000 mm: O7O+O7Q (maximum permissible distance with extended overhang on three-axle vehicle)

- ① For vehicles with two rear axles (three-axle or dual axle, code ZC4 or X5T), the distance in the X direction is measured from the center between the two axle centers.

c) Maximum vehicle/body width (without outside mirrors)

O-code	Width [mm]
O94	2100–2200
O95	2200–2300
O96	2300–2400
O9K	2400–2500

- ① The maximum vehicle/body width must always be documented in XENTRY. It serves as a reference for the Y-position of the radar sensors.

8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist

As of model year 2024, a new system generation of the multi purpose camera (MPC) is being installed as standard on all vehicles with a windshield at the lower windshield position; see position 2 of the illustration in 8.9.6 Automatic driving lights/rain sensor (→ page 340).

On vehicles without a windshield, the necessary components for a subsequent installation are supplied along with the vehicle; see following section and 7.14.6 Multi-purpose camera (MPC) on vehicles without windshield (→ page 236).

In combination with the front radar, See 8.9.3 Active Brake Assist/Active Distance Assist DISTRONIC (→ Page 314) and other sensors in the front of the vehicle, this camera ensures the functionality of various driving assistance systems.

The function of the MPC is ensured in combination with the windshields installed ex factory or as a Mercedes-Benz genuine part.

Driving assistance systems with MPC:

- High Beam Assist (code LA1)
- Active Lane Keeping Assist (code JB4)
- Traffic Sign Assist (code JA9)
- Intelligent Speed Limit Assist (code JS2)

As well as the driving assistance systems in chapter 8.9.3:

- Active Brake Assist (code BA3)
- Active Distance Assist DISTRONIC (code ET4)

The High Beam Assist (code LA1) automatically switches the high beam on and off according to the traffic situation and therefore relieves the driver of the need to brighten and dim the headlamps manually.

The Lane Keeping Assist (code JB4) can help to avoid accidents caused by the vehicle leaving the lane unintentionally.

The Traffic Sign Assist (code JA9) assists the driver in identifying traffic signs.

The intelligent Speed Limit Assist (code JS2) makes it easier for the driver to adhere to stipulated speed limits.

! NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

⚠ WARNING

Any changes to the MPC and the surrounding area (e.g. changes to the windshield, including the inclination of the windshield), as well as any bodies, installed parts or modifications impacting the function of the MPC, are not permitted.

The cutouts in black print on the windshield for the MPC must not be covered by film applications, See 3.13 Adhesive films on the vehicle exterior (→ Page 75).

Otherwise, driving assistance systems may no longer work as intended.

There is risk of an accident and danger to life and limb!

As exceptions for vehicles delivered without a windshield, observe the information in the following section.

! NOTE

In the event of changes to the vehicle that permanently modify the inclination in the longitudinal direction, e.g. bodies or conversions with an increase in weight, modification to the weight distribution, or modifications to the suspension (list not exhaustive), the camera must be readjusted.

Open model designations without CoC documents or with incomplete CoC documents (e.g. chassis vehicles ex factory) are delivered without calibration of the camera due to the lack of body.

Adjustment or calibration of the camera must be independently arranged by the bodybuilder manufacturer after the vehicle body has been constructed with a load typical for the body.

Have the readjustment or calibration carried out at a qualified specialist workshop possessing the required expertise and tools in order to perform the necessary work. It is recommended to employ the services of a Mercedes-Benz Service Partner for this purpose.

- i** More information is available from your Mercedes-Benz Service Partner or from the Technical Consultancy (→ page 18).

MPC on vehicles without a windshield

- i** Information and specifications for the MPC can be found in 7.14.6 Multi-purpose camera (MPC) on vehicles without windshield (→ page 236)
- i** Contact the Technical Consultancy during the planning phase (→ page 18).

8.9.6 Automatic driving lights/rain sensor

The automatic driving lights (code LA2) is installed as standard depending on the country or available as special equipment.

The rain sensor (code JF1) is already integrated and is added in its function depending on the country. A sensor for measuring the humidity and solar radiation is also integrated for automatic regulation of the air conditioning system.

! NOTE

The light/rain sensor must, with exception of the cab base vehicles (codes F50 and F92), only be used with the windshields installed ex factory or available as a Mercedes-Benz genuine part.

⚠ WARNING

Any changes to the light/rain sensor, its attachment to the vehicle and the surrounding area (e.g. changes to the windshield), as well as any bodies, installed parts or modifications impacting the function of the sensor, are not permitted.

The cutouts in black print on the windshield for the sensor must not be covered by film applications (including shatter protection films or infrared-absorbing films), See 3.13 Adhesive films on the vehicle exterior (→ Page 75).

Otherwise, driving assistance systems may no longer work as intended.

There is risk of an accident and danger to life and limb!

Installation on cowl cab base vehicles (code F50) and vehicles with omitted windshield (code F92)

Information and specifications on installation See 7.14.5 Connecting point for automatic driving lights (code LA2)/rain sensor (code JF1) on vehicles without windshield (→ Page 233).

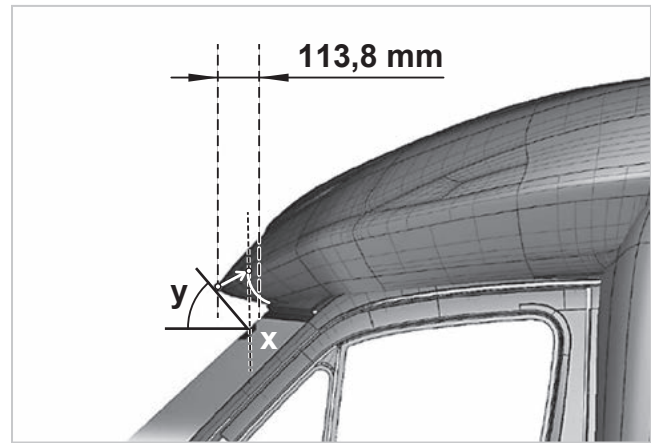
Installation of a light/light sensor can only be implemented using the standard sensor type, because the sensor communicates with the vehicle over the VCS LIN interface.

Vehicles with roof overhang (e.g. alcoves)

Proper functioning can be impaired by overhanging parts:

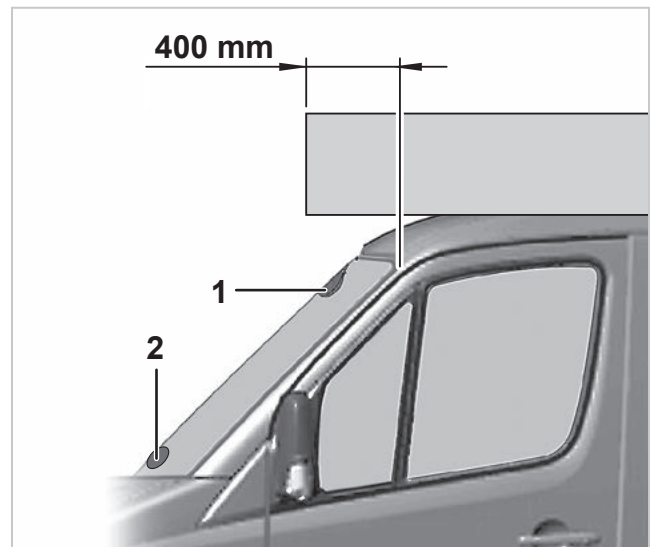
- Deviations from the standard equipment occur depending on the overhang. Please note that there may be other effects on the system depending on the driving situation, weather situation, shade, or seasons (height of the sun).
- The complete functionality of the light/rain sensor is ensured up to a small roof overhang according to the following illustration without limitations.
- With a roof overhang of over 200 mm in length, the following impairments can manifest more significantly:
 - Unnecessary activation of the driving light due to erroneous measurement of the light intensity
 - Lack of automatic activation of the windshield wipers as a result of insufficient detection of moisture
 - Manual intervention from the driver may be required depending on the situation.

i Further information on use of the driving assistance systems can be found in the operator's manual for your vehicle.



Torsioning play

- x Position of sensor
- y Torsioning play (ideally 90°)



Maximum roof overhang

- 1 Position of light/rain sensor
- 2 Position of multi purpose camera (as of model year 2024)

- With a roof overhang up to max. 400 mm (measured from the outer edge of the windshield, see picture), the fundamental functionality of the rain sensor (code JF1) and Headlamp Assist (code LA2) can still function correctly in accordance with the situation.

As of model year 2024, relocation of the light/rain sensor to the lower windshield position with the previously recognized special equipment code LA3 is no longer possible, since the multi purpose camera is installed at this position, See 8.9.5 High Beam Assist, Active Lane Keeping Assist, Traffic Sign Assist, Intelligent Speed Assist (→ Page 339).

! NOTE

On vehicles with automatic driving lights/rain sensor, the limit value of maximum 400 mm must therefore be adhered to for the roof overhang (measured from the outer edge of the windshield).

8.9.7 Tire pressure loss warning system

! WARNING

Do not make any modifications in the area under the vehicle floor, including the vehicle axles and the system components of the tire pressure monitoring system (see following illustration). Otherwise, the function of the tire pressure monitoring system may be compromised by the effects of electromagnetic reflections. This might result in the driver being unaware of any tire pressure loss; as a consequence, critical driving situations could occur.

There is risk of an accident and danger to life and limb!

Furthermore, the vehicle may no longer meet registration requirements.

Regulations (EU) 2019/2144 and UN regulation UN R 141 specify which vehicles must be equipped with a tire pressure loss warning system. Also observe Regulation (EU) 2018/858 with regard to certain special bodies.

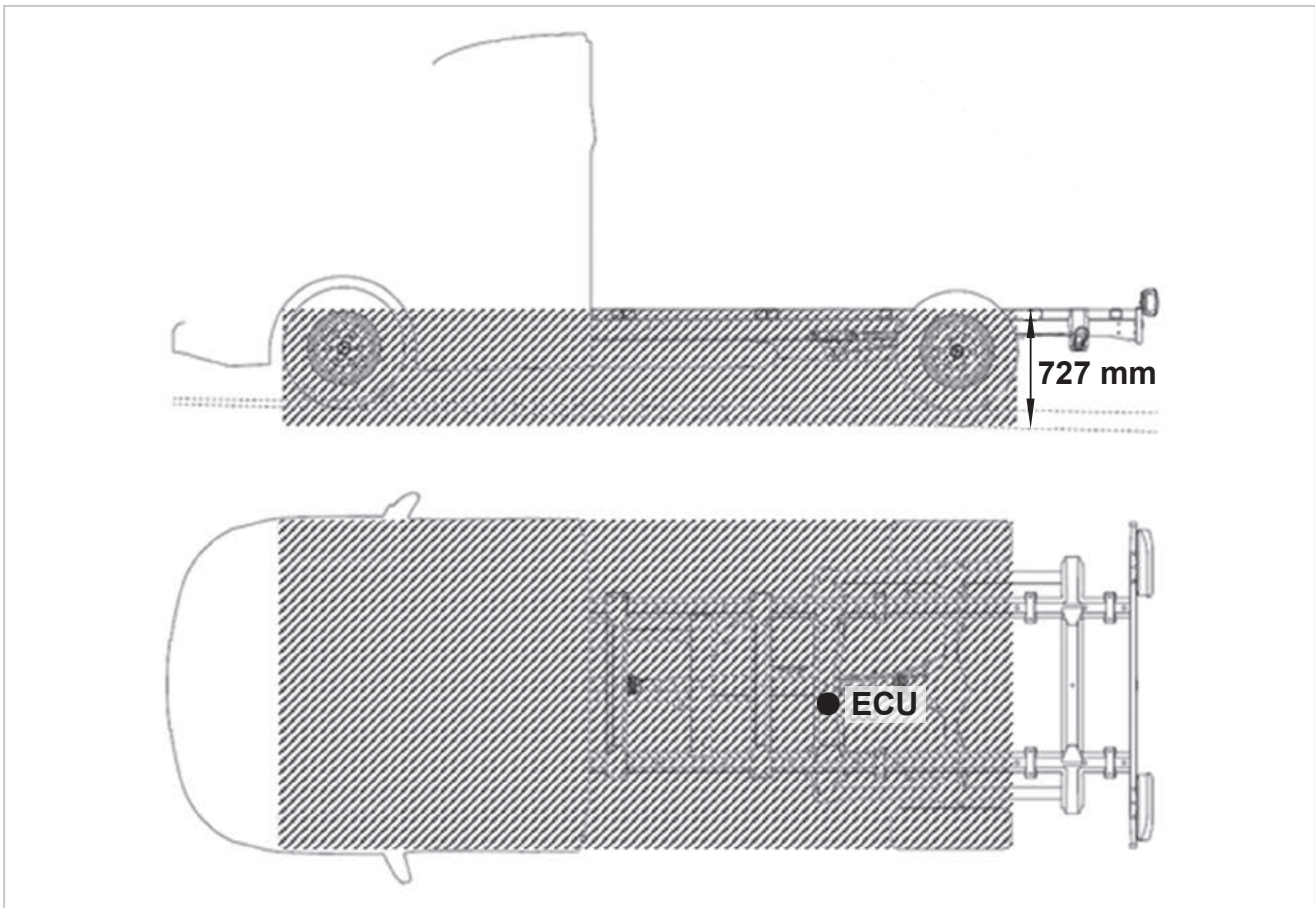
! NOTE

The approval of the tire pressure monitoring system must be observed as per UN R 141 in all countries in which EU regulations and UN regulations are applicable.

Also make sure to observe all national laws, directives and registration regulations.

These specifications apply to all mounted wheels, including in the event that a spare wheel is used, as well as for conversion to winter tires or other wheel types.

Non-adherence constitutes a deviation from the legal specifications and can result in the operating permit becoming invalid.



Prohibited areas for tire pressure loss warning system, applies to both versions of the system used in the Sprinter

The wireless tire pressure loss warning system on the front and rear axle (code RY2) installed ex factory complies with the specifications outlined previously.

The system includes a receiver module in the control unit (ECU), which is positioned on the underbody between the axles, and tire pressure sensors, which are screwed into the rims together with the valve and measure the tire pressure there. These transmit the measured pressure via 433 MHz radio to the control unit.

Screening plates and timed consumers or other detachable parts in the area of the underbody can impair the radio reception, potentially resulting in individual tire pressures not being displayed. A maximum of 4 sensors (6 sensors on twin tires at the rear axle) are assigned to the system and monitored simultaneously.

! NOTE

It is essential to observe the specifications in the above warning notice and the additional specifications in 8.2 Electromagnetic compatibility (EMC) (→ page 266).

If, despite this, body mounting work is planned for the vehicle underbody, the functioning of the system must be checked by the bodybuilder manufacturer. All criteria of UN R 64 or 141 must be fulfilled.

After a maximum of 10 minutes' driving (over 25 km/h), the tire pressure of all fitted wheels must be evident in the instrument cluster. If this is the case, then the control unit has received signals from all of the sensors. It is recommended that the vehicles be tested with maximum equipment. All electrical consumers should be switched on during the test.

- ❗ If you have any questions, contact the Technical Consultancy during the planning phase (→ page 18).

Additionally, after body modifications in this area, another inspection of the tire pressure loss warning system by an officially recognized testing organization may be required in accordance with UN R 64 or 141; the costs for this must be borne by the bodybuilder manufacturer. This serves to prove that the vehicle is eligible for registration as per the legal specifications because the tire pressure loss warning system has not been affected.

- ❗ Further information and specifications on the display and operation of the tire pressure loss warning system can be found in the operator's manual for your vehicle.

8.9.8 Parking packages/Moving-off information Assist

The parking packages with the Parking Assist can make things easier for the driver when moving in and out of parking spaces or maneuvering, by indicating obstacles in the immediate vicinity of the vehicle.

The Moving-off information Assist (code JF7, Moving-off Information System MOIS) can warn of a dangerous situation in the immediate vicinity in front of and to the side of the vehicle front when moving off at low speed.

This new driving assistance system is required as standard on all vehicles with N2 registration in countries with CoC documents due to legal requirements, and is therefore added. In other countries, this driving assistance system is available as special equipment.

- ❗ Contact your Technical Service or the relevant type-approval authority for information on the country-specific legal regulations regarding these driving assistance systems.

! NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

⚠ WARNING

Any modifications to the cameras and sensors, their attachment to the vehicle and their surrounding area, as well as any bodies or conversions with an impact on the function and detection ranges of the cameras and sensors, are not permissible.

Otherwise, driving assistance systems may no longer work as intended.

There is risk of an accident and danger to life and limb!

For planned bodies or conversions and in driving operation, ensure that the function and detection ranges are not restricted.

! NOTE

In special cases where the detection range of the cameras or sensors may be impaired by detachable parts required by law or authorities, contact the Technical Consultancy (→ page 18) to coordinate the type, size and position of the required detachable parts and, if necessary, their compatibility with the driving assistance system.

1) Parking packages for closed model designations

Code	Equipment ¹⁾
JB6	Parking package with 360° camera for closed model designations (FKA/FKB) without rear attachments
JB7	Parking package with reversing camera for closed model designations (FKA/FKB) without rear attachments

1) The description is intended as an explanation and is not the same as the equipment description.

The variants of the parking package systems require a vehicle equipped with the MBUX multimedia system, See 8.17.1 Infotainment systems (→ Page 382). The views and visual warnings are shown on the screen of the MBUX multimedia system.

If special equipment approved by Mercedes-Benz is subsequently installed on vehicles with code JB6 or JB7, the appropriate parameter set for the parking system must be coded at a Mercedes-Benz Service Partner.

- ❗ These special equipment options are not available for open model designations.
- ❗ For further information on the parking package, please refer to the relevant sales information and contact your Mercedes-Benz sales partner.

2) Ultrasonic sensors

! NOTE

Detachable parts fitted in the detection range of the ultrasonic sensors may impair operation of the Parking Assist or the Moving-off information Assist (e.g. warning signs, trailer coupling, body overhangs, wheel carriers, steps, impact protection).

Therefore, make sure to observe the warning and the information (→ page 345).

Bumpers with installed ultrasonic sensors must not be subsequently painted. The coat of paint impairs the emission and reception of the ultrasonic signals.

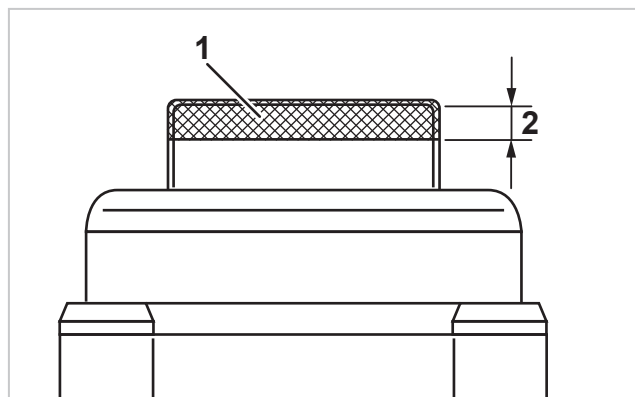
! NOTE

Sensors that have already been painted must not be repainted or touched up.

Unpainted sensors and sensors painted in a range of colors are available from your Mercedes-Benz Sales Partner.

The maximum thickness that the paint coat on the diaphragm can be without impairing sensor operation is 120 µm. This also includes repeated paint applications and the coat from cathodic electrodeposition (KTL coat). The KTL coat thickness is between 12 µm and 25 µm.

- ▶ To ensure proper functioning of the sensors, check the coating thickness at random points.
- ▶ When painting, it is essential that not only the diaphragm itself, but also the cylindrical edge of the sensor diaphragm is coated with paint evenly all the way round and covering at least 2 mm.



Area to be painted on sensor diaphragm cylindrical edge

- 1 Area to be painted
- 2 Maximum paint coat thickness 120 µm

! NOTE

The coat of paint must not be ground off mechanically. This could damage the cathodic dip coating or the sensor diaphragm.

! NOTE

If the surface has been primed with cathodic dip coating, the paint must not be removed by chemical means. This could damage the cathodic dip coating and a new coating would not be able to be applied afterward. Nor is it permitted to touch up damaged areas chemically or mechanically.

8.9.9 Reversing camera

Available special equipment

Code	Equipment ¹⁾
FR3	Analog reversing camera with inside rearview mirror display
FR8	Digital reversing camera with display via central display
FV1	Pre-installation for analog reversing camera with inside rearview mirror display
FR7	Pre-installation for digital reversing camera with display via central display

1) This description is intended as an explanation and is not the same as the code designation.

! NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

! NOTE

External contours of the vehicle body, detachable parts that protrude from the outer surface of the body, or retrofitted vehicle equipment can all lead to a restriction of the viewing area behind the vehicle or disturbances in the visual impression in the display, and thus result in limitations to the function of the driving assistance system.

! NOTE

Compliance with country-specific laws and guidelines on the viewing area behind the vehicle (e.g. UN R 158) and the fulfillment of the obligation to provide evidence for the registration and driving operation of the converted vehicle are the sole responsibility of the bodybuilder manufacturer.

! NOTE

Bodybuilder manufacturer-specific operator's manuals must contain appropriate warnings and information for the vehicle user, which at least refer to:

- Possible restriction due to retrofitted detachable parts or vehicle equipment that protrude into the viewing area of the reversing camera, e.g. bicycle rack at the rear of the vehicle
- Deactivation of pedestrian emergency braking when reversing after documentation of code O1N in the VeDoc vehicle data card (→ page 349)
- Possible impairment of the function of the driving assistance system if the reversing camera and its surroundings on the vehicle are damaged
- Obligation to have the function of the driving assistance system checked if damage is detected and after accidents affecting the reversing camera or its surroundings on the vehicle; it is recommended to have this check carried out by a Mercedes-Benz Service Partner.

1) Reversing camera ex factory

Two reversing camera systems are available ex factory, which, depending on the country-specific legal regulations, are either already installed in the vehicle as standard or are available as special equipment:

- Analog reversing camera with inside rearview mirror display (code FR3)
- Digital reversing camera with display via the central display (code FR8)

If the parking package special equipment (code JB6 or JB7, See 8.9.8 Parking packages/Moving-off information Assist (→ Page 344)) is not selected for closed model designations, the digital reversing camera (code FR8) is fitted as standard in markets with CoC documents. The camera is also installed in these markets on open model designations with a platform ex works (code P02).

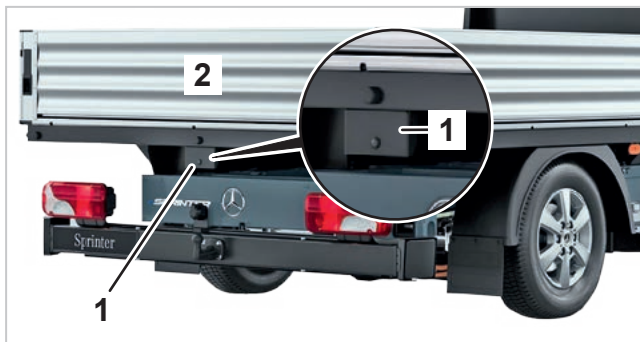
For other markets, the analog or the digital camera is available as special equipment depending on the country.

On closed model designations (panel van and Tourer), the installation is implemented with a holder on the rear roof frame; on open model designations with a platform ex factory (code P02) it is implemented with a holder under the platform.



Position of reversing camera on closed model designations

- 1 Holder for reversing camera on rear roof frame



Position of reversing camera for open model designations (example for vehicle with platform ex works)

- 1 Holder for reversing camera under platform
- 2 Platform ex works (code P02)

For open model designations without platform ex works (without code P02), pre-installations are available for the analog or digital reversing camera (code FV1 or FR7), see section 2 (→ page 349).

⚠ WARNING

Modifications to the camera systems and their surroundings as well as installation work affecting the function and detection range of the camera are not permissible.

Otherwise, driving assistance systems may no longer function as intended and legal requirements may no longer be met.

There is risk of an accident and danger to life and limb!

The bodybuilder manufacturer must clarify possible exceptions to detachable parts at the rear of the vehicle (e.g. bicycle rack) with the Technical Consultancy and/or the responsible registration authority with regard to the country-specific legal requirements (e.g. UN R 158).

In addition, it should be noted that detachable parts on the rear of the vehicle can lead to restrictions in the visibility area behind the vehicle and pedestrian emergency braking when reversing.

In the case of bodybuilder manufacturer-specific special attachments with a greater effect on the detection range of the camera, e.g. cargo liftgate, it is permissible to change the camera position in the longitudinal direction of the vehicle. The camera position and orientation can also be adapted within the limit values prescribed for the pre-installation (code FR7), see specifications in section 3 (→ page 351) and figures (→ page 354).

! NOTE

The guide lines of the digital reversing camera (code FR8) in the head unit display are designed exclusively for the basic vehicle ex factory.

If approved detachable parts are retrofitted to the rear of the vehicle, the appropriate parameter set must be coded in the reversing camera by your Mercedes-Benz Service Partner and the guide lines must be hidden.

The guide lines of the reversing camera must also be hidden in the following cases:

- if the vehicle is extended to the rear by bodybuilder manufacturer-specific detachable parts,
- if the camera position is changed for special bodies,
- if the camera is retrofitted with the pre-installation code FR7, see section 2b (→ page 351).

To do this, the code O1N "Activation of reversing camera without guide lines" must be documented in the VeDoc vehicle data card.

In addition, if the position of the camera changes, the new position information must be added to the vehicle data card via XENTRY with the corresponding O-codes; see section 5 (→ page 362). The reversing camera must then be recoded and calibrated via XENTRY; see "Commissioning of control units" guide (→ page 387).

⚠ WARNING

For the previously mentioned body situations, it is not possible to ensure that the pedestrian emergency braking function works properly when reversing.

There is risk of an accident and danger to life and limb!

For this reason, the documentation of code O1N deactivates pedestrian emergency braking when reversing.

The bodybuilder manufacturer's operator's manual must inform the vehicle user of the deactivation of this function (→ page 347).

2) Retrofitting of reversing camera

There are two options available for retrofitting a reversing camera on open model designations:

- Pre-installation for analog reversing camera with inside mirror display (code FV1)
- Pre-installation for digital reversing camera with display via the head unit display (code FR7)

! NOTE

Compliance with country-specific laws and guide-lines on the viewing area behind the vehicle (e.g. UN R 158) and the fulfillment of the obligation to provide evidence for the registration and driving operation of the converted vehicle are the sole responsibility of the bodybuilder manufacturer.

2a) Analog reversing camera: Pre-installation of reversing camera with inside rearview mirror display (code FV1)

It is recommended to use the analog camera approved by Mercedes-Benz AG. It is not supplied with the vehicle on delivery and is therefore available as an individual part with an adapter wiring harness from the Mercedes-Benz Service Partner:

Item number **A 000 905 68 15**

- ❗ Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.
- ❗ Information on component geometry and the camera's cone of view can be found under the item number in the 3D design data and 2D drawings. For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).
- ❗ For special bodies and questions, please contact us by e-mail in German or English, stating the vehicle model designation:

bb-3d_data-van@mercedes-benz.com



❗ NOTE

If you are using analog cameras other than those approved by Mercedes-Benz AG, these must comply with the same standard NTSC 60 Hz.

With this pre-installation, an interface for connecting the analog reversing camera is available for open model designations in the driver's seat box and for closed model designations in the area of the third brake lamp (roof in the upper rear area), see section 4 (→ page 357).

❗ NOTE

The reversing camera system is classified as ASIL Level A according to ISO 26262. For all the components involved, it must be ensured that no delayed or frozen images can occur, see section 5 (→ page 362). The bodybuilder manufacturer must provide proof of this on request.

2b) Digital reversing camera: Pre-installation for electric reversing aid (code FR7) for retrofitting of a digital reversing camera with display via head unit display

This special equipment is only available for open model designations. For closed model designations, the special equipment code FR8 must be used; see section 1 (→ page 347).

! NOTE

Only the digital reversing camera approved by Mercedes-Benz AG must be used for the digital reversing camera system.

Any other digital camera is not permitted, because it is not approved for combining with the other system components (e.g. the central display).

Code FR7 is only possible in combination with the MBUX multimedia system special equipment with 10.25-inch touchscreen (code E7M or E7H).

This pre-installation provides the interface for connecting the reversing camera in the driver's seat frame; see section 4 (→ page 357).

From model year 2024, a new system generation of the digital reversing camera (RVC5) and control unit will be used. The following new information and specifications apply.

- i If you have any questions, please contact the Technical Consultancy team (→ page 18).
- i If necessary, you can obtain information and specifications on vehicles with the special equipment code FR7 and the digital reversing camera prior to model year 2024 from previous mounting directives. These are still available in the Mounting Directives Archive in the Bodybuilder Portal.

3) Retrofitting the digital reversing camera with pre-installation code FR7

The digital reversing camera is not included when the vehicle is delivered and is therefore available (with hardware and software in the delivery package) without wiring harness via a Mercedes-Benz Service Partner:

Item number **A 000 900 85 50**

- i In the change information from 02/2024, the successor item number A 000 900 35 55 with new software was specified; this can also be used. Up to the final release for vehicles with code FR7, the software for this camera will be automatically adapted during commissioning. The release of the new software with modified installation specifications will be announced later in separately issued change information.
If you have any questions, please contact the Technical Consultancy team (→ page 18).

The 3D design data of the camera with the 3D cone of vision is available for the digital planning and design of a holder for attachment of the reversing camera to the body:

Item number **A 177 901 06 04**

- i Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.
- i Information on component geometry and the camera's cone of view can be found under the item number in the 3D design data and 2D drawings. For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).

- i** For special bodies and questions, please contact us by e-mail in German or English, stating the vehicle model designation:

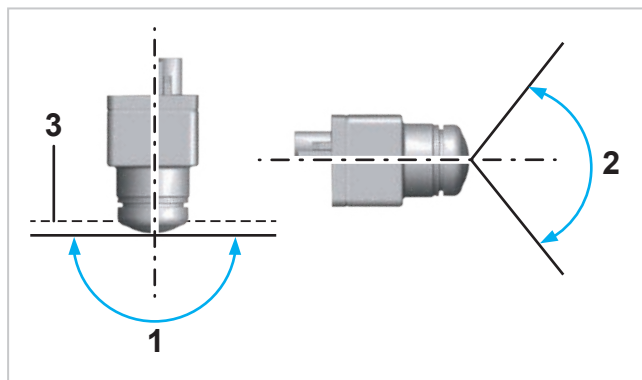
bb-3d_data-van@mercedes-benz.com



Specifications for the reversing camera holder

The bodybuilder manufacturer must produce a separate holder for positioning and installing the reversing camera, which must meet the following specifications:

- The holder must be specially adapted for installation on the bodybuilder manufacturer's specific body and be suitable for use of the camera while driving.
- The holder must not restrict the viewing area or the viewing angle limits of the camera. For this, it is essential to make the edge of the holder in the directly adjacent area of the camera lens without any protrusion; see following illustration. For the design of the holder, use the previously mentioned 3D design data of the cone of vision.



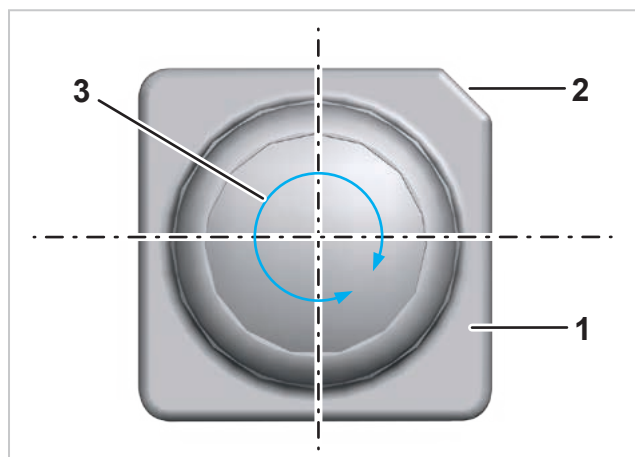
Horizontal **1** and vertical **2** viewing area or viewing angle limit and limit contour for camera holder **3** (schematic); see 3D design data for details of the viewing cone

- i** The holders of the reversing camera for closed and open model designations ex factory with code FR3 or FR8 (→ page 347) can potentially be used for installation on the bodybuilder-manufacturer-specific body or as a basis for the structural design of your own holder (→ page 355).

! NOTE

The camera must be attached with the holder so that the beveled corner is exactly at the top right (when viewed from behind), see the following illustration.

A maximum deviation in the installation position of $\pm 1^\circ$ (rotation around the center axis of the camera) is permitted.



Alignment of the reversing camera on the body, view from behind

- 1 Reversing camera (RCV5)
- 2 Beveled corner at top right
- 3 Permissible tolerance of the installation position around the axis of rotation: max. $\pm 1^\circ$

- For consistently good image quality, design the holder with high component rigidity to reduce camera vibrations as much as possible.
- Comply with IP protection class IP6K7K for the camera and electrical connections.

- Observe the electrical connection specifications (→ page 357) also when planning, manufacturing and installing the holder.
- The plugs of the cables must be located in the interior of the body.
- Take precautions to ensure that no liquid collects in the holder or in the cut-out in the body.
- Take suitable measures to protect the line from damage in the area of the cable run into the body interior. Also seal the cable run against water and dirt ingress. When using the adapter wiring harness, a variant with cable grommet is available, see section 4b (→ page 360). This seals the opening of a cable run with a sheet metal part.

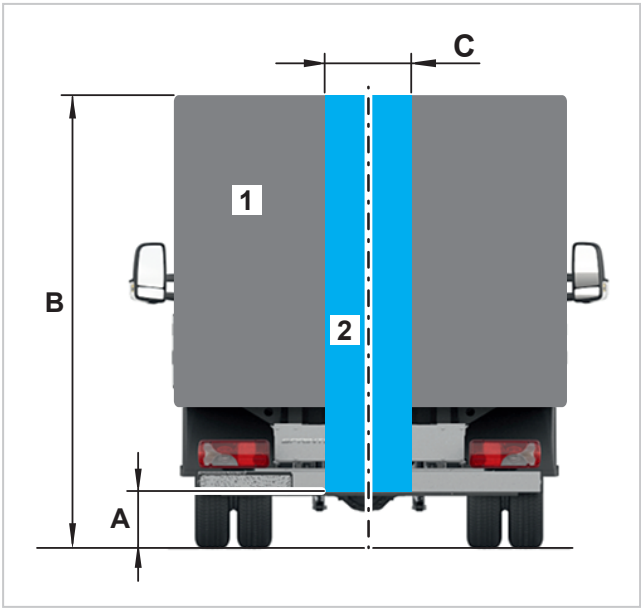
Positioning and installation specifications

! NOTE

The limit values for the installation position and alignment of the reversing camera on the vehicle rear must be observed.

In the event of deviations, a Letter of Compatibility is required (→ page 19); this can be requested via the Bodybuilder Portal (→ page 20). Also observe the information in section 2 (→ page 349).

- The reference point for positioning the camera is the center of the camera lens.
- Depending on the bodybuilder manufacturer-specific body, a position in the marked area **2** of the rear of the vehicle can be selected for the installation of the camera.
- Positioning and installation of the reversing camera on the upper roof frame is permitted if the limit value for the maximum height is not exceeded, see the following illustration (→ page 354).
- If the reversing camera is installed directly on the body in the case of bodybuilder manufacturer-specific special bodies and if there is a greater restriction of its recording range, which is no longer permissible according to the country-specific regulations, it is permissible to change the camera position in the longitudinal direction of the vehicle (X-axis on the vehicle side). An adapter component part for bridging the distance between the camera and the body must be sufficiently rigid and stable for driving operation. In the case of such special bodies, it is recommended to contact the Technical Consultancy team (→ page 18) during the planning phase in order to coordinate the planned body work and its compatibility with the driving assistance system. The guide lines of the reversing camera must then be hidden during commissioning with code O1N; see note at the end of section 1 (→ page 349).
- The installation position of the camera and the angle of inclination must be specified when commissioning the reversing camera in XENTRY to ensure correct coding of the control unit with O-codes and calibration of the camera; see the following illustration with details of the permissible range for the installation position and information in section 5 (→ page 362).

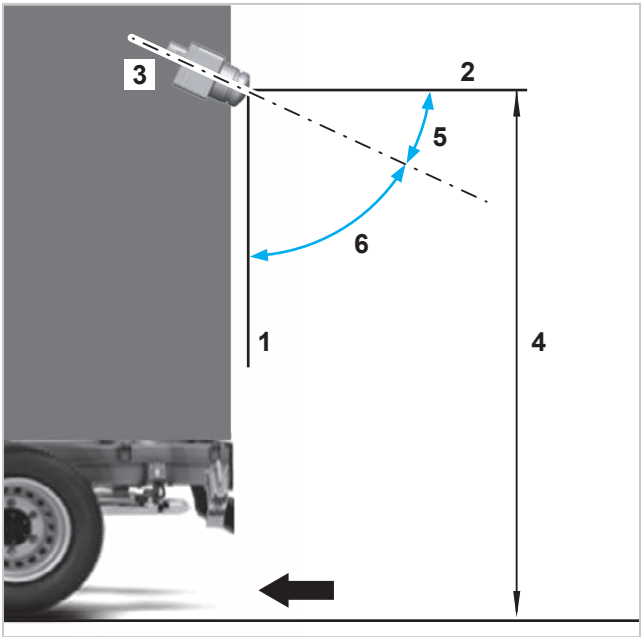


Permissible range for the installation position of the reversing camera in the Y and Z directions (schematic)

- 1 Example of any bodybuilder manufacturer-specific body
- 2 Permissible range for the installation position

Permissible range for the installation position		
	Installation position	Limit value [mm]
A	Height above roadway ¹⁾	At least 540 ²⁾
B	Height above roadway ¹⁾	Max. 3500
C	Deviation, related to vehicle center	Max. ±200

- 1) The height in the Z-direction above the roadway must be measured on the vehicle with a fully completed body in running order without load.
- 2) Depending on the type of body and the installation position of the camera, the legal requirements (e.g. as per UN R 158) may not be fulfilled at low altitudes. Check the prerequisites for compliance with legal requirements for the body-specific installation situation.



Alignment of the camera on the body (schematic)

- 1 Vertical Z-direction
- 2 Horizontal X-direction
- 3 Center axis of camera lens
- 4 Position in Z-direction: Installation height above roadway
- 5 Angle of inclination to the horizontal plane (for entry in XENTRY)
- 6 Angle of inclination to the vertical plane (for ease of measurement on the body)
- ➔ Direction of travel

- The angle of inclination must be specified for the calibration of the camera during commissioning in XENTRY; see section 5 (→ page 362). After changing the angle of inclination, the changed value must be entered again and the camera must be recalibrated.
- Vibrations and movements of the camera while driving can have a negative effect on the image quality and must therefore be reduced as much as possible. It is recommended that the holder with camera only be installed on structural components and/or in vehicle surroundings with high inherent rigidity.
- When positioning and installing the camera, ensure that the viewing area to the rear is not restricted. Observe the notes on impermissible restrictions of the viewing area – See 8.9.9 Reversing camera (→ Page 347) – and the legal requirements for the viewing area and permissible interfering contours in accordance with section 2 (→ page 349). With regard to this, also note the illustration (→ page 352).

Possible use of the standard holders (from code FR8)

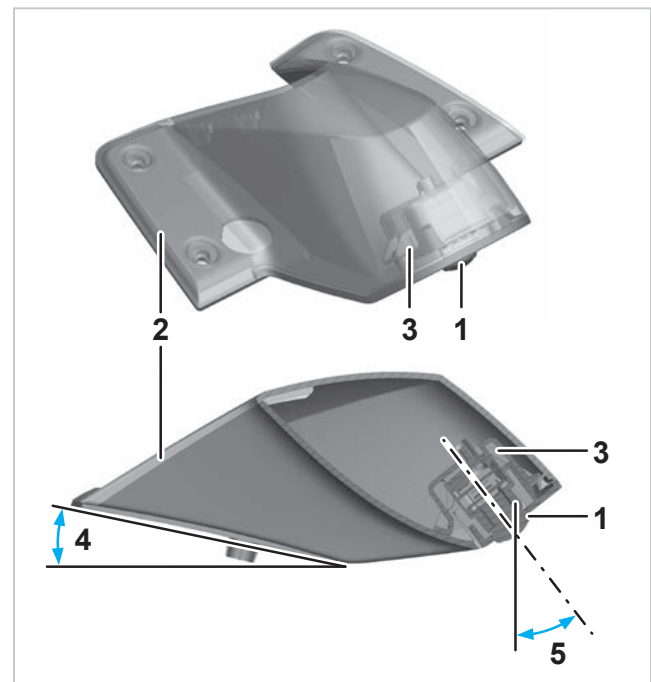
a) Closed model designations

The reversing camera (code FR8) is installed ex works on the closed model designation with a holder on the rear roof frame with roof height code LH2, see figure (→ page 348).

This holder can also be used for a bodybuilder manufacturer-specific body solution or as a basis for the design of a new holder.

Item number **A 907 820 93 01**

Note that this holder has an inclination **4** and a curvature on the underside due to the vehicle contour and that the camera is attached to the holder with a tilt to the vertical plane **5**. This must be taken into account when planning to use the holder.



Reversing camera holder (code FR8)

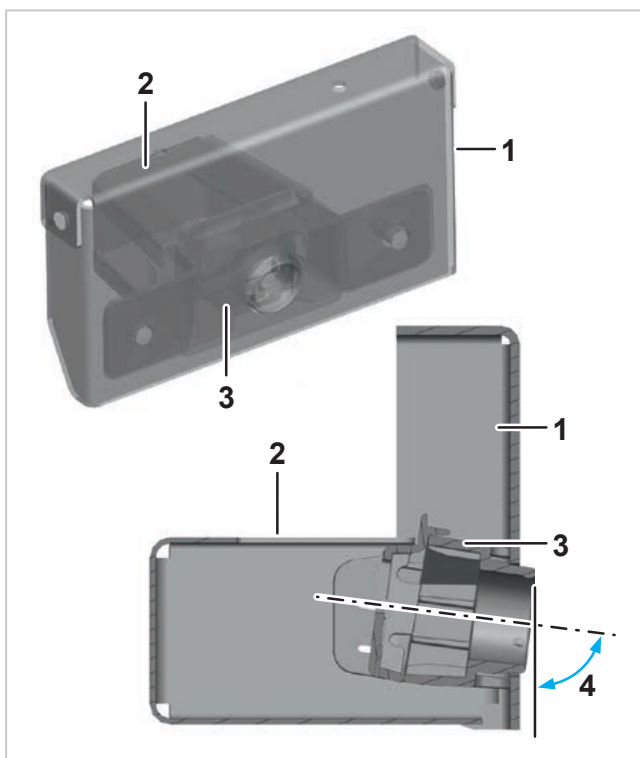
- 1 Reversing camera
- 2 Holder
- 3 Camera mounting
- 4 Inclination of holder approx. 13° in CAD design position
- 5 Camera tilt approx. 40° in CAD design position

b) Open model designations

Ex factory (code P02), the reversing camera (code FR8) is installed on a structural part (retaining plate) underneath the platform with fastening parts and covers on the open model designations; see illustration (→ page 348).

These component parts can be used for a bodybuilder manufacturer-specific body solution or as a basis for the design of a new holder.

Make sure that the camera is mounted in the holder with a tilt to the vertical **4**. This must be taken into account when planning to use the holder.



Mounting and cover parts for reversing camera

- 1 Holder plate under platform
- 2 Camera cover with electrical connection
- 3 Camera attachment (illustration without reversing camera)
- 4 Camera tilt approx. 82° in CAD design position

i Details of the holders of the reversing camera are available in the 3D design data and 2D drawings. For information on the provision of 3D design data in the Bodybuilder Portal, see (→ page 25).

i Item numbers for other versions of these holders, e.g. variants depending on the steering type or in combination with the rear-view camera for digital inside mirror (code F84/F85, See 8.9.10 Rearview camera for digital inside mirror (→ Page 364)) as well as for fastening parts and covers for the camera for open model designations with platform ex works (code P02) can be obtained from the Technical Consultancy (→ page 18).

i Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

4) Electrical connection

After installation, the reversing camera must be electrically connected directly to the vehicle's connecting point by a qualified electrician using a video line and, in the case of the analog camera, also using a power supply line.

! NOTE

For the production of a video line (high-frequency coaxial line), a wiring harness manufacturer is recommended that is qualified for FAKRA connections (plugs or sockets) and has the necessary production facilities.

The wiring harness must satisfy the following requirements:

- The wiring harness (cable with plug) made by a qualified electrician or wiring harness manufacturer must be checked for electrical connections and functionality before use (according to DS characteristic: Mandatory documentation relevant to safety). The test results must be submitted on request.
- An additional connecting point along the length of the line between the camera or adapter cable set (if used) and the vehicle's connecting point is not permitted.
- The bodybuilder manufacturer must ensure a water-tight connection of the lines to the camera.
- When laying the video cable, the minimum bending radius must not be less than $5 \times d$ (d = cable diameter).
- Ensure that the lines are routed correctly in the vehicle, See 8.4 Interfaces and electric lines (→ Page 271).
- The quality of the video/high-frequency line must correspond at least to the values specified by LEONI Dacar.
- The line quality was designed for the lengths of line specified in sections 4a (→ page 358) and 4b (→ page 360) from the connecting point to the camera. If a longer line is required, its functional capability must be assured by means of tests conducted at the bodybuilder manufacturer's expense. The test results must be submitted on request.

- The reversing camera system must not be disrupted by EMC influences, e.g. by 230 V systems, actuators etc.

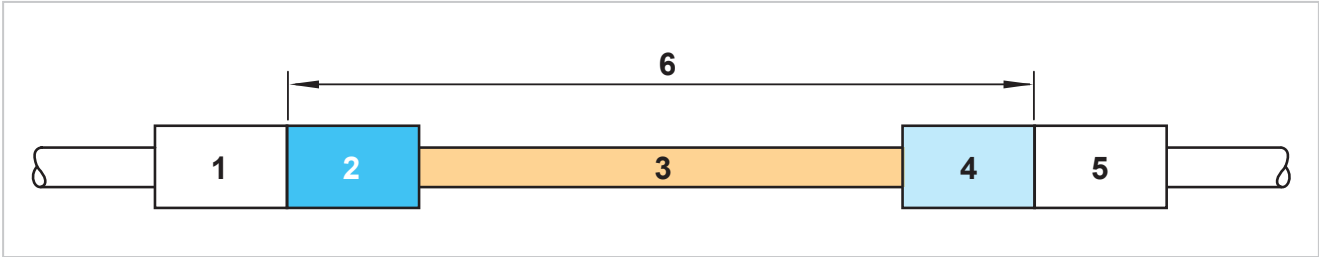
4a) Analog reversing camera (code FV1)

The position of the connecting point on the vehicle for

the electrical connection is located:

- on open model designations in the driver's seat box,
- on closed model designations in the area of the third brake light (roof area at the rear of the vehicle).

Observe the following information on the necessary parts for the production of a wiring harness:



Assignment of the line components to the following tables

For information only (white):

- 1 Connecting points on the vehicle
- 5 Connections on the adapter wiring harness of the camera

Required for the production of the wiring harness 6:

- 2 Connections to the connecting point (blue)
- 3 Electric line (orange)
- 4 Connections to the adapter wiring harness (light blue)

Electric line

	Type	Cross-section or specification	Manufacturer	Maximum permissible length
3	FLRY Voltage	2 × 0.5 mm ²	Commercial provider or cable manufacturer	10 m
	Video (coax)	RG179/Dacar 360-Coax-B (105)-75-1.68-2.7	Leoni	10 m

Observe all technical specifications and requirements in the manufacturer's data sheet.

Electrical contacts

	Position	Type	Type (pins)	Reference item number	Manufacturer	Manufacturer no. Coding, type
1	Connecting points on the vehicle	Voltage	Socket (4 pins)	A 037 545 65 28	TE connectivity	1-929170-1 Coding A, female
		Video (Fakra)	Socket	A 023 545 60 26	Rosenberger	59Z063-DC0-F Coding F, female
2	Connections to the connecting point	Voltage	Plug (4 pins ¹⁾)	A 056 545 40 28	TE connectivity	1-2112850-2 Coding A, male
			Pin contact	A 032 545 39 28		963730-1
		Video (Fakra)	Plug	A 053 545 40 28	Rosenberger	59Z064-DC0-F ²⁾ Coding F, male
			Pin contact	A 001 982 07 28		59S17E-1D8A4
4	Connection to the power line of the camera	Voltage	Socket (2 pins ¹⁾)	A 029 545 10 26	Hirschmann	805-120-511 Coding A, female
		Contact	Socket contact	A 003 982 74 26	Kostal Kontakt	22140734133
		Seal		A 001 545 50 80	Systeme ³⁾	10800507250
	Connection to the video line of the camera	Video (Fakra)	Socket	A 000 545 91 12	Rosenberger	59Z163-DC0-F ²⁾ Coding F, female
		Fakra crimp	Socket contact	A 001 982 10 26		59K17F-1D8A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
	Connections on the adapter wiring harness of the camera	Voltage	Plug (2 pins)	A 211 545 15 28	Hirschmann	872-863-501 Coding A, male
		Video (Fakra)	Plug	A 000 545 93 12	Rosenberger	59Z176-C01-F Coding F, male

- 1) Pin assignment: Pin 1 GND, pin 2 UBAT (tml. 15)
 2) Mercedes-Benz-specific parts with "DC0" (e.g. 59Z064-DC0-F) are available in neutral form from the manufacturer with "000" (e.g. 59Z064-000-F).
 3) Sale via distributor, e.g. Herth+Buss

i If you have any questions about the electrical connection, please contact the Technical Consultancy team (→ page 18).

i Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

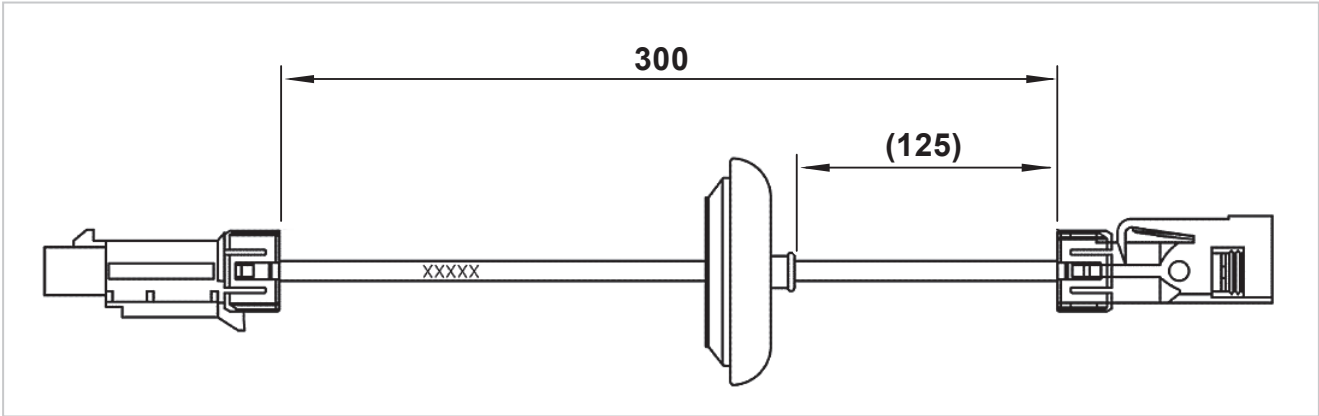
4b) Digital reversing camera (code FR7)

The following variants of a short, optional adapter wiring harness with a line length of 300 mm are available for easier assembly of the camera with holder on the body:

- With cable grommet:
Item number **A 907 820 83 01**
- Without cable grommet:
Item number **A 907 820 75 01**

i Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

The cable grommet can be slid on the cable, is suitable for sheet metal parts from 0.6 to 3 mm and requires a round opening with a diameter of 25 mm (see 2D drawing of cable grommet A 000 998 55 02).



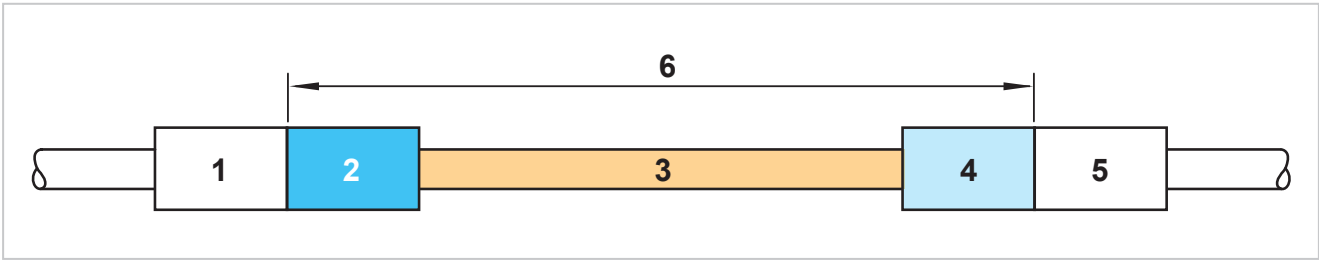
Adapter wiring harness for easier assembly (value in brackets depending on the sliding position of the cable grommet)

The position of the connecting point on the vehicle for the electrical connection is located:

- on open model designations in the driver's seat box.

For closed model designations, only the digital reversing camera is fitted ex works (code FR8), therefore no connecting point for pre-installation is available.

Observe the following information on the necessary parts for the production of a wiring harness:



Assignment of the line components to the following tables

For information only (**white**):

- 1** Vehicle-side connecting point
- 5** Connection on the adapter wiring harness or direct connection of the camera

Required for the production of the wiring harness **6**:

- 2** Connection to connecting point (**blue**)
- 3** Electric line (**orange**)
- 4** Connection to the adapter wiring harness or to the direct connection of the camera (**light blue**)

Electric line

	Type	Specification	Manufacturer	Maximum permissible length
3	Video (coax)	RG174/Dacar 462-Coax 50-1.52-2.8/T105	Leoni	14.5 m
	Alternative: Video (coax)	RTK031/Dacar 302-Coax-B(105)-50-2.1-3.3	Leoni	18 m

Observe all technical specifications and requirements in the manufacturer's data sheet.

Electrical contacts

	Position	Type	Type	Reference item number	Manufacturer	Manufacturer no. Coding, type
1	Vehicle-side connecting point	Video (Fakra)	Socket	A 023 545 60 26	Rosenberger	59Z063-DC0-F Coding F, female
2	Connection to connecting point	Video (Fakra)	Plug	A 053 545 40 28	Rosenberger	59Z064-DC0-F ²⁾ Coding F, male
	For line only RG174/Dacar 462	Fakra crimp parts set	Pin contact	A 000 982 02 28		59S100-102A4
	For line only RTK031/Dacar 302	Fakra crimp parts set	Pin contact	A 000 982 07 28		59S100-1M4A4
4	Connection to the adapter wiring harness or camera	Video (Fakra)	Socket (waterproof)	A 000 545 91 12	Rosenberger	59Z163-DC0-F ²⁾ Coding F, female
	For line only RG174/Dacar 462	Fakra crimp	Socket contact	A 000 982 47 26		59K130-102A4
			Seal	A 000 542 80 00		59K14B-102/50
			End cover	A 000 553 00 00		59Z163-000/51
	For line only RTK031/Dacar 302	Fakra crimp	Socket contact	A 000 982 73 26		59K130-1M4A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
5	Connection on the adapter wiring harness or direct connection of the camera ¹⁾	Video (Fakra)	Plug	A 000 545 93 12	Rosenberger	59Z176-C01-F Coding F, male

- 1) Integrated in camera
 2) Mercedes-Benz-specific parts with "DC0" (e.g. 59Z064-DC0-F) are available in neutral form from the manufacturer with "000" (e.g. 59Z064-000-F).

i If you have any questions about the electrical connection, please contact the Technical Consultancy team (→ page 18).

i Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

5) Commissioning of the reversing camera

! NOTE

After installing the digital reversing camera in combination with the code FR7, the O-code O1N "Activation of reversing camera without guide lines" must be coded into the vehicle and documented in the vehicle data card.

This is also absolutely necessary if the camera position is changed for code FR8, see note at the end of section 1 (→ page 347).

After installation of the reversing camera with new software as per section 3 (→ page 351) and its electrical connection to the connecting point on the vehicle, commissioning must be performed via the XENTRY Kit Mercedes-Benz diagnostic tool before the vehicle is put into circulation. In addition to activating the driving assistance system without guide lines with O-Code O1N, the installation position and the angle of inclination of the camera must be reported back in the bodybuilder manufacturer portal for the respective VIN. The corresponding O-codes are thus supplied to the vehicle data card by the system.

You can obtain access for entering these values in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ After logging into the Bodybuilder Portal, select the "Order data" button.
- ▶ Select the order for your vehicle using the order number to access the specific order overview.
- ▶ Select the "Feedback data for commissioning" button.
- ▶ When commissioning for the first time, select the sub-item "Activate pre-installation". There, you will find the pre-installations ordered for the vehicle and the field for entering the camera position and angle of inclination.

! NOTE

After the vehicle has been put into operation for the first time or after a subsequent change (e.g. to the position of the camera), adjustment via the Bodybuilder Portal is no longer possible.

In this case, the O-codes for specifying the camera position and angle of inclination must be adjusted manually via XENTRY in the vehicle data card; the affected control units must then be re-coded and the camera recalibrated in accordance with the "Commissioning of control units" guide.

- ① More information on the procedure can be found in the "Commissioning of control units" guide in the Bodybuilder Portal, See 8.19 Commissioning of control units (→ Page 387).
- ① For information on XENTRY, See 2.2.4 Diagnosis (→ Page 26).
- ① It is recommended to have the commissioning performed by a Mercedes-Benz Service Partner.

Overview of the O-codes for commissioning the reversing camera (code FR7)

The O-codes assigned when entering the camera position are listed in the following tables for planning the body and determining the installation position. The values specified in these tables apply to vehicles with a fully completed body in running order without load.

-  If you have any questions about the O-codes or if you cannot find the corresponding O-codes for your body situation in this overview, please contact the Technical Consultancy team (→ page 18).

a) Position of camera in Z direction

O-code	Height above roadway [mm]
OZ1	540–700
OZ2	701–1000
OZ3	1001–1500
OZ4	1501–2000
OZ5	2001–2500
OZ6	2501–3000
OZ7	3001–3500

b) Position of camera in X-direction

O-code	Distance from the center of the front axle [mm]
OX1	up to 5900
OX2	from 5901 onward

After commissioning and before the vehicle is placed on the market, the reversing camera system must be tested under real conditions (including the bodybuilder manufacturer's requirements) and must function without errors. In addition to country-specific requirements, e.g. according to UN R 158, the following requirements must also be met:

- When reverse gear is engaged, the picture from the reversing camera is displayed completely and without errors in the central display.
- No guide lines are displayed.
- The rear of the vehicle is clearly visible.

NOTE

The reversing camera system is classified as ASIL Level A according to ISO 26262. For all the components involved, it must be ensured that no delayed or frozen images can occur. The bodybuilder manufacturer must provide proof of this on request.

8.9.10 Rearview camera for digital inside mirror

Available special equipment

Code	Equipment
F84	Digital inside mirror for closed model designations (FKA/FKB)
F85	Digital inside mirror pre-installation for open model designations without platform (FHS/FHL)

1) Digital inside mirror

The digital inside mirror special equipment option (code F84) is available ex factory for closed model designations (FKA panel van or FKB crewbus) with normal roof (standard) and high roof (code D03) and is delivered fully assembled in the vehicle.

! NOTE

Do not make any changes to the camera systems installed at the factory.

The digital inside mirror can be combined with the optional reversing cameras (code FR3 or FR8) or parking package (code JB6 or JB7).

-  For further information on the digital inside mirror, please refer to the sales information and contact your Mercedes-Benz sales partner.
-  When using the digital inside mirror, observe the information and specifications in the operator's manual of your vehicle.

2) Digital inside mirror pre-installation

The digital inside mirror pre-installation special equipment (code F85) is only available from the factory for open model designations without platform (cab FHS and crewcab FHL).

This special equipment is not suitable for bodies with detachable parts in the rear area that restrict the field of view of the rearview camera or that require a deviation in positioning from the following specifications.

-  If you have any questions, please contact the Technical Consultancy team (→ page 18).

Depending on country-specific availability, the digital inside mirror pre-installation can be combined with the optional reversing camera pre-installations (code FV1 or FR7) or with the parking packages for closed model designations (code JB6 or JB7).

-  Information on the country-specific availability of this equipment can be obtained from your Mercedes-Benz sales partner.

When the vehicle is delivered, the digital inside mirror is installed in the vehicle in the same position as other inside mirrors (windshield centered at the top) and the necessary cable set is already routed up to the connecting point on the vehicle.

The reversing camera is supplied with the delivered vehicle without a camera housing:

Item number **A 910 905 35 00**

-  Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item number is invalid.

- i** Details on the geometry of the camera can be found in the 3D design data (→ page 25).

! NOTE

The bodybuilder manufacturer is solely responsible for attaching the rearview camera to the bodybuilder-manufacturer-specific vehicle body using a suitable camera holder and housing, connecting the camera to the connection point and commissioning the system.

The following information and specifications must be observed in this context.

! WARNING

If specifications are not observed or the system limits are exceeded, there is a risk of accidents and danger to life and limb!

! NOTE

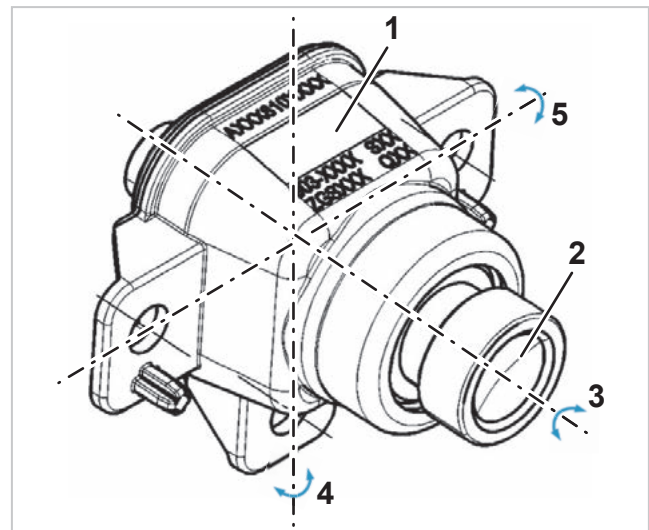
Before planning a vehicle body, the bodybuilder manufacturer must check whether all of the following specifications can be complied with. If this cannot be guaranteed, the use of this special equipment is not permissible.

- i** For information on alternative equipment to meet the legal requirement for inside mirrors, please contact your Mercedes-Benz sales partner.

2a) Mounting on the rear area of the vehicle

For correct image display in the digital inside mirror, attach the camera to the body so that none of the three screw points are at the top and the camera is instead positioned as shown in the following illustrations. Please note with regard to this:

- Horizontal alignment (rotation around camera longitudinal axis or center axis of camera lens **3**): maximum deviation $\pm 0.5^\circ$
- Alignment in longitudinal direction of vehicle (rotation around camera vertical axis **4**): maximum deviation $\pm 1.0^\circ$
- Inclination to the roadway and viewing angle (rotation around camera transverse axis **5**): see section 2c (→ page 367).



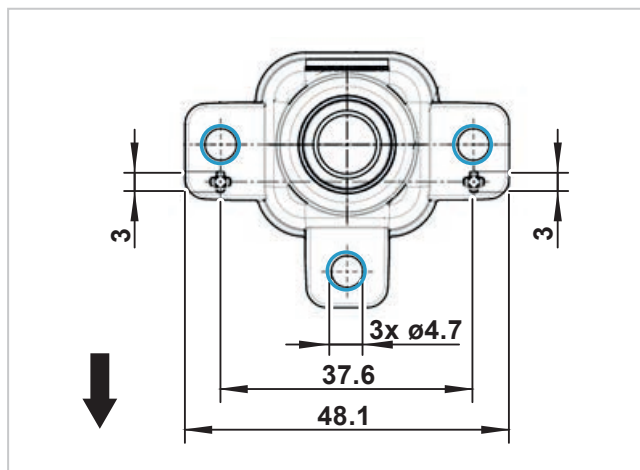
Rearview camera for digital inside mirror

- 1 Camera
- 2 Camera lens
- 3 Horizontal alignment
- 4 Alignment in longitudinal direction of vehicle
- 5 Inclination to the roadway and viewing angle

For fastening – depending on the body situation – select suitable screws with a diameter matching the screw hole in the camera housing.

! NOTE

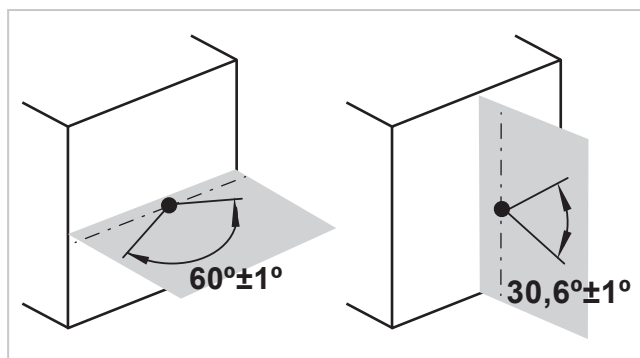
Modification of the camera (e.g. machining of the screw holes) is not permissible.



Camera with screw holes (marked blue)

➡ Direction towards the roadway (downwards)

For trouble-free operation of the rearview camera, pay attention to the viewing angle ranges and the immediate vehicle surroundings when designing a suitable camera holder and housing. The exact viewing angle cone is also available in the design data.



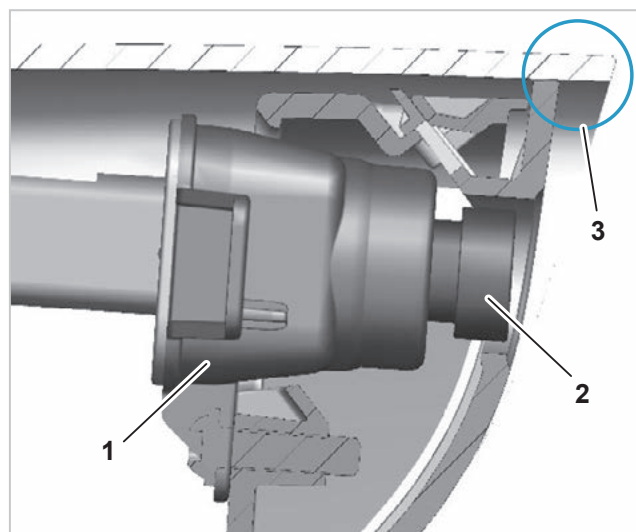
Horizontal (left) and vertical viewing angle ranges (right)

! NOTE

Possible detachable parts or other interfering contours of the body that protrude into the viewing angle range influence the picture area and can impair the function of the digital inside mirror.

Therefore, ensure the necessary viewing angle range without restrictions.

Design the camera housing so that there is a 5 mm overhang above the camera lens to limit direct sunlight from above and minimize stray light.



Example of camera housing with protrusion

- 1 Camera
- 2 Camera lens
- 3 Protrusion of camera housing of 5 mm

2b) Positioning at the rear area of the vehicle

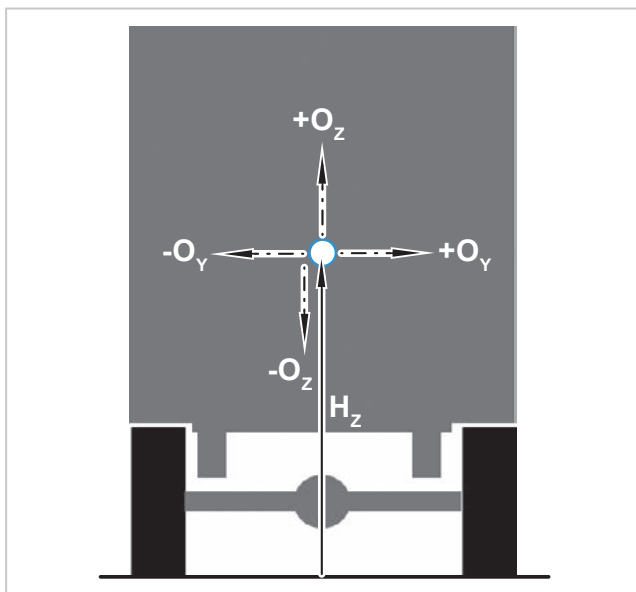
When positioning and tilting the reversing camera at the rear area of the vehicle, the reference point is the center point of the camera lens or the reference line is the center axis of the camera lens (see previous illustration).

For correct display of the image in the digital inside mirror, the rearview camera is coded at the factory to a nominal installation position.

Observe the maximum permissible deviations when positioning the camera at the rear of the vehicle:

Height above roadway	[mm]
Nominal installation height H_z	2732
Maximum vertical deviation O_z	±400

Position to center of vehicle	[mm]
Nominal position	0
Maximum horizontal deviation O_y	±200



Positioning of camera at vehicle rear (schematic)

The height of the camera position above the roadway must be measured on the vehicle with the complete body including a load typical for the body.

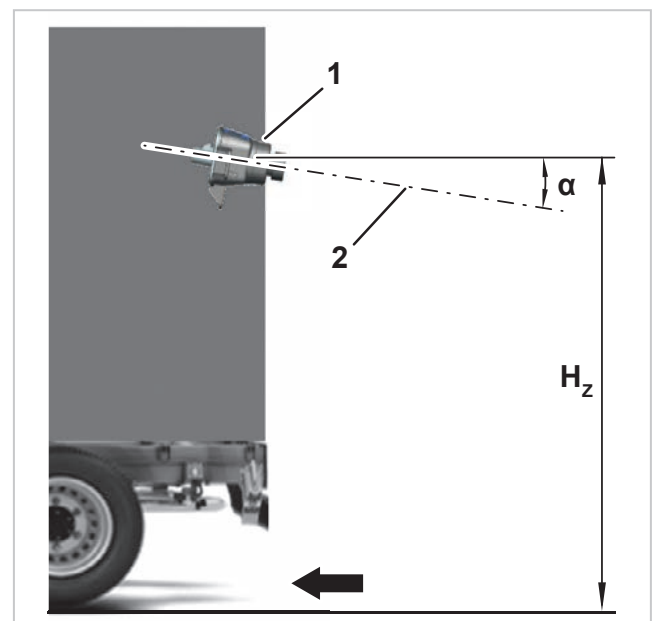
! NOTE

If the specified system limits are exceeded, the picture will no longer be transmitted without errors. Therefore, ensure compliance with the specifications for positioning.

2c) Inclination to the roadway and viewing angle

The viewing angle α i.e. the inclination of the camera to the roadway, depends on the height above the roadway H_z and is calculated as follows:

$$\alpha = \frac{H_z}{1000 \text{ mm}} + 1,65^\circ$$



Installation height and viewing angle for camera positioning in rear area

- 1 Camera
- 2 Center axis of camera lens
- ➔ Direction of travel

Installation height and viewing angle	Height H_z [mm]	Angle α
Maximum installation height	3132	4.8°
Nominal installation height	2732	4.4°
Minimum installation height	2332	4.0°

Observe maximum deviations of the viewing angle of $\pm 0.5^\circ$ when installing the camera.

! NOTE

If the specified system limits are exceeded, the picture will no longer be transmitted without errors. Therefore, ensure compliance with the specifications for the viewing angle.

2d) Electrical connection

After installation, the reversing camera must be electrically connected by a qualified electrician with a video line directly to the connecting point on the vehicle.

! NOTE

For the production of a video line (high-frequency coaxial line), a wiring harness manufacturer is recommended that is qualified for FAKRA connections (plugs or sockets) and has the necessary production facilities.

The wiring harness must satisfy the following requirements:

- The electrical connections and the functionality of the wiring harness manufactured by a wiring harness manufacturer must be fully checked prior to use (according to DS characteristic: Mandatory documentation relevant to safety). The test results must be submitted on request.
- An additional connecting point on the line length between the camera or adapter wiring harness (if present) and the vehicle's connecting point is not permissible.
- A watertight connection of the lines to the camera must be ensured by the bodybuilder manufacturer.
- When laying the video cable, the minimum bending radius must not be less than $5 \times d$ (d = cable diameter).
- Ensure that the lines are routed correctly in the vehicle, See 8.4 Interfaces and electric lines (→ Page 271).
- The quality of the video/high-frequency line must correspond at least to the values specified by LEONI Dacar.
- The line quality has been designed for the specified length of line from the connecting point to the camera. If a longer line is required, its functional capability must be assured by means of tests conducted at the bodybuilder manufacturer's expense. The test results must be submitted on request.
- The rearview camera must not be disrupted by EMC influences, e.g. by 230 V systems, actuators etc.

The position of the connecting point on the vehicle for the electrical connection is located:

- on open model designations in the driver's seat box

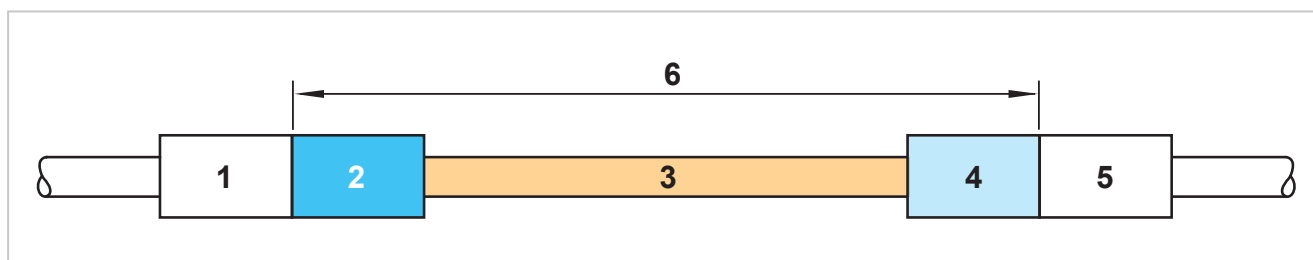
For closed model designations, only the complete system is installed ex works, therefore no connecting point is available for pre-installation.

For easier assembly, a short optional adapter wiring harness to the reversing camera with a line length of 284 mm will be available:

Item number **A 910 540 65 62**

- i** Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

For the production of a wiring harness, observe the following information on the necessary parts, as well as the distinction between the electrical connection to the adapter wiring harness **a** and the direct connection to the camera **b**:



Assignment of the line components to the following tables

For information only (**white**):

- 1** Vehicle-side connecting point
- 5** Connection to the adapter wiring harness **5a** or direct connection of the camera **5b**

Required for the production of the wiring harness **6**:

- 2** Connection to connecting point (**blue**)
- 3** Electric line (**orange**)
- 4** Connection to the adapter wiring harness **4a** or to the direct connection of the camera **4b** (**light blue**)

Electric line

	Type	Specification	Manufacturer	Maximum permissible length
3	Video (coax)	RG174/Dacar 462-Coax 50-1.52-2.8/T105	Leoni	14.5 m
	Alternative: Video (coax)	RTK031/Dacar 302-Coax-B(105)-50-2.1-3.3	Leoni	18 m

Observe all technical specifications and requirements in the manufacturer's data sheet.

Electrical contacts

	Position	Type	Type	Reference item number	Manufacturer	Manufacturer no. Coding, type
1	Vehicle-side connecting point	Video (Fakra)	Socket	A 023 545 55 26	Rosenberger	59Z063-DC0-A Coding A, female
2	Connection to connecting point	Video (Fakra)	Plug	A 053 545 35 28	Rosenberger	59Z064-DC0-A ²⁾ Coding A, male
	For line only RG174/Dacar 462	Fakra crimp parts set	Pin contact	A 000 982 02 28		59S100-102A4
	For line only RTK031/Dacar 302	Fakra crimp parts set	Pin contact	A 000 982 07 28		59S100-1M4A4
4a	Connection to the adapter wiring harness A 910 540 65 62	Video (Fakra)	Socket (waterproof)	A 000 545 63 46	Rosenberger	59Z163-DC0-C ²⁾ Coding C, female
	For line only RG174/Dacar 462	Fakra crimp	Socket contact	A 000 982 47 26		59K130-102A4
			Seal	A 000 542 80 00		59K14B-102/50
			End cover	A 000 553 00 00		59Z163-000/51
	For line only RTK031/Dacar 302	Fakra crimp	Socket contact	A 000 982 73 26		59K130-1M4A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
5a	Connection on adapter wiring harness	Video (Fakra)	Plug	A 000 545 25 48	Rosenberger	59Z176-C01-C Coding C, male
4b	Connection to direct connection of camera	Video (Fakra)	Socket (waterproof)	A 000 545 61 33	Rosenberger	59Z163-DC0-A ²⁾ Coding A, female
	For line only RG174/Dacar 462	Fakra crimp	Socket contact	A 000 982 47 26		59K130-102A4
			Seal	A 000 542 80 00		59K14B-102/50
			End cover	A 000 553 00 00		59Z163-000/51
	For line only RTK031/Dacar 302	Fakra crimp	Socket contact	A 000 982 73 26		59K130-1M4A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
5b	Direct connection of the camera	Video (Fakra)	Plug	A 910 905 35 00 ¹⁾	-	-

1) Integrated in camera

2) Mercedes-Benz-specific parts with "DC0" (e.g. 59Z064-DC0-A) are available in neutral form from the manufacturer with "000" (e.g. 59Z064-000-A).

i If you have any questions about the electrical connection, please contact the Technical Consultancy team (→ page 18).

i Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

2e) Commissioning of the rearview camera and the digital inside rearview mirror

After the rearview camera has been electrically connected to the digital inside mirror via the connecting point in the vehicle, the system performs an auto-calibration during driving operation. For this purpose, the horizon must be visible in the digital inside mirror when the driving terrain is level.

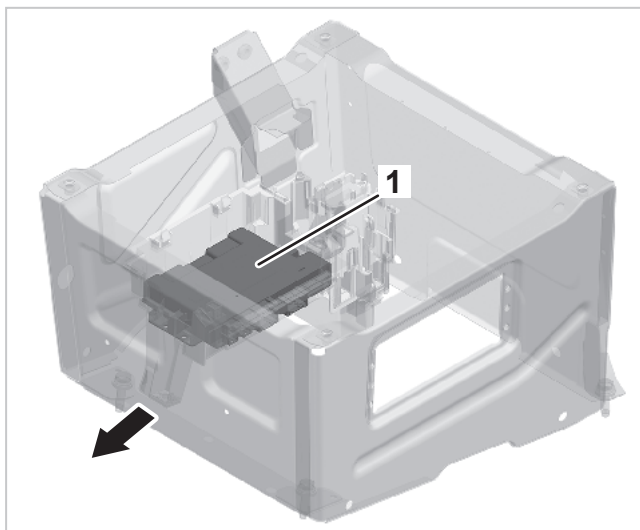
If blurred, dark areas can be seen at the side of the image, check whether there are any detachable parts or other interfering contours in the viewing angle range. Such restrictions of the rear viewing angle range must not be visible. Therefore, make sure that the previously mentioned specifications for the unobstructed viewing angle range are observed.

- ❗ For more information on the procedure for the commissioning of control units, See 8.19 Commissioning of control units (→ Page 387).
- ❗ For further information on the digital inside mirror, please refer to the sales information and contact your Mercedes-Benz sales partner.
- ❗ When using the digital inside mirror, observe the information and specifications in the operator's manual of your vehicle.

8.10 Parameterizable special module (PSM)

The parameterizable special module PSM (internal designation: Multi-Purpose Module MPM) is available as special equipment (code ED5).

The PSM is located in the driver's seat frame.



PSM installation position

- 1 Position of PSM in driver's seat box
 ➔ Direction of travel

The PSM forms an interface between the on-board electronics and the bodybuilder manufacturer's electrical installations and modifications that is clearly defined, capable of diagnosis and EMC-tested. For example, individual vehicle statuses can be read in and evaluated by the Body CAN bus or HMI2 bus, and then routed through to the CAN bus of the bodybuilder manufacturer (ABH CAN, as per ISO 11992-3).

i Model year 2024 marks the introduction of a new generation of PSM with a larger capacity (increased data storage, faster data processing), as well as greater parameterization possibilities and correspondingly adapted software.

Observe the information and specifications from the previous Body/Equipment Mounting Directives and Function Descriptions of the PSM for all vehicles with an older generation of PSM. These are still available in the Body/Equipment Mounting Directives Archive in the Bodybuilder Portal.

! NOTE

Parameterizations from the previous PSM cannot be transferred to the new PSM generation as of model year 2024. If you have any questions or require data conversion, contact the Technical Consultancy with regard to the PSM (→ page 18).

The PSM provides an interface, rather than a system. The validation of the PSM functionality available via the interface must be performed in accordance with ISO 26262. Furthermore, take into account applicable cyber security requirements, such as UN-R 155 and ISO 21434. Exceptions are listed in the "PSM function description" in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



The protection of functions in which the PSM is involved, and compliance with applicable standards (e.g. ISO 26262 on functional safety and UN R-155 and ISO 21434 on cyber security) and specifications is always the responsibility of the bodybuilder manufacturer.

Customer-specific requirements can take the form of special inputs, such as an external remote engine start and stop, or special outputs, such as pulse-pause-modulated engine speed or CAN-bus-compatible control units in bodies or trailers.

⚠ WARNING

The wiring of the vehicle must not be tampered with, as this may lead to fault messages and malfunctions in other control units.

There is risk of an accident and danger to life and limb!

Changes to the wiring of the on-board electronics are therefore not permitted.

The PSM is connected to the vehicle network via body CAN bus and HMI2 bus and therefore has access to specific messages sent by the linked control units (e.g. idle active, parking brake active, vehicle speed, engine speed). In contrast, individual signals can be monitored or generated at analog and digital inputs and outputs.

! NOTE

The near-engine mounted functions rpm limitation, torque limitation and speed limitation must be deactivated by ignition ON/OFF. These must not automatically become active after engine start-up.

The PSM coding must be designed so that additional actuation by the customer is required after ignition ON/OFF and/or after engine start in order to activate these.

Example

- The engine control unit sends the engine speed in a message, which can be read by the PSM. The PSM converts the engine speed information into a pulse-width-modulated signal and makes this available at an output.
- In the opposite direction, the PSM can increase or reduce the desired engine speed by evaluating sensors.

- ① The PSM is parameterized using XENTRY Kit (→ page 26).
- ① Information about possible parameterizations can be found in the detailed "PSM Function Description" in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ① You can obtain information about standard parameterization possibilities from your Mercedes-Benz Service Partner.
- ① Information on parameterization possibilities that are not covered by the "PSM function description" is available from the Technical Consultancy with regard to the PSM (→ page 18).
- ① Professional training on the PSM is available to you via our Mercedes-Benz training portal:

<https://learningsystem.mercedes-benz.com>



! NOTE

When writing a coding (e.g. rpm control), all previous parameterizations are erased. Prior backup of data is recommended.

For vehicles with a communication module installed ex factory:

! NOTE

Based on the Group-wide decision of the steering committee for "Autonomous driving ethics and law", Mercedes-Benz AG only makes vehicle data available via internal back ends. Internal back ends are Mercedes me connect or the Extended Vehicle Backend. External vehicle data transmission via the PSM is no longer available as standard.

Exceptions continue to exist for multi-brand applications, fleet applications, car sharing – these are also applications in which the PSM serves as the basis for an additional vehicle body and no vehicle data is sent to a third-party back end.

Mercedes-Benz AG will continue to support individual PSM queries for use in the vehicle environment. For more information, contact the Technical Consultancy with regard to the PSM (→ page 18).

In relation to data transfer to third-party back ends, observe:

! NOTE

Since 2006, data publication in the vehicle via the PSM or the FMS (Fleet Management Standards) interface has been a common feature for Mercedes-Benz. This data transfer to third parties is no longer possible.

It is important to check first whether the data can be used via Mercedes-Benz AG's alternative interfaces/ API (Application Programming Interface) solutions, e.g. Connect Your Fleet.

If data transfer via these interfaces is no longer possible, exceptions may be requested. For this, contact the Technical Consultancy (→ page 18).

Exceptions to this are the use cases in which the PSM serves as the basis for an additional vehicle body and no vehicle data is sent to a third-party back end.

Also excepted are vehicles >3.5 tonnes that use the PSM in order to follow the Fleet Management Standard (FMS).

8.10.1 PSM functions

Read-in of body CAN or HMI2 bus

- Vehicle status
 - Terminal 15
 - Terminal 61
 - secure exterior etc.
- Light status
 - Rotary light switch and steering column switch requests (e.g.: high beams, turn signal indicators, low beams, front fog lamps etc.)
 - Hazard warning light system, UCP
- Window status
 - Windshield wipers and rear window wipers
- Central locking system
 - Doors open/closed, unlocked/locked
- Engine CAN information
 - Wheel speed
 - Speed
 - Engine speed, etc.
 - Cruise control operation
 - Brakes operated, etc.
 - Transmission
 - Clutch information
 - Steering angle, etc.
 - Tachograph information as per FMS standard
- Equipment features
 - Door installation
 - Transmission, etc.

Output of body CAN or HMI2 bus

- Light control
 - Parking lights
 - Standing lights
 - Turn signal indicators
 - High beams
 - Front fog lamps, etc.
- Alarm functions
 - Alarm light functions
 - Hazard warning light system
 - Special signaling system
 - Horn
- Central locking function
 - Lock/unlock front, load compartment, and overall vehicle
- Windshield and rear window
 - Front windshield wipers and rear window wipers
- Various functions
 - Actuating the interior light
 - Triggering a sound signal and visual message in the instrument cluster
 - Loading active (blocking of engine start)
 - CAN gateway e.g. for retarder function
- Warning tones in the instrument cluster
 - PSM defective
 - Undervoltage

8.10.2 PSM interfaces

- I/O module
 - 25 parameterizable inputs
 - 25 parameterizable outputs, of which 2 outputs are for actuation of relay changeover contacts
- Logic functions (mini-PLC)

The mini-PLC (PLC: Programmable Logic Controller) is a module with freely programmable and freely interconnectable function blocks for creating any signal links:

 - 224 AND/NAND/OR/EXOR/NOR/EXNOR
 - 56 RS and D flip-flops
 - 84 retriggerable/non-retriggerable timer stages
 - 28 hysteresis elements with adjustable thresholds
 - 56 threshold switches with 4 stages
 - 28 counters
- Arithmetic unit
 - 40 computing blocks
 - 16 filters
 - 16 comparators
 - 8 characteristics
 - 32 non-volatile memories
 - 40 freely definable constants

8.10.3 Bodybuilder manufacturer CAN

A second CAN bus is available at the PSM.

This bodybuilder manufacturer CAN (ABH CAN or Multi-purpose CAN) contains:

- High-speed CAN Class C
- Extended CAN identifier (29-bit)
 - Baud rate can be changed between 500 kBit/s, 250 kBit/s and 125 kBit/s

! NOTE

When installing a retarder, the baud rate must absolutely remain set at 250 kBit/s. Changing the baud rate is not permissible in this case.

- Signal format: Intel (LSB first)
- All bus content can be activated separately and independently through parameterization:
 - FMS (Fleet Management System), transmission direction only
 - ISO 11992-2 and 3 (in parts)
 - Freely assignable messages (J1939)

i The variety of options of the PSM cannot be fully detailed within the scope of this Body/Equipment Mounting Directive, e.g. regarding interfaces in connection with Mercedes-Benz Advanced Control (MBAC, code JA3) or the Onboard Logic Unit (OLU, code JO2 to JO6).

i Further information can be found in the detailed "PSM Function Description" in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



i Information on parameterization possibilities that are not covered by the "PSM function description" is available from the Technical Consultancy with regard to the PSM (→ page 18).

8.11 Signal acquisition and actuation module SAM

The power circuit on model series 907 comprises the signal acquisition and actuation module (SAM) in connection with a fuse and relay block (SRB). These limit switches supply the systems and control units with power, depending on the function sequence. The requirements are relayed to the SAM either over CAN, LIN or via directly scanned switches and sensors. The fuses on the fuse and relay blocks also provide protection for individual components.

The fuse and relay box acts as a monitor for electromechanical switching units.

With the update to the existing Body/Equipment Mounting Directive to model year 2024, the overview of the functions in the SAM are omitted.

For more information, contact the Technical Consultancy (→ page 18).

8.12 Pre-installation for retarder

The retarder acts as an additional brake on the rear wheels. To prevent the rear wheels from locking up when braking, the ESP® must switch off the retarder if the wheels begin to lock up. For this purpose, the retarder must be connected up to the vehicle network via the PSM. The factory setting for the ESP® code is "Retarder not present". Both ESP® and PSM must be appropriately recoded after installing the retarder.

As a matter of principle, it is recommended to install the retarder type AF30-35 from Telma S.A., which is available ex factory with code PB2 in combination with KW codes (KW: customer request), See 6.5.5 Auxiliary brakes (retarders) (→ Page 176).

Special equipment code BR9 provides a pre-installation for retrofitting a retarder that is designed for this retarder.

! NOTE

It is not permissible to install a different retarder (of either a different type or from a different manufacturer).

i You can find further information on using a retarder in 6.5.5 Auxiliary brakes (retarders) (→ page 176).

The scope of the special equipment includes the wiring in the vehicle interior (CAN connection of the retarder control unit to the vehicle ABH-CAN/PSM) for the fault indication and the connection to the vehicle control units, the high current supply to the contact body (maximum 100 A) as well as the power supply to the retarder control units (maximum 5 A).

The PSM program will be passively coded when the pre-installation for the retarder is installed in the vehicle.

The PSM forwards the signals from the vehicle to the connecting point under the vehicle to enable communication with the retarder control unit. The warning lamp is actuated by the retarder control unit and the vehicle control units.

A terminal 30T connection and a retarder ground connection are provided on the substructure for the retarder power supply.

The power supply of the retarder control unit is provided via a terminal 15 connection on the substructure.

There is a PSM CAN connection on the substructure for the signal connection of the retarder control unit.

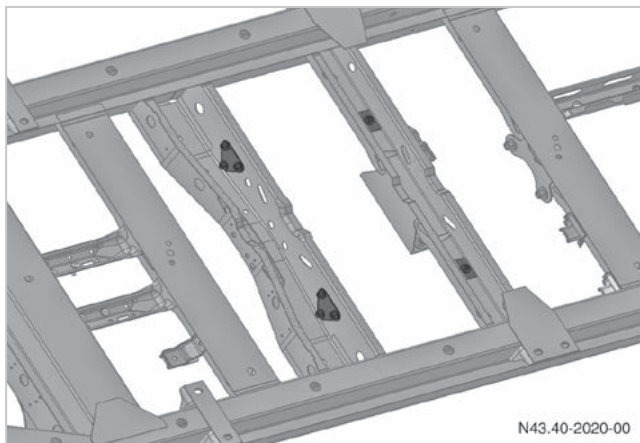
The bodybuilder manufacturer is responsible for any wiring still required under the vehicle (from the vehicle's PSM control unit and the terminal 15 power supply to the retarder) and for the positioning of components. An installation description and the wiring to the retarder control unit must be provided by the retarder manufacturer.

For an optional PSM connection (e.g. CAN), a plug connection can be ordered via code E5M; See 8.4.9 Interface overview (→ Page 281).

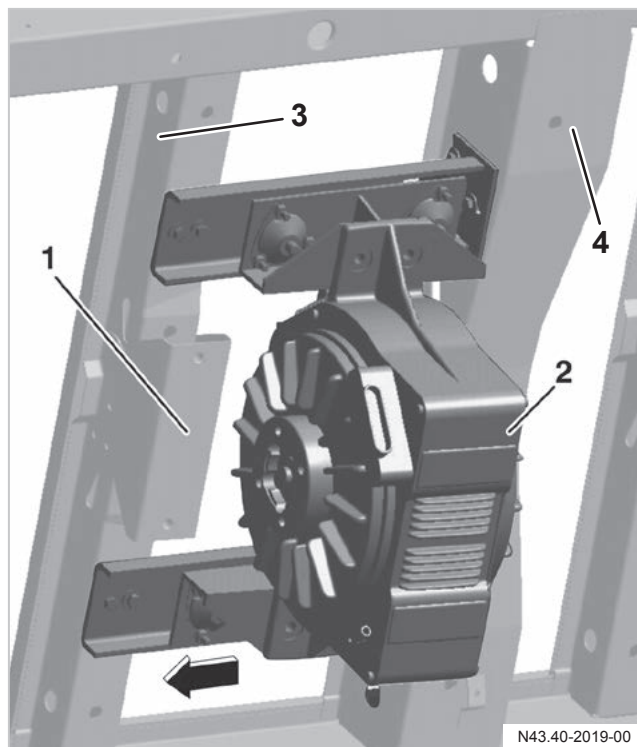
! NOTE

When installing a retarder, the baud rate at the PSM must absolutely remain set at 250 kBit/s. Changing the baud rate is not permissible in this case.

For information on mechanical connection; See 6.5.5 Auxiliary brakes (retarders) (→ Page 176).



Fastening points for pre-installation for retarder



Telma retarder installation

- 1 Suspension of propeller shaft intermediate bearing
- 2 Retarder
- 3 Connection for terminal 15, 30T, CAN
- 4 Ground connection
- ➡ Direction of travel

i More information on coding the PSM and ESP® is available from the Technical Consultancy with regard to the PSM (→ page 18) or from your Mercedes-Benz Service Partner. A "Retrofitting guide for retarder electronics for Sprinter - BR 907" is also available from the Technical Consultancy.

8.13 Pre-installation for liftgate

The special equipment "Liftgate pre-installation" (code EV3) contains the following items, among others:

- Control current pre-installation
- On/off switch in the cab, which closes or opens the control circuit of the liftgate
- Main current pre-installation
- 25 mm² ground line, secured to the vehicle frame, with a blue 1-pin ITT Cannon main current connector on the liftgate end
- 35 mm² positive line, with an 8 mm cable lug on the battery end for connecting in the main fuse box at the auxiliary battery main fuse location (terminal 30a), with a red 1-pin ITT Cannon high current connector on the liftgate end
- Both lines overhang the end of the right longitudinal member by 1000 mm. The free line lengths are tied back in the left longitudinal member.

! NOTE

If an electrohydraulic liftgate is fitted, use an alternator, a battery with higher capacity, and, in principle, an auxiliary battery.

! NOTE

Before the liftgate is used for the first time, the body-builder manufacturer must insert an electrical fuse in the appropriate location in the driver's seat base.

For the mechanical connection, see the section "Attachments on the rear frame section" (→ page 117) and 6.6.6 Liftgate (cargo liftgate) (→ page 190).

8.14 Continuous engine operation feature

The continuous engine operation feature (code MW1) is available ex factory for special-purpose vehicles, e.g. police cars, ambulances.

It is not possible to retrofit the engine run-on in the model series 907.

Ordering of the continuous engine operation feature is restricted. If you have any further questions, contact the Technical Consultancy (→ page 18).

⚠ WARNING

Retrofitting a continuous engine operation feature can cause critical vehicle issues, problems with the on-board electronics or fault messages.

There is danger to life and limb!

Retrofitting continuous engine operation feature is not permitted.

8.15 Automatic start/stop

The purpose of the automatic start/stop system (code MJ8) is to reduce the emission values. If the automatic start/stop system is used, an enhanced on-board electrical system is used for stabilizing the on-board electrical system voltage. It includes an additional voltage dip limiter and a modified fuse box.

If the system is permanently deactivated, the emission values in the General Operating Permit must be adapted to the values of the vehicle without automatic start/stop (code MJ8).

To avoid dropping below the defined charge capacity of the battery, the system is automatically deactivated if the power consumption exceeds 140 A or if the charge state of the battery is low.

Power take-offs

It is possible to use the power take-offs (See 6.5.3 Power take-offs (→ Page 169)) in combination with the automatic start/stop system. No automatic system deactivation takes place in this case. A system deactivation that is temporary and that only takes place as and when required is possible with PSM parameterization (setting the "blue light bit") in special cases.

 For more information, contact the Technical Consultancy (→ page 18).

Chassis

If additional possibilities for entering and exiting the vehicle are created for using the chassis, they must be integrated in the door monitoring system in the SAM. This prevents the vehicle from starting inadvertently when the driver exits the vehicle through the new doors.

Information on the integration of additional body-manufacturer-specific doors in the door monitoring can be found in 8.7.2 Central locking system and post-delivery integration of doors of bodybuilder manufacturer (→ page 301).

 For more information, contact the Technical Consultancy (→ page 18).

8.16 Electrical circuit diagrams

Electrical circuit diagrams can be found in the Workshop Information System (→ page 26).

You can also download the electrical circuit diagrams after logging into the Bodybuilder Portal:

- ▶ Select "Vans" on the home screen (dashboard),
- ▶ Click on the "Engineering and Information" tile, then select the vehicle image of the required model series,
- ▶ Select "⊕ Retrieve Technology".
- ▶ In "Technical Data", click on the "Electrical Circuit Diagrams" button and select the required data.
- ▶ After making your selection in the selection menu, the electrical circuit diagrams are available to you in "Circuit Diagrams".

8.17 Telematics and Infotainment

Abbreviations in the field of telematics:

Abbrevia- tion	Meaning/description
HU	Head unit
NTG7	Multimedia system with 10.25" display
DVI	Digital Visualization Interface
CAN	Controller Area Network
CAR PC	Control unit or PC; installed in the vehicle
TMDS	Transition-Minimized Differential Signaling
RGB signal	Red-Green-Blue signal
PSM	Parameterizable Special Module
DAB	Digital Audio Broadcasting
LTE	Long Term Evolution
AM/FM	Amplitude Modulation/Frequency Modulation
SDARS	Satellite Digital Audio Radio Services
HDMI	High-Definition Multimedia Interface
RHC	Right hand circulation polarization
MBUX	Mercedes-Benz User Experience (multimedia system based on NTG7)
AUB	Antenna switchover box
GPS	Global Positioning System
GNSS	Global Navigation Satellite System

8.17.1 Infotainment systems

The new MBUX multimedia system based on NTG7 is integrated in the Sprinter as of model year 2024. It provides the comfort known from model series in our Cars segment, offering various entertainment and user experience applications.

The following infotainment systems are available ex factory:

Infotainment system		Markets		Series		Extended equipment				
Code		Within CoC	Outside of CoC	Voice control "Hey, Mercedes"	Telephony, Bluetooth®	Smart-phone integration (Code E4S)	Naviga-tion (Code E1E)	Reversing camera (code FR8 ²⁾ /FR7), Parking Package RFK (code JB7), Parking Package 360° (code JB6/JB1-3)	Intelligent Speed Assist (code JS2) ³⁾	Interface to MBAC (code JA3)
E1O	MB audio system	–	SA	–	X	–	–	–	–	–
E7M ¹⁾	MBUX multi-media system	Series	SA	X	X	X	X	X	X	–
E7H ¹⁾	MBUX multi-media system HIGH	SA	SA	X	X	X	X	X	X	X

1) With 10.25" touchscreen

2) Standard equipment in CoC markets for closed model designations and market-dependent in country series

3) Standard equipment in CoC markets, outside of CoC: Traffic Sign Assist (code JA9) available depending on country

A radio pre-installation (codes ER0/E7A) is available as of model year 2024 outside of the CoC markets as standard equipment.

A pre-installation for Navigation (code E7B) is added for vehicles in markets with CoC registration and Mercedes me services (code JH3) as standard, so that Navigation (code E1E) can be subsequently ordered from the Mercedes me Store.

i You can find further information about the infotainment systems, additional special equipment for telematics and the equipment packages from the sales documents and your Mercedes-Benz AG Sales Partner.

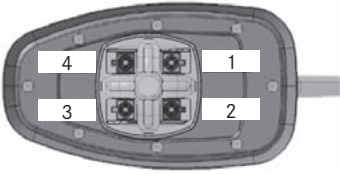
i The functions and the operation of the various items of equipment are described in the Operator's Manual for your vehicle.

i Observe the information and specifications on operating multimedia and vehicle systems in 7.15.6 Features for camper vans (→ page 255) and 8.3.1 On-board electrical system battery (→ page 267).

i For general questions regarding telematics and Infotainment, contact the Technical Consultancy (→ page 18).

8.17.2 Antennas

Antenna variants with equipment ex factory

Equipment, function	Antenna type (item number)	Explanation
		
Radio pre-installation Entry-level radio	Antenna for radio (A 907 905 17 02)	1 AM/FM1: FAKRA code A 2 FM2: FAKRA code E 3 Not provided 4 Not provided
Navigation	Antenna for radio TEL GNSS (A 907 905 18 02)	1 TEL: FAKRA code D 2 FM2: FAKRA code E 3 GNSS: FAKRA code C 4 AM/FM1: FAKRA code A
DAB	Antenna for radio TEL GNSS DAB (A 907 905 19 02)	1 TEL: FAKRA code D 2 FM2/DAB2: FAKRA code E 3 GNSS: FAKRA code C 4 AM/FM1/DAB1: FAKRA code B

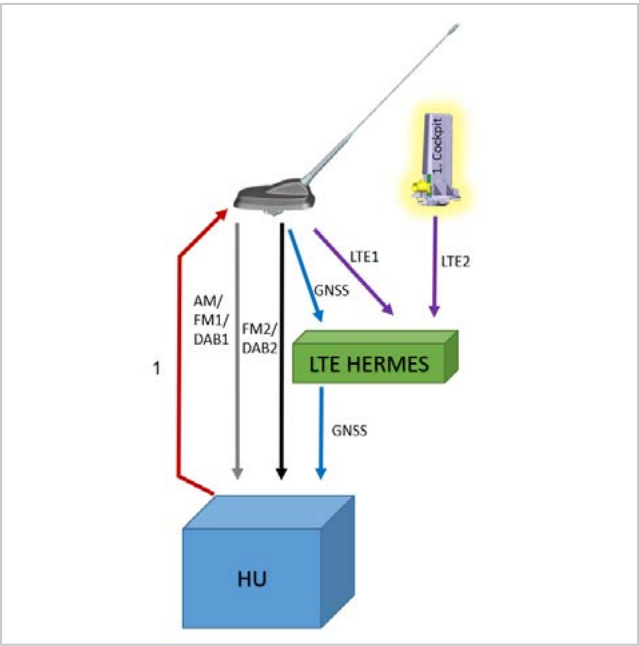
Antennas for vehicles with and without cab roof ex factory

For vehicles without a cab roof (cab base vehicles code F28 or F50), a pre-installation for antennas is available as special equipment (code E4A) instead of the ex-factory antenna, See 7.14.10 Connecting point for antenna switchover box for third-party antennas (code E4A) (→ Page 241).

This special equipment contains an antenna switchover box in the driver's seat box as an interface for an accessory antenna used by the bodybuilder manufacturer and the associated wiring harness for the power supply.

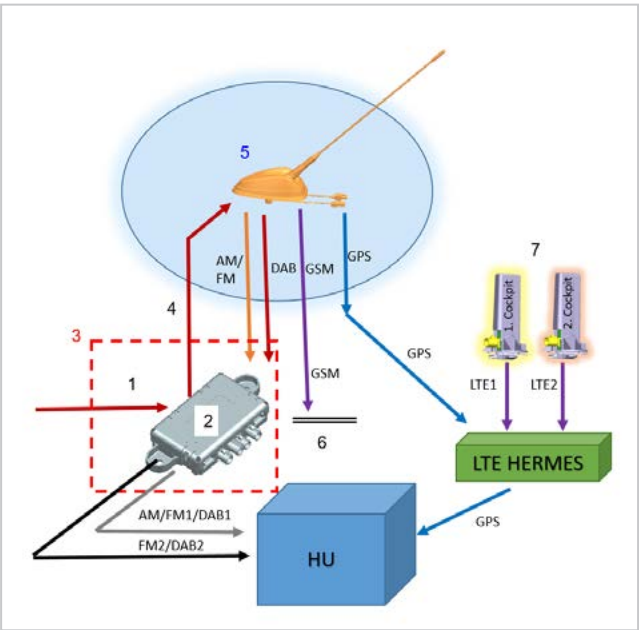
The antenna switchover box serves solely for the modulation of signals of an accessory antenna for the further signal processing in the Mercedes-Benz head unit. A connection of the antenna installed ex factory and available via Mercedes-Benz Service is not permissible with the antenna switchover box.

Observe the following illustrations of the connection schematics. Also observe the installation instructions and specifications of the accessory antenna manufacturer.



Connection scheme: Vehicles with antenna ex factory

- 1 Power supply (external feed) via head unit (HU)



Connection scheme: Vehicles with pre-installation for antenna (code E4A)

- 1 Power supply 12 V (terminal 30 from wiring harness)
2 Antenna switchover box (AUB)
3 Antenna bodybuilder manufacturer interface (code E4A)
4 Power supply 12 V via AUB
5 Accessory antenna
6 GSM interface in radio layer
7 Two LTE antennas in cockpit (second antenna always with code F28/F50)

Requirement for antenna switch box to bodybuilder-manufacturer-specific antenna

Supply the antenna switchover box with the following signals:

- FM (87–108 MHz)
- AM (531–1720 kHz and 5.9–26.1 MHz)
- DAB band III (174–240 MHz)
- Impedance 50 Ω

Global Navigation Satellite System (GNSS)

Supply the connection of the navigation system with the following signal:

Antenna	
Antenna type	Patch
Frequency	1575–1606 MHz
Gain in antenna	2 dBic, mean value Theta 0–20°
Polarization	RHC

Amplifier	
Amplification	28 dB core
Noise factor	≤ 1 dB (without filter)
Impedance	50 Ω

Power supply	
On-board electrical system voltage	4.5–5.5 V
Power consumption	20 mA ± 3 mA
External feed	Yes

Satellite radio for USA

For SDARS (Satellite Digital Audio Radio Services), use the antenna A 213 905 51 05 and the antenna dome A 238 827 01 00.

To procure these components, contact your Mercedes-Benz Service Partner. Refer to the current SiriusXM guidelines for design specifications, which you can request from this antenna manufacturer.

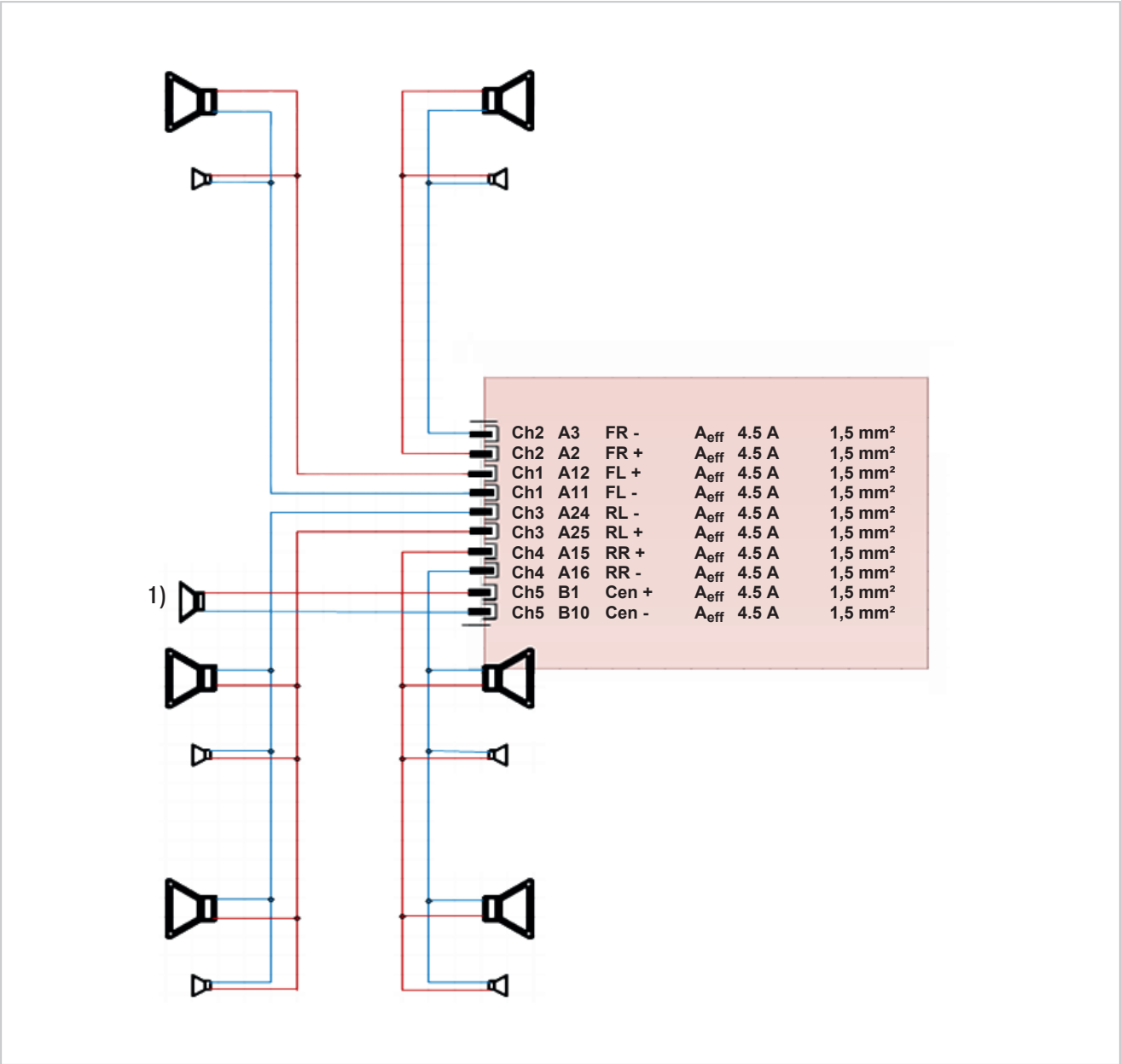
8.17.3 Speakers

NTG7 has 5 channels for connecting speakers to the system.

Two pairs of speakers (woofer/mid-range speaker and tweeter) are installed as standard in the doors at the front left (FL) and front right (FR). A mid-range speaker (Cen) ensures a better sound experience.

Additional speaker pairs are available for the panel van and crewbus model designations as special equipment in the rear passenger compartment (Code EL9). A fade-in/fade-out of the audio sources at the rear left (RL) and rear right (RR) is thus facilitated.

The total impedance per channel must not be less than 2 Ω.



Connections for speakers

1 Only with NTG7

8.18 Connectivity



Mercedes-Benz offers Mercedes me digital products that enhance the vehicle experience and make the customer's everyday life easier. With Mercedes me, customers stay connected to their vehicle – important information and practical functions concerning the Mercedes-Benz vehicle are available at any given time. This enables the driver to keep an eye on their vehicle, even when they are not near it. They can also use their smartphone to operate vehicle functions and modify their personal preferences locally or remotely.

The basic requirement for using this connectivity solution is the "Communication module for digital services" equipment in the vehicle (code JH3).

This communication module is installed as standard on a country-specific basis or it can be ordered as special equipment.

Mercedes me connect or individual Mercedes me connect services are not available in every country. Ask your Mercedes-Benz Service Partner whether these functions are available in your country.

- i Observe the terms of use for Mercedes me connect and other services.

They are available along with further information in the Mercedes me Portal:

<https://me.secure.mercedes-benz.com>



- i Information on operating this equipment is available in your vehicle's operator's manual.

8.19 Commissioning of control units

For all open model designations (FHS/FHL) without platform ex factory (code P02) with incomplete CoC documents (code XC1), commissioning of the control units is always required after complete assembly.

For this purpose, a commissioning process was introduced in XENTRY, which is documented with the code X96 "Mandatory commissioning by bodybuilder manufacturer" for all vehicles mentioned above (see bodybuilder manufacturer information 25/2023 and 06/2024). With this, upon delivery of the vehicle, the warning message "Commissioning by bodybuilder manufacturer incomplete" appear in the instrument cluster, which disappears again once the commissioning has been properly and fully implemented.

You can find the latest edition of the guide for the commissioning of the control units ("Commissioning Guide") with all relevant information and a description of the procedure from the Bodybuilder Portal:

- ▶ Select "Vans" on the home screen (dashboard),
- ▶ Click on the "Engineering and Information" button, and then the vehicle image of the required model series,
- ▶ Under "Technical data", click on the tile "Electrics and electronics",
- ▶ Select the guideline for the "Commissioning of control units" and select the desired language.

As the final build status, such as the wheelbase length or also the body width, is not yet known at the point in time of delivery, not all control units can be accordingly configured in the plant, and must instead be adjusted to the actual vehicle status after body or modification work carried out by the bodybuilder manufacturer.

Extensive previous knowledge and preparations are required so that the correct control unit configuration and flawless function of the individual E/E systems are ensured.

This includes checking the vehicle properties, including all equipment codes and vehicle dimensions, as well as the comprehensive identification of all control units that need to be adjusted due to the vehicle build status.

If SCN coding is performed for a control unit, the diagnostic system checks all recorded vehicle characteristics (codes) in the vehicle data card and, based on the codes, sets the recorded parameters from the Mercedes-Benz development system.

If the vehicle build status is changed or expanded by the bodybuilder manufacturer, the modification to the vehicle will need to be recorded via a code by amending the vehicle data card in the first step of control unit commissioning.

This is the only way in which the diagnostic system can determine the correct parameters and carry out the required SCN coding of the control unit properly.

- ❗ It is recommended that the completion of the vehicle is carried out by Mercedes-Benz Service. Alternatively, the bodybuilder manufacturer can implement the completion of the vehicle themselves. For this, observe the following information.

! NOTE

The bodybuilder manufacturer is responsible for ensuring that the O-codes documented in the vehicle data card that are necessary for the commissioning also correspond to the actual body status of the vehicle.

If commissioning is carried out by Mercedes-Benz Service, the bodybuilder manufacturer must first report the body status in the feedback screen in the Bodybuilder Portal.

The affected control units must then be selected using the commissioning guide and requested from Mercedes-Benz Service.

! NOTE

An essential prerequisite for independent commissioning of the control units is successful participation in the corresponding training at Mercedes-Benz and acquisition of the Mercedes-Benz diagnostic tool XENTRY Kit.

In addition, the bodybuilder manufacturer commits to comply with and fully implement the Mercedes-Benz specifications.

The Mercedes-Benz training portal can be accessed here:

<https://learningsystem.mercedes-benz.com>



i If you have any questions or require additional information, create a request via an XSF ticket in the XENTRY system.

! NOTE

An essential prerequisite for independent commissioning of the control units is successful participation in the corresponding training at Mercedes-Benz and acquisition of the Mercedes-Benz diagnostic tool XENTRY Kit.

In addition, the bodybuilder manufacturer commits to comply with and fully implement the Mercedes-Benz specifications.

! NOTE

After commissioning and before the vehicle is placed on the market, it must always be ensured that all existing vehicle and driving assistance systems are functioning flawlessly. For this, contact a technical service that has been designated by the responsible registration authority and that possesses the necessary knowledge and testing equipment.

Depending on the vehicle model and vehicle equipment, different control units can be affected by the commissioning:

 More information can be found in the current edition of the guide "Commissioning of control units".

System	Control unit designation in XENTRY	Commissioning scope (example)
Electronic Stability Program	N30/4 - ESP	Calibrate sensor for longitudinal and lateral acceleration. SCN coding for changes to the wheelbase or rear axle track width or when retrofitting a trailer coupling with code E57
Electronic ignition lock	N73/8 - EZS	SCN coding when retrofitting a trailer coupling with code E57
Drivetrain control unit	N127 - CPC	SCN coding when retrofitting a trailer coupling with code E57
Brake Assist and Active Distance Assist	B92/1 - DTR	SCN coding for change of body width and overhang with code BA3 or ET4
Blind Spot Assist	B92/21 and B92/24 - IRS-HRA/-HLA	SCN coding when changing the body width and the overhang with code J1V (additional coding of door control unit required for exit warning system; see table line "Door control unit")
Rain/light sensor	B38/2 - RGLS	SCN coding for code F50
Multifunction camera	B84/13 - MFK	SCN coding and initial calibration for code JW5, JB4, JA9, or LA1
Reversing camera	B84/3 - RFK	SCN coding for installation of a reversing camera with code FR7
Signal acquisition and actuation module	N10 - SAM	Carry out SCN coding in general.
Headlamp control unit	E1/2N9 and E2/2N9 - SG-LEDSW-L/-R	Zero-position calibration for Mercedes-Benz LED headlamps with code LG7
Tire pressure monitoring system	N88/2 - RDK	When installing tire pressure monitoring system code RY2, the tire pressure sensors must be activated.
MBUX multimedia system	A26/17 - Head unit	SCN coding when installing a reversing camera with code FR7 or trailer coupling with code E57
Airbag control unit	N5/2 - SRS	SCN coding must always be carried out for code F50. Always check the ACTUAL values for bodybuilder-manufacturer-specific seats with code F28.
Telematics services communication module	N112/9 - HERMES	Where a bodybuilder-manufacturer-specific GPS antenna (GPS reception and activation) is installed
Trailer coupling control unit	N28/1 - Trailer recognition	SCN coding when installing a trailer coupling with code E57
Door control unit	N69/1 and N69/2 - TSG-VL/-VR	SCN coding for code F50 and J1V

8.20 Tachograph

The following special equipment is available ex factory for compliance with legal obligations on the use of tachographs (or other control devices) as per Regulation (EU) 165/2014 or other country-specific laws and regulations, as well as for the automatic recording of road tolls:

Code	Equipment
JD6	EU intelligent tachograph
JV6	Pre-installation for tachograph
J4V	Pre-installation for EU intelligent tachograph
J92	Omission of tachograph
EQ3	Pre-installation for toll system

When ordering a pre-installation for tachographs (codes JV6 or J4V), the following special equipment is recommended for installation of a tachograph in the vehicle:

Code	Equipment
FF8	1 device installation frame at front under headliner
H1B	Operation of tachograph at front under headliner
FE9	1 device installation frame under cockpit on front passenger side
J2B	Operation of tachograph under cockpit on front passenger side

 You can find more information on this special equipment in the ADVANTAGES product information system (→ page 29) or from your Mercedes-Benz AG Sales Partner.

The bodybuilder manufacturer is responsible for selecting special equipment for the tachograph that is suitable for the specific body or modification.

In terms of possible installation positions, a 1-device installation frame at the front under the headliner (codes FF8 or H1B) or on the front passenger side under the cockpit (codes FE9 or J2B) are available as special equipment.

It is recommended that the pre-installations for the tachographs (codes JV6 or J4V) and for the toll system (code EQ3) are installed at a qualified Mercedes-Benz AG specialist workshop. Technical information on retrofitting can be found in the Workshop Information System (→ page 26).

Observe Implementing Order (EU) 2016/799 or other country-specific laws and regulations (e.g. as per Section 57b of the StVZO (German vehicle licensing regulations) for vehicle registration in Germany) when commissioning and calibrating the tachographs.

! NOTE

After completion of body mounting or modification work, the bodybuilder manufacturer is responsible for checking whether there is a need to calibrate or re-commission and for conducting this again depending on the result.

All body work that may lead to one of the following effects is relevant:

- Change to the distance pulse count
- Change to the effective tire circumference

Body work such as changes to the axle loads of the curb weight may be responsible for effects like these.

For cab base vehicles (codes F50 and F28), for this reason, no initial calibration of the tachographs is performed at the factory.

The proper commissioning and calibration of the tachograph is fundamentally the responsibility of the bodybuilder manufacturer. This must be performed after completion of the vehicle in a specialist workshop with officially recognized approval for the checking and calibration of tachographs, provided that the bodybuilder manufacturer themselves does not possess such an approval.

9.1 Vehicle center of gravity

General

The position of the vehicle center of gravity in the longitudinal direction of the vehicle (x-direction) is specified relative to the center of the front axle. The center of gravity height (z-direction) is specified with reference to the roadway.

It is recommended that you have the position of the center of gravity checked by a recognized and experienced testing institution (e.g. DEKRA, Technical Inspection Association (TÜV), or another organization). For support with regard to this, contact the Technical Consultancy (→ page 18).

If the vehicle center of gravity is determined by the bodybuilder manufacturer itself, the procedures in accordance with 9.1.1 Determination of the center of gravity in the x-direction (→ page 391) and 9.1.2 Determination of the vehicle center of gravity in the z-direction (→ page 393) must be followed, and appropriately qualified personnel must be used to ensure that realistic and useful results are achieved.

Measure the vehicle in a load condition appropriate for its intended purpose (see the following pages for an example).

! NOTE

Practical determination of the vehicle center of gravity position in the x-direction and z-direction must only be carried out by appropriately qualified staff using suitable and calibrated vehicle scales.

In order to reduce measuring errors, each measurement value should be determined at least three times and the average should be calculated from these three values.

! NOTE

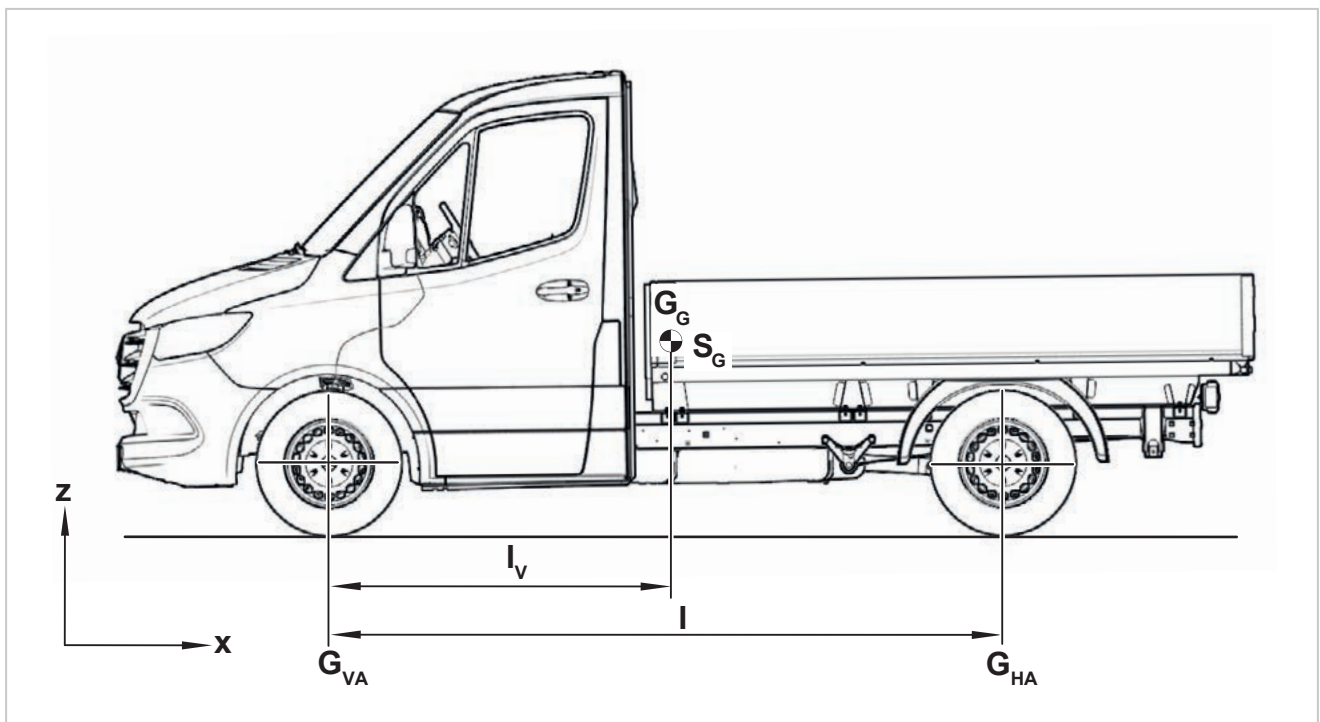
The calculated vehicle center of gravity position in the z-direction (CoG height) must not exceed the limit values specified under 4.1.2 Maximum permissible position of the vehicle center of gravity (→ page 78).

9.1.1 Determination of the center of gravity in the x-direction

Center of gravity coordinates in x-direction (front/rear axle load distribution)

Recommended procedure:

- ▶ In the first series of measurements, weigh the vehicle with the complete attachment or body and without payload.
- ▶ In a second series of measurements, weigh the vehicle with the complete attachment or body and with a payload appropriate for its intended purpose, taking into account the permissible gross mass and the permissible axle loads.
- ▶ For the measurements, set the tire pressure in all tires to 6 bar.
- ▶ Completely fill all liquid reservoirs for operating fluids (fuel tank, AdBlue® tank, washer fluid reservoirs and, if installed, hydraulic oil reservoir).
- ▶ Shut off the engine on the scales, shift the transmission to neutral position, and release the brakes.
- ▶ The vehicle must be parked horizontally for weighing.
- ▶ First weigh the individual axle loads (front and rear axle loads) and then the gross vehicle mass.
- ▶ Using these measurements, calculate the position of the center of gravity in the longitudinal direction of the vehicle using equations (3) and (4).
- ▶ Use equation (2) to check the results from (3) and (4).



Coordinates and dimensions

Calculate axle load

$$G_G = G_{HA} + G_{VA} \quad (1)$$

$$l = l_V + l_H \quad (2)$$

Calculate position of vehicle center of gravity in x-direction

$$l_V = \frac{G_{HA} \times l}{G_G} \quad (3)$$

$$l_H = \frac{G_{VA} \times l}{G_G} \quad (4)$$

Weights

- G_G Gross combination mass of vehicle
- G_{VA} Front axle load of vehicle
(specification or weighing of the basic vehicle)
- G_{HA} Rear axle load of the empty vehicle
(specification or weighing of the basic vehicle)

Dimensions

- l_V Distance of overall center of mass of vehicle to front axle
- l_H Distance of overall center of mass of vehicle to rear axle
- l Wheelbase
- S_G Overall vehicle center of gravity

i Wheelbase "l" is defined by the vehicle model designation (see order) or is determined by means of a length measurement according to DIN 70020, Part 1.

9.1.2 Determination of the vehicle center of gravity in the z-direction

Center of gravity coordinates in z-direction (height of center of gravity h_s for the overall vehicle)

For determination of the overall center of gravity height of the vehicle above the roadway h_s by the bodybuilder manufacturer, it is recommended to carry out the following procedure after completion of the vehicle:

Weigh the vehicle on a plate-type scale or suitable wheel load scale in two different drive positions in succession. For this, determine the axle loads measured when the vehicle stands horizontal and level (G_{VA} and G_{HA} ; See 9.1.1 Determination of the center of gravity in the x-direction (→ Page 391)) and the axle loads for an axle raised by h' (Q_{HA} or Q_{VA}). The lift height h' should be as large as possible in accordance with the front and rear overhang angle of the vehicle (also known as the angle of approach/departure). The target value is $h' > 1100$ mm (wheelbase 3665 mm) and 1400 mm (wheelbase 4325 mm).

In order to reduce measuring errors, perform at least six individual measurements for each vehicle axle when measuring the axle load: Three per axle when the vehicle is in a level state and three when an axle is raised. Based on the three measurements in each condition, calculate an average for each axle.

Use the equations (5) to (7) for this. Important: The measurement results must be close to each other. If individual measurements vary widely from each other, the deviating individual measurement must be repeated. Then calculate the total value between the raised front and rear axles.

In order to improve the accuracy of the final result, determine the axle load change both with a raised rear axle and with a raised front axle.

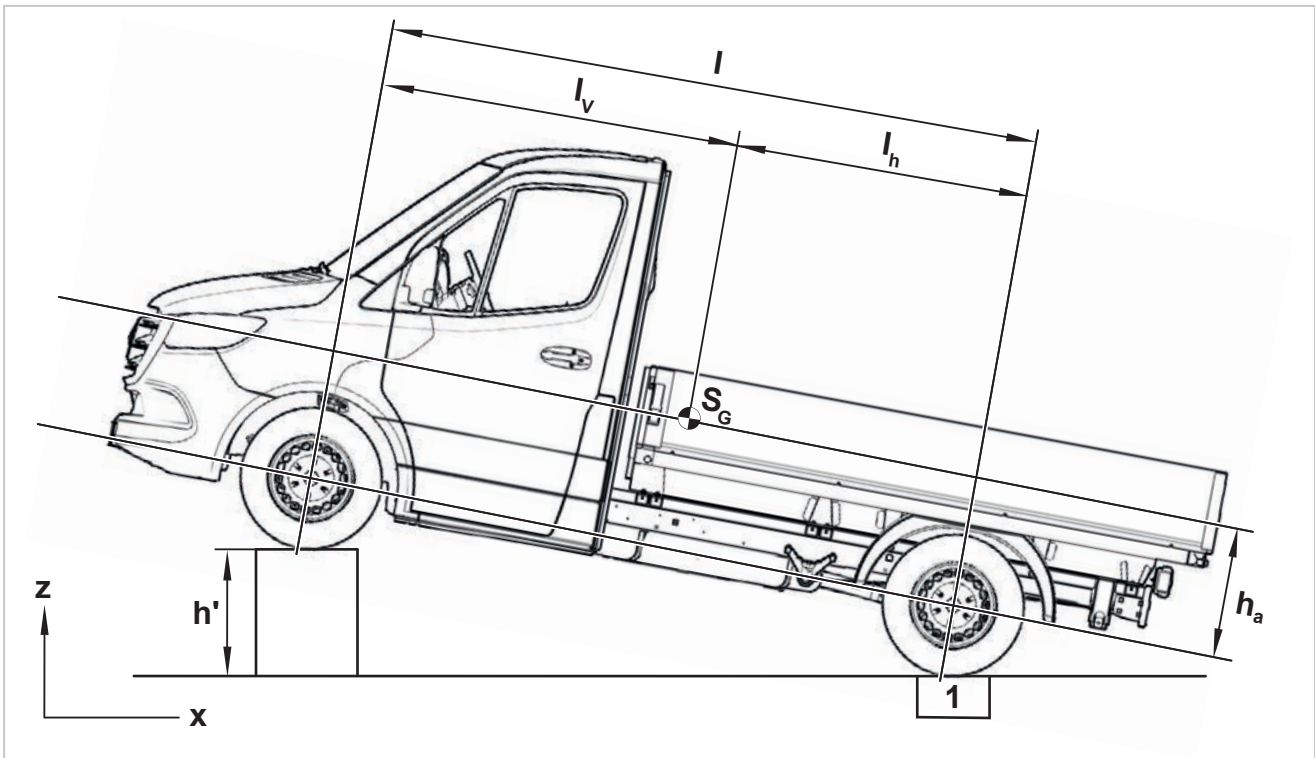
! NOTE

In order to avoid faulty measurements, please note:

- ▶ When weighing, the vehicle must be exactly horizontal. Compensate for any height differences between the axles caused by the scales accordingly.
- ▶ When raising to the required lift height, both axles must be blocked to prevent suspension jounce and rebound.
- ▶ When raising, the left and right sides must be raised in parallel.
- ▶ No part of the vehicle may bottom out when raising to the required lift height.
- ▶ All vehicle wheels must be able to roll: Gearshift system in neutral position, all brakes including parking brake released, chocks placed at a sufficient distance from the wheels if necessary.
- ▶ Set all tires to a tire pressure of 6 bar.
- ▶ Between the individual measurements and to turn it around (in order to weigh the other axle), move the vehicle under its own engine power so that any stresses in the vehicle/chassis are relaxed.
- ▶ Make sure that no objects inside the vehicle can move during the measurements.
- ▶ It must be ensured that all fluids (operating fluids and other media specified/required for the intended purpose, e.g. hydraulic oil, fresh water, etc.) and the gear and equipment required for operation as well as the necessary counterbal-lasts are all topped up and present.

Recommended procedure:

- ▶ In the first series of measurements, weigh the vehicle with the complete attachment or body and axle blocking without payload.
 - ▶ In the second series of measurements, weigh the vehicle with the complete attachment or body, axle blocking and with a payload appropriate for its intended purpose, taking into account the permissible gross mass and the permissible axle loads.
 - ▶ Inflate the tires up to a tire pressure of 6 bar.
 - ▶ Completely fill all liquid reservoirs for operating fluids (fuel tank, washer fluid reservoirs, hydraulic fluid reservoir, coolant reservoir, AdBlue® tank etc.) and load or attach all the gear and equipment required for operation as well as the necessary counterballasts.
 - ▶ Shut off the engine on the scales, shift the transmission to neutral position, and release the brakes.
 - ▶ Position the vehicle with the rear axle (HA) at a horizontal and level position on the scales and determine the axle load.
 - ▶ Raise the front axle (VA) by the lift height h' . Increasing the height h' while taking the other boundary conditions of the vehicle into account improves the final result. The value h' must be determined for all individual measurements with raised axle and should be identical wherever possible. As an alternative to measuring the raised height h' , the angle α between the wheel hubs can be determined.
 - ▶ Determine the resulting axle load shift Q_{HA} at the rear axle (HA) on the scales.
 - ▶ Lower the vehicle and perform lifting measurements 2 and 3. Between all the individual measurements and to turn it around (in order to weigh the other axle), move the vehicle under its own engine power so that any stresses in the vehicle/chassis are relaxed.
 - ▶ After the 3rd measurement, lower the vehicle, turn it around, and carry out corresponding measurements on the front axle (initially G_{FA} with the rear axle level, and then Q_{FA} with the rear axle raised by height h').
- ▶ Using the calculated values, calculate the height of the vehicle center of gravity using equations (5) to (7).
 - ▶ When calculating using the equations (3) to (9), use all length measurements in millimeters (mm) and all weights in decanewtons (1 daN = 10 N). $G = 1 \text{ daN} = 10 \text{ N}$ is the weight force corresponding to the mass $m = 1 \text{ kg}$.



Coordinates and dimensions

Determine height of vehicle center of gravity

$$h_S = h_a + r_{stat} \quad (5)$$

For raised front axle:

$$h_S = \left(\frac{l}{h'} \times \frac{Q_{HA} - G_{HA}}{G_G} \times \sqrt{l^2 - h'^2} \right) + r_{stat} \quad (6)$$

For raised rear axle:

$$h_S = \left(\frac{l}{h'} \times \frac{Q_{VA} - G_{VA}}{G_G} \times \sqrt{l^2 - h'^2} \right) + r_{stat} \quad (7)$$

r_{stat}	Loaded tire radius
Q_{VA}	Front axle load with vehicle raised at rear
Q_{HA}	Rear axle load with vehicle raised at front
h_S	Height of vehicle center of gravity above roadway
h_a	Height of vehicle center of gravity above wheel center
h'	Height by which the vehicle was raised
S_G	Overall vehicle center of gravity
1	Weighing device

i Wheelbase "l" is defined by the vehicle model designation (see order) or is determined by means of a length measurement according to DIN 70020, Part 1.

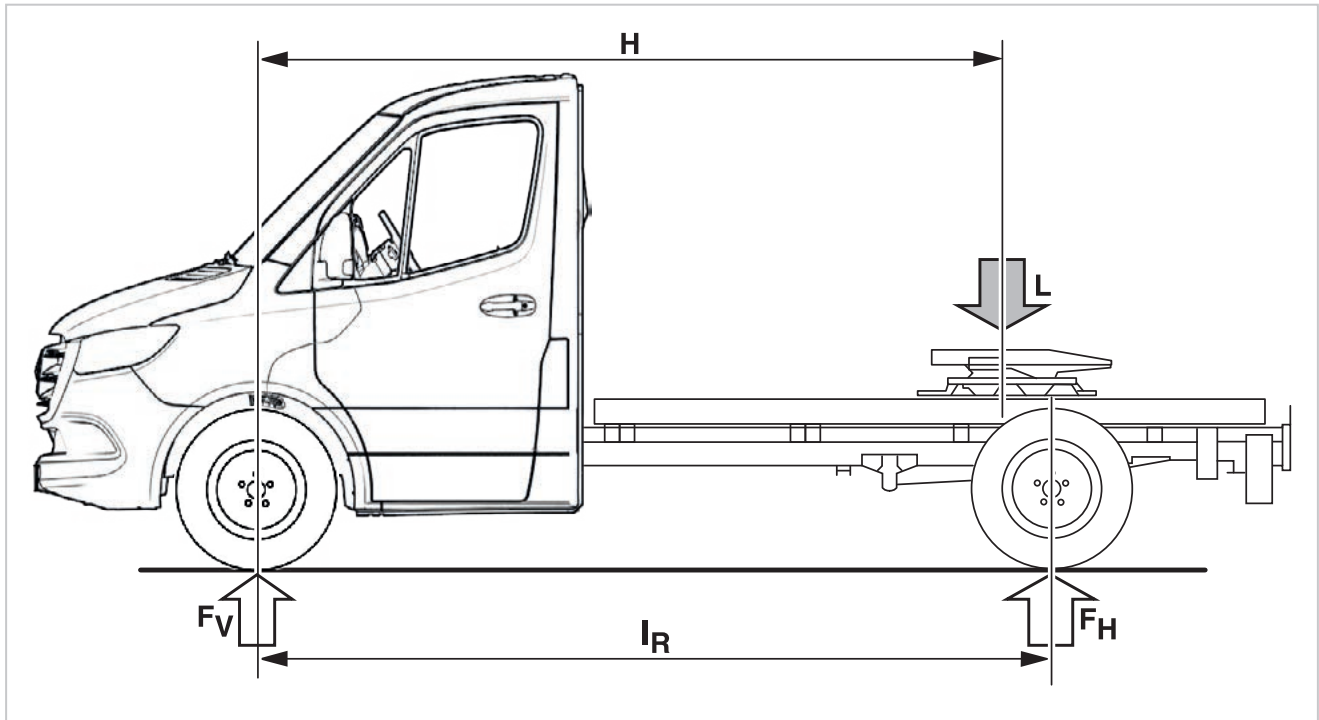
Examples of loads appropriate for the intended purpose:

- Camper vans
 - All reservoirs for operating fluids 100% full, including fuel tanks
 - All registered seats loaded with 80 kg
 - Fresh water tank empty if located in the underbody, full if located above the vehicle floor (depending on the layout of the vehicle)
 - Waste water container empty
 - All cabinets, storage boxes and the rear hold with typical camper van loads, if appropriate up to the maximum possible load capacity of the stowage compartments
 - Including all planned and available special equipment (e.g. awning, roof-mounted air conditioning system, bicycle rack, SAT system, solar system, pop-up roof, etc.)
 - Unavailable special equipment is to be simulated by appropriate dummy weights at the installation location.
- Box bodies
 - All reservoirs for operating fluids 100% full, including fuel tanks
 - 1 × 80 kg driver
 - The box body must be loaded appropriately for the intended purpose in terms of weight and load height.
 - Including the refrigeration unit, if planned for the body
- Touring coaches
 - All reservoirs for operating fluids 100% full, including fuel tanks
 - 1 × 80 kg driver
 - All registered seats and standing room occupied with 80 kg
 - The luggage compartment must be loaded appropriately for the intended purpose in terms of weight and load height.
 - Stowage areas above the passengers filled
 - For roof-mounted air conditioning systems, simulate the weight and installation location

9.2 Location of fifth-wheel coupling

Calculating the position of the fifth-wheel coupling

- i** For information on modifications to the semitrailer truck, See 7.10 Semitrailer trucks (→ Page 218).



Coordinates and dimensions

Fifth-wheel coupling D-value

$$D = \frac{0,6 \times 9,81 \times Z \times A}{Z + A - L} \quad (8)$$

To avoid exceeding the maximum axle loads, calculate the position of the fifth-wheel coupling as follows:

$$H = \frac{F_{H^*} \times I_R}{L} \quad (9)$$

$$F_{H^*} = F_H - F_{HL} \quad (10)$$

$$L = Z + A - \frac{0,6 \times 9,81 \times Z \times A}{D} \quad (11)$$

- A Permissible gross mass of semitrailer
- D Drawbar ratio of coupling
- F_H Maximum permissible rear axle load
- F_{HL} Rear axle load of unladen vehicle
- F_{H^*} Resulting max. rear axle load
- H Distance between front axle and fifth-wheel coupling
- I_R Wheelbase
- L Max. fifth wheel load
- Z Permissible gross mass of towing vehicle

Comply with the permissible axle loads at the front and rear axles.

10.1 Bulb ratings of tail lamps

The following tables show the different variants of outdoor lights in terms of lamp type and performance with the differences between closed model designations (gBM/FKA, FKB) and open model designations (oBM/FHS, FHL).

Observe these specifications when installing lighting systems; See 8.5 Lighting (→ Page 286).

Scope of special equipment "Deactivation of lamp failure indicator" (code JW2) – explanation of entries in the following tables:

- "Yes" means that, in the event of a malfunction of the lamp, e.g. due to bulb failure, there is no message in the instrument cluster and no entry in the fault memory of the diagnostic interface.
- "No" means that this function is not available and therefore there is still a message in the instrument cluster and an entry in the fault memory of the diagnostic interface.

Illuminants of the conventional exterior lighting are actuated via pulse width modulation. An effective value is always set. The frequency varies according to country, and is 100 Hz for all countries except for NAFTA (USA/Canada), where it is 120 Hz.

 Information on plugs and mating connectors on the basic vehicle is available from the Technical Consultancy (→ page 18) or in the Workshop Information System (→ page 26).

10.1.1 Conventional headlamps

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
High beams	H15	2 × 55	Operated via control unit BCMFA2	Yes
Low beams	H7	2 × 55	Operated via control unit BCMFA2	No
Daytime running lights	H15	2 × 21	Operated via control unit BCMFA2	Yes
Standing lights	W5 W	2 × 5	Operated via control unit BCMFA2	Yes
Turn signal indicators	PY21 W	2 × 21	Operated via control unit BCMFA2	No

In cab base vehicles (code F50, cowl), the omission of headlamps (code L92) is added automatically as standard.

Code L92 is not available for other vehicles.

When conventional headlamps are retrofitted by the bodybuilder manufacturer, adhere to the power values specified in the table.

10.1.2 Pre-installation for LED headlamps with code L93

Code L93 causes the headlamp to be omitted and also the coding in the body controller to be adapted to suit the LED loads.

However, the wiring harness is still identical to that of a conventional headlamp.

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
High beams	LED	2 × 55 ¹⁾	Operated via control unit BCMFA2	Yes
Low beams	LED	2 × 55 ¹⁾	Operated via control unit BCMFA2	No
Daytime running lights	LED	2 × 21 ²⁾	Operated via control unit BCMFA2	Yes
Standing lights	LED	2 × 5 ²⁾	Operated via control unit BCMFA2	Yes
Turn signal indicators	LED	2 × 21 ²⁾	Operated via control unit BCMFA2	No

1) The current dropping below a minimum value of 200 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.

2) The current dropping below a minimum value of 100 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.

10.1.3 Conventional tail lamps

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
Turn signal indicators	PY21 W	2 × 21	Standard in tail lamp, closed/open model des.	No
Tail lights	P21 W	2 × 21	Standard in tail lamp, closed model des.	No
	R5 W	4 × 5	Standard in tail lamp, open model des.	No
Reversing light	P21 W	2 × 21	Standard in tail lamp, closed/open model des.	Yes
Reversing light (code L21) ²⁾	LED	2 × 1.7	Standard for defined vehicles with code L21 ²⁾	Yes
Rear fog lamp	P21 W	1 × 21	Standard in tail lamp, closed/open model des.	Yes
Brake light	P21 W	2 × 21	Standard in tail lamp, closed/open model des.	No
Rear license plate lamp	–	–	Standard for open model designation: Illuminated via windows in the bottom of the left tail lamp Closed model designation: See table of additional lighting functions	Code JW2 not available
Side marker lamp	R5 W	2 × 5	Standard in tail lamp, open model des. Actuation only with use of code LB1 ¹⁾ or LB2 ¹⁾	Code LB1: yes Code LB2: no

1) Side marker lamps (code LB1) or pre-installation for side marker lamps (code LB2)

2) For more information on the reversing light with code L21, See 8.5.3 Tail lamps (→ Page 287)

10.1.4 Conventional tail lamps on open model designations with code L90

Code L90 causes the tail lamp to be omitted and also the coding in the body controller to be adapted to suit the installation of conventional tail lamps (bulbs) on open model designations.

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
Turn signal indicators	PY21 W	2 × 21 ²⁾		No
Tail lights	R5 W	2 × 5 ²⁾		No
Reversing light	P21 W	2 × 21 ²⁾		Yes
Rear fog lamp	P21 W	1 × 21 ²⁾		Yes
Brake light	P21 W	2 × 21 ²⁾		No
Rear license plate lamp ¹⁾	–	–		Code JW2 not available
Side marker lamp	R5 W	2 × 5 ²⁾		Yes

- 1) On open model designations, it is recommended to supply additionally installed license plate lighting via the ducts of the side marker lamps in the tail lamp. The power of the license plate lamps must not exceed 5 W per channel. Note that actuation only occurs with the use of the side marker lamp (code LB1) or pre-installation for side marker lamp (code LB2).
- 2) The current dropping below a minimum value of 60 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.

10.1.5 LED tail lamps on open model designations with code L91

Code L91 causes the tail lamp to be omitted and also the coding in the body controller to be adapted to suit the installation of LED tail lamps on FHS/FHL models.

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
Turn signal indicators	LED	2 × 6 ²⁾		No
Tail lights	LED	2 × 9 ²⁾		No
Reversing light	LED	2 × 3.5 ²⁾		Yes
Rear fog lamp	LED	1 × 3.5 ²⁾		Yes
Brake light	LED	2 × 6 ²⁾		No
Rear license plate lamp ¹⁾	LED	–		Yes
Side marker lamp	LED	2 × 1 ²⁾	Standard in tail lamp, open model des. Actuation only with use of code LB1 ³⁾ or LB2 ³⁾	Code LB1: yes Code LB2: no

- 1) On open model designations, it is recommended to supply additionally installed license plate lighting via the ducts of the side marker lamps in the tail lamp. The power of the license plate lamps must not exceed 5 W per channel. Note that actuation only occurs with the use of the side marker lamp (code LB1) or pre-installation for side marker lamp (code LB2).
- 2) The current dropping below a minimum value of 60 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.
- 3) Side marker lamps (code LB1) or pre-installation for side marker lamps (code LB2).

10.1.6 Additional lighting functions

Function	Type	Power [W]	Remarks	Deactivation of lamp failure indicator Code JW2
Third brake light	LED	1 × 1.3	Standard for closed model designation, pre-installation for third brake lamp, code LV6, available for open model designations ¹⁾	No
Clearance lamps	LED	2 × 1	Open model designation, special equipment code L07	Yes
Side marker lamp	LED	6 × 1	Closed/open model designation, code LB1 Pre-installation for side marker lamps, code LB2	Yes
Rotating beacon	H1	2 × 55	Closed model designation, code LB7, rotating beacon, yellow, front left/rear right Open model designation, code LB8, rotating beacon, yellow, front left Closed/open model designation, code LV1, pre-installation for rotating beacon	Code JW2 not available ⁷⁾
Fog light with cornering light function	H11	2 × 55	Closed/open model designation, integrated in bumper, code L13 Pre-installation for fog light, code LV4 ²⁾ LED fog light with code L93 ³⁾	Yes
	LED	2 × 55		Yes
Fog light without cornering light function	H11	2 × 55	Closed/open model designation, integrated in bumper, code L16 Pre-installation is not available LED fog light in combination with code L93 ³⁾	Yes
	LED	2 × 55 ³⁾		Yes
Additional turn signal lights	LED	2 × 5	Open model designation, special equipment, additional turn signal lamps, code L77 ⁴⁾	Code JW2 not available ⁷⁾
	W16W	2 × 16	Closed/open model designation, standard equipment in outside mirror ⁵⁾	No
	LED	2 × 3.8	Closed model designation, special equipment, additional turn signal lamps on roof, code LB3 ⁶⁾	Code JW2 not available ⁷⁾
	P21 W	2 × 21	Closed/open model designation, turn signal indicators on fender with code L44 (without turn signal indicator in outside mirror) ⁵⁾	No
Rear license plate lamp	W5W	2 × 5	Standard in license plate holder, closed model	Yes

- 1) Note that the fault monitoring for the pre-installation is always active, so the installation and use of a 3rd brake light are required.
- 2) The current dropping below a minimum value of 500 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.
- 3) The current dropping below a minimum value of 100 mA results in fault entries and fault messages in the instrument cluster if code JW2 is not used.
- 4) Additional turn signal lights with code L77 can only be ordered on open model designations. They are actuated via the additional turn signal module. There is no failure monitoring of the additional turn signal indicators.

- 5) A turn signal indicator can be actuated either in the outside mirror or on the fender. The same connection on the door control unit is used. Therefore, it is not possible to combine turn signal indicators in the outside mirror and on the fender.
- 6) Additional turn signal lights with code LB3 can only be ordered on closed model designations. These are fitted at the rear on the roof. They are actuated in parallel with the turn signal indicators in the tail lamp. Due to the failure monitoring of the turn signal indicators in the tail lamp, the specified power rating of the additional turn signal lights must not be exceeded. There is no failure monitoring of the additional turn signal lights.
- 7) Fault monitoring does not occur.

10.2 Trailer coupling hole patterns

On open and closed model designations from model series 907, the reinforcement of the trailer coupling in the left/right rear longitudinal member (code Q11) is installed in the vehicle as standard.

If you do not wish to have this reinforcement, it can be deselected for closed model designations with code QW1 when ordering the vehicle. Code QW1 is not available for open model designations. If these reinforcements were not installed at the factory, it is not permissible to fit a trailer coupling.

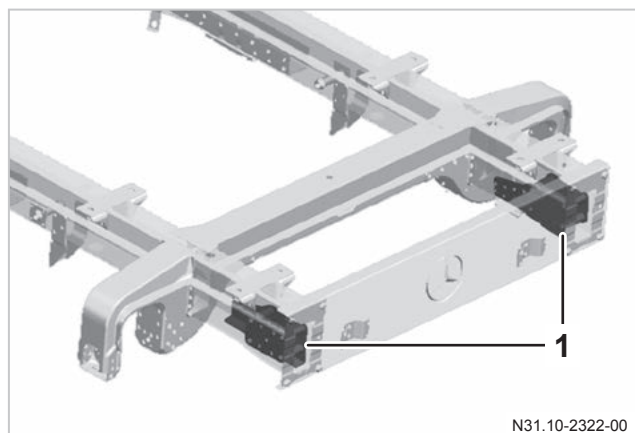
⚠ WARNING

Risk of accident from the unapproved installation of a trailer hitch!

If the retrofitting of a trailer hitch is not permitted and you install a trailer hitch or other component parts, the longitudinal frame member will be weakened and could break. If this happens, the trailer can detach from the vehicle.

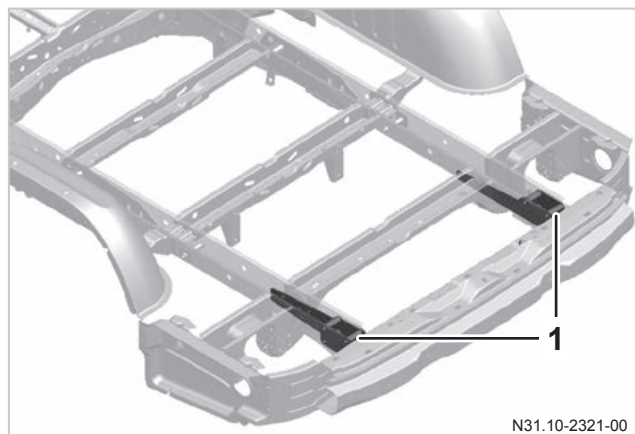
There is risk of an accident and danger to life and limb!

Only retrofit a trailer hitch if this is permitted.



Longitudinal member, model series 907 open model designation with code Q11

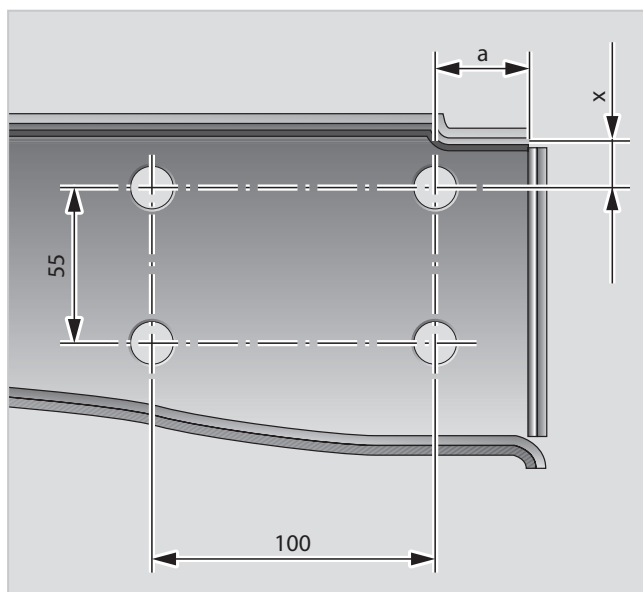
1 Trailer coupling insert code Q11



Longitudinal member, model series 907 closed model designation with code Q11

1 Trailer coupling insert code Q11

10.2.1 Installation dimensions, version 1

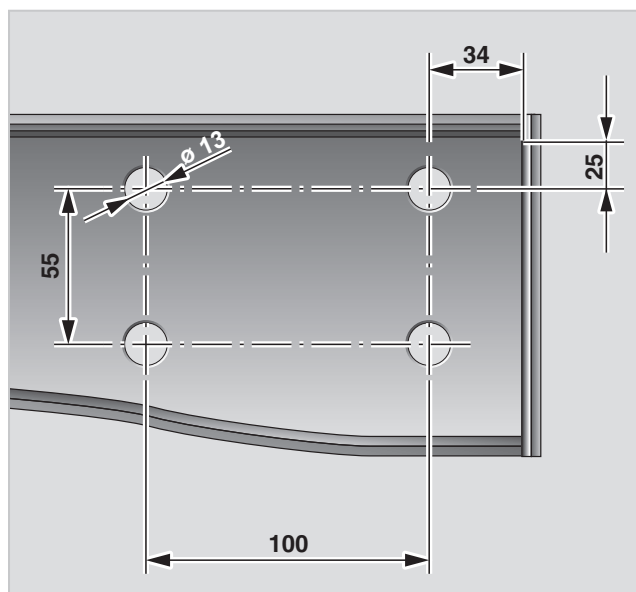


Installation dimensions for closed model designation (FKA/FKB) 3.0–5.0 t

Vehicle model	Wheelbase [mm]	a [mm]	x [mm]	Overhang ¹⁾ [mm]
FKA/FKB 3.0–5.0 t	3665 (A2)	35	26	1246
	4325 (A3)	35	26	1621
	4325 (A3)	35	26	2021

1) Overhang = vehicle overhang dimension

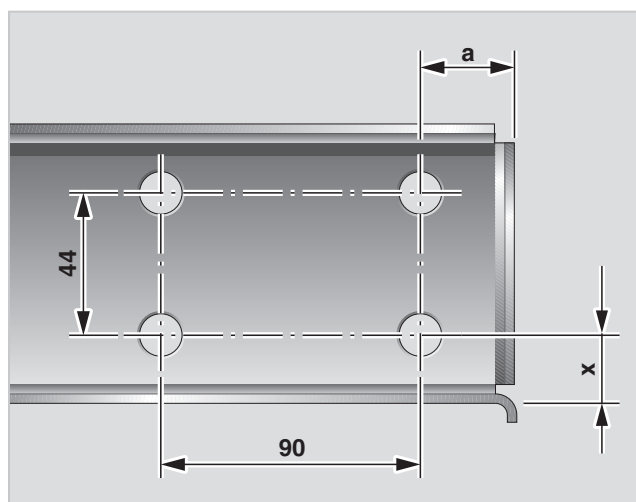
10.2.2 Installation dimensions, version 2



Installation dimensions for open model designation (FHS/FHL); basic variants 3.5 t (2.8–4.1 t)

Overhang = vehicle overhang dimension incl. tail lamp

10.2.3 Installation dimensions, version 3



Installation dimensions for open model designation (FHS/FHL) 5.0 t

Vehicle model	Wheelbase [mm]	a [mm]	x [mm]	Overhang ¹⁾ [mm]
FHS/FHL 5.0 t	3665 (A2)	27	34	1418
	4325 (A3)	27	34	1518

1) Overhang = vehicle overhang dimension incl. tail lamp

A

Accident prevention	43
Active Brake Assist.....	314
Active Distance Assist DISTRONIC	314
AdBlue®	139
Additional assemblies	97, 166
Adhesive films	75
ADVANTAGES	29
ADVANTAGES product information system	29
Advice for bodybuilder manufacturers.....	18
Airbag control unit.....	153
Airbags.....	96, 156
Air conditioning system	166
Air deflectors	177
Air intake.....	122, 147
Air suspension	110
Alternator.....	176, 276
Antenna connection	299
Antennas.....	383
Anticorrosion protection.....	70, 102
Applying for Letters of Compatibility	20
Area to be painted	346
Assembly frame.....	200
Assembly frame as floor assembly.....	208
Assembly frame profile dimensions	203
Assistance systems.....	307
Attachment above cab	177
Attachments	177
Attachment to the frame.....	204
Automatic driving lights	232, 340
Automatic start/stop	380
Auxiliary battery	269
Auxiliary brakes	176
Auxiliary heating.....	167
Auxiliary lamps	287
Axle loads.....	82

B

Battery	267
Blind Spot Assist.....	318
Bodies on chassis with base.....	227
Bodies on chassis with lowered roof.....	257
Body	112
Bodybuilder manufacturers outside the sales network	25
Bodybuilder Portal.....	24
Body-in-white	85, 112
Body limit values	98
Box bodies	214
Brake hoses	100
Brake system	84, 108
Bulb ratings of exterior lighting	398
Buses.....	263

C

Cab	122
Cab base vehicles.....	227
Cable extension	273
Cables and lines	100
Cab roof	123
Camper vans.....	84, 223
CAN bus	97, 271
Cargo liftgate	190
CB radio	299
Center of gravity calculation.....	391
Central locking system.....	301
CERON.....	27
Commissioning of control units.....	387
Concept Letter of Compatibility.....	20
Connectivity.....	386
Continuous engine operation feature.....	379
Conventions	12
Copyright	17
Cowl	128
Crosswind Assist	309
Current collector	277
Cyber Security	37

D		F	
Damage prevention	99	Fender	127
Damper	84, 108	Fifth-wheel coupling	221, 397
Deactivating transport mode	73	Frame end crossmember	127
Deep discharge protection.....	285	Front axle	106
Design data	25	Front axle load	78
Design of bodies	200	Fuel.....	136
Diagnosis	26	Fuel filler neck	112
Diagnostics connection.....	15	Fuel system.....	136
Digital inside mirror.....	364	Functional safety	307
Dimensions	59	Fuses	272
Disconnecting the on-board electrical system voltage.....	271	G	
Doors.....	301, 305	Ground bolt.....	284
Doors and flaps.....	125	H	
Drilling work on the body-in-white.....	115	Handover.....	73
Drivetrain	136	Headlamps	286
Drivetrain limit values	91	High beam and Lane Keeping Assist	339
Driving assistance systems	307	High Beam Assist	339
Dumper bodies	216	I	
E		Identification data	61
EC Certificate of Conformity (CoC).....	28	Identification plate	61
Electrical circuit diagrams	381	Individual Letter of Compatibility	20
Electric lines	272	Infotainment	382
Electrics and electronics.....	265	Intelligent Speed Limit Assist	339
Electromagnetic compatibility (EMC)	266	Interface overview.....	281
Electronic ignition starter switch (EZS).....	300	Interfaces.....	271
Electronic Stability Program (ESP®).....	308	Interior	209
Elevating work platform	223	Interior lamps	296
Engine air intake.....	147	Internal noises	162
Engine compartment flap.....	228	J	
Engine cooling	92, 147	Jaw.....	195
Engine peripherals.....	91, 136		
Engine speed regulation.....	150		
European type approval (ETA)	28		
Exhaust system	141		
Exit warning	318		

L		N	
Lane Keeping Assist	339	Noise level	162
Letter of Compatibility	19		
Liftgate	190, 379	O	
Lifting roof	131	On-board electrical system battery	267
Lighting	286	Operational safety	14
Limit values for additional assemblies	97	Outside mirror	306
Limit values for attachments	97	Overhang	88
Limit values for electrics and electronics	97	Overhang extension	119
Limit values for engine peripherals and drivetrain	91	P	
Limit values for the basic vehicle	78	Painting work	104
Limit values for the body	98	Panel van	212
Limit values for the body-in-white	85	Panel van and crewbus roof	128
Limit values for the interior	96	Parameterizable special module PSM	372
Limit values for the suspension	80	Parking packages	344
Limit values for tire sizes	83	Permissible axle loads	82
Load compartment sliding door	305	Permissible position of the vehicle center of gravity	78
Load compartment trim parts	164	Permissible vehicle dimensions	79
Loading crane	188	Platform bodies	214
Load rails at the side wall ex factory	179	Post-delivery integration of doors	301
Lowered roof	123, 257	Power take-offs	169
Lower wheel well on rear axle	85	Pre-installation for shelf system	186
		Preservation work	104
M		Product liability	37
Maintenance	71	Product safety	37
Major assemblies	148	Programmable logic controller (PLC)	376
Marker lamps	290	Propeller shafts	148
Material quality	201	PSM functions	375
Mini PLC	376	PSM interfaces	376
Mobile communication systems	97, 299	Pyrotechnical components	158
Model designation	47, 49		
Model overview	49	Q	
Modifications to seats	96	QR code rescue sticker	62
Modifications to the basic vehicle	106	Qualified specialist workshop	16
Modifications to the cab	122	Quality system	45
Modifications to the interior	209		
Modifications to the panel van	212		
Modifications to the vehicle	16		
Moving-off information Assist	344		

R		
Radio	299	
Rain and light sensor	340	
Rain sensor	232, 340	
Rear Cross Traffic Alert	318	
Rear seats	210	
Rearview camera	364	
Recovery vehicles	222	
Recycling	43	
Refrigerated vehicles	215	
Repair	71	
Reprocessing of component parts	43	
Rescue card	62	
Retarder	176, 377	
Retrofitting an air conditioning system	166	
Retrofitting load rails at the roof	184	
Retrofitting load rails at the side wall	180	
Retrofitting of electrical devices	275	
Reversing camera	347	
Roof hatch	131	
Roof load	90	
Roof luggage racks	178	
S		
Safety equipment	152	
Safety notes	14	
SCR exhaust gas aftertreatment system	139	
Seat belts	154	
Seat covers	162	
Seats	160, 209	
Selecting the basic vehicle	53	
Self-supporting bodies	208	
Semi-integrated bodies	259	
Semitrailer trucks	218	
Shelves in the load compartment	178	
Shielding gas plug welding	69	
Sideguard Assist	318	
Side marker lamps	97	
Side wall	123	
Side window heater	306	
Signal acquisition and actuation module (SAM)	377	
Sliding door to load compartment	305	
Software Update Management System (SUMS)	37, 411	
Soundproofing	70	
Spare wheel	111	
Speakers	385	
Special equipment	74	
Specialist workshop	16	
Speed regulation	150	
Speed signal	283	
Stabilizers	108	
Steerability	78	
Steering geometry	84	
Storage and handover	105	
Storage of batteries	72	
Storing the vehicle	72	
Suspension	84, 106, 108	
Suspension limit values	80	
T		
Tachograph	390	
Tack welding	70	
Tail lamps	287, 291	
Technical Compliance Awareness Package	46	
Technical Consultancy	18	
Technical limit values for planning	78	
Telematics	381	
Telephone	299	
Thorax/pelvis side airbags	156	
Threaded connections	66	
Tie-down eyes	178	
Tire load capacity	60	
Tire pressure loss warning system	342	
Tires	65, 111	
Tire sizes	83	
Torsionally stiff body types	222	
Traceability	39	
Trademarks	39	

Traffic Sign Assist	339
Trailer coupling	117, 193, 196
Trailer coupling hole patterns	403
Transmitter key	300
Transport mode	73
Travel distance signal	283

U

Ultrasonic sensors	346
Underfloor paneling	134
Underride guard	198

V

Vehicle center of gravity	78
Vehicle front	93
Vehicle front position lamps	97
Vehicle identification data	61
Vehicle markings with warning signs	77
Vehicle modifications	57
Vehicle safety	13
Vehicle stability	64
Ventilation	163

W

Weight specifications	59
Welding work	101
Welding work on the body-in-white	114
Weld joints	68
Wheel alignment	84
Wheelbase modifications	89
Wheels	111
Wheel wells	127
Window airbag	158
Window lifters and window hinges	305
Windows	124, 305
Windshield wipers	306
Windshield wiper system	228
Wiring harness connecting points	231
WLTP	29
Work before the handover	73
Workshop Information System (WIS)	26

X

XENTRY	26
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List of abbreviations

Abbreviation	Explanation
ABA	Active Brake Assist
ABH	Bodybuilder manufacturer
ABS	Anti-Lock Braking System
ADR	European Agreement Concerning the International Carriage of Dangerous Goods by Road
AeJ	Modification year
AGM	Absorbent Glass Mat, design of lead-acid batteries
API	Application Programming Interface
ARL	Body/Equipment Mounting Directive(s)
ASR	Anti-slip control
AWP	Awareness Package for Bodybuilder Manufacturers
BC	Body Controller
BM	Model designation
BR	Model series
BSM	Blind Spot Monitoring/Blind Spot Assist
BTS	Battery disconnect switch
CAN	Controller Area Network
CERON	Certification Online, database system for filing and exchange between state authorities (e.g. Federal Office for Motor Vehicles) and vehicle manufacturers
CoC	Certificate of Conformity
DBE	Overhead control panel (OCP)
DIN	Deutsches Institut für Normung/"Deutsche Industrie-Norm" (German Standards Institute/"German Industry Standard")
DSP	Digital Signal Processor
ATA	Electronic anti-theft alarm system
EIS	Electronic ignition switch (ignition starter switch), see also EZS (German abbreviation)
EMC	Electromagnetic Compatibility
EN	European Standard
ERG	Single-invoice transaction
ESP®	Electronic Stability Program
ETA	European Type Approval
EZS	Electronic ignition switch (ignition starter switch), see also EIS (English abbreviation)
FMS	Fleet Management Standard
FMVSS	Federal Motor Vehicle Safety Standards
gBM	Closed model designation
GFK	Glass-fiber-reinforced plastic
GGVSEB	German regulation concerning the transport of dangerous goods by road, rail and inland waterways
GSR	General Safety Regulation
ISO	International Organization for Standardization
CSC	Customer Support Center
KTL	Cathodic Electrodeposition
KW	Customer request
LDS	Rotary light switch
LSS	Combination switch

Abbreviation	Explanation
MBN	Mercedes-Benz Standard (company standard)
MFL	Manual driving light
MOIS	Moving-off Information System/Moving-off information Assist
MPC	Multi Purpose Camera/Multifunction camera
NA	Power take-off (PTO)
OBD	On-Board Diagnostics
OBF	Upper control panel (UCP)
PSM	Parameterizable Special Module
PWM	Pulse Width Modulation
QR Code	Quick Response Code, brand term for two-dimensional binary code
RCTA	Rear Cross Traffic Alert
RoW	Rest of World
RSC	Roll Stability Control
RSP	Belt tensioner
RWD	Rear Wheel Drive
SAM	Signal acquisition and actuation module
SCN	Software Calibration Number
SML	Side marker lamps
SPS	Programmable Logic Controller (PLC)
STL	Standing lights
StVZO	German Regulations Authorizing the Use of Vehicles for Road Traffic
SUMS	Software Update Management System
SWA	Crosswind Assist
tCMS	Technical Compliance Management System (management system for ensuring compliance in collaboration with bodybuilder manufacturers)
TPMS	Tire Pressure Monitoring System
TSG	Door control unit
UVV	Accident Prevention Regulation
LOC	Letter of Compatibility
VDA	German Association of the Automotive Industry
VIN	Vehicle Identification Number
WIS	Workshop Information System
WLTP	Worldwide Harmonized Light Vehicles Test Procedure for determining level of exhaust gas emissions
ZFZR	Central Vehicle Register
Max. GVW	Max. gross vehicle weight/Permissible gross mass
ZRG	Dual-invoice transaction
ZV	Central locking system

